The Igusa Square Root

The Borcherds determinant of $\phi_{0,1}$, its denominator algebra, and the $K3 \times E$ realization problem

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April 2026

Abstract

For $X = K3 \times E$ the physical input used here is the protected index. Right-moving localization in the K3 Ramond sector gives

$$Z_{\mathrm{K3}} = 2\phi_{0,1}, \qquad \phi_{0,1} = \sum f(n,l) q^n r^l.$$

Replacing the protected K3 index by its Borcherds half produces only a virtual K_0-determinant package. It does not produce a microscopic Hilbert space. Unitarity is used to localize the right-moving K3 supertrace before the index is taken; the index itself is a charge-graded virtual object that no longer requires Hilbert positivity. For virtual super vector spaces $\mathbb{U}_{n,l}$ with $\operatorname{sdim} \mathbb{U}_{n,l} = f(n,l)$, set $\mathcal{V}_{(n,l,m)} = \mathbb{U}_{nm,l}$. In the Borcherds cusp chamber the determinant is

$$\mathcal{D}_X(Z) = 64 q^{1/2} r^{1/2} s^{1/2} \prod_{(n,l,m)\in\Gamma_{\mathrm{eff}}} (1 - q^n r^l s^m)^{f(nm,l)} = \Delta_5(Z).$$

The constant 64 is the theta-constant normalization $[q^{1/2} r^{1/2} s^{1/2}]\Delta_5 = 64$; hence the monic Borcherds product is $D_5 = 64^{-1}\Delta_5$.

The same coefficients are the signed root supermultiplicities in the Gritsenko–Nikulin Borcherds–Kac–Moody denominator algebra. Under

$$\alpha(n,l,m) = 2nf_2 - lf_3 + 2mf_{-2}$$

one has

$$\operatorname{den}(\mathfrak{g}_{\Delta_5}) = 64^{-1}\Delta_5(2Z), \qquad \operatorname{smult} \mathfrak{g}_{\Delta_5,\alpha(n,l,m)} = f(nm,l)$$

on the active support. Thus the denominator algebra supplies the algebraic square root target and its target-internal Borcherds bracket. It does not supply a compact BPS Hilbert space, compact Hall correspondences, an orientation, or a BPS operator product/local collision map. Equivalently, one may package finite windows of this imported target as an algebraic Dirac–Igusa target reference tower. Such a tower has PBW bases, the standard target invariant form, the target radical quotient, formal first-order blocks, and a formal Pfaffian product by construction from $\mathfrak{g}_{\Delta_5}$. It is not a compact $K3 \times E$ Hall source.

The square root in the title is meant in Dirac's sense. The primitive OP/DT scalar branch $Z^X_{\mathrm{OP/DT}} = -4096\,\Delta_5^{-2}$ is orientation-forgetful: the sign character is squared away. The missing compact object is a first-order protected operator/algebra $\mathfrak{D}_X$ whose protected Pfaffian, after a supplied Pfaffian-to-automorphic line comparison, is Δ_5, and whose scalar trace $\operatorname{Tr}_{\mathrm{prot}}((-1)^F \mathfrak{D}_X^{-2}) = -4096\,\Delta_5^{-2}$; the orientation/Pfaffian datum is the algebraic input that distinguishes Δ_5 from $-\Delta_5$

and that the squared scalar cannot recover. A compact realization must construct $\mathfrak{D}_X$, the orientation system, Weyl transport, and the type-II Pfaffian wall atlas, and must prove

$$\epsilon_o = \nu_{\Delta_5}$$

on $W^{(2)}(\Lambda_{II}^{2,1})$.

The scalar Behrend-weighted branch is imported from Oberdieck–Pixton, Oberdieck–Pandharipande, and Oberdieck. In OP's normalization $\chi_{10}^{\mathrm{OP}} = D_5^2$, so

$$\chi_{10}^{\mathrm{OP}} = (64^{-1}\Delta_5)^2 = 4096^{-1}\Delta_5^2, \qquad Z_{\mathrm{DT}}^X = Z_{\mathrm{OP}}^X = -\left(\chi_{10}^{\mathrm{OP}}\right)^{-1} = -4096\,\Delta_5^{-2}.$$

The factor $4096 = 64^2$ is a normalization conversion; the minus sign is the OP scalar convention. Neither fixes the automorphic reflection character or the compact orientation monodromy.

A compact realization would have to begin with an additive formal dyonic charge lattice Γ_X^{form}, its algebraic/effective Hall support, the Gram map $\Pi_X : \Gamma_X^{\mathrm{form}} \to \Gamma_{\mathrm{gram}}$, and a holomorphic E_3-factorization algebra on $K3 \times E$. The formal lattice is $\Lambda_S \oplus \Lambda_S$. The algebraic even Mukai sector relevant to the OP/DT D6–D2–D0 branch, and its stability-dependent effective semigroup, are smaller inputs; the present construction does not assert that every formal full-Mukai class is a compact Hall state. The triple $(n, l, m) \in \Gamma_{\mathrm{gram}}$ records Gram-degree invariants $(Q^2/2, Q \cdot P, P^2/2)$ of a formal pair $c = (Q, P)$; it is not the additive Hall grading itself. The Gram map is quadratic:

$$\Pi_X(c + c') = \Pi_X(c) + \Pi_X(c') + B(c, c').$$

Thus strict fixed-lift raw Gram descent cannot carry the type-II real-root brackets. The first structural correction is the normal-ordered extension

$$\widehat{\Gamma}_X = \Gamma_X^{\mathrm{form}} \oplus_B \Gamma_{\mathrm{gram}}, \qquad \overline{\Pi}_X(c, T) = \Pi_X(c) - T,$$

for which $\overline{\Pi}_X$ is additive. A compact realization would then require a specified formality point, model-dependent homotopy/framing datum, quantum master equation, anomaly control, descent, and finite-first HN completion. It would then specialize along the K3 direction to a chiral object on $\mathrm{Ran}(E)$, solve the $\mathrm{Ran}(E)$-support-locality obstruction in the K3-integrated chiral shadow by mixed local/wrapped correspondences, and identify the Gram-descended protected primitive bracket with $\mathfrak{g}_{\Delta_5}$. Equivalently, the compact realization problem is finite-stage: construct geometric Hall–Dirac source data and a compatible functor into the algebraic target reference tower, preserving degrees, Hall products and coproducts, orientation-gerbe twists, source coalgebras, first-order blocks, Pfaffian lines, primitive representatives, radical/PBW data, and transitions. This paper computes the virtual determinant and identifies it with Δ_5 by imported Borcherds theory; proves the formal lattice normal-ordering theorem and the strict fixed-lift raw-descent obstruction; records the imported scalar and denominator theorems; constructs only the target-side reference tower; and leaves the compact Hall/factorization realization as the remaining geometric problem. The final part records the eight Cléry–Gritsenko diagonal-divisor rows and the data required before any boundary row becomes a physical host.

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1 Introduction

The local object comes first. A compact BPS theory on $X = K3 \times E$, if it exists in the form needed here, should produce protected local observables before any trace is taken. After twisting and passing to the protected index, the positive Hilbert space has collapsed to a charge-graded virtual object, but locality has not disappeared: disjoint insertions must still factorize and collisions must remember extensions. Throughout the compact-realization problem, S is a smooth complex K3 surface and E is a smooth elliptic curve; singular K3 limits, nodal or cuspidal elliptic limits, imprimitive curve-class contributions, the $s = 0$ compact Hall cusp, multi-component charge closures, and boundary closures of Hall correspondences are deferred boundary problems, not standing hypotheses.

The calculation begins with the protected K3 index, passes through the Borcherds product and the Gritsenko–Nikulin denominator algebra, and then asks for a compact $K3 \times E$ Hall object whose primitive bracket descends to that denominator algebra. The scalar OP/DT equality is only the square of such a first-order Pfaffian object. It is not an input to the compact Hall source, the Pfaffian line, the orientation character, or primitive recognition.

1.0.0.1 Protected locality. Right-moving localization of the unitary K3 sigma model gives

$$Z_{\mathrm{K3}} = 2\phi_{0,1}, \qquad \phi_{0,1} = \sum f(n,l) q^n r^l.$$

The half-index used by the Borcherds lift is represented only in $K_0(\mathrm{sVect}_{\mathrm{fd}})$. Its coefficients are signed superdimensions, not dimensions of a constructed compact D-brane Hilbert space.

1.0.0.2 Compact and hybrid factorization. A compact realization would first supply the formal dyonic lattice, the algebraic/effective Hall support inside it, the non-additive Gram map $\Pi_X : \Gamma_X^{\mathrm{form}} \to \Gamma_{\mathrm{gram}}$, and a charge-completed holomorphic E_3-factorization algebra

$$\mathcal{A}_X^{E_3}, \qquad X = K3 \times E,$$

then a chain-level $K3$-to-E specialization

$$\mathcal{F}_X^{(0)} \simeq \mathrm{Sp}^{\mathrm{ch}}_{K3,E}(\mathcal{A}_X^{E_3}) \in \mathrm{Fact}(\mathrm{Ran}(E)).$$

This projection-finite chiral shadow is not reconstructed from the scalar partition function or from the Borcherds product. The full object is the hybrid $\mathcal{F}_X^{\mathrm{hyb}}$ only after the finite local/wrapped correspondence data exist. There is also a geometric obstruction: a one-dimensional subscheme with positive elliptic degree projects onto all of E. Hence a naive support-local object on $\mathrm{Ran}(E)$ sees only projection-finite sectors. The full Igusa s-direction requires finite-stage local/wrapped correspondence data and a separated degree convention: the geometric elliptic-degree map b_R^{geom}, the Gram trace exponent $m_R = \mathrm{pr}_3 \overline{\Pi}_X$, rigidified wrapped prequotients with anchors, local-local, ordered local/wrapped, and wrapped-wrapped extension stacks, the eight-word two-step binary flag atlas, quotient-after-correspondence descent, and protected integration with trace degree s^{m_R}. A detached global Hecke/Fock factor is only a determinant-level repair. It becomes relevant to the compact realization only after it is represented by those wrapped prequotients and acts through mixed Hall correspondences before the reduced E-quotient. This is open.

1.0.0.3 Primitive recognition. Until Definition 6.1, the notation Γ_X^{phys} denotes the same formal full-Mukai lattice later written Γ_X^{form}; the superscript is mnemonic and not the four-dimensional $\mathcal{N} = 4$ electric–magnetic charge lattice. The microscopic algebra sought from such a compact realization is first the upstairs protected primitive bracket

$$[\ ,\ ] : \mathrm{Prim}_c^{\mathrm{prot}}(\mathcal{F}_X) \otimes \mathrm{Prim}_{c'}^{\mathrm{prot}}(\mathcal{F}_X) \longrightarrow \mathrm{Prim}_{c+c'}^{\mathrm{prot}}(\mathcal{F}_X), \qquad c, c' \in \Gamma_X^{\mathrm{phys}}.$$

The desired comparison is a theorem only after the following geometric objects have been constructed: an oriented Hall object, protected integration, cocycle-corrected Gram descent $\overline{\Pi}^{\Theta}_{X,*}$, a conilpotent source coalgebra C_X with finite source filtration and collision coproduct, the bar comparison b_X, the cobar map Θ_{Kos}, Koszul Mittag–Leffler exactness for the completed bar/cobar inverse system, presentation-level

primitive Lie comparison, no-extra-relations theorem, completed PBW compatibility, exact triangular completion/$\lim^1$-vanishing, and radical quotient. With these data one obtains

$$H^0 \operatorname{Prim}_{\mathrm{prot}}(\overline{\Pi}^{\Theta}_{X,*}\mathcal{F}_X)/\overline{\operatorname{Rad}\langle\,,\,\rangle_X} \cong \mathfrak{g}_{\Delta_5}$$

and the chiral cobar comparison identifies $\Omega^{\mathrm{ch}}_E C_X$ with the formal current envelope $\mathsf{Den}_{\Delta_5,E}$. The determinant alone does not construct any of these objects.

1.0.0.4 Borcherds denominator. The target algebra is imported from the Borcherds–Gritsenko–Nikulin construction. With

$$\alpha(n,l,m) = 2nf_2 - lf_3 + 2mf_{-2},$$

the visible active coefficients satisfy

$$\operatorname{smult} \mathfrak{g}_{\Delta_5,\alpha(n,l,m)} = f(nm,l), \qquad \operatorname{den}(\mathfrak{g}_{\Delta_5}) = 64^{-1}\Delta_5(2Z).$$

This determines signed root supermultiplicities and the denominator identity. It does not determine compact Hall constants.

1.0.0.5 OP scalar square. The primitive reduced scalar branch is imported from Oberdieck–Pixton, Oberdieck–Pandharipande, and Oberdieck:

$$Z^X_{\mathrm{DT}}(T,Q,P) = Z^X_{\mathrm{OP}}(P,Q,T) = -4096\,\Delta_5(Z)^{-2}.$$

The constants are fixed once. The theta-constant form satisfies

$$[q^{1/2}r^{1/2}s^{1/2}]\Delta_5 = 64, \qquad D_5 = 64^{-1}\Delta_5.$$

OP's monic product is $\chi^{\mathrm{OP}}_{10} = D_5^2$, hence

$$\chi^{\mathrm{OP}}_{10} = 4096^{-1}\Delta_5^2, \qquad Z^X_{\mathrm{OP}} = -(\chi^{\mathrm{OP}}_{10})^{-1} = -4096\Delta_5^{-2}.$$

The number 64 is a theta-leading coefficient. The number -4096 is the OP scalar normalization. Neither proves the compact orientation character. The comparison

$$\epsilon_o = \nu_{\Delta_5}$$

between geometric orientation monodromy and the automorphic reflection character remains a geometric condition on the compact realization.

1.0.0.6 Eight-row boundary rows. The diagonal-divisor boundary problem concerns the eight Cléry–Gritsenko forms

$$\Delta_5, \Delta_2, \Delta_1, \Delta_{1/2}, \nabla_3, \nabla_2, \nabla_{3/2}, Q_1.$$

Rows 1–4 have Gritsenko–Nikulin denominator theorems. Rows 5–8 have Cléry–Gritsenko products and Govindarajan CHL/WKB denominator theorems; $F_7 = \nabla_{3/2}$ is on the required μ_4-multiplier cover. A physical host for any boundary row requires data fixing the modular group or cover, character, divisor, cusp Jacobi inputs, product lattice, Weyl vector, real roots, reflection character, physical charge lattice, protected Gram-index, orientation, reduced scalar theorem if one is claimed, and primitive algebra

comparison. Without those data the row is only an automorphic product plus any separately proved denominator theorem.

All super vector spaces in the representation-theoretic constructions are over $\mathbb{C}$. Real vector spaces occur only after tensoring lattices with $\mathbb{R}$ to discuss cones, chambers, and period domains. Products may be written over Γ_{eff}; the active factors are exactly the triples in Γ_{act}, where the exponent $f(nm, l)$ is non-zero.

Part I

The automorphic square root

2 The physical question

A charged BPS state is not, in the first instance, a local field. It lies in a superselection sector. Its charge is measured at infinity, and its existence depends on a stability chamber. Thus the ordinary local operator algebra of an untwisted four-dimensional theory is not the primary object of this paper.

At the level of the protected index the situation is different. After taking the cohomology of the chosen supercharge, long multiplets cancel. The object which remains is a charge-graded virtual object, not a positive Hilbert space, whose Euler characteristic is stable under deformation. The index remembers the cancellation law of a unitary theory, not its positive inner product.

For $X = K3 \times E$ the expected local object in a compact realization is a supplied holomorphic E_3-factorization algebra

$$\mathcal{A}_X^{E_3}, \qquad X = K3 \times E.$$

The auxiliary notation $\Phi_3^{\mathrm{FA}}(X)$, when used, is only a placeholder for this source datum; no such functor is constructed in this paper. No external Φ_3^{FA} formalism is used here as an existence theorem unless it supplies the chain-level local observables, factorization maps, descent, anomaly/QME control, and compact charge completion required below. In a compact realization $\mathcal{A}_X^{E_3}$ is the object whose disjoint holomorphic insertions factorize. The elliptic curve E enters only after a supplied chain-level $K3$-specialization whose output is denoted

$$\mathcal{F}_X \in \mathrm{Fact}(\mathrm{Ran}(E)).$$

This is the first compact-realization obstruction. The projection-finite sector, where $p_E(\mathrm{Supp}\, A)$ is a finite subset of E, is naturally local on $\mathrm{Ran}(E)$. A one-dimensional sector of positive elliptic degree is not. If $C \subset K3 \times E$ has positive elliptic degree, then $(p_E)_*[C]$ is a positive multiple of $[E]$; hence $p_E(C)$ is a one-dimensional closed subset of the irreducible curve E, and so $p_E(C) = E$. Such a class is wrapped or global over E, not localized at finitely many points of E.

This is the $\mathrm{Ran}(E)$-support-locality obstruction in the K3-integrated chiral shadow. Naive support-local factorization on $\mathrm{Ran}(E)$ sees projection-finite sectors, but by itself it misses the positive s-direction of the Igusa product, the direction which records elliptic degree after the OP/Borcherds variable conversion. A compact realization must therefore choose at least one finite categorical repair: projection-finite local factorization, rigidified wrapped prequotients, local-local, ordered local/wrapped, and wrapped-wrapped extension correspondences, the eight-word two-step binary flag atlas with higher factorization separate, quotient-after-correspondence descent, and protected integration whose s-degree is the trace exponent $m_R = \mathrm{pr}_3\, \overline{\Pi}_X$; the geometric elliptic degree records wrapped support rather than the Borcherds exponent. A separate global Fock/Hecke mode is insufficient unless it is realized by these wrapped prequotients and coupled to the local sector by mixed Hall actions before the reduced E-quotient. This is a realization obstruction. It is not a defect in the Borcherds determinant or in the Gritsenko–Nikulin denominator identity.

Thus, in a realization, the curve-level rule is not imposed externally. It is the one-dimensional chiral shadow of the holomorphic E_3-observable: well-separated protected insertions on E multiply, while collisions remember extension and bound-state data. In primitive form this is the expected bracket

$$[\ ,\ ] : \mathrm{Prim}_c^{\mathrm{prot}}(\mathcal{F}_X) \otimes \mathrm{Prim}_{c'}^{\mathrm{prot}}(\mathcal{F}_X) \longrightarrow \mathrm{Prim}_{c+c'}^{\mathrm{prot}}(\mathcal{F}_X), \qquad c, c' \in \Gamma_X^{\mathrm{phys}}.$$

The Igusa square-root problem is to construct this primitive protected algebra and its Gram descent. The determinant below gives the decategorified signed Gram-index shadow that such a realization would have, not a source-derived multiplicity:

$$\operatorname{sdim} \operatorname{Prim}^{\mathrm{prot}}_{\gamma}(\overline{\Pi}^{\Theta}_{X,*}\mathcal{F}_X) = f(nm, l), \qquad \gamma = (n, l, m).$$

The Borcherds–Kac–Moody denominator algebra gives the algebraic candidate:

$$H^{\circ} \operatorname{Prim}_{\mathrm{prot}}(\overline{\Pi}^{\Theta}_{X,*}\mathcal{F}_X) \overset{?}{\cong} \mathfrak{g}_{\Delta_5}.$$

The determinant and denominator comparison are established below. The displayed isomorphism is the compact E_3/Hall/chiral realization problem.

The square root is therefore not an analytic ambiguity in Δ_5^2. It is the normalized automorphic determinant of the virtual protected one-particle index. The scalar square forgets the automorphic reflection character; a compact Hall orientation realizing that character is part of the open Dirac–Igusa datum.

Schema of the local protected observable algebra

Definition 2.1 *(Local protected observable algebra schema).* For an open $U \subset X = K3 \times E$, let $\Gamma^{\mathrm{eff}}_{X,\sigma}(U) \subset \Gamma^{\mathrm{alg}}_X \subset \Gamma^{\mathrm{form}}_X$ be the supplied algebraic/effective charge support. For $c \in \Gamma^{\mathrm{eff}}_{X,\sigma}(U)$, let $\mathfrak{M}_c(U)$ denote the supplied derived moduli stack of compactly supported B-branes on U of Mukai charge c. The PTVV theorem supplies the ambient (-1)-shifted symplectic structure on the CY3 perfect-complex moduli problem [69]; the compactly supported, reduced, equivariant Hall object used here is additional protected-sector input. Let Φ_c denote the BBDJS perverse sheaf $P^{\bullet}_{\mathfrak{M}_c,s_c}$ on an oriented d-critical truncation, when such a d-critical structure and square-root orientation have been supplied [11, Theorem 6.9]. Let o_c denote the orientation line, a square root of the determinant line of the cotangent complex of $\mathfrak{M}_c(U)$, supplied as part of the Hall datum. The *local protected observable algebra of X on U* is the source object

$$\mathcal{A}_X(U) := \bigoplus_{c \in \Gamma^{\mathrm{eff}}_{X,\sigma}(U)} H^{\mathrm{BM}}_*\big(\mathfrak{M}_c(U),\, \Phi_c \otimes o_c\big),$$

graded by that supplied support, not by the full formal lattice. For disjoint $U_1, \dots, U_k \subset X$, the proposed factorization map is

$$\mathcal{A}_X(U_1) \otimes \cdots \otimes \mathcal{A}_X(U_k) \longrightarrow \mathcal{A}_X(U_1 \sqcup \cdots \sqcup U_k),$$

defined chain-level by Massey–Saito Thom–Sebastiani for vanishing cycles

$$\Phi_{F_1 \oplus \cdots \oplus F_k} \simeq \Phi_{F_1} \boxtimes \cdots \boxtimes \Phi_{F_k}$$

on disjoint d-critical charts [58, 71], together with the Ext-vanishing $\operatorname{Ext}^*(\mathcal{F}, \mathcal{G}) = 0$ for branes $\mathcal{F}, \mathcal{G}$ with disjoint supports. This is the local protected-observable pre-interface schema for the present paper. It becomes a compact E_3 source $\mathcal{A}^{E_3}_X$ only after the finite witnesses of Definition 10.8 are supplied; the disjoint-supports factorization on opens of $X = K3 \times E$ is only the first holomorphic E_3-factorization test.

Remark 2.2. The components are external assumptions: PTVV (−1)-shifted symplectic structure on derived moduli of B-branes [69]; BBDJS vanishing-cycle perverse sheaf and its Thom–Sebastiani isomorphism (extending Massey 2001 and Saito 1990 to the d-critical setting [58, 71]) [11, Theorem 6.9]; orientation lines and Joyce–Upmeier-compatible multiplicative orientation data supplied separately [46, Theorem 4.1] or Kontsevich–Soibelman [51]. The factorization map relies on disjoint-supports Ext-vanishing for compactly supported branes (absent for non-compact supports, where one recovers chiral algebra structure, not commutative factorization). Compactness on $K3 \times E$ is open because the relevant moduli of branes are typically non-quasicompact in Λ_S-charge; the sectorial truncation of Proposition 10.53 is the candidate truncation.

3 The one-particle index and its determinant

For the determinant calculation set

$$Z = \begin{pmatrix} z_1 & z_2 \\ z_2 & z_3 \end{pmatrix}, \qquad q = e^{2\pi i z_1}, \qquad r = e^{2\pi i z_2}, \qquad s = e^{2\pi i z_3}.$$

The Gram-index lattice is

$$\Gamma_{\text{ind}} = \mathbb{Z}^3, \qquad \gamma = (n, l, m), \qquad x^\gamma = q^n r^l s^m.$$

The standard cusp polarization is the ordered limit

$$0 < |s| \ll |q| \ll |r|^{-1} \ll 1.$$

It determines the semigroup

$$\Gamma_{\text{eff}} = \{(n, l, m) \mid m > 0,\ n \geq 0,\ l \in \mathbb{Z}\} \cup \{(n, l, 0) \mid n > 0,\ l \in \mathbb{Z}\} \cup \{(0, l, 0) \mid l < 0\}.$$

$m > 0$ is positive because s is the smallest parameter; if $m = 0$, then $n > 0$ is positive because q is next; if $m = n = 0$, positivity is $l < 0$ because $|r|^{-1} \ll 1$. This is a chamber convention for the completed product, not a microscopic charge positivity statement.

Definition 3.1 *(Ramond-sector K3 index and its half).* Let $\mathcal{H}_{\text{RR}}(S)$ be the Ramond–Ramond Hilbert space of a unitary $\mathcal{N} = (4, 4)$ sigma model on a K3 surface S. Its central charges are

$$c = \overline{c} = 6.$$

This Hilbert space is a supplied worldsheet input for the protected K3 index. It is not a construction of the compact $K3 \times E$ D-brane Hilbert space. Choose a right-moving Ramond supercharge Q_R with

$$Q_R^2 = 0, \qquad \{Q_R, Q_R^\dagger\} = 2\left(\overline{L}_0 - \frac{\overline{c}}{24}\right).$$

The K3 elliptic genus is the protected supertrace

$$Z_{\text{K3}}(\tau, z) = \operatorname{Tr}_{\mathcal{H}_{\text{RR}}(S)} (-1)^{F_L + F_R} q^{L_0 - c/24} \overline{q}^{\overline{L}_0 - \overline{c}/24} r^{J_0}.$$

The BPS space in degree (n, l) is

$$\mathcal{H}^{\text{BPS}}_{n,l}(S) := H^\bullet(\{\psi \mid (L_0 - c/24)\psi = n\psi,\ J_0\psi = l\psi\}, Q_R).$$

Lemma 3.2 *(Right-moving localization).* Then

$$Z_{\mathrm{K3}}(\tau, z) = \sum_{n\geq 0,\, l\in\mathbb{Z}} \operatorname{sdim} \mathcal{H}^{\mathrm{BPS}}_{n,l}(S) q^n r^l.$$

Proof. If $\overline{L}_0 - \overline{c}/24 > 0$, the displayed anticommutator pairs states with opposite $(-1)^{F_R}$ parity. Their supertrace cancels. The unpaired states are the Q_R-harmonic representatives, hence the graded superdimension of Q_R-cohomology gives the elliptic genus. □

Remark 3.3 *(Where unitarity enters).* Unitarity is used here before the index is taken: it gives the adjoint relation which pairs positive right-moving energy states and makes the supertrace localize. After passing to the protected index, the object is virtual. A coefficient $f(n, l)$ is a superdimension, not a positive dimension. Negative coefficients record an odd excess in the protected supertrace, or in the chosen K_0-representative. They do not determine the two parity dimensions without an additional Hilbert-space or denominator-algebra parity datum.

Definition 3.4 *(Grothendieck half-index representative).* Eichler–Zagier normalization [27]:

$$Z_{\mathrm{K3}}(\tau, z) = 2\phi_{0,1}(\tau, z),$$
$$\phi_{0,1}(\tau, z) = \sum_{n\geq 0,\, l\in\mathbb{Z}} f(n, l) q^n r^l.$$

Thus

$$\operatorname{sdim} \mathcal{H}^{\mathrm{BPS}}_{n,l}(S) = 2f(n, l).$$

The arbitrary Grothendieck representative

$$\mathbb{U}_{n,l} \in K_0(\mathrm{sVect}_{\mathrm{fd}}), \qquad \operatorname{sdim} \mathbb{U}_{n,l} = f(n, l),$$

is the normalized half-index class used by the Borcherds lift. It is not asserted to be a canonical half of $\mathcal{H}^{\mathrm{BPS}}_{n,l}(S)$ as a Hilbert space. As a K_0-model one may take $\mathbb{U}_{n,l}$ to be $\mathbb{C}^{f(n,l)}_{\bar{0}}$ when $f(n, l) \geq 0$, and $\mathbb{C}^{-f(n,l)}_{\bar{1}}$ when $f(n, l) < 0$. No canonical choice of basis, parity lift, or operator product is implied.

$$\phi_{0,1} = r^{-1} + 10 + r + O(q), \qquad \frac{f(0, 0)}{2} = 5.$$

Lemma 3.5 *(Fock trace and superdeterminant).* Let

$$V = \bigoplus_{\lambda\in P} V_\lambda$$

be a P-graded super vector space with finite-dimensional homogeneous pieces and let $a_\lambda = \operatorname{sdim} V_\lambda$. Put

$$\mathcal{F}(V) = \mathrm{Sym}(V_{\bar{0}}) \otimes \bigwedge(V_{\bar{1}}).$$

As a formal character,

$$\operatorname{sch} \mathcal{F}(V) := \sum_{\mu\in P} \operatorname{sdim} \mathcal{F}(V)_\mu\, x^\mu = \prod_{\lambda\in P} (1 - x^\lambda)^{-a_\lambda}.$$

The inverse character is the superdeterminant

$$\operatorname{sdet}_V(1 - x) = \prod_{\lambda\in P} (1 - x^\lambda)^{a_\lambda}.$$

Proof.

$$V_\lambda = \mathbb{C}_{\overline{0}}: \qquad \operatorname{sch}\operatorname{Sym}(V_\lambda) = (1 - x^\lambda)^{-1}.$$
$$V_\lambda = \mathbb{C}_{\overline{1}}: \qquad \operatorname{sch}\bigwedge(V_\lambda) = 1 - x^\lambda.$$

Tensor product over a homogeneous basis proves the product. □

When $P = \Gamma_{\mathrm{eff}}$, all products below are taken in the cusp-completed semigroup algebra determined by $0 < |s| \ll |q| \ll |r|^{-1} \ll 1$. Every coefficient receives only finitely many contributions in this completion.

$$x^{(n,l,m)} = q^n r^l s^m.$$

Form now the virtual Borcherds determinant package. The third index m is the second-quantized/Borcherds degree. The positive m sector carries the Hecke–DMVV exponents $f(nm, l)$. The $m = 0$ sector is a cusp/Weyl boundary package carrying the repeated boundary indices $f(0, l)$. Neither sector is a microscopic one-particle Hilbert space. The compact Hall $s = 0$ cusp is therefore a separate boundary datum: it is controlled by the Borcherds chamber data only after the corresponding source-side boundary operations are supplied.

$$\mathcal{V}^{\mathrm{bulk}} := \bigoplus_{\substack{(n,l,m)\in\Gamma_{\mathrm{eff}}\\ m>0}} \mathbb{U}_{nm,l}, \qquad \mathcal{V}^{\mathrm{cusp}} := \bigoplus_{(n,l,0)\in\Gamma_{\mathrm{eff}}} \mathbb{U}_{0,l}.$$

$$\mathcal{V} = \mathcal{V}^{\mathrm{bulk}} \oplus \mathcal{V}^{\mathrm{cusp}}, \qquad \mathcal{V}_{(n,l,m)} = \mathcal{V}_{n,l,m} := \mathbb{U}_{nm,l}, \qquad \operatorname{sdim}\mathcal{V}_{n,l,m} = f(nm, l).$$

This is a definition in K_0, not a microscopic state space. The $m > 0$ sector is the K_0-level virtual half of the DMVV product [25], obtained by replacing $Z_{\mathrm{K3}} = 2\phi_{0,1}$ by $\phi_{0,1}$. The normalized Fock character in the Borcherds chamber is

$$\operatorname{sch}\mathcal{F}(\mathcal{V}) = \prod_{\gamma\in\Gamma_{\mathrm{eff}}} (1 - x^\gamma)^{-\operatorname{sdim}\mathcal{V}_\gamma} = \prod_{(n,l,m)\in\Gamma_{\mathrm{eff}}} (1 - q^n r^l s^m)^{-f(nm,l)}.$$

$$\operatorname{sdet}_{\mathcal{V}}(1 - x) = \prod_{\gamma\in\Gamma_{\mathrm{eff}}} (1 - x^\gamma)^{\operatorname{sdim}\mathcal{V}_\gamma} = \prod_{(n,l,m)\in\Gamma_{\mathrm{eff}}} (1 - q^n r^l s^m)^{f(nm,l)}.$$

The sector $m > 0$ is the normalized half of the second-quantized DMVV index. The terms with $m = 0$ are cusp and Weyl-chamber boundary factors of the Borcherds product; the determinant reads only their signed indices $f(0, l)$. The normalized cusp-form section below proves the leading coefficient which fixes the Weyl monomial

$$v_0(Z) = 64 q^{1/2} r^{1/2} s^{1/2}.$$
$$v_0 \notin \bigoplus_{\gamma\in\Gamma_{\mathrm{eff}}} \mathcal{V}_\gamma.$$

Definition 3.6 (Normalized K_0-determinant).

$$\mathcal{D}_X(Z) = v_0(Z)\operatorname{sdet}_{\mathcal{V}}(1 - x) = 64 q^{1/2} r^{1/2} s^{1/2} \prod_{(n,l,m)\in\Gamma_{\mathrm{eff}}} (1 - q^n r^l s^m)^{f(nm,l)}.$$

$$\mathcal{Z}^{\mathrm{vir}}_{\mathrm{Borch}}(Z) = \mathcal{D}_X(Z)^{-1} = \frac{1}{64} q^{-1/2} r^{-1/2} s^{-1/2} \prod_{(n,l,m)\in\Gamma_{\mathrm{eff}}} (1 - q^n r^l s^m)^{-f(nm,l)}.$$

The object constructed here is the virtual K_0-determinant; compact BPS meaning is part of the separate realization problem.

Proposition 3.7 *(Determinant normalization).*

$$[q^{1/2}r^{1/2}s^{1/2}]\,\mathcal{D}_X = 64,$$

$$\mathcal{Z}^{\mathrm{vir}}_{\mathrm{Borch}} = \frac{1}{64}q^{-1/2}r^{-1/2}s^{-1/2}\,\mathrm{sch}\,\mathcal{F}(\mathcal{V}),$$

$$\mathcal{D}_X = 64q^{1/2}r^{1/2}s^{1/2}\,\mathrm{sdet}_{\mathcal{V}}(1-x).$$

Proof.

$$\mathrm{sch}\,\mathcal{F}(\mathcal{V}) = \prod_{(n,l,m)\in\Gamma_{\mathrm{eff}}}(1-q^n r^l s^m)^{-f(nm,l)}, \qquad \mathrm{sdet}_{\mathcal{V}}(1-x) = \prod_{(n,l,m)\in\Gamma_{\mathrm{eff}}}(1-q^n r^l s^m)^{f(nm,l)}.$$

$$v_0(Z)\,\mathrm{sdet}_{\mathcal{V}}(1-x) = \mathcal{D}_X, \qquad v_0(Z)^{-1}\,\mathrm{sch}\,\mathcal{F}(\mathcal{V}) = \mathcal{Z}^{\mathrm{vir}}_{\mathrm{Borch}}.$$

□

Proposition 3.8 *(Virtuality of the square root).* The displayed construction produces a class

$$[\mathcal{V}] \in K_0(\Gamma_{\mathrm{eff}}\text{-graded finite-dimensional super vector spaces})$$

together with its Fock determinant. By itself this determinant does not construct a Hilbert space, a vertex algebra, a holomorphic E_3-factorization algebra, or a chiral factorization algebra whose character is $\mathcal{Z}^{\mathrm{vir}}_{\mathrm{Borch}}$.

All denominator statements use only the equalities

$$\mathrm{sdim}\,\mathcal{V}_{(n,l,m)} = f(nm,l), \qquad \mathcal{D}_X = v_0\,\mathrm{sdet}_{\mathcal{V}}(1-x).$$

The denominator algebra and formal current envelope are constructed later from the Borcherds root datum, not from a canonical half of the K3 Hilbert space.

Proof. By definition, $\mathcal{V}_{(n,l,m)} = \mathbb{U}_{nm,l}$ is specified only through its class in $K_0(\mathrm{sVect}_{\mathrm{fd}})$ and through the integer $\mathrm{sdim}\,\mathbb{U}_{nm,l} = f(nm,l)$. The Fock trace and superdeterminant lemma depends only on these signed dimensions. No operation in the construction supplies composition maps, operator products, locality, or a Hilbert completion. □

Proposition 3.9 *(Dirac/Pfaffian recognition target).* The class $[\mathcal{V}] \in K_0$ and its Fock determinant $\mathcal{D}_X = v_0\,\mathrm{sdet}_{\mathcal{V}}(1-x) = \Delta_5$ record signed Gram-degree multiplicities and a formal infinite-product; they do not produce an operator with a spectrum. The protected-locality side of the constellation requires the following construction:

> construct from the compact $K3 \times E$ Dirac–Igusa datum of Definition 10.65 a first-order protected operator/algebra $\mathfrak{D}_X$ acting on a chiral protected module, whose charge-graded Pfaffian, after the supplied Pfaffian-to-automorphic line comparison, satisfies
>
> $$\iota_{\mathrm{aut}}(\mathrm{Pf}_{\mathrm{prot}}(\mathfrak{D}_X)) = \mathcal{D}_X = \Delta_5.$$
>
> Its scalar trace, in the OP normalization of Definition 11.3 below, satisfies
>
> $$\mathrm{Tr}_{\mathrm{prot}}\big((-1)^F(\mathfrak{D}_X)^{-2}\big) = -4096\,\Delta_5^{-2}.$$

Existence of $\mathfrak{D}_X$ is open; it would convert the K_0 identity of Proposition 3.8 into a first-order protected statement. The target carries the sign-square ambiguity of Conjecture 5.3 and the orientation monodromy of Open Problem 5.7; choosing a square root of $(\mathfrak{D}_X)^{-2}$ requires the orientation datum of that open problem and is not fixed by the OP scalar.

Definition 3.10 *(Finite generic type-II local-sign datum).* A finite generic local-sign datum for the type-II Pfaffian model consists, for each of the three simple type-II walls $\mathbb{H}_{\delta_i}$, $i = 1, 2, 3$, of the following data.

(Pf1) A generic wall point x_i in the reduced compact Hall sector and an s_{δ_i}-equivariant real analytic slice (U_i, u_i) through x_i, with

$$\mathbb{H}_{\delta_i} \cap U_i = \{u_i = 0\}, \qquad s_{\delta_i}(u_i) = -u_i.$$

(Pf2) A splitting of the reduced determinant complex on the slice into its tangent determinant factor and a normal factor

$$K^{\mathrm{red}}|_{U_i} \simeq K_i^{\mathrm{tan}} \oplus K_i^{\mathrm{nor}},$$

with s_{δ_i} preserving the tangent determinant factor.

(Pf3) An oriented quasi-isomorphism of the normal factor to a finite sum of odd rank-one Pfaffian crossings

$$K_i^{\mathrm{nor}} \simeq \bigoplus_{j=1}^{N_i^{\mathrm{Pf}}} \left[\mathbb{R} \xrightarrow{u_i} \mathbb{R}\right].$$

In the induced trivialization the normal Pfaffian section is $u_i^{N_i^{\mathrm{Pf}}}$ times a unit.

(Pf4) The three normal-mode ranks $N_i^{\mathrm{Pf}} = N_{\delta_i}^{\mathrm{Pf}}$ are

$$N_1^{\mathrm{Pf}} = N_2^{\mathrm{Pf}} = N_3^{\mathrm{Pf}} = 1.$$

This is a finite generic local-sign datum: three wall charts, three anti-invariance checks $s_{\delta_i}(u_i) = -u_i$, three tangent-factor compatibility checks, and three odd Pfaffian normal-mode ranks. These ranks are local Dirac/Pfaffian data; they are not read from the automorphic divisor. These local data construct neither the global reduced orientation line nor the Weyl lift, and are not the full O2 wall atlas with component, boundary, and overlap compatibility. It uses no scalar normalization of Δ_5, no factor 64, and no OP/DT scalar sign.

Definition 3.11 *(Rank-one type-II Pfaffian crossing).* Fix a simple type-II root δ. Let T_δ be a real analytic tangent chart with trivial s_δ-action, and let

$$U_\delta = T_\delta \times (-\varepsilon, \varepsilon), \qquad s_\delta(t, u) = (t, -u).$$

The wall is $T_\delta \times \{0\}$. Let $\mathscr{L}_\delta^{\mathrm{tan}}$ be an oriented tangent Pfaffian line on T_δ, pulled back to U_δ, with s_δ-invariant unit v_δ. The normal rank-one odd Pfaffian crossing is the skew family

$$D_u = \begin{pmatrix} 0 & u \\ -u & 0 \end{pmatrix}$$

on $\mathbb{R}^2$, equivalently the normal determinant factor

$$K_\delta^{\mathrm{cross}} = \left[\mathbb{R} \xrightarrow{u} \mathbb{R}\right]$$

placed in odd Pfaffian parity. Its Pfaffian line is trivialized by the standard orientation and its Pfaffian section is u. The total local Pfaffian section is

$$\mathrm{pf}_\delta = v_\delta u.$$

Thus

$$s_\delta^* \mathrm{pf}_\delta = -\mathrm{pf}_\delta, \qquad s_\delta^*(\mathrm{pf}_\delta^2) = \mathrm{pf}_\delta^2 .$$

Proposition 3.12 *(The rank-one crossing gives the type-II local sign).* For $i = 1, 2, 3$, choose a tangent chart T_{δ_i}, an invariant tangent Pfaffian unit v_{δ_i}, and the rank-one crossing of Definition 3.11. Then the three charts form the finite generic type-II local-sign datum of Definition 3.10 with

$$N_{\delta_1}^{\mathrm{Pf}} = N_{\delta_2}^{\mathrm{Pf}} = N_{\delta_3}^{\mathrm{Pf}} = 1.$$

At finite HN height R, suppose the successor maps preserve the tangent charts, the wall coordinate $u_{\delta_i,R}$, and the invariant units $v_{\delta_i,R}$. Then the O2 residual entries

$$\mathfrak{o}_{\delta_i,R}^{\mathrm{chart}}, \quad \mathfrak{o}_{\delta_i,R}^{\mathrm{wall}}, \quad \mathfrak{o}_{\delta_i,R}^{\mathrm{split}}, \quad \mathfrak{o}_{\delta_i,R}^{\mathrm{unit}}, \quad \mathfrak{o}_{\delta_i,R}^{\mathrm{rank}}, \quad \mathfrak{o}_{\delta_i,R'R}^{\mathrm{tr}}$$

vanish for these local models, and

$$\varepsilon_{\delta_i,R}^{\mathrm{loc}} = -1.$$

Proof. The fixed locus of $s_\delta(t, u) = (t, -u)$ is $u = 0$, so the wall and the tangent fixed locus agree. The determinant factor is already split: the tangent line is $\mathscr{L}_\delta^{\mathrm{tan}}$, and the normal factor is $K_\delta^{\mathrm{cross}}$. The unit v_δ is invariant by construction. The skew matrix $\left(\begin{smallmatrix} 0 & u \\ -u & 0 \end{smallmatrix}\right)$ has Pfaffian u, so the normal mode rank is one. Reflection sends u to $-u$ and fixes v_δ; hence the local Pfaffian changes sign and its square is fixed. The stated transition hypothesis says exactly that the same coordinate, splitting, unit, and rank-one crossing restrict from height R' to height R. □

Proposition 3.13 *(Local obstruction datum for* (O2)*).* Fix one of the three simple type-II roots $\delta \in \{\delta_1, \delta_2, \delta_3\}$. A geometric proof of the rank-one Pfaffian model at $\mathbb{H}_\delta$ is not a consequence of $(\delta, \delta) = 2$, of the automorphic divisor order, or of the Maass character. In the reduced compact Hall framework it must supply the following local deformation-obstruction calculation.

(L1) Let $x_\delta = [C_\delta]$ be a generic wall object. An s_δ-equivariant real analytic d-critical/Kuranishi chart

$$(U_\delta, x_\delta, u_\delta, K_\delta^{\mathrm{red}}) \longrightarrow \mathfrak{M}_{C_\delta}^{\mathrm{red}}$$

is supplied, where K_δ^{red} is the real form of the cosection-reduced self-Ext complex

$$\mathrm{RHom}_{\mathrm{red}}(C_\delta, C_\delta)$$

in that chart, the wall is exactly $U_\delta \cap \{u_\delta = 0\}$, and

$$s_\delta(u_\delta) = -u_\delta, \qquad T_{x_\delta}(U_\delta \cap \mathbb{H}_\delta) = T_{x_\delta} U_\delta^{s_\delta}.$$

(L2) A splitting in the s_δ-equivariant derived category

$$K_\delta^{\mathrm{red}} \simeq K_\delta^{\mathrm{tan}} \oplus K_\delta^{\mathrm{nor}}$$

whose tangent factor carries the wall tangent determinant and whose normal factor is the anti-invariant normal determinant factor. Equivalently, the extension class

$$\eta_\partial \in \mathrm{Ext}^1_{D^{s_\partial}(U_\partial)}(K^{\mathrm{nor}}_\partial, K^{\mathrm{tan}}_\partial)$$

vanishes after the reduced E-quotient and the chosen quotient orientation trivializations are imposed.

(L3) In an oriented trivialization of the normal Pfaffian line,

$$\mathrm{Pf}(K^{\mathrm{nor}}_\partial) = v_\partial\, u_\partial^{N^{\mathrm{Pf}}_\partial}, \qquad v_\partial(x_\partial) \neq 0, \qquad s_\partial(v_\partial) = v_\partial,$$

and $K^{\mathrm{nor}}_\partial$ is oriented quasi-isomorphic to

$$\bigoplus_{j=1}^{N^{\mathrm{Pf}}_\partial} \left[\mathbb{R} \xrightarrow{u_\partial} \mathbb{R}\right].$$

The rank-one assertion used in (O2) is the further equality

$$N^{\mathrm{Pf}}_\partial = 1.$$

If (L1)–(L3) hold, then the local Hall/Pfaffian sign is

$$s_\partial^* \mathrm{Pf}(K^{\mathrm{nor}}_\partial) = (-1)^{N^{\mathrm{Pf}}_\partial} \mathrm{Pf}(K^{\mathrm{nor}}_\partial),$$

while the squared determinant section is fixed. If the unit v_∂ has a nontrivial reflection character, that character multiplies the displayed sign and belongs to the orientation-transport datum $(\mathrm{O1})^+$. If the splitting obstruction η_∂ does not vanish, the tangent and normal determinants cannot be separated and no rank-one normal sign has been computed. If $N^{\mathrm{Pf}}_\partial \neq 1$, the order-one automorphic divisor does not supply the local Hall model required by (O2). Thus the exact geometric task is to identify the reduced normal self-Ext factor with the rank-one crossing of Definition 3.11: construct the three charts, the three reduced self-Ext complexes, the three equivariant splittings, the three invariant Pfaffian units, and the three rank computations $N^{\mathrm{Pf}}_{\partial_i} = 1$. In any compact source construction these local data must also come from the same finite hybrid correspondence system that supplies the positive elliptic-degree sector; this is the compatibility obstruction of Proposition 10.32.

Theorem 3.14 *(Finite type-II local-sign computation).* Fix a finite HN height R and a simple type-II wall $\mathbb{H}_{\partial_i}$, $i \in \{1, 2, 3\}$. The computation applies after the residual tuple below has been trivialized. A finite geometric computation of the generic local sign required by (O2) at this wall is the vanishing of the residual

$$\mathfrak{O}^{\mathrm{O2}}_{\partial_i,R} = \left(\mathfrak{o}^{\mathrm{chart}}_{\partial_i,R}, \mathfrak{o}^{\mathrm{wall}}_{\partial_i,R}, \mathfrak{o}^{\mathrm{split}}_{\partial_i,R}, \mathfrak{o}^{\mathrm{unit}}_{\partial_i,R}, \mathfrak{o}^{\mathrm{rank}}_{\partial_i,R}, \mathfrak{o}^{\mathrm{tr}}_{\partial_i,R'R}\right)$$

for every $R' \geq R$. These entries mean, respectively: existence of a wall object and s_{∂_i}-equivariant real d-critical chart; equality of the fixed tangent with the wall tangent and anti-invariance $s_{\partial_i}(u_{\partial_i,R}) = -u_{\partial_i,R}$; vanishing of the tangent/normal extension class; invariance of the normal Pfaffian unit after quotient orientation trivialization; the rank-one normal quasi-isomorphism

$$K^{\mathrm{nor}}_{\partial_i,R} \simeq \left[\mathbb{R} \xrightarrow{u_{\partial_i,R}} \mathbb{R}\right];$$

this is the rank-one crossing of Definition 3.11; and compatibility of these choices under the HN transition $R' \to R$.

If $\mathfrak{O}^{\mathrm{O2}}_{\partial_i,R} = 0$, then in the reduced chart

$$\mathrm{Pf}(K^{\mathrm{nor}}_{\partial_i,R}) = v_{\partial_i,R} u_{\partial_i,R}, \qquad s_{\partial_i}(v_{\partial_i,R}) = v_{\partial_i,R},$$

and therefore

$$s^*_{\partial_i}\,\mathrm{Pf}(K^{\mathrm{nor}}_{\partial_i,R}) = -\,\mathrm{Pf}(K^{\mathrm{nor}}_{\partial_i,R}), \qquad s^*_{\partial_i}\,\mathrm{Pf}(K^{\mathrm{nor}}_{\partial_i,R})^2 = \mathrm{Pf}(K^{\mathrm{nor}}_{\partial_i,R})^2.$$

Vanishing of these residuals for $i = 1, 2, 3$ gives the three finite generic type-II local-sign data at height R, and the computed local wall-chart signs are

$$\varepsilon^{\mathrm{loc}}_{\partial_i,R} = -1, \qquad i = 1, 2, 3.$$

The theorem computes only the sign from the local chart after the residuals vanish. It does not construct the wall charts, the reduced self-Ext complexes, the splittings, the quotient orientation trivializations, or the transition null-homotopies, and no global character is defined by this theorem. The full O2 wall-atlas datum also requires the component, boundary, and overlap data of Proposition 10.32. If, in addition, (O1) descends the reduced orientation and $(\mathrm{O1})^+$ supplies an honest HN-compatible Weyl action on the quotient orientation, these local signs induce the global values $\epsilon_o(s_{\partial_i}) = -1$. The automorphic divisor order $d^{\mathrm{aut}}_{\partial_i} = 1$ is the independent divisor computation recorded in Lemma 3.18.

Proof. The vanishing of $\mathfrak{o}^{\mathrm{chart}}_{\partial_i,R}$ and $\mathfrak{o}^{\mathrm{wall}}_{\partial_i,R}$ gives the local coordinate $u_{\partial_i,R}$ and the reflection rule $s_{\partial_i}(u_{\partial_i,R}) = -u_{\partial_i,R}$. The splitting residual separates the tangent determinant from the anti-invariant normal determinant. The unit residual removes an extra reflection character from the normal Pfaffian unit. The rank residual identifies the normal factor with the odd rank-one crossing of Definition 3.11. Hence the normal Pfaffian is $v_{\partial_i,R} u_{\partial_i,R}$, and reflection multiplies it by -1. Squaring removes this sign. The transition residual says that the same computation survives restriction from R' to R. □

Theorem 3.15 *(Dirac–Pfaffian normalization and orientation character).* Suppose the following data items are granted on the reduced compact critical Hall/CoHA sector of $X = K3 \times E$ of Definition 5.4.

(D0) *Compact first-order datum.* A finite-stage first-order D_0-datum in the sense of Definition 10.54 is supplied. Thus, for every R in the chosen cofinal HN subsystem, the datum contains a finite Hall/factorization stage $D_{0,R}$ on the finite support $\widehat{\Gamma}_R$, a finite target test window Γ^{test}_R, successor transition maps whose composites give the smaller-stage maps, source/observable comparison maps, and a vanishing stage residual tuple $\mathfrak{O}_{D_0,R}$. Only after these finite data, transition cocycles, and support/parity residuals vanish may one form the HN-completed state object. The completed object is part of the supplied datum: it carries its charge completion by $\widehat{\Gamma}_X$, a source observable shadow on $\mathrm{Ran}^{\mathrm{hyb}}(E)$, a protected first-order operator/algebra $\mathfrak{D}_X$, a Pfaffian line with normalized section, and a Θ_Π-corrected descent whose primitive algebra is Γ_{gram}-graded with signed superdimensions

$$\mathrm{sdim}\, P^{\Pi}_{X,(n,l,m)} = f(nm, l) \qquad ((n, l, m) \in \Gamma_{\mathrm{eff}}).$$

This is an assumption of the theorem. The proved lattice part is the normal-ordered homomorphism $\overline{\Pi}_X$; the imported target part is the Borcherds–Gritsenko–Nikulin denominator algebra; the compact state-side construction remains open unless the finite stages, transition maps, source/observable comparison, and residual trivializations of Definition 10.54 are supplied.

(O1) *Strong multiplicative reduced orientation datum.* There exists a Joyce–Upmeier-type orientation line [46]

$$o_C \in \mathrm{Pic}(\mathfrak{M}_X^{or}), \qquad o_C^{\otimes 2} \simeq \det \mathrm{RHom}_{\mathrm{red}}(C, C),$$

in the sense of Definition 4.3, satisfying

$$o_{C\oplus C'} \simeq o_C \otimes o_{C'} \quad \text{compatibly with the reduced extension correspondences,}$$

and equipped with the quotient-descent package for the reduced E-quotient. The Weyl-equivariant lift is a separate additional datum, isolated in $(\mathrm{O1})^+$, and is transported by Gram-projection $\Pi_X : \Gamma_X^{\mathrm{form}} \to \Gamma_{\mathrm{gram}}$. Here descent under the reduced E-quotient is part of the datum and means the simultaneous vanishing, on every stratum used by the Hall correspondences, of

$$\alpha_C^{\mathrm{red}} \in H^2(\mathfrak{M}_C^{\mathrm{red}}; \mathbb{F}_2), \qquad \alpha_C^{E,\mathrm{free}} = a_1(C)u_1 + a_2(C)u_2 \in H^2(BE; \mathbb{F}_2),$$

and of the finite-stabilizer classes

$$\beta_C^{E,N} \in H^2(BE[N]; \mathbb{F}_2).$$

The displayed finite-stabilizer groups are the point-stabilizer restrictions of the actual Borel equivariant obstruction classes. For a finite-stabilizer stratum S with stabilizer $E[N]$, the primary class is

$$\widetilde{\beta}_{R,S}^{E,N} \in H^2_{E[N]}(S; \mathbb{F}_2).$$

A point-stabilizer calculation closes (O1) only after the mixed Borel terms, the stabilizer action on $H^\bullet(S; \mathbb{F}_2)$, and the spectral-sequence differentials on the corresponding stratum are shown to vanish. These are degree-two gerbe obstruction classes. Their vanishing is not the same datum as a trivial finite-stabilizer action on an already linearized line. After the degree-two class has been null-trivialized, the residual choices of finite-stabilizer linearization form an $H^1(BE[N]; \mathbb{F}_2)$-torsor; the trivial quotient orientation requires the zero point of this torsor. This H^1-choice is part of the quotient-orientation package transported in $(\mathrm{O1})^+$. Thus (O1) is not merely an ordinary Picard-line assertion: it consists of an ordinary square root in Pic, null-trivializations of the reduced and equivariant degree-two gerbes, and an equivariant line whose finite-stabilizer degree-one characters

$$\lambda_C^{E,N} \in H^1(BE[N]; \mathbb{F}_2)$$

are all chosen to be zero after the corresponding gerbes have vanished. On finite compact/wrapped HN stages this is precisely the quotient-orientation character system of Proposition 5.15. For $N = 2$, with $E[2] = \langle e_1, e_2 \rangle$ and $e_3 = e_1 + e_2$, this is the Klein-four condition

$$\beta_C^{E,E[2]} = b_{20}x_1^2 + b_{11}x_1x_2 + b_{02}x_2^2, \qquad (b_{20}, b_{11}, b_{02}) = (r_1, r_1 + r_2 + r_3, r_2),$$

where

$$\mathrm{res}_{\langle e_i \rangle} \beta_C^{E,E[2]} = r_i \tau_i^2.$$

If this class vanishes, an $E[2]$-linearization still has character

$$\lambda_C^{E[2]} = \lambda_1 x_1 + \lambda_2 x_2 \in H^1(BE[2]; \mathbb{F}_2),$$

with cyclic restrictions

$$\rho_1 = \lambda_1, \qquad \rho_2 = \lambda_2, \qquad \rho_3 = \lambda_1 + \lambda_2.$$

Trivial $E[2]$-linearization means $\rho_1 = \rho_2 = \rho_3 = 0$. The bits r_i live in degree two; the bits ρ_i live in degree one. For even N with $2^a \parallel N$, the mod-2 residual is a class on the two-primary stabilizer

$$E[N]_{(2)} \simeq (\mathbb{Z}/2^a)^2.$$

The case $a = 1$ is the displayed Klein-four calculation. For $a \geq 2$,

$$H^*(B(\mathbb{Z}/2^a)^2; \mathbb{F}_2) = \Lambda(x_1, x_2) \otimes \mathbb{F}_2[y_1, y_2], \qquad |x_i| = 1, \quad |y_i| = 2, \quad x_i^2 = 0,$$

and hence

$$H^2(B(\mathbb{Z}/2^a)^2; \mathbb{F}_2) = \mathbb{F}_2\langle y_1, x_1x_2, y_2\rangle.$$

Writing

$$\beta^{E,N}_{C,(2)} = A_1^{(N)} y_1 + A_{12}^{(N)} x_1x_2 + A_2^{(N)} y_2,$$

the order-two cyclic restrictions see

$$\mathrm{res}_{\langle h_1\rangle} \beta^{E,N}_{C,(2)} = A_1^{(N)} \tau_1^2, \quad \mathrm{res}_{\langle h_2\rangle} \beta^{E,N}_{C,(2)} = A_2^{(N)} \tau_2^2, \quad \mathrm{res}_{\langle h_3\rangle} \beta^{E,N}_{C,(2)} = (A_1^{(N)} + A_2^{(N)})\tau_3^2,$$

where $h_i = 2^{a-1} g_i$ for a basis g_1, g_2 of $(\mathbb{Z}/2^a)^2$ and $h_3 = h_1 + h_2$. The coefficient $A_{12}^{(N)}$ is invisible to these cyclic order-two restrictions. Thus the $E[2]$ vector (r_1, r_2, r_3) detects only the two-torsion residual; it is not a computation of $\beta^{E,N}_C$ for even $N \geq 4$. For $a \geq 2$, even after $\beta^{E,N}_{C,(2)} = 0$, the remaining two-primary linearization choices form

$$H^1(B(\mathbb{Z}/2^a)^2; \mathbb{F}_2) = \mathbb{F}_2\langle x_1, x_2\rangle,$$

and the trivial linearization requires that character to vanish on the chosen two-primary generators. Invariance of the underlying line under translations of E is not a substitute for these quotient-linearization data.

(O1)$^+$ *Weyl-equivariant determinant-line transport.* The orientation line of (O1) is equipped with coherent lifts of the three type-II wall transports

$$\tau_i : o_C \longrightarrow s^*_{\partial_i} o_C, \qquad i = 1, 2, 3,$$

but the word "coherent" is load-bearing. At finite height R, the first object is the finite partial action groupoid $\mathcal{G}_R$ whose objects are retained oriented quotient strata and whose arrows are the retained type-II wall transports. A choice of determinant-line lifts gives

$$c_{o,R} \in Z^2(\mathcal{G}_R; \underline{\mathbb{F}}_2).$$

The finite obstruction is $[c_{o,R}] \in H^2(\mathcal{G}_R; \underline{\mathbb{F}}_2)$, together with the quotient-torsor defects. Only on a complete $W^{(2)}(\Lambda^{2,1}_{II})$-orbit with constant coefficients does this reduce to the group cohomology class below. A choice of lifts over words in the simple reflections is then first only a projective determinant-line action: for $w, w' \in W^{(2)}(\Lambda^{2,1}_{II})$ one has a composition defect

$$w'^* \tau_w \circ \tau_{w'} = (-1)^{c_o(w,w')} \tau_{ww'}, \qquad c_o \in Z^2\Big(W^{(2)}(\Lambda^{2,1}_{II}); \mathbb{F}_2\Big).$$

Its class

$$[c_0] \in H^2\Big(W^{(2)}(\Lambda_{II}^{2,1}); \mathbb{F}_2\Big)$$

is the Weyl determinant-line cocycle obstruction. The datum $(O1)^+$ requires $[c_0] = 0$, a choice of 1-cochain killing c_0, and normalization $\tau_i^2 = 1$ for $i = 1, 2, 3$. Only then do the lifts satisfy the Coxeter presentation of $W^{(2)}(\Lambda_{II}^{2,1})$, or equivalently give an honest action on the oriented Pfaffian line over the wall-complement of the reduced compact Hall sector. This is a geometric determinant-line monodromy datum. It does not include a scalar formula; τ_i must also transport the quotient-orientation package

$$\mathfrak{q}_C := \Big(\alpha_C^{\mathrm{red}}, \alpha_C^{E,\mathrm{free}}, \{(\beta_C^{E,N}, \lambda_C^{E,N})\}_{N\geq 2}\Big)$$

and the chosen trivializations of these classes between the charge strata lying over Weyl-related Gram degrees. Equivalently, after the degree-two classes in $\mathfrak{q}_C$ vanish and the zero finite characters $\lambda_C^{E,N} = 0$ are selected, choose null-trivializations

$$\theta_C = \Big(\theta_C^{\mathrm{red}}, \theta_C^{E,\mathrm{free}}, \{\theta_C^{E,N}\}_{N\geq 2}\Big).$$

For finite stabilizers $\theta_C^{E,N}$ includes the null-trivialization of the finite Borel class and, after the mixed Borel terms and differentials vanish, its point-stabilizer restriction $\beta_C^{E,N} \in H^2(BE[N])$, together with the choice of the zero $H^1(BE[N])$-linearization character. For a simple type-II reflection s_{∂_i}, the failure of τ_i to preserve the quotient trivialization is the torsor difference

$$\omega_{i,C} \in H^1(\mathfrak{M}^{\mathrm{red}}_{s_{\partial_i}C}; \mathbb{F}_2) \oplus \bigoplus_{N\geq 2} H^1(BE[N]; \mathbb{F}_2).$$

Here $s_{\partial_i}C$ denotes the chosen target stratum over the Weyl-reflected Gram degree. The connected BE-linearization contributes no H^1-term because $H^1(BE; \mathbb{F}_2) = 0$. The datum $(O1)^+$ includes the vanishing of all $\omega_{i,C}$ and the Coxeter coherence of the resulting transports. Without this compatibility the reflection sign is a sign of an ordinary square-root line, not a sign on the reduced E-quotient Hall sector. The passage from three generator signs to a character is then the pure group-theoretic Lemma 3.22, after the projective and quotient-torsor obstruction package of Lemma 3.24 has vanished. At finite HN height this is the cocycle system of Corollary 3.25; the global datum is the compatible cofinal inverse system of those splittings.

(O2) *Pfaffian local model and wall atlas on type-II walls.* The local normal model is the rank-one crossing of Definition 3.11, embedded in the reduced Hall chart as in Definition 3.10. Proposition 3.13 states the deformation-obstruction calculation which identifies the reduced self-Ext normal factor with that crossing. The local wall-chart signs which feed the geometric reflection character

$$\epsilon_0 : W^{(2)}(\Lambda_{II}^{2,1}) \longrightarrow \{\pm 1\}$$

of Open Problem 5.7 are computed from local Pfaffian normal forms at generic points of the type-II walls $\mathbb{H}_\partial$. Namely, the reduced determinant complex is assumed to split, after removing the tangent determinant factor, as

$$K_\partial^{\mathrm{nor}} \simeq \bigoplus_{j=1}^{N_\partial^{\mathrm{Pf}}} \big[\mathbb{R} \xrightarrow{u_\partial} \mathbb{R}\big],$$

where u_∂ is a real local equation of the wall, $s_\partial(u_\partial) = -u_\partial$, and the Pfaffian section is $u_\partial^{N_\partial^{\mathrm{Pf}}}$ times a unit. For the three simple type-II walls the rank-one crossing gives

$$N_{\partial_1}^{\mathrm{Pf}} = N_{\partial_2}^{\mathrm{Pf}} = N_{\partial_3}^{\mathrm{Pf}} = 1.$$

The Borcherds divisor order and the Maass character are independent automorphic computations; they do not construct the Hall orientation line and do not use scalar normalization. At finite height R, once (O1) supplies the quotient orientation line, $(\mathrm{O1})^+$ supplies a normalized lift $\tau_{i,R,S} : o_{R,S} \to s_{\partial_i}^* o_{R,S}$, and the O2/hybrid residual for ∂_i vanishes, the supplied finite-stage sign is the monodromy of the Pfaffian section along the wall-reflection loop. In a local chart with

$$K_{\partial_i,R}^{\mathrm{nor}} \simeq [\mathbb{R} \xrightarrow{u_i} \mathbb{R}], \qquad \mathrm{Pf}(K_{\partial_i,R}^{\mathrm{nor}}) = v_i u_i, \qquad s_{\partial_i}(v_i) = v_i,$$

let $\ell_{\partial_i,R,S}$ be the path in the reduced quotient wall-complement carrying $u_i = \epsilon$ to $u_i = -\epsilon$, closed by the Weyl lift $\tau_{i,R,S}$. Then

$$M_{\ell_{\partial_i,R,S}}(v_i u_i) = v_i(-u_i) = -v_i u_i,$$

so

$$\epsilon_{o,R}(s_{\partial_i}) = (-1)^{N_{\partial_i,R}^{\mathrm{Pf}}}.$$

For the supplied rank-one type-II normal models this gives the datum implication $\epsilon_{o,R}(s_{\partial_i}) = -1$. It becomes compact Hall monodromy only after the quotient-orientation classes, the projective cocycle $[c_o]$, the torsor defects $\omega_{i,C}$, the finite-stabilizer characters $\lambda^{E,N}$, the O2/hybrid residuals, and $\mathfrak{O}_{\mathrm{hyb},R}$ vanish compatibly in R. For the global orientation-character conclusion below, (O2) also includes the component, boundary, overlap, quotient, and protected integration compatibility package of Proposition 10.32; the generic wall sign by itself is not a global Weyl-transport datum.

Then:

(i) (*Pfaffian normalization.*) After the supplied Pfaffian-to-automorphic comparison

$$\iota_{\mathrm{aut}} : \mathscr{L}_{\mathrm{Pf},X} \xrightarrow{\sim} \mathcal{L}^5 \otimes \nu_{\Delta_5},$$

compatible with cusp trivialization and squaring, the supplied first-order object has protected Pfaffian equal to the theta-normalized Borcherds determinant:

$$\iota_{\mathrm{aut}}(\mathrm{Pf}_{\mathrm{prot}}(\mathfrak{D}_X)) \;=\; \mathcal{D}_X \;=\; \Delta_5.$$

Without ι_{aut}, this is only equality of normalized formal Borcherds products at $\mathfrak{c}_\infty$. Its scalar trace, in the OP normalization of Definition 11.3, satisfies

$$\mathrm{Tr}_{\mathrm{prot}}\big((-1)^F\, \mathfrak{D}_X^{-2}\big) \;=\; -4096\, \Delta_5^{-2}.$$

(ii) (*Three-sign identification and Weyl character.*) For the supplied orientation character and for the three type-II simple reflections ∂_i of the Weyl chamber $\mathcal{P}_{II}$ of Section The Weyl chamber and the Weyl vector (notation $\delta_1 = 2f_2 - f_3$, $\delta_2 = 2f_{-2} - f_3$, $\delta_3 = f_3$),

$$\epsilon_o(s_{\partial_i}) \;=\; -1 \;=\; \nu_{\Delta_5}(s_{\partial_i}), \qquad i = 1, 2, 3,$$

and therefore

$$\epsilon_o\big|_{W^{(2)}(\Lambda_{II}^{2,1})} \;=\; \nu_{\Delta_5}\big|_{W^{(2)}(\Lambda_{II}^{2,1})}.$$

The three $\mathrm{Aut}(\mathcal{P}_{II}) \simeq S_3$ chamber-permuting reflections $s_{f_{-2}-f_2}, s_{f_2-f_3}, s_{f_2+f_3}$ have automorphic Maass value +1, but no Hall orientation sign on them is asserted unless the oriented Hall action is separately extended from $W^{(2)}(\Lambda^{2,1}_{II})$ to the semidirect product $W^{(2)}(\Lambda^{2,1}_{II}) \rtimes \mathrm{Aut}(\mathcal{P}_{II})$.

(iii) (*Categorical refinement.*) Under (D0), (O1), $(\mathrm{O1})^+$, and (O2), the supplied chiral protected first-order theory refines the virtual K_0-determinant package $[\mathcal{V}], \mathcal{D}_X$ of Proposition 3.8. Its decategorification is the Borcherds product

$$\iota_{\mathrm{aut}}\big(\mathrm{Pf}_{\mathrm{prot}}(\mathfrak{D}_X)\big)\big|_{\mathfrak{c}_\infty} = 64\, q^{1/2} r^{1/2} s^{1/2} \prod_{(n,l,m)\in\Gamma_{\mathrm{eff}}} (1 - q^n r^l s^m)^{f(nm,l)} = \Delta_5,$$

and its orientation monodromy character is the Maass character ν_{Δ_5} restricted to $W^{(2)}(\Lambda^{2,1}_{II})$.

Definition 3.16 *(Finite orientation construction target).* Fix a finite HN height R. A finite construction of (O1), $(\mathrm{O1})^+$, and (O2) is the vanishing of

$$\mathfrak{O}^{\mathrm{or}}_R = \Big(\mathfrak{O}^{\mathrm{O1}}_R, \mathfrak{O}^{\mathrm{O1+}}_R, \{\mathfrak{O}^{\mathrm{O2/hyb}}_{\delta_i,R}\}_{i=1}^3, \mathfrak{o}^{\mathrm{or,tr}}_{R'R}\Big).$$

Here $\mathfrak{O}^{\mathrm{O1}}_R = 0$ means that, on every object, extension, and flag stratum S,

$$\alpha^{\mathrm{red}}_{R,S} = 0, \qquad \alpha^{E,\mathrm{free}}_{R,S} = 0, \qquad \widetilde{\beta}^{E,N}_{R,S} = 0, \qquad \lambda^{E,N}_{R,S} = 0,$$

with the odd, $N = 2$, and higher even $E[N]$-cases computed as in Corollary 5.14, and with all Thom–Sebastiani orientation transports and flag coherences trivialized. The condition $\mathfrak{O}^{\mathrm{O1+}}_R = 0$ means that the retained type-II wall transports define a finite action-groupoid cocycle $c_{o,R}$, that

$$c_{o,R} = \delta a^{\mathrm{or}}_R, \qquad \widetilde{\tau}^2_{i,R} = 1, \qquad \omega_{i,R,S} = 0,$$

and that these choices are compatible with successor maps $R' \to R$. For each i, the condition $\mathfrak{O}^{\mathrm{O2/hyb}}_{\delta_i,R} = 0$ means that the wall object lies in the same finite hybrid correspondence system, that the reduced wall complex splits

$$K^{\mathrm{red}}_{\delta_i,R} \simeq K^{\mathrm{tan}}_{\delta_i,R} \oplus K^{\mathrm{nor}}_{\delta_i,R}, \qquad K^{\mathrm{nor}}_{\delta_i,R} \simeq [\mathbb{R} \xrightarrow{u_{\delta_i,R}} \mathbb{R}],$$

and that

$$s_{\delta_i}(u_{\delta_i,R}) = -u_{\delta_i,R}, \qquad \mathrm{Pf}(K^{\mathrm{nor}}_{\delta_i,R}) = v_{\delta_i,R} u_{\delta_i,R}, \qquad s_{\delta_i}(v_{\delta_i,R}) = v_{\delta_i,R}.$$

The wall charts must cover every retained component and boundary stratum of $\mathbb{H}_{\delta_i}$, and on overlaps the tangent/normal splittings, normal Pfaffian units, quotient orientations, and protected integration maps must agree after a zero Cech sign class. The middle wall $\delta_2 \leftrightarrow (0,1,1)$ must be represented by a retained wrapped wall object

$$x_{\delta_2,R} \in \mathcal{M}^{\mathrm{wr,rig}}_{\eta,R}, \qquad b^{\mathrm{geom}}_R(\eta) > 0, \qquad \overline{\Pi}_X(\eta) = (0,1,1),$$

together with its rigidifying anchor, unreduced and reduced deformation complexes, quotient-compatible tangent/normal splitting, and protected integration compatibility in the sense of Proposition 10.32. No Maass character or scalar normalization is part of this construction.

Lemma 3.17 *(Local Pfaffian wall sign).* Assume the δ_i-chart in Definition 3.16. Then

$$s^*_{\delta_i}\,\mathrm{Pf}(K^{\mathrm{nor}}_{\delta_i,R}) = -\,\mathrm{Pf}(K^{\mathrm{nor}}_{\delta_i,R}), \qquad s^*_{\delta_i}\,\mathrm{Pf}(K^{\mathrm{nor}}_{\delta_i,R})^2 = \mathrm{Pf}(K^{\mathrm{nor}}_{\delta_i,R})^2.$$

Proof. The invariant unit is fixed by s_{δ_i}, while $u_{\delta_i,R}$ is anti-invariant. Hence

$$s_{\delta_i}^*(v_{\delta_i,R}u_{\delta_i,R}) = -v_{\delta_i,R}u_{\delta_i,R}.$$

Squaring removes the sign. □

D_0 contains the normal-ordered lattice part and the imported denominator target. On the reduced $K3 \times E$ sector its compact finite stages, transition maps, and obstruction trivializations must be constructed. The orientation clauses (O1), $(\mathrm{O1})^+$, and (O2) require the corresponding geometric construction; on Calabi–Yau threefolds (O1) is Joyce–Song [44], the multiplicative orientation cocycle on $\mathrm{Perf}(X)$ is Joyce–Upmeier [46] and Joyce–Tanaka–Upmeier [45], on Calabi–Yau 4-fold analogues the unreduced orientation exists by Borisov–Joyce [10], Cao–Gross–Joyce [15], and Bojko's quasi-projective Borisov–Joyce extension [6]; the virtual-cycle constructions of Oh–Thomas [68], [67] and rank-one Pin^c-orientation sign conventions in DT/PT Hilbert-scheme models are compatible with the BBDJS/Joyce–Upmeier framework and with the Cao–Kool–Monavari Calabi–Yau 4-fold orientation package [16]; non-compact and Joyce–Song extensions are Bojko [6]. These sources supply the orientation technology and partial local inputs. They do not by themselves prove the cosection-reduced $K3 \times E$ quotient orientation, its $E[2]$-linearization, or its type-II Weyl-equivariant lift. The cosection-induced parity flip $\mathrm{vd}^{\mathrm{red}} = \mathrm{vd}^{\mathrm{std}} + 1$ of the Maulik–Toda / Oberdieck reduced obstruction theory [59, 64] is absorbed into (O1) as a $\mathbb{Z}/2$-grading shift $o_C^{\mathrm{red}} = o_C^{\mathrm{std}} \otimes \Pi$ of the BBDJS [11] line under cosection contraction of the PTVV [69] (-1)-shifted symplectic form by σ_{K3} (clause (P4) of Open Problem 5.7). This parity shift is part of the reduced orientation datum and may be compatible with the fermionic grading used in the protected supertrace. It is not the source of the OP scalar minus sign and it does not determine the Hall reflection character. The latter must be computed from monodromy of the orientation line around the type-II walls. The theorem therefore has datum form: granted (D0), (O1), $(\mathrm{O1})^+$, and (O2), the normalization and character follow.

Proof of Theorem 3.15. Step 1 (Pfaffian product from the supplied first-order datum). By (D0), the compact first-order object $\mathfrak{D}_X$ and its normal-ordered primitive multiplicities are supplied. By Definition 4.3, the orientation line of a stable charge-c object satisfies $o_C^{\otimes 2} \simeq \det \mathrm{RHom}_{\mathrm{red}}(C, C)$. The hypothesis (O1) promotes this to an extension-multiplicative system: on every direct sum $C \oplus C'$, $o_{C\oplus C'} \simeq o_C \otimes o_{C'}$, compatibly with the Hall extension stack and descended under the reduced E-quotient. Thus the supplied $\mathfrak{D}_X$ has a protected Pfaffian product whose exponent in normal-ordered Gram degree (n, l, m), by the supplied D_0 signed-superdimension residual, is

$$\mathrm{sdim}\, P^{\Pi}_{X,(n,l,m)} = f(nm, l).$$

With the leading cusp normalization fixed by the theta coefficient, the graded product is

$$\mathrm{Pf}_{\mathrm{prot}}(\mathfrak{D}_X) = v_0\ \mathrm{sdet}_{\mathcal{V}}(1 - x) \quad \text{with} \quad v_0 = 64 q^{1/2} r^{1/2} s^{1/2},$$

where sdet is the superdeterminant of Proposition 3.8. The theorem does not construct $\mathfrak{D}_X$; it identifies the Pfaffian of a supplied object satisfying (D0) and (O1).

Step 2 (Comparison with the automorphic section). Proposition 4.5 identifies $v_0\ \mathrm{sdet}_{\mathcal{V}}(1 - x)$ with the Igusa cusp form Δ_5 at the level of Fourier coefficients. To state an equality of sections, the supplied D_0-Pfaffian datum must also contain an isomorphism

$$\iota_{\mathrm{aut}} : \mathscr{L}_{\mathrm{Pf},X} \xrightarrow{\sim} \mathcal{L}^5 \otimes \nu_{\Delta_5}$$

compatible with the cusp trivialization and with squaring to $\mathcal{L}^{10}$. Under this supplied comparison, $\iota_{\mathrm{aut}}(\mathrm{pf}_X)$ and Δ_5 have the same Borcherds expansion at $\mathfrak{c}_\infty$, hence are equal. By the q-expansion principle for holomorphic genus-two Siegel modular forms with Maass character, after the supplied line comparison and cusp trivialization, equality of this cusp expansion gives equality of sections. Without ι_{aut}, the conclusion is only equality of normalized formal products.

Step 3 (Scalar square in OP normalization). Squaring forgets the orientation. Under $\iota_{\mathrm{aut}}^{\otimes 2}$,

$$\iota_{\mathrm{aut}}^{\otimes 2}\big(\mathrm{Pf}_{\mathrm{prot}}(\mathfrak{D}_X)^2\big) = \det \mathrm{RHom}_{\mathrm{red}}(C, C)|_{\mathrm{tot}} = \Delta_5^2,$$

as sections of $\mathcal{L}^{10}$ (the squared automorphic line is the trivially-charactered determinant, since $\Delta_5^2 = \Delta_{10}$ is scalar by Section The normalized cusp form). The protected scalar trace is then $\mathrm{Tr}_{\mathrm{prot}}((-1)^F \mathfrak{D}_X^{-2}) = C_{\square,X}\,\Delta_5^{-2}$ for some normalization scalar $C_{\square,X}$; the explicit $C_{\square,X} = -4096$ follows from Proposition 11.1 of Oberdieck–Pixton, which gives $Z_{\mathrm{OP}}^X = -(\chi_{10}^{\mathrm{OP}})^{-1} = -4096\,\Delta_5^{-2}$, together with the arithmetic $-4096 = -(64)^2$ of the squared theta-leading coefficient and the OP scalar sign convention recorded in Definition 11.3.

Step 4 (Orientation-character identity (ii)). By (O2), after the full wall-atlas compatibility package has been supplied, each type-II simple reflection s_{δ_i} acts on o_C by the parity of its oriented real normal modes. Lemma 3.18 is the Hall-side local calculation: under the supplied Pfaffian normal form $N_{\delta_i}^{\mathrm{Pf}} = 1$, hence $\epsilon_o(s_{\delta_i}) = (-1)^{N_{\delta_i}^{\mathrm{Pf}}} = -1$. Lemma 8.2 is the independent automorphic calculation: $\nu_{\Delta_5}(s_{\delta_i}) = -1$ for the same three simple reflections, and $+1$ on the three chamber-permuting $\mathrm{Aut}(\mathcal{P}_{II})$-reflections. The latter $+1$-values are automorphic target data only; they are not Hall orientation signs unless a semidirect-product Hall lift is supplied.

The passage from the three simple Hall signs to a character on the full type-II Weyl group is purely group-theoretic only after $(\mathrm{O1})^+$ has killed the projective determinant-line cocycle and the quotient trivialization torsors of Lemma 3.24. Then $(\mathrm{O1})^+$ gives an honest lifted Coxeter action on the reduced quotient orientation. By Lemma 3.22,

$$W^{(2)}(\Lambda_{II}^{2,1})_{\mathrm{ab}} \simeq (\mathbb{Z}/2)^3,$$

with one generator for each s_{δ_i}. Therefore two $\{\pm 1\}$-valued characters of $W^{(2)}(\Lambda_{II}^{2,1})$ that agree on $s_{\delta_1}, s_{\delta_2}, s_{\delta_3}$ agree everywhere. Hence $\epsilon_o = \nu_{\Delta_5}$ on $W^{(2)}(\Lambda_{II}^{2,1})$.

Step 5 (Categorical refinement (iii)). The chiral protected theory is supplied by (D0). Steps 1–4 identify its decategorification and character: the class $[\mathcal{V}] \in K_0(\mathrm{sVect}_{\mathrm{fd}})$ is the decategorified primitive content, the formal product $v_0\ \mathrm{sdet}_{\mathcal{V}}(1-x)$ is the Pfaffian section $\mathrm{Pf}_{\mathrm{prot}}(\mathfrak{D}_X)$, and the formal character equality is the orientation-character identity of (ii). At the chosen isotropic cusp $\mathfrak{c}_\infty$ this decategorification is the Borcherds product. The supplied O1/O1$^+$/O2 character is ν_{Δ_5} on the type-II Weyl group. Without those supplied data, the Maass character remains an automorphic character computation. □

Lemma 3.18 *(Pfaffian signs on the type-II simple walls).* Under the compact first-order datum (D0), the strong orientation datum (O1), the Weyl-equivariant extension $(\mathrm{O1})^+$, and the Pfaffian local model (O2) of Theorem 3.15, for each $\delta_i \in \{\delta_1, \delta_2, \delta_3\}$ the orientation-monodromy reflection sign is

$$\epsilon_o(s_{\delta_i}) = (-1)^{N_{\delta_i}^{\mathrm{Pf}}} = -1, \qquad N_{\delta_i}^{\mathrm{Pf}} = 1.$$

Here $N_{\delta_i}^{\mathrm{Pf}}$ is the number of supplied odd real normal modes in the local Pfaffian normal form. It is not the automorphic divisor order, although the independent target order agrees in the three simple type-II cases. The lemma makes no Hall-orientation claim for the three chamber-permuting reflections in $\mathrm{Aut}(\mathcal{P}_{II})$.

Proof. Local Pfaffian calculation. Hypothesis (O2) supplies, at a generic point of each $\mathbb{H}_{\delta_i}$, a real local equation $u_i = u_{\delta_i}$ with $s_{\delta_i}(u_i) = -u_i$ and a normal determinant-complex splitting

$$K^{\mathrm{nor}}_{\delta_i} \simeq \left[\mathbb{R} \xrightarrow{u_i} \mathbb{R}\right].$$

The normal Pfaffian section is u_i times an invariant unit after quotient-orientation trivialization. Transport by the reflection sends u_i to $-u_i$, so it multiplies the Pfaffian line by -1; after squaring, the determinant section u_i^2 is fixed. Hence one supplied odd real normal mode gives

$$\epsilon_0(s_{\delta_i}) = (-1)^{N^{\mathrm{Pf}}_{\delta_i}} = (-1)^1 = -1, \qquad i = 1, 2, 3.$$

Independent automorphic target labels. The three roots $\delta_1 = 2f_2 - f_3, \delta_2 = 2f_{-2} - f_3, \delta_3 = f_3$ have Gram pairing $(\delta_i, \delta_i) = 2$, $(\delta_i, \delta_j) = -2$ for $i \neq j$, by the matrix displayed in Section The Weyl chamber and the Weyl vector. All three are real simple roots of the Lorentzian Kac–Moody algebra $\mathfrak{g}_{\Delta_5}$. In the Gram-index chart of Definition 8.3 (using $\alpha(\gamma) = 2nf_2 - lf_3 + 2mf_{-2}$), the three simple walls correspond to the lattice triples

$$\delta_1 \;\leftrightarrow\; (n, l, m) = (1, 1, 0), \qquad \delta_2 \;\leftrightarrow\; (0, 1, 1), \qquad \delta_3 \;\leftrightarrow\; (0, -1, 0),$$

matching the K3 elliptic-genus support $f(0, \pm 1) = 1, f(0, 0) = 10, f(nm, l) = 0$ for $4nm - l^2 < -1$. Thus the automorphic divisor order

$$d^{\mathrm{aut}}_{\delta_i} := \mathrm{ord}_{\mathbb{H}_{\delta_i}}(\Delta_5)$$

is 1 for $i = 1, 2, 3$. This target order is a check against the supplied local model; it is not a construction of the compact Hall orientation line.

The computation is:

δ_i	Gram label	$N^{\mathrm{Pf}}_{\delta_i}$	$d^{\mathrm{aut}}_{\delta_i}$	Hall sign from the supplied model
$\delta_1 = 2f_2 - f_3$	$(n, l, m) = (1, 1, 0), f(0, 1) = 1$	1	1	-1
$\delta_2 = 2f_{-2} - f_3$	$(n, l, m) = (0, 1, 1), f(0, 1) = 1$	1	1	-1
$\delta_3 = f_3$	$(n, l, m) = (0, -1, 0), f(0, -1) = 1$	1	1	-1

□

Remark 3.19 *(Maass character on the type-II Weyl group).* The argument below verifies the target character $\nu_{\Delta_5}|_{W^{(2)}(\Lambda^{2,1}_{II})}$ using only the Maass multiplier and the Borcherds product. It does not construct the Hall orientation character ϵ_0. The comparison $\epsilon_0 = \nu_{\Delta_5}$ remains a geometric statement supplied by (O1), (O1)$^+$, and (O2). No scalar normalization enters this sign comparison: replacing Δ_5 by $c\Delta_5$, $c \in \mathbb{C}^\times$, changes neither the Maass character nor the divisor orders. The factor 64 only names the monic Borcherds product $D_5 = 64^{-1}\Delta_5$, and the OP/DT scalar -4096 is unused.

Step 1 (Maass-side coboundary). Define the automorphic parity cochain $\varepsilon^{\mathrm{aut}}_\Delta \colon W^{(2)}(\Lambda^{2,1}_{II}) \to \mathbb{Z}/2$ by $\varepsilon^{\mathrm{aut}}_\Delta(g) := \frac{1}{2}(1 - \nu_{\Delta_5}(g))$, so $(-1)^{\varepsilon^{\mathrm{aut}}_\Delta(g)} = \nu_{\Delta_5}(g)$. Maass's character is a homomorphism [56], [37, Section 3], [36], hence

$$c^{\mathrm{aut}}_\Delta(g, h) := \varepsilon^{\mathrm{aut}}_\Delta(g) + \varepsilon^{\mathrm{aut}}_\Delta(h) - \varepsilon^{\mathrm{aut}}_\Delta(gh) \equiv 0 \pmod 2.$$

The displayed value of ν_{Δ_5} on the six generators is Lemma 8.2: -1 on $s_{\delta_1}, s_{\delta_2}, s_{\delta_3}$, $+1$ on $s_{f_{-2}-f_2}, s_{f_2-f_3}, s_{f_2+f_3}$; equivalently $\varepsilon^{\mathrm{aut}}_{\Delta}(s_{\delta_i}) = 1$ and $\varepsilon^{\mathrm{aut}}_{\Delta}(s_{f_{-2}-f_2}) = \varepsilon^{\mathrm{aut}}_{\Delta}(s_{f_2-f_3}) = \varepsilon^{\mathrm{aut}}_{\Delta}(s_{f_2+f_3}) = 0$.

Step 2 (Verification on the nine pairs of three-wall generators). For the three-wall computation the only nontrivial check is on the 3×3 table of pairs $(g, h) \in \{s_{\delta_1}, s_{\delta_2}, s_{\delta_3}\}^{\times 2}$. The relations used here are only $s_{\delta_i}^2 = 1$ and the fact that ν_{Δ_5} is a character. For $i \neq j$ the Coxeter exponent is $m_{ij} = \infty$, so there is no braid relation to impose:

- Diagonal $(g, g) = (s_{\delta_i}, s_{\delta_i})$ for $i = 1, 2, 3$ (three pairs): $gh = 1$, $\varepsilon^{\mathrm{aut}}_{\Delta}(gh) = 0$, so $c^{\mathrm{aut}}_{\Delta}(g, h) = 1 + 1 - 0 = 2 \equiv 0 \pmod 2$.
- Off-diagonal $(g, h) = (s_{\delta_i}, s_{\delta_j})$, $i \neq j$ (six pairs): since ν_{Δ_5} is a character, $\nu_{\Delta_5}(gh) = (-1)(-1) = +1$, giving $\varepsilon^{\mathrm{aut}}_{\Delta}(gh) = 0$ and $c^{\mathrm{aut}}_{\Delta}(g, h) = 1 + 1 - 0 = 2 \equiv 0 \pmod 2$.

All nine pairs vanish modulo 2; $c^{\mathrm{aut}}_{\Delta}$ is a coboundary on the three-wall block, with primitive $\varepsilon^{\mathrm{aut}}_{\Delta}$. This is an automorphic calculation only.

Step 3 (Divisor exponent). The Gritsenko–Nikulin specialization of Borcherds' product theorem [8], [39, Theorem 2.1, (2.1)–(2.3), Example 2.4], together with the theta normalization [37, Theorem 4.1], writes

$$D_5 = 64^{-1}\Delta_5 = \mathrm{Borch}(\phi_{0,1}) = q^{1/2} r^{1/2} s^{1/2} \prod_{(n,l,m)\in\Gamma_{\mathrm{eff}}} (1 - q^n r^l s^m)^{f(nm,l)}$$

in the type-II cusp polarization $(q, r, s) = (e^{2\pi i z_1}, e^{2\pi i z_2}, e^{2\pi i z_3})$. This fixes the automorphic divisor orders

$$d^{\mathrm{aut}}_{\delta} := \mathrm{ord}_{\mathbb{H}_\delta}(\Delta_5).$$

For the three type-II roots these orders are

$$d^{\mathrm{aut}}_{\delta_1} = f(0, 1) = 1, \qquad d^{\mathrm{aut}}_{\delta_2} = f(0, 1) = 1, \qquad d^{\mathrm{aut}}_{\delta_3} = f(0, -1) = 1.$$

Together with Lemma 8.2, this gives the automorphic target signs $\nu_{\Delta_5}(s_{\delta_i}) = -1 = (-1)^{d^{\mathrm{aut}}_{\delta_i}}$. The symbol $d^{\mathrm{aut}}_{\delta}$ is deliberately distinct from the Pfaffian normal-mode rank N^{Pf}_{δ} in (O2).

The chamber-permuting roots

$$\alpha_1 = f_{-2} - f_2, \qquad \alpha_2 = f_2 - f_3, \qquad \alpha_3 = f_2 + f_3$$

are type-I roots in $\Lambda^{2,1} \setminus \Lambda^{2,1}_{II}$. In the type-II parametrization $\alpha(n, l, m) = 2nf_2 - lf_3 + 2mf_{-2}$, the coefficients of f_2 and f_{-2} must be even. Thus $k = 1$ does not contribute to any α_i. For $2\alpha_1$ one obtains either $(-1, 0, 1)$ or $(1, 0, -1)$, outside Γ_{eff}. For $2j\alpha_2$ one obtains $(j, 2j, 0)$, and for $2j\alpha_3$ one obtains $(j, -2j, 0)$; the q^0-row of $\phi_{0,1}$ has support only at $l = 0, \pm 1$, so $f(0, \pm 2j) = 0$ for $j \geq 1$ [27, §2.2]. Hence

$$\mathrm{ord}_{\mathbb{H}_{\alpha_i}}(\Delta_5) = 0, \qquad i = 1, 2, 3,$$

and Lemma 8.2 gives automorphic monodromy $+1$ around these chamber-permuting reflections.

What this check proves. It proves the automorphic target values

$$\nu_{\Delta_5}(s_{\delta_i}) = -1, \qquad \nu_{\Delta_5}(s_{f_{-2}-f_2}) = \nu_{\Delta_5}(s_{f_2-f_3}) = \nu_{\Delta_5}(s_{f_2+f_3}) = +1.$$

It does not prove that a compact Hall orientation line transports with these signs. That bridge is exactly the geometric content of the Pfaffian local model and Weyl-equivariant orientation cocycle.

Remark 3.20 *(Comparison of the two proofs).* The proof of Lemma 3.18 and the Maass-character check have different logical roles. The first computes the Hall orientation signs from the supplied Pfaffian local model (O2). The second computes the automorphic target signs of ν_{Δ_5}. The equality $\epsilon_o = \nu_{\Delta_5}$ follows only after the local Dirac model supplies the Pfaffian normal-mode parity and the Weyl-equivariant orientation cocycle transports it as a character. The Borcherds product and Maass character supply the target character but not the Hall orientation character.

Remark 3.21 *(Imaginary roots are not Weyl reflections).* The Weyl group of the Borcherds–Kac–Moody datum is generated by real simple roots only. Imaginary simple roots contribute to the Weyl–Kac–Borcherds denominator through the additive correction coefficients $m(a)$ and $\tau(a)$ of Gritsenko–Nikulin [39, Theorem 2.1, (2.1)–(2.3)], [37, Sections 3–4]; they do not define reflections s_a in $W^{(2)}(\Lambda^{2,1}_{II})$. Thus the orientation character

$$\epsilon_o : W^{(2)}(\Lambda^{2,1}_{II}) \to \{\pm 1\}$$

is a real-root reflection datum. A compact Hall theory may also have local divisor monodromy around imaginary components of the Borcherds divisor, but that monodromy is not a Weyl character and is not asserted here. In particular, no formula $\epsilon_o(s_a) = (-1)^{m(a)}$ is available for imaginary a, and the coefficients $m(a)$ must not be identified with the visible product exponents $f(nm, l)$ without the Gritsenko–Nikulin additive-side computation.

Lemma 3.22 *(Pure group-theoretic Weyl sign extension).* For any signs $\epsilon_i \in \{\pm 1\}$, $i = 1, 2, 3$, there is a unique character

$$\chi_\epsilon : W^{(2)}(\Lambda^{2,1}_{II}) \longrightarrow \{\pm 1\}$$

with $\chi_\epsilon(s_{\delta_i}) = \epsilon_i$. Hence a supplied Weyl-equivariant determinant-line transport datum $(\mathrm{O1})^+$, together with local Pfaffian signs $\epsilon_i = (-1)^{N^{\mathrm{Pf}}_{\delta_i}}$ from (O2), induces a unique orientation character. For $N^{\mathrm{Pf}}_{\delta_1} = N^{\mathrm{Pf}}_{\delta_2} = N^{\mathrm{Pf}}_{\delta_3} = 1$, this character is the determinant character on $W^{(2)}(\Lambda^{2,1}_{II})$.

Proof. The Cartan matrix of the type-II simple reflections, by $(\delta_i, \delta_i) = 2$ and $(\delta_i, \delta_j) = -2$ for $i \neq j$ (Section The Weyl chamber and the Weyl vector), is

$$\begin{pmatrix} 2 & -2 & -2 \\ -2 & 2 & -2 \\ -2 & -2 & 2 \end{pmatrix}.$$

Thus $A_{ij}A_{ji} = 4$ for $i \neq j$, so the Coxeter exponents are $m_{ij} = \infty$. The chamber calculation in Section The Weyl chamber and the Weyl vector identifies $\{\delta_1, \delta_2, \delta_3\}$ as the primitive walls of the fundamental type-II chamber $\mathcal{P}_{II}$, and the Gritsenko–Nikulin reflection-group input supplies the Coxeter presentation from these walls. Hence the type-II reflection group is the universal rank-three Coxeter group

$$W^{(2)}(\Lambda^{2,1}_{II}) \simeq \langle s_{\delta_1}, s_{\delta_2}, s_{\delta_3} \mid s^2_{\delta_1} = s^2_{\delta_2} = s^2_{\delta_3} = 1 \rangle.$$

Its abelianization is

$$W^{(2)}(\Lambda^{2,1}_{II}) / [W^{(2)}, W^{(2)}] \simeq (\mathbb{Z}/2)^3,$$

one $\mathbb{Z}/2$ per Coxeter generator s_{δ_i}.

The only defining relations are $s^2_{\delta_i} = 1$. Any assignment $s_{\delta_i} \mapsto \epsilon_i \in \{\pm 1\}$ respects these relations, since $\epsilon_i^2 = 1$, and therefore extends uniquely to a $\{\pm 1\}$-valued character, equivalently to a homomorphism

factoring through $(\mathbb{Z}/2)^3$. This is pure group theory. Applied to a supplied determinant-line action from $(\mathrm{O1})^+$, the scalars computed by (O2) on the three generators give the orientation character. If all three $\epsilon_i = -1$, the character sends every Coxeter generator to -1, hence is the determinant character. □

Proposition 3.23 *(Local Weyl transport from the three rank-one crossings).* Let

$$U^\circ = \prod_{i=1}^{3} \big(T_{\partial_i} \times ((-\varepsilon, \varepsilon) \setminus \{0\})\big)$$

be the product of the three rank-one type-II crossing complements of Definition 3.11. Put

$$\mathscr{L}_{\mathrm{loc}} = \bigotimes_{i=1}^{3} \mathscr{L}^{\tan}_{\partial_i} \otimes \bigotimes_{i=1}^{3} \mathrm{Pf}\big(K^{\mathrm{cross}}_{\partial_i}\big), \qquad \mathrm{pf}_{\mathrm{loc}} = \prod_{i=1}^{3} v_{\partial_i} u_i.$$

Define $\tau_i : s^*_{\partial_i}\mathscr{L}_{\mathrm{loc}} \to \mathscr{L}_{\mathrm{loc}}$ by the identity on every tangent factor and on every normal factor except the i-th crossing, where it is the standard pullback under $u_i \mapsto -u_i$. Then:

$$\tau_i^2 = 1, \qquad \tau_i(s^*_{\partial_i}\,\mathrm{pf}_{\mathrm{loc}}) = -\,\mathrm{pf}_{\mathrm{loc}}, \qquad i = 1, 2, 3.$$

Since

$$W^{(2)}(\Lambda^{2,1}_{II}) \simeq \langle s_{\partial_1}, s_{\partial_2}, s_{\partial_3} \mid s^2_{\partial_i} = 1\rangle,$$

the three maps τ_i extend uniquely to an honest action of $W^{(2)}(\Lambda^{2,1}_{II})$ on $\mathscr{L}_{\mathrm{loc}}$. The associated projective cocycle is zero and the local orientation character is

$$\epsilon^{\mathrm{loc}}_o(s_{\partial_i}) = -1, \qquad i = 1, 2, 3.$$

Proof. For the i-th crossing, $s^*_{\partial_i} u_i = -u_i$ and $s^*_{\partial_i} v_{\partial_i} = v_{\partial_i}$. The map τ_i therefore sends the i-th Pfaffian factor $v_{\partial_i} u_i$ to $-v_{\partial_i} u_i$ and fixes the other two factors. Hence $\mathrm{pf}_{\mathrm{loc}}$ changes sign under τ_i, while $\mathrm{pf}^2_{\mathrm{loc}}$ is fixed. Applying s_{∂_i} twice sends u_i back to u_i, and the standard Pfaffian trivialization is unchanged, so $\tau_i^2 = 1$. There are no braid relations among the three type-II generators; the Coxeter presentation has only $s^2_{\partial_i} = 1$. Therefore the universal property of the free product of three order-two groups gives a unique honest Weyl action with these generator maps. Since the action is honest, the composition defect $c_o(w, w')$ is zero. The displayed values of $\epsilon^{\mathrm{loc}}_o$ are exactly the scalar action on the local Pfaffian section. □

Lemma 3.24 *(Weyl quotient-orientation obstruction package).* Let $W = W^{(2)}(\Lambda^{2,1}_{II})$, and let $\mathcal{G}_W$ be the action groupoid of the retained reduced quotient strata under the supplied type-II wall transports. On a complete W-orbit with constant coefficients this groupoid cohomology reduces to ordinary group cohomology of W. Suppose (O1) supplies an ordinary orientation line together with the full reduced quotient package

$$\theta_C = \Big(\theta^{\mathrm{red}}_C, \theta^{E,\mathrm{free}}_C, \{(\theta^{E,N}_C, \lambda^{E,N}_C = 0)\}_{N\geq 2}\Big)$$

on every charge stratum used by the compact/wrapped Hall correspondences. A choice of simple wall transports τ_i gives an honest Weyl-equivariant orientation on the reduced quotient if and only if the following two independent obstructions vanish.

$$[c_o] = 0 \in H^2(\mathcal{G}_W; \underline{\mathbb{F}}_2)$$

with a killing groupoid cochain $b \in C^1(\mathcal{G}_W; \underline{\mathbb{F}}_2)$ for which the renormalized lifts $\widetilde{\tau}_w = (-1)^{b(w)}\tau_w$ satisfy $\widetilde{\tau}_i^2 = 1$ for the three simple reflections; and

$$\omega_{i,C} = 0 \in H^1(\mathfrak{M}^{\mathrm{red}}_{s_{\delta_i}C}; \mathbb{F}_2) \oplus \bigoplus_{N \geq 2} H^1(BE[N]; \mathbb{F}_2)$$

for every simple reflection and every relevant charge stratum, with the word defects obeying their natural torsor cocycle law. Since the type-II Weyl group has no braid relations among the three simple reflections, the generator-square constraints are the only Coxeter relation constraints after the projective cocycle is killed. Killing $[c_o]$ gives an action on the ordinary determinant line; killing the $\omega_{i,C}$ gives an action on the quotient orientation with its finite-stabilizer zero characters. Neither condition is implied by the Maass character ν_{Δ_5}.

Proof. The chosen transports over words in the generators define a central $\{\pm 1\}$-extension of the retained action groupoid precisely when their composition defect is a 2-cocycle

$$w'^* \tau_w \circ \tau_{w'} = (-1)^{c_o(w,w')} \tau_{ww'}.$$

Changing the word lifts by $(-1)^{b(w)}$ replaces c_o by $c_o + \delta b$. Hence the projective determinant-line action becomes an honest action on the ordinary orientation line exactly when $[c_o]$ is zero and the chosen splitting is normalized on the relations. By Lemma 3.22, the type-II Coxeter presentation is the free product of three copies of $\mathbb{Z}/2$. Thus the only relation-level normalizations are $\widetilde{\tau}_i^2 = 1$ on a complete W-orbit; on a truncated finite system this is the corresponding statement in the retained action groupoid.

After the ordinary line action is honest, the quotient null-trivializations form torsors under

$$H^1(\mathfrak{M}^{\mathrm{red}}_C; \mathbb{F}_2) \oplus \bigoplus_{N \geq 2} H^1(BE[N]; \mathbb{F}_2).$$

The difference between the transported trivialization and the chosen trivialization on the reflected stratum is exactly $\omega_{i,C}$. Composition of transports gives the usual torsor cocycle law $\omega_{ww',C} = w'^* \omega_{w,C} + \omega_{w',C}$. Therefore vanishing on the retained generators, together with the square coherences already imposed above, propagates to every word in the retained action groupoid, and to every word of W only after a complete orbit has been supplied. Conversely, any honest Weyl action on the reduced quotient orientation preserves the chosen null-trivializations and zero finite characters, so both c_o and all $\omega_{i,C}$ vanish after the corresponding choices. The construction uses only determinant-line and quotient-torsor data, not the scalar automorphic character. □

Corollary 3.25 *(Finite-stage Weyl cocycle over the quotient character system).* Fix a finite compact/wrapped HN stage R, a finite test set $\mathcal{O}_R$ of reduced charge strata closed under the simple type-II wall transports retained at height R, and a quotient-orientation character system on $\mathcal{O}_R$ in the sense of Proposition 5.15. Choose determinant-line lifts $\tau_{i,R,C}$ for the simple type-II reflections whenever both C and $s_{\delta_i}C$ lie in $\mathcal{O}_R$. For words whose intermediate strata remain in $\mathcal{O}_R$, choose word lifts $\tau_{w,R,C}$ and define

$$w'^* \tau_{w,R,w'C} \circ \tau_{w',R,C} = (-1)^{c_{o,R}(w,w';C)} \tau_{ww',R,C}.$$

Then $c_{o,R}$ is a 2-cocycle for the finite action groupoid generated by these retained transports; on any genuine W-orbit it is the corresponding group 2-cocycle with values in locally constant $\mathbb{F}_2$-functions on

O_R. Changing the word lifts by a 1-cochain changes $c_{o,R}$ by a coboundary. The quotient-trivialization defect

$$\omega_{w,R,C} \in H^1(\mathfrak{M}^{\mathrm{red}}_{wC}; \mathbb{F}_2) \oplus \bigoplus_{N \geq 2} H^1(BE[N]; \mathbb{F}_2)$$

satisfies

$$\omega_{ww',R,C} = w'^* \omega_{w,R,C} + \omega_{w',R,C}.$$

Thus the finite-stage Weyl lift exists on O_R if and only if $c_{o,R}$ is killed by a cochain normalized on the relations $s^2_{\partial_i} = 1$, and the defects $\omega_{i,R,C}$ vanish for the three simple reflections. A cofinal family of such finite-stage splittings, compatible with the HN transition maps, is exactly the Weyl-equivariant lift $(\mathrm{O1})^+$ on the compact/wrapped sector. A non-zero $c_{o,R}$, a failed square normalization, or a non-zero $\omega_{i,R,C}$ at one finite stage obstructs $(\mathrm{O1})^+$.

Proof. The first display is the definition of the composition defect for the chosen word lifts. Associativity of determinant-line composition gives the cocycle identity. Multiplying $\tau_{w,R,C}$ by $(-1)^{b(w,C)}$ replaces $c_{o,R}$ by $c_{o,R} + \delta b$, so an honest determinant-line action is the same as a splitting of this central extension, normalized on the Coxeter relations. For the type-II subgroup the only Coxeter relations are $s^2_{\partial_i} = 1$.

The quotient-orientation character system is a torsor under the displayed H^1-groups. The difference between the transported trivialization and the chosen trivialization on the target stratum is therefore the class $\omega_{w,R,C}$, and functoriality of transport gives the displayed torsor cocycle law. Vanishing on the retained simple generators, together with the square normalizations, propagates to every retained word in the finite action groupoid; on a genuine W-orbit this is the free product $C_2 * C_2 * C_2$. Compatibility in R is exactly the finite-stage inverse-system requirement in the compact/wrapped datum. No Maass character or scalar normalization enters the construction. □

Corollary 3.26 *(Semidirect wall-transport obstruction).* Let

$$G = W^{(2)}(\Lambda^{2,1}_{II}) \rtimes \mathrm{Aut}(\mathcal{P}_{II})$$

and suppose (O1) has been supplied on the reduced quotient strata. Let $\mathcal{G}_G$ be the action groupoid of the retained strata under the chosen G-transports; on a complete G-orbit with constant coefficients this reduces to ordinary group cohomology of G. An extension of the type-II Weyl orientation transport from W to G is an additional datum. It exists on the reduced quotient orientation if and only if the projective determinant-line cocycle for the chosen lifts to $\mathcal{G}_G$,

$$c^G_o \in Z^2(\mathcal{G}_G; \underline{\mathbb{F}}_2),$$

has trivial class in $H^2(\mathcal{G}_G; \underline{\mathbb{F}}_2)$, the chosen killing cochain normalizes the Coxeter relations in W and the S_3-relations in $\mathrm{Aut}(\mathcal{P}_{II})$, and the quotient-torsor defects

$$\omega_{g,C} \in H^1(\mathfrak{M}^{\mathrm{red}}_{gC}; \mathbb{F}_2) \oplus \bigoplus_{N \geq 2} H^1(BE[N]; \mathbb{F}_2)$$

vanish for the chosen generators g of G. The S_3-part must transport the full quotient package

$$\left(\alpha^{\mathrm{red}}, \alpha^{E,\mathrm{free}}, \{(\beta^{E,N}, \lambda^{E,N} = 0)\}_{N \geq 2}\right)$$

between chamber-permuted strata.

Consequently the Maass values

$$\nu_{\Delta_5}(s_{f_{-2}-f_2}) = \nu_{\Delta_5}(s_{f_2-f_3}) = \nu_{\Delta_5}(s_{f_2+f_3}) = +1$$

are only automorphic target values. They do not construct a Hall orientation sign on the semidirect-product generators. Without the vanishing above, the manuscript asserts the Hall orientation character only on $W^{(2)}(\Lambda^{2,1}_{II})$, under (O1), $(\mathrm{O1})^+$, and (O2).

Proof. The proof is the same obstruction calculation as Lemma 3.24, with $\mathcal{G}_W$ replaced by $\mathcal{G}_G$. A projective action on the determinant line is a central $\mathbb{F}_2$-extension of the retained action groupoid, hence is classified by $[c_o^G]$. On a complete G-orbit this is the usual group-cohomology class, and a splitting must respect all relations of the semidirect presentation, not only the three generator squares in W. After the ordinary determinant-line action is honest, the chosen quotient trivializations remain torsors under the displayed H^1-groups; their failure to be preserved is exactly $\omega_{g,C}$. Vanishing on a generating set, with the relation coherences, propagates through the retained action groupoid, and to all of G only after a complete G-orbit has been supplied. The Maass character is a scalar automorphic character and has no map to these determinant-line or quotient-torsor classes. □

Remark 3.27. The table values of ν_{Δ_5} on the six generators (Lemma 8.2), the determinant identity $\mathcal{D}_X = \Delta_5$ (Proposition 4.5), and the OP scalar $-4096\,\Delta_5^{-2}$ (Proposition 11.1) are independent of the compact Hall hypotheses; they are imported or proved under the cited source hypotheses, normalizations, and chamber conventions. The Pfaffian recognition (i), the orientation-character identity (ii), and the categorical upgrade (iii) use (O1), $(\mathrm{O1})^+$, and (O2). Lemma 3.22 identifies the strongest algebraic consequence of a supplied Weyl-equivariant lift: the sign factors through the $(\mathbb{Z}/2)^3$ abelianization of $W^{(2)}(\Lambda^{2,1}_{II})$. Lemma 3.24 separates this group-theoretic sign extension from the projective determinant-line and quotient-torsor obstructions that must vanish before there is an honest Weyl action on the reduced quotient orientation. Open: (O1) for the reduced E-quotiented Weyl-equivariant compact critical $K3 \times E$ Hall/CoHA sector. Joyce–Upmeier [46], Joyce–Tanaka–Upmeier [45], and Borisov–Joyce–Bojko [10] cover the unreduced $\mathrm{Perf}(X)$, and [46] covers shifted symplectic stacks, but the reduced E-quotient with type-II Weyl-equivariant lift is not in the literature; (O2) is the supplied Pfaffian local normal form on type-II walls of that sector, not a consequence of the automorphic product. The local obstruction datum of Proposition 3.13 names the missing chart, splitting, invariant-unit, and rank-one normal-mode computations. The squared scalar branch $Z^X_{\mathrm{OP}} = -4096\,\Delta_5^{-2}$ is orientation-forgetful; rescaling Δ_5 by a nonzero scalar leaves ν_{Δ_5} invariant. The three type-II rows match Borcherds automorphic divisor orders $d^{\mathrm{aut}}_{\partial_i} = 1$ via $f(nm, l)$ at $(1,1,0)$, $(0,1,1)$, $(0,-1,0)$; the independent automorphic check at the three chamber-permuting type-I roots yields divisor order 0 from the explicit support of $\phi_{0,1}$ [27, §2.2].

Open Problem 3.28 *(Protected one-particle lift).* Construct, if it exists, a protected one-particle object with a Γ_{eff}-graded super-vector-space decomposition

$$\mathcal{H}^{\mathrm{1p,ch}} = \bigoplus_{\gamma \in \Gamma_{\mathrm{eff}}} \mathcal{H}^{\mathrm{1p,ch}}_{\gamma}$$

with protected superdimensions

$$\mathrm{Tr}_{\mathcal{H}^{\mathrm{1p,ch}}_{(n,l,m)}} (-1)^F = f(nm, l).$$

The determinant construction uses only the virtual classes $\mathbb{U}_{n,l}$ and their signed dimensions. The algebraic object constructed from the same charge grading and the same signed multiplicities is the Borcherds denominator algebra. Such a lift would still be weaker than the holomorphic E_3-factorization algebra, because it would not by itself supply the collision law.

4 The normalized cusp form

Freitag [29], Van der Geer [77], and Igusa [42, 43]: the even-weight scalar subring is

$$M_*^{\mathrm{even}}(\mathbf{Sp}_4(\mathbb{Z})) = \mathbb{C}[E_4, E_6, \Delta_{10}, \Delta_{12}],$$

$$\mathrm{wt}(E_4, E_6, \Delta_{10}, \Delta_{12}) = (4, 6, 10, 12).$$

$$\Delta_{10} = \Delta_5^2, \qquad \mathrm{wt}(\Delta_5) = 5, \qquad \Delta_5|_5 g = \nu_{\Delta_5}(g)\Delta_5.$$

$$\Delta_5 = \prod_{\substack{a,b\in(\mathbb{Z}/2\mathbb{Z})^2\\ {}^t ab\equiv 0 \mod 2}} \nu_{a,b},$$

$$\nu_{a,b}(z) = \sum_{l\in\mathbb{Z}^2} \exp\bigl(\pi i(z[l+\tfrac{1}{2}a] + {}^t bl)\bigr), \qquad z[x] = {}^t xzx.$$

For $M = \begin{pmatrix} A & B \\ C & D \end{pmatrix} \in \mathbf{Sp}_4(\mathbb{Z})$, Maass's character [56]

$$\nu_{\Delta_5} : \mathbf{Sp}_4(\mathbb{Z}) \to \mathbb{C}^\times, \qquad |\nu_{\Delta_5}| = 1,$$

is determined on the standard generators by

$$\nu_{\Delta_5}\begin{pmatrix} 0 & \mathbf{I}_2 \\ -\mathbf{I}_2 & 0 \end{pmatrix} = 1,$$

$$\nu_{\Delta_5}\begin{pmatrix} \mathbf{I}_2 & B \\ 0 & \mathbf{I}_2 \end{pmatrix} = (-1)^{b_1+b_2+b_3},$$

$$\nu_{\Delta_5}\begin{pmatrix} {}^t A^{-1} & 0 \\ 0 & A \end{pmatrix} = (-1)^{(1+a_1+a_4)(1+a_2+a_3)+a_1a_4}$$

with $A = \begin{pmatrix} a_1 & a_2 \\ a_3 & a_4 \end{pmatrix} \in \mathbf{GL}_2(\mathbb{Z})$ and $B = \begin{pmatrix} b_1 & b_2 \\ b_2 & b_3 \end{pmatrix} \in M_{2\times 2}(\mathbb{Z})$.

Δ_5 is a weight-5 form with character ν_{Δ_5}, and $\Delta_5^2 = \Delta_{10}$ is scalar.

For

$$Z = \begin{pmatrix} z_1 & z_2 \\ z_2 & z_3 \end{pmatrix},$$

$$\Delta_5(Z) = \sum_{\substack{n,l,m\equiv 1 \mod 2\\ 4nm-l^2>0\\ n,m>0}} c_\Delta(n,l,m)\exp\bigl(\pi i(nz_1 + lz_2 + mz_3)\bigr),$$

with

$$c_\Delta(1,1,1) = 64, \qquad c_\Delta(n,l,m) \in 64\mathbb{Z}.$$

Proposition 4.1 *(Isotropic cusp coefficient).*

$$1 + \frac{1}{64}\sum_{t\geq 1} c_\Delta(1+2t,1,1)q^t = \prod_{k\geq 1}(1-q^k)^9.$$

Proof.

$$\Delta_5(Z) = \sum_{\substack{m>0\\ m\equiv 1 \bmod 2}} \phi_{5,m}(z_1,z_2)\exp(\pi i m z_3).$$

$$\phi_{5,1} \in J_{5,\frac{1}{2}}(\nu_{\Delta_5}).$$

$$\nu_{\mathrm{II}}(z_1,z_2) = \sum_{n\in\mathbb{Z}}(-1)^n \exp\left(\frac{\pi i}{4}(2n+1)^2 z_1 + \pi i(2n+1)z_2\right).$$

$$\nu_{\mathrm{II}} = -q^{1/8}r^{-1/2}\prod_{n\geq 1}(1-q^{n-1}r)(1-q^n r^{-1})(1-q^n),$$

$$q = \exp(2\pi i z_1), \qquad r = \exp(2\pi i z_2).$$

$$\psi_{5,\frac{1}{2}} = \eta(z_1)^9\nu_{\mathrm{II}}(z_1,z_2), \qquad \eta(z_1) = q^{1/24}\prod_{n\geq 1}(1-q^n).$$

$$\frac{1}{64}\phi_{5,1}(z_1,z_2) = \psi_{5,\frac{1}{2}}(z_1,z_2) = -q^{1/2}r^{-1/2}\prod_{n\geq 1}(1-q^{n-1}r)(1-q^n r^{-1})(1-q^n)^{10}.$$

$$1 + \frac{1}{64}\sum_{t\geq 1} c_\Delta(1+2t,1,1)q^t = \prod_{k\geq 1}(1-q^k)^9.$$

□

Definition 4.2 (Massless divisor and determinant sign).

$$\mathcal{H}_{\mathrm{diag}} = \left\{\begin{pmatrix} a & 0\\ 0 & b\end{pmatrix} \mid a,b\in\mathbb{H}_1\right\}.$$

Cléry–Gritsenko [35], $N = 1$:

$$\operatorname{supp}\operatorname{div}(\Delta_5) = \mathbf{Sp}_4(\mathbb{Z})\cdot\mathcal{H}_{\mathrm{diag}}, \qquad \operatorname{mult}_{\mathcal{H}_{\mathrm{diag}}}(\Delta_5) = 1.$$

$$\mathcal{M}_{\mathrm{massless}} := \operatorname{supp}\operatorname{div}(\Delta_5), \qquad \epsilon_{\det} := \nu_{\Delta_5}.$$

$$\Delta_5|_5 g = \nu_{\Delta_5}(g)\Delta_5, \qquad \Delta_5(s_{\delta_i}Z) = -\Delta_5(Z).$$

The displayed sign on the type-II diagonal reflections s_{δ_i}, $i = 1,2,3$, is the Maass-character verification carried out in Lemma 8.2 below; the $\nu_{\Delta_5}(s_{\delta_i}) = -1$ line of that lemma is the input used here.

Definition 4.3 (Orientation line of the protected sector). Let C be a stable object in the reduced compact critical Hall/CoHA sector envisaged in Definition 5.4, with derived self-Ext line after cosection reduction

$$\det \mathrm{RHom}_{\mathrm{red}}(C,C) \in \mathrm{Pic}(\mathfrak{M}_X^{or}).$$

By an *orientation line* of the protected sector is meant a line bundle

$$o_C \in \mathrm{Pic}(\mathfrak{M}_X^{or})$$

together with the squaring isomorphism

$$o_C^{\otimes 2} \simeq \det \mathrm{RHom}_{\mathrm{red}}(C, C),$$

fitting into a strong orientation datum compatible with the extension correspondences of Definition 5.4. This is the existence frame for orientations on critical and (-1)-shifted symplectic moduli; on Calabi–Yau threefolds the orientation-data formalism is Joyce–Song [44], and on Calabi–Yau 4-folds it is the Borisov–Joyce / Oh–Thomas construction [10], [68], [67]. The $K3 \times E$ case is neither of these: its reduced compact critical sector is not constructed in this paper, and the existence of a square-root line o_C for that sector is one of the unresolved data items of Open Problem 5.7. No such line is constructed here.

Remark 4.4 *(Polarization is independent of orientation).* The identity

$$B(c, c) = 2\Pi_X(c)$$

is the polarization identity for the quadratic Gram map Π_X. It is independent of the parity split of a protected cohomology representative and independent of the choice of an orientation line. Orientation data enter later, when one tries to promote a determinant section to a Pfaffian section and to compute its monodromy under wall reflections. It does not change the Gram polarization. In particular no Bott-periodicity convention, Grothendieck–Witt sign convention, or choice of Bott element is used to prove $B(c, c) = 2\Pi_X(c)$. Such conventions may rigidify an already constructed orientation theory; they do not supply the lattice polarization, the reduced E-quotient linearization, or the compact Pfaffian sign.

Proposition 4.5 *(Borcherds–Gritsenko–Nikulin identification).* By the Borcherds–Gritsenko–Nikulin product formula for $\phi_{0,1}$,

$$\mathcal{D}_X(Z) = \Delta_5(Z).$$

Proof. The general Borcherds product mechanism is [8, Theorem 10.1]; no scalar normalization is taken from that general theorem. The exact genus-two specialization used here is Gritsenko–Nikulin: [39, Theorem 2.1 and Example 2.4, especially (2.15)–(2.16)] gives the monic product

$$D_5(Z) = q^{1/2} r^{1/2} s^{1/2} \prod_{(n,l,m) \in \Gamma_{\mathrm{eff}}} (1 - q^n r^l s^m)^{f(nm,l)}.$$

GNII denotes this monic product by Δ_5. This paper reserves Δ_5 for the theta-constant normalization. In that convention, [37, Theorem 4.1] gives

$$D_5(Z) = 64^{-1} \Delta_5(Z).$$

Hence

$$64 q^{1/2} r^{1/2} s^{1/2} \prod_{(n,l,m) \in \Gamma_{\mathrm{eff}}} (1 - q^n r^l s^m)^{f(nm,l)} = \Delta_5(Z).$$

$$\mathcal{D}_X(Z) = \Delta_5(Z).$$

□

Remark 4.6 *(Normalization of the square root).* The constants in the square-root product are fixed by the following chain:

Object	Fixed value
Fourier variables	$q = e^{2\pi i z_1},\ r = e^{2\pi i z_2},\ s = e^{2\pi i z_3}$
Borcherds input	$Z_{\mathrm{K3}} = 2\phi_{0,1}$
Borcherds product normalization	$D_5^{\mathrm{mon}} = D_5^{\mathrm{GNII}} = 64^{-1}\Delta_5^{\theta}$, and this paper writes $\Delta_5 = \Delta_5^{\theta}$
Weight	$\mathrm{wt}\ \mathrm{Borch}_\theta(\phi_{0,1}) = f(0,0)/2 = 5$
Igusa leading coefficient	$[q^{1/2}r^{1/2}s^{1/2}]\Delta_5 = 64$
Monic product	$D_5 = 64^{-1}\Delta_5 = q^{1/2}r^{1/2}s^{1/2}\prod(1-q^n r^l s^m)^{f(nm,l)}$
Determinant normalization	$\mathcal{D}_X = \Delta_5, \quad v_0 = 64q^{1/2}r^{1/2}s^{1/2}$
Virtual inverse	$\mathcal{Z}_{\mathrm{Borch}}^{\mathrm{vir}} = 64^{-1}q^{-1/2}r^{-1/2}s^{-1/2}\,\mathrm{sch}\,\mathcal{F}(\mathcal{V})$
BKM denominator	$\mathrm{den}(\mathfrak{g}_{\Delta_5}) = D_5(2Z) = 64^{-1}\Delta_5(2Z)$
Scalar square	$Z_{\mathrm{OP}}^X = -4096\Delta_5^{-2}$; in general $Z_\square^X = C_\square\Delta_5^{-2}$, hence $(\mathcal{Z}_{\mathrm{Borch}}^{\mathrm{vir}})^2 = Z_\square^X/C_\square$

Proposition 4.7 *(Borcherds–Gritsenko weight identity at $N = 1$).* Let $\phi_{0,1} \in J_{0,1}^{\mathrm{weak}}$ be the weight-zero index-one weak Jacobi form with Fourier expansion $\phi_{0,1}(z_1, z_2) = \sum_{n,l} f(n,l)\, q^n r^l$. The theta-normalized Borcherds lift at $N = 1$ is Δ_5, and the Borcherds weight reads

$$\mathrm{wt}(\Delta_5) = \frac{f(0,0)}{2} = 5 = \frac{c_1(0)}{2},$$

with $c_1(0) := f(0,0) = 10$. Since the weight is integral, no metaplectic cover is needed for the weight factor.

Proof. The Borcherds singular-theta regularisation of $\phi_{0,1}$ produces a meromorphic Siegel modular form of weight $f(0,0)/2$; at $N = 1$ the paramodular group is $\mathbf{Sp}_4(\mathbb{Z})$ by Gritsenko–Nikulin [37]. The input Jacobi form has $f(0,0) = 10$ (Eichler–Zagier [27], elliptic genus of K3), hence the output weight is 5. Identifying $\mathrm{Borch}_\theta(\phi_{0,1})$ with Δ_5 via the Igusa identification ($\mathcal{D}_X = \Delta_5$) fixes $\mathrm{wt}(\Delta_5) = 5$. Setting $c_1(0) := f(0,0)$ exhibits the row ($N = 1,\ \mathrm{wt} = 5,\ c_1(0) = 10$) of the Cléry–Gritsenko catalogue [35]; the remaining integral and half-integral Cléry–Gritsenko rows are listed in [35, Theorem 1.4]. Integrality $\mathrm{wt}(\Delta_5) \in \mathbb{Z}$ means that no metaplectic cover is required for the weight factor. The Hodge automorphic line $\mathcal{L}^5$ remains nontrivial, and the Maass character ν_{Δ_5} is a finite-order character of $\mathbf{Sp}_4(\mathbb{Z})$. $\square$

Part II

Physical charge and normal-ordered descent

5 D6–D2–D0 PROTECTED-INDEX DICTIONARY

Only the OP/DT scalar branch requires an elliptically fibered K3 surface with a section. The Borcherds determinant above used only the universal K3 elliptic genus.

Let S be an elliptically fibered K3 surface

$$\pi : S \to \mathbb{P}^1$$

with a section s_1, and let E be an elliptic curve. Set

$$X = S \times E.$$

Let $F \in \mathrm{Pic}(S)$ be the fibre class and put

$$\beta_h = s_1 + hF, \qquad h \geq 0.$$

Type IIA D6–D2–D0 charge:

$$\gamma(h, d, n) = (1;\ (\beta_h, d);\ n).$$

Write $\Omega_X(h, d, n)$ for the putative protected D6–D2–D0 index attached to this charge in a compact physical realization. The microscopic trace is not constructed here; the desired datum would be:

$$\overset{?}{\exists}\ \mathcal{H}_{\gamma(h,d,n)}, \qquad \mathrm{Tr}_{\mathcal{H}_{\gamma(h,d,n)}}(-1)^F = \Omega_X(h, d, n).$$

$$\mathrm{Hilb}^n(X, (\beta_h, d)) = \{Y \subset X \mid [Y] = (\beta_h, d),\ \chi(\mathcal{O}_Y) = n\}.$$

$$E \curvearrowright X, \qquad (x, e') \mapsto (x, e + e').$$

THEOREM 5.1 *(D6–D2–D0 Mukai–Gram dictionary).* Let $X = S \times E$ with S K3 and E elliptic. Let $Y \subset X$ be a one-dimensional subscheme with class $[Y] = (\beta, d) \in H_2(S, \mathbb{Z}) \oplus H_2(E, \mathbb{Z})$ and $\chi(\mathcal{O}_Y) = n \in \mathbb{Z}$. Let $\mathcal{I}_Y \in D^b(X)$ be its ideal sheaf. Decompose the Mukai vector

$$v_X(\mathcal{I}_Y) = \mathrm{ch}(\mathcal{I}_Y) \cdot \sqrt{\mathrm{td}(X)}$$

against the Künneth basis $\widetilde{H}(S, \mathbb{Z}) \otimes \{1_E, \omega_E\}$, where $\widetilde{H}(S, \mathbb{Z}) = H^0(S, \mathbb{Z}) \oplus H^2(S, \mathbb{Z}) \oplus H^4(S, \mathbb{Z})$ carries the Mukai pairing

$$(r, D, s) \cdot (r', D', s') = D \cdot D' - rs' - r's$$

and $\omega_E \in H^2(E, \mathbb{Z})$ is the fundamental class. Then

$$v_X(\mathcal{I}_Y) = (1,\ 0,\ 1 - d) \otimes 1_E + (0,\ -\beta,\ -n) \otimes \omega_E.$$

With

$$P_Y = (1, 0, 1 - d), \qquad Q_Y = (0, -\beta, -n),$$

the Gram-degree triple is

$$\left(\frac{Q_Y^2}{2},\ Q_Y \cdot P_Y,\ \frac{P_Y^2}{2}\right) = \left(\frac{\beta^2}{2},\ n,\ d - 1\right).$$

In particular, for $\beta = \beta_h = s_1 + hF$ with $\beta_h^2 = 2h - 2$,

$$\Pi_X(Q_Y, P_Y) = (h - 1, n, d - 1).$$

Proof. Since $X = S \times E$ is a Calabi–Yau threefold,

$$\mathrm{ch}(\mathcal{I}_Y) = 1 - \mathrm{ch}(\mathcal{O}_Y).$$

For a one-dimensional sheaf on a Calabi–Yau threefold,

$$\mathrm{ch}_2(\mathcal{O}_Y) = \mathrm{PD}[Y], \qquad \int_X \mathrm{ch}_3(\mathcal{O}_Y) = \chi(\mathcal{O}_Y) = n.$$

The Todd square root factorizes as

$$\sqrt{\mathrm{td}(X)} = \pi_S^* \sqrt{\mathrm{td}(S)}\, \pi_E^* \sqrt{\mathrm{td}(E)} = \pi_S^*(1, 0, 1) \cup 1,$$

since $\sqrt{\mathrm{td}(E)} = 1$ and $\sqrt{\mathrm{td}(S)} = (1, 0, 1)$ in Mukai notation on a K3 surface. Decomposing the curve class, the K3-vertical component $\beta \times \{\mathrm{pt}\}$ contributes $\beta^\vee \otimes \omega_E$, while the elliptic component $d\{\mathrm{pt}\} \times E$ contributes $d[\mathrm{pt}_S] \otimes 1_E$. Thus

$$\mathrm{ch}(\mathcal{O}_Y) = (0, 0, d) \otimes 1_E + (0, \beta, n) \otimes \omega_E.$$

Multiplying

$$1 - \mathrm{ch}(\mathcal{O}_Y) = (1, 0, -d) \otimes 1_E + (0, -\beta, -n) \otimes \omega_E$$

by $(1, 0, 1) \otimes 1_E$ gives

$$v_X(\mathcal{I}_Y) = (1, 0, 1 - d) \otimes 1_E + (0, -\beta, -n) \otimes \omega_E,$$

because $\beta^\vee[\mathrm{pt}_S] = 0$ on the K3 surface.

The Mukai pairing on $\widetilde{H}(S, \mathbb{Z})$ is

$$(r, D, s) \cdot (r', D', s') = D \cdot D' - rs' - r's.$$

Therefore

$$\frac{Q_Y^2}{2} = \frac{\beta^2}{2}, \qquad Q_Y \cdot P_Y = (-\beta) \cdot 0 - 0(1 - d) - 1(-n) = n,$$

and

$$\frac{P_Y^2}{2} = \frac{-2(1)(1 - d)}{2} = d - 1.$$

For $\beta = \beta_h$, $\beta_h^2 = 2h - 2$, this gives $\Pi_X(Q_Y, P_Y) = (h - 1, n, d - 1)$. □

The Gram-degree triple $(Q^2/2,\ Q \cdot P,\ P^2/2) = (h - 1,\ n,\ d - 1)$ is the Gram-index shadow of the Hilbert/PT charge. It is not itself an additive Hall charge in the Mukai lattice of X. This is the Gram-index data used by the Borcherds product and by the OP normalization of Theorem 11.2. The hypothesis $\beta_h^2 = 2h - 2$ of Theorem 11.2 is exactly the $Q^2/2 = h - 1$ line of this theorem.

Proposition 5.2 *(Reduced scalar quotient integration).* The reduced DT/PT side used in this paper is the coefficientwise scalar Behrend-weighted quotient integration

$$\mathbf{DT}^X_{n,(\beta,d)} = \int_{\mathrm{Hilb}^n(X,(\beta,d))/E} \nu_{\mathrm{Hilb}}\, de,$$

and

$$\mathsf{P}^X_{n,(\beta,d)} = \int_{P_n(X,(\beta,d))/E} \nu_P\, de$$

for the stable-pair moduli space $P_n(X,(\beta,d))$. It lands in numbers, coefficient by coefficient. Oberdieck's reduced stable-pair quotient invariant agrees with the reduced virtual-class incidence definition, and his reduced DT/PT theorem identifies the two reduced quotient series for nonzero K3 class [64, Theorem 3(i)]. This is scalar integration. It is not a Hall product, not a CoHA integration map, and not an orientation character.

Proof. Behrend's constructible function gives the weighted Euler characteristic used in Donaldson–Thomas theory [4], [74]. The E-action on $X = S \times E$ forces the ordinary unreduced invariant to vanish, so the reduced theory integrates over the quotient by E. Bryan's Hilbert-scheme normalization and Oberdieck's stable-pair normalization are exactly the quotient Behrend-weighted Euler characteristics [13], [64]. Oberdieck proves the quotient stable-pair definition equals the reduced incidence invariant [64, Theorem 1] and proves the reduced DT/PT comparison for nonzero K3 class [64, Theorem 3(i)]. These statements produce integers indexed by (β, d, n); no algebra object or primitive projection is constructed. □

Conjecture 5.3 *(Protected D-brane index).* For the D6–D2–D0 charge

$$\gamma(h,d,n) = (1;\ (\beta_h, d);\ n),$$

if such a compact physical Hilbert space is supplied, the conjectural enumerative dictionary identifies its protected trace with the Behrend-weighted Hilbert-scheme quotient invariant:

$$\Omega_X(h,d,n) := \mathrm{Tr}_{\mathcal{H}_{\gamma(h,d,n)}}(-1)^F \overset{\text{phys}}{=} \mathbf{DT}^X_{n,(\beta_h,d)}.$$

This physical identification is not used in the scalar theorem below.

Definition 5.4 *(Protected Hall integration criterion).* A protected Hall or critical CoHA integration map refining Proposition 5.2 would have to supply the following data before comparison with $\mathfrak{g}_{\Delta_5}$: an extension-closed compact Hall/CoHA sector for the reduced E-quotient, oriented critical extension correspondences, vanishing-cycle coefficient systems, HN-finite sector completions, protected integration compatible with the completed Hall product and wall crossing, a primitive projection, and a lifted Weyl action on the oriented completed theory.

If these data exist, the test is rigid: protected primitives must have signed dimensions

$$\mathrm{sdim}\ \mathrm{Prim}_{(n,l,m)} = f(nm, l),$$

their bracket must be the Borcherds–Gritsenko–Nikulin bracket, and the induced orientation character must restrict on the type-II Weyl group as

$$\epsilon_0|_{W^{(2)}(\Lambda^{2,1}_{II})} = \nu_{\Delta_5}|_{W^{(2)}(\Lambda^{2,1}_{II})}.$$

General CoHA technology supplies a language for such conditions [51], [23]; it does not construct this compact $K3 \times E$ datum.

LEMMA 5.5 *(Brauer-class parity criterion for Pin^c-lift gerbes).* Let C be a protected Mukai charge on the reduced $K3 \times E$ sector, and let α^{red}_C and α^E_C be the reduced Pin^c-lift gerbe class and the E-equivariant descent obstruction of Open Problem 5.7. On any projection-finite stratum on which:

(i) a GRR/Stiefel–Whitney comparison identifies the reduced orientation gerbe of the self-Ext complex with the numerical parity class $c_1(L)^2/2 \bmod 2$ of a K3 chamber polarization line L, and

(ii) $c_1(L)^2/2 \equiv 0 \pmod 2$, and

(iii) the E-descent parity $c_1(\text{degree}) \cup c_1(\text{level})$ is even,

one has

$$\alpha_C^{\text{red}} = 0, \qquad \alpha_C^E = 0$$

on that stratum. In particular this proves the claimed vanishing on rank-two chambers satisfying the displayed parity and on HN strata whose graded factors reduce to rank-one/rank-two pieces on which the ordinary and quotient orientation classes have already vanished. It does not prove the length-one stable higher-rank case.

Proof. The BBDJS orientation gerbe of the reduced self-Ext complex is the second Stiefel–Whitney class of the determinant line, equivalently the mod-two Brauer class of the square-root gerbe [11]. On a stratum satisfying the first hypothesis the supplied GRR/Stiefel–Whitney comparison represents this class by

$$\frac{c_1(L)^2}{2} \bmod 2.$$

The second hypothesis therefore gives $\alpha_C^{\text{red}} = 0$. The equivariant Serre spectral sequence for the free E-translation quotient identifies the connected descent obstruction with $c_1(\text{degree}) \cup c_1(\text{level}) \bmod 2$, so the third hypothesis gives $\alpha_C^E = 0$. Without the first comparison, the displayed parity is only a numerical diagnostic, not a proof of gerbe triviality. Additivity of orientation gerbes on HN direct sums, from the Joyce–Upmeier / Joyce–Tanaka–Upmeier multiplicative orientation formalism [45, 46], then transports the vanishing to HN strata whose graded factors all lie in the proved rank-one/rank-two sectors; off-diagonal self-Ext blocks occur in dual pairs and contribute even Stiefel–Whitney class. No induction over a stable length-one higher-rank factor is used. □

Remark 5.6. Lemma 5.5 reads the parity off the Chern integrand of the chamber polarization; Lemma 5.18 reads it off the HN-filtration data. Both are reduction criteria: they agree with the rank-one base case (P1) and with each other on the rank-two/HN-reducible strata after the rank-one finite-data checks of (P1) have been performed, but neither closes stable higher-rank length-one strata.

Open Problem 5.7 *(Orientation monodromy character).* The geometric orientation monodromy character

$$\epsilon_o : W^{(2)}(\Lambda_{II}^{2,1}) \longrightarrow \{\pm 1\},$$

expected to satisfy

$$\epsilon_o|_{W^{(2)}(\Lambda_{II}^{2,1})} = \nu_{\Delta_5}|_{W^{(2)}(\Lambda_{II}^{2,1})},$$

is independent of the normalization of Δ_5. Multiplying Δ_5 by a nonzero scalar leaves its Maass character unchanged. The character is obtained by transporting the square root of the determinant line around the simple diagonal wall reflections; the expected local model is Pfaffian, in the sense made precise as hypothesis (O2) of Theorem 3.15: a simple real-root reflection reverses one oriented normal mode, so its square is scalar while the square root changes sign.

(P1) *Rank-one D6–D2–D0 stratum: ordinary Pin^c input and quotient checks.* On the rank-one D6–D2–D0 stratum $\mathfrak{M}_C^{\text{red},(1)}$ of physical Mukai charge $C = (1, \beta, n)$, the desired reduced orientation line $o_C^{\text{red},(1)} \in \text{Pic}(\mathfrak{M}_C^{\text{red},(1)})$ is reduced to four pieces of data: an ordinary Pin^c lift of the cosection-reduced self-Ext frame bundle, a connected BE-equivariant linearization class, finite-stabilizer gerbe classes on the small-orbit strata, and the residual finite-stabilizer H^1-linearization characters after those gerbes

have been null-trivialized. The standard orientation technologies [11, 16, 45, 46] provide the local Pin^c frame and sign-rigidification model used in Lemma 5.16. They do not, without the following quotient computations, construct the reduced E-quotient orientation. The ordinary obstruction class is

$$\alpha_C^{\mathrm{red}} := w_2\big(\det \mathrm{RHom}_{\mathrm{red}}(C,C)\big) + \mathrm{cs}^* w_2\big(\det(\mathrm{coker}\ \mathrm{cs})\big) \ \in\ H^2(\mathfrak{M}_C^{\mathrm{red}};\mathbb{Z}/2)$$

of the reduced orientation gerbe. The connected quotient class has the form

$$\alpha_C^{E,\mathrm{free}} = a_1(C)u_1 + a_2(C)u_2 \in H^2(BE;\mathbb{F}_2),$$

with u_1, u_2 as in Lemma 5.8. On the $N = 2$ small-orbit stratum the finite residual is

$$\beta_C^{E,E[2]} = b_{20}(C)x_1^2 + b_{11}(C)x_1x_2 + b_{02}(C)x_2^2 \in H^2(BE[2];\mathbb{F}_2).$$

After $\beta_C^{E,E[2]} = 0$, the chosen two-torsion linearization has a degree-one character

$$\lambda_C^{E[2]} = \lambda_1(C)x_1 + \lambda_2(C)x_2 \in H^1(BE[2];\mathbb{F}_2).$$

Thus rank-one (O1) through the two-torsion small-orbit boundary is proved exactly when

$$\alpha_C^{\mathrm{red}} = 0, \qquad a_1(C) = a_2(C) = 0, \qquad b_{20}(C) = b_{11}(C) = b_{02}(C) = 0, \qquad \lambda_1(C) = \lambda_2(C) = 0.$$

The first equality is the ordinary Pin^c orientation calculation. The remaining bits are quotient gerbe and linearization data. The three finite-sector bits r_1, r_2, r_3 of Proposition 5.12 are equivalent to the three coefficients b_{20}, b_{11}, b_{02}, but they do not see the connected BE-class $a_1u_1 + a_2u_2$ or the degree-one finite character $\lambda_C^{E[2]}$. Thus the three cyclic degree-two bits cannot by themselves be promoted to a proof of rank-one (O1). Rank-one (O1) through the two-torsion boundary is proved only after the ordinary class, the two connected coefficients, the three finite cyclic degree-two residuals, and the finite degree-one character have all been computed to vanish. None of these vanishing statements is implied by ordinary translation invariance or by the OP scalar square. Full rank-one descent through all small-orbit strata also requires $\beta_C^{E,N} = 0$ and the trivial $H^1(BE[N];\mathbb{F}_2)$ linearization character for every even $N \geq 4$, with the two-primary residual described in Lemma 5.13; the $E[2]$ bits do not imply those vanishings. Higher-rank semistable strata remain open except where the HN criteria below reduce them to already-oriented factors.

(P2) *Type-II diagonal reflections: wall-character formula.* Assume (P1), the multiplicative orientation cocycle of Joyce–Upmeier [46] and Joyce–Tanaka–Upmeier [45] extending equivariantly across the reduced E-quotient, and the Pfaffian local model (O2) of Theorem 3.15, equivalently the three local calculations isolated in Proposition 3.13, the induced orientation character satisfies $\epsilon_o(s_{\partial_i}) = -1 = \nu_{\Delta_5}(s_{\partial_i})$, $i = 1, 2, 3$, so $\epsilon_o = \nu_{\Delta_5}$ on $W^{(2)}(\Lambda_{II}^{2,1})$. The three $\mathrm{Aut}(\mathcal{P}_{II}) \simeq S_3$ chamber-permuting reflections have Maass value $+1$, but Hall signs on them require a separately supplied semidirect-product lift. This is exactly conclusion (ii) of Theorem 3.15, under (O1), $(\mathrm{O1})^+$, and (O2), with the explicit row-by-row calculation in Lemma 3.18.

(P3) *Higher-rank obstruction.* The required task is to compute the obstruction classes

$$\alpha_C^{\mathrm{red}} \ \in\ H^2(\mathfrak{M}_C^{\mathrm{red}};\mathbb{Z}/2), \qquad \alpha_C^{E} \ \in\ H_E^2(\mathfrak{M}_C^{\mathrm{red}};\mathbb{F}_2),$$

where α_C^{red} is the Pin^c-lift gerbe class of the reduced derived self-Ext frame bundle on $\mathfrak{M}_C^{\mathrm{red}}$, and α_C^E is the equivariant-linearization obstruction to descent of the orientation along the free E-translation quotient.

On the rank-one stratum it should be computed, not inferred from the OP scalar, by a direct Chern-class calculation in the $E_2^{2,0}$-term of the equivariant Serre spectral sequence $\mathfrak{M}_C^{\mathrm{red}} \to \mathfrak{M}_C^{\mathrm{red}}/E \to BE$ of Lemma 5.8. One must then prove their joint vanishing on the Harder–Narasimhan-completed semistable stack $\mathfrak{M}_C^{\mathrm{red,ss,HN}}$. The small-orbit gerbe contribution $\overline{\beta}_C^E$ of Lemma 5.8(iv) from the $N \geq 2$ torsion locus $K3 \times (E/E[N]) \subset X$ is governed by the finite-stabilizer criterion of Lemma 5.9: the band at $N = 2$ is the Klein-four $BE[2]$ (not cyclic $B\mu_2$), and vanishing requires a trivial $E[2]$-equivariant linearization of the orientation line, detected on the three nontrivial cyclic subgroups. Granted the ordinary Pin^c gerbe vanishing, the connected BE-linearization vanishing, and this finite-stabilizer vanishing, (O1) would follow from the Joyce–Upmeier [46] multiplicative cocycle plus the supplied Pin^c lift; granted (O2), the wall-character formula of (P2) would follow by Theorem 3.15. On higher-rank semistable strata the joint vanishing of α_C^{red} and α_C^E is reduced by Lemma 5.18 to HN strata whose graded factors are already oriented; stable higher-rank length-one strata and even finite-stabilizer strata not covered by a trivial linearization remain open. The Pin^c-lift program of [46], [45], [16], [6], [11] supplies the analytic technology on the cosection-twisted reduced E-quotient; non-compact and Joyce–Song extensions are Bojko [6].

(P4) *Cosection-induced parity flip from σ_{K_3}:* $\mathrm{vd}^{\mathrm{red}} = \mathrm{vd}^{\mathrm{std}} + 1$*; absorbed into (O1) as a $\mathbb{Z}/2$-grading shift on o_C.* The Maulik–Toda / Oberdieck reduced obstruction theory [59, 64] contracts the PTVV [69] (-1)-shifted symplectic form $\omega_{\mathfrak{M}}$ on $\mathfrak{M}_C(X)$ against the cosection $\mathrm{cs}\colon \mathrm{Obs}_{\mathfrak{M}_C} \to \mathcal{O}_{\mathfrak{M}_C}$ generated by Mukai dilation along σ_{K_3}, and the restriction $\omega_{\mathfrak{M}}^{\mathrm{red}}$ to the cosection vanishing locus $\mathfrak{M}_C^{\mathrm{red}} \hookrightarrow \mathfrak{M}_C$ is non-degenerate on the OP/DT D6–D2–D0 stratum, with virtual dimension shifted by one, $\mathrm{vd}_C^{\mathrm{red}} = \mathrm{vd}_C^{\mathrm{std}} + 1$. After a reduced square-root orientation datum has been supplied, this dimension shift would propagate as a $\mathbb{Z}/2$-grading shift on the orientation line. This is the analogue, in the reduced $K3 \times E$ rank-one sector, of the square-root sign choice in the BBDJS oriented d-critical formalism [11, Theorem 6.9], together with a separately supplied Joyce–Upmeier-compatible multiplicative orientation datum [46, Theorem 4.1]:

$$o_C^{\mathrm{red}} \;=\; o_C^{\mathrm{std}} \otimes \Pi,$$

where Π is the parity-flip line generated by the cosection cokernel. The strong orientation datum (O1) includes the coherence datum that makes this shift commute with the extension correspondences of Definition 5.4 and preserves the Joyce–Upmeier multiplicative cocycle of (P2). Oberdieck proves the quotient/incidence comparison in [64, Theorem 1] and the reduced DT/PT comparison for nonzero K3 class in [64, Theorem 3(i)]. This fixes the scalar reduced PT/DT comparison, not an orientation branch. The parity-flip is absorbed into (O1): the strong orientation line o_C demanded by Theorem 3.15 is, on the reduced sector, o_C^{red}, not the unreduced BBDJS line, and the squaring isomorphism is to the reduced derived self-Ext $\det \mathrm{RHom}_{\mathrm{red}}(C,C) = \mathrm{cofib}(\sigma_{K_3}\colon \mathrm{RHom}(C,C)[-2] \to \mathrm{RHom}(C,C))$. On $X = S \times E$, with $\sigma = \mathrm{pr}_S^* \sigma_S$, the chain-level cosection on a perfect object F is the semiregularity map

$$\mathrm{cs}_F(\xi) = \int_X \sigma \wedge \mathrm{Tr}\big((\xi \otimes 1_{\Omega^1_X}) \circ \mathrm{At}(F)\big), \qquad \xi \in \mathrm{Ext}^2(F,F),$$

or equivalently a degree-one map

$$\mathrm{CS}_F : R\Gamma R\mathcal{H}om(F,F)[1] \longrightarrow \mathbb{C}[1].$$

For a universal exact triangle $A \to C \to B \to A[1]$, this cosection is additive on the filtered Hall deformation complex:

$$q^* \,\mathrm{CS}_C = p_1^* \,\mathrm{CS}_A + p_2^* \,\mathrm{CS}_B\,.$$

Consequently the cosection part of reduced Thom–Sebastiani descent vanishes on the actual exact-triangle and two-step flag stacks. This is not an equality on the unrestricted obstruction complex of the middle object. With a local splitting of the extension,

$$\mathrm{At}(C) = \begin{pmatrix} \mathrm{At}(A) & \nabla e \\ 0 & \mathrm{At}(B) \end{pmatrix},$$

a lower off-filtration obstruction $\zeta \in \mathrm{Ext}^2(A, B)$ gives the residual

$$\mathcal{R}_{A,B}(\zeta) = \int_X \sigma \wedge \mathrm{Tr}(\zeta \circ \nabla e).$$

For $A = B = k_x$, an E-direction extension class e, and K3-plane ζ with $\sigma_S(\zeta) = 1$, this residual is nonzero. Thus reduced TS is healed precisely by restricting to the filtered Hall correspondence; an unrestricted middle-object statement is false. Operationally this parity shift belongs to the reduced orientation datum and to any eventual Berezinian convention for a compact first-order object. The overall minus in $Z^X_{\mathrm{OP}} = -4096\,\Delta_5^{-2}$ is the scalar OP/DT branch convention. It may be compatible with a chosen fermionic grading, but it is not evidence for the Hall orientation character. The reflection character ϵ_o must be computed from monodromy of the orientation line around the type-II walls. The (P3) residual is the joint vanishing of $\alpha^{\mathrm{red}}_C + \alpha^E_C$ on the HN-completed semistable stack with $\mathrm{rk}\, C \geq 2$ and on the small-orbit $BE[N]$ finite-stabilizer strata. Lemma 5.18 and Lemma 5.5 reduce the computation for the HN-reducible and rank-two parity pieces; Lemma 5.9 gives the $N = 2$ Klein-four detection criterion. Stable higher-rank length-one strata and nontrivial finite-stabilizer linearizations remain genuine open obligations.

LEMMA 5.8 *(Connected and finite-stabilizer E-descent of the orientation line).* Let $C \in \Gamma^{\mathrm{phys}}_X$ and let $\mathfrak{M}^{\mathrm{red}}_C$ be the cosection-reduced critical Hall stack of Definition 5.4. The translation action of E on $X = K3 \times E$ lifts to an action on $\mathfrak{M}^{\mathrm{red}}_C$, free outside the small-orbit locus

$$\mathfrak{M}^{\mathrm{red,small}}_C = \bigcup_{N \geq 2} \mathfrak{M}^{\mathrm{red}}_C\big(K3 \times (E/E[N])\big) \;\subset\; \mathfrak{M}^{\mathrm{red}}_C,$$

and on the free locus produces a quotient stack $\overline{\mathfrak{M}}^{\mathrm{red}}_C = \mathfrak{M}^{\mathrm{red}}_C / E$. Suppose the unreduced orientation o^{red}_C of (P1) of Open Problem 5.7 exists on the free locus. Its descent to a line $\overline{o}^{\mathrm{red}}_C \in \mathrm{Pic}(\overline{\mathfrak{M}}^{\mathrm{red}}_C)$ is governed by the equivariant Serre spectral sequence

$$E_2^{p,q} = H^p(BE;\, H^q(\mathfrak{M}^{\mathrm{red}}_C; \mathbb{Z}/2)) \implies H_E^{p+q}(\mathfrak{M}^{\mathrm{red}}_C; \mathbb{Z}/2)$$

attached to the fibration $\mathfrak{M}^{\mathrm{red}}_C \to \overline{\mathfrak{M}}^{\mathrm{red}}_C \to BE$. The connected E-quotient and the finite stabilizer strata have different cohomology and must be treated separately.

(i) (*Connected free E-descent.*) Topologically $E^{\mathrm{top}} \simeq T^2$, hence

$$BE \simeq BT^2 \simeq K(\mathbb{Z}^2, 2), \qquad H^*(BE; \mathbb{F}_2) = \mathbb{F}_2[u_1, u_2], \quad |u_i| = 2.$$

Therefore

$$H^1(BE; \mathbb{F}_2) = 0, \qquad H^2(BE; \mathbb{F}_2) = \mathbb{F}_2 u_1 \oplus \mathbb{F}_2 u_2.$$

In particular there is no connected-torus obstruction in bidegree

$$E_2^{1,1} = H^1(BE;\mathbb{F}_2) \otimes H^1(\mathfrak{M}_C^{\mathrm{red}};\mathbb{F}_2) = 0.$$

The degree-two obstruction on the free locus has only the two standard components

$$E_2^{2,0} = H^2(BE;\mathbb{F}_2) \otimes H^0(\mathfrak{M}_C^{\mathrm{red}};\mathbb{F}_2)$$

and

$$E_2^{0,2} = H^0(BE;\, H^2(\mathfrak{M}_C^{\mathrm{red}};\mathbb{F}_2)).$$

Thus the free-locus descent obstruction is a class

$$\alpha_C^{E,\mathrm{free}} \in H_E^2(\mathfrak{M}_C^{\mathrm{red}};\mathbb{F}_2)$$

whose connected-torus part lies in $\mathbb{F}_2 u_1 \oplus \mathbb{F}_2 u_2$, not in a rank-three $\langle \alpha^2, \alpha\beta, \beta^2\rangle$ group.

(ii) (*Rank-one free-locus criterion.*) On the rank-one stratum $\mathfrak{M}_C^{\mathrm{red},(1)}$, $C = (1, \beta, n)$, descent of $o_C^{\mathrm{red},(1)}$ through the free E-quotient is equivalent to the vanishing of the two components in (i): the ordinary reduced orientation gerbe in $E_2^{0,2}$, and the connected-torus equivariant linearization class in $E_2^{2,0}$. If a direct Chern-class calculation identifies the latter with a class of the form

$$c_1(\mathrm{degree})u_1 + c_1(\mathrm{level})u_2 \in H^2(BE;\mathbb{F}_2),$$

then the class vanishes exactly when those two mod-2 coefficients vanish. The OP scalar normalization by itself does not prove this equivariant linearization statement; it is a separate orientation calculation.

(iii) (*Finite stabilizer descent.*) On the small-orbit locus $\mathfrak{M}_C^{\mathrm{red}}(K3 \times (E/E[N]))$ the E-quotient is the stacky orbifold $[\mathfrak{M}_C^{\mathrm{red}}(K3 \times (E/E[N]))/(E/E[N])]$ banded by $BE[N]$, not by $B\mu_N$ unless a cyclic subgroup has been chosen. The orientation line may pick up a finite-stabilizer descent class

$$\widetilde{\beta}_{C,S}^{E,N} \in H_{E[N]}^2(S;\mathbb{F}_2).$$

Its point-stabilizer restriction is a class in $H^2(BE[N];\mathbb{F}_2)$ only after the mixed Borel terms, stabilizer action on $H^\bullet(S;\mathbb{F}_2)$, and spectral-sequence differentials have vanished. For $N = 2$,

$$E[2] \simeq (\mathbb{Z}/2)^2, \qquad H^*(BE[2];\mathbb{F}_2) = \mathbb{F}_2[x_1, x_2], \quad |x_i| = 1,$$

so

$$H^2(BE[2];\mathbb{F}_2) = \mathbb{F}_2\langle x_1^2, x_1x_2, x_2^2\rangle.$$

This rank-three group is the point-stabilizer reduction of the finite two-torsion calculation; it is not the full Borel descent class and it is not connected BE. If N is odd, the positive-degree mod-2 cohomology of $BE[N]$ vanishes because $E[N]$ has odd order. For even N, the residual class is detected on the 2-primary part and must be computed stratum by stratum from the equivariant linearization of the orientation line. More precisely, if $2^a \parallel N$ and $a \geq 2$, then

$$H^*(B(\mathbb{Z}/2^a)^2;\mathbb{F}_2) = \Lambda(x_1, x_2) \otimes \mathbb{F}_2[y_1, y_2], \qquad |x_i| = 1, \quad |y_i| = 2, \quad x_i^2 = 0,$$

so

$$\beta_{C,(2)}^{E,N} = A_1^{(N)} y_1 + A_{12}^{(N)} x_1x_2 + A_2^{(N)} y_2.$$

This is not determined by the $N = 2$ Klein-four vector.

(iv) (*Combined criterion.*) The orientation line descends over the full reduced quotient only if the ordinary reduced orientation gerbe, the connected-torus equivariant linearization class, and every finite Borel class $\widetilde{\beta}^{E,N}_{C,S}$ vanish on the relevant strata. After each finite Borel class has been null-trivialized, the chosen stabilizer linearization has a residual character

$$\lambda^{E,N}_{C} \in H^1(BE[N];\mathbb{F}_2),$$

and the quotient orientation requires this character to be zero. Translation invariance of the underlying ordinary line is not enough: a fixed line can carry a non-trivial equivariant character.

Proof. The cohomology calculation in (i) is the standard classifying space computation for a compact torus:

$$BT^2 = K(\mathbb{Z}^2, 2), \qquad H^*(BT^2;\mathbb{F}_2) = \mathbb{F}_2[u_1, u_2], \quad |u_i| = 2.$$

Hence $H^1(BE;\mathbb{F}_2) = 0$, and the bidegree $(1,1)$ term in the equivariant Serre spectral sequence vanishes. The only total-degree two terms on the free connected quotient are $(2,0)$ and $(0,2)$, as displayed. This proves the connected separation.

For (ii), descent of a square-root line through a free group quotient is exactly an equivariant linearization problem together with the ordinary gerbe-trivialization problem. The spectral sequence in (i) records these two classes. A proposed rank-one Chern-class formula is therefore a criterion: once the two coefficients of u_1, u_2 and the ordinary gerbe class vanish, the line descends on that stratum. The scalar identity $Z^X_{\mathrm{OP/DT}} = -4096\Delta_5^{-2}$ has no information about a choice of E-equivariant linearization, so it cannot by itself prove the criterion.

For (iii), the stabilizer of the N-fold small-orbit stratum is the finite group $E[N] \simeq (\mathbb{Z}/N)^2$. At $N = 2$ this is the Klein-four group. Its mod-2 cohomology is the polynomial ring on the two degree-one character classes, hence its degree-two part has basis x_1^2, x_1x_2, x_2^2. If N is odd, transfer kills positive-degree $\mathbb{F}_2$-cohomology because $|E[N]| = N^2$ is invertible in $\mathbb{F}_2$. Even N reduces to its 2-primary stabilizer. This proves the finite-stabilizer typing and separates it from connected BE.

Clause (iv) is the standard descent criterion for a line on a stack quotient: ordinary square-root gerbe, connected equivariant linearization, finite-stabilizer Borel degree-two classes, and the residual degree-one stabilizer characters must all vanish. A line fixed as an ordinary line can still carry a non-trivial character of the stabilizer group, so ordinary translation invariance does not imply equivariant descent. □

LEMMA 5.9 *(Klein-four criterion for the small-orbit gerbe residual).* Let $C \in \Gamma^{\mathrm{phys}}_X$ and adopt the notation of Lemma 5.8. The two-torsion subgroup $E[2] \subset E$ is the Klein-four group $\mathbb{Z}/2 \times \mathbb{Z}/2$, generated by the three nontrivial two-torsion points $e_1, e_2, e_3 \in E[2]$ (any two are independent). The small-orbit stratum at $N = 2$ is $\mathfrak{M}^{\mathrm{red}}_C(K3 \times (E/E[2]))$, banded by $BE[2]$, not $B\mu_2$; the cohomological residual of clause (iv) of Lemma 5.8 at $N = 2$ is the class

$$\beta^{E,E[2]}_{C} \;\in\; H^2(BE[2];\mathbb{Z}/2) \;\cong\; (\mathbb{Z}/2)^3,$$

with cohomology basis

$$H^*(BE[2];\mathbb{Z}/2) \;=\; \mathbb{Z}/2[\,c_1(L_1), c_1(L_2)\,], \quad H^2(BE[2];\mathbb{Z}/2) \;=\; \mathbb{Z}/2\langle c_1(L_1)^2\rangle \oplus \mathbb{Z}/2\langle c_1(L_2)^2\rangle \oplus \mathbb{Z}/2\langle c_1(L_1) \smile c_1(L_2)$$

where L_1, L_2 are the two character lines of the dual $E[2]^\vee \simeq E[2]$, and the three nontrivial cyclic subgroups $\mu_2^{(i)} = \langle e_i\rangle \subset E[2], i = 1,2,3$, restrict the basis to the three generators of $H^2(B\mu_2^{(i)};\mathbb{Z}/2) \cong \mathbb{Z}/2$ componentwise. Write

$$\beta^{E,E[2]}_{C} = b_{20}x_1^2 + b_{11}x_1x_2 + b_{02}x_2^2, \qquad x_j = c_1(L_j),$$

and let τ_i be the degree-one generator of $H^*(B\mu_2^{(i)}; \mathbb{Z}/2)$. Define the three cyclic residual bits $r_i \in \mathbb{Z}/2$ by

$$\mathrm{res}_{\mu_2^{(i)}}\big(\beta_C^{E,E[2]}\big) = r_i \tau_i^2.$$

With $e_3 = e_1 + e_2$,

$$r_1 = b_{20}, \qquad r_2 = b_{02}, \qquad r_3 = b_{20} + b_{11} + b_{02},$$

hence

$$b_{20} = r_1, \qquad b_{02} = r_2, \qquad b_{11} = r_1 + r_2 + r_3.$$

Thus the $N = 2$ finite-stabilizer residual vanishes if and only if the three cyclic residual bits r_1, r_2, r_3 vanish. On the rank-one D6–D2–D0 input $C = (1, \beta, n)$, the following hypotheses are sufficient, and in the finite $E[2]$-sector necessary, for the $N = 2$ residual to vanish and for the chosen $E[2]$-linearization to be trivial:

(K1) (*Künneth split.*) The reduced orientation line factors as a tensor product $o_C^{\mathrm{red}} = o_{K3} \otimes o_E$ on the protected sector, where o_{K3} is the Pin^c orientation of the K3 PT factor and o_E is the Pin^c orientation of the elliptic fibration factor, by the Künneth factorization of the cosection-twisted Joyce–Tanaka–Upmeier multiplicative cocycle [46],[45] on $\mathfrak{M}_C^{\mathrm{red}}$ viewed as a relative Hilbert scheme over the elliptic base, and the Cao–Kool–Monavari Pin^c lift on the K3 factor [16].

(K2) (*K3 factor.*) The action of $E[2]$ on $X = K3 \times E$ is fibrewise translation on the second factor and trivial on the first. The supplied factorwise equivariant lift on the K3 orientation factor is required to be the trivial $E[2]$-lift; this removes no elliptic finite-stabilizer obstruction.

(K3) (*Elliptic factor: finite gerbe and character data.*) Translation by $e_i \in E[2]$ fixes the underlying elliptic orientation line o_E as an ordinary line. This fixedness does not decide the equivariant lift. First one must compute the three degree-two gerbe bits

$$r_i(C) = \mathrm{res}_{\mu_2^{(i)}}(\beta_C^{E,E[2]})/\tau_i^2 \in \mathbb{Z}/2, \qquad i = 1, 2, 3.$$

The degree-two vanishing hypothesis is

$$r_1(C) = r_2(C) = r_3(C) = 0.$$

After this vanishing has supplied a null-trivialized gerbe, the set of $E[2]$-linearizations is an

$$H^1(BE[2]; \mathbb{F}_2) = \mathbb{F}_2\langle x_1, x_2\rangle$$

torsor. For a supplied linearization write its character as

$$\lambda_C^{E[2]} = \lambda_1 x_1 + \lambda_2 x_2,$$

or, on the three cyclic inertia sectors,

$$\mathrm{res}_{\mu_2^{(i)}}\, \lambda_C^{E[2]} = \rho_i \tau_i, \qquad \rho_1 = \lambda_1, \quad \rho_2 = \lambda_2, \quad \rho_3 = \lambda_1 + \lambda_2.$$

The required trivial-linearization hypothesis is

$$\rho_1(C) = \rho_2(C) = \rho_3(C) = 0.$$

Equivalently, the Pin^c sign character

$$w_1(o_E^{E[2]})\colon E[2] \longrightarrow \mathbb{Z}/2$$

vanishes on e_1, e_2, e_3. This is a degree-one equivariant-linearization hypothesis, distinct from the degree-two gerbe bits r_i, and not a consequence of ordinary translation invariance alone.

(K4) If (K1), (K2), and (K3) hold, then the degree-two bits $r_i(C)$ and the degree-one character bits $\rho_i(C)$ vanish for every $\mu_2^{(i)} \subset E[2]$. Since

$$b_{20} = r_1, \qquad b_{02} = r_2, \qquad b_{11} = r_1 + r_2 + r_3,$$

the three coefficients of $\beta_C^{E,E[2]}$ vanish, and therefore

$$\beta_C^{E,E[2]} = 0 \ \in \ H^2(BE[2]; \mathbb{Z}/2).$$

The additional equations $\rho_i = 0$ select the trivial point of the $H^1(BE[2]; \mathbb{F}_2)$-torsor of linearizations.

Thus $\beta_C^{E,E[2]} = 0$ and the finite $E[2]$-linearization is trivial under the stated hypotheses. For odd N, the finite-stabilizer obstruction vanishes cohomologically by Lemma 5.8. For even $N \geq 4$, the obstruction is detected on the 2-primary finite stabilizer and remains subject to the same equivariant linearization calculation; it is not closed by ordinary translation invariance.

Proof. The small-orbit stratum at $N = 2$ is the locus on which the E-action factors through the quotient $E \to E/E[2]$, so its isotropy is exactly $E[2]$ and the orbifold quotient $[\mathfrak{M}_C^{\mathrm{red}}(K3 \times (E/E[2]))/(E/E[2])]$ is banded by $BE[2]$ (the inertia of the residual stabiliser), *not* by $B\mu_2$. Since $E[2] \simeq (\mathbb{Z}/2)^2$ is the Klein-four group, the cohomology $H^*(BE[2]; \mathbb{F}_2)$ has two independent degree-one generators, whereas $H^*(B\mu_2; \mathbb{F}_2)$ has one. The two stacks therefore cannot be equivalent. The mod-2 cohomology is the polynomial ring on two degree-one generators $c_1(L_1), c_1(L_2)$ with $L_1, L_2 \in \widehat{E[2]}$ the two characters (Cartan–Eilenberg folklore for the cohomology of an elementary abelian 2-group), recovering $H^2(BE[2]; \mathbb{Z}/2) \cong (\mathbb{Z}/2)^3$ as stated.

The three nontrivial cyclic subgroups $\mu_2^{(i)} = \langle e_i \rangle$ are detected as follows: writing $E[2]$ additively with basis $\{e_1, e_2\}$ and $e_3 = e_1 + e_2$, the inclusion $\mu_2^{(1)} \hookrightarrow E[2]$ restricts L_1 to the sign character of $\mu_2^{(1)}$ and L_2 to the trivial character; the inclusion $\mu_2^{(2)} \hookrightarrow E[2]$ restricts L_2 to the sign character of $\mu_2^{(2)}$ and L_1 to the trivial character; the inclusion $\mu_2^{(3)} \hookrightarrow E[2]$ restricts both L_1, L_2 to the sign character of $\mu_2^{(3)}$, so $c_1(L_1)$ and $c_1(L_2)$ restrict to the same generator and $c_1(L_1) \smile c_1(L_2)$ restricts to its square. The three restriction maps are therefore jointly injective on the rank-three space $H^2(BE[2]; \mathbb{Z}/2)$, so a class in $H^2(BE[2]; \mathbb{Z}/2)$ vanishes if and only if its image in each $H^2(B\mu_2^{(i)}; \mathbb{Z}/2)$ vanishes. This is the standard Klein-four detection by cyclic restriction.

Equivalently, for $\beta = b_{20}x_1^2 + b_{11}x_1x_2 + b_{02}x_2^2$, the three restrictions are

$$\mathrm{res}_1(\beta) = b_{20}\tau_1^2, \qquad \mathrm{res}_2(\beta) = b_{02}\tau_2^2, \qquad \mathrm{res}_3(\beta) = (b_{20} + b_{11} + b_{02})\tau_3^2.$$

This gives the displayed conversion between the coefficient vector (b_{20}, b_{11}, b_{02}) and the cyclic residual vector (r_1, r_2, r_3).

(K1). The Künneth factorization $o_C^{\mathrm{red}} = o_{K3} \otimes o_E$ is the underlying splitting of the multiplicative orientation cocycle of [46],[45] along the product factorization $X = K3 \times E$ on the protected sector,

where the moduli of stable pairs decompose as a relative Hilbert scheme over the elliptic base. The factorwise Pin^c lift on the K3 factor is a supplied input, modeled on the BBDJS/Joyce–Upmeier and Cao–Kool–Monavari orientation packages [11, 16, 46]; the elliptic ordinary factor is the orientation of the cotangent bundle Ω^1_E, which is trivial of rank one and trivialized by the flat translation-invariant one-form $\mathrm{d}z$.

(K2). The translation $t_{e_i} : X \to X$ acts as the identity on the K3 factor and as translation by e_i on the elliptic factor. Pulling back along t_{e_i} does not move the K3 factor. The trivial $E[2]$-lift on o_{K_3} is therefore a supplied factorwise linearization choice, not a consequence that settles the elliptic quotient descent.

(K3). The cotangent bundle Ω^1_E is trivial, trivialized by $\mathrm{d}z$; translation acts trivially on $\mathrm{d}z$. This proves only fixedness of the underlying ordinary line. It does not determine the equivariant lift. There are two separate finite calculations: the degree-two gerbe residual $\operatorname{res} \beta_C^{E,E[2]} = r_i \tau_i^2$, and, only after this residual has been null-trivialized, the degree-one character $\operatorname{res} \lambda_C^{E[2]} = \rho_i \tau_i$ of the chosen line linearization. Hypothesis (K3) is the explicit finite computation $r_i = 0$ and $\rho_i = 0$ for all three cyclic inertia sectors.

(K4). Combining (K1), (K2), (K3), the three cyclic degree-two residuals and degree-one character bits vanish. The coefficient conversion above then gives $b_{20} = b_{11} = b_{02} = 0$, hence $\beta_C^{E,E[2]} = 0 \in H^2(BE[2]; \mathbb{Z}/2)$; the equations $\rho_i = 0$ give the trivial $E[2]$-linearization. The odd-N cases vanish for the cohomological reason of Lemma 5.8. Even-N cases for $N \geq 4$ require the corresponding 2-primary equivariant linearization calculation. □

Remark 5.10 *(Orbifold-stratum subtlety).* The Künneth factorization (K1) is constructed on the smooth locus of $\mathfrak{M}_C^{\mathrm{red}}$; on the orbifold stratum $K3 \times \{e_i\} \subset X$ it requires the orbifold extension of the Joyce–Tanaka–Upmeier multiplicative cocycle to the inertia stack of the Klein-four banded quotient. Joyce–Tanaka–Upmeier [45] provide the orientation technology for finite-band quotient stacks. In the present Klein-four case the required check is concrete: the inertia decomposition into the three nontrivial cyclic sectors must carry the same trivial equivariant linearization. Equivalently, the degree-two equalities $r_i(C) = 0$ and the degree-one equalities $\rho_i(C) = 0$ of clause (K4) must both hold. Once that is verified, the orbifold stratum introduces no additional obstruction beyond the cyclic-sector criterion.

Remark 5.11 *(Two-torsion routes which do not replace the finite criterion).* There are three natural attempts to turn the $N = 2$ small-orbit calculation into a direct vanishing theorem. They are useful checks, but none replaces Lemma 5.9 and Corollary 5.14.

Hyperelliptic route. One may try to identify the stabilizer with the hyperelliptic involution $[-1]$ on E and prove vanishing from a trivial action on Picard data. This is not a theorem in the present generality. The action of $[-1]$ on $\mathrm{Pic}^0(E)$ is inversion, not pointwise triviality, and ordinary fixedness of the line does not select the equivariant lift. The route becomes valid only after one supplies the actual $E[2]$-linearized orientation line and proves that its degree-one character $\lambda_C^{E[2]} \in H^1(BE[2]; \mathbb{F}_2)$ is zero. That is precisely the $\rho_i = 0$ part of the finite criterion.

Wu-class route. A direct Stiefel–Whitney computation can identify the degree-two gerbe class with the w_2-class of the finite stabilizer representation, and Wu's formula expresses it as

$$w_2 = v_2 + Sq^1 v_1.$$

This gives an independent route to the bits (r_1, r_2, r_3) once the reduced tangent/self-Ext complex and its $E[2]$-linearization have been constructed. It still does not compute the residual $H^1(BE[2]; \mathbb{F}_2)$-character of the chosen linearization. Hence it can prove a degree-two gerbe vanishing, not strong reduced orientation.

Orbifold Pin^c route. One can attempt to build the lift on the inertia stack of the four $E[2]$-fixed sectors by combining the K3 Pin^c package of Cao–Kool–Monavari [16], the Joyce–Upmeier/Joyce–Tanaka–Upmeier orientation cocycle [45, 46], and Toën's orbifold Riemann–Roch [75]. This route requires an inertia-stack Pin^c lift on each cyclic sector, an age-correction parity calculation, multiplicative gluing across the smooth and orbifold strata, and trivial residual character under the whole Klein-four stabilizer. The $N = 2$ age calculation does not imply the higher even $E[N]$ calculation; for $N \geq 4$ the mixed two-primary class $A_{12}^{(N)} x_1 x_2$ and the degree-one characters $\lambda_i^{(N)}$ remain separate obstructions.

Thus the stable theorem in the manuscript is the finite Klein-four criterion: degree-two gerbe bits and degree-one linearization characters are both named and must both vanish. The three routes above are cross-checks or construction programmes for those entries. They are not used as vanishing theorems.

PROPOSITION 5.12 *(Rank-one two-torsion gerbe and linearization package).* On the rank-one D6–D2–D0 stratum $\mathfrak{M}^{\mathrm{red},(1)}_{(1,\beta,n)}$, assume the ordinary reduced Pin^c orientation line has been constructed. The remaining $N = 2$ finite-stabilizer gerbe obstruction to descent of this line through the small-orbit quotient is the vector of three degree-two bits

$$r_i = \mathrm{res}_{\mu_2^{(i)}}(\beta_C^{E,E[2]})/\tau_i^2 \in \mathbb{Z}/2, \qquad i = 1, 2, 3.$$

Equivalently,

$$\beta_C^{E,E[2]} = 0 \quad \Longleftrightarrow \quad (r_1, r_2, r_3) = (0, 0, 0).$$

After this class has been null-trivialized, the triviality of the $E[2]$-linearization is a separate degree-one question. The set of linearizations is an $H^1(BE[2]; \mathbb{F}_2)$-torsor; for a supplied linearization $\lambda_C^{E[2]}$, write

$$\mathrm{res}_{\mu_2^{(i)}} \lambda_C^{E[2]} = \rho_i \tau_i.$$

Trivial $E[2]$-linearization means $(\rho_1, \rho_2, \rho_3) = (0, 0, 0)$. The r_i are not the ρ_i: they live in different cohomological degrees.

(C) *Cyclic-sector character, after null-trivialization.* For a genuine $E[2]$-linearized orientation line on the inertia sector,

$$\rho_i = \frac{1 - \chi_i}{2}, \qquad \chi_i = \text{the scalar by which } e_i \text{ acts on } o_C^{\mathrm{red}}|_{I_{e_i}\mathfrak{M}}.$$

This computes the degree-one linearization character, not the degree-two gerbe class.

(W) *Stiefel–Whitney/Wu.* The degree-two bit is the coefficient of τ_i^2 in the equivariant reduced obstruction class:

$$\mathrm{res}_{\mu_2^{(i)}} w_2^{E[2]}\left(\mathbb{T}_C^{\mathrm{red}}\right) = r_i \tau_i^2, \qquad w_2 = v_2 + Sq^1 v_1,$$

where $\mathbb{T}_C^{\mathrm{red}}$ is the reduced virtual tangent/self-Ext complex on the cyclic fixed sector and v_j are its Wu classes.

(P) *Inertia-stack Pin^c.* The degree-two bit is the Pin^c central holonomy corrected by the mod-2 orbifold Riemann–Roch age:

$$r_i = \mathrm{hol}_{\ker(\mathrm{Pin}^c \to SO)}\left(\widetilde{P}^{\mathrm{red}}_{\mathrm{Pin}^c}|_{I_{e_i}\mathfrak{M}}\right)(e_i) + \mathrm{age}^{\mathrm{red}}_{e_i} \pmod 2.$$

Thus the rank-one two-torsion small-orbit sector is closed only after both vectors vanish:

$$(r_1, r_2, r_3) = (0, 0, 0), \qquad (\rho_1, \rho_2, \rho_3) = (0, 0, 0).$$

The present manuscript computes neither vector; the proposition is the finite obstruction and linearization target, not a vanishing theorem.

Proof. Lemma 5.9 identifies

$$H^2(BE[2]; \mathbb{F}_2) = \mathbb{F}_2\langle x_1^2, x_1x_2, x_2^2\rangle$$

and proves that restriction to the three cyclic subgroups $\mu_2^{(i)} = \langle e_i\rangle$ is jointly injective in degree two. Hence $\beta_C^{E,E[2]}$ vanishes if and only if its three restrictions $r_i\tau_i^2$ vanish.

For a line over the quotient stack, after the degree-two gerbe class has been null-trivialized, the restriction to $\mu_2^{(i)}$ is a one-dimensional equivariant line on $[I_{e_i}\mathfrak{M}/\mu_2^{(i)}]$. The character $\chi_i = \pm 1$ is trivial exactly when the degree-one coefficient ρ_i is zero; this is the formula in (C). The degree-two residual is a different class. It is the second Stiefel–Whitney obstruction of the equivariant reduced tangent/self-Ext complex, and Wu's identity gives $w_2 = v_2 + Sq^1 v_1$, giving (W). Finally, a Pin^c lift of the cyclic-sector frame records the central $\mathbb{Z}/2$-holonomy of $\ker(\mathrm{Pin}^c \to SO)$; orbifold Riemann–Roch contributes the age correction on the fixed inertia sector. This is (P). The formulae (W) and (P) compute the residual degree-two class $\mathrm{res}_{\mu_2^{(i)}} \beta_C^{E,E[2]}$, while (C) computes the residual degree-one choice after that class has been null-trivialized. □

Lemma 5.13 *(Even finite-stabilizer residuals beyond two torsion).* Let N be even and write $2^a \parallel N$. On the N-fold small-orbit stratum of the reduced E-quotient, the positive-degree mod-2 finite-stabilizer obstruction is the two-primary class

$$\beta_{C,(2)}^{E,N} \in H^2(B(\mathbb{Z}/2^a)^2; \mathbb{F}_2).$$

For $a = 1$ this is the Klein-four class $\beta_C^{E,E[2]}$ of Lemma 5.9. For $a \geq 2$,

$$H^*(B(\mathbb{Z}/2^a)^2; \mathbb{F}_2) = \Lambda(x_1, x_2) \otimes \mathbb{F}_2[y_1, y_2], \qquad |x_i| = 1, \quad |y_i| = 2, \quad x_i^2 = 0,$$

and therefore

$$H^2(B(\mathbb{Z}/2^a)^2; \mathbb{F}_2) = \mathbb{F}_2\langle y_1, x_1x_2, y_2\rangle.$$

Write

$$\beta_{C,(2)}^{E,N} = A_1^{(N)} y_1 + A_{12}^{(N)} x_1x_2 + A_2^{(N)} y_2.$$

Let g_1, g_2 be the standard generators of $(\mathbb{Z}/2^a)^2$, put $h_i = 2^{a-1}g_i$, $i = 1, 2$, and $h_3 = h_1 + h_2$. For the three order-two cyclic subgroups $\langle h_i\rangle$,

$$\mathrm{res}_{\langle h_1\rangle} \beta_{C,(2)}^{E,N} = A_1^{(N)}\tau_1^2, \qquad \mathrm{res}_{\langle h_2\rangle} \beta_{C,(2)}^{E,N} = A_2^{(N)}\tau_2^2,$$

$$\mathrm{res}_{\langle h_3\rangle} \beta_{C,(2)}^{E,N} = (A_1^{(N)} + A_2^{(N)})\tau_3^2.$$

The mixed coefficient $A_{12}^{(N)}$ is invisible to all cyclic restrictions, not only to the three order-two subgroups above. It is detected by the rank-two quotient $(\mathbb{Z}/2^a)^2 \twoheadrightarrow (\mathbb{Z}/2)^2$, which pulls the mixed class $\bar{x}_1\bar{x}_2$ back to x_1x_2. Therefore the three $E[2]$ bits r_1, r_2, r_3 do not determine the even $N \geq 4$ residual. Vanishing for

$N \geq 4$ is a separate 2^a-primary equivariant-linearization calculation on the small-orbit stratum. After $\beta^{E,N}_{C,(2)} = 0$, triviality of the chosen two-primary linearization is a further degree-one condition:

$$\lambda^{E,N}_C = \lambda^{(N)}_1 x_1 + \lambda^{(N)}_2 x_2 \in H^1(B(\mathbb{Z}/2^a)^2; \mathbb{F}_2)$$

must vanish. This character is detected on the cyclic subgroups generated by g_1 and g_2, not by the order-two subgroups $\langle h_i \rangle$.

Proof. The odd part of $E[N] \simeq (\mathbb{Z}/N)^2$ has order invertible in $\mathbb{F}_2$, so transfer identifies the positive-degree $\mathbb{F}_2$-cohomology with that of the two-primary factor $(\mathbb{Z}/2^a)^2$. If $a = 1$, this is $E[2]$ and the cohomology ring is $\mathbb{F}_2[x_1, x_2]$, $|x_i| = 1$, giving the Klein-four basis $\langle x_1^2, x_1x_2, x_2^2 \rangle$. If $a \geq 2$, the cyclic calculation gives

$$H^*(B\mathbb{Z}/2^a; \mathbb{F}_2) \cong \Lambda(x) \otimes \mathbb{F}_2[y], \qquad |x| = 1, \quad |y| = 2, \quad x^2 = 0.$$

Kunneth for the product gives the degree-two basis $\langle y_1, x_1x_2, y_2 \rangle$.

For the restriction formulas, restrict first to the order-two subgroup $\langle h_1 \rangle = \langle 2^{a-1} g_1 \rangle$. The degree-one class $x_1 \in H^1(B\mathbb{Z}/2^a; \mathbb{F}_2)$ restricts to zero because the generator h_1 maps to $2^{a-1} \equiv 0 \pmod 2$ under the mod-two character, while the degree-two cyclic extension class y_1 restricts to the generator $\tau_1^2 \in H^2(B\langle h_1 \rangle; \mathbb{F}_2)$. The same calculation holds for h_2. For $h_3 = 2^{a-1}(g_1 + g_2)$, both y_1 and y_2 restrict to τ_3^2, while x_1 and x_2 restrict to zero. Hence x_1x_2 restricts to zero on all three order-two cyclic subgroups, and the displayed formulas follow.

The quotient $(\mathbb{Z}/2^a)^2 \twoheadrightarrow (\mathbb{Z}/2)^2$ identifies $H^1(B(\mathbb{Z}/2)^2; \mathbb{F}_2)$ with the span of x_1, x_2; hence $\bar{x}_1\bar{x}_2$ inflates to x_1x_2. The mixed coefficient is therefore a genuine rank-two equivariant-linearization obstruction. It is invisible on every cyclic subgroup: for a cyclic subgroup of order 2, the degree-one classes x_1, x_2 restrict to zero as above; for a cyclic subgroup of order 2^b with $b \geq 2$, the restrictions of x_1, x_2 are scalar multiples of the single degree-one generator x, and x_1x_2 restricts to a scalar multiple of $x^2 = 0$ in $H^2(B\mathbb{Z}/2^b; \mathbb{F}_2)$. This is a different stabilizer calculation from the $N = 2$ Klein-four residual, so the vanishing of (r_1, r_2, r_3) supplies no vanishing theorem for $N \geq 4$. Finally, when the degree-two class has vanished, the residual two-primary linearisations are classified by

$$H^1(B(\mathbb{Z}/2^a)^2; \mathbb{F}_2) = \mathbb{F}_2\langle x_1, x_2 \rangle.$$

The order-two subgroups generated by $h_i = 2^{a-1} g_i$ do not detect this character, since $x_j(h_i) = 0$. Triviality must be checked on the full two-primary generators, for instance g_1, g_2. □

COROLLARY 5.14 *(Finite-stabilizer obstruction to* (O1)*).* Fix a small-orbit stratum S of the reduced E-quotient with stabilizer $E[N]$. Assume that the ordinary reduced square-root gerbe α^{red}_C and the connected free quotient class $\alpha^{E,\mathrm{free}}_C$ have already been null-trivialized on this stratum. Then the finite-stabilizer part of the strong reduced orientation (O1) exists with trivial quotient linearization if and only if the Borel descent class $\widetilde{\beta}^{E,N}_{C,S}$ and the selected residual character vanish. After the Borel mixed terms, stabilizer action on $H^\bullet(S; \mathbb{F}_2)$, and spectral-sequence differentials have been killed, this condition is computed by the following finite point-stabilizer tuple.

(i) If N is odd, there is no positive-degree mod-2 finite-stabilizer obstruction:

$$H^{>0}(BE[N]; \mathbb{F}_2) = 0.$$

(ii) If $N = 2$, the tuple is

$$\mathfrak{o}_C^{E[2]} = \big(r_1, r_2, r_3; \rho_1, \rho_2, \rho_3\big),$$

where

$$\operatorname{res}_{\langle e_i\rangle} \beta_C^{E,E[2]} = r_i \tau_i^2, \qquad \operatorname{res}_{\langle e_i\rangle} \lambda_C^{E[2]} = \rho_i \tau_i.$$

Equivalently,

$$\beta_C^{E,E[2]} = 0 \quad \text{and} \quad \lambda_C^{E[2]} = 0.$$

(iii) If N is even and $2^a \parallel N$ with $a \geq 2$, the tuple is

$$\mathfrak{o}_C^{E[N]} = \big(A_1^{(N)}, A_{12}^{(N)}, A_2^{(N)}; \lambda_1^{(N)}, \lambda_2^{(N)}\big),$$

where

$$\beta_{C,(2)}^{E,N} = A_1^{(N)} y_1 + A_{12}^{(N)} x_1 x_2 + A_2^{(N)} y_2$$

and

$$\lambda_C^{E,N} = \lambda_1^{(N)} x_1 + \lambda_2^{(N)} x_2 \in H^1(B(\mathbb{Z}/2^a)^2; \mathbb{F}_2).$$

Thus the $E[2]$ three-bit calculation proves only the point-stabilizer reduction of the $N = 2$ degree-two gerbe vanishing. It does not prove the higher even finite-stabilizer class $A_{12}^{(N)} x_1 x_2$, and it does not prove the residual degree-one linearization character. Any non-zero entry of $\mathfrak{o}_C^{E[N]}$ obstructs (O1) on that small-orbit stratum. No scalar automorphic character, Borcherds divisor order, or OP/DT scalar square changes this finite quotient obstruction.

Proof. For a finite stabilizer $G = E[N]$, the Borel descent spectral sequence

$$E_2^{p,q} = H^p(BG;\, H^q(\mathfrak{M}_C^{\mathrm{red}}; \mathbb{F}_2)) \Longrightarrow H_G^{p+q}(\mathfrak{M}_C^{\mathrm{red}}; \mathbb{F}_2)$$

puts the finite square-root obstruction $\widetilde{\beta}_{C,S}^{E,N}$ in total degree two. After that class is null-trivialized, linearizations of the resulting line form an $H^1(BG; \mathbb{F}_2)$-torsor. Hence descent with the trivial quotient orientation is equivalent to the simultaneous vanishing of the degree-two finite Borel class and the selected degree-one character. The displayed point-stabilizer formulas are reductions of this Borel statement after the mixed terms and differentials vanish. The odd case is transfer. The $N = 2$ case is Lemma 5.9 together with Proposition 5.12. The higher even case is Lemma 5.13. □

Proposition 5.15 *(Finite-stage quotient-orientation character system).* Fix a finite HN height R in the compact/wrapped datum of Definition 10.27. Let Str_R^{or} be the finite set of reduced object, extension, and flag strata of $\mathsf{Corr}_R^{E,\mathrm{hyb}}$ on which vanishing-cycle coefficients and orientation lines are pulled, pushed, or compared. For $S \in \mathsf{Str}_R^{or}$, write $C(S)$ for its charge label and $\mathcal{N}_R(S)$ for the finite stabilizers $E[N]$ appearing on its small-orbit components.

A finite-stage quotient-orientation character system is the datum

$$\mathfrak{Q}_R^{or} = \Big(\{o_{R,S}\}_{S \in \mathsf{Str}_R^{or}}, \{\theta_{R,S}^{\mathrm{red}}\}, \{\theta_{R,S}^{E,\mathrm{free}}\}, \{\theta_{R,S}^{E,N},\ \lambda_{R,S}^{E,N} = 0\}_{N \in \mathcal{N}_R(S)}, \{\mu_{R,e}^{or}\}, \{\Theta_{R,\mathrm{flag}}^{or}\}\Big).$$

Here $o_{R,S}^{\otimes 2}$ is identified with the reduced determinant line on object strata and with the corresponding pulled-back coefficient determinant on extension and flag strata; $\theta_{R,S}^{\mathrm{red}}$ null-trivializes the ordinary reduced gerbe, $\theta_{R,S}^{E,\mathrm{free}}$ null-trivializes the connected BE-class, and $\theta_{R,S}^{E,N}$ null-trivializes the finite Borel class

$$\widetilde{\beta}_{R,S}^{E,N} \in H^2_{E[N]}(S; \mathbb{F}_2).$$

Its point-stabilizer restriction is a class $\beta^{E,N}_{R,S} \in H^2(BE[N];\mathbb{F}_2)$ only after the mixed Borel terms, stabilizer action on $H^\bullet(S;\mathbb{F}_2)$, and spectral-sequence differentials have been killed. After this finite Borel class has been null-trivialized, the chosen finite-stabilizer linearization has character

$$\lambda^{E,N}_{R,S} \in H^1(BE[N];\mathbb{F}_2),$$

and the quotient system selects the zero character. The maps $\mu^{or}_{R,e}$ are the Thom–Sebastiani orientation transports along the local-local, local/wrapped, and wrapped-wrapped correspondence generators e, and $\Theta^{or}_{R,\mathrm{flag}}$ records their two-step flag coherences.

Such a system exists on the finite stage if and only if every stratum $S \in \mathsf{Str}^{or}_R$ has

$$\alpha^{\mathrm{red}}_{R,S} = 0, \qquad \alpha^{E,\mathrm{free}}_{R,S} = 0,$$

and its finite-stabilizer obstruction tuple vanishes as follows:

$$N \text{ odd}: \quad H^{>0}(BE[N];\mathbb{F}_2) = 0,$$

$$N = 2: \quad (r_1, r_2, r_3; \rho_1, \rho_2, \rho_3) = 0,$$

$$2^a \parallel N,\ a \geq 2: \quad \left(A^{(N)}_1, A^{(N)}_{12}, A^{(N)}_2; \lambda^{(N)}_1, \lambda^{(N)}_2\right) = 0,$$

with the notation of Corollary 5.14, and all orientation-transport defects along the correspondence generators, flag stacks, and $R' \geq R$ transition maps vanish. If the systems $\mathfrak{Q}^{or}_R$ exist for a cofinal family of finite stages and are compatible with the transition maps, their inverse limit is exactly the strong reduced orientation datum (O1) on the compact/wrapped Hall sector. Conversely, any (O1) datum restricts to such a compatible finite-stage system. No automorphic scalar, Borcherds divisor order, or Maass target character detects any of the finite $H^1(BE[N])$ characters.

Proof. For each stratum S, the ordinary and connected E-quotient parts are the descent classes of Lemma 5.8. For every finite small-orbit stabilizer $E[N]$, the degree-two finite gerbe and the residual degree-one character are precisely the obstruction tuple of Corollary 5.14. Thus the displayed stratum-wise vanishings are necessary and sufficient for the coefficient orientation line to descend with trivial finite-stabilizer character on each object, extension, and flag stratum.

The remaining conditions are functorial, not cohomological. A Hall operation at height R is the pull–Thom–Sebastiani–push map attached to a generator of $\mathsf{Corr}^{E,\mathrm{hyb}}_R$. The orientation transports $\mu^{or}_{R,e}$ must identify the descended coefficient systems on the source and target of this generator. The two ways to compose such identifications over a two-step flag stack differ by a finite defect; vanishing of all flag defects is exactly the strict pentagon condition at height R. The same statement for successor maps is the displayed transition compatibility. Since the stage is finite, these are finitely many equalities. Passing to a cofinal inverse system gives (O1), and restriction of (O1) to a finite height gives the listed data. The final sentence follows because the obstruction groups are the displayed Borel-equivariant H^2 and H^1 groups, whereas a scalar automorphic character is not a class in them. □

LEMMA 5.16 *(Rank-one orientation-sign rigidification from a supplied lift).* Assume that, after the ordinary reduced gerbe class has vanished, a square-root orientation lift of $\det \mathrm{RHom}_{\mathrm{red}}(C, C)$ on $\mathfrak{M}^{\mathrm{red},(1)}_{(1,\beta,n)}$ has been constructed in the reduced $K3 \times E$ Hall datum. This is the analogue, in the reduced rank-one sector, of the square-root sign choice in the BBDJS oriented d-critical formalism [11, Theorem 6.9], together with a separately supplied Joyce–Upmeier-compatible multiplicative orientation

datum [46, Theorem 4.1]; it is not supplied by those theorems alone. The lift determines the orientation only up to a global $\mathbb{Z}/2$ ambiguity $\eta_C \in H^0(\mathfrak{M}^{\mathrm{red},(1)}_{(1,\beta,n)}; \mathbb{Z}/2)$. On the rank-one D6–D2–D0 stratum, this residual sign is rigidified by the following data, jointly:

(S1) the Joyce–Upmeier orientability cocycle [46], [45] on Hall extension correspondences, which imposes multiplicative compatibility when rank-one factors are used as graded pieces of HN or direct-sum strata. This is a compatibility constraint for extending signs beyond the rank-one components; it is not a proof that the rank-one locus itself is generated by one component under direct sum;

(S2) the requirement that, after the supplied comparison ι_{aut}, the Pfaffian section $\mathrm{Pf}_{\mathrm{prot}}(\mathfrak{D}_X)$ is read as a section of $\mathcal{L}^5 \otimes \nu_{\Delta_5}$ and restricts at the unique top-cohomology generator of the isotropic-cusp expansion to the monic Borcherds normalization $D_5 = 64^{-1}\Delta_5$ of Proposition 4.5; equivalently, the leading Fourier coefficient of $\mathrm{Pf}_{\mathrm{prot}}$ at the isotropic cusp $\mathfrak{c}_\infty$ equals +1 (positive orientation, monic);

(S3) scalar consistency, after squaring, with the OP/DT scalar sign $C_{\square,X} = -4096$ of Proposition 11.1, i.e. $(\mathrm{Pf}_{\mathrm{prot}})^2|_{\text{rank-one}} = \Delta_5^2$ with positive leading coefficient $c_\Delta(1,1,1)^2 = 64^2 = 4096$, the absolute value of $C_{\square,X}$, and the OP scalar sign $\mathrm{sign}(C_{\square,X}) = -1$ absorbed into the fermionic shift $(-p_{\mathrm{DT}})^n$ of the OP/DT chamber, not into the orientation line; this is a scalar compatibility check, not a rigidifying input for η_C.

Then, on any connected rank-one component for which the supplied ι_{aut} identifies the Pfaffian section with the monic cusp branch, the residual sign η_C is fixed to +1 for that component. This fixes a chosen global sign there. It does not prove that the same choice extends over all rank-one charge components, over HN sums, or through the reduced E-quotient; those compatibilities are part of (O1) and $(\mathrm{O1})^+$, including the connected BE-linearization and finite $E[N]$-residual vanishings.

Proof. On a connected component the supplied square-root orientation lift is unique up to multiplying by a constant sign; on the disconnected components of $\mathfrak{M}^{\mathrm{red},(1)}_{(1,\beta,n)}$ (indexed by the discrete Mukai-charge label $(1,\beta,n)$, connected within each fixed charge by the Beauville–Mukai moduli construction [41, 60, 61]) the ambiguity is the function $\eta\colon \pi_0\mathfrak{M}^{\mathrm{red},(1)} \to \{\pm 1\}$.

(S1) fixes the multiplicative compatibility that any extension of η to Hall direct-sum or HN correspondence strata must satisfy: the supplied Joyce–Upmeier-type multiplicative orientation isomorphism on the retained exact-triangle correspondence is required to have the Mukai parity sign

$$(-1)^{\langle C, C'\rangle_{\mathrm{Mukai}}}.$$

This is a convention and compatibility condition in the reduced Hall datum, not a consequence of Joyce–Upmeier alone on the $K3 \times E$ quotient. The BBDJS oriented d-critical formalism and the cosection-twist compatibility of [11, 16] are background orientation technologies; the reduced quotient multiplicative sign is still part of the supplied (O1) datum. Hence an already supplied extension of η must be multiplicative modulo the Mukai parity. This does not show that all rank-one charge components are generated by one component.

(S2) fixes η on any connected component on which the supplied ι_{aut} compares the Pfaffian section with the monic cusp branch: the Pfaffian section is a section of $\mathcal{L}^5 \otimes \nu_{\Delta_5}$, and Proposition 4.5 together with the square-root normalization table of Remark Normalization of the square root require the leading Fourier coefficient of $\mathrm{Pf}_{\mathrm{prot}}$ at $\mathfrak{c}_\infty$ to be +1 (the monic $D_5 = 64^{-1}\Delta_5$ being the canonical Borcherds

output). The coefficient 64 is the theta-normalization converting the monic Borcherds product D_5 to Δ_5; it is not the square of an orientation line. The orientation line chooses the sign of a supplied first-order Pfaffian section, not the scalar prefactor. This pins $\eta_C \in \{+1, -1\}$ to a specific value on that component.

(S3) verifies consistency: squaring an orientation line forgets η, so the only sign data surviving after squaring are $(c_\Delta(1,1,1))^2 = 4096$ (positive, automatic) and the fermionic-shift sign $\mathrm{sign}(C_{\square,X}) = -1$, which lives in the OP/DT chamber of Proposition 11.1 (the $(-p_{\mathrm{DT}})^n$ factor) and does not constrain η. So (S3) is consistent with (S2) but does not refine it. Combining (S1) and (S2), $\eta_C = +1$ on each supplied connected component whose Pfaffian branch has been compared with the monic cusp branch; any extension to other rank-one charge components or to HN sums remains a separate orientation datum.

The choice is determinate once the ordinary lift and quotient linearizations have been supplied. This is a reduced Hall orientation sign convention informed by the BBDJS oriented d-critical formalism [11, Theorem 6.9] and by a separately supplied Joyce–Upmeier-compatible multiplicative orientation datum [46, Theorem 4.1]. It is not a proof that $\alpha_C^{E,\mathrm{free}}$ or $\beta_C^{E,E[2]}$ vanishes. □

Remark 5.17. The same data (S1), (S2), (S3) rigidify η_C on a higher-rank semistable stratum only after the Pin^c-lift gerbe class α_C^{red} of (P3) and the required quotient linearization classes have already vanished.

LEMMA 5.18 *(HN-reducible Pin^c-lift gerbe criterion).* Let C be a higher-rank protected charge and let $\tau = (C_1, \dots, C_\ell)$ be an HN-type of length $\ell \geq 2$ in the reduced semistable stack. On the corresponding HN stratum, the reduced orientation gerbe satisfies

$$\alpha_C^{\mathrm{red}}|_\tau = \sum_i \alpha_{C_i}^{\mathrm{red}} \bmod 2,$$

with off-diagonal self-Ext blocks contributing even class. Therefore $\alpha_C^{\mathrm{red}}|_\tau = 0$ on every HN stratum whose graded factors lie in the already-oriented rank-one sector or in a rank-two sector covered by Lemma 5.5. If, in addition, the E-equivariant parity

$$\sum_i d_E(C_i)\,\mathrm{rk}(\det C_i)$$

is even, then the joint class $\alpha_C^{\mathrm{red}} + \alpha_C^E$ vanishes on that HN stratum, subject to the finite-stabilizer criterion of Lemma 5.9 on any small-orbit boundary. Stable higher-rank length-one HN strata are not covered by this lemma.

Proof. The HN stratification expresses a length-ℓ stratum as an extension stack over the product of the moduli of its graded factors. The Joyce–Upmeier and Joyce–Tanaka–Upmeier multiplicative orientation formalism [45, 46] identifies the orientation gerbe of the direct-sum target with the tensor product of the gerbes of the graded pieces. The off-diagonal reduced self-Ext blocks occur in dual pairs under the shifted-symplectic pairing, so their Stiefel–Whitney contribution is even. This gives the displayed additivity formula. The stated vanishing follows immediately when each graded factor lies in a sector whose orientation gerbe has already been trivialized.

The connected E-descent obstruction is computed by the equivariant Serre spectral sequence of the free translation quotient; on an HN stratum its parity is the sum of the elliptic rotation numbers of the graded factors. If that sum is even, the connected obstruction vanishes. The remaining torsion part is the small-orbit finite stabilizer gerbe; the rank-one $N = 2$ detection criterion is isolated in Lemma 5.9. No induction on stable higher-rank length-one factors is used. □

6 Physical charge and normal-ordered Gram descent

The Hall charge is additive. The Igusa degree is not. This section isolates the exact defect and removes it by normal ordering. The result is a lattice theorem, not yet a chain-level Hall strictification.

Definition 6.1 (Dyonic Mukai lattice and quadratic Gram projection). Let S be a complex K3 surface and put

$$\Lambda_S := H^*(S,\mathbb{Z}) = H^0(S,\mathbb{Z}) \oplus H^2(S,\mathbb{Z}) \oplus H^4(S,\mathbb{Z}),$$

the Mukai lattice in the sense of Mukai [60, 61]: even, unimodular, signature $(4,20)$, rank 24, equipped with the Mukai pairing $\langle (r,c,s),(r',c',s')\rangle_{\mathrm{Muk}} = c\cdot c' - rs' - r's$. The formal full-Mukai dyonic lattice is the additive abelian group

$$\Gamma_X^{\mathrm{form}} := \Lambda_S \oplus \Lambda_S, \qquad c = (Q,P) \in \Gamma_X^{\mathrm{form}},$$

graded electric $\oplus$ magnetic. Throughout the formulae below in this paper,

$$\Gamma_X^{\mathrm{phys}} := \Gamma_X^{\mathrm{form}}.$$

In legacy lattice formulae below the superscript phys is only mnemonic for this formal even Mukai sector Γ_X^{form}. It is not the full four-dimensional $\mathcal{N}=4$ electric–magnetic charge lattice and not compact Hall support. It is the D6–D2–D0/OP branch on which the Mukai Gram map is tested before imposing algebraicity, effectivity, stability, or nonemptiness of compact Hall moduli. For a chosen stability sector the actual Hall support is a semigroup $\Gamma_{X,\sigma}^{\mathrm{eff}} \subset \Gamma_X^{\mathrm{alg}} \subset \Gamma_X^{\mathrm{form}}$; membership in that semigroup is not proved by the formal lattice arithmetic. The Gram-index lattice is

$$\Gamma_{\mathrm{gram}} := \mathbb{Z}^3.$$

The Gram map is the quadratic projection

$$\Pi_X : \Gamma_X^{\mathrm{phys}} \longrightarrow \Gamma_{\mathrm{gram}}, \qquad (Q,P) \longmapsto \left(\tfrac{(Q,Q)}{2}, (Q,P), \tfrac{(P,P)}{2}\right).$$

The bilinear obstruction to additivity of Π_X is the symmetric cocycle

$$B : \Gamma_X^{\mathrm{phys}} \times \Gamma_X^{\mathrm{phys}} \longrightarrow \Gamma_{\mathrm{gram}}, \qquad B(c,c') = \big((Q,Q'),(Q,P')+(Q',P),(P,P')\big),$$

satisfying $\Pi_X(c+c') = \Pi_X(c) + \Pi_X(c') + B(c,c')$ and $B(c,c) = 2\Pi_X(c)$. As an ordinary additive cochain identity,

$$B = -\delta\Pi_X, \qquad (\delta q)(c,c') = q(c) + q(c') - q(c+c').$$

The obstruction used below is therefore not a nonzero ordinary cohomology class; it is the absence of an additive linear trivialization compatible with raw bracket degrees. The central extension

$$\widehat{\Gamma}_X := \Gamma_X^{\mathrm{phys}} \oplus_B \Gamma_{\mathrm{gram}}, \qquad (c,T)\star(c',T') := \big(c+c',\ T+T'+B(c,c')\big),$$

has the normal-ordered Gram homomorphism

$$\overline{\Pi}_X(c,T) = \Pi_X(c) - T.$$

It is the additive abelian group on which a Hall product graded by formal charge and normal-ordered Gram invariants can be formulated. A compact Hall source must still restrict this formal group to its algebraic and effective support.

Remark 6.2. The quadratic-projection identity $\Pi_X(c+c') = \Pi_X(c) + \Pi_X(c') + B(c,c')$ expands $(Q+Q', Q+Q')/2, (Q+Q', P+P'), (P+P', P+P')/2$. Cohomologically the central extension is split ($[B] = 0$, Remark 6.10), but the splitting requires the quadratic primitive Π_X, not a linear cochain. Dyonic charges with distinct (Q, P) but identical (n, l, m) sit in the same Gram sector; the Hall bracket distinguishes them through the physical grading on Γ_X^{phys} before passing to Gram invariants.

The preceding determinant was constructed in one cusp chamber. Duality does not preserve the displayed chamber inequalities in the same coordinates; it transports the chamber, the Fourier expansion, and the automorphic line together. The invariant object is

$$\mathcal{D}_X = \Delta_5 \in M_5(\mathbf{Sp}_4(\mathbb{Z}), \nu_{\Delta_5}),$$

with cusp expansion

$$\mathcal{D}_X|_{\mathfrak{c}_\infty} = 64 q^{1/2} r^{1/2} s^{1/2} \prod_{(n,l,m)\in\Gamma_{\mathrm{eff}}} (1 - q^n r^l s^m)^{f(nm,l)}.$$

For

$$Z = \begin{pmatrix} z_1 & z_2 \\ z_2 & z_3 \end{pmatrix},$$

put

$$q = e^{2\pi i z_1}, \qquad r = e^{2\pi i z_2}, \qquad s = e^{2\pi i z_3}.$$

For

$$T(n,l,m) = \begin{pmatrix} n & l/2 \\ l/2 & m \end{pmatrix}, \qquad n, l, m \in \mathbb{Z},$$

put

$$(T(n,l,m), Z)_{\mathrm{tr}} = \mathrm{tr}(T(n,l,m)Z) = nz_1 + lz_2 + mz_3.$$

Then

$$x^{(n,l,m)} = \exp(2\pi i (T(n,l,m), Z)_{\mathrm{tr}}) = q^n r^l s^m.$$

Definition 6.3 *(Gram-index notation).*

$$\Gamma_{\mathrm{gram}} := \Gamma_{\mathrm{ind}} := \mathbb{Z}^3, \qquad \Gamma_{\mathrm{act}} \subset \Gamma_{\mathrm{eff}} \subset \Gamma_{\mathrm{gram}}.$$

Here Γ_{eff} is the cusp-positive semigroup and Γ_{act} is the non-zero Jacobi-support locus. This is notation for Gram invariants and Fourier indices. It is not the compact Hall support; the formal dyonic lattice is $\Gamma_X^{\mathrm{phys}} = \Gamma_X^{\mathrm{form}} = \Lambda_S \oplus \Lambda_S$ of Definition 6.1, and the Gram-index lattice is the image of the quadratic projection $\Pi_X : \Gamma_X^{\mathrm{phys}} \to \Gamma_{\mathrm{gram}}$. A Hall algebra cannot be graded by Γ_{ind} directly because Π_X is quadratic; the correct grading lattice is the central extension $\widehat{\Gamma}_X = \Gamma_X^{\mathrm{phys}} \oplus_B \Gamma_{\mathrm{gram}}$ (Definition 6.1). The grading on $\mathcal{A}_{\mathrm{den}}$ by Γ_{ind} is then the normal-ordered image of a $\widehat{\Gamma}_X$-grading on a hypothetical Hall source whose actual support lies in an algebraic/effective semigroup. The chain-level lift of this image is part of the open Hall datum.

Proposition 6.4 *(Gram matrix convention for protected charges).* Assume a protected charge grading with primitive electric-magnetic pair Q, P in an even lattice. Set

$$n = \frac{(Q,Q)}{2}, \qquad l = (Q,P), \qquad m = \frac{(P,P)}{2}.$$

Then

$$\begin{pmatrix} (Q,Q) & (Q,P) \\ (P,Q) & (P,P) \end{pmatrix} = 2T(n,l,m),$$

and the Gram discriminant is

$$\det(2T(n,l,m)) = 4nm - l^2.$$

The genus-two chemical potential Z pairs with the same matrix:

$$\exp(2\pi i \ \mathrm{tr}(TZ)) = q^n r^l s^m.$$

Thus the Borcherds monomial records the Gram invariants of the protected charge grading, once such a physical grading has been supplied.

Proof. By hypothesis the compactification charge lattice is even, so $(Q,Q)/2$ and $(P,P)/2$ are integral and $(Q,P) \in \mathbb{Z}$. The matrix identity and its determinant are immediate. The final equality is the Fourier character attached to the chemical potential Z. The Harvey–Moore and Dijkgraaf–Verlinde–Verlinde interpretation of this Fourier character as a physical charge grading is additional physical input [40], [26]. □

Definition 6.5 *(Formal additive even Mukai sector and Gram shadow).* Let $X = S \times E$, with S a K3 surface, and let

$$\Lambda_S := \widetilde{H}(S,\mathbb{Z}) = H^0(S,\mathbb{Z}) \oplus H^2(S,\mathbb{Z}) \oplus H^4(S,\mathbb{Z})$$

with Mukai pairing

$$(r,D,s)\cdot(r',D',s') = D\cdot D' - rs' - r's.$$

In the formal even $K3 \times E$ Mukai sector, every class decomposes as

$$v_X = P \otimes 1_E + Q \otimes \omega_E, \qquad Q,P \in \Lambda_S,$$

where $\int_E \omega_E = 1$. The formal additive dyonic lattice is

$$\Gamma_X^{\mathrm{form}} := \Lambda_S \oplus \Lambda_S, \qquad \Gamma_X^{\mathrm{phys}} := \Gamma_X^{\mathrm{form}}, \qquad c = (Q,P).$$

The superscript phys is retained only as a mnemonic in older formulas for this formal sector. For exact triangles or Hall extensions inside a supplied algebraic effective source sector the class c is additive. The Igusa variables are the Gram shadow

$$\Pi_X : \Gamma_X^{\mathrm{form}} \longrightarrow \Gamma_{\mathrm{gram}}, \qquad \Pi_X(Q,P) = \left(\frac{Q^2}{2},\ Q\cdot P,\ \frac{P^2}{2}\right).$$

The map Π_X is an invariant map to Fourier/root degrees; it is not the compact Hall support.

Lemma 6.6 *(Formal primitive torsion-one lift of every Gram triple).* Suppose Λ_S contains two orthogonal hyperbolic planes

$$U_1 \oplus U_2 = \mathbb{Z}e_1 \oplus \mathbb{Z}f_1 \oplus \mathbb{Z}e_2 \oplus \mathbb{Z}f_2,$$

with $e_i^2 = f_i^2 = 0$, $e_i \cdot f_i = 1$, and $U_1 \perp U_2$. Then every Gram triple $(n,l,m) \in \mathbb{Z}^3$ has a primitive saturated formal full-Mukai lift $c = (Q,P) \in \Gamma_X^{\mathrm{phys}}$ with

$$\Pi_X(Q,P) = (n,l,m).$$

One may take

$$Q = e_1 + nf_1, \qquad P = lf_1 + e_2 + mf_2.$$

For this displayed lift the standard wedge torsion invariant

$$I(Q, P) := \gcd\{\text{coordinates of } Q \wedge P \text{ in a } \mathbb{Z}\text{-basis of } \bigwedge^2 \Lambda_S\}$$

is equal to 1.

Proof. The hyperbolic-plane pairings give

$$Q^2 = 2n, \qquad P^2 = 2m, \qquad Q \cdot P = l.$$

Therefore $\Pi_X(Q, P) = (n, l, m)$. The sublattice $\mathbb{Z}Q + \mathbb{Z}P \subset \Lambda_S$ is primitive because the coefficient matrix of Q, P in the ordered basis (e_1, f_1, e_2, f_2) contains the minor

$$\det\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = 1$$

using the e_1, e_2 rows. Hence the lift is primitive saturated. Moreover

$$Q \wedge P = l\, e_1 \wedge f_1 + e_1 \wedge e_2 + m\, e_1 \wedge f_2 + n\, f_1 \wedge e_2 + nm\, f_1 \wedge f_2.$$

The coefficient of $e_1 \wedge e_2$ is 1, so the gcd of all coordinates of $Q \wedge P$ is 1. Thus $I(Q, P) = 1$ for the displayed formal lift. □

Remark 6.7. The obstruction in the compact Hall problem is therefore not pointwise existence of formal charge planes with a prescribed Igusa degree. Every formal Borcherds degree has such a primitive saturated lift, and the displayed lift is torsion-one in the wedge sense above. This does not prove algebraicity, effectivity, nonempty compact moduli, duality orbit descent, or source-index descent. The obstruction is functorial compatibility with Hall addition and the primitive bracket. No Nikulin orbit-uniqueness theorem, T-duality descent of protected indices, or DVV primitive-sector formula is used here. The lemma proves only the displayed formal torsion-one lift.

Lemma 6.8 *(Gram additivity defect).* For $c_i = (Q_i, P_i) \in \Gamma_X^{\mathrm{phys}}$,

$$\Pi_X(c_1 + c_2) = \Pi_X(c_1) + \Pi_X(c_2) + B(c_1, c_2),$$

where

$$B(c_1, c_2) = \big(Q_1 \cdot Q_2,\ Q_1 \cdot P_2 + Q_2 \cdot P_1,\ P_1 \cdot P_2\big) \in \Gamma_{\mathrm{gram}}.$$

Proof. Expand

$$\frac{(Q_1 + Q_2)^2}{2}, \qquad (Q_1 + Q_2) \cdot (P_1 + P_2), \qquad \frac{(P_1 + P_2)^2}{2}.$$

The three cross terms are exactly the three components of $B(c_1, c_2)$. □

Lemma 6.9 *(Symmetric bilinear 2-cocycle and on-the-nose associativity of the Gram-corrected charge product).* Let S be a complex K3 surface and $\Lambda_S = H^*(S, \mathbb{Z})$ the Mukai lattice (even unimodular, signature $(4, 20)$). Let $\Gamma_X^{\mathrm{form}} := \Lambda_S \oplus \Lambda_S$ and $\Gamma_X^{\mathrm{phys}} := \Gamma_X^{\mathrm{form}}$, $\Gamma_{\mathrm{gram}} := \mathbb{Z}^3$, and let $\Pi_X : \Gamma_X^{\mathrm{phys}} \to \Gamma_{\mathrm{gram}}$, $B : \Gamma_X^{\mathrm{phys}} \times \Gamma_X^{\mathrm{phys}} \to \Gamma_{\mathrm{gram}}$ be as in Definition 6.1, namely, with $c = (Q, P)$,

$$\Pi_X(Q, P) = \big(\tfrac{(Q,Q)}{2},\ (Q, P),\ \tfrac{(P,P)}{2}\big), \qquad B((Q, P), (Q', P')) = \big((Q, Q'), (Q, P') + (Q', P), (P, P')\big).$$

Then:

(i) B is symmetric and $\mathbb{Z}$-bilinear, and takes integral values: $B \in \mathrm{Hom}_{\mathbb{Z}}\big(S^2\Gamma_X^{\mathrm{phys}}, \Gamma_{\mathrm{gram}}\big) \subset \mathrm{Hom}_{\mathbb{Z}}\big(\Gamma_X^{\mathrm{phys}} \otimes_{\mathbb{Z}} \Gamma_X^{\mathrm{phys}}, \Gamma_{\mathrm{gram}}\big)$.

(ii) (Polarisation.) $B(c, c) = 2\Pi_X(c)$ for every $c \in \Gamma_X^{\mathrm{phys}}$.

(iii) (Bilinear cocycle identity.) B is normalised, $B(c, 0) = B(0, c) = 0$, and the additive coboundary

$$(\delta B)(a, b, c) := B(b, c) - B(a + b, c) + B(a, b + c) - B(a, b)$$

vanishes identically. We use this identity to define the extension below; no nonzero ordinary cohomology class is asserted.

(iv) (Star-associativity on the nose.) The product on $\widehat{\Gamma}_X := \Gamma_X^{\mathrm{phys}} \oplus \Gamma_{\mathrm{gram}}$,

$$(c, T) \star (c', T') := \big(c + c',\ T + T' + B(c, c')\big),$$

is associative and commutative; the unit is $(0, 0)$ and the inverse of (c, T) is $(-c,\ -T + B(c, c)) = (-c,\ -T + 2\Pi_X(c))$. Thus $(\widehat{\Gamma}_X, \star)$ is an abelian group, and the sequence

$$0 \longrightarrow \Gamma_{\mathrm{gram}} \xrightarrow{T \mapsto (0,T)} \widehat{\Gamma}_X \xrightarrow{\mathrm{pr}_1} \Gamma_X^{\mathrm{phys}} \longrightarrow 0$$

is a central extension of abelian groups with cocycle B.

(v) (Normal-ordered Gram map.) The map

$$\overline{\Pi}_X : \widehat{\Gamma}_X \longrightarrow \Gamma_{\mathrm{gram}}, \qquad \overline{\Pi}_X(c, T) = \Pi_X(c) - T,$$

is a homomorphism from $(\widehat{\Gamma}_X, \star)$ to the additive group Γ_{gram}.

(vi) (Raw placement and splitting.) There are two maps from the physical charge group into the normal-ordered extension:

$$i_0(c) := (c, 0), \qquad s(c) := (c, \Pi_X(c)).$$

The map i_0 is the raw Hall placement of a generator with visible Gram degree $\Pi_X(c)$; it is not additive for $\star$ unless $B \equiv 0$. The map s is the split additive section: it is a homomorphism for $\star$, but

$$\overline{\Pi}_X(i_0(c)) = \Pi_X(c), \qquad \overline{\Pi}_X(s(c)) = 0.$$

Equivalently $B = -\delta\Pi_X$ for the quadratic cochain Π_X. The extension is split as an abstract extension of abelian groups, and $(c, T) \mapsto (c, T - \Pi_X(c))$ is an isomorphism $\widehat{\Gamma}_X \xrightarrow{\sim} \Gamma_X^{\mathrm{phys}} \oplus \Gamma_{\mathrm{gram}}$ of abelian groups carrying $\star$ to componentwise addition.

(vii) (Quadratic obstruction.) B is not the coboundary of any *linear* 1-cochain $L : \Gamma_X^{\mathrm{phys}} \to \Gamma_{\mathrm{gram}}$: if $B(c, c') = L(c) + L(c') - L(c + c')$ for an additive L, then setting $c' = c$ and using (ii) gives $2\Pi_X(c) = 2L(c) - L(2c) = 0$, i.e. $\Pi_X \equiv 0$, contradiction. Trivialising $[B]$ requires the quadratic refinement Π_X; the defect is a polarization-correction class, not a separate Heisenberg extension datum.

Proof. Write $c = (Q, P)$, $c' = (Q', P')$, $c'' = (Q'', P'')$ with $Q, Q', Q'', P, P', P'' \in \Lambda_S$, and abbreviate $(\,\cdot\,,\,\cdot\,) : \Lambda_S \times \Lambda_S \to \mathbb{Z}$ by juxtaposition, $QQ' = (Q, Q')$, so that $B(c, c') = (QQ',\ QP' + Q'P,\ PP')$.

(i) Symmetry, bilinearity, integrality. The components QQ' and PP' are the symmetric Mukai pairing on Λ_S, so each is symmetric and $\mathbb{Z}$-bilinear. The middle component is the polarisation $(c, c') \mapsto QP' + Q'P$ of the symmetric biadditive form $(c, c') \mapsto QP' + Q'P$; swapping $(Q, P) \leftrightarrow (Q', P')$ returns $Q'P + QP' = QP' + Q'P$, so it is symmetric, and $\mathbb{Z}$-bilinear in each slot by bilinearity of $(\,\cdot\,,\,\cdot\,)$. Integrality of every component follows from the unimodular even structure on Λ_S.

(ii) Polarisation. Set $c' = c$: the cross terms reduce to QQ, $QP + QP = 2QP$, PP, so $B(c, c) = (Q^2, 2QP, P^2) = 2\Pi_X(c)$.

(iii) Cocycle. Bilinearity gives normalisation $B(c, 0) = B(0, c) = 0$ immediately. For the cocycle identity, observe that any symmetric $\mathbb{Z}$-bilinear form $\beta : G \times G \to A$ on an abelian group G with values in a trivial G-module A satisfies, by bilinearity in the first slot,

$$\beta(a + b, c) = \beta(a, c) + \beta(b, c), \qquad \beta(a, b + c) = \beta(a, b) + \beta(a, c).$$

Substituting into $(\delta\beta)(a, b, c) = \beta(b, c) - \beta(a + b, c) + \beta(a, b + c) - \beta(a, b)$,

$$(\delta\beta)(a, b, c) = \beta(b, c) - \beta(a, c) - \beta(b, c) + \beta(a, b) + \beta(a, c) - \beta(a, b) = 0.$$

Applied componentwise to B, this yields $\delta B = 0$. Hence B is a normalized symmetric bilinear additive 2-cocycle with values in Γ_{gram} in the displayed sense.

(iv) Associativity, identity, inverse. Compute the two bracketings of $(c, T) \star (c', T') \star (c'', T'')$:

$$\big((c, T) \star (c', T')\big) \star (c'', T'') = \big(c + c' + c'',\ T + T' + T'' + B(c, c') + B(c + c', c'')\big),$$
$$(c, T) \star \big((c', T') \star (c'', T'')\big) = \big(c + c' + c'',\ T + T' + T'' + B(c', c'') + B(c, c' + c'')\big).$$

The associator is the difference of the central components, $\mathrm{Ass}(c, c', c'') = B(c, c') + B(c + c', c'') - B(c', c'') - B(c, c' + c'')$, which equals $-(\delta B)(c, c', c'') = 0$ by (iii); equivalently, it vanishes by bilinearity of B. Commutativity follows from symmetry of B. The identity computation $(c, T) \star (0, 0) = (c, T + B(c, 0)) = (c, T)$ and the inverse computation $(c, T) \star (-c, -T + B(c, c)) = (0, B(c, c) - B(c, c)) = (0, 0)$ (using $B(c, -c) = -B(c, c)$ by bilinearity) finish the abelian-group structure. That B is the cocycle of the displayed central extension is direct from the formula for $\star$ and normalisation $B(c, 0) = B(0, c) = 0$.

(v) Normal-ordered Gram map. Using Lemma 6.8,

$$\begin{aligned}
\overline{\Pi}_X((c, T) \star (c', T')) &= \overline{\Pi}_X(c + c', T + T' + B(c, c')) \\
&= \Pi_X(c + c') - T - T' - B(c, c') \\
&= \Pi_X(c) + \Pi_X(c') - T - T' \\
&= \overline{\Pi}_X(c, T) + \overline{\Pi}_X(c', T').
\end{aligned}$$

Thus $\overline{\Pi}_X$ is additive.

(vi) Splitting. By Lemma 6.8, $\Pi_X(c+c') = \Pi_X(c) + \Pi_X(c') + B(c,c')$, so

$$s(c) \star s(c') = (c+c',\ \Pi_X(c) + \Pi_X(c') + B(c,c')) = (c+c',\ \Pi_X(c+c')) = s(c+c').$$

Thus s is a homomorphism, and $B(c,c') = \Pi_X(c+c') - \Pi_X(c) - \Pi_X(c') = -(\partial\Pi_X)(c,c')$ shows $B = -\partial\Pi_X$, so $[B] = 0$. The change of variables $(c,T) \mapsto (c, T - \Pi_X(c))$ is bijective on the underlying set with inverse $(c,T) \mapsto (c, T + \Pi_X(c))$, and carries $\star$ to componentwise addition:

$$(c, T - \Pi_X(c)) + (c', T' - \Pi_X(c')) = (c+c',\ T + T' - \Pi_X(c) - \Pi_X(c')) = (c+c',\ (T + T' + B(c,c')) - \Pi_X(c+c')),$$

matching the change of variables applied to $(c+c', T+T'+B(c,c')) = (c,T) \star (c',T')$.

(vii) Quadratic obstruction. Suppose $B = \partial L$ for some additive $L : \Gamma_X^{\mathrm{phys}} \to \Gamma_{\mathrm{gram}}$, i.e. $B(c,c') = L(c) + L(c') - L(c+c')$. Setting $c' = c$, $B(c,c) = 2L(c) - L(2c)$; but L additive gives $L(2c) = 2L(c)$, so $B(c,c) = 0$, contradicting $B(c,c) = 2\Pi_X(c)$ for any $c = (Q,P)$ with $(Q,Q), (Q,P), (P,P)$ not all zero. Hence no linear trivialisation exists; the quadratic primitive Π_X is necessary.

Scope of the chain-level claim. Clauses (i)–(vii) prove an ordinary set-level and H^0-level polarization correction. They do not prove that the protected Hall chain complex has no higher associators. In particular no acyclicity of a bar complex $B_*(\widehat{\Gamma}_X)$ is asserted: the bar construction of a discrete abelian group records group homology, and cannot be used to kill Stasheff coherences. Chain-level strictness of the Hall product therefore remains part of the orientation/coherence hypotheses in Theorem 10.39; what this lemma supplies is only the explicit quadratic polarization correction $B = -\partial\Pi_X$ and the split degree monoid on which a later cocycle-corrected descent can act. □

Remark 6.10. The preceding lemma records the additive degree correction. It proves that the Gram defect is the coboundary of the quadratic refinement Π_X:

$$d_{\mathrm{Hoch}}(-\Pi_X) = B$$

in Γ_{gram}-valued additive cochains. The no-go statement below is not a claim that B defines a nonzero ordinary cohomology class. It is the narrower statement that raw bracket recognition needs an additive degree target, while $c \mapsto \Pi_X(c)$ is genuinely quadratic and has no additive linear trivialization. The lemma does not strictify the Hall chain complex. A chain-level Hall product still requires the multiplicative orientation, cyclic equivariance, and associator data listed in Theorem 10.39; absent those data the lift is only A_∞-coherent, and the obstruction is tracked by the supplied cochain Θ, not by acyclicity of a group bar complex.

Definition 6.11 *(Normal-ordered Gram-corrected charge extension).* Set

$$\widehat{\Gamma}_X := \Gamma_X^{\mathrm{phys}} \oplus \Gamma_{\mathrm{gram}}.$$

For $c, c' \in \Gamma_X^{\mathrm{phys}}$ and $T, T' \in \Gamma_{\mathrm{gram}}$, define

$$(c,T) \star (c',T') := (c+c',\ T + T' + B(c,c')).$$

The section

$$s : \Gamma_X^{\mathrm{phys}} \longrightarrow \widehat{\Gamma}_X, \qquad s(c) = (c, \Pi_X(c))$$

is additive for $\star$. The raw placement

$$i_0 : \Gamma_X^{\mathrm{phys}} \longrightarrow \widehat{\Gamma}_X, \qquad i_0(c) = (c, 0)$$

is the placement of a generator before composite normal-ordering counterterms have been accumulated. Thus

$$\overline{\Pi}_X(i_0(c)) = \Pi_X(c), \qquad \overline{\Pi}_X(s(c)) = 0.$$

The normal-ordered Gram map is

$$\overline{\Pi}_X : \widehat{\Gamma}_X \longrightarrow \Gamma_{\mathrm{gram}}, \qquad \overline{\Pi}_X(c, T) = \Pi_X(c) - T.$$

Thus the compact Hall object, if constructed, is graded upstairs by Γ_X^{phys}, or by the refinement $\widehat{\Gamma}_X$ after the normal-ordering counterterm is recorded. The Γ_{gram}-graded Igusa object is obtained only after normal-ordered Gram pushforward along $\overline{\Pi}_X$, or by a chain-level categorification of the same subtraction.

Lemma 6.12 *(Group law of the normal-ordered charge extension).* The operation $\star$ of Definition 6.11 makes $\widehat{\Gamma}_X$ an abelian group with identity $(0, 0)$ and inverse

$$(c, T)^{-1} = (-c, -T + B(c, c)).$$

The section $s(c) = (c, \Pi_X(c))$ is additive for $\star$, and $\overline{\Pi}_X$ is a homomorphism.

Proof. By Lemma 6.9(i),(iii),(iv), $\star$ is associative and commutative, the identity is $(0, 0)$, and the inverse of (c, T) is $(-c, -T + B(c, c))$; the section s is the splitting of clause (vi). Clause (v) of the same lemma proves that $\overline{\Pi}_X$ is a homomorphism. □

Lemma 6.13 *(Normal-ordering flag formula).* For $c_1, \ldots, c_k \in \Gamma_X^{\mathrm{phys}}$, multiplication of raw placements in $\widehat{\Gamma}_X$ gives

$$i_0(c_1) \star \cdots \star i_0(c_k) = \left(\sum_i c_i, \sum_{i<j} B(c_i, c_j) \right).$$

Consequently

$$\overline{\Pi}_X(i_0(c_1) \star \cdots \star i_0(c_k)) = \sum_i \Pi_X(c_i).$$

Proof. Induct on k. The case $k = 2$ is the definition of $\star$. The induction step adds

$$B\left(\sum_{i \le k} c_i, c_{k+1} \right) = \sum_{i \le k} B(c_i, c_{k+1})$$

by bilinearity. Applying $\overline{\Pi}_X$ subtracts the accumulated pairwise B-term from $\Pi_X(\sum_i c_i) = \sum_i \Pi_X(c_i) + \sum_{i<j} B(c_i, c_j)$. □

Definition 6.14 *(Raw homogeneous and fixed-lift Π_X-pushforwards).* Let $P = \bigoplus_{c \in \Gamma_X^{\mathrm{phys}}} P_c$ be a Γ_X^{phys}-graded primitive object. The raw homogeneous Π_X-pushforward attempts to grade by the quadratic value alone:

$$\left(\Pi_{X,*}^{\mathrm{raw}} P \right)_\gamma := \bigoplus_{\Pi_X(c) = \gamma} P_c.$$

It records no T-coordinate and no normal-ordering counterterm. A strict fixed-lift raw pushforward is the narrower operation obtained by choosing one lift $L(\gamma) \in \Gamma_X^{\mathrm{phys}}$ for every retained γ and setting

$$\left(\Pi_{X,*}^{\mathrm{raw},L} P\right)_\gamma := P_{L(\gamma)}.$$

THEOREM 6.15 *(Raw homogeneous fixed-lift Π_X-pushforward cannot realize the type-II real-root strings).* Let

$$P = \bigoplus_{c \in \Gamma_X^{\mathrm{phys}}} P_c$$

be a Γ_X^{phys}-graded Lie superalgebra with

$$[P_c, P_{c'}] \subset P_{c+c'}.$$

Suppose a strict fixed-lift raw pushforward along the quadratic map $\Pi_X : \Gamma_X^{\mathrm{phys}} \to \Gamma_{\mathrm{gram}}$ were a Γ_{gram}-graded Lie superalgebra structure on the displayed bracket channels of chosen charge lifts. Then every non-zero bracket channel $[P_c, P_{c'}]$ contributing to that strict raw pushforward must satisfy

$$B(c, c') = 0.$$

For the type-II real simple roots of the Gritsenko–Nikulin algebra, this condition is incompatible with the non-zero second real-root strings $2\delta_i + \delta_j$, $i \neq j$. Hence these type-II real-root brackets cannot be obtained by strict fixed-lift raw Π_X-descent. They require the normal-ordered extension $\widehat{\Gamma}_X = \Gamma_X^{\mathrm{phys}} \oplus_B \Gamma_{\mathrm{gram}}$ and the homomorphism $\overline{\Pi}_X(c, T) = \Pi_X(c) - T$.

Proof. If the raw pushforward is Γ_{gram}-graded, then a non-zero bracket of classes of raw Gram degrees $\Pi_X(c)$ and $\Pi_X(c')$ must land in raw Gram degree $\Pi_X(c) + \Pi_X(c')$. But the upstairs bracket lands in physical degree $c + c'$, whose raw Gram shadow is

$$\Pi_X(c + c') = \Pi_X(c) + \Pi_X(c') + B(c, c').$$

Thus every non-zero bracket channel visible in a raw pushforward must have $B(c, c') = 0$.

Let c_i, c_j be physical lifts of two distinct type-II real simple root degrees γ_i, γ_j, so

$$\Pi_X(c_i) = \gamma_i, \qquad \Pi_X(c_j) = \gamma_j, \qquad \gamma_i \neq 0.$$

The Gritsenko–Nikulin real simple roots have Cartan matrix

$$\begin{pmatrix} 2 & -2 & -2 \\ -2 & 2 & -2 \\ -2 & -2 & 2 \end{pmatrix}.$$

The real-root string through distinct simple roots contains non-zero brackets in degrees $\gamma_i + \gamma_j$ and $2\gamma_i + \gamma_j$; this is the low-degree real-root information recorded in Proposition 10.79. The first non-zero bracket would force

$$B(c_i, c_j) = 0.$$

The second non-zero bracket is obtained by bracketing once more with the c_i-degree generator, so raw Gram additivity would force

$$B(c_i, c_i + c_j) = 0.$$

By bilinearity and polarization,

$$B(c_i, c_i + c_j) = B(c_i, c_i) + B(c_i, c_j) = 2\Pi_X(c_i) + 0 = 2\gamma_i.$$

Since $\gamma_i \neq 0$ in the torsion-free lattice Γ_{gram}, this is non-zero. Contradiction.

The contradiction uses the fixed chosen lifts c_i, c_j and the raw quadratic map Π_X. It does not exclude special isolated B-isotropic bracket channels, and it does not decide a mixed-lift fibre-summed raw construction in which several lifts of a single Gram degree are combined before taking brackets. It excludes the strict homogeneous fixed-lift raw pushforward needed by this proof. The normal-ordered extension removes exactly this obstruction because $\overline{\Pi}_X$ is additive on $\widehat{\Gamma}_X$. □

Remark 6.16. The remaining raw problem is the fibre-summed version: one may try to sum several charge lifts over a fixed Gram degree before bracketing. Theorem 6.15 neither proves nor disproves such a construction.

Remark 6.17 *(Normal-ordered primitive recognition target).* The interface with the denominator algebra is therefore not the raw map Π_X. It is the corrected route

$$\Gamma_X^{\text{phys}} \longrightarrow \widehat{\Gamma}_X \xrightarrow{\overline{\Pi}_X} \Gamma_{\text{gram}} \xrightarrow{\alpha} \Lambda_{II}^{2,1}.$$

At chain level this becomes the descent $\overline{\Pi}^{\Theta}_{X,*}$ of Open Problem 10.34. The primitive recognition problem is to construct a protected primitive Hall algebra whose normal-ordered descended bracket is the Borcherds–Kac bracket of $\mathfrak{g}_{\Delta_5}$. Raw Gram pushforward is not an admissible recognition target.

Definition 6.18 *(Gram-index pairing).* For

$$\gamma = (n, l, m), \qquad \gamma' = (n', l', m'),$$

define

$$\langle \gamma, \gamma' \rangle_{\text{ind}} = 2(nm' + n'm) - ll'.$$

The Gram discriminant is

$$\Delta_{\text{gram}}(\gamma) = \langle \gamma, \gamma \rangle_{\text{ind}} = 4nm - l^2.$$

Remark 6.19 *(Chamber transport).* The pairing $\langle\ ,\ \rangle_{\text{ind}}$ is the pairing on Gram invariants. The order on Γ_{eff} is a Fock polarization at one cusp. A duality element transports this polarization to the corresponding product chamber at the transported cusp.

Part III

Denominator algebra and compact Dirac–Igusa realization

7 The exterior-square model of the automorphic group

The form Δ_5 is an automorphic form on the Siegel space. The physical U-duality interpretation of the same arithmetic group is external input from the $K3 \times E$ compactification; it is not needed for the lattice calculation below. The calculation records the first-principles bridge from the Siegel variables to the type-IV Lorentzian lattice on which the denominator algebra is written. $\mathbf{Sp}_4(\mathbb{Z})$ is the arithmetic symmetry of the genus-two chemical potential and of the invariant Gram plane on $\Lambda^{3,2}$; it is not asserted to be the full T-duality or U-duality group of the $K3 \times T^2$ compactification, which is larger.

$$\Lambda^4 = \mathbb{Z}e_1 \oplus \mathbb{Z}e_2 \oplus \mathbb{Z}e_3 \oplus \mathbb{Z}e_4.$$

$$g \in \mathbf{SL}_4(\mathbb{Z}) : \qquad \wedge^2 g : \wedge^2\Lambda^4 \longrightarrow \wedge^2\Lambda^4.$$

$$u \wedge v = (u, v)e_1 \wedge e_2 \wedge e_3 \wedge e_4.$$

$$\big(\wedge^2\Lambda^4, (\ ,\ )\big) \simeq \mathrm{II}_{3,3}.$$

Fix

$$q_0 = e_1 \wedge e_3 + e_2 \wedge e_4.$$

For $x, y \in \Lambda^4$, define

$$-x \wedge y \wedge q_0 = B_{q_0}(x, y)e_1 \wedge e_2 \wedge e_3 \wedge e_4.$$

$$\{g \in \mathbf{SL}(\Lambda^4) \mid (\wedge^2 g)q_0 = q_0\} \simeq \mathbf{Sp}_4(\mathbb{Z}).$$

Set

$$\Lambda^{3,2} = q_0^{\perp} \subset \wedge^2\Lambda^4.$$

$$f_1 = e_1 \wedge e_2, \quad f_2 = e_2 \wedge e_3, \quad f_3 = e_1 \wedge e_3 - e_2 \wedge e_4, \quad f_{-2} = e_4 \wedge e_1, \quad f_{-1} = e_4 \wedge e_3.$$

$$(f_1, f_{-1}) = (f_2, f_{-2}) = -1, \qquad (f_3, f_3) = 2.$$

$$\Lambda^{3,2} \simeq \Lambda^{(1,1)} \oplus \Lambda^{(1,1)} \oplus [2], \qquad \Lambda^{(1,1)} = \begin{pmatrix} 0 & -1 \\ -1 & 0 \end{pmatrix}.$$

$$\mathbb{H}^{\mathbf{IV}} = \{Z \in \mathbb{P}(\Lambda^{3,2} \otimes \mathbb{C}) \mid (Z, Z) = 0,\ (Z, \overline{Z}) < 0\}.$$

$$\mathbb{H}^{\mathbf{IV}}_{+} = \{{}^t(z_2^2 - z_1 z_3,\ z_3,\ z_2,\ z_1,\ 1)z_0 \in \mathbb{H}^{\mathbf{IV}} \mid \mathrm{Im} z_1 > 0\}.$$

Writing $y_i = \mathrm{Im} z_i$,

$$(Z, \overline{Z}) < 0 \quad \Longleftrightarrow \quad y_1 y_3 - y_2^2 > 0.$$

$$\mathbb{H}^{\mathbf{IV}}_{+} \simeq \left\{ \begin{pmatrix} z_1 & z_2 \\ z_2 & z_3 \end{pmatrix} \in \mathbb{H}_2 \right\}.$$

Let $\mathbf{O}_{\mathbb{R}}(\Lambda^{3,2})_+$ be the index-two subgroup preserving $\mathbb{H}^{\mathbf{IV}}_{+}$. Its kernel on $\mathbb{H}^{\mathbf{IV}}_{+}$ is $\{\pm\mathbf{I}_5\}$, and

$$\mathbf{O}_{\mathbb{R}}(\Lambda^{3,2})_+ = \{\pm\mathbf{I}_5\}\mathbf{SO}_{\mathbb{R}}(\Lambda^{3,2})_+,$$

where $\mathbf{SO}_{\mathbb{R}}(\Lambda^{3,2})_+$ has real spin-norm 1. Put

$$\mathbf{O}(\Lambda^{3,2})_+ = \mathbf{O}_{\mathbb{R}}(\Lambda^{3,2})_+ \cap \mathbf{O}(\Lambda^{3,2})$$

and

$$\mathbf{SO}(\Lambda^{3,2})_+ = \mathbf{SO}_{\mathbb{R}}(\Lambda^{3,2})_+ \cap \mathbf{O}(\Lambda^{3,2}).$$

$$M = \begin{pmatrix} m_1 & m_2 \\ m_2 & m_3 \end{pmatrix},$$

$$\wedge^2(g_0) = \wedge^2\left(\begin{pmatrix} 0 & \mathbf{I}_2 \\ -\mathbf{I}_2 & 0 \end{pmatrix}\right) = \begin{pmatrix} 0 & 0 & 0 & 0 & -1 \\ 0 & 0 & 0 & -1 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & -1 & 0 & 0 & 0 \\ -1 & 0 & 0 & 0 & 0 \end{pmatrix},$$

$$\wedge^2(g_M) = \wedge^2\left(\begin{pmatrix} \mathbf{I}_2 & M \\ 0 & \mathbf{I}_2 \end{pmatrix}\right) = \begin{pmatrix} 1 & -m_1 & 2m_2 & -m_3 & m_2^2 - m_1 m_3 \\ 0 & 1 & 0 & 0 & m_3 \\ 0 & 0 & 1 & 0 & m_2 \\ 0 & 0 & 0 & 1 & m_1 \\ 0 & 0 & 0 & 0 & 1 \end{pmatrix},$$

$$\wedge^2(g_A) = \wedge^2\left(\begin{pmatrix} {}^tA^{-1} & 0 \\ 0 & A \end{pmatrix}\right) = \det(A)\begin{pmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & a_1^2 & -2a_1a_2 & a_2^2 & 0 \\ 0 & -a_1a_3 & a_1a_4 + a_2a_3 & -a_2a_4 & 0 \\ 0 & a_3^2 & -2a_3a_4 & a_4^2 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

Lemma 7.1. The exterior-square representation is

$$\wedge^2 : \mathbf{Sp}_4(\mathbb{Z})/\{\pm\mathbf{I}_4\} \to \mathbf{SO}_+(\Lambda^{3,2}) \simeq \mathbf{O}(\Lambda^{3,2})_+/\{\pm\mathbf{I}_5\}.$$

For every $g \in \mathbf{Sp}_4(\mathbb{Z})$,

$$\begin{tikzcd} \mathbb{H}_2 \arrow[r] \arrow[d] & \mathbb{H}_2 \arrow[d] \\ \mathbb{H}_+^{\mathbf{IV}} \arrow[r, "g\wedge g"] & \mathbb{H}_+^{\mathbf{IV}} \end{tikzcd}$$

Proof. The Pfaffian pairing is preserved by $\wedge^2\mathbf{SL}_4$. The subgroup fixing

$$q_0 = e_1 \wedge e_3 + e_2 \wedge e_4$$

is the group preserving the alternating form B_{q_0}, namely $\mathbf{Sp}_4(\mathbb{Z})$. Its action preserves $q_0^\perp = \Lambda^{3,2}$ and the displayed component of the type-**IV** domain. The kernel on $q_0^\perp$ is $\{\pm\mathbf{I}_4\}$. The exceptional isomorphism $\mathbf{PSp}_4(\mathbb{R}) \simeq \mathbf{SO}_{\mathbb{R}}(3,2)_+$ identifies the image with $\mathbf{SO}_+(\Lambda^{3,2})$. The displayed matrices are the integral generators on both sides. Substitution of

$${}^t(z_2^2 - z_1z_3, z_3, z_2, z_1, 1)z_0$$

into the two actions produces the commutative square. □

Definition 7.2 (Automorphic duality datum).

$$G_{\mathrm{aut}} = \mathbf{PSp}_4(\mathbb{Z}) \simeq \mathbf{SO}_+(\Lambda^{3,2}).$$

The virtual determinant $\mathcal{D}_X = \Delta_5$ is not an invariant scalar function on $\mathbb{H}_2$. It is a section of the automorphic line

$$\mathcal{L}^5 \otimes \nu_{\Delta_5}$$

with transformation law

$$\Delta_5(g \cdot Z) = \nu_{\Delta_5}(g) \det(CZ + D)^5 \Delta_5(Z), \qquad g = \begin{pmatrix} A & B \\ C & D \end{pmatrix} \in \mathbf{Sp}_4(\mathbb{Z}).$$

The scalar square is a section of $\mathcal{L}^{10}$:

$$\Delta_5^2(g \cdot Z) = \det(CZ + D)^{10} \Delta_5^2(Z).$$

For every scalar-square branch,

$$Z_{\square}^X(g \cdot Z) = \det(CZ + D)^{-10} Z_{\square}^X(Z).$$

8 The Weyl chamber and the Weyl vector

Fix the primitive hyperbolic sublattice

$$\Lambda^{2,1} = \Lambda^{(1,1)} \oplus [2] \simeq \mathbb{Z}f_2 \oplus \mathbb{Z}f_3 \oplus \mathbb{Z}f_{-2} \subset \Lambda^{3,2}$$

and its type-II sublattice

$$\Lambda_{II}^{2,1} = \{mf_2 + lf_3 + nf_{-2} \in \Lambda^{2,1} \mid m \equiv n \equiv 0 \pmod 2\}.$$

Change polarization on the same Gram-index lattice:

$$\Gamma_{\mathrm{ind}} \xrightarrow{\alpha} \Lambda_{II}^{2,1}, \qquad \alpha(n, l, m) = 2nf_2 - lf_3 + 2mf_{-2}.$$

The cusp polarization gives the Fock determinant. The denominator polarization gives the Borcherds chamber.

Extend $\phi \in \mathbf{O}(\Lambda^{2,1})$ by the identity on $(\Lambda^{2,1})^{\perp}$:

$$\mathbf{O}(\Lambda^{2,1}) \hookrightarrow \mathbf{O}(\Lambda^{3,2}).$$

For the component-preserving subgroup attached to the positive cone below,

$$\Delta_5 \in M_5(\mathbf{O}(\Lambda^{2,1})_+, \nu_{\Delta_5}).$$

For a primitive element $\alpha \in \Lambda^{2,1}$ satisfying $(\alpha, \alpha) > 0$ and

$$(\alpha, \alpha) \mid 2(\Lambda^{2,1}, \alpha),$$

Kac reflection [48]:

$$s_\alpha : x \mapsto x - 2\frac{(x, \alpha)}{(\alpha, \alpha)}\alpha.$$

$$s_\alpha(\alpha) = -\alpha, \qquad s_\alpha|_{\alpha^\perp} = \mathbf{I}.$$

For $(\alpha, \alpha) = 2$:

$$s_\alpha : x \mapsto x - (x, \alpha)\alpha.$$

Lemma 8.1. Let $(\alpha, \alpha) = 2$. For

$$\delta_1 = 2f_2 - f_3, \qquad \delta_2 = 2f_{-2} - f_3, \qquad \delta_3 = f_3,$$

[37, Theorem 2.1 and Section 4]; equivalently [39, Section 5.1.1, case (t=1,II,0)].

$$\Delta_5(s_\alpha Z) = -\Delta_5(Z)$$

when $\alpha \in \{\delta_1, \delta_2, \delta_3\}$. For

$$\alpha \in \{f_{-2} - f_2,\ f_2 - f_3,\ f_2 + f_3\},$$
$$\Delta_5(s_\alpha Z) = \Delta_5(Z)$$

Proof. Let

$$\overline{U} = \wedge^2 \begin{pmatrix} {}^tU^{-1} & 0 \\ 0 & U \end{pmatrix}, \qquad U \in \mathbf{GL}_2(\mathbb{Z}).$$

Required reflections:

$$s_{f_{-2}-f_2} = -\overline{\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}}, \qquad s_{f_3} = -\overline{\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}}, \qquad s_{f_2-f_3} = -\overline{\begin{pmatrix} 1 & -1 \\ 0 & -1 \end{pmatrix}}.$$

Maass's character has value -1 on s_{δ_i} and $+1$ on the three other reflection classes. □

The time-like cone is

$$C(\Lambda^{2,1}) = \{x \in \Lambda^{2,1} \otimes \mathbb{R} | (x, x) < 0\}.$$

Let $C(\Lambda^{2,1})_+$ be the component for which

$$\Omega(C(\Lambda^{2,1})_+) := \Lambda^{2,1} \otimes \mathbb{R} + iC(\Lambda^{2,1})_+ \subset \mathbb{H}_+^{\mathbf{IV}}.$$

For

$$z = z_3 f_2 + z_2 f_3 + z_1 f_{-2} \in \Omega(C(\Lambda^{2,1})_+),$$

the type **IV** point is

$$Z = {}^t(z_2^2 - z_1 z_3,\ z_3,\ z_2,\ z_1,\ 1) z_0.$$

Let $\mathbf{O}(\Lambda^{2,1})_+$ be the index-two subgroup fixing $C(\Lambda^{2,1})_+$. It acts discretely on

$$C(\Lambda^{2,1})_+ / \mathbb{R}_{>0}.$$

For $(\alpha, \alpha) > 0$,

$$\mathbb{H}_\alpha = \{\mathbb{R}_{>0} x \in C(\Lambda^{2,1})_+ / \mathbb{R}_{>0} \mid (x, \alpha) = 0\},$$

and

$$\mathbb{H}_{\alpha,+} = \{\mathbb{R}_{>0} x \in C(\Lambda^{2,1})_+ / \mathbb{R}_{>0} \mid (x, \alpha) \leq 0\}.$$

Let $W(\Lambda^{2,1})$ be the subgroup generated by these positive-norm reflections:

$$W(\Lambda^{2,1}) \subset \mathbf{O}(\Lambda^{2,1})_+.$$

Nikulin's theorem for $\Lambda^{(1,1)} \oplus [2]$ [63]:

$$\mathbf{O}(\Lambda^{2,1})_+ = W^{(2)}(\Lambda^{2,1}),$$

where $W^{(2)}$ is generated by reflections in square-2 vectors. For $k > 0$ put

$$\Delta^{(k)}(\Lambda^{2,1}) = \{\delta \in \Lambda^{2,1} \mid \delta \text{ primitive}, \ (\delta, \delta) = k, \ s_\delta \in \mathbf{O}(\Lambda^{2,1})\}.$$

Each $\delta \in \Delta^{(2)}(\Lambda^{2,1})$ has one of two divisibilities:

- Type I: $(\delta, \Lambda^{2,1}) = \mathbb{Z}$
- Type II: $(\delta, \Lambda^{2,1}) = 2\mathbb{Z}$

The type sublattices are

$$\Lambda_I^{2,1} = \{mf_2 + lf_3 + nf_{-2} \in \Lambda^{2,1} \mid m + l + n \equiv 0 \pmod 2\},$$

and the already defined $\Lambda_{II}^{2,1}$. Type I roots lie in $\Lambda_I^{2,1}$; type II roots lie in $\Lambda_{II}^{2,1}$. The reflection subgroups are normal, with indices

$$[W^{(2)}(\Lambda^{2,1}) : W^{(2)}(\Lambda_I^{2,1})] = 2, \qquad [W^{(2)}(\Lambda^{2,1}) : W^{(2)}(\Lambda_{II}^{2,1})] = 6.$$

The chambers are

$$\mathcal{P}' = \bigcap_{\delta \in \mathcal{P}'_{prim}} \mathbb{H}_{\delta,+}, \qquad \mathcal{P}' \in \{\mathcal{P}, \mathcal{P}_I, \mathcal{P}_{II}\},$$

with walls

$$\mathcal{P}_{prim} = \{f_2 - f_3, \ f_{-2} - f_2, \ f_3\},$$
$$\mathcal{P}_{I,prim} = \{f_2 - f_3, \ f_{-2} - f_2, \ f_2 + f_3\},$$
$$\mathcal{P}_{II,prim} = \{\delta_1, \delta_2, \delta_3\}.$$

For

$$\mathrm{Aut}(\mathcal{P}') = \{g \in \mathbf{O}(\Lambda^{2,1})_+ \mid g\mathcal{P}' = \mathcal{P}'\},$$
$$\mathrm{Aut}(\mathcal{P}) = 1, \qquad \mathrm{Aut}(\mathcal{P}_I) = \langle s_{f_3} \rangle, \qquad \mathrm{Aut}(\mathcal{P}_{II}) \simeq S_3,$$

with $\mathrm{Aut}(\mathcal{P}_{II})$ generated by $s_{f_2 - f_3}$ and $s_{f_{-2} - f_2}$. The chamber decomposition is

$$\mathbf{O}(\Lambda^{2,1})_+ \simeq W^{(2)}(\Lambda^{2,1}) \simeq W^{(2)}(\Lambda_I^{2,1}) \rtimes \mathrm{Aut}(\mathcal{P}_I) \simeq W^{(2)}(\Lambda_{II}^{2,1}) \rtimes \mathrm{Aut}(\mathcal{P}_{II}).$$

Lemma 8.2 *(Multiplier restriction).* • for $w \in W^{(2)}(\Lambda_I^{2,1})$ and $a \in \mathrm{Aut}(\mathcal{P}_I)$,

$$\Delta_5(w \cdot aZ) = \det(a)\Delta_5(Z)$$

- for $w \in W^{(2)}(\Lambda_{II}^{2,1})$ and $a \in \mathrm{Aut}(\mathcal{P}_{II})$,

$$\Delta_5(w \cdot aZ) = \det(w)\Delta_5(Z)$$

Equivalently,

$$\nu_{\Delta_5}(s_{\delta_i}) = -1, \qquad i = 1, 2, 3,$$

and

$$\nu_{\Delta_5}(s_{f_{-2} - f_2}) = \nu_{\Delta_5}(s_{f_2 - f_3}) = \nu_{\Delta_5}(s_{f_2 + f_3}) = +1.$$

Proof. Maass's multiplier formula for the product of the ten even theta constants is recorded in [37, (1.3)–(1.5)], citing [56]. Under the exterior-square identification $\mathbf{PSp}_4(\mathbb{Z}) \simeq \mathbf{SO}_+(\Lambda^{3,2})$, Gritsenko–Nikulin compute the reflection values in [37, Proposition 2.1] and the type-II chamber restriction in [37, Proposition 2.2]: it is -1 on the three type-II simple reflections and trivial on the three $\mathrm{Aut}(\mathcal{P}_{II})$-reflections. This is a character computation, not a normalization computation: multiplying Δ_5 by any nonzero scalar leaves the displayed values unchanged. Thus the six generator values are exactly

$$\nu_{\Delta_5}(s_{\delta_i}) = -1, \qquad \nu_{\Delta_5}(s_{f_{-2}-f_2}) = \nu_{\Delta_5}(s_{f_2-f_3}) = \nu_{\Delta_5}(s_{f_2+f_3}) = +1.$$

For the type-II chamber decomposition

$$W^{(2)}(\Lambda^{2,\mathrm{I}}_{II}) \rtimes \mathrm{Aut}(\mathcal{P}_{II}),$$

the first line says that the restriction to $W^{(2)}(\Lambda^{2,\mathrm{I}}_{II})$ is the determinant character, while the second says that the $\mathrm{Aut}(\mathcal{P}_{II})$-factor acts trivially. The type-I statement is the parallel restriction in the decomposition

$$W^{(2)}(\Lambda^{2,\mathrm{I}}_{I}) \rtimes \mathrm{Aut}(\mathcal{P}_{I}),$$

where the nontrivial chamber automorphism contributes $\det(a)$. □

Set

$$\Delta(\Lambda^{2,\mathrm{I}}_{II})_+ = \mathbb{R}_{\geq 0}\delta_1 + \mathbb{R}_{\geq 0}\delta_2 + \mathbb{R}_{\geq 0}\delta_3$$

and

$$\Delta(\Lambda^{2,\mathrm{I}}_{II})^*_+ = \{x \in \Lambda^{2,\mathrm{I}} \otimes \mathbb{R} | (x, \delta_i) \leq 0\}.$$

For the finite-volume chamber $\mathcal{P}_{II}$, the dual cone obeys

$$\Delta(\Lambda^{2,\mathrm{I}}_{II})^*_+ \subset \overline{C(\Lambda^{2,\mathrm{I}})_+} \subset \Delta(\Lambda^{2,\mathrm{I}}_{II})_+.$$

The Weyl vector is determined by

$$(\rho, \delta_i) = -\frac{(\delta_i, \delta_i)}{2} = -1, \qquad i = 1, 2, 3.$$

$$(\delta_i, \delta_j) = \begin{pmatrix} 2 & -2 & -2 \\ -2 & 2 & -2 \\ -2 & -2 & 2 \end{pmatrix}.$$

$$\rho = \frac{1}{2}\delta_1 + \frac{1}{2}\delta_2 + \frac{1}{2}\delta_3 = f_2 - \frac{1}{2}f_3 + f_{-2}.$$

Also

$$\rho \in C(\Lambda^{2,\mathrm{I}})_+ = C(\Lambda^{2,\mathrm{I}}_{II})_+.$$

Fix

$$z = z_1 f_{-2} + z_2 f_3 + z_3 f_2 \in \Lambda^{2,\mathrm{I}} \otimes \mathbb{R} + iC(\Lambda^{2,\mathrm{I}})_+ = \Lambda^{2,\mathrm{I}}_{II} \otimes \mathbb{R} + iC(\Lambda^{2,\mathrm{I}}_{II})_+.$$

The dual lattice is

$$(\Lambda^{2,\mathrm{I}}_{II})^* = \mathbb{Z}\frac{1}{2}f_2 + \mathbb{Z}\frac{1}{2}f_3 + \mathbb{Z}\frac{1}{2}f_{-2} = \frac{1}{2}\Lambda^{2,\mathrm{I}}.$$

Definition 8.3 *(Gram-index degrees in the Weyl chamber).* For $\gamma = (n, l, m) \in \Gamma_{\mathrm{ind}}$ set

$$\alpha(\gamma) = \alpha(n, l, m) = 2nf_2 - lf_3 + 2mf_{-2} \in \Lambda_{II}^{2,1}.$$

With

$$z = z_1 f_{-2} + z_2 f_3 + z_3 f_2,$$

the character of the Gram-index degree is

$$x^\gamma = q^n r^l s^m = \exp(-\pi i(\alpha(\gamma), z)).$$

For $\gamma' = (n', l', m')$,

$$(\alpha(\gamma), \alpha(\gamma')) = -2\langle \gamma, \gamma' \rangle_{\mathrm{ind}}, \qquad (\alpha(\gamma), \alpha(\gamma)) = -2(4nm - l^2).$$

Thus the Borcherds degree is the Lorentzian image of the Gram-index grading. The Weyl chamber and real walls are part of the Gritsenko–Nikulin/Nikulin root datum. The cusp polarization fixed in Section 6 selects the corresponding product expansion, and the active Jacobi support gives the nonzero visible product exponents in that chamber.

$$E_0 := \rho.$$

$$q^{1/2} r^{1/2} s^{1/2} = \exp(-\pi i(\rho, z)).$$

$$\exp(-2\pi i(\alpha(\gamma), z)), \qquad \exp(-2\pi i(\rho, z)).$$

Definition 8.4 *(Active support and Borcherds cone).* In the standard cusp chamber, the active support is

$$\Gamma_{\mathrm{act}} = \{\gamma = (n, l, m) \in \Gamma_{\mathrm{eff}} \mid f(nm, l) \neq 0\}.$$

The inactive triples in $\Gamma_{\mathrm{eff}} \setminus \Gamma_{\mathrm{act}}$ are ordering data only; their product exponents are zero. The product does not prove absence of root spaces with zero signed superdimension. The proof-grade Borcherds cone is the real cone generated by the active positive Gram-index degrees after the Lorentzian degree map:

$$C_{\mathrm{Borch}} := \mathbb{R}_{\geq 0}\, \alpha(\Gamma_{\mathrm{act}}) \subset \Lambda_{II}^{2,1} \otimes \mathbb{R}.$$

The real-root hyperplanes are the walls used by the denominator identity in this polarization.

Lemma 8.5 *(Active support gives the type-II chamber cone).* For $\gamma = (n, l, m) \in \Gamma_{\mathrm{act}}$,

$$\alpha(\gamma) = n\delta_1 + m\delta_2 + (n + m - l)\delta_3,$$

with

$$n \geq 0, \qquad m \geq 0, \qquad n + m - l \geq 0.$$

Consequently

$$C_{\mathrm{Borch}} = \mathbb{R}_{\geq 0}\delta_1 + \mathbb{R}_{\geq 0}\delta_2 + \mathbb{R}_{\geq 0}\delta_3.$$

Proof. The displayed decomposition is

$$n(2f_2 - f_3) + m(2f_{-2} - f_3) + (n + m - l)f_3 = 2nf_2 - lf_3 + 2mf_{-2}.$$

The effective chamber gives $n, m \geq 0$. Since $\phi_{0,1}$ is weak of index one, $f(nm, l) \neq 0$ implies $4nm - l^2 \geq -1$. If $n = 0$ or $m = 0$, this gives $l^2 \leq 1$, and the boundary cases in Γ_{eff} give $l \leq n + m$. If $n, m > 0$ and $l \geq n + m + 1$, then

$$l^2 \geq (n + m + 1)^2 = 4nm + (n - m)^2 + 2(n + m) + 1 > 4nm + 1,$$

contradicting $l^2 \leq 4nm + 1$. Hence $n + m - l \geq 0$. The reverse inclusion follows from the active triples

$$(1, 0, 0), \qquad (0, 0, 1), \qquad (0, -1, 0),$$

whose degrees are $\delta_1 + \delta_3$, $\delta_2 + \delta_3$, and δ_3, and from the active triples

$$(1, 1, 0), \qquad (0, 1, 1),$$

whose degrees are δ_1 and δ_2. □

For $n, l, m \in \mathbb{Z}$ with $n \equiv m \equiv l \equiv 1 \mod 2$, $n, m > 0$ and $4nm - l^2 > 0$, set

$$a = (n - 1)f_2 - \frac{l - 1}{2} f_3 + (m - 1)f_{-2}, \qquad m(a) = -\frac{1}{64} c_\Delta(n, l, m).$$

[37, Theorem 2.1]

$$\rho + a = nf_2 - \frac{l}{2} f_3 + mf_{-2},$$

and

$$\begin{aligned} &\frac{1}{64} c_\Delta(n, l, m) \exp(\pi i(nz_1 + lz_2 + mz_3)) \\ &= \frac{1}{64} c_\Delta(n, l, m) \exp\left(-\pi i\left(nf_2 - \frac{l}{2} f_3 + mf_{-2}, z\right)\right) \\ &= -m(a) \exp(-\pi i(\rho + a, z)), \end{aligned}$$

with

$$a \in (\Lambda^{2,1})^* = \frac{1}{2} \Lambda^{2,1}_{II}.$$

The normalization $c_\Delta(1, 1, 1) = 64$ fixes $m(0) = -1$. Since $c_\Delta(n, l, m) \in 64\mathbb{Z}$, all $m(a)$ are integral. The inequality $4nm - l^2 > 0$ implies

$$\rho + a \in C(\Lambda^{2,1})_+.$$

Let $\lambda = \rho + a$. For nonzero coefficients in the chosen chamber,

$$\lambda \in (\Lambda^{2,1}_{II})^* \cap \mathbb{R}_+ \mathcal{P}_{II}, \qquad (\lambda, \delta_i) \leq 0.$$

If $(\lambda, \delta_i) = 0$, then

$$c_\lambda = -c_\lambda = 0.$$

For nonzero terms, $(\lambda, \delta_i) < 0$. Since $(\rho, \delta_i) = -1$,

$$(a, \delta_i) \leq 0, \qquad a \in \Delta(\Lambda^{2,1}_{II})^*_+.$$

$$a \in \mathbb{R}_{\geq 0} \Delta(\Lambda^{2,1}_{II})^*_+ \cap \overline{C(\Lambda^{2,1})_+} = \mathbb{R}_{\geq 0} \mathcal{P}_{II}.$$

The congruences $m \equiv n \equiv l \equiv 1 \pmod 2$ imply

$$a \in 2(\Lambda^{2,1})^* = \Lambda^{2,1}_{II}.$$

$$a = 0 \qquad \text{or} \qquad a \in \Lambda^{2,1}_{II} \cap \mathbb{R}_{>0} \mathcal{P}_{II}.$$

Lemma 8.6.

$$\frac{1}{64}\Delta_5(Z) = \sum_{w\in W^{(2)}(\Lambda_{II}^{2,1})} \det(w)(\exp(-\pi i(w(\rho), z)) - \sum_{a\in\Lambda_{II}^{2,1}\cap\mathbb{R}_{>0}\mathcal{P}_{II}} m(a)\exp(-\pi i(w(\rho+a), z)))$$

Proof. Let c_λ be the coefficient of $e^{-\pi i(\lambda,z)}$ in $64^{-1}\Delta_5$. The leading term has

$$\lambda = \rho, \qquad c_\rho = 1.$$

The other exponents in the chosen chamber have

$$\lambda = \rho + a, \qquad a \in \Lambda_{II}^{2,1} \cap \mathbb{R}_{>0}\mathcal{P}_{II}, \qquad c_{\rho+a} = -m(a).$$

Anti-invariance:

$$c_{w\lambda} = \det(w)c_\lambda.$$

If λ lies on a simple wall, then

$$s_{\delta_i}\lambda = \lambda, \qquad c_\lambda = -c_\lambda = 0.$$

Summing the signed Weyl translates proves the formula. □

The primitive isotropic rays of $\overline{\mathcal{P}_{II}}$ are generated by

$$2f_2, \qquad 2f_{-2}, \qquad 2f_2 - 2f_3 + 2f_{-2}.$$

The group $\mathrm{Aut}(\mathcal{P}_{II}) \simeq S_3$ is transitive on these three vectors. For $a_0 = 2f_2$:

$$1 + \frac{1}{64}\sum_{t\geq 1} c_\Delta(1+2t, 1, 1)q^t = \prod_{k\geq 1}(1-q^k)^9$$

In denominator notation,

$$1 - \sum_{t\geq 1} m(ta_0)q^t = \prod_{t\geq 1}(1-q^t)^{\tau(ta_0)}, \qquad \tau(ta_0) = 9 \quad (t \geq 1).$$

By S_3-symmetry,

$$\tau(ta) = 9 \qquad (t \geq 1)$$

for every primitive isotropic ray

$$\mathbb{Z}_{>0}a \subset \overline{\mathcal{P}_{II}} \cap \Lambda_{II}^{2,1}.$$

On isotropic rays, the coefficients $m(ta)$ are determined by the eta-product expansion; the signed simple-root supermultiplicity is $\tau(ta)$. The imaginary simple roots of $\mathfrak{g}_{\Delta_5}$ are these isotropic and timelike rays in $\overline{\mathcal{P}_{II}} \cap \Lambda_{II}^{2,1}$, with multiplicities $\tau(a)$ on isotropic rays and $m(a)$ on negative-norm rays [37, Theorem 2.1], [39, (5.1), Section 5.1.1].

The real root datum is the wall set

$$\mathcal{P}_{II,prim} = \{\delta_1, \delta_2, \delta_3\}$$

with Gram matrix [37, Theorem 4.1] (and [39, Section 5.1.1, case (t=1,II,0)] as secondary)

$$(\delta_i, \delta_j) = \begin{pmatrix} 2 & -2 & -2 \\ -2 & 2 & -2 \\ -2 & -2 & 2 \end{pmatrix}$$

Definition 8.7 *(Simple real roots and perturbative walls).*

$$\Delta^{re}_{\mathrm{simp}} = \{\delta_1, \delta_2, \delta_3\}, \qquad \Delta^{re} = W^{(2)}(\Lambda^{2,1}_{II})\Delta^{re}_{\mathrm{simp}},$$

$$\mathcal{W}_{\mathrm{pert}} = \bigcup_{\delta \in \Delta^{re}_{\mathrm{simp}}} \mathbb{H}_\delta.$$

9 The denominator algebra

$$A = ((\delta_i, \delta_j))_{1 \le i,j \le 3} = \begin{pmatrix} 2 & -2 & -2 \\ -2 & 2 & -2 \\ -2 & -2 & 2 \end{pmatrix}$$

$$A \text{ symmetric generalized Cartan}, \qquad \mathfrak{g}(A) \text{ Kac–Moody}.$$

$$\mathcal{A}_{\mathrm{den}} := \mathfrak{g}_{\Delta_5}.$$

This is the target-internal algebraic denominator square root. It is the automorphic correction of $\mathfrak{g}(A)$ constructed by Gritsenko–Nikulin [37, Sections 3–4], with root spaces graded by the Gram-degree map

$$\gamma = (n, l, m) \longmapsto \alpha(\gamma) = 2nf_2 - lf_3 + 2mf_{-2}.$$

The Borcherds correction character in the coefficient chamber is

$$C_{\mathrm{im}}(z) = 1 - \sum_{a \in \Lambda^{2,1}_{II} \cap \mathbb{R}_{>0}\mathcal{P}_{II}} m(a) e^{-2\pi i (a,z)}.$$

$$e^{-2\pi i(\rho,z)} C_{\mathrm{im}}(z).$$

Signed root-superdimension convention; Harvey–Moore is the external physical interpretation [40]:

$$\mathrm{smult}(\alpha) = \dim \mathfrak{g}_{\alpha,\bar{0}} - \dim \mathfrak{g}_{\alpha,\bar{1}}.$$

Let $\mathcal{R}$ be the root set of $\mathfrak{g}_{\Delta_5}$, and let $\mathcal{R}_+$ be the positive root set in the $\mathcal{P}_{II}$-chamber.

Definition 9.1 *(Denominator Lie superalgebra).* For $\gamma \in \Gamma_{\mathrm{ind}}$ set

$$\mathcal{A}_{\mathrm{den},\gamma} := \begin{cases} \Lambda^{2,1}_{II} \otimes \mathbb{C}, & \gamma = 0, \\ \mathfrak{g}_{\Delta_5,\alpha(\gamma)}, & \alpha(\gamma) \in \mathcal{R}, \\ 0, & \alpha(\gamma) \notin \mathcal{R}. \end{cases}$$

The zero degree is the Cartan subalgebra and is not governed by $f(0,0)$. The bracket is the Borcherds–Kac–Moody bracket of $\mathfrak{g}_{\Delta_5}$:

$$[\ ,\ ] : \mathcal{A}_{\mathrm{den},\gamma} \otimes \mathcal{A}_{\mathrm{den},\gamma'} \longrightarrow \mathcal{A}_{\mathrm{den},\gamma+\gamma'}.$$

Theorem 9.2 *(Denominator algebra identity).* The denominator Lie superalgebra has the following properties.

$$\mathcal{A}_{\mathrm{den}} = \mathcal{A}_{\mathrm{den},0} \oplus \bigoplus_{\substack{\gamma \in \Gamma_{\mathrm{ind}} \\ \alpha(\gamma) \in \mathcal{R}}} \mathcal{A}_{\mathrm{den},\gamma} = \mathcal{A}_{\mathrm{den},0} \oplus \mathcal{A}^{\mathrm{bulk}}_{\mathrm{den}} \oplus \mathcal{A}^{\mathrm{cusp}}_{\mathrm{den}}, \qquad [\mathcal{A}_{\mathrm{den},\gamma}, \mathcal{A}_{\mathrm{den},\gamma'}] \subset \mathcal{A}_{\mathrm{den},\gamma+\gamma'},$$

where $\mathcal{A}_{\mathrm{den}}^{\mathrm{bulk}} = \bigoplus_{\alpha(\gamma)\in\mathcal{R},\, m(\gamma)>0} \mathcal{A}_{\mathrm{den},\gamma}$ records the $m > 0$ Borcherds bulk and $\mathcal{A}_{\mathrm{den}}^{\mathrm{cusp}} = \bigoplus_{\alpha(\gamma)\in\mathcal{R},\, m(\gamma)=0} \mathcal{A}_{\mathrm{den},\gamma}$ records the $m = 0$ Weyl-chamber cusp boundary, in parallel with the $\mathcal{V} = \mathcal{V}_{\mathrm{bulk}} \oplus \mathcal{V}_{\mathrm{cusp}}$ splitting at the determinant level above. Its visible positive signed Gram-degree multiplicities are the K3 half-index coefficients:

$$\mathrm{smult}(\alpha(n,l,m)) = f(nm,l) \qquad ((n,l,m) \in \Gamma_{\mathrm{act}}).$$

Its Weyl–Kac–Borcherds denominator is

$$\mathrm{den}(\mathcal{A}_{\mathrm{den}}) = 64^{-1}\Delta_5(2Z).$$

The theorem concerns the Borcherds denominator algebra. A microscopic BPS Hilbert space and its operator product require separate construction.

Proof. Gritsenko–Nikulin [39, Theorem 2.1, (2.1)–(2.3)] construct $\mathfrak{g}_{\Delta_5}$ as the generalized Borcherds–Kac–Moody superalgebra whose real simple roots are $\delta_1, \delta_2, \delta_3$ and whose imaginary simple-root multiplicities are read from the Borcherds correction character. A product exponent gives the signed multiplicity of a visible positive root; the simple imaginary-root datum is Gritsenko–Nikulin's $m(a)$ for negative-norm a and $\tau(a)$ on isotropic rays. These are different pieces of the denominator construction. A root-graded Lie superalgebra has

$$[\mathfrak{g}_{\Delta_5,\alpha}, \mathfrak{g}_{\Delta_5,\beta}] \subset \mathfrak{g}_{\Delta_5,\alpha+\beta}.$$

Here $\alpha \in \Lambda_{II}^{2,1} \otimes \mathbb{C}$, and the addition $\alpha + \beta$ is linear inside the Cartan; this is the linear root grading of a Borcherds–Kac superalgebra. The Gram-degree map

$$\gamma = (n,l,m) \longmapsto \alpha(\gamma) = 2nf_2 - lf_3 + 2mf_{-2} \in \Lambda_{II}^{2,1} \otimes \mathbb{C}$$

is the inverse correspondence of root indexing: it is additive in (n,l,m) and recovers the linear addition in $\Lambda_{II}^{2,1} \otimes \mathbb{C}$. This linear additivity is the relevant statement for the denominator algebra of Δ_5, where the grading lattice is $\Lambda_{II}^{2,1}$ and $\Gamma_{\mathrm{ind}} = \mathbb{Z}^3$ is the Gram parametrization of that grading. In a hypothetical Hall source, the underlying additive lattice is the formal Mukai lattice $\Gamma_X^{\mathrm{form}} = \Lambda_S \oplus \Lambda_S$, restricted further by actual algebraic support. The comparison with the denominator grading is made only after passing to the normal-ordered extension $\widehat{\Gamma}_X$ and the additive homomorphism $\overline{\Pi}_X$, or on isolated raw bracket channels where the B-defect vanishes. The raw quadratic map Π_X is not itself a bracket-recognition grading target. The denominator identity reads

$$\frac{1}{64}\Delta_5(2Z) = e^{-2\pi i(\rho,z)} \prod_{\alpha\in\mathcal{R}_+} (1 - e^{-2\pi i(\alpha,z)})^{\mathrm{smult}(\alpha)}.$$

Comparing this with the Borcherds product gives

$$\mathrm{smult}(\alpha(n,l,m)) = f(nm,l) \qquad ((n,l,m) \in \Gamma_{\mathrm{act}}).$$

□

10 Compact Dirac–Igusa realization problem

The denominator algebra is the target. A compact $K3 \times E$ theory would be a source. The distinction is between observables and states. The formal current envelope below is an observable algebra attached to the imported Borcherds–Kac–Moody target; it is not a compact BPS state space and it contains no Hall extension correspondences. Its target-side bar–cobar counit remains internal to that observable algebra. A genuine Dirac–Igusa realization must supply the state-side Hall object, the orientation/Pfaffian line, the normal-ordered Gram descent, and the chiral Koszul comparison before one may identify its primitives with $\mathfrak{g}_{\Delta_5}$. Throughout this compact realization problem, S is a smooth complex K3 surface and E is a smooth elliptic curve; singular K3 limits, nodal or cuspidal elliptic degenerations, imprimitive curve-class contributions, the $s = 0$ compact Hall cusp, and multi-component charge closures belong to separate boundary data, not to the standing compact source datum.

Definition 10.1 *(Formal current envelope).* Let C be a smooth complex curve. Put

$$\mathrm{Cur}_C(\mathcal{A}_{\mathrm{den}}) := \mathcal{A}_{\mathrm{den}} \otimes \mathcal{O}_C$$

with the level-zero current bracket

$$[a \otimes f, b \otimes g] = [a, b] \otimes f g.$$

Its chiral enveloping algebra in the sense of Beilinson–Drinfeld [5] is

$$\mathsf{Den}_{\Delta_5,C} := U_C^{\mathrm{ch}}(\mathrm{Cur}_C(\mathcal{A}_{\mathrm{den}})).$$

For the compact realization problem one specializes $C = E$, the elliptic curve in $X = S \times E$:

$$\mathrm{Cur}_E(\mathcal{A}_{\mathrm{den}}) := \mathcal{A}_{\mathrm{den}} \otimes \mathcal{O}_E$$

and

$$\mathsf{Den}_{\Delta_5,E} := U_E^{\mathrm{ch}}(\mathrm{Cur}_E(\mathcal{A}_{\mathrm{den}})).$$

Proposition 10.2 *(Formal current envelope).* $\mathsf{Den}_{\Delta_5,C}$ is the formal completed ind Beilinson–Drinfeld current envelope on C obtained from the denominator Lie superalgebra. For $C = E$ this gives the target chiral object used in the $K3 \times E$ realization problem. Its constant-current Lie superalgebra is $\mathcal{A}_{\mathrm{den}}$, and the denominator of that constant-current bracket is

$$64^{-1}\Delta_5(2Z).$$

A compact E_3/Hall/chiral BPS realization requires additional microscopic input.

Proof. In the completed ind setting, the current construction sends a Lie superalgebra to a Lie-* algebra on a smooth curve, and the chiral enveloping construction sends a Lie-* algebra to a chiral algebra [5]; equivalently, in vertex-algebra language, this is the universal current vertex algebra at level zero [47]. Constant sections of $\mathcal{A}_{\mathrm{den}} \otimes \mathcal{O}_E$ recover the original Lie superalgebra. The denominator statement is exactly Theorem 9.2. □

Definition 10.3 *(Finite algebraic Dirac–Igusa target reference tower).* Fix a finite nonzero cusp-positive active window $A_R \subset \Gamma_{\mathrm{gram}}$ in the positive/root side of the Borcherds product, downward-saturated only with respect to the fixed finite cofinal target-window order used below, and a group homomorphism

$$L : \Gamma_{\mathrm{gram}} \longrightarrow \Gamma_X^{\mathrm{form}}$$

used only to choose normal-ordered target labels

$$s_L(\gamma) := \big(L\gamma, \Pi_X(L\gamma) - \gamma\big) \in \widehat{\Gamma}_X.$$

Let

$$\mathfrak{g}_R^{\mathrm{tar}} := \bigoplus_{\gamma \in A_R} \mathfrak{g}_{\Delta_5,\alpha(\gamma)}$$

be the corresponding finite degree-selected target piece of the imported Gritsenko–Nikulin denominator algebra, with the globally defined target invariant form, radical quotient, and Borcherds–Kac bracket restricted to the selected degrees. Let

$$\mathsf{U}^{\mathrm{PBW}}_{\Delta_5,R} \subset U(\mathfrak{g}_{\Delta_5})$$

be the finite positive-window PBW filtration piece spanned by monomials in allowed nonzero root letters from A_R, with Cartan powers and positive-negative cancellations excluded from this finite truncation piece. Its role is target filtration data, not a finite Hopf quotient of the full enveloping algebra. Multiplication is a tower operation

$$\mathsf{U}^{\mathrm{PBW}}_{\Delta_5,R} \otimes \mathsf{U}^{\mathrm{PBW}}_{\Delta_5,R'} \longrightarrow \mathsf{U}^{\mathrm{PBW}}_{\Delta_5,R+R'},$$

not necessarily an endomorphism of one fixed window.

For $\gamma = (n, l, m) \in A_R$, choose a finite formal super vector space V_γ with $\operatorname{sdim} V_\gamma = f(nm, l)$ and put $x^\gamma = q^n r^l s^m$. This choice records only the signed target exponent in the algebraic reference tower; it is not a compact primitive space, parity lift, Pfaffian line, or source root representative. The formal algebraic first-order block attached to γ is the odd endomorphism of

$$K_\gamma := V_\gamma \oplus \Pi V_\gamma^\vee$$

given, after choosing the displayed splitting, by

$$D_\gamma = \begin{pmatrix} 0 & 1 - x^\gamma \\ 1 & 0 \end{pmatrix}, \qquad D_\gamma^2 = (1 - x^\gamma)\,\mathrm{id}_{K_\gamma}.$$

After a finite algebraic super-Pfaffian convention is fixed, define the formal finite product

$$\mathrm{sPf}_R^{\mathrm{alg}}(D) := \prod_{\gamma \in A_R} (1 - x^\gamma)^{\operatorname{sdim} V_\gamma}.$$

The data

$$\mathsf{DI}^{\mathrm{alg}}_{\Delta_5,E,L;R} := \Big(\mathfrak{g}_R^{\mathrm{tar}},\ \mathsf{U}^{\mathrm{PBW}}_{\Delta_5,R},\ s_L,\ \{K_\gamma, D_\gamma\}_{\gamma \in A_R},\ \mathrm{sPf}_R^{\mathrm{alg}}\Big)$$

are the finite algebraic Dirac–Igusa target reference tower. They are constructed from the target algebra. They are not a compact $K3 \times E$ Hall source, not a geometric orientation gerbe, not a source coalgebra, and not protected integration.

Proposition 10.4 *(Target reference normalization).* For the positive-window pro-target reference tower $\mathsf{DI}^{\mathrm{alg}}_{\Delta_5,E,L} := \{\mathsf{DI}^{\mathrm{alg}}_{\Delta_5,E,L;R}\}_R$, the normalized formal product is

$$64 q^{1/2} r^{1/2} s^{1/2} \lim_R \mathrm{sPf}_R^{\mathrm{alg}}(D) = \Delta_5(Z),$$

in the Borcherds cusp chamber, whenever the windows exhaust the active support.

Proof. By construction,

$$\mathrm{sPf}_R^{\mathrm{alg}}(D) = \prod_{\gamma=(n,l,m)\in A_R} (1 - q^n r^l s^m)^{f(nm,l)}.$$

Passing through an exhausting compatible sequence of finite windows gives the monic Borcherds product $D_5 = 64^{-1}\Delta_5$ by the imported Borcherds–Gritsenko–Nikulin product formula. Multiplying by the theta-normalizing Weyl monomial $64q^{1/2}r^{1/2}s^{1/2}$ gives Δ_5.

This is a target-reference normalization. It proves no compact $K3 \times E$ Hall source, no geometric Pfaffian line, no orientation monodromy, no source coalgebra, and no primitive recognition theorem. □

Open Problem 10.5 *(Compact realization of the target reference tower).* Construct finite geometric Hall–Dirac source data $\mathfrak{H}^{\mathrm{geom}}_{X,R}$ and compatible realization functors

$$\mathsf{Real}_{X,R} : \mathfrak{H}^{\mathrm{geom}}_{X,R} \longrightarrow \mathsf{DI}^{\mathrm{alg}}_{\Delta_5,E,L;R}$$

preserving the normal-ordered $\widehat{\Gamma}_X$-degree, Hall product and coproduct correspondences, quotient-after-correspondence data, orientation-gerbe twists, hybrid local/wrapped operations, source coalgebras, first-order blocks, Pfaffian lines with leading cusp trivialization, primitive representatives, Hopf pairings, radical kernels, PBW filtrations, and finite-window transitions. The existence of such compatible functors, and of the geometric source domain itself, is the compact $K3 \times E$ Dirac–Igusa realization problem.

Definition 10.6 *(Holomorphic E_3 quantum upgrade).* The determinant and denominator results above are protected Euler-characteristic statements. The expected quantum upgrade does not start from the scalar determinant. It would start from a supplied holomorphic E_3-factorization algebra on the Calabi–Yau threefold

$$\mathcal{A}_X^{E_3}, \qquad X = K3 \times E.$$

By a holomorphic E_3-factorization algebra is meant here: a factorization algebra on X in the sense of Costello–Gwilliam ([19], [20] esp. vol. II §10–11), locally constant in real directions and holomorphic in three complex directions, whose factorization structure maps are governed by holomorphic polydiscs in $\mathbb{C}^3$; equivalently a factorization algebra in the holomorphic E_3-operadic sense, together with whatever homotopy/framing datum is required by the chosen operadic model. No identification with an $SO(6)$ or $SO(3)$ framed topological operad is asserted here. This symbol, and the placeholder $\Phi_3^{\mathrm{FA}}(X)$, denote the source object required in the Calabi–Yau-to-chiral construction; they are not a constructed functor evaluated on X. The candidate concrete realization, supplied as a schema, is the local protected observable algebra $\mathcal{A}_X(U)$ of Definition 2.1. A compact holomorphic-Chern–Simons/BV model, if constructed with the required descent, quantum master equation, and anomaly cancellation, would be a candidate source for this datum in the completed charge category. Such a source also requires a chosen E_3-formality point, a chain-level homotopy/framing datum for the chosen operadic model, compact-support control, and finite-stage HN quotients on which all operations are defined before passing to the completed inverse system.

The curve-level BPS algebra would be obtained only after a supplied chain-level specialization along the $K3$-direction. Split the triple identification into three named pieces, each carrying a separately labelled comparison map:

$$\mathcal{A}_X^{E_3} \xrightarrow{\mathrm{Sp}^{\mathrm{ch}}_{K3,E}} \mathcal{F}_X^{(0)} \xrightarrow{\kappa_E} \mathcal{F}_X^{(1)} \xrightarrow{\pi_*^{\mathrm{ch}}} (p_E)_!^{\mathrm{fact}}\mathcal{A}_X^{E_3}.$$

Here: $\mathrm{Sp}^{\mathrm{ch}}_{K_3,E}$ is required to be a chain-level K3-integration/factorization-pushforward datum along the $K3$-fibration $p_E : K3 \times E \to E$, comparing the relevant Beilinson–Drinfeld and Costello–Gwilliam models. Its source category, target category, support condition, charge grading, and functorial transition squares are part of the supplied datum; its output $\mathcal{F}_X^{(0)}$ is intended to be a chiral algebra on E (equivalently a factorization algebra on $\mathrm{Ran}(E)$). κ_E is the chiral-to-ordinary pushforward/formality comparison $\mathcal{F}_X^{(0)} \to \mathcal{F}_X^{(1)}$, not the source cobar comparison Θ_{Kos}, where $\mathcal{F}_X^{(1)}$ is the candidate identification of the chiral pushforward with the ordinary factorization pushforward $(p_E)_!^{\mathrm{fact}}\mathcal{A}_X^{E_3}$ regarded as a sheaf-of-cochain factorization algebra on $\mathrm{Ran}(E)$; ordinary $*$- and exceptional !-pushforward agree only after the finite constructibility and proper-support defects vanish. Identification of chiral and ordinary factorization pushforward is the open rectification comparison. π_*^{ch} is required to be the closing identification of $\mathcal{F}_X^{(1)}$ with $(p_E)_!^{\mathrm{fact}}\mathcal{A}_X^{E_3}$ as factorization data, comparing the Costello–Gwilliam push along a proper holomorphic submersion with the topological pushforward; this comparison is required to reduce to the standard projection-finite comparison and is open in the wrapped sectors. The words "chiral algebra on E, equivalently factorization algebra on $\mathrm{Ran}(E)$" always mean a chosen Beilinson–Drinfeld / Costello–Gwilliam / formality comparison inside the supplied model categories, not an unqualified equality of theories. The convenient shorthand $\mathcal{F}_X$ refers to any one of $\mathcal{F}_X^{(0)}$, $\mathcal{F}_X^{(1)}$, or $(p_E)_!^{\mathrm{fact}}\mathcal{A}_X^{E_3}$ once κ_E and π_*^{ch} have been supplied. Without the rectification comparison and the factorization-pushforward identification, the three names are distinct objects. At finite height this means a diagram, for every successor $R^+ \to R$,

$$\begin{tikzcd}
\mathcal{A}^{E_3}_{X,R^+} \arrow[r, "\mathrm{Sp}_{R^+}"] \arrow[d, "\rho_{R^+R}"'] & \mathcal{F}^{(0)}_{X,R^+} \arrow[r, "\kappa_{E,R^+}"] \arrow[d, "\tau^{(0)}_{R^+R}"'] & \mathcal{F}^{(1)}_{X,R^+} \arrow[r, "\pi^{\mathrm{ch}}_{*,R^+}"] \arrow[d, "\tau^{(1)}_{R^+R}"'] & (p_E)_!^{\mathrm{fact}}\mathcal{A}^{E_3}_{X,R^+} \arrow[d, "(p_E)_!^{\mathrm{fact}}\rho_{R^+R}"] \\
\mathcal{A}^{E_3}_{X,R} \arrow[r, "\mathrm{Sp}_{R}"] & \mathcal{F}^{(0)}_{X,R} \arrow[r, "\kappa_{E,R}"] & \mathcal{F}^{(1)}_{X,R} \arrow[r, "\pi^{\mathrm{ch}}_{*,R}"] & (p_E)_!^{\mathrm{fact}}\mathcal{A}^{E_3}_{X,R}
\end{tikzcd}$$

commuting up to specified homotopy. Concretely, the finite datum must fix dg/enriched model categories, or stable ∞-categories with an equivalent rectification calculus,

$$C_R^X, \qquad C_R^{\mathrm{o}}, \qquad C_R^{\mathrm{I}}, \qquad C_R^{\mathrm{hyb}},$$

for the height-R holomorphic E_3-source, the BD/CG projection-finite chiral shadow on $\mathrm{Ran}(E)$, the ordinary factorization pushforward, and the hybrid correspondence-module category with $Q_{E,R}$, together with chosen cofibrant-fibrant representatives in the model-category case. The displayed finite diagram category must carry a projective/Reedy-type model structure, or the stable ∞-categorical replacement must supply the same rectification and coherence calculus. The transition identities

$$\tau^{(0)}_{R^+R}\mathrm{Sp}_{R^+} = \mathrm{Sp}_R\rho_{R^+R}, \qquad \tau^{(1)}_{R^+R}\kappa_{E,R^+} = \kappa_{E,R}\tau^{(0)}_{R^+R},$$

$$(p_E)_!^{\mathrm{fact}}\rho_{R^+R}\,\pi^{\mathrm{ch}}_{*,R^+} = \pi^{\mathrm{ch}}_{*,R}\tau^{(1)}_{R^+R}$$

and the product and unit squares for Sp, κ_E, and π_*^{ch} are then part of one rectified finite diagram, not separate informal homotopies. Equivalently the finite obstruction tuple

$$\mathfrak{O}_R^{K_3\to E} = (\mathfrak{o}_R^{\mathrm{Sp}},\ \mathfrak{o}_R^{\kappa},\ \mathfrak{o}_R^{\pi},\ \mathfrak{o}_R^{\mu,u},\ \mathfrak{o}_{R^+R}^{\mathrm{tr}})$$

must be null-trivialized. Only after this null-trivialization may the displayed zigzag be read as a strict finite zigzag in the projection-finite sector; in positive elliptic degree it is replaced by the hybrid datum $\mathfrak{O}_{\mathrm{hyb},R} = 0$. The forward map $\kappa_{E,R}$ is especially not the canonical bar–cobar counit: the canonical source-side map has the direction

$$\varepsilon^{\mathrm{src}}_{\mathrm{bar},R} : \Omega^{\mathrm{ch}}_E \bar{B}^{\mathrm{ch}}_E(\mathcal{F}^{(0)}_{X,R}) \longrightarrow \mathcal{F}^{(0)}_{X,R}.$$

A forward $\kappa_{E,R}$ requires a chosen cofibrant/fibrant model and homotopy inverse in the Francis–Gaitsgory bar–cobar domain. Turning the displayed homotopy-commutative rectangle into strict finite maps requires a model structure on the full finite diagram category and a coherent rectification compatible with HN transitions, quotient orientation, and wrapped correspondences. Such a rectification is not supplied by cofibrant/fibrant replacement alone: replacement can choose a homotopy inverse after the hybrid object is already in the bar–cobar domain, but it does not construct the wrapped prequotient, anchors, mixed correspondences, or the $(0,1,1)$ primitive. The first multiplicative square to prove is already

$$\mathrm{Sp}_R\, m_{\mathcal{U},R} = \mu^{\mathrm{loc}}_R(\mathrm{Sp}_R \otimes \mathrm{Sp}_R)$$

on the projection-finite local-local sector, and the first wrapped test is the $m > 0$ degree $(0,1,1)$. Restrictions to chambers, pushforwards to E, and chiral Koszul dualities are then operations on $\mathcal{A}^{E_3}_X$ and on its E_1-chiral shadow $\mathcal{F}_X$, not replacements for the E_3-source.

After these data have been constructed, the required Igusa statement is the pair of comparisons

$$H^0\,\mathrm{Prim}_{\mathrm{prot}}(\overline{\Pi}^{\Theta}_{X,*}\mathcal{F}_X) \overset{?}{\cong} \mathfrak{g}_{\Delta_3}, \qquad \Omega^{\mathrm{ch}}_E C_X \overset{?}{\simeq} \mathsf{Den}_{\Delta_3,E}.$$

Here C_X is the source coalgebra later formalized in Definition 10.19; in wrapped degree it is attached to the hybrid object $\mathcal{F}^{\mathrm{hyb}}_X$, not merely to the projection-finite Ran shadow. The first comparison would be the bracket theorem. The second would be the chiral Koszul theorem. Both require microscopic data beyond the scalar square and beyond the formal current envelope.

Definition 10.7 *(Rectified finite $K3$-to-E specialization).* Fix a finite HN height R. A rectified finite $K3$-to-E specialization is not the displayed zigzag alone, but a homotopy-coherent finite diagram together with a chosen rectification: either a specified projective/Reedy-type model structure and a cofibrant-fibrant strict representative presenting that diagram, or a stable ∞-categorical rectification calculus supplying the same coherences. The finite diagram contains

$$\mathcal{A}^{E_3}_{X,R}, \qquad \mathcal{F}^{(0)}_{X,R}, \qquad \mathcal{F}^{(1)}_{X,R}, \qquad (p_E)^{\mathrm{fact}}_{!}\mathcal{A}^{E_3}_{X,R}.$$

In the projection-finite sector $\mathrm{Sp}^{\mathrm{ch}}_{K3,E,R}$ is defined by the supplied finite closed-configuration !-pushforward along the $K3$-fibre incidence stacks, with the local-local Hall correspondences supplying the factorization products. The map $\kappa_{E,R}$ is a chosen Beilinson–Drinfeld/Costello–Gwilliam rectification equivalence between the chiral and ordinary factorization presentations. The map $\pi^{\mathrm{ch}}_{*,R}$ is the comparison from this ordinary presentation to the finite factorization pushforward $(p_E)^{\mathrm{fact}}_{!}\mathcal{A}^{E_3}_{X,R}$.

The obstruction tuple $\mathfrak{O}^{K3\to E}_R$ records precisely the failure of these three arrows, their product and unit squares, and their HN transition squares to be null-homotopic in this finite diagram category. Thus the displayed zigzag is a constructed finite specialization only after $\mathfrak{O}^{K3\to E}_R = 0$.

For positive elliptic degree this construction has no target in ordinary $\mathrm{Fact}(\mathrm{Ran}(E))$. The replacement datum is the hybrid correspondence-module object

$$\mathcal{B}_R^{E,\mathrm{hyb}} : \mathsf{Corr}_R^{E,\mathrm{hyb}} \longrightarrow \mathsf{Ch}_{\mathbb{C}}$$

together with $Q_{E,R}$, and the additional condition is $\mathfrak{O}_{\mathrm{hyb},R} = 0$.

Definition 10.8 *(Compact holomorphic E_3 source interface).* Fix a stability sector (σ, S) and the finite HN height quotients of Proposition 10.53. A compact holomorphic E_3 source interface for $X = K3 \times E$ is not, by itself, a compact quantum field theory. It is the following finite-first chain datum:

$$\mathbb{A}^{E_3}_{X,\sigma,S} = \left\{ \mathcal{A}^{E_3}_{X,\sigma,S,\leq R},\ \rho_{R'R},\ m_{\mathcal{U},R},\ S_R, \{-,-\}_R, \Delta^{\mathrm{BV}}_R,\ \eta_{\mathcal{V},R}, \omega_{\mathrm{QME},R}, \omega_{\mathrm{anom},R},\ \varphi^{E_3}_R, \tau^{\mathrm{fr}}_R,\ \mathrm{Sp}^{\mathrm{ch}}_{K3,E,R} \right\}_{R>0,\ R'\geq R}.$$

Here $\mathcal{A}^{E_3}_{X,\sigma,S,\leq R}$ is a charge-supported cochain-level holomorphic E_3-factorization object with charges in $\Gamma^{HN}_{\sigma,S}(R)$; $\rho_{R'R}$ are transition maps for the HN inverse system; $m_{\mathcal{U},R}$ are the structure maps attached to disjoint holomorphic polydiscs $\mathcal{U} \subset X$; S_R, $\{-,-\}_R$, Δ^{BV}_R are the finite BV action, antibracket, and BV Laplacian conventions used to formulate the finite classical or quantum master equation; $\eta_{\mathcal{V},R}$ are Cech–Weiss descent cochains for the chosen holomorphic-polydisc cover $\mathcal{V}$; $\omega_{\mathrm{QME},R}$ and $\omega_{\mathrm{anom},R}$ are the finite QME and anomaly cocycles with specified null-homotopies; $\varphi^{E_3}_R$ is the chosen Tamarkin–Kontsevich E_3-formality torsor point and τ^{fr}_R is the chain-level framing/homotopy witness for the holomorphic operadic model; and $\mathrm{Sp}^{\mathrm{ch}}_{K3,E,R}$ is the finite-stage $K3$-to-E specialization datum, including source, target, support, charge grading, wrapped-sector compatibility, and transition squares. The interface is required to carry:

(i) descent, QME, and anomaly-cancellation witnesses at every finite height R. Concretely, a finite BV presentation consists, at least locally on $\mathcal{V}$, of a finite charge-supported cyclic L_∞-algebra

$$(L_R, \ell_{k,R}, \langle -,- \rangle_R), \qquad E_R = L_R[1], \qquad \mathrm{Obs}_R = \widehat{\mathrm{Sym}}(E_R^\vee),$$

and a total BV complex

$$C^{\mathrm{BV}}_R = \mathrm{Tot}\, C^\bullet(\mathcal{V}, \mathrm{Obs}_R[[\hbar]]).$$

The pairing gives the bracket and, after a Berezinian/orientation choice, Δ^{BV}_R. The classical action is

$$S^{(0)}_R(a) = \sum_{k\geq 1} \frac{1}{(k+1)!} \langle a, \ell_{k,R}(a,\dots,a)\rangle_R,$$

and cyclicity plus the L_∞-relations give the finite classical master equation

$$d_R S^{(0)}_R + \frac{1}{2}\{S^{(0)}_R, S^{(0)}_R\}_R = 0.$$

The quantum equation is the recursive equation for $S_R = S^{(0)}_R + \hbar S^{(1)}_R + \cdots$,

$$d_R S_R + \frac{1}{2}\{S_R, S_R\}_R \quad \text{or} \quad d_R S_R + \frac{1}{2}\{S_R, S_R\}_R + \hbar \Delta^{\mathrm{BV}}_R S_R$$

is killed by the supplied finite homotopy $\omega_{\mathrm{QME},R}$. The first anomaly is

$$\mathfrak{a}_R := \Delta^{\mathrm{BV}}_R S^{(0)}_R \in Z^1(C^{\mathrm{BV}}_R, Q_R), \qquad Q_R = d_R + \{S^{(0)}_R, -\}_R,$$

with obstruction class

$$[\mathfrak{a}_R] \in H^1(C_R^{\mathrm{BV}}, Q_R).$$

An anomaly cancellation cochain is $S_R^{(1)}$ satisfying $Q_R S_R^{(1)} + \mathfrak{a}_R = 0$; in the dg Lie case this includes the modular/unimodularity class

$$\chi_R(x) = \mathrm{str}_{L_R}(\mathrm{ad}_x), \qquad [\chi_R] \in H^1_{\mathrm{CE}}(L_R).$$

All of these data are required to be compatible with $\rho_{R'R}$;

(ii) a fixed E_3-formality point and a homotopy/framing witness for the chosen holomorphic E_3-operadic model, treated as a chosen torsor point rather than a canonical equality;

(iii) compatibility of $m_{\mathcal{U},R}$, specialization, and the transition maps $\rho_{R'R}$;

(iv) compact-support control and continuity for the HN inverse system, with $\mathcal{A}^{E_3}_{X,\sigma,S,\leq R}$ a quotient by a factorization ideal killing outputs of height $> R$, and with every structure map $m_{\mathcal{U},R}$ descended from higher stages;

(v) the rule that the $K3$-to-E specialization is an ordinary $\mathrm{Ran}(E)$-factorization object only on the projection-finite part; positive elliptic degree must be routed through the hybrid wrapped datum of Definition 10.27.

If all finite stages and transition maps are supplied, the completed source is the inverse limit

$$\mathcal{A}^{E_3}_X := \varprojlim_R \mathcal{A}^{E_3}_{X,\sigma,S,\leq R}.$$

This equation is a construction from the displayed finite data, not an existence theorem for those data.

Lemma 10.9 *(Formal finite BV skeleton versus compact source).* In a finite window containing the three real simple roots there is a nonempty formal BV chain skeleton. For example, put

$$\Gamma_R^{\mathrm{win}} = \{0, \delta_1, \delta_2, \delta_3\}, \qquad L_R = V \oplus V^\vee[-1], \qquad V = \mathbb{C}x_1 \oplus \mathbb{C}x_2 \oplus \mathbb{C}x_3,$$

with x_i even, $\deg x_i = \delta_i$, the standard odd symplectic pairing $\langle x_i, \xi_j \rangle = \delta_{ij}$, and all L_∞-brackets equal to zero. Then

$$S_R^{(0)} = 0, \qquad d_R S_R + \frac{1}{2}\{S_R, S_R\}_R + \hbar \Delta_R^{\mathrm{BV}} S_R = 0, \qquad \mathfrak{a}_R = \Delta_R^{\mathrm{BV}} S_R^{(0)} = 0,$$

so the finite QME and anomaly cocycles are killed by the zero homotopies, after choosing the same E_3-formality and framing torsor data as in Definition 10.8.

This proves only nonemptiness of the formal finite chain language. It does not construct the compact Hall/hybrid source. In the first Igusa window

$$\delta_1 \leftrightarrow (1,1,0), \qquad \delta_2 \leftrightarrow (0,1,1), \qquad \delta_3 \leftrightarrow (0,-1,0),$$

one has $m_R(\delta_1) = m_R(\delta_3) = 0$ and $m_R(\delta_2) = 1$, where $m_R = \mathrm{pr}_3 \overline{\Pi}_X$ is the Borcherds trace exponent. In the rank-one D6–D2–D0 dictionary this is geometric elliptic degree $d = m_R + 1 = 2$, so a geometric representative of the middle wall cannot live in the projection-finite Ran sector. A compact source datum must instead exhibit

$$x_{\delta_2,R} \in \mathcal{M}^{\mathrm{wr,rig}}_{\eta,R}, \qquad \overline{\Pi}_X(x_{\delta_2,R}) = (0,1,1),$$

together with its anchor, unreduced and quotient deformation complexes, mixed local/wrapped correspondences, s_{∂_2}-equivariant d-critical chart, normal Pfaffian wall, and quotient-orientation trivializations. Without this wrapped datum the O2/hybrid comparison square is not defined, and

$$\mathfrak{o}_R^{\lambda,\mathrm{loss}} \neq 0 \quad \Longrightarrow \quad \mathfrak{O}_{\partial_2,R}^{\mathrm{O2/hyb}} \neq 0.$$

Definition 10.10 *(Finite compact E_3-source obstruction datum).* Fix (σ, S, R). A finite compact E_3-source candidate is the finite datum of Definition 10.8, together with the finite Hall support, finite-type substacks, quotient-orientation data, and $K3$-to-E specialization source and target categories required there. Its typed obstruction tuple is

$$\mathfrak{O}_R^{E_3-\mathrm{src}} = \left(\mathfrak{o}_R^{\mathrm{fin}},\ \mathfrak{o}_R^{\mathrm{desc}},\ \mathfrak{o}_R^{\mathrm{QME}},\ \mathfrak{o}_R^{\mathrm{anom}},\ \mathfrak{o}_R^{E_3-\mathrm{form}},\ \mathfrak{o}_R^{\mathrm{fr}},\ \mathfrak{o}_R^{\mathrm{HN}},\ \mathfrak{o}_R^{\mathrm{Sp}},\ \mathfrak{O}_{\mathrm{hyb},R},\ \mathfrak{o}_{R^+R}^{\mathrm{tr}}\right).$$

The entries record, respectively, finite-type Hall support and finite-dimensional protected chain spaces; Cech–Weiss descent for local observables; the finite classical/quantum master equation; anomaly cancellation; the chosen E_3-formality point; the chain-level framing/homotopy witness for the holomorphic operadic model; finite-first HN quotient and transition control; the finite $K3$-to-E specialization diagram; the full hybrid local/wrapped obstruction tuple of Definition 10.27; and successor transition compatibility. The zero groupoid

$$Z(\mathfrak{O}_R^{E_3-\mathrm{src}})$$

is the full subgroupoid of finite candidates for which every entry vanishes with a specified null-trivialization compatible with the successor transition maps. This is a typed datum zero locus, not a single derived obstruction complex; no such complex is asserted before the finite Hall/stability/moduli problem has itself been solved.

Proposition 10.11 *(Zero-source equivalence).* For fixed (σ, S, R), the zero groupoid $Z(\mathfrak{O}_R^{E_3-\mathrm{src}})$ is equivalent to the groupoid of finite compact holomorphic E_3-source interfaces at height R with the finite Hall, orientation, hybrid, and specialization data required in Definition 10.8. A compatible cofinal system of such zeroes, satisfying the Mittag–Leffler exactness condition of Definition 10.54, constructs the completed source

$$\mathcal{A}_X^{E_3} = \varprojlim_R \mathcal{A}_{X,\sigma,S,\leq R}^{E_3}.$$

This identifies the finite entries whose simultaneous vanishing gives the interface. It is not a proof that the zero groupoid is nonempty.

Proof. The definition of $\mathfrak{O}_R^{E_3-\mathrm{src}}$ lists exactly the finite entries whose vanishing turns the candidate data into the interface of Definition 10.8: finite Hall support, descent, BV master equation, anomaly null-homotopy, formality/framing, HN continuity, specialization, hybrid local/wrapped operations, and transition compatibility. Thus an object in the zero groupoid is precisely a finite interface with those null-trivializations, and any finite interface determines such a zero by reading off its defining witnesses. The inverse-limit assertion is the finite-first construction of Definition 10.8, with Mittag–Leffler exactness added to prevent the completed comparison from weakening to a pro-comparison. □

Corollary 10.12 *(No nonemptiness from the finite source datum).* The formal BV skeleton of Lemma 10.9 and the zero-source equivalence of Proposition 10.11 do not imply that $Z(\mathfrak{O}_R^{E_3-\mathrm{src}})$ is

nonempty for any finite height R, and do not construct a cofinal compatible system of zeroes. Consequently they do not construct $\mathcal{A}_X^{E_3}$. They define the finite obstruction problem whose solution is Open Problem 10.14.

Proposition 10.13 *(Finite-R compact E_3-source construction protocol).* Fix (σ, S, R). A nonempty compact holomorphic E_3-source interface at height R is constructed only after the following finite source-side data are constructed and their residuals vanish:

$$\mathfrak{S}_R, \quad \mathcal{F}^{\mathrm{hyb}}_{X,\sigma,S,\leq R}, \quad Q_{E,R}, \quad I_R^{\mathrm{prot}}, \quad (S_R, \{-,-\}_R, \Delta_R^{\mathrm{BV}}), \quad \eta_{\mathcal{V},R},$$

$$\omega_{\mathrm{QME},R}, \quad \omega_{\mathrm{anom},R}, \quad \varphi_R^{E_3}, \quad \tau_R^{\mathrm{fr}}, \quad \mathrm{Sp}^{\mathrm{ch}}_{K3,E,R},$$

together with successor transition maps. The data must include the wrapped primitive in degree $(0, 1, 1)$, its rigidified anchor, unreduced and reduced deformation complexes, mixed local/wrapped correspondences, quotient-orientation trivializations, and the finite O2/hybrid comparison square. If the displayed data are supplied as components of the full finite interface datum of Definition 10.8, including the finite E_3-source object, the product maps $m_{\mathcal{U},R}$, finite-type, properness, and admissibility witnesses, and if every entry of $\mathfrak{D}_R^{E_3-\mathrm{src}}$ is supplied with a compatible null-trivialization, then they determine an object of $Z(\mathfrak{D}_R^{E_3-\mathrm{src}})$. Conversely, every object of this zero groupoid forgets to such source-side data, together with the additional finite interface, admissibility, product, and transition coherences recorded in the definition.

Open Problem 10.14 *(Compact holomorphic E_3 source).* Construct a nonempty interface $\mathbb{A}^{E_3}_{X,\sigma,S}$ in the sense of Definition 10.8, and hence the completed source object $\mathcal{A}_X^{E_3}$, for $X = K3 \times E$. A candidate compact holomorphic-Chern–Simons/BV construction must prove, before any comparison with Δ_5, the following finite-chain data:

(i) descent for local observables on X and a quantum master equation after compact charge completion;
(ii) anomaly cancellation, a fixed E_3-formality point in the sense of Tamarkin–Kontsevich formality for E_n-operads [50, 52, 73], and a chain-level homotopy/framing datum for the chosen holomorphic E_3-operadic model; compact-support control;
(iii) a finite-first sectorial completion compatible with Harder–Narasimhan quotients, with all operations defined on finite stages before completion;
(iv) the chain-level $K3$-to-E specialization $\mathrm{Sp}^{\mathrm{ch}}_{K3,E}$;
(v) the projection-finite local sector and the hybrid wrapped repair for positive elliptic degree;
(vi) a strong orientation, protected integration, and primitive recognition datum.

Existing BV, CoHA, and chiral-algebra formalisms give the language for these requirements. They do not, by themselves, construct this compact source for $K3 \times E$. The external open data are: the E_3-formality point (Tamarkin–Kontsevich formality at a chosen associator, regarded here as a chosen torsor point [50, 52, 73]; rationality of formality is open over $\mathbb{Q}$ but known over $\mathbb{R}$), and the chain-level homotopy/framing datum for the chosen holomorphic E_3-operadic model; chain-level constructibility for the candidate concrete realization $\mathcal{A}_X(U)$ of Definition 2.1 is open).

Remark 10.15 *(Where the quantum comparison would live).* Assume the source algebra $\mathcal{A}_X^{E_3}$, the specialization $\mathrm{Sp}^{\mathrm{ch}}_{K_3,E}$, a conilpotent coalgebra C_X with supplied comparison

$$b_X : C_X \xrightarrow{\simeq} \bar{B}^{\mathrm{ch}}_E(\mathcal{F}_X^{\mathrm{hyb}}),$$

and a quasi-isomorphism

$$\Theta_{\mathrm{Kos}} : \Omega^{\mathrm{ch}}_E C_X \xrightarrow{\simeq} \mathsf{Den}_{\Delta_5,E}$$

have all been supplied. Then $\mathcal{F}_X^{\mathrm{hyb}}$ is the place where the protected primitive bracket and the chiral Koszul comparison must be formulated. Costello–Gwilliam locality explains why a holomorphic quantum theory would produce factorization products before any trace or determinant is taken [18], [19], [20]; Beilinson– Drinfeld chiral operations are the curve-level algebraic form of collision maps [5]. None of this is obtained from the scalar square or from the denominator identity alone.

Remark 10.16 *(Pushforward, restriction, Koszul duality).* The second step is specialization, not inversion. A restriction to a chamber or sector is a restriction of $\mathcal{A}_X^{E_3}$ before, or of $\mathcal{F}_X$ after, the $K3$-specialization. Pushforward along p_E is the geometric notation for the same specialization. A compact Koszul theorem would require a supplied conilpotent coalgebra C_X, an identification

$$C_X \overset{?}{\simeq} \bar{B}^{\mathrm{ch}}_E(\mathcal{F}_X^{\mathrm{hyb}}),$$

and a further quasi-isomorphism

$$\Omega^{\mathrm{ch}}_E C_X \overset{?}{\simeq} \mathsf{Den}_{\Delta_5,E}.$$

These are not procedures for reconstructing the holomorphic E_3-source from the denominator alone.

Proposition 10.17 *(Target-internal bar–cobar counit).* Choose an augmented filtered model of the formal target $\mathsf{Den}_{\Delta_5,E}$ whose finite charge truncations lie in the conilpotent chiral setting where Francis–Gaitsgory chiral Koszul duality applies. Then the target has its own bar–cobar counit:

$$\Omega^{\mathrm{ch}}_E \bar{B}^{\mathrm{ch}}_E(\mathsf{Den}_{\Delta_5,E}) \longrightarrow \mathsf{Den}_{\Delta_5,E}.$$

The filtration used here is target-internal: it is the charge filtration on the imported current envelope, together with the chosen augmentation. It is not the source filtration $F^{\mathrm{co}}_{R,\bullet}$, and its bar coalgebra is not $C_{X,R}$. Cf. Kac [47] for the underlying vertex algebra structure, Beilinson–Drinfeld [5, §3.7] for the chiral bar–cobar formalism, and Francis–Gaitsgory [28, Theorem 6.2.2] for chiral Koszul duality in the conilpotent setting. The counit reconstructs the already defined target from its target bar coalgebra. It is not the compact Koszul comparison

$$\Omega^{\mathrm{ch}}_E C_X \overset{?}{\simeq} \mathsf{Den}_{\Delta_5,E}$$

unless C_X has first been constructed from the compact $K3 \times E$ factorization object and identified with $\bar{B}^{\mathrm{ch}}_E(\mathcal{F}_X^{\mathrm{hyb}})$. The missing datum is the source coalgebra, its collision coproduct, and the source comparison map, not a homotopy inverse to the target-side counit.

Proof. Beilinson–Drinfeld bar–cobar formalism is an adjunction on chiral algebras and conilpotent chiral coalgebras on the curve [5, §3.7]. Applying this formalism to the known target $\mathsf{Den}_{\Delta_5,E}$, after choosing the displayed augmented filtered model, produces a tautological counit map back to that

target. The finite charge filtration is used only to place this target calculation in the conilpotent formal category. No moduli stack of compact objects, source collision coproduct, orientation system, protected integration map, or primitive representative enters this construction. Those are precisely the extra data in Definition 10.65. □

Lemma 10.18 *(Source–target separation for the Koszul lane).* Fix a finite height R. The target counit

$$\varepsilon_{\Delta,R} : \Omega_E^{\mathrm{ch}} \bar{B}_E^{\mathrm{ch}}(\mathsf{Den}_{\Delta_5,E,\leq R}) \longrightarrow \mathsf{Den}_{\Delta_5,E,\leq R}$$

is a morphism internal to the imported current envelope. A source Koszul datum is different data: it must first construct

$$\mathcal{F}^{\mathrm{hyb}}_{X,\sigma,S,\leq R}, \qquad C_{X,R}, \qquad F^{\mathrm{co}}_{R,\bullet}, \qquad \Delta^{\mathrm{ch}}_R,$$

from the finite hybrid correspondence category $\mathsf{Corr}_R^{E,\mathrm{hyb}}$, its quotient functor $Q_{E,R}$, orientations, the eight-word two-step binary flag atlas, and the separate higher colored configuration residual; only then may it compare

$$b_{X,R} : C_{X,R} \simeq \bar{B}_E^{\mathrm{ch}}(\mathcal{F}^{\mathrm{hyb}}_{X,\sigma,S,\leq R}, \varepsilon^{\mathrm{hyb}}_{X,R}), \qquad \Theta_{\mathrm{Kos},R} : \Omega_E^{\mathrm{ch}} C_{X,R} \simeq \mathsf{Den}_{\Delta_5,E,\leq R}.$$

Thus setting $C_{X,R} = \bar{B}_E^{\mathrm{ch}}(\mathsf{Den}_{\Delta_5,E,\leq R})$, or choosing a homotopy inverse to $\varepsilon_{\Delta,R}$, does not produce a source datum. It sets the separation residual $\mathfrak{o}_R^{\mathrm{sep}}$ to zero only after an independent source construction of $C_{X,R}$, $F^{\mathrm{co}}_{R,\bullet}$, Δ^{ch}_R, and $b_{X,R}$ has been supplied and identified with the target-side model.

Proof. The domain of $\varepsilon_{\Delta,R}$ contains only the target augmentation, target charge filtration, and target chiral operations. It does not contain the rigidified wrapped prequotients, anchors, mixed local/wrapped correspondences, quotient-after-correspondence functor, orientation transport, protected integration, or primitive representatives. These are the data from which the source coalgebra and collision coproduct are required to be built. Therefore a target bar–cobar inverse can be compared with a source coalgebra after that coalgebra exists; it cannot define the source coalgebra or its primitive restriction. □

Definition 10.19 *(Chiral Koszul source datum).* Fix (σ, S, R), the finite HN support $\widehat{\Gamma}_R$, and the Gram image Γ_R^{Π}. Let

$$\mathsf{Den}_{\Delta_5,E,\leq R}$$

denote the charge-truncation of the formal current envelope $\mathsf{Den}_{\Delta_5,E}$ to the finite set of Gram degrees Γ_R^{Π}. A finite-stage chiral Koszul source datum is a datum

$$\mathfrak{K}^{\mathrm{Kos}}_{X,R} = \left(\mathcal{F}^{\mathrm{hyb}}_{X,\sigma,S,\leq R}, C_{X,R}, F^{\mathrm{co}}_{R,\bullet}, \Delta^{\mathrm{ch}}_R, b_{X,R}, \Theta_{\mathrm{Kos},R}\right)$$

with the following properties. Every entry is source-side finite data except for the terminal comparison with the imported target $\mathsf{Den}_{\Delta_5,E,\leq R}$. In particular, neither $\bar{B}_E^{\mathrm{ch}}(\mathsf{Den}_{\Delta_5,E,\leq R})$ nor a homotopy inverse to the target counit is allowed to define $C_{X,R}$, $F^{\mathrm{co}}_{R,\bullet}$, or Δ^{ch}_R. This is the source/target separation of Lemma 10.18.

(i) *Source chiral object.* $\mathcal{F}^{\mathrm{hyb}}_{X,\sigma,S,\leq R}$ is the finite hybrid object of Definition 10.27. Its $b_R^{\mathrm{geom}} = 0$ part comes from the $K3$-specialized E_3-source; its $b_R^{\mathrm{geom}} > 0$ part is present only through the rigidified wrapped prequotient, the ordered two-sided mixed operations, the full flag atlas, quotient-after-correspondence descent, and protected integration. Equivalently, the hybrid residual $\mathfrak{O}_{\mathrm{hyb},R}$ has been trivialized at this height. A detached global Fock or Hecke factor does not supply this entry unless it acts through the displayed mixed correspondences before the reduced E-quotient.

(ii) *Conilpotent chiral coalgebra.* $C_{X,R}$ is a coaugmented conilpotent chiral coalgebra on E, with coaugmentation $\eta_R^C : k \to C_{X,R}$ and counit $\varepsilon_R^C : C_{X,R} \to k$, filtered by a finite increasing source charge filtration

$$0 = F^{\mathrm{co}}_{R,-1} \subset F^{\mathrm{co}}_{R,0} \subset \cdots \subset F^{\mathrm{co}}_{R,N_R} = C_{X,R}$$

whose associated graded pieces are finite dimensional in each Γ_R^{Π}-degree. Its reduced iterated coproduct is nilpotent on each finite stage. The chiral coproduct Δ_R^{ch} is the collision coproduct attached to the finite Hall/factorization correspondences, including local-local, all ordered local-wrapped, wrapped-wrapped, and eight-word flag stacks. It is a finite pull-push construction on the collision correspondence before $Q_{E,R}$, and after quotient it satisfies the coalgebra identities

$$(\Delta_R^{\mathrm{ch}} \otimes 1)\Delta_R^{\mathrm{ch}} = (1 \otimes \Delta_R^{\mathrm{ch}})\Delta_R^{\mathrm{ch}}, \qquad (\varepsilon_R^C \otimes 1)\Delta_R^{\mathrm{ch}} = (1 \otimes \varepsilon_R^C)\Delta_R^{\mathrm{ch}} = \mathrm{id}.$$

For $R' \geq R$ the transition map $\rho^C_{R'R} : C_{X,R'} \to C_{X,R}$ preserves the filtration and makes the finite coproduct square commute:

$$\begin{tikzcd} C_{X,R'} \arrow[r, "\Delta_{R'}^{\mathrm{ch}}"] \arrow[d, "\rho^C_{R'R}"'] & C_{X,R'} \otimes C_{X,R'} \arrow[d, "\rho^C_{R'R} \otimes \rho^C_{R'R}"] \\ C_{X,R} \arrow[r, "\Delta_R^{\mathrm{ch}}"'] & C_{X,R} \otimes C_{X,R}. \end{tikzcd}$$

This finite diagram is the source of the completed coproduct on $C_X = \varprojlim_R C_{X,R}$. The coproduct is not transported from $\bar{B}_E^{\mathrm{ch}}(\mathsf{Den}_{\Delta_\varsigma,E,\leq R})$ along $\Theta_{\mathrm{Kos},R}$, and no completed tensor statement is made before these finite squares commute.

(iii) *Bar comparison.* There is a charge-preserving quasi-isomorphism of conilpotent chiral coalgebras

$$b_{X,R} : C_{X,R} \xrightarrow{\simeq} \bar{B}_E^{\mathrm{ch}}(\mathcal{F}^{\mathrm{hyb}}_{X,\sigma,S,\leq R}, \varepsilon^{\mathrm{hyb}}_{X,R}).$$

It is compatible with primitive projection, the completed Hopf pairing, the reduced E-quotient, and the normal-ordered descent $\overline{\Pi}^{\Theta}_{X,*}$. This is a source-side comparison map, not the tautological bar construction of the target envelope. Its target is the reduced bar construction of the supplied augmented hybrid source object, with length pieces generated by the augmentation ideal, not the bar construction of $\mathsf{Den}_{\Delta_\varsigma,E,\leq R}$.

(iv) *Cobar comparison with the target envelope.* There is a charge-preserving quasi-isomorphism of augmented chiral algebras

$$\Theta_{\mathrm{Kos},R} : \Omega_E^{\mathrm{ch}} C_{X,R} \xrightarrow{\simeq} \mathsf{Den}_{\Delta_\varsigma,E,\leq R}.$$

On constant-current primitives, after the Hopf-radical quotient, this map restricts to the primitive recognition morphism

$$\iota_R^{\mathrm{Prim}} : H^0 \operatorname{Prim}_{\mathrm{prot}}(\overline{\Pi}^{\Theta}_{X,*}\mathcal{F}^{\mathrm{hyb}}_{X,\sigma,S,\leq R}) \longrightarrow \mathfrak{g}_{\Delta_\varsigma,\leq R}.$$

Here $\mathfrak{g}_{\Delta_\varsigma,\leq R}$ is the same finite Γ_R^{Π}-truncation on constant currents. The target is the imported Beilinson–Drinfeld current envelope of the Gritsenko–Nikulin denominator algebra. The map does not construct the compact source and is not obtained from the scalar square. Its direction is

source-to-target. A chosen homotopy inverse, if one exists after localization or completion, is not part of the datum and cannot be used to define the source coalgebra or its primitive representatives. The displayed primitive restriction is a condition on already supplied source primitives after the closed Hopf-radical quotient.

(v) *Finite residual.* The exact Koszul obstruction tuple at height R is

$$\mathfrak{O}_{\mathrm{Kos},R} = \Big(\mathfrak{o}_R^{C}, \mathfrak{o}_R^{\mathrm{fil}}, \mathfrak{o}_R^{\mathrm{conil}}, \mathfrak{o}_R^{\Delta,\mathrm{ch}}, \{\mathfrak{o}_{R'R}^{C,\mathrm{tr}}\}_{R'\geq R}, \mathfrak{o}_R^{b}, \mathfrak{o}_R^{\Omega}, \mathfrak{o}_R^{\mathrm{hyb}}, \mathfrak{o}_R^{\Pi}, \mathfrak{o}_R^{\mathrm{sep}}, \mathfrak{o}_R^{\mathrm{Prim}}\Big).$$

Here the finite coalgebra defects are

$$\mathfrak{o}_R^{\mathrm{fil}} := \Big(F_{R,-1}^{\mathrm{co}},\ C_{X,R}/F_{R,N_R}^{\mathrm{co}},\ \{d_C(F_{R,p}^{\mathrm{co}}) \to C_{X,R}/F_{R,p}^{\mathrm{co}}\}_p,\ \{\Delta_R^{\mathrm{ch}}(F_{R,p}^{\mathrm{co}}) \to (C_{X,R}\otimes C_{X,R})/\sum_{i+j\leq p} F_{R,i}^{\mathrm{co}}\otimes F_{R,j}^{\mathrm{co}}\}_p\Big),$$

$$\overline{\Delta}_R^{\mathrm{ch}} := \Delta_R^{\mathrm{ch}} - \big((\eta_R^C\varepsilon_R^C)\otimes \mathrm{id} + \mathrm{id}\otimes(\eta_R^C\varepsilon_R^C)\big),$$

and

$$\mathfrak{o}_R^{\mathrm{conil}} := (\overline{\Delta}_R^{\mathrm{ch}})^{N_R+1},$$

$$\mathfrak{o}_R^{\Delta,\mathrm{ch}} := \Big((\Delta_R^{\mathrm{ch}}\otimes 1)\Delta_R^{\mathrm{ch}} - (1\otimes\Delta_R^{\mathrm{ch}})\Delta_R^{\mathrm{ch}},\ (\varepsilon_R^C\otimes 1)\Delta_R^{\mathrm{ch}} - \mathrm{id},\ (1\otimes\varepsilon_R^C)\Delta_R^{\mathrm{ch}} - \mathrm{id}\Big),$$

and

$$\mathfrak{o}_{R'R}^{C,\mathrm{tr}} := \Big((\rho_{R'R}^C\otimes\rho_{R'R}^C)\Delta_{R'}^{\mathrm{ch}} - \Delta_R^{\mathrm{ch}}\rho_{R'R}^C,\ \{\rho_{R'R}^C(F_{R',p}^{\mathrm{co}}) \to C_{X,R}/F_{R,p}^{\mathrm{co}}\}_p\Big).$$

Here d_C is the finite-stage coalgebra differential, and a displayed map to a quotient is zero exactly when the corresponding subobject inclusion holds. The entries record, respectively: existence of the source coalgebra; bounded exhaustive source filtration; conilpotence of the reduced iterated coproduct; finite collision-coproduct construction with coassociativity and counit; transition-strict compatibility of $(C_{X,R}, F_{R,\bullet}^{\mathrm{co}}, \Delta_R^{\mathrm{ch}})$; the bar quasi-isomorphism $b_{X,R}$; the cobar quasi-isomorphism $\Theta_{\mathrm{Kos},R}$; compatibility with the hybrid local/wrapped operations, meaning $\mathfrak{O}_{\mathrm{hyb},R} = 0$; compatibility with normal-ordered descent and the reduced E-quotient; source/target separation, meaning that the source coalgebra, filtration, coproduct, and bar comparison are supplied before the target comparison and are not reconstructed from the target counit; and the primitive restriction after the closed Hopf-radical quotient. A full Koszul datum is the compatible vanishing and trivialization of $\mathfrak{O}_{\mathrm{Kos},R}$ for all R, followed by passage to the HN inverse limit.

Proposition 10.20 *(Canonical finite source-bar coalgebra).* Fix R and assume the finite hybrid residual $\mathfrak{O}_{\mathrm{hyb},R}$ vanishes, with the transition and properness data of Definition 10.27. Assume also that the finite hybrid stage supplies an integer N_R such that every nonzero ordered word of non-vacuum hybrid inputs of total HN height $\leq R$ has length at most N_R. Equivalently, the HN height is positive on every non-vacuum colour which survives in the reduced bar construction; if this bounded-length condition fails, the filtration and conilpotence residuals below are not killed. Finally assume a supplied augmentation $\varepsilon_{X,R}^{\mathrm{hyb}} : \mathcal{F}_{X,\sigma,S,\leq R}^{\mathrm{hyb}} \to k\mathbf{1}$, and write

$$\overline{\mathcal{F}}_{X,R} := \ker(\varepsilon_{X,R}^{\mathrm{hyb}}).$$

Under these hypotheses there is a canonical source-side choice of the finite coalgebra in Definition 10.19:

$$C_{X,R}^{\mathrm{bar}} := \bar{B}_E^{\mathrm{ch}}(\mathcal{F}_{X,\sigma,S,\leq R}^{\mathrm{hyb}}, \varepsilon_{X,R}^{\mathrm{hyb}}).$$

This is the reduced chiral bar coalgebra of the supplied hybrid source, not the bar coalgebra of $\mathsf{Den}_{\Delta_5,E,\le R}$. Equivalently, its length p summand is generated by ordered p-fold inputs from the augmentation ideal $\overline{\mathcal{F}}_{X,R}$ whose total HN height is $\le R$, with all outputs of height $> R$ set to zero:

$$C^{\mathrm{bar}}_{X,R} = k\mathbf{1} \oplus \bigoplus_{1\le p\le N_R} \bar{B}^{\mathrm{ch},p}_E(\overline{\mathcal{F}}_{X,R}).$$

The finite source filtration is the bar-length filtration

$$F^{\mathrm{co}}_{R,p}C^{\mathrm{bar}}_{X,R} := k\mathbf{1} \oplus \bigoplus_{1\le j\le p} \bar{B}^{\mathrm{ch},j}_E(\overline{\mathcal{F}}_{X,R}), \qquad 0 \le p \le N_R,$$

where N_R is the supplied bounded-length constant. The coaugmentation is the inclusion of the vacuum summand and the counit is projection to that summand. For a homogeneous bar generator $x = [x_1|\cdots|x_p]$, the collision coproduct is

$$\Delta^{\mathrm{bar}}_R(x) = \mathbf{1}\otimes x + x\otimes \mathbf{1} + \sum_{i=1}^{p-1}[x_1|\cdots|x_i]\otimes[x_{i+1}|\cdots|x_p],$$

with the displayed cuts realized by the corresponding finite collision and flag pull-push maps in $\mathsf{Corr}^{E,\mathrm{hyb}}_R$. The eight-word flag coherences and the four-input pentagon give coassociativity; the vacuum collision flags give the counit identities; and, for the reduced coproduct obtained by removing the two vacuum terms, the bar-length filtration gives

$$\left(\overline{\Delta}^{\mathrm{bar}}_R\right)^{N_R+1} = 0.$$

Thus, after replacing $C_{X,R}$ by $C^{\mathrm{bar}}_{X,R}$, the finite residuals

$$\mathfrak{o}^{C}_R, \qquad \mathfrak{o}^{\mathrm{fil}}_R, \qquad \mathfrak{o}^{\mathrm{conil}}_R, \qquad \mathfrak{o}^{\Delta,\mathrm{ch}}_R$$

vanish. The bar comparison $b_{X,R}$ may be chosen to be the identity of $\bar{B}^{\mathrm{ch}}_E(\mathcal{F}^{\mathrm{hyb}}_{X,\sigma,S,\le R}, \varepsilon^{\mathrm{hyb}}_{X,R})$. For $R' \ge R$, let

$$\tau^{E,\mathrm{hyb}}_{R'R} : \mathcal{F}^{\mathrm{hyb}}_{X,\sigma,S,\le R'} \longrightarrow \mathcal{F}^{\mathrm{hyb}}_{X,\sigma,S,\le R}$$

be the finite hybrid transition. Define the source coalgebra transition

$$\rho^{\mathrm{bar}}_{R'R} := \bar{B}^{\mathrm{ch}}_E(\overline{\tau}^{E,\mathrm{hyb}}_{R'R}) \quad \text{followed by the height-}R\text{ truncation.}$$

Here $\overline{\tau}^{E,\mathrm{hyb}}_{R'R}$ is the induced map on augmentation ideals. Then $\rho^{\mathrm{bar}}_{R'R}$ preserves the bar-length filtration. The transition residual $\mathfrak{o}^{C,\mathrm{tr}}_{R'R}$ vanishes exactly when the finite hybrid transition functor carries ordered collision flags, quotient compatibilities, and the bar-length filtration at height R' to the corresponding data at height R.

Proof. The construction applies the Beilinson–Drinfeld chiral bar construction to the already supplied source chiral object $\mathcal{F}^{\mathrm{hyb}}_{X,\sigma,S,\le R}$. Since the finite HN quotient kills charges of height $> R$, and since non-vacuum words have length bounded by N_R, only finitely many non-vacuum bar lengths occur. The reduced bar construction has a vacuum coaugmentation, a projection counit, and the standard deconcatenation coproduct. In the present correspondence model, each deconcatenation is implemented by cutting the ordered collision flag; admissibility of the pull-push maps and Thom–Sebastiani transport

are part of $\mathfrak{O}_{\mathrm{hyb},R} = 0$. The associativity and pentagon axioms for the finite flag stacks are precisely the equality of the two iterated cut maps. Hence the displayed coproduct is coassociative and counital. The reduced coproduct sends a word of length p to tensor words whose individual lengths are $< p$; after at most p reduced iterations every branch has length one, and the next reduced coproduct is zero. Since $p \leq N_R$, the $(N_R + 1)$-fold reduced iterate is zero. The construction of $\rho^{\mathrm{bar}}_{R'R}$ is functorial from $\tau^{E,\mathrm{hyb}}_{R'R}$, and the displayed transition square is exactly the assertion that this functorial bar map respects the finite collision flags after quotient and truncation. The construction has used only the source hybrid correspondence category, not the target current envelope. □

Corollary 10.21 *(Remaining source-to-target cobar obstruction).* Under the hypotheses of Proposition 10.20, the source coalgebra, finite filtration, collision coproduct, conilpotence, and bar comparison are no longer the Koszul obstruction. A cobar quasi-isomorphism to the target still requires two independent finite assertions.

First, the supplied hybrid source must itself lie in the Beilinson–Drinfeld/Francis–Gaitsgory bar–cobar domain. Thus it must be augmented and placed in the completed or coderived conilpotent category in which the source-internal counit

$$\varepsilon^{\mathrm{src}}_{\mathrm{bar},R} : \Omega^{\mathrm{ch}}_E C^{\mathrm{bar}}_{X,R} \longrightarrow \mathcal{F}^{\mathrm{hyb}}_{X,\sigma,S,\leq R}$$

is a quasi-isomorphism. This is a source-internal assertion about the supplied hybrid object, not the target counit.

Second, one must construct a source-to-target chiral algebra quasi-isomorphism

$$\vartheta_R : \mathcal{F}^{\mathrm{hyb}}_{X,\sigma,S,\leq R} \xrightarrow{\simeq} \mathsf{Den}_{\Delta_S,E,\leq R}.$$

If both assertions hold, then

$$\Theta_{\mathrm{Kos},R} : \Omega^{\mathrm{ch}}_E C^{\mathrm{bar}}_{X,R} \xrightarrow{\simeq} \mathsf{Den}_{\Delta_S,E,\leq R}$$

is the composite

$$\Theta_{\mathrm{Kos},R} := \vartheta_R \circ \varepsilon^{\mathrm{src}}_{\mathrm{bar},R}.$$

Equivalently, the following triangle commutes in the finite homotopy category:

$$\Theta_{\mathrm{Kos},R} = \vartheta_R \circ \varepsilon^{\mathrm{src}}_{\mathrm{bar},R}.$$

For this source-bar model the exact finite defects are

$$\mathfrak{o}^{\varepsilon,\mathrm{src}}_R := \Big(\mathfrak{o}^{\mathrm{aug}}_R,\ \mathfrak{o}^{\mathrm{FG}}_R,\ H^\bullet \operatorname{Cone}(\varepsilon^{\mathrm{src}}_{\mathrm{bar},R})\Big),$$

and

$$\mathfrak{o}^{\vartheta}_R := \Big(d_\Delta \vartheta_R - \vartheta_R d_{\mathrm{hyb}},\ \vartheta_R \mu^{\mathrm{hyb}}_R - \mu_{\Delta,R}(\vartheta_R \otimes \vartheta_R),\ \vartheta_R \eta^{\mathrm{hyb}}_R - \eta_{\Delta,R},\ \vartheta_R \circ \tau^{E,\mathrm{hyb}}_{R'R} - \pi^{\Delta}_{R'R} \circ \vartheta_{R'},\ H^\bullet \operatorname{Cone}(\vartheta_R),\ \vartheta_R|_{\mathrm{Prim}} -$$

Thus

$$\mathfrak{o}^{\Omega}_R := \Big(\mathfrak{o}^{\varepsilon,\mathrm{src}}_R,\ \mathfrak{o}^{\vartheta}_R\Big)$$

is the surviving cobar residual. Its vanishing is exactly the assertion that the source bar-cobar counit is admissible and a quasi-isomorphism and that the compact hybrid source is quasi-isomorphic, with transition and primitive compatibility, to the imported denominator current envelope.

Thus the finite source-bar construction kills the coalgebra part of $\mathfrak{D}_{\mathrm{Kos},R}$ after the hybrid source is supplied, but it does not identify compact primitives with the imported Gritsenko–Nikulin current envelope. That identification remains a source-to-target theorem, not a target-internal counit. No homotopy inverse to $\Omega_E^{\mathrm{ch}}\bar{B}_E^{\mathrm{ch}}(\mathsf{Den}_{\Delta_s,E,\leq R}) \to \mathsf{Den}_{\Delta_s,E,\leq R}$ enters this construction.

Proposition 10.22 *(Consequence of the chiral Koszul source datum).* Assume the finite-stage data $\mathfrak{K}_{X,R}^{\mathrm{Kos}}$ are supplied for all R, with compatible transition maps, in particular with $\mathfrak{o}_{R'R}^{C,\mathrm{tr}} = 0$ for every $R' \geq R$. Assume also the Koszul Mittag–Leffler condition: $R^1 \varprojlim = 0$ for the inverse systems of source coalgebras, completed chiral bar and cobar constructions, target truncations, and the cones of $b_{X,R}$ and $\Theta_{\mathrm{Kos},R}$ [79, §3.5]. Then the inverse limit

$$C_X := \varprojlim_R C_{X,R}$$

is a completed pro-conilpotent chiral coalgebra attached to the hybrid compact source, conilpotent on every finite HN quotient, and the maps $b_{X,R}$ and $\Theta_{\mathrm{Kos},R}$ induce

$$b_X : C_X \xrightarrow{\simeq} \bar{B}_E^{\mathrm{ch}}(\mathcal{F}_X^{\mathrm{hyb}}), \qquad \Theta_{\mathrm{Kos}} : \Omega_E^{\mathrm{ch}} C_X \xrightarrow{\simeq} \mathsf{Den}_{\Delta_s,E}.$$

This identifies the supplied source cobar algebra with the formal target current envelope. It does not construct the E_3-source, the hybrid wrapped Hall correspondences, the orientation line, the primitive recognition data, or C_X itself from the target-internal counit.

Proof. The inverse limit exists by the transition compatibility, bounded exhaustive finite charge filtration, finite coassociative and counital collision coproduct, and continuity hypotheses in the datum. The coradical filtration is finite on every Γ_R^{Π}-truncation and continuous in the HN completion; the limit is therefore pro-conilpotent, with finite-stage conilpotence preserved. The coproduct on C_X is the inverse limit of the finite Δ_R^{ch}. The quasi-isomorphisms $b_{X,R}$ and $\Theta_{\mathrm{Kos},R}$ commute with the transition maps. The Koszul Mittag–Leffler condition kills the $R^1 \varprojlim$ obstruction on their cones, hence these compatible finite quasi-isomorphisms pass to completed quasi-isomorphisms. Without that exactness one obtains only a compatible pro-quasi-isomorphism. The final sentence is the definition of the datum: the source object, mixed correspondences, orientation, primitive presentation, and source/target-separation datum are separate entries of $\mathfrak{K}_X$. □

Remark 10.23 *(Locality).* In a compact realization, the required local object would be a finite-stage holomorphic E_3-factorization algebra on X, and its $K3$-specialization would be a finite-stage chiral factorization algebra on $\mathrm{Ran}(E)$. This assertion is local only for projection-finite sectors over E. The automorphic product is a protected global trace after charge completion, wall-chamber choice, and summation over compact $K3$-modes. Duality monodromy, elliptic-degree wrapping, and the Borcherds chamber are therefore global features of the trace. They do not replace the local factorization maps on disjoint opens. Singular $K3$ surfaces, nodal or cuspidal elliptic curves, imprimitive curve-class contributions, the $s = 0$ compact Hall cusp, and multi-component charge closures require boundary-degeneration data; the Ran-local statement here is the smooth-S, smooth-E compact-stage assertion.

Definition 10.24 *(Projection-finite, wrapped, and hybrid Ran strata).* Let $p_E : K3 \times E \to E$ be projection. A compact Hall or chiral sector is projection-finite over E if every object A in it satisfies

$$p_E(\mathrm{Supp}\, A) \subset F$$

for some finite subset $F \subset E$. A one-dimensional sector is wrapped over E if its curve class has positive elliptic degree, i.e. if the pushforward of its one-cycle to E is $b[E]$ with $b > 0$. For such one-dimensional Hall sectors write

$$b(A) = \deg(p_E)_*[\mathrm{Supp}_1 A] \in \mathbb{Z}_{\geq 0}.$$

The same letter $b(\gamma)$ denotes the elliptic-degree component of a charge γ. Then projection-finite objects have $b(A) = 0$, while wrapped objects have $b(A) > 0$.

The hybrid base used below is a finite-stage coloured correspondence base, not an ordinary Ran space with one scalar attached. At height R, after the datum below is supplied, an object consists of a finite projection-local support $F \subset E$, finitely many wrapped colours $\eta_j \in \Gamma_R^{\mathrm{wr}}$, and rigidified wrapped objects W_j carrying anchor values $\lambda_{\eta_j,R}(W_j) \in E$. Its arrows are generated by finite collisions of F, by extension of local colours, by mixed local/wrapped correspondences, and by wrapped/wrapped correspondences. The notation $\mathrm{Ran}^{\mathrm{hyb}}(E)$ denotes this finite-coloured system through the inverse system in R. Its underlying degree shadow is the displayed disjoint union

$$\mathrm{Ran}^{\mathrm{hyb}}(E) = \bigsqcup_{b \geq 0} \mathrm{Ran}(E)_{[b]}, \qquad \mathrm{Ran}(E)_{[b]} := \mathrm{Ran}(E) \times \{b\}.$$

The stratum $b = 0$ is the ordinary finite-support Ran stratum. A stratum $b > 0$ records finite local collision points together with a global wrapped elliptic leg of degree b; it is not an assertion that the wrapped support is finite over E.

Lemma 10.25 *(Projection-to-E support-locality obstruction).* Let $C \subset K3 \times E$ be a one-dimensional subscheme whose curve class has positive elliptic degree:

$$(p_E)_*[C] = b[E], \qquad b > 0.$$

Then $p_E(C) = E$. In particular, C is not localized at finitely many points of E, and it is not contained in $p_E^{-1}(U)$ for any proper open $U \subsetneq E$.

Proof. If $p_E(C)$ were zero-dimensional, then the pushforward of the one-cycle $[C]$ to the curve E would vanish. This contradicts $(p_E)_*[C] = b[E]$ with $b > 0$. Hence $p_E(C)$ is a one-dimensional closed subset of the irreducible curve E, since p_E is proper, and therefore $p_E(C) = E$. □

Corollary 10.26 *(Naive Ran locality misses the elliptic-degree direction).* A support-local factorization algebra on $\mathrm{Ran}(E)$, defined only by separating supports over disjoint opens of E, sees the projection-finite sector. It does not by itself realize the full s-direction of the Igusa product.

Proof. The variable s is the Borcherds variable corresponding to the OP elliptic-degree direction after the $m = d - 1$ variable shift of Proposition 11.1. By Lemma 10.25, positive elliptic degree projects onto all of E. Such objects cannot be separated by choosing disjoint opens in E. Therefore ordinary support-local factorization on $\mathrm{Ran}(E)$ has no locality mechanism for the wrapped elliptic-degree sector. □

Definition 10.27 *(Finite hybrid local/wrapped factorization datum).* Fix (σ, S, R) as in Definition 10.8. A finite-stage hybrid local/wrapped factorization datum is a datum

$$\mathcal{F}^{\mathrm{hyb}}_{X,\sigma,S,\leq R} = \Big(\mathcal{C}_{\sigma,S,\leq R},\ b_R^{\mathrm{geom}},\ \mathcal{F}_R^{\mathrm{loc}},\ \mathcal{G}_R^{\mathrm{wr}},\ \mathfrak{E}_R^{\mathrm{loc}},\ \mathfrak{E}_R^{\mathrm{hyb}},\ \mathfrak{E}_R^{\mathrm{wr}},\ \mathfrak{Flag}_R^{\mathrm{hyb}},\ \mathsf{Corr}_R^{E,\mathrm{hyb}},\ \lambda_R,\ Q_{E,R},\ \theta_R^Q,\ o_R,\ I_R^{\mathrm{prot}}\Big)$$

with the following finite data and axioms. Every stack, sheaf, operation, and coherence named below lives at the fixed height R; the inverse limit appears only after all finite stages and transition maps have been supplied.

(i) *Finite Hall stage and elliptic degree.* $C_{\sigma,S,\leq R}$ is a supplied extension-closed finite charge-support HN quotient of the candidate compact Hall category, under the finite-type hypothesis [K3×E-fin], with finite-type charge substacks $\mathfrak{M}_{\gamma,R}$ for $\gamma \in \Gamma^{HN}_{\sigma,S}(R)$, vanishing-cycle coefficients $\Phi_{\gamma,R}$, and orientation lines $o_{\gamma,R}$. The map

$$b^{\mathrm{geom}}_R : \Gamma^{HN}_{\sigma,S}(R) \longrightarrow \mathbb{Z}_{\geq 0}$$

is the geometric elliptic-degree component. Write $\Gamma^{\mathrm{loc}}_R = \{\alpha \mid b^{\mathrm{geom}}_R(\alpha) = 0\}$ and $\Gamma^{\mathrm{wr}}_R = \{\eta \mid b^{\mathrm{geom}}_R(\eta) > 0\}$. The ideal of height $> R$ is already killed at this stage, and all operations below are understood to be zero when their output has height $> R$. The map b^{geom}_R is additive on extensions that remain in the quotient:

$$b^{\mathrm{geom}}_R(0) = 0, \qquad b^{\mathrm{geom}}_R(\gamma + \gamma') = b^{\mathrm{geom}}_R(\gamma) + b^{\mathrm{geom}}_R(\gamma')$$

whenever $\gamma, \gamma', \gamma + \gamma' \in \Gamma^{HN}_{\sigma,S}(R)$; if the sum has height $> R$, the corresponding operation is zero in the finite quotient.

(ii) *Projection-finite local sector through closed configurations.* The finite local datum is not the open-support substack alone. For a finite set I, let

$$Z_I \subset E \times E^I, \qquad Z_I = \bigcup_{i \in I} \{(e, (e_j)) \mid e = e_i\}.$$

For $\alpha \in \Gamma^{\mathrm{loc}}_R$ the stage supplies a closed-support incidence stack

$$\mathfrak{M}^{\mathrm{loc,cl}}_{\alpha,R,I} = \big\{(A, \mathbf{e}) \in \mathfrak{M}_{\alpha,R} \times E^I \;\big|\; p_E(\mathrm{Supp}\, A) \subset Z_I(\mathbf{e})\big\},$$

compatible with diagonals $E^J \to E^I$, symmetric-group descent, and the d-critical/vanishing-cycle structure. The open notation

$$\mathcal{M}^{\mathrm{loc}}_{\alpha,R}(U) = \{A \in \mathfrak{M}_{\alpha,R} \mid p_E(\mathrm{Supp}\, A) \subset U\}$$

is recovered only after finite arity and support comparison have been supplied:

$$\mathcal{M}^{\mathrm{loc}}_{\alpha,R}(U) \simeq \operatorname*{colim}_I \Big(\mathfrak{M}^{\mathrm{loc,cl}}_{\alpha,R,I} \times_{E^I} U^I\Big).$$

Let

$$\pi_{\alpha,R,I} : \mathfrak{M}^{\mathrm{loc,cl}}_{\alpha,R,I} \longrightarrow E^I$$

be the configuration map and put

$$\mathscr{H}^{\mathrm{loc}}_{\alpha,R,I} := (\pi_{\alpha,R,I})_!(\Phi^{\mathrm{loc}}_{\alpha,R,I} \otimes o^{\mathrm{loc}}_{\alpha,R,I}).$$

Here $(\pi_{\alpha,R,I})_!$ is the exceptional direct image in the chosen compact-support or Borel–Moore six-functor theory; it is not licensed by the displayed formula until the finite constructibility and proper-support defects vanish. For $U \subset E$ open,

$$\mathcal{F}^{\mathrm{loc}}_{\alpha,R}(U) := \bigoplus_{I/S_I} R\Gamma_c\Big(U^I,\ j^!_{U,I}\mathscr{H}^{\mathrm{loc}}_{\alpha,R,I}\Big),$$

and the direct sum over $\alpha \in \Gamma^{\mathrm{loc}}_R$ is $\mathcal{F}^{\mathrm{loc}}_R(U)$. Thus locality is !-restriction on closed configuration spaces; the old open-support Borel–Moore formula is a theorem to be proved from this atlas, not the primary finite datum.

(iii) *Wrapped prequotient sector and rigidification.* For $\eta \in \Gamma_R^{\mathrm{wr}}$, a prequotient stack

$$\mathcal{M}_{\eta,R}^{\mathrm{wr,rig}} \xrightarrow{\rho_{\eta,R}} \mathcal{M}_{\eta,R}^{\mathrm{wr}} \subset \mathfrak{M}_{\eta,R}$$

is supplied, together with a fixed origin $o_E \in E$ and an E-equivariant anchor

$$\lambda_{\eta,R} : \mathcal{M}_{\eta,R}^{\mathrm{wr,rig}} \longrightarrow E, \qquad \lambda_{\eta,R}(t \cdot W) = t + \lambda_{\eta,R}(W),$$

where $+$ is the group law on E with respect to the chosen origin. The origin o_E is part of the supplied datum and is fixed once globally in the datum; a different choice of origin shifts $\lambda_{\eta,R}$ by a constant and the anchor descends to a $\mathrm{Pic}^0(E)$-torsor whose zero point is the fixed datum. Its objects have support dominating E. The pair $(\rho_{\eta,R}, \lambda_{\eta,R})$ is a supplied rigidification, not a canonical construction. A natural candidate is the normalized determinant

$$\lambda(A) = \det Rp_{E,*}A \otimes \mathcal{O}_E(-\chi(A) \cdot o_E) \in \mathrm{Pic}^0(E) \simeq E.$$

This candidate has translation weight $\chi(A)$:

$$\lambda(t \cdot A) = \lambda(A) + \chi(A)t.$$

Thus it gives the required unit-weight anchor only on strata where this weight has been rigidified, divided by a chosen finite cover, or replaced by an equivalent Abel–Jacobi/framing datum. In particular, on $\chi(A) = 0$ strata it does not by itself remember the relative E-position. Nor does a finite divisor or Abel–Jacobi shadow by itself remember the wrapped Hall geometry. If $i_s : E \hookrightarrow K3 \times E$ is the elliptic fibre over $s \in K3$, then

$$W_0 = i_{s,*}(\mathcal{O}_E \oplus \mathcal{O}_E), \qquad W_L = i_{s,*}(L \oplus L^{-1}), \quad L \in \mathrm{Pic}^0(E), \quad L^2 \neq \mathcal{O}_E,$$

can have the same rank, determinant, charge shadow, and prescribed finite divisor anchor, while their wrapped self-extension dimensions on the elliptic factor differ:

$$\dim \mathrm{Ext}_E^1(\mathcal{O}_E \oplus \mathcal{O}_E, \mathcal{O}_E \oplus \mathcal{O}_E) = 4, \qquad \dim \mathrm{Ext}_E^1(L \oplus L^{-1}, L \oplus L^{-1}) = 2.$$

Thus any anchor advertised as lossless must remember enough of the wrapped correspondence data to recover these Ext stacks, not only a finite multiset on E. The anchor residual is therefore a typed finite datum, not a name:

$$\mathfrak{o}_R^{\lambda} = \left(\mathfrak{o}_R^{\lambda,\mathrm{ex}},\ \mathfrak{o}_R^{\lambda,\mathrm{unit}},\ \mathfrak{o}_R^{\lambda,\mathrm{loss}},\ \mathfrak{o}_R^{\lambda,\mathrm{multi}},\ \mathfrak{o}_{R'R}^{\lambda,\mathrm{tr}}\right).$$

These components assert, respectively, existence of the rigidified prequotient and anchor; unit translation law, possibly after a specified finite cover; losslessness of the anchor on the relative E-position needed by wrapped correspondences; compatibility of the input, intermediate, and output anchors on mixed and wrapped flag stacks; and compatibility under successor HN transitions. The datum requires the chosen anchor, its translation law, and the vanishing of all components of $\mathfrak{o}_R^{\lambda}$ as finite data. Put

$$\mathcal{G}_{\eta,R}^{\mathrm{wr}} := H_*^{\mathrm{BM}}\left(\mathcal{M}_{\eta,R}^{\mathrm{wr,rig}}, \Phi_{\eta,R}^{\mathrm{wr}} \otimes o_{\eta,R}^{\mathrm{wr}}\right), \qquad \mathcal{G}_R^{\mathrm{wr}} := \bigoplus_{\eta \in \Gamma_R^{\mathrm{wr}}} \mathcal{G}_{\eta,R}^{\mathrm{wr}}.$$

If no such prequotient memory is supplied, the wrapped part of the datum is absent.

(iv) *Local, mixed, and wrapped extension correspondences.* For disjoint opens $(U_i) \subset E$, local charges α_i, and wrapped charges η, η_1, η_2, the finite stage contains correspondences

$$\prod_i \mathcal{M}^{\mathrm{loc}}_{\alpha_i,R}(U_i) \xleftarrow{p} \mathfrak{E}^{\mathrm{loc}}_{\alpha_1,\dots,\alpha_r,R}(U_1,\dots,U_r) \xrightarrow{q} \mathcal{M}^{\mathrm{loc}}_{\sum_i \alpha_i,R}\Big(\bigcup_i U_i\Big),$$

$$\prod_i \mathcal{M}^{\mathrm{loc}}_{\alpha_i,R}(U_i) \times \mathcal{M}^{\mathrm{wr,rig}}_{\eta,R} \xleftarrow{p} \mathfrak{E}^{\mathrm{hyb}}_{\alpha_1,\dots,\alpha_r;\eta,R}(U_1,\dots,U_r) \xrightarrow{q} \mathcal{M}^{\mathrm{wr,rig}}_{\eta+\sum_i \alpha_i,R},$$

and

$$\mathcal{M}^{\mathrm{wr,rig}}_{\eta_1,R} \times \mathcal{M}^{\mathrm{wr,rig}}_{\eta_2,R} \xleftarrow{p} \mathfrak{E}^{\mathrm{wr}}_{\eta_1,\eta_2,R} \xrightarrow{q} \mathcal{M}^{\mathrm{wr,rig}}_{\eta_1+\eta_2,R}.$$

The second display is only the one-sided shorthand. Since Hall convolution is ordered, the datum contains, for every ordered mixed word with local inputs on both sides of one wrapped input, correspondences

$$\prod_{i\in I_-} \mathcal{M}^{\mathrm{loc}}_{\alpha_i,R}(U_i) \times \mathcal{M}^{\mathrm{wr,rig}}_{\eta,R} \times \prod_{j\in I_+} \mathcal{M}^{\mathrm{loc}}_{\beta_j,R}(V_j) \xleftarrow{p} \mathfrak{E}^{\mathrm{hyb}}_{\alpha_-;\eta;\beta_+,R}((U_i),(V_j)) \xrightarrow{q} \mathcal{M}^{\mathrm{wr,rig}}_{\eta+\sum_{i\in I_-}\alpha_i+\sum_{j\in I_+}\beta_j,R}.$$

The displayed mixed correspondence above is the case $I_+ = \varnothing$. The case $I_- = \varnothing$ is the right local action on a wrapped input. Both are part of the datum. Similarly $\mathfrak{E}^{\mathrm{wr}}_{\eta_1,\eta_2,R}$ is ordered in (η_1,η_2); no symmetry of wrapped inputs is assumed unless an extra symmetry datum is supplied. The middle stacks classify exact triangles or exact sequences with the displayed associated graded pieces. They are constructed in the unreduced E-equivariant theory. In the local legs the displayed open correspondences are shorthand for the closed-configuration correspondences over E^I, E^J, and $E^{I\sqcup J}$, followed by !-restriction to the chosen opens. The stage supplies closed-support finite-type witnesses, exceptional-pushforward admissibility witnesses for $q_!$, pullback/Gysin admissibility witnesses for p, and Thom–Sebastiani identifications for the coefficient systems. Properness is allowed as the special case $q_! = q_*$ on the protected support; affine Ext directions are handled only through compact-support integration in the fixed six-functor formalism. For every generator e among the local-local, ordered mixed, and wrapped-wrapped correspondences, write

$$\mathfrak{o}^{\mathrm{prop}}_{e,R}, \qquad \mathfrak{o}^{\mathrm{adm}}_{e,R}, \qquad \mathfrak{o}^{\mathrm{TS}}_{e,R}$$

for the defects of closed-configuration/proper-support control, pull-push admissibility, and Thom–Sebastiani compatibility. Their aggregate defects in the three families are

$$\mathfrak{o}^{\mathrm{prop,LL}}_{R}, \qquad \mathfrak{o}^{\mathrm{prop,mix}}_{R}, \qquad \mathfrak{o}^{\mathrm{prop,WW}}_{R}.$$

They vanish exactly when all pull-push maps used below are honest finite-stage operations, not formal correspondences without a defined pushforward. The mixed stacks use the anchor $\lambda_{\eta,R}$ to remember the relative position between the finite local supports U_i and the wrapped leg; replacing this memory by a scalar Fock coefficient is not this datum. The wrapped-wrapped stacks are likewise anchored: before quotienting they carry the map

$$\mathfrak{E}^{\mathrm{wr}}_{\eta_1,\eta_2,R} \longrightarrow E\times E\times E, \qquad \xi \longmapsto \big(\lambda_{\eta_1,R}(W_1), \lambda_{\eta_2,R}(W_2), \lambda_{\eta_1+\eta_2,R}(W)\big),$$

where W_1, W_2, W are the two inputs and the output of ξ. The finite correspondence keeps this relation, not only the total geometric degree $b^{\mathrm{geom}}_R(\eta_1+\eta_2)$.

These object stacks, extension stacks, and flag stacks generate a finite E-equivariant correspondence category

$$\mathsf{Corr}_R^{E,\mathrm{hyb}}.$$

Its objects are the finitely many local and rigidified wrapped charge stacks at height R. Its arrows are generated by the displayed correspondences, with composition by derived fibre product and with every summand of height $> R$ identified with zero. The flag stacks below are the specified 2-morphisms witnessing equality of the two composites. The finite operation maps are the displayed pull-push maps

$$\mu_R^\bullet = q_! \circ \mathrm{TS}_R^\bullet \circ p^*,$$

namely

$$\mu^{\mathrm{loc}}_{\alpha_1,\dots,\alpha_r,R} : \bigotimes_i \mathcal{F}^{\mathrm{loc}}_{\alpha_i,R}(U_i) \longrightarrow \mathcal{F}^{\mathrm{loc}}_{\sum_i \alpha_i,R}\left(\bigcup_i U_i\right),$$

$$\mu^{\mathrm{hyb}}_{\alpha_1,\dots,\alpha_r;\eta,R} : \bigotimes_i \mathcal{F}^{\mathrm{loc}}_{\alpha_i,R}(U_i) \otimes \mathcal{G}^{\mathrm{wr}}_{\eta,R} \longrightarrow \mathcal{G}^{\mathrm{wr}}_{\eta+\sum_i \alpha_i,R},$$

and

$$\mu^{\mathrm{wr}}_{\eta_1,\eta_2,R} : \mathcal{G}^{\mathrm{wr}}_{\eta_1,R} \otimes \mathcal{G}^{\mathrm{wr}}_{\eta_2,R} \longrightarrow \mathcal{G}^{\mathrm{wr}}_{\eta_1+\eta_2,R}.$$

Equivalently, the unreduced finite hybrid object at height R is the correspondence-module functor

$$\mathcal{B}_R^{E,\mathrm{hyb}} : \mathsf{Corr}_R^{E,\mathrm{hyb}} \longrightarrow \mathrm{Ch}_{\mathbb{C}}$$

sending

$$\mathcal{M}^{\mathrm{loc}}_{\alpha,R}(U) \longmapsto \mathcal{F}^{\mathrm{loc}}_{\alpha,R}(U), \qquad \mathcal{M}^{\mathrm{wr,rig}}_{\eta,R} \longmapsto \mathcal{G}^{\mathrm{wr}}_{\eta,R},$$

and sending a generator $e = (X \xleftarrow{p} \mathfrak{E}_e \xrightarrow{q} Y)$ to

$$\mathcal{B}_R^{E,\mathrm{hyb}}(e) = q_! \circ \mathrm{TS}_{e,R} \circ p^*.$$

This functor is defined only after the properness/admissibility defects above vanish. The reduced finite object is the quotient-after- correspondence module

$$\mathcal{B}_R^{\mathrm{red,hyb}} := Q_{E,R}\mathcal{B}_R^{E,\mathrm{hyb}}.$$

The two-sided mixed correspondences give, in the same notation,

$$\mu^{\mathrm{hyb}}_{\alpha_-;\eta;\beta_+,R} : \bigotimes_{i\in I_-} \mathcal{F}^{\mathrm{loc}}_{\alpha_i,R}(U_i) \otimes \mathcal{G}^{\mathrm{wr}}_{\eta,R} \otimes \bigotimes_{j\in I_+} \mathcal{F}^{\mathrm{loc}}_{\beta_j,R}(V_j) \longrightarrow \mathcal{G}^{\mathrm{wr}}_{\eta+\sum_i \alpha_i+\sum_j \beta_j,R}.$$

An output with at least one wrapped input is wrapped. The mixed correspondence is also required to record the surviving terms in $\mathrm{RHom}(A, W)$ and $\mathrm{RHom}(W, A)$ for local A and wrapped W; replacing it by a scalar global mode loses this data.

(v) *Full flag associativity and coherence.* Write L for a projection-finite local input and W for a rigidified wrapped input. For every word

$$w = (\epsilon_1, \epsilon_2, \epsilon_3) \in \{\mathrm{L}, \mathrm{W}\}^3$$

and every compatible ordered triple of charges and local opens, the datum contains a two-step flag stack

$$\mathfrak{Flag}^{w}_{\xi_1,\xi_2,\xi_3,R}$$

classifying filtrations with the prescribed associated graded pieces. Here $\mathfrak{E}_{\xi_i,\xi_j,R}$ means the corresponding ordered local-local, mixed local/wrapped, or wrapped-wrapped extension stack from the previous clause. The flag stack carries comparison maps to the two iterated correspondence fibre products

$$\mathfrak{E}_{\xi_1,\xi_2,R} \times_{\mathcal{M}_{\xi_1+\xi_2,R}} \mathfrak{E}_{\xi_1+\xi_2,\xi_3,R}, \qquad \mathfrak{E}_{\xi_2,\xi_3,R} \times_{\mathcal{M}_{\xi_2+\xi_3,R}} \mathfrak{E}_{\xi_1,\xi_2+\xi_3,R}.$$

The eight words

$$\mathrm{LLL},\ \mathrm{LLW},\ \mathrm{LWL},\ \mathrm{WLL}, \mathrm{LWW},\ \mathrm{WLW},\ \mathrm{WWL},\ \mathrm{WWW}$$

are all part of the datum. In the mixed and wrapped words the flag stack retains the input, intermediate, and output anchors before quotienting by E. The two parenthesizations give the same map in the finite quotient:

$$\mu_{\xi_1+\xi_2,\xi_3,R} \circ (\mu_{\xi_1,\xi_2,R} \otimes 1) = \mu_{\xi_1,\xi_2+\xi_3,R} \circ (1 \otimes \mu_{\xi_2,\xi_3,R}).$$

Here μ means one of the displayed local, mixed, or wrapped maps. The equality is required after vanishing cycles, Thom–Sebastiani transport, orientation signs, and the quotient by height $> R$. Equivalently, the two pull-push maps are induced by the same Borel–Moore class on the corresponding two-step flag stack. This is a finite-stage coherence axiom; no associativity is imported from the completed inverse limit. The defect class of this equality for the word w is denoted

$$\mathfrak{o}^{\mathrm{assoc}}_{w,R}.$$

The datum requires $\mathfrak{o}^{\mathrm{assoc}}_{w,R} = 0$ for all eight words, with the chosen 2-morphisms satisfying the usual four-input pentagon on the finite flag-of-flags stacks. This eight-word atlas gives binary associativity. Full hybrid factorization still requires the higher colored configuration atlas, units, symmetry or ordering conventions, refinement maps, descent, and overlap coherences for all finite HN windows. None of these follows from the eight displayed length-three words.

(vi) *Reduced E-quotient order.* All stacks and correspondences above are E-equivariant before quotienting. Let $\mathsf{Corr}^{E,\mathrm{hyb}}_R$ and $\mathsf{Corr}^{\mathrm{red,hyb}}_R$ denote the finite unreduced and reduced correspondence bicategories generated by these stacks, correspondences, and flag diagrams. The quotient datum first supplies a pseudofunctor between these bicategories, with coherent 2-morphisms for composition, base-change, and Thom–Sebastiani transport. Applying vanishing-cycle Borel–Moore chains, with the orientation lines included, gives the finite correspondence module categories $\mathsf{BM}^{E,\mathrm{hyb}}_R$ and $\mathsf{BM}^{\mathrm{red,hyb}}_R$. The symbol $Q_{E,R}$ denotes the induced supplied functor

$$Q_{E,R} : \mathsf{BM}^{E,\mathrm{hyb}}_R \longrightarrow \mathsf{BM}^{\mathrm{red,hyb}}_R$$

from the finite unreduced equivariant vanishing-cycle/Borel–Moore category generated by the stacks above to the finite reduced category. It is not only a map on object stacks. It is allowed only after the local, wrapped, and mixed extension correspondences have been formed and the chosen

rigidification has either descended or has been integrated out with a specified reduced kernel. For every generator e of $\mathsf{Corr}_R^{E,\mathrm{hyb}}$, the quotient package contains the reduced correspondence $\overline{e}$, the quotient maps from e to $\overline{e}$, admissibility witnesses for $\overline{p}^*$ and $\overline{q}_!$, proper-support witnesses for $\overline{q}$, and orientation-descent isomorphisms for the coefficient systems. If the anchor is integrated out rather than descended, the datum also contains the reduced kernel $K_{\lambda,R}$ and its compatibility with composition. The quotient construction is therefore a functor on the finite correspondence category, not a quotient of the object spaces followed by a new Hall product. For each operation $\mu_R^\bullet = q_!\,\mathrm{TS}\,p^*$ the datum contains a specified comparison isomorphism

$$\theta^Q_{\mu,R}:\ Q_{E,R}q_!\,\mathrm{TS}\,p^* \xrightarrow{\sim} \overline{q}_!\overline{\mathrm{TS}}\,\overline{p}^*(Q_{E,R}^{\boxtimes k}),$$

which is compatible with the flag-stack coherences. The finite-stabilizer gerbe classes needed for the orientation line must vanish on every stack entering these squares. For each two-step flag, the two composites of $\theta^Q_{\mu,R}$ agree with the descended flag 2-morphism; for $R' \geq R$, the transition functors commute with $Q_{E,R}$ through a specified natural isomorphism. A quotient-first construction is not this datum. The four quotient residuals are

$$\mathfrak{o}_R^{Q,\mathrm{form}},\qquad \mathfrak{o}_R^{Q,\mathrm{adm}},\qquad \mathfrak{o}_R^{Q,\mathrm{flag}},\qquad \mathfrak{o}_{R'R}^{Q,\mathrm{tr}}.$$

They record, respectively, existence of the functor $Q_{E,R}$ on the finite correspondence category, admissibility/properness after quotient, compatibility with all flag 2-morphisms, and compatibility with HN transition maps. The datum requires their vanishing and chosen null-homotopies.

(vii) *Protected integration and the Igusa s-direction.* The finite stage carries a protected integration map

$$I_R^{\mathrm{prot}} : Q_{E,R}\mathcal{F}^{\mathrm{hyb}}_{X,\sigma,S,\leq R} \longrightarrow \mathbb{C}[\Gamma_{\mathrm{eff}}]_{\leq R}$$

where the subscript denotes the image of the finite h_S-box. The map is compatible with the three reduced products $\overline{\mu}_R^\bullet$, coproducts, primitive projection, and transition maps. This means a chain-level character of the finite reduced correspondence algebra, not only the assignment of an s-degree. Let $m_{\Gamma,R}$, $\Delta_{\Gamma,R}$, and $P^{\mathrm{Prim}}_{\Gamma,R}$ denote the finite target multiplication, coproduct, and primitive projector on the chosen decategorified monomial model. The integration defects are

$$\mathfrak{o}_R^{I,\mathrm{ch}} := I_R^{\mathrm{prot}} d_R,$$

$$\mathfrak{o}_{w,R}^{I,\mathrm{prod}} := I_R^{\mathrm{prot}}\overline{\mu}_R^w - m_{\Gamma,R}(I_R^{\mathrm{prot}})^{\otimes k},\qquad w \in \{\mathrm{L},\mathrm{W}\}^k,$$

$$\mathfrak{o}_R^{I,\mathrm{co}} := \Delta_{\Gamma,R}I_R^{\mathrm{prot}} - (I_R^{\mathrm{prot}}\otimes I_R^{\mathrm{prot}})\overline{\Delta}_R,\qquad \mathfrak{o}_R^{I,\mathrm{Prim}} := P^{\mathrm{Prim}}_{\Gamma,R}I_R^{\mathrm{prot}} - I_R^{\mathrm{prot}}P_R^{\mathrm{Prim}}.$$

They are required to vanish, compatibly with $Q_{E,R}$, the orientation trivializations, and transitions in R. Equivalently, for every reduced operation $\overline{\mu}_R^w$ and homogeneous inputs x_i of charges γ_i,

$$I_R^{\mathrm{prot}}\big(\overline{\mu}_R^w(x_1,\dots,x_k)\big) = \prod_i I_R^{\mathrm{prot}}(x_i)$$

whenever the product is defined at height R; if the output charge has height $> R$, both sides are zero in the finite quotient. If a homogeneous class x has charge γ, and $\mathcal{F}^{\mathrm{hyb}}_{\gamma,R}$ denotes the

corresponding local or wrapped summand, then every nonzero monomial in $I_R^{\mathrm{prot}}(Q_{E,R}x)$ has s-degree $m_R(\gamma)$:

$$I_R^{\mathrm{prot}}(Q_{E,R}x) \in s^{m_R(\gamma)}\mathbb{C}[q^{\pm 1}, r^{\pm 1}]$$

under the OP/Borcherds variable convention of Proposition 11.1. More precisely, I_R^{prot} is homogeneous for the full Gram label $\overline{\Pi}_X(\gamma) = (n, l, m)$, and the displayed condition records the Gram trace exponent $m_R(\gamma) = m$, while $b_R^{\mathrm{geom}}(\gamma)$ records geometric elliptic support degree. Thus positive powers of the Igusa s-variable require positive Gram trace exponent, and the wrapped colour records the geometric support needed to realize such sectors. This is a source-support condition, not a consequence of the target product. If a target window contains a degree with $m > 0$, and in particular the test degree $\delta_2 \leftrightarrow (0, 1, 1)$, the finite stage must contain a wrapped source summand mapping to that degree under $\overline{\Pi}_X$. Otherwise the coefficient lies outside the verified compact source image. For products, the displayed character law and additivity of m_R give

$$s^{m_R(\gamma_1+\cdots+\gamma_k)} = \prod_i s^{m_R(\gamma_i)}$$

inside the finite HN quotient. Hence the s-direction is coupled to the actual local/wrapped operation, not appended as an external Fock determinant. For $R' \geq R$, let $\tau_{R'R}^{E,\mathrm{hyb}}$ be the unreduced finite hybrid transition, let

$$\bar{\tau}_{R'R}^{\mathrm{hyb}} : Q_{E,R'}\mathcal{F}_{X,\sigma,S,\leq R'}^{\mathrm{hyb}} \longrightarrow Q_{E,R}\mathcal{F}_{X,\sigma,S,\leq R}^{\mathrm{hyb}}$$

be the induced reduced chain map, and let $\pi_{R'R}^{\Gamma}$ be the induced truncation of the monomial target. The integration transition defect is

$$\mathfrak{o}_{R'R}^{I,\mathrm{tr}} := I_R^{\mathrm{prot}}\bar{\tau}_{R'R}^{\mathrm{hyb}} - \pi_{R'R}^{\Gamma} I_{R'}^{\mathrm{prot}} Q_{E,R'}.$$

It is required to vanish. Thus protected integration is compatible with the HN inverse system, not merely with one finite trace. The coefficient of a monomial $q^n r^l s^{m_R(\gamma)}$ is a protected finite-stage trace coefficient. Equality of these coefficients with the Borcherds exponents $f(nm, l)$, and the primitive Borcherds–Kac–Moody bracket test, are additional recognition requirements, not part of the definition.

(viii) *Hybrid residual tuple.* The exact finite obstruction tuple for the hybrid lane is

$$\mathfrak{O}_{\mathrm{hyb},R} = \Big(\mathfrak{o}_R^{\mathrm{stage}}, \mathfrak{o}_R^{\lambda}, \mathfrak{o}_R^{\mathrm{corr,LL}}, \mathfrak{o}_R^{\mathrm{corr,mix}}, \mathfrak{o}_R^{\mathrm{corr,WW}}, \mathfrak{o}_R^{\mathrm{conf}}, \mathfrak{o}_R^{\mathrm{inert,obj}}, \mathfrak{o}_R^{\mathrm{inert,corr}}, \mathfrak{o}_R^{\mathrm{inert,flag}}, \mathfrak{o}_R^{\mathrm{conf\text{-}prop}}, \mathfrak{o}_R^{\mathrm{prop,LL}}, \mathfrak{o}_R^{\mathrm{prop,mix}}, \mathfrak{o}_R^{\mathrm{pro}}$$

These entries record finite HN existence, anchor rigidification, local-local correspondences, all ordered mixed correspondences, wrapped-wrapped correspondences, the closed finite-configuration atlas, finite residual inertia for object, correspondence, and flag stacks, closed-support Hall admissibility over configuration spaces, properness/admissibility and Thom–Sebastiani validity for the pull-push operations, the eight-word two-step binary flag coherence, the higher colored configuration residual, quotient formation/admissibility/flag and transition descent, chain/product/colored-refinement/coproduct/primitive integration laws, integration transition compatibility, and HN transition compatibility. A finite hybrid datum means the compatible vanishing and trivialization of this tuple for every $R' \geq R$. Without $\mathfrak{O}_{\mathrm{hyb},R} = 0$, the notation $\mathcal{F}_{X,\sigma,S,\leq R}^{\mathrm{hyb}}$ names an obstruction problem, not a source object. The inertia entries are not formalities: diagonal scalar rigidification

removes at most the common $\mathbb{G}_m$, while direct sums and flags can retain $\mathbb{G}_m$, PGL_n, or parabolic stabilizers. Their vanishing is the finite-stack condition used later in Proposition 10.29. The entry $\mathfrak{o}_R^{\mathrm{col}}$ records the higher colored configuration atlas, unit and vacuum maps, symmetry or ordering conventions, refinement maps, descent, and overlap coherences not detected by the eight length-three words. If this entry is not supplied, vanishing of the remaining binary terms gives only the two-step Hall/module datum, not a full hybrid factorization object. The entry $\mathfrak{o}_R^{I,\mathrm{col}}$ records compatibility of protected integration with the higher colored configuration atlas and its refinement maps; binary product compatibility does not imply this colored-refinement integration law.

(ix) *Finite HN completion.* The construction is made only for charges in $\Gamma^{HN}_{\sigma,S}(R)$. For $R' \geq R$, transition maps

$$\tau^{E,\mathrm{hyb}}_{R'R} : \mathcal{F}^{\mathrm{hyb}}_{X,\sigma,S,\leq R'} \longrightarrow \mathcal{F}^{\mathrm{hyb}}_{X,\sigma,S,\leq R}$$

must be induced by functors $\mathsf{Corr}^{E,\mathrm{hyb}}_{R'} \to \mathsf{Corr}^{E,\mathrm{hyb}}_{R}$ compatible with all three kinds of correspondences, with the flag associativity constraints, with the rigidifying anchors, with $Q_{E,R}$, with the comparison isomorphisms θ^Q_R, and with protected integration. They preserve geometric elliptic degree: if a homogeneous summand of charge γ' at height R' maps to a nonzero summand of charge γ at height R, then

$$b^{\mathrm{geom}}_R(\gamma) = b^{\mathrm{geom}}_{R'}(\gamma').$$

The transition residual $\mathfrak{o}^{\mathrm{tr}}_{\mathrm{hyb},R'R}$ is the joint defect of object truncation, correspondence base-change, flag/pentagon coherence, anchor compatibility, and the induced chain map

$$\bar{\tau}^{\mathrm{hyb}}_{R'R} : \mathcal{B}^{\mathrm{red,hyb}}_{R'} \longrightarrow \mathcal{B}^{\mathrm{red,hyb}}_{R}.$$

When all finite stages and transitions have been supplied, the completed hybrid object is

$$\mathcal{F}^{\mathrm{hyb}}_X := \varprojlim_R \mathcal{F}^{\mathrm{hyb}}_{X,\sigma,S,\leq R}.$$

It is the charge-completed hybrid source specified by those finite data over $\mathrm{Ran}^{\mathrm{hyb}}(E)$. Its restriction to $\mathrm{Ran}(E)_{[0]}$ is ordinary projection-finite Ran factorization. Its $b > 0$ strata are wrapped modules acted on by the local sector through the mixed Hall correspondences. This inverse-limit formula is a construction from the displayed finite data; it is not an existence theorem for those data.

Proposition 10.28 *(Closed-configuration local-local properness in a coherent heart).* Fix R, local charges $\alpha, \beta \in \Gamma^{\mathrm{loc}}_R$, and finite sets I, J. Suppose the finite local sector is realized inside a noetherian abelian Hall heart

$$\mathcal{A}_{\sigma,R} \subset D^b(\mathrm{Coh}(K_3 \times E))$$

whose finite HN substacks $\mathfrak{M}_{\alpha,R}$ are finite type, specialization closed, closed under the subobjects and quotients entering the local-local Hall correspondence, and have a bounded nilpotent thickness in the E-direction on the finite window. Then the closed configuration incidence stack

$$\mathfrak{M}^{\mathrm{loc,cl}}_{\alpha,R,I} = \{(A, \mathbf{e}) \mid p_E(\mathrm{Supp}\, A) \subset Z_I(\mathbf{e})\}$$

is a closed finite-type substack of $\mathfrak{M}_{\alpha,R} \times E^I$. The local-local extension stack of short exact sequences

$$0 \to A \to B \to C \to 0, \qquad A \in \mathfrak{M}^{\mathrm{loc,cl}}_{\alpha,R,I}, \quad C \in \mathfrak{M}^{\mathrm{loc,cl}}_{\beta,R,J}, \quad B \in \mathfrak{M}^{\mathrm{loc,cl}}_{\alpha+\beta,R,I\sqcup J},$$

has

$$q : \mathfrak{E}^{\mathrm{loc,cl}}_{\alpha,\beta,R,I,J} \longrightarrow \mathfrak{M}^{\mathrm{loc,cl}}_{\alpha+\beta,R,I\sqcup J}$$

projective, hence proper and finite type. Over a fixed B this fibre is a closed subfunctor of the relative Quot scheme of B; it need not be finite. For instance subobjects $\mathcal{O}_x \subset \mathcal{O}_x^{\oplus 2}$ form $\mathbb{P}^1$. With BBDJS coefficients restricted by !-pullback to the closed incidence stacks, proper base-change gives the local-local closed-support Hall pull-push, still subject to the independent Thom–Sebastiani and orientation coherences.

The same assertion is false for raw exact triangles in $\mathrm{Perf}(K3 \times E)$. For $x \in K3 \times \{e\}$ and a DVR R_0 with uniformizer t, the map

$$t : \mathcal{O}_x \otimes R_0 \longrightarrow \mathcal{O}_x \otimes R_0$$

is an isomorphism on the generic fibre, so its cone is the vacuum, while the special fibre has cone

$$\mathcal{O}_x \oplus \mathcal{O}_x[1].$$

If the finite window excludes nonzero zero-charge objects, the valuative criterion for the corresponding q-map fails. Closed support does not repair this; an abelian Quot compactification, or an explicit compact-support six-functor replacement for constructible coefficients, is the required input.

Proposition 10.29 *(Finite d-critical Hall product).* Fix R and closed-configuration substacks $\mathfrak{M}^{\mathrm{loc,cl}}_{c,R,I}$ in a noetherian abelian Hall heart as in Proposition 10.28. Assume, in addition, the following finite data:

(i) finite-type quasi-smooth derived Artin enhancements of the selected object, extension, and two-step flag stacks, with perfect obstruction theories;
(ii) finite residual inertia after scalar, E-translation, and wrapped rigidifications;
(iii) finite-type extension and flag stacks for the retained charges, with p^* and $q_!$ admissible;
(iv) the PTVV (-1)-shifted symplectic structures restrict to these finite substacks, so that their classical truncations carry Joyce d-critical structures [11, 69];
(v) the K3 semiregularity cosections are surjective on the reduced strata and satisfy the displayed filtered Hall additivity of (P4);
(vi) BBDJS orientation lines and Joyce–Upmeier orientation transports are supplied, with all finite-stabilizer characters trivialized [46];
(vii) Thom–Sebastiani transports for the reduced vanishing-cycle sheaves satisfy the two-step flag pentagons and the higher colored configuration coherences.

Then the finite reduced local-local product

$$m_R^{\mathrm{red}} = q_! \circ \mathrm{TS}_R^{\mathrm{red}} \circ p^*$$

is an honest morphism on

$$H_c^\bullet\Big(\mathfrak{M}^{\mathrm{loc,cl}}_{c,R,I}, \Phi^{\mathrm{red}}_{c,R,I} \otimes o^{\mathrm{red}}_{c,R,I}\Big),$$

and is associative on the finite stage. The first obstruction to removing the hypotheses is finite inertia: proper finite-type correspondences do not by themselves eliminate residual $\mathbb{G}_m$, direct-sum, or parabolic automorphisms, and a residual $\mathbb{G}_m$ already contributes $H^\bullet(B\mathbb{G}_m) = \mathbb{C}[u]$ to protected cohomology. Here "finite stage" means finite charge support, finite-type stacks, and finite-dimensional protected

cohomology after the stated quotients and admissibility hypotheses. It does not mean that the Hall correspondence maps are finite morphisms: even in the local-local abelian Quot model the q-fibres may be positive-dimensional, as in the $\mathbb{P}^1$-family of subobjects $\mathcal{O}_x \subset \mathcal{O}_x^{\oplus 2}$.

Proposition 10.30 *(Finite consequences of the hybrid datum).* Assume the finite-stage data of Definition 10.27 are supplied for all R, with compatible transition maps. Then:

(i) the $b = 0$ restriction is an ordinary support-local factorization object on $\mathrm{Ran}(E)$;

(ii) every operation with a wrapped input lands in a wrapped stratum;

(iii) the displayed ordered mixed maps, including the two-sided maps $\mu^{\mathrm{hyb}}_{\alpha_-;\eta;\beta_+,R}$, define the only available Hall actions of projection-finite insertions on wrapped elliptic-degree sectors in $\mathsf{Corr}^{E,\mathrm{hyb}}_R$ before the reduced E-quotient;

(iv) vanishing of the closed-configuration and properness/admissibility defects $\mathfrak{o}^{\mathrm{conf}}_R$, $\mathfrak{o}^{\mathrm{conf\text{-}prop}}_R$, $\mathfrak{o}^{\mathrm{prop,LL}}_R$, $\mathfrak{o}^{\mathrm{prop,mix}}_R$, and $\mathfrak{o}^{\mathrm{prop,WW}}_R$ makes the Borel–Moore pull-push functor $\mathcal{B}^{E,\mathrm{hyb}}_R$ and its reduced quotient $\mathcal{B}^{\mathrm{red,hyb}}_R$ well-defined;

(v) vanishing of the eight classes $\mathfrak{o}^{\mathrm{assoc}}_{w,R}$ gives an associative finite-stage reduced Hall product after vanishing cycles, orientations, HN quotienting, the quotient functor, and the comparison isomorphisms $\theta^{Q}_{\mu,R}$;

(vi) a protected trace or primitive bracket obtained from this object is compatible with reduced E-integration only through the specified $Q_{E,R}$-compatibility isomorphisms, their flag coherences, finite-stabilizer conditions, and the vanishing of the integration defects $\mathfrak{o}^{I,\mathrm{ch}}_R$, $\mathfrak{o}^{I,\mathrm{prod}}_{w,R}$, $\mathfrak{o}^{I,\mathrm{col}}_R$, $\mathfrak{o}^{I,\mathrm{co}}_R$, and $\mathfrak{o}^{I,\mathrm{Prim}}_R$, compatibly with $\mathfrak{o}^{I,\mathrm{tr}}_{R'R} = 0$;

(vii) for homogeneous $x \in Q_{E,R}\mathcal{F}^{\mathrm{hyb}}_{\gamma,R}$,

$$I^{\mathrm{prot}}_R(x) \in s^{m_R(\gamma)}\mathbb{C}[q^{\pm 1}, r^{\pm 1}],$$

hence the positive s-direction cannot be supplied by the projection-finite Ran stratum; on products this law is multiplicative because m_R is additive under $\overline{\Pi}_X$ inside the finite quotient.

A decoupled global Fock or Hecke factor without $\mathsf{Corr}^{E,\mathrm{hyb}}_R$, anchored mixed correspondences, proper pull-push operations, the eight-word two-step binary flag atlas with the higher colored residual $\mathfrak{o}^{\mathrm{col}}_R$, quotient residuals, and protected-integration residuals may reproduce a determinant-level product, but it does not define the local-wrapped primitive brackets needed for recognition of $\mathfrak{g}_{\Delta_s}$.

Proof. The first two assertions are built into the elliptic-degree stratification and Lemma 10.25. The third assertion is the definition of the mixed operation maps: before the reduced quotient they are the only displayed operations remembering relative E-position between finite local support and a wrapped leg, including the left and right ordered actions. The properness assertion is exactly the vanishing of the generator-wise proper-support, admissibility, and Thom–Sebastiani defects. The flag assertion is exactly the eight-word two-step binary flag axiom transported through $\mathsf{Corr}^{E,\mathrm{hyb}}_R$, with higher colored factorization kept in the separate residual $\mathfrak{o}^{\mathrm{col}}_R$. The quotient assertion is the quotient-order axiom, the coherent comparison isomorphism $\theta^{Q}_{\mu,R}$, and the vanishing of the quotient residuals. The last statement is the degree law and character property in the protected integration clause, together with the vanishing of the named integration defects and their transition residual. No compact source object, orientation system, quotient functor, or protected integration map is constructed in this proposition. □

Corollary 10.31 *(No quotient-first universal hybrid repair).* Fix R. A candidate finite correspondence category for the wrapped sector gives the source object $\mathcal{F}^{\mathrm{hyb}}_{X,\sigma,S,\leq R}$ only if it contains the unreduced local, wrapped, and ordered mixed source stacks of Definition 10.27, their anchors, the eight-word flag atlas, the quotient functor $Q_{E,R}$ formed after those correspondences, the comparison maps $\theta^{Q}_{\mu,R}$, and protected integration with all entries of $\mathfrak{D}_{\mathrm{hyb},R}$, including its transition entries, trivialized. A quotient of object stacks formed before the mixed and wrapped correspondences, or a determinant/Fock category carrying only the trace degree s^{m_R}, does not define the hybrid source.

Proof. By Lemma 10.25, a positive elliptic-degree sector is not a finite-support Ran sector. The only displayed mechanism retaining its relative E-position is the rigidified wrapped prequotient with anchor, together with the unreduced mixed and wrapped correspondences. If one quotients first, these anchors are no longer available as source data for $\mathfrak{E}^{\mathrm{hyb}}_{R}$, the two-sided maps $\mu^{\mathrm{hyb}}_{\alpha_-;\eta;\beta_+,R}$, or the flag 2-morphisms. Then $Q_{E,R}$, $\theta^{Q}_{\mu,R}$, and the product, coproduct, primitive, and transition integration residuals are not defined as finite source data. A scalar determinant or Fock trace can record coefficients after decategorification; it cannot supply the correspondence-level primitive brackets required by the datum. □

Proposition 10.32 *((O2)/hybrid compatibility obstruction).* Fix a finite HN height R and one of the three type-II simple walls. Under the degree map $\alpha(n,l,m) = 2nf_2 - lf_3 + 2mf_{-2}$ the wall labels are

wall	(n,l,m)	elliptic-degree shadow
$\delta_1 = 2f_2 - f_3$	$(1,1,0)$	$m = 0$
$\delta_2 = 2f_{-2} - f_3$	$(0,1,1)$	$m = 1$
$\delta_3 = f_3$	$(0,-1,0)$	$m = 0.$

A finite compact construction which claims both the hybrid source $\mathcal{F}^{\mathrm{hyb}}_{X,\sigma,S,\leq R}$ of Definition 10.27 and the type-II Pfaffian wall model of Definition 3.10 must supply a compatibility package at the same height R. It consists of:

(i) a wall object $x_{\delta_i,R}$ in the finite local or rigidified wrapped moduli used by $\mathsf{Corr}^{E,\mathrm{hyb}}_{R}$, mapping to Gram degree δ_i;

(ii) a finite restriction of the unreduced hybrid deformation theory to the wall object, producing a self-Ext complex $K^{E}_{\delta_i,R}$;

(iii) a quotient-compatible comparison diagram

$$\begin{tikzcd}
\mathsf{BM}^{E,\mathrm{hyb}}_{R} \arrow[r, "\ell_{\delta_i,R}"] \arrow[d, "Q_{E,R}"'] & K^{E}_{\delta_i,R} \arrow[r, "\mathrm{split}_{\delta_i,R}"] \arrow[d, "Q^{K}_{\delta_i,R}"] & K^{E,\mathrm{tan}}_{\delta_i,R} \oplus K^{E,\mathrm{nor}}_{\delta_i,R} \arrow[d, "Q^{\mathrm{tan}}_{\delta_i,R} \oplus Q^{\mathrm{nor}}_{\delta_i,R}"] \\
\mathsf{BM}^{\mathrm{red,hyb}}_{R} \arrow[r, "\bar{\ell}_{\delta_i,R}"'] & K^{\mathrm{red}}_{\delta_i,R} \arrow[r, "\overline{\mathrm{split}}_{\delta_i,R}"'] & K^{\mathrm{tan}}_{\delta_i,R} \oplus K^{\mathrm{nor}}_{\delta_i,R},
\end{tikzcd}$$

where the bottom row is the reduced wall chart of Proposition 3.13;

(iv) an invariant normal Pfaffian unit and rank-one normal model

$$\mathrm{Pf}(K^{\mathrm{nor}}_{\delta_i,R}) = v_{\delta_i,R} u_{\delta_i,R}, \qquad s_{\delta_i}(v_{\delta_i,R}) = v_{\delta_i,R},$$

after the quotient orientation trivializations used by $Q_{E,R}$;

(v) protected integration compatibility: if $\mathrm{pf}_{\delta_i,R}$ is the local Pfaffian class of the normal crossing, then every nonzero monomial of

$$I_R^{\mathrm{prot}}(Q_{E,R}\,\mathrm{pf}_{\delta_i,R})$$

is supported on the full monomial $q^n r^l s^m$ for the displayed Gram label (n, l, m), and the quotient-oriented Pfaffian sign is preserved under s_{δ_i}. This support and sign equality is compatible with the transition residual $\mathfrak{o}^{I,\mathrm{tr}}_{R'R}$.

Equivalently, the finite residual

$$\mathfrak{O}^{\mathrm{O2/hyb}}_{\delta_i,R} = \left(\mathfrak{o}^{\mathrm{wall}}_{\delta_i,R}, \mathfrak{o}^{\mathrm{Ext\text{-}split}}_{\delta_i,R}, \mathfrak{o}^{\mathrm{unit}}_{\delta_i,R}, \mathfrak{o}^{\mathrm{rank}}_{\delta_i,R}, \mathfrak{o}^{\mathrm{atlas,comp}}_{\delta_i,R}, \mathfrak{o}^{\mathrm{atlas,bdy}}_{\delta_i,R}, \mathfrak{o}^{\mathrm{atlas,ov}}_{\delta_i,R}, \mathfrak{o}^{Q,\mathrm{wall}}_{\delta_i,R}, \mathfrak{o}^{I,\mathrm{wall}}_{\delta_i,R}\right)$$

must be trivialized. The δ_2 row has $m = 1$, hence its wall object cannot be supplied by the projection-finite Ran stratum alone. It must pass through the rigidified wrapped prequotient and the ordered mixed correspondences before the reduced E-quotient. A local rank-one chart detached from $\mathsf{Corr}^{E,\mathrm{hyb}}_R$, or a Fock/Hecke factor carrying only the scalar s-degree, does not make this residual vanish. For O2 this compatibility must be imposed as a wall atlas: at finite height R, each retained component and boundary stratum of $\mathbb{H}_{\delta_i}$ needs a chart of the above form, and the normal Pfaffian units, tangent/normal splittings, quotient orientations, and protected integration maps must agree on chart overlaps. A single generic chart computes only a local sign. The atlas entries record, respectively, charts on every retained wall component, charts on retained boundary strata, and agreement of normal Pfaffian units, tangent/normal splittings, quotient orientations, and protected integration maps on chart overlaps, compatibly with HN transitions. For the middle wall this means, concretely, the datum must contain a finite-height object

$$x_{\delta_2,R} \in \mathcal{M}^{\mathrm{wr,rig}}_{\eta,R}, \qquad \overline{\Pi}_X(x_{\delta_2,R}) = (0,1,1), \qquad b^{\mathrm{geom}}_R(\eta) > 0,$$

with a unit-weight lossless anchor; an unreduced hybrid deformation complex $K^E_{\delta_2,R}$, K3 semiregularity cosection, and reduced quotient complex $K^{\mathrm{red}}_{\delta_2,R}$; an s_{δ_2}-equivariant real d-critical chart

$$(U_{\delta_2,R}, x_{\delta_2,R}, u_{\delta_2,R}, K^{\mathrm{red}}_{\delta_2,R}), \qquad U_{\delta_2,R} \cap \mathbb{H}_{\delta_2} = \{u_{\delta_2,R} = 0\}, \qquad s_{\delta_2}(u_{\delta_2,R}) = -u_{\delta_2,R};$$

a quotient-compatible splitting

$$K^{\mathrm{red}}_{\delta_2,R} \simeq K^{\mathrm{tan}}_{\delta_2,R} \oplus K^{\mathrm{nor}}_{\delta_2,R}, \qquad K^{\mathrm{nor}}_{\delta_2,R} \simeq [\mathbb{R} \xrightarrow{u_{\delta_2,R}} \mathbb{R}];$$

an invariant unit $\mathrm{Pf}(K^{\mathrm{nor}}_{\delta_2,R}) = v_{\delta_2,R} u_{\delta_2,R}$; and vanishing of the ordinary reduced orientation gerbe, the connected BE-class, all finite Borel classes, and all residual $H^1(BE[N])$ orientation characters on the object, extension, and flag strata entering this chart. Without this list the δ_2 wall is not a constructed rank-one Pfaffian wall.

Proof. The three Gram labels are the inverse calculation recorded in the type-II root section:

$$\delta_1 \leftrightarrow (1,1,0), \qquad \delta_2 \leftrightarrow (0,1,1), \qquad \delta_3 \leftrightarrow (0,-1,0).$$

The local Pfaffian proposition requires a reduced wall chart, a reduced self-Ext complex, a tangent/normal splitting, an invariant normal unit, and $N^{\mathrm{Pf}}_{\delta_i} = 1$. The hybrid datum requires unreduced local, wrapped, and ordered mixed correspondences, followed only then by $Q_{E,R}$ and by protected integration. If both claims are claims about the same compact source, the wall complex must be the localization of the finite

hybrid deformation theory and its splitting, unit, quotient descent, and protected trace must commute with the hybrid operations. The displayed diagram and residual tuple are precisely this finite compatibility requirement.

For δ_2 the target monomial has s-degree one. By Lemma 10.25, positive elliptic degree is not supported at finitely many points of E. Thus an ordinary projection-finite Ran object cannot supply the wall object whose local Pfaffian class integrates to the δ_2 label. The wrapped prequotient, its anchor, the mixed correspondences, and the quotient-after-correspondence functor are not optional decoration; they are the data needed even to state the compatibility of the local Pfaffian calculation with the hybrid s-direction. □

Open Problem 10.33 *(Hybrid wrapped factorization).* Construct the finite-stage data $\mathcal{F}^{\mathrm{hyb}}_{X,\sigma,S,\leq R}$ of Definition 10.27 and prove that their inverse limit exists in the completed Hall/factorization category. The exact obstruction is the simultaneous construction of:

(i) the compact extension-closed category and stability/HN sector whose local and wrapped moduli are finite type in bounded h_S-height;

(ii) a rigidified wrapped prequotient $\mathcal{M}^{\mathrm{wr,rig}}_{\eta,R} \to \mathcal{M}^{\mathrm{wr}}_{\eta,R}$ retaining relative E-position without inserting a positive elliptic-degree object at a finite Ran point;

(iii) local-local, all ordered two-sided local-wrapped, and wrapped-wrapped extension stacks with admissible vanishing-cycle pull-push functors and proper-support witnesses for the q-maps, equivalently vanishing of $\mathfrak{o}^{\mathrm{prop,LL}}_R$, $\mathfrak{o}^{\mathrm{prop,mix}}_R$, and $\mathfrak{o}^{\mathrm{prop,WW}}_R$;

(iv) Thom–Sebastiani-compatible strong orientations on all three kinds of correspondences and on the eight-word two-step binary flag atlas, proving the displayed parenthesization identities and the four-input pentagon in finite HN quotients, with higher colored factorization kept as a separate obligation;

(v) descent through the reduced E-quotient after forming correspondences, including the finite-stabilizer gerbe checks and the $Q_{E,R}$-compatibility isomorphisms, with $\mathfrak{o}^{Q,\mathrm{form}}_R = \mathfrak{o}^{Q,\mathrm{adm}}_R = \mathfrak{o}^{Q,\mathrm{flag}}_R = \mathfrak{o}^{Q,\mathrm{tr}}_{R'R} = 0$;

(vi) protected integration whose decategorified trace has s-degree $m_R = \mathrm{pr}_3\,\overline{\Pi}_X$, while b^{geom}_R records geometric wrapped support, recovers the Igusa s-direction coefficientwise, vanishes on the chain/product/coproduct/primitive defects $\mathfrak{o}^{I,\bullet}_R$, has $\mathfrak{o}^{I,\mathrm{tr}}_{R'R} = 0$ for all $R' \geq R$, and whose primitive bracket passes the Borcherds–Kac–Moody presentation test.

Equivalently, one must trivialize $\mathfrak{O}_{\mathrm{hyb},R}$ for every finite HN height R, including the quotient, integration, and HN transition components for every $R' \geq R$. Until these six items are supplied, $\mathcal{F}^{\mathrm{hyb}}_X$ is a finite categorical datum whose existence remains open, not a compact QFT constructed in this paper and not a detached determinant factor. Ordinary disjoint-open factorization on $\mathrm{Ran}(E)$ is only the projection-finite $b = 0$ part.

Open Problem 10.34 *(Normal-ordered descent of the protected bracket).* Let $\Gamma^{\mathrm{form}}_X = \Lambda_S \oplus \Lambda_S$, $\Pi_X : \Gamma^{\mathrm{form}}_X \to \Gamma_{\mathrm{gram}}$, $B : \Gamma^{\mathrm{form}}_X \times \Gamma^{\mathrm{form}}_X \to \Gamma_{\mathrm{gram}}$ be as in Definition 6.1. Suppose a Hall lift $\widetilde{P}_X$ of the protected primitive algebra is supplied with a $\Gamma^{\mathrm{eff}}_{X,\sigma}$-graded bracket, for a supplied semigroup $\Gamma^{\mathrm{eff}}_{X,\sigma} \subset \Gamma^{\mathrm{alg}}_X \subset \Gamma^{\mathrm{form}}_X$. Construct a cocycle-corrected descent $\overline{\Pi}^{\Theta}_{X,*}$ categorifying the normal-ordered homomorphism $\overline{\Pi}_X$:

$\widehat{\Gamma}_X \to \Gamma_{\mathrm{gram}}$. The raw Hall bracket is concentrated in additive charge $c + c'$ inside the supplied effective support, whose raw Gram shadow is

$$\Pi_X(c + c') = \Pi_X(c) + \Pi_X(c') + B(c, c').$$

The coefficient category, central translations, Hochschild cocycle B_{ch}, and cochain $\Theta_\Pi(c) = \mathsf{T}_{-\Pi_X(c)}$ are those of Definition 10.35. The datum Θ_Π must lift the lattice cochain identity

$$d_{\mathrm{Hoch}}(-\Pi_X) = B$$

in Γ_{gram}-valued additive cochains to the Picard dg chain identity

$$d_{\mathrm{Hoch}}\Theta_\Pi = B_{\mathrm{ch}}$$

in the Picard dg groupoid of central translations. Its triple homotopies must be compatible with the Hall extension correspondences, the Joyce–Upmeier orientation cocycle, the Hopf pairing, the reduced E-quotient, the finite fibre products over $\overline{\Pi}_X$, the completed HN tensor topology, and the S^1-action on the cyclic complex. Without such a lift, the raw bracket has the wrong Gram degree and the ordinary pushforward along Π_X is not a recognition target.

Definition 10.35 *(Normal-ordering coefficient dg category).* Fix a finite HN quotient R of the protected Hall sector, before passing to the inverse limit. Let

$$\mathsf{Ch}^{\Gamma_{\mathrm{gram}}}_{k,R,E}$$

be the dg category of Γ_{gram}-graded complexes of protected vanishing-cycle sheaves after the cosection-reduced E-quotient, with differential preserving the Gram degree and with support projected to the finite test window of the stage. The transition functors for $R' \geq R$ are required to be strict continuous maps for the HN filtration. They define the completed separated dg category

$$\mathsf{Ch}^{\Gamma_{\mathrm{gram}}}_{k,\widehat{\mathrm{HN}},E} := \varprojlim_R \mathsf{Ch}^{\Gamma_{\mathrm{gram}}}_{k,R,E}.$$

Every completed tensor product in this section is taken in this topology:

$$V \widehat{\otimes} W := \varprojlim_R (V_R \otimes W_R)$$

for compatible finite stages. No tensor product or coproduct is formed directly on an unfiltered infinite direct sum. For $\eta \in \Gamma_{\mathrm{gram}}$ define the central translation dg autoequivalence

$$(\mathsf{T}_\eta V)_\gamma := V_{\gamma+\eta}, \qquad d_{\mathsf{T}_\eta V} = d_V.$$

Then $\mathsf{T}_0 = \mathrm{id}$ and $\mathsf{T}_\eta \mathsf{T}_{\eta'} = \mathsf{T}_{\eta+\eta'}$ strictly. Let $\mathsf{Pic}^{\mathrm{dg}}_{\Gamma_{\mathrm{gram}}}$ denote the Picard dg groupoid generated by these translations and their chain homotopies.

For formal charges $c, c' \in \Gamma_X^{\mathrm{form}}$ put

$$B_{\mathrm{ch}}(c, c') := \mathsf{T}_{B(c,c')}.$$

This is the translation-valued Hochschild 2-cocycle representing the normal-ordering defect. A chain-level normal-ordering datum is a normalized 1-cochain

$$\Theta_\Pi(c) = \mathsf{T}_{-\Pi_X(c)} \in C^1_{\mathrm{Hoch}}(\Gamma_X^{\mathrm{phys}}; \mathsf{Pic}^{\mathrm{dg}}_{\Gamma_{\mathrm{gram}}})$$

together with natural quasi-isomorphisms

$$\vartheta_{c,c'} : \mathsf{T}_{-\Pi_X(c)}\mathsf{T}_{-\Pi_X(c')}\mathsf{T}_{\Pi_X(c+c')} \xrightarrow{\sim} \mathsf{T}_{B(c,c')}.$$

Equivalently,

$$d_{\mathrm{Hoch}}\Theta_\Pi = B_{\mathrm{ch}}.$$

The translation cochain itself is formal once the finite coefficient dg category exists. The non-formal datum is its realization on the compact Hall correspondence targets, with orientation, quotient, cyclic, and HN transition compatibility. For a finite Γ_X^{phys}-supported complex $V = \bigoplus_c V_c$, the uncompleted shorthand is

$$(\overline{\Pi}^{\Theta}_{*,\mathrm{fin}} V)_\gamma = \prod_{\Pi_X(c)=\gamma} V_c,$$

with the product finite. Its product and bracket structure is transported by the Θ_Π-quasi-isomorphisms. Thus a raw bracket output in charge $c + c'$, whose Gram shadow is $\Pi_X(c + c')$, is read after $\mathsf{T}_{B(c,c')}$ in degree

$$\Pi_X(c + c') - B(c, c') = \Pi_X(c) + \Pi_X(c').$$

At a finite stage the actual coefficient object is defined on $\widehat{\Gamma}_R$. If

$$V_R = \bigoplus_{(c,T)\in\widehat{\Gamma}_R} (V_R)_{(c,T)}$$

is a finite $\widehat{\Gamma}_R$-graded complex, set

$$(\overline{\Pi}^{\Theta}_{R,*} V_R)_\gamma = \prod_{\substack{(c,T)\in\widehat{\Gamma}_R \\ \overline{\Pi}_X(c,T)=\gamma}} (V_R)_{(c,T)}.$$

This product is finite; when the fibre is empty it is 0. The Γ_X^{phys}-graded formula above is only the special case $T = 0$ before the counterterm has been recorded. For a compatible inverse system $V = \varprojlim_R V_R$, the completed descent is

$$(\overline{\Pi}^{\Theta}_{X,*} V)_\gamma := \varprojlim_R (\overline{\Pi}^{\Theta_R}_{R,*} V_R)_\gamma.$$

Thus a product over a fibre of $\overline{\Pi}_X$ means an inverse limit of finite fibre products. The notation is not a licence to multiply over an uncontrolled infinite fibre. Products, brackets, coproducts, pairings, and radical quotients after $\overline{\Pi}^{\Theta}_{X,*}$ are defined only when their finite-stage maps commute with the transition maps and are continuous for the displayed HN completion.

Lemma 10.36 *(Finite Picard normal-ordering lift).* Fix a finite HN stage R for which the coefficient dg category of Definition 10.35 and the finite set $\widehat{\Gamma}_R$ are supplied. Put

$$\Theta^{\mathrm{phys}}_{\Pi,R}(c) := \mathsf{T}_{-\Pi_X(c)} \qquad (c \in \Gamma_X^{\mathrm{phys}})$$

and, after the counterterm T has been recorded,

$$\widehat{\Theta}_{\Pi,R}(c,T) := \mathsf{T}_{-\overline{\Pi}_X(c,T)} = \mathsf{T}_{T-\Pi_X(c)} \qquad ((c,T) \in \widehat{\Gamma}_R).$$

Then:

(i) $B_{\mathrm{ch},R}(c,c') = \mathsf{T}_{B(c,c')}$ is a normalized Hochschild 2-cocycle in the Picard dg groupoid of central translations, and

$$d_{\mathrm{Hoch}}\Theta^{\mathrm{phys}}_{\Pi,R} = B_{\mathrm{ch},R}.$$

(ii) $\widehat{\Theta}_{\Pi,R}$ is a strict Picard character on every allowed finite normal-ordered charge channel $(c,T) \star (c',T') \in \widehat{\Gamma}_R$:

$$d_{\mathrm{Hoch}}\widehat{\Theta}_{\Pi,R} = 0.$$

Equivalently, the $\widehat{\Gamma}_R$-grading has already absorbed the Gram defect, and $\overline{\Pi}_X$ is additive on $\widehat{\Gamma}_X$.

(iii) If the successor transition ρ_{R^+R} is the degree-truncation map of (Do-F) and commutes with central translations, then the finite fibre square (NO1) commutes for every finite $\widehat{\Gamma}_{R^+}$-graded complex V_{R^+}.

Thus the Picard part of normal ordering is formal after the finite coefficient dg category is supplied. What is not formal is the action of these translations on Hall pull-push correspondences, orientation lines, cyclic complexes, and the product/coproduct/pairing transition maps.

Proof. The translation functors satisfy $\mathsf{T}_\eta\mathsf{T}_{\eta'} = \mathsf{T}_{\eta+\eta'}$ and $\mathsf{T}_0 = \mathrm{id}$ strictly. Therefore

$$(d_{\mathrm{Hoch}}\Theta^{\mathrm{phys}}_{\Pi,R})(c,c') = \mathsf{T}_{-\Pi_X(c)-\Pi_X(c')+\Pi_X(c+c')} = \mathsf{T}_{B(c,c')},$$

by Lemma 6.8. The same bilinear cocycle identity for B gives $d_{\mathrm{Hoch}}B_{\mathrm{ch},R} = 0$, and normalization follows from $B(c,0) = B(0,c) = 0$.

For $\widehat{\Theta}_{\Pi,R}$, compute with $(c,T) \star (c',T') = (c+c', T+T'+B(c,c'))$:

$$\begin{aligned} &-\overline{\Pi}_X(c,T) - \overline{\Pi}_X(c',T') + \overline{\Pi}_X((c,T)\star(c',T')) \\ &\quad = -\Pi_X(c) + T - \Pi_X(c') + T' + \Pi_X(c+c') - T - T' - B(c,c') = 0. \end{aligned}$$

Hence its Hochschild boundary is the identity translation.

Finally, the fibre of $\overline{\Pi}_X : \widehat{\Gamma}_R \to \Gamma_{\mathrm{gram}}$ over a fixed γ is finite because $\widehat{\Gamma}_R$ is finite. Since $\widehat{\Gamma}_R \subset \widehat{\Gamma}_{R^+}$, the two composites in (NO1) both first select the finite summands with $\overline{\Pi}_X(c,T) = \gamma$ and then kill exactly those summands not lying in $\widehat{\Gamma}_R$. Commutation with central translations identifies the coefficient maps. Thus the square commutes. □

Lemma 10.37 *(Kunneth-pure multiplicative orientation criterion).* Adopt the notation of Definition 6.5 and Definition 4.3. Suppose a projection-finite Kunneth-pure substack of the reduced compact protected derived moduli stack of $K3 \times E$ is supplied, closed under the Hall extension correspondences under consideration. Suppose also that on this substack:

(i) the full self-Ext determinant admits a functorial Kunneth decomposition, including the cross terms for direct sums;

(ii) the elliptic factor carries a chosen Costello–Li flat trivialization compatible with the reduced E-quotient [21, 22];

(iii) the K3 factors carry supplied Joyce–Upmeier multiplicative orientation isomorphisms with strict pentagon; and
(iv) these choices are S^1-equivariant on the cyclic complexes.

Then the tensor product of these supplied data gives multiplicative orientation isomorphisms

$$\mu_{c,c'} : o_{c+c'} \xrightarrow{\sim} o_c \otimes o_{c'}$$

on that Kunneth-pure sector, natural for the restricted Hall correspondences and compatible with the reduced E-quotient. No assertion is made here for the full compact or wrapped $K3 \times E$ protected sector.

Proof. Tensor product. On the supplied Kunneth-pure substack, the full self-Ext determinant has the assumed factorization into K3 factors and the flat elliptic factor, including all direct-sum cross terms. Tensor the supplied factorwise Joyce–Upmeier multiplicative orientation maps with the chosen Costello–Li trivialization [21, 22] on the elliptic factor. This gives the displayed maps $\mu_{c,c'}$ on precisely that restricted sector.

Naturality, the strict pentagon, and reduced- E-compatibility follow from the corresponding supplied properties on the factors. The S^1-statement is exactly the assumed equivariance of the factorwise cyclic data.

The proof uses no global product decomposition of the full compact moduli stack and no projection formula for wrapped branes. Therefore it proves (O) only on the explicitly supplied Kunneth-pure sector. The full compact/wrapped orientation problem remains the open obligation recorded in Open Problem 5.7. □

Remark 10.38 *(Orientation comparison on the supplied sector).* The criterion above explains how a multiplicative orientation would factor after one has already supplied the projection-finite Kunneth-pure sector and all factorwise orientation data. It does not construct the reduced compact $K3 \times E$ orientation gerbe, its cyclic equivariance, or its HN-completed higher-rank extension. Those remain part of the Dirac–Igusa realization datum.

Theorem 10.39 *(Normal-ordered Hall descent datum and obstruction).* Adopt the notation of Definition 6.1 and Definition 6.5: formal lattice $\Gamma_X^{\mathrm{form}} = \Lambda_S \oplus \Lambda_S$, Gram shadow $\Pi_X : \Gamma_X^{\mathrm{form}} \to \Gamma_{\mathrm{gram}} = \mathbb{Z}^3$, symmetric quadratic cocycle $B(c, c') = (Q \cdot Q',\ Q \cdot P' + Q' \cdot P,\ P \cdot P')$ satisfying $\Pi_X(c + c') = \Pi_X(c) + \Pi_X(c') + B(c, c')$. Suppose: In the Hall-geometric clauses below, quantifiers over charges mean charges in the supplied effective support or in the finite normal-ordered support of the relevant stage. The formal lattice Γ_X^{form} is only the ambient lattice for the obstruction; it is not asserted to be fully realized by compact moduli. At each finite stage the theorem uses the supplied normal-ordered refinement of the primitive layer,

$$\widetilde{P}_R = \bigoplus_{(c,T) \in \widehat{\Gamma}_R} \widetilde{P}_{R,(c,T)},$$

whose bracket lands in

$$[\widetilde{P}_{R,(c,T)}, \widetilde{P}_{R,(c',T')}] \subset \widetilde{P}_{R,(c,T)\star(c',T')}.$$

The no-overcount residual of Definition 10.54 is required to vanish, so these labels refine rather than duplicate the c-graded primitive source. The completed object is the inverse limit of these finite normal-ordered stages.

(P) *(PTVV host.)* The protected primitive layer $\widetilde{P}_X = \bigoplus_c \widetilde{P}_{X,c}$ is realized as $\widetilde{P}_X = H^0\,\mathrm{Prim}_{\mathrm{prot}}(\mathcal{F}_X)$ for an oriented critical Hall/factorization object $\mathcal{F}_X$ on a (-1)-shifted symplectic derived moduli stack $\mathfrak{M}_{X,c}$ of compactly supported B-branes on $K3 \times E$ in the sense of Pantev–Toen–Vaquie–Vezzosi [69] (cf. the local schema of Definition 2.1), graded by supplied algebraic/effective charge $c \in \Gamma^{\mathrm{eff}}_{X,\sigma} \subset \Gamma^{\mathrm{alg}}_X \subset \Gamma^{\mathrm{form}}_X$, with reduced E-quotient compatible with the projection $p_E : K3 \times E \to E$ of Definition 5.4. The factorization product on $\mathcal{F}_X$ is A_∞ (homotopy E_3): the BBDJS [11] Thom–Sebastiani isomorphism on disjoint d-critical charts is a derived natural isomorphism, and downstream consumption acts on the $H^0\,\mathrm{Prim}_{\mathrm{prot}}$ truncation, on which the bracket descends strictly. Strict E_3 rectification of the protected chain-level product is a separate residual obligation: it would require coherent strictification of the Massey–Saito/BBDJS Thom–Sebastiani product together with the multiplicative orientation cocycle. It is not used by the denominator identity or the primitive-recognition theorem, which consume only the strict H^0-level Lie superalgebra. The binary Hall operations are represented, before the reduced quotient, by correspondences

$$\mathfrak{M}_{X,c} \times \mathfrak{M}_{X,c'} \xleftarrow{p_{c,c'}} \mathfrak{E}_{c,c'} \xrightarrow{q_{c,c'}} \mathfrak{M}_{X,c+c'},$$

and the two parenthesizations of a triple product are compared on the two-step flag stack $\mathfrak{Flag}_{a,b,c}$. These correspondences are part of the input; the Borcherds product does not construct them.

(E) *(Reduced E-quotient and completion.)* The Hall correspondences, orientation lines, Hopf pairing, and central translation functors T_η are constructed before quotienting by translations of E, then descend to the cosection-reduced E-quotient and to the sectorial HN inverse system. In particular T_η commutes with the reduced E-quotient, with HN completion, and with the local-local, wrapped-wrapped, and mixed local/wrapped extension correspondences of Open Problem 10.33. The completed primitive object is the inverse limit of finite stages:

$$P^{\Pi}_X = \varprojlim_R P^{\Pi}_R, \qquad P^{\Pi}_X \widehat{\otimes} P^{\Pi}_X = \varprojlim_R (P^{\Pi}_R \otimes P^{\Pi}_R).$$

The maps m_R, $[\,\cdot\,,\,\cdot\,]_R$, Δ_R, $\langle\,\cdot\,,\,\cdot\,\rangle_R$, and $\overline{\Pi}^{\Theta}_{R,*}$ commute with all transition maps. Thus every product over a fibre of $\overline{\Pi}_X$, and every completed tensor product used by the Hopf structure, is the inverse limit of finite operations. This is a hypothesis; it is not a formal consequence of the lattice homomorphism $\overline{\Pi}_X$.

(O) *(S^1-equivariant multiplicative orientation cocycle.)* The collection $\{\mu_{c,c'}\}$ is an S^1-equivariant multiplicative orientation cocycle on (-1)-shifted symplectic stacks satisfying the strict pentagon. Concretely: an orientation datum $\{o_c\}_{c \in \Gamma^{\mathrm{eff}}_{X,\sigma}}$ in the sense of Definition 4.3 is supplied with *Joyce–Upmeier multiplicative isomorphisms* [46]

$$\mu_{c,c'} \;:\; o_{c+c'} \xrightarrow{\sim} o_c \otimes o_{c'}, \qquad c, c' \in \Gamma^{\mathrm{eff}}_{X,\sigma},$$

satisfying the strict cocycle pentagon $\mu_{c+c',c''} \circ \mu_{c,c'} = \mu_{c,c'+c''} \circ \mu_{c',c''}$ on the Cartesian extension correspondence, compatible with the reduced E-quotient and with the HN/sectorial completion of Lemma 10.53, and *equivariant under the loop-rotation S^1-action on the cyclic complex of* $\mathfrak{M}_{X,c}$. This last condition lifts the orientation cocycle from a Hochschild trivialization to a *cyclic* trivialization, and is exactly the kind of orientation technology Joyce–Upmeier [46] supply once the reduced compact sector and its cyclic model have been constructed. The S^1-equivariance is not inferred from the scalar determinant; it is an input on the Hall side. Equivalently, the collection

$\{o_c\}$ is an S^1-equivariant multiplicative section of the determinant gerbe over the moduli of objects in the protected sector.

(Θ) *(Cocycle-corrected degree shifts.)* A coherent family of central degree-shift data is supplied on the primitive bracket targets in the coefficient dg category of Definition 10.35. The chain representative of the Gram defect is the normalized translation-valued Hochschild 2-cocycle

$$B_{\mathrm{ch}}(c, c') := \mathsf{T}_{B(c,c')} \in Z^2_{\mathrm{Hoch}}(\Gamma_X^{\mathrm{phys}}; \mathsf{Pic}^{\mathrm{dg}}_{\Gamma_{\mathrm{gram}}}).$$

A normalized 1-cochain

$$\Theta_\Pi \in C^1_{\mathrm{Hoch}}(\Gamma_X^{\mathrm{phys}}; \mathsf{Pic}^{\mathrm{dg}}_{\Gamma_{\mathrm{gram}}}), \qquad \Theta_\Pi(c) = \mathsf{T}_{-\Pi_X(c)}$$

is supplied with coherent natural quasi-isomorphisms

$$d_{\mathrm{Hoch}}\Theta_\Pi = B_{\mathrm{ch}}.$$

The formula for Θ_Π is the formal central translation cochain. The hypothesis is stronger: these translations must act on the actual Hall correspondence targets and must be compatible with the orientation, reduced quotient, cyclic model, and finite HN transitions. Equivalently, for every pair c, c' the raw bracket target of Gram degree $\Pi_X(c + c')$ is translated by $\mathsf{T}_{B(c,c')}$ to the additive Gram degree $\Pi_X(c) + \Pi_X(c')$. For triples a, b, c, the two composites over $\mathfrak{Flag}_{a,b,c}$,

$$(\vartheta_{a,b} \otimes 1) \circ \vartheta_{a+b,c}, \qquad (1 \otimes \vartheta_{b,c}) \circ \vartheta_{a,b+c},$$

agree after the Hall associator, Thom–Sebastiani transport, and orientation signs. Their quotient in $\pi_1\mathsf{Pic}^{\mathrm{dg}}_{\Gamma_{\mathrm{gram}}}$ is the triple obstruction

$$\mathrm{Pent}_\Theta(a, b, c) := \big[(\vartheta_{a,b} \otimes 1) \circ \vartheta_{a+b,c}\big] - \big[(1 \otimes \vartheta_{b,c}) \circ \vartheta_{a,b+c}\big],$$

with the Hall associator and orientation transport inserted in the middle. Hypothesis (Θ) requires $\mathrm{Pent}_\Theta(a, b, c) = 0$ for every two-step Hall flag. On degrees this is exactly

$$B(a, b) + B(a + b, c) = B(b, c) + B(a, b + c),$$

i.e. $d_{\mathrm{Hoch}}B_{\mathrm{ch}} = 0$, but the displayed equality is required as a chain homotopy, not only as a lattice identity.

(F) *(Frobenius pairing.)* The protected Hall bracket $[\,\cdot\,,\,\cdot\,]$ and the protected Hopf pairing $\langle\,\cdot\,,\,\cdot\,\rangle$ on $\widetilde{P}_X$ (the latter furnished by the squaring isomorphism $o_c^{\otimes 2} \simeq \det \mathrm{RHom}(C, C)$ of Definition 4.3 and the PTVV [69] (-1)-shifted symplectic form on $\mathfrak{M}_{X,c}$, trivialized along the reduced E-quotient via the Costello–Li flat Calabi–Yau pairing [18, 21, 22]) satisfy the graded-Frobenius identity

$$\langle [x, y], z\rangle + (-1)^{|x|\cdot|y|}\langle y, [x, z]\rangle = 0$$

for homogeneous $x, y, z \in \widetilde{P}_X$ of additive formal charges c_x, c_y, c_z with $c_x + c_y + c_z = 0$. After applying $\overline{\Pi}^{\Theta}_{X,*}$ the same identity is imposed for the transported bracket $[\,\cdot\,,\,\cdot\,]_\Theta$.

(Cyc) *(Negative-cyclic lift.)* The Hochschild cochain Θ_Π has an S^1-equivariant lift Θ_Π^- in the negative-cyclic complex of the protected Hall sector, and the lifted equation

$$(b + uB_{\mathrm{C}})\Theta_\Pi^- = B_{\mathrm{ch}}^-$$

holds, where B_{C} is the Connes operator and B_{ch}^- is a negative-cyclic representative whose forgetful image in Hochschild cochains is B_{ch}. The symbol B_{C} is unrelated to the Gram-defect cocycle B; the former is the mixed-complex operator and the latter is the symmetric bilinear normal-ordering defect $(c, c') \mapsto (Q \cdot Q', Q \cdot P' + Q' \cdot P, P \cdot P')$. The forgetful map sends the cyclic equation to the Hochschild equation in (Θ), so the cyclic lift is a refinement, not a replacement, of normal ordering.

Then, writing $P_X^\Pi := \overline{\Pi}_{X,*}^\Theta \widetilde{P}_X$, for finite normal-ordered labels (c, T), (c', T') whose summands are supplied, the finite-stage normal-ordered lift of the raw physical-charge bracket

$$[\widetilde{P}_{R,(c,T)}, \widetilde{P}_{R,(c',T')}] \subset \widetilde{P}_{R,(c,T)\star(c',T')}$$

induces, after passage to the compatible inverse limit, a cocycle-corrected Gram-graded bracket

$$[\,\cdot\,,\,\cdot\,]_\Theta : (P_X^\Pi)_{\overline{\Pi}_X(c,T)} \otimes (P_X^\Pi)_{\overline{\Pi}_X(c',T')} \longrightarrow (P_X^\Pi)_{\overline{\Pi}_X(c,T)+\overline{\Pi}_X(c',T')}.$$

Thus B is not erased; it is the coherent degree-shift by which the raw target $\Pi_X(c + c') = \Pi_X(c) + \Pi_X(c') + B(c, c')$ is transported to the additive normal-ordered Gram target $\overline{\Pi}_X((c, T) \star (c', T'))$. This object

$$P_X^\Pi$$

is therefore a Γ_{gram}-graded Lie superalgebra at the stated H^0 level, and hypothesis (1b) of Theorem 10.66 is satisfied with the Θ-corrected descent. This conclusion asserts what follows after the source Hall stages, finite transition compatibilities, Frobenius and cyclic lifts, and radical ideal/coideal stability have been supplied; it is not a construction of those source data.

If the finite-fibre HN topology of (E), (Θ), its triple coherence, (F), (Cyc), and radical ideal/coideal stability are not supplied, the minimal computable obstruction is the finite-HN inverse system of classes

$$\mathfrak{o}_\Theta^{\mathrm{top}} := \left\{\mathfrak{o}_{\Theta,R'R}^{\mathrm{fib}}, \mathfrak{o}_{\Theta,R'R}^{m}, \mathfrak{o}_{\Theta,R'R}^{\Delta}, \mathfrak{o}_{\Theta,R'R}^{\langle\,,\,\rangle}\right\}_{R' \geq R},$$

where $\rho_{R'R}$ is the source transition, $\rho_{R'R}^\Pi$ is the descended primitive transition, and these are the finite Hom-space defects

$$\mathfrak{o}_{\Theta,R'R}^{\mathrm{fib}} := \rho_{R'R}^\Pi \overline{\Pi}_{R',*}^{\Theta_{R'}} - \overline{\Pi}_{R,*}^{\Theta_R} \rho_{R'R},$$

$$\mathfrak{o}_{\Theta,R'R}^{m} := \rho_{R'R}^\Pi m_{R'}^\Theta - m_R^\Theta(\rho_{R'R}^\Pi \otimes \rho_{R'R}^\Pi), \qquad \mathfrak{o}_{\Theta,R'R}^{\Delta} := (\rho_{R'R}^\Pi \otimes \rho_{R'R}^\Pi)\Delta_{R'}^\Theta - \Delta_R^\Theta \rho_{R'R}^\Pi,$$

$$\mathfrak{o}_{\Theta,R'R}^{\langle\,,\,\rangle} := \langle\,\cdot\,,\,\cdot\,\rangle_R(\rho_{R'R}^\Pi \otimes \rho_{R'R}^\Pi) - \langle\,\cdot\,,\,\cdot\,\rangle_{R'},$$

$$\mathfrak{o}_\Theta^{\mathrm{Hoch}} := [B_{\mathrm{ch}} - d_{\mathrm{Hoch}}\Theta_\Pi] \in HH^2_{\mathrm{red},E}(\widetilde{P}_X^{\mathrm{br}}; \mathsf{Pic}^{\mathrm{dg}}_{\Gamma_{\mathrm{gram}}}),$$

$$\mathfrak{o}_\Theta^{\mathrm{tri}} := [\mathrm{Pent}_\Theta] \in HH^3_{\mathrm{red},E}(\widetilde{P}_X^{\mathrm{br}}; \pi_1\mathsf{Pic}^{\mathrm{dg}}_{\Gamma_{\mathrm{gram}}}),$$

$$\mathfrak{o}_\Theta^{\mathrm{Jac}} := [\mathrm{Jac}_\Theta] \in C^3_{\mathrm{CE,red},E}(\overline{\Pi}_{X,*}^\Theta \widetilde{P}_X; \overline{\Pi}_{X,*}^\Theta \widetilde{P}_X),$$

$$\mathfrak{v}_F := \big[(x,y,z) \mapsto \langle [x,y]_\Theta, z\rangle + (-1)^{|x||y|}\langle y, [x,z]_\Theta\rangle\big] \in C^3_{\mathrm{cyc,red},E}\big(\overline{\Pi}^{\Theta}_{X,*}\widetilde{P}_X; k\big),$$

$$\mathfrak{v}^{\mathrm{cyc}}_\Theta := [B^-_{\mathrm{ch}} - (b + uB_{\mathrm{C}})\Theta^-_\Pi] \in HC^{-2}_{\mathrm{red},E}\big(\widetilde{P}^{\mathrm{br}}_X; \mathsf{Pic}^{\mathrm{dg}}_{\Gamma_{\mathrm{gram}}}\big),$$

$$\mathfrak{v}^{\mathrm{rad}}_\Theta := \big(\mathfrak{v}^{\mathrm{rad,Lie}}_\Theta, \mathfrak{v}^{\mathrm{rad,coid}}_\Theta\big),$$

defined on the HN-completed reduced E-quotient of the protected bracket category, where

$$P^\Pi_X = \overline{\Pi}^{\Theta}_{X,*}\widetilde{P}_X, \qquad \mathrm{Rad}^\Pi_X = \{r \in P^\Pi_X \mid \langle r, w\rangle = 0 \text{ for all } w \in P^\Pi_X\}.$$

The two components are

$$\mathfrak{v}^{\mathrm{rad,Lie}}_\Theta := \big[(x,r,z) \mapsto \langle [x,r]_\Theta, z\rangle\big] \in \mathrm{Hom}_{\mathrm{red},E}\big(P^\Pi_X \otimes \mathrm{Rad}^\Pi_X \otimes P^\Pi_X, k\big),$$

and, for the quotient map $\bar{q}_\Pi : P^\Pi_X \to P^\Pi_X/\overline{\mathrm{Rad}^\Pi_X}$,

$$\mathfrak{v}^{\mathrm{rad,coid}}_\Theta(r) := (\bar{q}_\Pi \widehat{\otimes} \bar{q}_\Pi)\Delta_\Theta(r), \qquad r \in \mathrm{Rad}^\Pi_X\,.$$

Here Δ_Θ is the descended completed Hopf coproduct. The condition $\mathfrak{v}^{\mathrm{rad,coid}}_\Theta = 0$ says exactly that

$$\Delta_\Theta(\mathrm{Rad}^\Pi_X) \subset \overline{\mathrm{Rad}^\Pi_X \widehat{\otimes} P^\Pi_X + P^\Pi_X \widehat{\otimes} \mathrm{Rad}^\Pi_X}.$$

Here $\mathfrak{v}^{\mathrm{top}}_\Theta$ is the assertion that $\overline{\Pi}^{\Theta}_*$, the transported product, the transported coproduct, and the pairing are strict morphisms of the finite HN inverse system; its vanishing is what makes the completed fibre products and $\widehat{\otimes}$-operations well defined. The class $\mathrm{Pent}_\Theta(a,b,c)$ is the difference of the two triple composites over $\mathfrak{Flag}_{a,b,c}$, and Jac_Θ is the signed Jacobiator of the transported bracket. The Hochschild shadow is computed on homogeneous charges by the explicit formula for $B(c,c')$; the triple shadow is the displayed cocycle identity for B, hence zero at the lattice level; the cyclic component is computed by applying B_{C} to the chosen cyclic orientation representative. The class $\mathfrak{v}^{\mathrm{rad}}_\Theta$ vanishes exactly when the Hopf radical is both a Lie ideal for the transported bracket and a coideal for the completed descended coproduct. Closedness of the radical in the HN topology is still a separate topological hypothesis.

Proof. Step 1 (The coefficient category). The coefficient system for normal ordering is the Picard dg groupoid of central translations of $\mathsf{Ch}^{\Gamma_{\mathrm{gram}}}_{k,\widehat{\mathrm{HN}},E}$, as in Definition 10.35. That definition constructs $\overline{\Pi}^{\Theta}_{X,*}$ as the inverse limit of the finite fibre products $(\overline{\Pi}^{\Theta}_{R,*}V_R)_\gamma$. The finite-transition part of hypothesis (E), encoded by $\mathfrak{v}^{\mathrm{top}}_\Theta = 0$, is what permits the transported product, coproduct, and pairing to pass from finite stages to the completed HN object. With the convention $(\mathsf{T}_\eta V)_\gamma = V_{\gamma+\eta}$, applying $\mathsf{T}_{B(c,c')}$ to a raw target of degree $\Pi_X(c+c')$ displays it in degree

$$\Pi_X(c+c') - B(c,c') = \Pi_X(c) + \Pi_X(c').$$

Thus $B_{\mathrm{ch}}(c,c') = \mathsf{T}_{B(c,c')}$ is the exact translation-valued Hochschild 2-cocycle measuring failure of raw Gram additivity.

Step 2 (The boundary equation). For the standard additive Hochschild differential on normalized cochains,

$$(d_{\mathrm{Hoch}}\Theta_\Pi)(c,c') = \Theta_\Pi(c) + \Theta_\Pi(c') - \Theta_\Pi(c+c').$$

The underlying degree of the translation $\Theta_\Pi(c) = \mathsf{T}_{-\Pi_X(c)}$ gives

$$d_{\mathrm{Hoch}}\Theta_\Pi(c,c') = -\Pi_X(c) - \Pi_X(c') + \Pi_X(c+c') = B(c,c').$$

Hypothesis (Θ) is precisely the statement that this arithmetic boundary is represented by actual central translation quasi-isomorphisms on the protected Hall bracket targets.

Step 3 (Triples, Jacobi, and Frobenius). For three charges a, b, c, the two ways of translating the target of a two-step bracket differ by

$$B(a,b) + B(a+b,c) - B(b,c) - B(a,b+c) = -d_{\mathrm{Hoch}}B(a,b,c) = 0$$

by Lemma 6.9. The supplied pentagon homotopies in (Θ) lift this equality from degrees to the Hall chain category: the obstruction is precisely $\mathrm{Pent}_\Theta(a,b,c)$. If it vanishes, the normal-ordered product is associative on H^0, and the supercommutator bracket inherits the Jacobi identity from the upstairs Hall bracket. The corresponding signed Jacobiator is

$$\mathrm{Jac}_\Theta(x,y,z) = [x,[y,z]_\Theta]_\Theta - [[x,y]_\Theta,z]_\Theta - (-1)^{|x||y|}[y,[x,z]_\Theta]_\Theta;$$

its degree shadow is zero by the same cocycle identity, and its chain class is the image of the Hall associator and Pent_Θ. Hypothesis (F) is then the separate vanishing of

$$\mathfrak{o}_F(x,y,z) = \langle [x,y]_\Theta, z\rangle + (-1)^{|x||y|}\langle y, [x,z]_\Theta\rangle.$$

It is not implied by the degree calculation.

Step 4 (Orientation, reduced quotient, and radical quotient). Hypothesis (O) transports the orientation lines through the same extension correspondences used by the Hall product; hypothesis (E) asserts that the translation action, orientation transport, and mixed local/wrapped correspondences descend through the reduced E-quotient and HN completion. Let Rad_X^Π be the radical of the descended Hopf pairing. The exact obstruction to Lie-ideal stability is

$$\mathfrak{o}_\Theta^{\mathrm{rad,Lie}}(x,r,z) = \langle [x,r]_\Theta, z\rangle, \qquad r \in \mathrm{Rad}_X^\Pi .$$

If (F) holds, this expression equals $-(-1)^{|x||r|}\langle r, [x,z]_\Theta\rangle = 0$, so the radical is a Lie ideal. This does not prove coideal stability. Let $\bar{q}_\Pi : P_X^\Pi \to P_X^\Pi/\overline{\mathrm{Rad}_X^\Pi}$. The exact coideal obstruction is

$$\mathfrak{o}_\Theta^{\mathrm{rad,coid}}(r) = (\bar{q}_\Pi \widehat{\otimes} \bar{q}_\Pi)\Delta_\Theta(r).$$

Its vanishing is equivalent to $\Delta_\Theta(r)$ lying in the completed sum

$$\overline{\mathrm{Rad}_X^\Pi \widehat{\otimes} P_X^\Pi + P_X^\Pi \widehat{\otimes} \mathrm{Rad}_X^\Pi}.$$

Closedness of Rad_X^Π in the Hall topology remains a separate input. Only after ideal, coideal, and closedness conditions does the quotient carry the same Γ_{gram}-graded Hopf–Lie structure.

Step 5 (Cyclic lift and obstruction). The Hochschild construction controls the displayed H^0 bracket. For the full chain-level descent, the degree-shift cochain must be fixed by loop rotation. This is exactly hypothesis (Cyc): $(b + uB_{\mathrm{C}})\Theta_\Pi^- = B_{\mathrm{ch}}^-$. If such a negative-cyclic lift is absent, or if the finite-HN topology, triple/Jacobi/Frobenius/radical coherence fails, the seven classes displayed in the theorem are the obstruction. They are minimal for this model: vanishing of $\mathfrak{o}_\Theta^{\mathrm{top}}$ is equivalent to the finite-fibre

descent and completed tensor operations commuting with the HN inverse system; vanishing of $\mathfrak{o}^{\mathrm{Hoch}}_{\Theta}$ is equivalent to existence of the degree-shift trivialization on bracket targets; vanishing of $\mathfrak{o}^{\mathrm{tri}}_{\Theta}$ is equivalent to compatibility of that trivialization with the two-step Hall flag correspondences; vanishing of $\mathfrak{o}^{\mathrm{Jac}}_{\Theta}$ is the Jacobi compatibility of the transported bracket; vanishing of $\mathfrak{o}_F$ is the Frobenius/Hopf invariance of the transported bracket; vanishing of $\mathfrak{o}^{\mathrm{cyc}}_{\Theta}$ is equivalent to choosing the trivialization compatibly with the S^1-equivariant orientation data; and vanishing of $\mathfrak{o}^{\mathrm{rad}}_{\Theta}$, together with closedness in the HN topology, is the condition for passage to the Hopf radical quotient. □

Proposition 10.40 *(Finite verification diagrams for normal ordering).* Fix a successor arrow $R^+ \to R$ in the cofinal HN subsystem of Definition 10.54. Let $P^{\Pi}_R = \overline{\Pi}^{\Theta_R}_{R,*}\widetilde{P}_R$ be a finite normal-ordered primitive stage, with transition $\rho^{\Pi}_{R^+R} : P^{\Pi}_{R^+} \to P^{\Pi}_R$. A supplied finite system kills the seven normal-ordering obstructions of Theorem 10.39 at this successor arrow exactly when the following finite maps commute, or are supplied with the displayed null-homotopies, in the indicated finite Hom complexes.

(NO1) Finite fibres.

$$\begin{tikzcd} V_{R^+} \arrow[r, "\overline{\Pi}^{\Theta_{R^+}}_{R^+,*}"] \arrow[d, "\rho_{R^+R}"'] & \overline{\Pi}^{\Theta_{R^+}}_{R^+,*}V_{R^+} \arrow[d, "\rho^{\Pi}_{R^+R}"] \\ V_R \arrow[r, "\overline{\Pi}^{\Theta_R}_{R,*}"'] & \overline{\Pi}^{\Theta_R}_{R,*}V_R. \end{tikzcd}$$

This is $\mathfrak{o}^{\mathrm{fib}}_{\Theta,R^+R} = 0$; it says that fibre products over $\overline{\Pi}_X$ are finite before completion and transition-compatible before the inverse limit is taken.

(NO2) Product, coproduct, pairing.

$$\begin{tikzcd} P^{\Pi}_{R^+} \otimes P^{\Pi}_{R^+} \arrow[r, "m^{\Theta}_{R^+}"] \arrow[d, "\rho^{\Pi}_{R^+R}\otimes\rho^{\Pi}_{R^+R}"'] & P^{\Pi}_{R^+} \arrow[d, "\rho^{\Pi}_{R^+R}"] \\ P^{\Pi}_R \otimes P^{\Pi}_R \arrow[r, "m^{\Theta}_R"'] & P^{\Pi}_R \end{tikzcd} \qquad \begin{tikzcd} P^{\Pi}_{R^+} \arrow[r, "\Delta^{\Theta}_{R^+}"] \arrow[d, "\rho^{\Pi}_{R^+R}"'] & P^{\Pi}_{R^+} \otimes P^{\Pi}_{R^+} \arrow[d, "\rho^{\Pi}_{R^+R}\otimes\rho^{\Pi}_{R^+R}"] \\ P^{\Pi}_R \arrow[r, "\Delta^{\Theta}_R"'] & P^{\Pi}_R \otimes P^{\Pi}_R. \end{tikzcd}$$

Together with

$$\langle \rho^{\Pi}_{R^+R}x, \rho^{\Pi}_{R^+R}y\rangle_R = \langle x, y\rangle_{R^+},$$

these are $\mathfrak{o}^{m}_{\Theta,R^+R} = \mathfrak{o}^{\Delta}_{\Theta,R^+R} = \mathfrak{o}^{\langle\,,\,\rangle}_{\Theta,R^+R} = 0$.

(NO3) Hochschild and negative-cyclic equations. For every finite physical charge pair c, c' whose normal-ordered channels occur in $\widehat{\Gamma}_{R^+}$, the physical cochain has the finite equations

$$d_{\mathrm{Hoch}}\Theta^{\mathrm{phys}}_{\Pi,R}(c,c') = B_{\mathrm{ch},R}(c,c'), \qquad (b + uB_{\mathrm{C}})\Theta^{-}_{\Pi,R} = B^{-}_{\mathrm{ch},R}.$$

Equivalently, on a normal-ordered label (c, T) one may use

$$\widehat{\Theta}_{\Pi,R}(c,T) = \mathsf{T}_{T-\Pi_X(c)}, \qquad d_{\mathrm{Hoch}}\widehat{\Theta}_{\Pi,R} = 0,$$

and not mix this domain with the physical $B_{\mathrm{ch},R}$-cochain. They kill $\mathfrak{o}^{\mathrm{Hoch}}_{\Theta,R}$ and $\mathfrak{o}^{\mathrm{cyc}}_{\Theta,R}$. The second equation is imposed in the negative-cyclic complex; it is not obtained by forgetting to Hochschild cochains.

(NO4) Triple flag. For every two-step finite Hall flag over $\mathfrak{Flag}_{a,b,c,R}$, the two translated products must agree after inserting the Hall associator and orientation transport:

$$(\mathsf{A}_{a,b,c} \circ (\vartheta_{a,b} \otimes 1) \circ \vartheta_{a+b,c}) - ((1 \otimes \vartheta_{b,c}) \circ \vartheta_{a,b+c}) = 0.$$

This is the finite representative of $\mathrm{Pent}_\Theta(a, b, c)$. Its degree shadow is the bilinear identity for B; the chain equality is additional data.

(NO5) Jacobi. The transported bracket must have zero finite Jacobiator

$$\mathrm{Jac}_{\Theta,R}(x, y, z) = 0$$

for all homogeneous $x, y, z \in P_R^\Pi$.

(NO6) Frobenius. The finite Frobenius defect

$$\mathfrak{o}_{F,R}(x, y, z) = \langle [x, y]_\Theta, z\rangle_R + (-1)^{|x||y|}\langle y, [x, z]_\Theta\rangle_R$$

must vanish.

(NO7) Hopf radical. Finally, for $\mathrm{Rad}_R^\Pi = \{r \in P_R^\Pi \mid \langle r, w\rangle_R = 0\ \forall w \in P_R^\Pi\}$ (equivalently the two-sided radical when the finite Hopf pairing is graded symmetric) and $\bar{q}_{\Pi,R} : P_R^\Pi \to P_R^\Pi/\mathrm{Rad}_R^\Pi$, one must have both

$$\langle [x, r]_\Theta, z\rangle_R = 0 \qquad (x, z \in P_R^\Pi,\ r \in \mathrm{Rad}_R^\Pi)$$

and the finite coideal square

$$\begin{tikzcd}
\mathrm{Rad}_R^\Pi \arrow[r, "\Delta_R^\Theta"] \arrow[d, "0"'] & P_R^\Pi \otimes P_R^\Pi \arrow[d, "\bar{q}_{\Pi,R}\otimes\bar{q}_{\Pi,R}"] \\
0 \arrow[r] & (P_R^\Pi/\mathrm{Rad}_R^\Pi) \otimes (P_R^\Pi/\mathrm{Rad}_R^\Pi).
\end{tikzcd}$$

These are the finite Lie-ideal and coideal parts of $\mathfrak{o}_{\Theta,R}^{\mathrm{rad}} = 0$. Closedness of the inverse limit radical is the additional HN topological assertion that the kernels Rad_R^Π form a transition-compatible closed subsystem. Equivalently, on any downward-saturated finite root window W, after choosing homogeneous bases of the finite primitive pieces, (NO7) is the following matrix computation. Let

$$G_\beta : P_{R,\beta}^\Pi \times P_{R,-\beta}^\Pi \to \mathbb{C}, \qquad K_\beta = \ker G_\beta \cap \ker G_{-\beta}^t, \qquad Q_\beta : P_{R,\beta}^\Pi \to P_{R,\beta}^\Pi/K_\beta$$

be the positive-negative pairing, two-sided radical block, and quotient map. Let $B_{\alpha,\beta}$ be the bracket matrix, let $M_{\alpha,\beta}$ be the product matrix, and let $D_\rho^{\mu,\nu}$ be the coproduct matrix $P_{R,\rho}^\Pi \to P_{R,\mu}^\Pi \otimes P_{R,\nu}^\Pi$. The finite test is:

$$\langle [x, y]_\Theta, z\rangle_R + (-1)^{|x||y|}\langle y, [x, z]_\Theta\rangle_R = 0,$$

for all retained triples; the Hopf adjointness matrices

$$\langle M_{\alpha,\beta}(x \otimes y), z\rangle_R = \langle x \otimes y, D_{\alpha+\beta}^{\alpha,\beta} z\rangle_R;$$

the quotient-tensor nondegeneracy of $\bar{G}_\mu \otimes \bar{G}_\nu$; and the direct containments

$$Q_{\alpha+\beta}B_{\alpha,\beta}(P_{R,\alpha}^\Pi \otimes K_\beta) = 0, \qquad Q_{\alpha+\beta}B_{\alpha,\beta}(K_\alpha \otimes P_{R,\beta}^\Pi) = 0, \qquad (Q_\mu \otimes Q_\nu)D_\rho^{\mu,\nu}K_\rho = 0,$$

together with transition preservation $\rho^{P}_{R^+R}(K_{R^+,\beta}) \subset K_{R,\beta}$. Frobenius gives the Lie-ideal half; Hopf adjointness plus quotient-tensor nondegeneracy gives the coideal half. The scalar determinant supplies none of these matrices.

Consequently, if (NO1)–(NO7) hold for every successor arrow and the non-successor transition maps are their composites, the inverse limit kills

$$\mathfrak{o}^{\mathrm{top}}_{\Theta}, \quad \mathfrak{o}^{\mathrm{Hoch}}_{\Theta}, \quad \mathfrak{o}^{\mathrm{tri}}_{\Theta}, \quad \mathfrak{o}^{\mathrm{Jac}}_{\Theta}, \quad \mathfrak{o}_{F}, \quad \mathfrak{o}^{\mathrm{cyc}}_{\Theta}, \quad \mathfrak{o}^{\mathrm{rad}}_{\Theta}.$$

If one displayed finite diagram fails, its homotopy class is the surviving obstruction. Neither non-degeneracy of the pairing nor the scalar Borcherds determinant removes (NO3), (NO6), or the coideal part of (NO7).

Lemma 10.41 *(Hopf–Frobenius datum and obstruction).* Adopt hypotheses (P) and (O) of Theorem 10.39, and grant the Hopf-pairing/radical hypothesis of Theorem 10.66: the protected Hopf pairing $\langle\,\cdot\,,\,\cdot\,\rangle$ on the protected primitive layer $\widetilde{P}_X = H^0\,\mathrm{Prim}_{\mathrm{prot}}(\mathcal{F}_X)$ is non-degenerate after the standard radical quotient. Non-degeneracy alone does not imply the graded Frobenius identity. The missing datum is the vanishing of the trilinear obstruction

$$\mathfrak{o}_F(x,y,z) := \langle [x,y], z\rangle + (-1)^{|x|\cdot|y|}\langle y, [x,z]\rangle, \qquad x \in \widetilde{P}_{X,c_x},\ y \in \widetilde{P}_{X,c_y},\ z \in \widetilde{P}_{X,c_z}, \quad c_x + c_y + c_z = 0,$$

viewed as a class

$$\mathfrak{o}_F \in C^3_{\mathrm{cyc}}\big(\widetilde{P}_X;\, k\big)$$

after the reduced E-quotient and radical quotient. Hypothesis (F) of Theorem 10.39 is equivalent to $\mathfrak{o}_F = 0$. It is supplied, for example, by the CY-3 Serre-functor criterion of Lemma 10.46.

Proof. Step 1 (Exact defect). For homogeneous x, y, z with $c_x + c_y + c_z = 0$, the displayed cochain $\mathfrak{o}_F(x,y,z)$ is precisely the left side of the graded-Frobenius identity. Hence hypothesis (F) is equivalent, by definition, to $\mathfrak{o}_F = 0$ after the reduced E-quotient and the chosen radical quotient. No determinant or non-degeneracy statement alone forces this vanishing.

Step 2 (What non-degeneracy detects). After quotienting by the radical, non-degeneracy identifies a bilinear operation $D(x,y)$ with the functional $z \mapsto \langle D(x,y), z\rangle$. Thus if the Frobenius defect is known to vanish after pairing against every z, non-degeneracy forces the corresponding operation-level defect to vanish in the quotient. This is detection, not construction: the cyclic or Serre-functor input must still supply the equality being detected.

Step 3 (Radical consequence). Let

$$\mathrm{Rad}^{\Pi}_X = \{r \mid \langle r, w\rangle = 0 \text{ for all } w\}.$$

If $\mathfrak{o}_F = 0$, then for every $r \in \mathrm{Rad}^{\Pi}_X$

$$\langle [x,r], z\rangle = -(-1)^{|x||r|}\langle r, [x,z]\rangle = 0.$$

Therefore Rad^{Π}_X is a Lie ideal. The assertion that it is closed and is also a coideal is the separate Hopf-radical hypothesis in Theorem 10.66. Equivalently, the coproduct must separately satisfy

$$(\bar{q}_{\Pi}\widehat{\otimes}\bar{q}_{\Pi})\Delta_{\Theta}(r) = 0 \qquad (r \in \mathrm{Rad}^{\Pi}_X),$$

with $\bar{q}_{\Pi}$ the quotient by the closed radical. Lemma 10.46 is one source for $\mathfrak{o}_F = 0$; it is not a source for coideal stability unless the cyclic/Serre datum is also proved compatible with the Hopf coproduct. □

Lemma 10.42 *(Finite Hopf-adjointness radical test).* Let P_R^Π be a finite normal-ordered primitive stage with transported product m_R^Θ, coproduct Δ_R^Θ, and graded symmetric Hopf pairing $\langle\,\cdot\,,\,\cdot\,\rangle_R$. Let

$$\mathrm{Rad}_R^\Pi = \{r \mid \langle r, w\rangle_R = \langle w, r\rangle_R = 0 \text{ for all } w \in P_R^\Pi\}.$$

Assume:

(H1) m_R^Θ and Δ_R^Θ are adjoint for the finite Hopf pairing:

$$\langle m_R^\Theta(x, y), z\rangle_R = \langle x \otimes y, \Delta_R^\Theta z\rangle_R \qquad (x, y, z \in P_R^\Pi),$$

with the tensor pairing induced from $\langle\,\cdot\,,\,\cdot\,\rangle_R$.

(H2) The induced tensor pairing on

$$(P_R^\Pi/\mathrm{Rad}_R^\Pi) \otimes (P_R^\Pi/\mathrm{Rad}_R^\Pi)$$

is non-degenerate.

Together with the supplied finite Hopf pairing, coproduct, and radical matrices, this is the algebraic finite-stage test for (NO7). Then the finite coideal square in (NO7) commutes:

$$(\bar{q}_{\Pi,R} \otimes \bar{q}_{\Pi,R})\Delta_R^\Theta(r) = 0 \qquad (r \in \mathrm{Rad}_R^\Pi).$$

If, in addition, the finite Frobenius defect $\mathfrak{o}_{F,R}$ vanishes, then Rad_R^Π is also a Lie ideal for the transported bracket. Thus finite (NO7) is killed by $\mathfrak{o}_{F,R} = 0$ plus Hopf adjointness and finite quotient non-degeneracy. The closed inverse-limit radical still requires the transition-compatible closedness condition of Theorem 10.66.

Proof. For $r \in \mathrm{Rad}_R^\Pi$ and $x, y \in P_R^\Pi$, Hopf adjointness gives

$$\left\langle x \otimes y, \Delta_R^\Theta(r)\right\rangle_R = \langle m_R^\Theta(x, y), r\rangle_R = 0,$$

because r lies in the right radical. Hence the class $(\bar{q}_{\Pi,R} \otimes \bar{q}_{\Pi,R})\Delta_R^\Theta(r)$ pairs to zero against every elementary tensor in the quotient. By finite non-degeneracy of the quotient tensor pairing, the class itself is zero. This is exactly the coideal square of (NO7).

If $\mathfrak{o}_{F,R} = 0$, then for $r \in \mathrm{Rad}_R^\Pi$

$$\langle [x, r]_\Theta, z\rangle_R = -(-1)^{|x||r|}\langle r, [x, z]_\Theta\rangle_R = 0$$

for every x, z, so $[x, r]_\Theta$ lies in the radical. This is the finite Lie-ideal half. □

Proposition 10.43 *(Negative-cyclic route to the normal-ordering cochain).* Assume, in addition to hypothesis (P) of Theorem 10.39, that the protected Hall sector has a cyclic model A_X^{prot} with mixed complex $(C_*(A_X^{\mathrm{prot}}), b, B_{\mathrm{C}})$, a cyclic class ϖ_X representing the PTVV form, and a Connes-contraction cochain $\iota_{\varpi_X}\Pi_X$ satisfying

$$(b + uB_{\mathrm{C}})(\iota_{\varpi_X}\Pi_X) = B_{\mathrm{ch}}^-,$$

where B_{ch}^- is a negative-cyclic lift whose Hochschild image is B_{ch}, and B_{ch} maps to the Gram-defect cocycle B under the Loday–Quillen–Tsygan comparison

$$HC^2_{\mathrm{strict}} \longrightarrow HH^2$$

[55], [76], [54, §2]. If the induced two-step flag homotopies have $\mathrm{Pent}_\Theta = 0$, then the cochain $\Theta_\Pi^- := \iota_{\varpi_X}\Pi_X$ supplies the negative-cyclic part of (Cyc) and the Hochschild part of hypothesis (Θ). It supplies none of the Frobenius identity, the finite-HN topology, or the Hopf-radical ideal/coideal condition.

Proof. After forgetting to Hochschild cochains, the displayed equation becomes $d_{\mathrm{Hoch}}\Theta_{\Pi} = B_{\mathrm{ch}}$, whose degree shadow is B, and the cocycle identity $\delta B = 0$ gives the degree shadow of the triple pentagon. The actual Hall flag coherence is still the vanishing of Pent_{Θ}. This route is a second construction of the Θ-datum once the cyclic model and the triple homotopies are available. It is not a consequence of (P) alone. It does not assert compatibility of fibre products over $\overline{\Pi}_X$ with the HN transitions, and it contains no information about the completed coproduct on a radical element. □

Remark 10.44 *(Comparison of the two proofs).* $[B] = [B_{\mathrm{ch}}]$ under the Loday–Quillen–Tsygan comparison $HC^*_{\mathrm{strict}} \to HH^*$ of [55], [76], [54, §2], with witnessing cochains Θ (PTVV–Joyce) and $\iota_{\varpi_X}\Pi_X$ (cyclic homology), whenever the cyclic model and comparison cochain are supplied.

Remark 10.45 *(Descent criterion).* Theorem 10.39 uses the PTVV host (P), the reduced quotient and HN completion (E), including finite-fibre compatibility of $\overline{\Pi}^{\Theta}_*$ and continuity of $\widehat{\otimes}$, the supplied S^1-equivariant multiplicative orientation cocycle (O), the degree-shift cochain Θ, the Frobenius pairing (F), the negative-cyclic lift (Cyc), and Hopf-radical stability. Lemma 10.37 constructs (O) only on a supplied projection-finite Kunneth-pure sector; it does not construct the full compact or wrapped reduced $K3 \times E$ protected sector. Thus the remaining obligations are geometric and chain-level: construct (P), (E), and (O) beyond the Kunneth-pure sector including the HN/higher-rank extension, prove that the supplied Θ identifies the raw Gram defect B with canonical degree shifts in the protected primitive bracket, kill $\mathfrak{o}^{\mathrm{top}}_{\Theta}$, $\mathfrak{o}_F$, and prove closed Hopf-radical coideal stability. The scalar Borcherds determinant and the denominator algebra do not discharge these obligations.

Lemma 10.46 *(Graded Frobenius identity from CY-3 Serre functor).* Suppose that the protected observable algebra $\mathcal{A}_X$ of Definition 2.1, after the reduced E-quotient and the Costello–Li flat trivialization [18, 21, 22] along the elliptic factor, is realized as a Calabi–Yau A_∞-category of dimension $d = 3$, i.e. it carries a Serre functor $S \simeq [3]$ and a non-degenerate cyclically invariant pairing

$$\langle\,\cdot\,,\,\cdot\,\rangle_S : \mathrm{Hom}_{\mathcal{A}_X}(X, Y) \otimes \mathrm{Hom}_{\mathcal{A}_X}(Y, X[3]) \longrightarrow k,$$

in the sense of Kontsevich–Soibelman [51] (cyclic A_∞-structure) and PTVV [69]/BBDJS [11] (orientation data realising the shift [3] on the moduli of objects of $\mathcal{A}_X$). Let $\widetilde{P}_X \subset H_*(\mathcal{A}_X)$ be the protected primitive subspace, equipped with the Hall bracket $[\,\cdot\,,\,\cdot\,]$ and the Hopf pairing $\langle\,\cdot\,,\,\cdot\,\rangle$ of Theorem 10.66. Suppose, moreover, that at every finite HN stage the primitive Hall bracket is represented by the antisymmetrisation of the finite product $m_{2,R}$, the cyclic trace descends through the reduced E-quotient, and the trace, pairing, and primitive projection commute with the transition maps. Then the graded Frobenius identity holds:

$$\langle [x, y], z\rangle + (-1)^{|x|\cdot|y|}\langle y, [x, z]\rangle = 0, \qquad x, y, z \in \widetilde{P}_X \text{ homogeneous},$$

i.e. hypothesis (F) of Theorem 10.39 is met.

Proof. Step 1 (Serre pairing and cyclic trace). A CY-d A_∞-category $\mathcal{A}_X$ is, by [51] (§ 10), an A_∞-category equipped with a quasi-isomorphism $S \xrightarrow{\sim} [d]$ of bimodules and a non-degenerate cyclically symmetric pairing $\langle\,\cdot\,,\,\cdot\,\rangle_S$ on $\mathrm{Hom}_{\mathcal{A}_X}(-,-)$, of total degree $-d$. Equivalently, there is a chain-level cyclic trace $\mathrm{tr}_S : \mathrm{Hom}_{\mathcal{A}_X}(X, X) \to k[-d]$ satisfying $\mathrm{tr}_S(f \circ g) = (-1)^{|f|\cdot|g|}\,\mathrm{tr}_S(g \circ f)$ for composable homogeneous f, g (loc. cit., Lemma 10.2.4). PTVV [69] produce the underlying $(-d)$-shifted symplectic form on $\mathfrak{M}_{X,c}$; BBDJS [11] extend the orientation/d-critical datum that realises $S \simeq [d]$ at chain level on the perfect-complex moduli. Costello–Li [18, 21, 22] flat trivialization along the elliptic factor reduces

$d = 4$ to $d = 3$, so the discussion takes $d = 3$ and the identification $\langle x, y\rangle = \mathrm{tr}_S(x \cdot y)$, where $\cdot$ is the cup product on the protected primitive subspace.

Step 2 (Frobenius from cyclic invariance). Specialise to homogeneous primitives $x, y, z \in \widetilde{P}_X$ and expand the Hall bracket as the graded commutator $[x, y] = x \cdot y - (-1)^{|x|\cdot|y|} y \cdot x$ (the bracket on primitives in a connected graded Hopf algebra is the antisymmetrised product, as part of the primitive Lie-superalgebra structure assumed in Theorem 10.66). Then

$$\begin{aligned}\langle [x, y], z\rangle &= \mathrm{tr}_S\big((x \cdot y - (-1)^{|x|\cdot|y|} y \cdot x) \cdot z\big)\\ &= \mathrm{tr}_S(x \cdot y \cdot z) - (-1)^{|x|\cdot|y|}\,\mathrm{tr}_S(y \cdot x \cdot z).\end{aligned}$$

Cyclic invariance of tr_S applied to the second term gives $\mathrm{tr}_S(y \cdot x \cdot z) = (-1)^{|y|\cdot(|x|+|z|)}\,\mathrm{tr}_S(x \cdot z \cdot y)$, and applied to the first gives $\mathrm{tr}_S(x \cdot y \cdot z) = (-1)^{|x|\cdot(|y|+|z|)}\,\mathrm{tr}_S(y \cdot z \cdot x)$. Substituting and re-collecting,

$$\langle [x, y], z\rangle = -(-1)^{|x|\cdot|y|}\,\mathrm{tr}_S\big(y \cdot (x \cdot z - (-1)^{|x|\cdot|z|} z \cdot x)\big) = -(-1)^{|x|\cdot|y|}\langle y, [x, z]\rangle,$$

which is (F).

Step 3 (Finite-stage chain-level lift). The cyclic trace tr_S is a strict invariant on the A_∞-homotopy category by [51], § 10.2; on d-critical thickenings it lifts to perfect-complex chain level by [11] (the cyclic structure on the vanishing-cycle dg algebra) and PTVV [69] (the shifted symplectic form descends to the cotangent complex of $\mathfrak{M}_{X,c}$). By the finite-stage hypotheses, this cyclic trace, the product, the pairing, and the primitive projector all descend through the reduced E-quotient and commute with HN transitions. The identity of Step 2 therefore holds at each finite R and passes to the completed protected primitive subspace $\widetilde{P}_X$. A cyclic trace on an auxiliary model would not prove the Frobenius identity for the Hall bracket unless these finite-stage compatibilities were supplied. □

Remark 10.47. Lemma 10.46 derives (F) from the CY-3 Serre functor of $\mathcal{A}_X$, parallel to the derivation through the squaring isomorphism $o_c^{\otimes 2} \simeq \det \mathrm{RHom}(C, C)$ of Definition 4.3 and the Hopf pairing of Theorem 10.66. Both derivations meet at $\langle\,\cdot\,,\,\cdot\,\rangle = \mathrm{tr}_S(\,\cdot\,\cdot\,\cdot\,)$, the BBDJS [11] cyclic upgrade of the PTVV [69] (-1)-shifted symplectic form, and produce the same sign convention $(-1)^{|x|\cdot|y|}$ on the Gram degree $c_x + c_y + c_z = 0$. It proves only $\mathfrak{v}_F = 0$; it does not prove $\mathfrak{v}_\Theta^{\mathrm{top}} = 0$, coideal stability of the Hopf radical, a no-extra-relations theorem, or PBW compatibility. It is therefore one input to the full Dirac–Igusa realization datum, not a substitute for it.

Proposition 10.48 *(Finite normal-ordering closure criterion).* Fix a successor arrow $R^+ \to R$. Suppose a finite Hall–Dirac atlas $\mathfrak{A}_R$ and its successor transition are supplied through the normal-ordering lane, and suppose the following finite data hold.

(N1) The coefficient dg categories, central translations, and degree-truncation transitions satisfy Lemma 10.36.

(N2) The Joyce–Kontsevich–Soibelman pull-push product, coproduct, and finite Hopf pairing are strictly compatible with the successor transition after Θ_Π-transport.

(N3) The S^1-equivariant lift $\Theta^-_{\Pi,R}$ satisfies

$$(b + uB_{\mathrm{C}})\Theta^-_{\Pi,R} = B^-_{\mathrm{ch},R},$$

and its Hochschild shadow is $d_{\mathrm{Hoch}}\Theta_{\Pi,R} = B_{\mathrm{ch},R}$.

(N4) The two translated products over every finite two-step Hall flag agree after inserting the Hall associator and Joyce–Upmeier orientation transport.

(N5) The transported bracket is the graded commutator of the finite product on H°, so the Jacobiator is the image of the associator and the Θ-pentagon.

(N6) The finite cyclic trace satisfies the CY-3 Serre-functor criterion of Lemma 10.46; equivalently $\mathfrak{o}_{F,R} = 0$.

(N7) The finite Hopf pairing is product-coproduct adjoint, its quotient tensor pairing is non-degenerate, and the radical kernels are preserved by the successor transition, as in Lemma 10.42.

Then the finite normal-ordering diagrams (NO1)–(NO7) of Proposition 10.40 commute at the arrow $R^+ \to R$. Hence the seven finite obstruction classes

$$\mathfrak{o}^{\mathrm{top}}_{\Theta,R^+R}, \quad \mathfrak{o}^{\mathrm{Hoch}}_{\Theta,R}, \quad \mathfrak{o}^{\mathrm{tri}}_{\Theta,R}, \quad \mathfrak{o}^{\mathrm{Jac}}_{\Theta,R}, \quad \mathfrak{o}_{F,R}, \quad \mathfrak{o}^{\mathrm{cyc}}_{\Theta,R}, \quad \mathfrak{o}^{\mathrm{rad}}_{\Theta,R}$$

vanish at that finite stage. Conversely, the first failed item among (N1)–(N7) is the corresponding obstruction component unless an independent finite null-homotopy for the same (NO) diagram is supplied.

Proof. By Lemma 10.36, (N1) kills the finite fibre square (NO1) and the Hochschild part of (NO3). Hypothesis (N2) is exactly the product, coproduct, and pairing square (NO2). Hypothesis (N3) kills the negative-cyclic part of (NO3), as in Proposition 10.43. Hypothesis (N4) is (NO4). Under (N5), associativity and (NO4) imply the finite Jacobi identity, so (NO5) vanishes. Hypothesis (N6) is exactly (NO6) by Lemma 10.46. Finally, Lemma 10.42 proves the coideal part of (NO7), while (NO6) proves the Lie-ideal part. Transition preservation of the finite radical kernels is the finite closedness condition needed before passing to the inverse limit. □

Remark 10.49 *(Unitarity).* This paper uses the protected index, the virtual determinant, and the Borcherds denominator algebra. A holomorphic E_3-factorization algebra in cochains is part of the open Dirac–Igusa realization datum, not a positive Hilbert-space construction already in hand. Negative coefficients $f(nm, l)$ record an odd excess in a protected supertrace or in a K_0-representative; actual parity dimensions require an additional Hilbert-space or denominator-algebra parity datum. If the untwisted compactification is unitary, the statement that survives the twist is right-moving cohomology, protected integration, and the signed index. Reflection positivity is not a hypothesis of the Borcherds denominator theorem.

Lemma 10.50 *(Cyclotomic-spectrum construction of the multiplicative orientation cocycle; alternative source for hypothesis (O) of Theorem* 10.39*).* Let $\mathcal{F}_X$ be the protected observable algebra of Definition 2.1, graded by supplied algebraic/effective charge $c \in \Gamma^{\mathrm{eff}}_{X,\sigma} \subset \Gamma^{\mathrm{alg}}_X \subset \Gamma^{\mathrm{form}}_X$, and let $\{o_c\}_{c\in\Gamma^{\mathrm{eff}}_{X,\sigma}}$ be the orientation lines of Definition 4.3. Suppose:

(C1) *(Cyclotomic host.)* Each charged sector $\mathcal{F}_{X,c}$ is realised as a connective cyclotomic spectrum in the sense of Nikolaus–Scholze [62]; the p-typical cyclotomic structure map $\varphi_p : \mathcal{F}_{X,c} \to (\mathcal{F}_{X,c})^{tC_p}$ is the Frobenius inherited from the chiral product on $\mathrm{Ran}(E)$. Equivalently, the underlying spectrum $HH(\mathcal{F}_{X,c})$ carries the natural S^1-action by the Connes operator B of Goodwillie [30], with genuine S^1-fixed points $HC^-(\mathcal{F}_{X,c}) := HH(\mathcal{F}_{X,c})^{hS^1}$.

(C2) *(Determinant track.)* The orientation line o_c lifts to a unit $\widehat{o}_c \in \pi_0 HC^-(\mathcal{F}_{X,c})$ representing the graded determinant $\det \mathrm{RHom}(\mathcal{C},\mathcal{C})^{1/2}$ of Definition 4.3 in the S^1-equivariant periodic Picard groupoid, in the sense of Joyce–Upmeier [46].

(C3) *(Strict tensor compatibility.)* The external product on $\{\mathcal{F}_{X,c}\}$ is symmetric monoidal as a map of cyclotomic spectra, $\mathcal{F}_{X,c} \boxtimes \mathcal{F}_{X,c'} \to \mathcal{F}_{X,c+c'}$, intertwining φ_p on each side.

Then there exists a multiplicative orientation cocycle

$$\mu_{c,c'} \;:\; o_{c+c'} \xrightarrow{\sim} o_c \otimes o_{c'}, \qquad c, c' \in \Gamma^{\mathrm{eff}}_{X,\sigma},$$

with the following properties:

(i) (Strict pentagon.) $\mu_{c+c',c''} \circ \mu_{c,c'} = \mu_{c,c'+c''} \circ \mu_{c',c''}$ as maps of HC^--modules, with no higher coherence remainder.

(ii) (S^1-equivariance.) Each $\mu_{c,c'}$ is an S^1-equivariant map of cyclotomic spectra; the Connes operator B intertwines source and target.

(iii) (PTVV compatibility.) The Tate construction $(-)^{tC_p}$ carries $\mu_{c,c'}$ to a comparison of Frobenius twists $\mathcal{F}^{(p)}_{X,c} \otimes \mathcal{F}^{(p)}_{X,c'} \to \mathcal{F}^{(p)}_{X,c+c'}$ compatible with the PTVV [69] (-1)-shifted symplectic form on $\mathfrak{M}_{X,c}$, in the sense that $\mu_{c,c'}$ restricts on the underlying graded determinants to the symplectic-Lagrangian comparison map of [46].

In particular, hypothesis (O) of Theorem 10.39 is satisfied, and the construction of $\{\mu_{c,c'}\}$ is independent of the Costello–Li flat Calabi–Yau pairing and of any Künneth decomposition of $\mathcal{F}_X$ along $K3 \times E$.

Proof. Step 1 (Construction of $\mu_{c,c'}$ in HC^-). Apply the S^1-fixed-points functor $(-)^{hS^1}$ to the symmetric monoidal external product of (C3): the resulting map

$$HC^-(\mathcal{F}_{X,c}) \otimes HC^-(\mathcal{F}_{X,c'}) \;\longrightarrow\; HC^-(\mathcal{F}_{X,c+c'})$$

is a strict map of $\mathbb{E}_\infty$-algebras in cyclotomic spectra by Nikolaus–Scholze [62, Cor. II.1.7, Thm. II.6.9], since $(-)^{hS^1}$ is lax symmetric monoidal and the cyclotomic Frobenius φ_p is itself symmetric monoidal. Take $\widehat{o}_c$ and $\widehat{o}_{c'}$ as in (C2); their product under this map lifts the right-hand side of the desired isomorphism $o_c \otimes o_{c'}$, and divides the lift $\widehat{o}_{c+c'}$ of $o_{c+c'}$ to define $\mu_{c,c'} := \widehat{o}_{c+c'} \cdot (\widehat{o}_c \boxtimes \widehat{o}_{c'})^{-1}$. The Connes operator B acts trivially on units of π_0, so $\mu_{c,c'}$ is S^1-equivariant on the nose.

Step 2 (Strict pentagon from $\mathbb{E}_\infty$-associativity). The pentagon $\mu_{c+c',c''} \circ \mu_{c,c'} = \mu_{c,c'+c''} \circ \mu_{c',c''}$ is the π_0-shadow of the strict associator of the $\mathbb{E}_\infty$-algebra structure on $\bigoplus_c HC^-(\mathcal{F}_{X,c})$ in cyclotomic spectra. Strictness follows because Nikolaus–Scholze [62, Thm. I.1.13] present the cyclotomic ∞-category as a strict $\mathbb{E}_\infty$-monoidal limit along the Frobenius diagram; the homotopy-coherent pentagon collapses to an on-the-nose identity at the level of π_0-units. No higher coherence datum is required, because $\pi_0 HC^-$ is a discrete commutative monoid and units satisfy strict associativity in any commutative monoid.

Step 3 (S^1-equivariance built in). By construction $HC^- = HH^{hS^1}$ is the genuine S^1-fixed-points spectrum, so every map produced by Step 1 is S^1-equivariant. The Connes operator B is the generator of the S^1-action on HH [30], and B acts as a derivation on HC^-; it kills π_0-units, hence kills $\mu_{c,c'}$ as a higher homotopy.

Step 4 (Tate-construction comparison with PTVV). Apply $(-)^{tC_p}$ to the equation defining $\mu_{c,c'}$. The Frobenius φ_p intertwines source and target by (C1) and (C3), so $\mu^{tC_p}_{c,c'}$ is the comparison of Frobenius twists $\mathcal{F}^{(p)}_{X,c} \otimes \mathcal{F}^{(p)}_{X,c'} \to \mathcal{F}^{(p)}_{X,c+c'}$. Restrict to the underlying graded determinants: the symplectic structure on $\mathfrak{M}_{X,c}$ of PTVV [69] is the (-1)-shifted symplectic form whose square root is o_c (Definition 4.3); under $(-)^{tC_p}$ this form is sent to the Tate-Frobenius pairing on $\det \mathrm{RHom}(\mathcal{C}, \mathcal{C})^{(p)}$, which by Joyce–Upmeier [46, multiplicative orientation construction] agrees with the symplectic-Lagrangian comparison map. Compatibility (iii) follows.

The data $\{\mu_{c,c'}\}$ is a multiplicative orientation cocycle satisfying (i)–(iii), so hypothesis (O) of Theorem 10.39 holds. The construction factors through the cyclotomic-spectrum machinery of Nikolaus–Scholze [62] and the Connes-B calculus of Goodwillie [30]; it does not invoke Costello–Li chiral pairings, Künneth decompositions of $\mathcal{F}_X$ along $K3 \times E$, or any compactness datum beyond (C1)–(C3). □

Remark 10.51. Hypotheses (C1), (C2), (C3) are open for $K3 \times E$: (C1) asks $\mathcal{F}_{X,c}$ to carry a cyclotomic structure in the Nikolaus–Scholze sense [62]; (C2) asks orientation lines to lift to $\pi_0 HC^-$ units, refining Definition 4.3; (C3) asks symmetric monoidality of $\{\mathcal{F}_{X,c}\}$ at the cyclotomic level. The cocycle $\mu_{c,c'}$ is built from the Hochschild–cyclic machinery of $\mathcal{F}_X$ alone, supplying an alternative construction of hypothesis (O) disjoint from the Costello–Li flat Calabi–Yau pairing and from any Künneth decomposition.

Definition 10.52 *(Sectorial Hall truncation).* Suppose an extension-closed compact Hall category graded by the additive formal lattice Γ_X^{form} of Definition 6.5, with actual charges in a supplied algebraic/effective support semigroup, together with the Gram map Π_X, a stability/HN/support datum $\sigma = (Z_\sigma, \mathcal{P}_\sigma)$, and semistable stacks $\mathfrak{M}^{ss}_{\sigma,S,c}$ for $c \in \Gamma^{\mathrm{eff}}_{X,\sigma} \subset \Gamma^{\mathrm{alg}}_X \subset \Gamma^{\mathrm{form}}_X$. Fix a strict sector $S \subset \mathbb{C}^\times$ of angular width $< \pi$, with no active semistable ray on its boundary. Choose ϑ so that

$$h_S(c) := \Re(e^{-i\vartheta} Z_\sigma(c)) > 0$$

on every nonzero semistable class with central charge in S. For $R > 0$, set

$$F_{\sigma,S}(R) = \{c \mid \mathfrak{M}^{ss}_{\sigma,S,c} \neq \varnothing,\ 0 < h_S(c) \leq R\},$$

and

$$\Gamma^{HN}_{\sigma,S}(R) = \left\{ c_1 + \cdots + c_k \;\middle|\; c_i \in F_{\sigma,S}(R),\ \sum_i h_S(c_i) \leq R \right\}.$$

If an oriented critical Hall object $\mathcal{H}^{or}_{\sigma,S}$ has been constructed, put

$$\mathcal{H}^{>R}_{\sigma,S} = \prod_{h_S(c)>R} \mathcal{H}^{or}_{\sigma,S,c}, \qquad \mathsf{F}_R\mathcal{H}^{or}_{\sigma,S} = \mathcal{H}^{or}_{\sigma,S}/\mathcal{H}^{>R}_{\sigma,S}.$$

Lemma 10.53 *(Finite charge-support Hall quotient).* A nonzero additive heart cannot have finite nonzero charge support in Γ^{form}_X. The pushed-forward active support $\Gamma_{\mathrm{act}} \subset \Gamma_{\mathrm{gram}}$ is not additive. The finite charge-support Hall object is therefore not a finite active-support heart.

Assume the support property, the HN property, finite type of the semistable stacks in bounded h_S-height, and the oriented critical Hall correspondences. Then $F_{\sigma,S}(R)$ and $\Gamma^{HN}_{\sigma,S}(R)$ are finite, $\mathcal{H}^{>R}_{\sigma,S}$ is a two-sided Hall ideal, and $\mathsf{F}_R\mathcal{H}^{or}_{\sigma,S}$ is a finite charge-support Hall quotient. The compact sectorial Hall object is the completed inverse system

$$\widehat{\mathcal{H}}^{or}_{\sigma,S} = \varprojlim_R \mathsf{F}_R\mathcal{H}^{or}_{\sigma,S}.$$

Proof. If a heart contains an object of nonzero charge c, it contains all finite direct sums of that object, hence all charges kc. Since Γ^{form}_X is torsion-free, the charge support is infinite. Also $(0,-1,0) \in \Gamma_{\mathrm{act}}$, while $(0,-2,0) \notin \Gamma_{\mathrm{act}}$, so active support is not additive.

For semistable classes in S, the strict-sector condition gives $h_S(c) \geq \kappa|Z_\sigma(c)|$ for some $\kappa > 0$. The support property and discreteness of the numerical lattice make $F_{\sigma,S}(R)$ finite. The function h_S is

positive and additive on extensions in the sector; hence HN sums of total height $\leq R$ form a finite set and the subspace of height $> R$ is a two-sided Hall ideal. The inverse limit is the h_S-adic Hall completion. This supplies the charge-index finiteness part of the finite Hall package; the full datum also includes finite-type d-critical substacks, finite-dimensional protected cohomology, and proper/admissible pull-push geometry. □

Definition 10.54 *(First-order Dirac–Igusa D_0-datum).* Fix a stability/HN sector (σ, S) as in Definition 10.52. Fix also a countable cofinal HN subsystem

$$\mathcal{R}_{\mathrm{HN}} = \{R_0 < R_1 < R_2 < \cdots\} \subset \mathbb{R}_{>0}, \qquad R_\nu \to \infty.$$

The datum operates under three explicit standing hypotheses:

[K3×E-fin] *Finite-type semistable substacks at bounded HN height.* For every $R > 0$, the substack of σ-semistable objects in $\mathsf{D}^b(\mathrm{Coh}(K3 \times E))$ of HN height $\leq R$ and active charge in $F_{\sigma,S}(R)$ is of finite type over $\mathbb{C}$. K3-surface finite-wall and HN technology motivates this hypothesis; the $K3 \times E$ threefold case is not in the cited literature and is taken as part of the supplied datum. Any proof of this datum must control the base E-direction, not only the fibrewise K3 moduli problem. For example, if A is a stable K3 object and the proposed height or charge forgets base line-bundle degree, the objects

$$p_S^* A \otimes p_E^* L, \qquad L \in \mathrm{Pic}(E),$$

have constant K3 fibre and constant coarse map to the K3 moduli space while carrying unbounded Picard data. In the semistable case $p_S^* A \otimes p_E^* V$, with V a vector bundle on E, imports unbounded $\mathrm{Bun}_r(E)$-HN data unless it is bounded, rigidified, or included in the numerical charge. Thus [K3×E-fin] includes one of the following explicit controls: the full $K3 \times E$ numerical charge and central charge bound the base Picard/Bun HN types; or the finite stage quotients and rigidifies these factors with a finite degree window; or an equivalent closed finite-type replacement is supplied. A precise candidate for the first alternative is the Abramovich–Polishchuk/Liu product stability construction [1, 53]. For a rational K3 stability condition $\sigma_S = (\mathcal{A}_S, Z_S)$, a degree-one line bundle H on E, and F in the AP global heart, Liu writes [53, §3]

$$L_F(n) = Z_S\big(R p_{S,*}(F \otimes p_E^* H^n)\big) = a(F)n + b(F) + i(c(F)n + d_L(F)),$$

and, for $s, t > 0$,

$$Z_{s,t}(F) = c(F)s + b(F) + i(-a(F)t + d_L(F)).$$

For a product object $F = A \boxtimes V$, with $Z_S(A) = x + iy$, $\mathrm{rk}\, V = r$, and $\deg V = d$, this gives

$$Z_{s,t}(A \boxtimes V) = r(ys - ixt) + d(x + iy).$$

The determinant of the map from (r, d) to this complex number is $sy^2 + tx^2$, nonzero for $A \neq 0$ after the usual rotation. Thus the full Liu charge sees the elliptic degree. Moreover a subbundle $W \subset V$ has slope term

$$\nu_t(A \boxtimes W) = -x/y + \mu_E(W)/t,$$

so the product stability detects the E-bundle HN slopes. This heals the Pic/Bun counterfamily only if a boundedness theorem for $\sigma_{s,t}$-semistable objects of fixed Liu charge is supplied; Piyaratne–Toda boundedness for their threefold Bridgeland/BMT setting [70] is not automatically this theorem. The

exact missing input is the following product-boundedness theorem: for S a complex K3 surface, E a smooth elliptic curve, a rational K3 stability condition σ_S, and Liu's product stability $\sigma_{s,t} = (\mathcal{A}^t_E, Z_{s,t})$ on $D^b(\mathrm{Coh}(S \times E))$, the stack

$$\mathfrak{M}^{ss}_{\sigma_{s,t}}(\gamma)$$

of $\sigma_{s,t}$-semistable objects of every fixed full Liu numerical class $\gamma = (v_1, v_2) \in \Lambda_S \oplus \Lambda_S$, or of any stronger full numerical class with finite fibres over the active charge, is an algebraic stack of finite type over $\mathbb{C}$. Together with Liu's support property, this fixed-class boundedness would imply [K3×E-fin] for bounded h_S-windows because $F_{\sigma,S}(R)$ is finite. Fibrewise Bayer–Macri boundedness by itself is not this theorem.

[K3×E-atlas] *Finite d-critical/cosection Hall atlas.* For every $R \in \mathcal{R}_{\mathrm{HN}}$ and retained full numerical class $\widehat{c}$, the finite stage supplies finite-type quasi-smooth derived Artin stacks

$$\mathfrak{M}_{R,\widehat{c}}, \qquad \mathfrak{E}_{R,\widehat{c},\widehat{c}'}, \qquad \mathfrak{F}^{(2)}_{R,\widehat{c},\widehat{c}',\widehat{c}''}$$

of objects, extensions, and two-step flags; finite residual inertia after scalar, E-translation, and wrapped rigidifications; BBDJS vanishing-cycle complexes; orientation lines; and surjective K3 semiregularity cosections. On the extension and flag stacks the cosections satisfy the filtered Hall additivity equation

$$q^* \mathrm{CS}_C = p_1^* \mathrm{CS}_A + p_2^* \mathrm{CS}_B \,.$$

The maps $p^*, q_!, p_!$ are admissible in the chosen compact-support six-functor theory, the reduced Thom–Sebastiani transports satisfy the two-step and colored flag coherences, and the reduced E-quotient is formed as a functor on the finite correspondence category. All finite-stabilizer gerbes are killed and all characters $\lambda^{E,N}$ are chosen zero. Under these hypotheses the finite reduced product is

$$m^{\mathrm{red}}_R = q_! \circ \mathrm{TS}^{\mathrm{red}}_R \circ p^*.$$

This is an atlas hypothesis, not a consequence of [K3×E-fin].

[ML] *Mittag–Leffler descent for the HN inverse system.* The transition maps $\rho_{R'R}$ of clauses (Do-F)–(Do-S) below satisfy the finite-slice Mittag–Leffler datum of Definition 10.55 on every finite Γ^{test}_R-supported piece. Equivalently, the relevant $\lim^1$ terms vanish there. The operative finite-dimensional criterion is Mittag–Leffler: for a countable cofinal subsystem $R_0 \leq R_1 \leq \cdots$, and for every fixed finite charge/root window W, the descending chain

$$\mathrm{im}\big(V_{R_m,W} \to V_{R_n,W}\big), \qquad m \geq n,$$

eventually stabilizes for each n. Uniform surjectivity is a sufficient strict model, and every ML system is pro-isomorphic to one with surjective transition maps after replacing stages by stable images, but surjectivity is not part of the definition. In particular, once the finite Hall kernels supply finite-dimensional protected Hall, primitive, chiral bar/cobar, cone, radical, quotient, PBW-graded, kernel, cokernel, and parity pieces on each W, and the transition maps are compatible with the product, coproduct, pairing, bracket, normal ordering, source chiral coproduct, bar/cobar comparison, and closed radical data, the ML condition follows on that window. If any object called a finite window still contains an infinite fibre or infinite bar/cobar piece, the stable-image argument does not apply; the model tower $\mathbb{C}[x] \xleftarrow{x} \mathbb{C}[x] \xleftarrow{x} \cdots$ has images $x^m\mathbb{C}[x]$ that never stabilize. Strictness of the PBW filtration and closedness of the completed radicals remain additional topological requirements. Under these hypotheses the completed Dirac–Igusa state is a Lie object in the strict sense rather than only a pro-Lie object.

The recognition theorem of Theorem 3.15 reads as a strict isomorphism only after all three standing hypotheses are granted. If [ML] fails, the recognition statement weakens to a pro-comparison.

Write $\Gamma_R^{HN} := \Gamma_{\sigma,S}^{HN}(R)$. For $c \in \Gamma_R^{HN}$, put

$$\mathcal{T}_R(c) = \left\{ \sum_{1 \le i < j \le k} B(c_i, c_j) \;\middle|\; c = c_1 + \cdots + c_k,\ c_i \in F_{\sigma,S}(R),\ \sum_i h_S(c_i) \le R \right\}.$$

The empty decomposition is allowed for $c = 0$, giving $T = 0$. The finite normal-ordered support at height R is

$$\widehat{\Gamma}_R = \{(c, T) \mid c \in \Gamma_R^{HN},\ T \in \mathcal{T}_R(c)\} \subset \widehat{\Gamma}_X, \qquad \Gamma_R^{\Pi} := \overline{\Pi}_X(\widehat{\Gamma}_R) \subset \Gamma_{\mathrm{gram}}.$$

The label T is not an extra brane charge and not a new moduli factor. It is finite normal-ordering coefficient data attached to an HN word, equivalently a central-translation label for the coefficient system. The same underlying total charge c may appear with several such labels before a no-overcount theorem is supplied. Thus $\widehat{\Gamma}_R$ is finite coefficient-system data; functorial descent of these labels through the finite Hall correspondences is part of the D_0-datum. The coefficient label must also be lossless. For a label (c, T), let $\mathcal{W}_R(c, T)$ be the finite set of HN words whose normal-ordering label is (c, T), let $\mathcal{H}_{R,w}^{or}$ denote the iterated extension-correspondence object attached to w, and let $\mathcal{H}_{R,(c,T)}^{or}$ be the summand labelled by (c, T) when the finite source kernel is supplied. The no-overcount residual

$$\mathfrak{o}_{R,c,T}^{\mathrm{no}} := \mathrm{Cone}\left(\bigoplus_{w \in \mathcal{W}_R(c,T)} \mathcal{H}_{R,w}^{or} \longrightarrow \mathcal{H}_{R,(c,T)}^{or} \right)$$

is an object of the finite derived category of source coefficient complexes. Its acyclicity records that the normal-ordering labels neither duplicate nor lose source summands. It is finite-stage source data, not a consequence of the lattice identity for $\overline{\Pi}_X$.

For nonzero $\gamma = (n, l, m) \in \Gamma_{\mathrm{gram}}$, set the finite target exponent

$$a_\Delta(\gamma) = \begin{cases} f(nm, l), & \gamma \in \Gamma_{\mathrm{eff}}, \\ 0, & \gamma \notin \Gamma_{\mathrm{eff}}. \end{cases}$$

Fix also a proper height

$$h_\Delta : \Gamma_{\mathrm{gram}} \longrightarrow \mathbb{Z}_{\ge 0}, \qquad h_\Delta(n, l, m) = |n| + |l| + |m|$$

on the Gram-index lattice. The D_0 finite-window construction also fixes a nondecreasing cofinal target-window schedule

$$N_R^\Delta \in \mathbb{Z}_{\ge 0}, \qquad R' \ge R \Rightarrow N_{R'}^\Delta \ge N_R^\Delta, \qquad \sup_R N_R^\Delta = \infty.$$

This schedule is a finite-window choice, not a relation between HN height and automorphic height. To make each transition-obstruction stage a finite datum, take all inverse limits below over the fixed cofinal subsystem $\mathcal{R}_{\mathrm{HN}}$. Cofinality does not change the completed HN object or the target exhaustion; the finite-type substack and proper-correspondence assertions are the separate geometric entries of the same datum. For $R = R_\nu$ write $R^+ = R_{\nu+1}$. Transition maps for non-successor arrows are composites of the

successor maps. Thus the independent transition obstruction at a stage is a finite obstruction, not an infinite product of defects over all larger heights. In what follows, R denotes an element of this cofinal subsystem.

Define the finite target and test windows

$$\Gamma_{\Delta,R} := \{\gamma \in \Gamma_{\mathrm{eff}} \setminus \{0\} \mid h_\Delta(\gamma) \le N_R^\Delta\}, \qquad \Gamma_R^{\mathrm{test}} := (\Gamma_R^\Pi \setminus \{0\}) \cup \Gamma_{\Delta,R}.$$

$$\Gamma_R^{\Pi,+} := \Gamma_R^\Pi \cap \Gamma_{\mathrm{eff}}, \qquad \Gamma_R^{\mathrm{test},+} := (\Gamma_R^{\Pi,+} \setminus \{0\}) \cup \Gamma_{\Delta,R}.$$

The superscript + denotes the cusp-positive Pfaffian half; the full Borcherds–Kac–Moody double also has the opposite root spaces and is tested separately in the primitive-recognition theorem. Write

$$A_R := \{\gamma \in \Gamma_R^{\mathrm{test},+} \mid a_\Delta(\gamma) \ne 0\}, \qquad Z_R := \{\gamma \in \Gamma_R^{\mathrm{test},+} \mid a_\Delta(\gamma) = 0\}.$$

The inclusion $A_R \subset \Gamma_R^{\Pi,+}$ is not built into the target schedule; it is a finite source-coverage residual tested below. The set $\widehat{\Gamma}_R$, and hence Γ_R^Π, $\Gamma_{\Delta,R}$, and Γ_R^{test}, is finite by Lemma 10.53 and properness of h_Δ. Cofinality of N_R^Δ makes the target windows exhaust $\Gamma_{\mathrm{eff}} \setminus \{0\}$. For $R' \ge R$ the inclusions

$$\widehat{\Gamma}_R \subset \widehat{\Gamma}_{R'}, \qquad \Gamma_R^\Pi \subset \Gamma_{R'}^\Pi, \qquad \Gamma_{\Delta,R} \subset \Gamma_{\Delta,R'}$$

hold. A first-order Dirac–Igusa D_0-datum for $X = K3 \times E$ is the choice of the schedule N_R^Δ above, together with a compatible inverse system of finite-stage state/observable packages

$$D_{0,R} = \big(\mathcal{H}_R^{or}, \mathcal{F}_R^{D_0}, K_R^\Pi, P_R^\Pi, m_R, \Delta_R, [\,\cdot\,,\,\cdot\,]_R, I_R^{\mathrm{prot}}, \Theta_{\Pi,R}, \Theta_{\Pi,R}^-, \mathfrak{D}_{X,R}, \mathscr{L}_{\mathrm{Pf},R}, \mathrm{pf}_{X,R}, \iota_R^{obs}\big), \qquad R \in \mathcal{R}_{\mathrm{HN}},$$

together with successor transition maps whose composites give every non-successor arrow. If the transition residuals below vanish and the composite transition identities hold, the inverse limit is denoted

$$D_0(X;\sigma,S) = \big(\widehat{\mathcal{H}}_{\sigma,S}^{or}, \mathcal{F}_X^{D_0}, K_X^\Pi, P_X^\Pi, \mathfrak{D}_X, \Theta_\Pi, \Theta_\Pi^-, \mathscr{L}_{\mathrm{Pf},X}, \mathrm{pf}_X, \iota_X^{obs}\big).$$

The clauses are the following.

(Do-F) *Finite-stage maps and transitions.* At height R, all state, observable, primitive, Pfaffian, and comparison data are defined on the finite support $\widehat{\Gamma}_R$ and its Gram image Γ_R^Π. The stage carries Hall product, coproduct, primitive bracket, and protected integration maps

$$m_R, \qquad \Delta_R, \qquad [\,\cdot\,,\,\cdot\,]_R, \qquad I_R^{\mathrm{prot}},$$

and a charge-preserving observable comparison

$$\iota_R^{obs} : \mathcal{F}_R^{D_0} \longrightarrow \mathsf{Den}_{\Delta_5,E}.$$

For the successor arrow $R^+ \to R$ there are transition maps

$$\rho_{R^+R}^H : \mathcal{H}_{R^+}^{or} \to \mathcal{H}_R^{or}, \qquad \rho_{R^+R}^F : \mathcal{F}_{R^+}^{D_0} \to \mathcal{F}_R^{D_0}, \qquad \rho_{R^+R}^K : K_{R^+}^\Pi \to K_R^\Pi, \qquad \rho_{R^+R}^P : P_{R^+}^\Pi \to P_R^\Pi,$$

and compatible maps $\rho_{R^+R}^{\mathrm{Pf}}$ on Pfaffian lines. These maps are degree-truncation maps: after projection to $\widehat{\Gamma}_R$, Γ_R^Π, and Γ_R^{test}, all summands of HN height $> R$ are zero. For a non-successor $R' \ge R$,

the map $\rho_{R'R}$ is the composite of the successor maps. Let $\pi^{\Delta}_{R^+R}$ denote the finite target truncation on $\mathsf{Den}_{\Delta_s,E}$ and on $\mathbb{C}[\Gamma_{\Delta,R^+}] \to \mathbb{C}[\Gamma_{\Delta,R}]$. The transition residual

$$\mathfrak{o}^{\mathrm{tr}}_{R^+R}$$

is the finite tuple of defect maps, in the corresponding finite Hom spaces over Γ^{test}_R, asserting compatibility of ρ_{R^+R} with

$$m, \quad \Delta, \quad [\,\cdot\,,\,\cdot\,], \quad I^{\mathrm{prot}}, \quad \Theta_{\Pi}, \quad \Theta^{-}_{\Pi}, \quad \mathfrak{D}, \quad \iota^{obs}, \quad \mathrm{pf}\,.$$

For example it includes the defects

$$\rho^F_{R^+R} m_{R^+} - m_R(\rho^F_{R^+R} \otimes \rho^F_{R^+R}), \qquad \rho^P_{R^+R}[\,\cdot\,,\,\cdot\,]_{R^+} - [\,\rho^P_{R^+R}(\cdot), \rho^P_{R^+R}(\cdot)\,]_R,$$
$$I^{\mathrm{prot}}_R \rho^F_{R^+R} - \pi^{\Delta}_{R^+R} I^{\mathrm{prot}}_{R^+}, \qquad \iota^{obs}_R \rho^F_{R^+R} - \pi^{\Delta}_{R^+R}\iota^{obs}_{R^+}, \qquad \rho^K_{R^+R}\mathfrak{D}_{X,R^+} - \mathfrak{D}_{X,R}\rho^K_{R^+R}, \qquad \rho^{\mathrm{pf}}_{R^+R}(\mathrm{pf}_{X,R^+}) - \mathrm{pf}_{X,R}\,.$$

The datum requires $\mathfrak{o}^{\mathrm{tr}}_{R^+R} = 0$ for every successor arrow. The zero means a chosen finite equality or null-homotopy in the indicated Hom complex. Non-successor compatibility is then the corresponding composite equality; if non-successor maps are recorded separately, the evident cocycle identity for $R'' \geq R' \geq R$ is also part of the finite transition datum. These equalities are finite-stage data, not formal consequences of the inverse-limit notation.

(Do-S) *State object and completed inverse system.* A datum supplies an oriented critical Hall/factorization inverse system whose completed state side is

$$\widehat{\mathcal{H}}^{or}_{\sigma,S} = \varprojlim_R \mathsf{F}_R \mathcal{H}^{or}_{\sigma,S}$$

formed first on the finite HN quotients of Definition 10.52. Each finite quotient carries finite-type semistable substacks, extension correspondences, vanishing-cycle coefficients, orientation lines, Hall product and coproduct maps, and protected integration. All products, coproducts, brackets, pairings, and comparison maps are defined on the finite quotients and are continuous for the inverse limit. Lemma 10.53 proves the finite-first formal shape once the Hall object and the HN/support hypotheses are supplied.

(Do-Q) *Reduced quotient and orientation obstruction classes.* Let St_R be the finite stable-stratum index set in the chosen finite-type atlas at height R. For every $C \in \mathsf{St}_R$, with stable stratum $\mathfrak{M}^{\mathrm{st}}_{R,C} \subset \mathcal{H}^{or}_R$, the datum includes the computed obstruction tuple

$$\mathfrak{o}^{or}_{R,C} = \Big(\alpha^{\mathrm{red}}_{R,C},\ \alpha^{E,\mathrm{free}}_{R,C},\ \{(\widetilde{\beta}^{E,N}_{R,C}, \lambda^{E,N}_{R,C})\}_{N\in\mathcal{N}_R(C)}\Big),$$

where

$$\alpha^{\mathrm{red}}_{R,C} \in H^2(\mathfrak{M}^{\mathrm{red}}_{R,C};\mathbb{F}_2), \qquad \alpha^{E,\mathrm{free}}_{R,C} \in H^2(BE;\mathbb{F}_2),$$

and

$$\widetilde{\beta}^{E,N}_{R,C} \in H^2_{E[N]}(\mathfrak{M}^{\mathrm{st}}_{R,C};\mathbb{F}_2)$$

is present for each finite stabilizer $E[N]$ appearing at height R. Its point-stabilizer restriction is a class $\beta^{E,N}_{R,C} \in H^2(BE[N];\mathbb{F}_2)$ only after the mixed Borel terms, stabilizer action on stratum cohomology, and spectral-sequence differentials have vanished. After $\widetilde{\beta}^{E,N}_{R,C}$ has been null-trivialized, the datum also records the finite linearization character

$$\lambda^{E,N}_{R,C} \in H^1(BE[N];\mathbb{F}_2)$$

and requires the zero character. The datum supplies null-trivializations of all degree-two classes and zero finite degree-one characters, compatible with the transition maps. These are exactly the ordinary reduced gerbe, connected BE-linearization, finite-stabilizer gerbe, and finite-stabilizer H^1-character obstructions of Lemma 5.8. Their vanishing gives the reduced quotient and orientation line needed for the D_0 Pfaffian. Equivalently, (Do-Q) supplies the finite-stage quotient-orientation character system of Proposition 5.15 for the local, wrapped, mixed, and flag strata at height R. It does not compute the Weyl monodromy character; that is the separate content of $(\mathrm{O1})^+$ and (O2) in Theorem 3.15.

(Do-C) *Charge completion; proved lattice input.* The finite quotients are graded by the normal-ordered charge extension

$$\widehat{\Gamma}_X = \Gamma_X^{\mathrm{phys}} \oplus_B \Gamma_{\mathrm{gram}}, \qquad (c,T) \star (c',T') = (c+c', T+T'+B(c,c')),$$

and the observable Gram degree is

$$\overline{\Pi}_X(c,T) = \Pi_X(c) - T.$$

A semistable one-particle generator of charge c is placed in $(c,0)$; composite HN words are labelled by the corresponding $(c,T) \in \widehat{\Gamma}_R$. Thus $T=0$ is not asserted for every normal-ordered composite. The bracket of homogeneous states satisfies

$$[\widetilde{P}_{X,(c,T)}, \widetilde{P}_{X,(c',T')}] \subset \widetilde{P}_{X,(c,T)\star(c',T')}.$$

At height R this statement is read inside $\widehat{\Gamma}_R$, with targets of height $> R$ killed by the quotient. The finite-support assertion follows from Lemma 10.53; the additivity of $\overline{\Pi}_X$ follows from Lemma 6.9; no chain-level Hall strictification is claimed by this lattice statement.

(Do-O) *State/observable interface with imported target.* A datum supplies, at finite height R, source observable shadows $\mathcal{F}_R^{D_0}$. Their compatible inverse limit $\mathcal{F}_X^{D_0}$ is obtained from $\widehat{\mathcal{H}}^{or}_{\sigma,S}$ by the supplied $K3$-to-E specialization and, in positive elliptic degree, by a hybrid wrapped repair. It is a source object. The target observable algebra is the target current envelope $\mathsf{Den}_{\Delta_5,E}$ of Proposition 10.2. The interface is a charge-preserving comparison morphism

$$\iota_X^{obs} : \mathcal{F}_X^{D_0} \longrightarrow \mathsf{Den}_{\Delta_5,E}$$

whose constant-current primitive restriction is

$$\iota_X^{\mathrm{Prim}} : H^0\,\mathrm{Prim}_{\mathrm{prot}}(\overline{\Pi}^{\Theta}_{X,*}\mathcal{F}_X^{D_0}) \longrightarrow \mathfrak{g}_{\Delta_5}.$$

The target side is imported from Gritsenko–Nikulin; the source and the comparison morphism are supplied data. The finite observable residual

$$\mathfrak{o}_R^{obs}$$

is the finite tuple of defects measuring whether ι_R^{obs} is a morphism for the finite product, coproduct, primitive bracket, protected trace, and constant-current primitive comparison after projection to Γ_R^{test}. Here m_{Δ_5}, Δ_{Δ_5}, $[\,\cdot\,,\,\cdot\,]_{\Delta_5}$, and tr_{Δ_5} denote the corresponding finite target operations induced from $\mathsf{Den}_{\Delta_5,E}$. Thus $\mathfrak{o}_R^{obs}$ includes

$$\iota_R^{obs} m_R - m_{\Delta_5}(\iota_R^{obs} \otimes \iota_R^{obs}), \qquad \Delta_{\Delta_5}\iota_R^{obs} - (\iota_R^{obs} \otimes \iota_R^{obs})\Delta_R,$$

$$\iota_R^{\mathrm{Prim}}([x,y]_R) - [\iota_R^{\mathrm{Prim}} x, \iota_R^{\mathrm{Prim}} y]_{\Delta_5}, \qquad I_R^{\mathrm{prot}} - \mathrm{tr}_{\Delta_5}\, \iota_R^{obs}.$$

The datum requires $\mathfrak{o}_R^{obs} = 0$. This is a finite comparison with the imported observable target. It is not the presentation-level primitive isomorphism, injectivity, surjectivity, or no-extra-relations theorem; those belong to the full recognition datum. No target truncation, target bar–cobar counit, or homotopy inverse to ι_R^{obs} is allowed to define $\mathcal{F}_R^{D_\circ}$. The source object is supplied first, and the arrow points source-to-target.

(Do-N) *Exact normal-ordered descent; finite-topology, Hochschild, triple, Jacobi, Frobenius, cyclic, and radical trivializations.* At finite height R, the datum $\Theta_{\Pi,R}^{\mathrm{phys}}$ is the central translation cochain

$$\Theta_{\Pi,R}^{\mathrm{phys}}(c) = \mathsf{T}_{-\Pi_X(c)}$$

in the coefficient dg category of Definition 10.35, and $\Theta_{\Pi,R}^{-}$ is its negative-cyclic lift. On normal-ordered labels we use the induced character

$$\widehat{\Theta}_{\Pi,R}(c,T) = \mathsf{T}_{T-\Pi_X(c)}, \qquad d_{\mathrm{Hoch}}\widehat{\Theta}_{\Pi,R} = 0.$$

For $(c,T),(c',T') \in \widehat{\Gamma}_R$, it gives coherent degree-shift quasi-isomorphisms

$$\Theta_{(c,T),(c',T')} : \Pi_X(c+c') - T - T' - B(c,c') \longrightarrow \big(\Pi_X(c) - T\big) + \big(\Pi_X(c') - T'\big)$$

on primitive bracket targets. Equivalently, Θ_Π categorifies the equality

$$\overline{\Pi}_X((c,T) \star (c',T')) = \overline{\Pi}_X(c,T) + \overline{\Pi}_X(c',T')$$

and turns the upstairs physical Hall bracket into a Γ_{gram}-graded bracket after $\overline{\Pi}_{X,*}^{\Theta}$. This is exactly the input of Theorem 10.39; constructing Θ_Π on the reduced compact/wrapped sector is open. The finite coefficient pieces are

$$(\overline{\Pi}_{R,*}^{\Theta} V_R)_\gamma = \prod_{\substack{(c,T)\in\widehat{\Gamma}_R \\ \overline{\Pi}_X(c,T)=\gamma}} (V_R)_{(c,T)},$$

and this product is finite. For the successor arrow $R^+ \to R$, the transition maps must commute with these finite fibre products, with m_R^{Θ}, with Δ_R^{Θ}, and with the finite pairing. The corresponding transition defect is

$$\mathfrak{o}_{\Theta,R^+R}^{\mathrm{top}} = \Big(\mathfrak{o}_{\Theta,R^+R}^{\mathrm{fib}}, \mathfrak{o}_{\Theta,R^+R}^{m}, \mathfrak{o}_{\Theta,R^+R}^{\Delta}, \mathfrak{o}_{\Theta,R^+R}^{\langle\,,\,\rangle}\Big),$$

with the components defined in Theorem 10.39. Its vanishing is the finite statement that the completed fibre products and $\widehat{\otimes}$-operations exist as inverse limits. Equivalently, with

$$B_{\mathrm{ch},R}(c,c') := \mathsf{T}_{B(c,c')},$$

the datum supplies trivializations

$$d_{\mathrm{Hoch}}\Theta_{\Pi,R} = B_{\mathrm{ch},R}, \qquad (b + uB_{\mathrm{C}})\Theta_{\Pi,R}^{-} = B_{\mathrm{ch},R}^{-},$$

and the triple and Jacobi coherences of Theorem 10.39. Thus the finite-stage obstruction classes

$$\mathfrak{o}_{\Theta,R^+R}^{\mathrm{top}}, \qquad \mathfrak{o}_{\Theta,R}^{\mathrm{Hoch}}, \qquad \mathfrak{o}_{\Theta,R}^{\mathrm{tri}}, \qquad \mathfrak{o}_{\Theta,R}^{\mathrm{Jac}}, \qquad \mathfrak{o}_{F,R}, \qquad \mathfrak{o}_{\Theta,R}^{\mathrm{cyc}}, \qquad \mathfrak{o}_{\Theta,R}^{\mathrm{rad}}$$

vanish in the finite-topology, Hochschild, triple-coherence, Chevalley–Eilenberg, cyclic-Frobenius, negative-cyclic, and Hopf-radical ideal/coideal groups displayed there. The radical class is read as the ordered pair $(\mathfrak{o}^{\mathrm{rad,Lie}}_{\Theta,R}, \mathfrak{o}^{\mathrm{rad,coid}}_{\Theta,R})$. Equivalently, every successor arrow must satisfy the finite tests (NO1)–(NO7) of Proposition 10.40. Proposition 10.48 records the finite closure proved by the preceding finite tests: the Picard/Hochschild and finite-fibre parts are formal after the coefficient dg category and degree-truncation topology are supplied, while the cyclic lift, two-step flag coherence, Frobenius trace, and Hopf-adjoint radical tests remain source-side data.

(Do-D) *First-order object.* There are perfect finite-stage Γ^{Π}_R-graded supercomplexes K^{Π}_R, compatible under $\rho^K_{R^+R}$, with finite-dimensional parity pieces. At each finite stage there is a $\mathbb{Z}/2$-odd skew-adjoint first-order operator/algebra

$$\mathfrak{D}_{X,R} : K^{\Pi}_R \longrightarrow K^{\Pi}_R$$

whose primitive symbol is the finite normal-ordered Hall bracket on P^{Π}_R. The successor transition compatibility

$$\rho^K_{R^+R}\mathfrak{D}_{X,R^+} = \mathfrak{D}_{X,R}\rho^K_{R^+R}$$

is one component of $\mathfrak{o}^{\mathrm{tr}}_{R^+R} = 0$. The inverse limits are

$$K^{\Pi}_X = \varprojlim_R K^{\Pi}_R, \qquad \mathfrak{D}_X = \varprojlim_R \mathfrak{D}_{X,R}.$$

Their protected primitive cohomology is

$$P^{\Pi}_X = H^0\,\mathrm{Prim}_{\mathrm{prot}}(\overline{\Pi}^{\Theta}_{X,*}\mathcal{F}^{D_0}_X),$$

and the primitive symbol of $\mathfrak{D}_X$ is the normal-ordered Hall bracket. The term "first-order" means the Pfaffian/primitive object before scalar squaring; it does not assert an analytic Dirac operator on an already constructed Hilbert space. Here "primitive symbol" means the principal symbol with respect to the finite CE/Koszul filtration on K^{Π}_R; the bracket carries the Koszul signs and Γ^{Π}_R-degree convention of the finite Hall product in (Do-F).

(Do-Pf) *Pfaffian line, finite support, and normalization with imported automorphic target.* For every R there is a Pfaffian line $\mathscr{L}_{\mathrm{Pf},R}$ attached to $\mathfrak{D}_{X,R}$ and to the finite reduced orientation line of the stage, with a finite squaring isomorphism

$$\mathscr{L}^{\otimes 2}_{\mathrm{Pf},R} \simeq \mathscr{L}_{\det,R}.$$

The successor maps $\rho^{\mathrm{Pf}}_{R^+R}$ commute with these square isomorphisms, as part of $\mathfrak{o}^{\mathrm{tr}}_{R^+R} = 0$, and identify the inverse limit with a Pfaffian line

$$\mathscr{L}_{\mathrm{Pf},X}, \qquad \mathscr{L}^{\otimes 2}_{\mathrm{Pf},X} \simeq \mathscr{L}_{\det,X},$$

attached to $\mathfrak{D}_X$ and the orientation line on the state object, together with a normalized section $\mathrm{pf}_{X,R}$ at height R and $\mathrm{pf}_X = \varprojlim_R \mathrm{pf}_{X,R}$. Let $x^\gamma = q^n r^l s^m$ for $\gamma = (n, l, m)$. Projected to the finite monomial span of $\Gamma^{\Pi,+}_R$, the Borcherds-cusp product formula below is not a construction of the compact source. It is the finite determinant calculation after the primitive spaces, the first-order block decomposition, and the Pfaffian line have been supplied. For $\gamma \in \Gamma^{\Pi,+}_R$ write $P^{\Pi,+}_{R,\gamma} := P^{\Pi}_{R,\gamma}$,

and for $\gamma \notin \Gamma_R^{\Pi,+}$ set $P_{R,\gamma}^{\Pi,+} = 0$. For a homogeneous primitive mode in cusp-positive degree $\gamma \in \Gamma_R^{\Pi,+}$, the elementary two-term Pfaffian/Koszul block is the skew block

$$J_\gamma = \begin{pmatrix} 0 & 1 - x^\gamma \\ -(1 - x^\gamma) & 0 \end{pmatrix},$$

acting on $P_{R,\gamma}^{\Pi,+} \oplus (P_{R,\gamma}^{\Pi,+})^\vee[1]$; its Pfaffian contribution, with the supplied Pfaffian orientation convention, is $(1 - x^\gamma)^{\operatorname{sdim} P_{R,\gamma}^{\Pi,+}}$. Equivalently, an even basis vector contributes $(1 - x^\gamma)$, an odd basis vector contributes $(1 - x^\gamma)^{-1}$, and the exponent is $\dim P_{R,\gamma,\bar{0}}^{\Pi,+} - \dim P_{R,\gamma,\bar{1}}^{\Pi,+}$. Thus the expansion is

$$\operatorname{pf}_{X,R}|_{\mathfrak{c}_\infty} = 64 q^{1/2} r^{1/2} s^{1/2} \prod_{\substack{\gamma \in \Gamma_R^{\Pi,+} \\ \gamma \neq 0}} (1 - x^\gamma)^{\operatorname{sdim} P_{R,\gamma}^{\Pi,+}}.$$

The Pfaffian half is not the full BKM double: it records the cusp-positive product factors. Negative root representatives and the Cartan/imaginary completion enter the Lie recognition statement after the primitive bracket and pairing are supplied. Write

$$p_{R,\gamma}^{\bar{0}} := \dim P_{R,\gamma,\bar{0}}^{\Pi,+}, \qquad p_{R,\gamma}^{\bar{1}} := \dim P_{R,\gamma,\bar{1}}^{\Pi,+}.$$

The finite-stage support, signed-parity, and leading-coefficient defects

$$\mathfrak{o}_{R,\gamma}^{\mathrm{supp}} := \begin{cases} P_{R,\gamma}^{\Pi,+}, & a_\Delta(\gamma) = 0, \\ 0, & a_\Delta(\gamma) \neq 0, \end{cases} \qquad \mathfrak{o}_{R,\gamma}^{\mathrm{par}} := p_{R,\gamma}^{\bar{0}} - p_{R,\gamma}^{\bar{1}} - a_\Delta(\gamma) \in \mathbb{Z},$$

$$\mathfrak{o}_R^{\mathrm{lead}} := \operatorname{lc}_{\mathfrak{c}_\infty}(\operatorname{pf}_{X,R}) - 64 \in \mathbb{C}$$

are required to satisfy

$$\mathfrak{o}_{R,\gamma}^{\mathrm{supp}} = \mathfrak{o}_{R,\gamma}^{\mathrm{par}} = 0 \quad (\gamma \in \Gamma_R^{\mathrm{test},+}), \qquad \mathfrak{o}_R^{\mathrm{lead}} = 0.$$

The support residual is an actual super-vector-space residual, not only a K_0 class: its vanishing excludes invisible even–odd cancelling primitive pieces in target-zero degrees. The parity residual is a signed superdimension residual; the Borcherds product fixes this integer, not a canonical split of the nonzero multiplicity into even and odd compact states. For a fixed R, these residuals decide only the supplied finite test window. The inclusion of active target degrees in the source image is itself a residual condition, not a consequence of the automorphic product. Thus the target windows do not prove compact source coverage; they test it inside the source-covered finite stages. If the residuals vanish compatibly for all R, and the cofinal windows are source-covered, the limit section has the prescribed expansion

$$\operatorname{pf}_X|_{\mathfrak{c}_\infty} = 64 q^{1/2} r^{1/2} s^{1/2} \prod_{\gamma = (n,l,m) \in \Gamma_{\mathrm{eff}}} (1 - q^n r^l s^m)^{\operatorname{sdim} P_{X,\gamma}^{\Pi}},$$

with

$$\operatorname{sdim} P_{X,(n,l,m)}^{\Pi} = f(nm, l).$$

No source-window theorem is asserted here. The schedule N_R^{Δ} is an external finite target test; it becomes a source coverage theorem only after the supplied finite Hall stages prove $A_R \subset \Gamma_R^{\Pi,+}$ compatibly in R. The equality of the resulting product with Δ_5 is imported from the Borcherds–Gritsenko–Nikulin product. The compact parity pieces realizing these signed dimensions are supplied data, not a consequence of the scalar OP square. No source computation of the primitive exponents is claimed here. The equalities $\operatorname{sdim} P^{\Pi}_{X,(n,l,m)} = f(nm, l)$ are part of the supplied D_0-datum, tested against the imported Borcherds product. A source computation would derive these signed dimensions from the compact Hall correspondences, orientation data, and protected integration before comparison with the target denominator algebra. A section-level statement $\mathrm{pf}_X = \Delta_5$ also requires a supplied isomorphism from the Pfaffian line $\mathscr{L}_{\mathrm{Pf},X}$ to the pullback of the automorphic weight-five line, compatible with the cusp trivialization and the squaring maps. The finite product formula alone is only a recognition test after such line-comparison data have been supplied.

Definition 10.55 *(Finite-slice Mittag–Leffler datum).* Fix a countable cofinal HN subsystem $\mathcal{R}_{\mathrm{HN}} = \{R_0 < R_1 < \cdots\}$. For each finite downward-saturated root window W, each cohomological degree j, and each tower V_R occurring among the supplied source coalgebras, bar/cobar constructions, target truncations, kernels, cokernels, radicals, PBW associated gradeds, and cones of $b_{X,R}$ and $\Theta_{\mathrm{Kos},R}$, put

$$V^j_{\nu,W} = (V_{R_\nu})^j_W.$$

A finite-slice Mittag–Leffler datum consists of finite-dimensional vector spaces $V^j_{\nu,W}$, transition maps $\rho_{\mu\nu} : V^j_{\mu,W} \to V^j_{\nu,W}$, and stable images: for every ν, W, j there is $\mu_0 \geq \nu$ such that

$$\operatorname{im}\Big(\rho_{\mu\nu} : V^j_{\mu,W} \to V^j_{\nu,W}\Big) = \operatorname{im}\Big(\rho_{\mu_0\nu} : V^j_{\mu_0,W} \to V^j_{\nu,W}\Big) \qquad (\mu \geq \mu_0).$$

Strictness for the HN topology and closedness of radical images are part of the datum. By the standard Mittag–Leffler criterion [79, §3.5], this finite-dimensional stable-image condition implies $R^1 \varprojlim_\nu V^j_{\nu,W} = 0$. It does not construct the towers V_R; it is a condition on the finite Hall, Koszul, radical, and PBW systems after those systems have been supplied.

Definition 10.56 *(Matrix form of finite-slice Mittag–Leffler data).* For each declared tower $\mathcal{V}$, finite window W, and cohomological degree j, the finite-slice Mittag–Leffler datum includes bases of the finite-dimensional spaces

$$V^{\mathcal{V},j}_{\nu,W}$$

and transition matrices

$$M^{\mathcal{V},j}_{\mu\nu,W} : V^{\mathcal{V},j}_{\mu,W} \longrightarrow V^{\mathcal{V},j}_{\nu,W} \qquad (\mu \geq \nu),$$

satisfying the cocycle identities

$$M^{\mathcal{V},j}_{\lambda\nu,W} = M^{\mathcal{V},j}_{\mu\nu,W} M^{\mathcal{V},j}_{\lambda\mu,W} \qquad (\lambda \geq \mu \geq \nu).$$

The stable-image entry is the finite linear-algebra assertion that for each $(\mathcal{V}, W, j, \nu)$ there is $\mu_0 \geq \nu$ such that

$$\operatorname{im} M^{\mathcal{V},j}_{\mu\nu,W} = \operatorname{im} M^{\mathcal{V},j}_{\mu_0\nu,W} \subset V^{\mathcal{V},j}_{\nu,W} \qquad (\mu \geq \mu_0).$$

Equivalently, after fixing the codomain basis, the column spaces agree; in matrix form this can be recorded by

$$\operatorname{rank} M^{\mathcal{V},j}_{\mu\nu,W} = \operatorname{rank} M^{\mathcal{V},j}_{\mu_0\nu,W} = \operatorname{rank}\Big[M^{\mathcal{V},j}_{\mu\nu,W}\ M^{\mathcal{V},j}_{\mu_0\nu,W}\Big].$$

For filtered, radical, and PBW towers the bases are filtration-adapted, and strictness is recorded by the corresponding block-rank equalities. This matrix datum is void until the finite source spaces and transition matrices have actually been supplied.

Remark 10.57 *(Chamber dependence and wall-crossing).* The datum is indexed by a Bridgeland-type supplied stability/HN sector (σ, S), in the K3-surface lineage of [3, 12]. The lattice cocycle $B(c, c')$ is chamber-independent: it records only Mukai pairings of fixed lattice vectors. The compact Hall/factorization side $D_0(X; \sigma, S)$ is chamber-dependent: the finite-type semistable substacks change under wall-crossing, and the Kontsevich–Soibelman wall-crossing formula relates Hall counts in adjacent chambers.

If the datum $D_0(X; \sigma, S)$ is supplied in chamber σ, then the normal-ordered map $\overline{\Pi}_X$ remains a homomorphism and the recognition target is $\mathfrak{g}_{\Delta_5}$ provided the chamber identifications hold for the OP/DT scalar import of Proposition 11.1; that the chosen σ lies in the OP/DT chamber, or that the wall-crossing transport is compatible with $\overline{\Pi}_X$, is a separate conjectural input. The manuscript does not assert chamber-independence of the recognition target; that is an open conjecture.

Proposition 10.58 *(Finite Pfaffian window obstruction).* Fix $R \in \mathcal{R}_{\mathrm{HN}}$, and put

$$A_R := \{\gamma \in \Gamma_R^{\mathrm{test},+} \mid a_\Delta(\gamma) \neq 0\}, \qquad Z_R := \{\gamma \in \Gamma_R^{\mathrm{test},+} \mid a_\Delta(\gamma) = 0\}.$$

For any supplied finite source package $D_{0,R}$, vanishing of $\mathfrak{o}^{\mathrm{supp}}_{R,\gamma}$ and $\mathfrak{o}^{\mathrm{par}}_{R,\gamma}$ on $\Gamma_R^{\mathrm{test},+}$ is equivalent to the two finite conditions

$$P^{\Pi,+}_{R,\gamma} = 0 \quad (\gamma \in Z_R), \qquad \gamma \in \Gamma_R^{\Pi,+}, \quad p^{\bar{0}}_{R,\gamma} - p^{\bar{1}}_{R,\gamma} = a_\Delta(\gamma) \quad (\gamma \in A_R).$$

In particular every active target degree in the cofinal window must already occur in the source image:

$$A_R \subset \Gamma_R^{\Pi,+}.$$

The strongest object constructed from the automorphic input alone at this stage is the finite K_0-Pfaffian shadow

$$[\mathsf{P}^{K_0}_{\Delta,R}] := \sum_{\gamma \in \Gamma_R^{\mathrm{test},+}} a_\Delta(\gamma)[\mathbb{C}_\gamma] \in K_0\Big(\Gamma_R^{\mathrm{test},+}\text{-graded sVect}_{\mathrm{fd}}\Big)$$

and the finite formal section

$$\mathrm{pf}^{K_0}_{\Delta,R} := 64 q^{1/2} r^{1/2} s^{1/2} \prod_{\gamma \in \Gamma_R^{\mathrm{test},+}} (1 - x^\gamma)^{a_\Delta(\gamma)}.$$

This shadow is a target invariant. It is not a compact Hall object, not a Pfaffian line on the reduced quotient, and not a first-order operator. For $\gamma \in A_R$, the pair of compact parity dimensions is not determined by the shadow: adding a finite vector space W_γ in both parities replaces

$$(P^{\Pi}_{R,\gamma,\bar{0}}, P^{\Pi}_{R,\gamma,\bar{1}}) \quad \text{by} \quad (P^{\Pi}_{R,\gamma,\bar{0}} \oplus W_\gamma, P^{\Pi}_{R,\gamma,\bar{1}} \oplus W_\gamma)$$

and leaves the finite product, support test, signed-parity test, and leading coefficient unchanged. Thus compact parity pieces on the cofinal target windows are extra source data. They cannot be constructed from Δ_5, from the denominator product, or from the OP scalar square.

Proof. If $\gamma \in Z_R$, the support residual is $\mathfrak{o}^{\mathrm{supp}}_{R,\gamma} = P^{\Pi,+}_{R,\gamma}$; its vanishing is exactly the first displayed condition. If $\gamma \in A_R$ and $\gamma \notin \Gamma^{\Pi,+}_R$, the convention $P^{\Pi,+}_{R,\gamma} = 0$ gives $\mathfrak{o}^{\mathrm{par}}_{R,\gamma} = -a_\Delta(\gamma) \neq 0$. Hence vanishing forces $\gamma \in \Gamma^{\Pi,+}_R$, and then the parity residual is exactly

$$p^{\bar{0}}_{R,\gamma} - p^{\bar{1}}_{R,\gamma} - a_\Delta(\gamma) = 0.$$

This proves the equivalence and the active-window coverage.

The finite formal section is obtained by applying the superdeterminant formula to the K_0-class $[\mathsf{P}^{K_0}_{\Delta,R}]$. This construction takes place in the Grothendieck group of finite graded super vector spaces. It contains no extension correspondences, no orientation line, and no skew-adjoint finite operator. Finally, the replacement by $W_\gamma \oplus \Pi W_\gamma$ has zero class in K_0, so it does not change $p^{\bar{0}}_{R,\gamma} - p^{\bar{1}}_{R,\gamma}$ or any product depending only on signed superdimension. The actual parity pieces are therefore invisible to the finite Pfaffian shadow. □

Open Problem 10.59 *(Source computation of the Igusa exponents).* Derive the equalities

$$\operatorname{sdim} P^{\Pi,+}_{R,\gamma} = a_\Delta(\gamma), \qquad \gamma \in \Gamma^{\mathrm{test},+}_R,$$

from the compact Hall correspondences, orientation data, quotient descent, and protected integration at finite height R, before comparison with the imported Borcherds product. In the first wrapped channel

$$\delta_2 = 2f_{-2} - f_3 \leftrightarrow (0,1,1),$$

this asks for a wrapped charge $\eta_2 \in \Gamma^{\mathrm{wr}}_R$ with

$$b^{\mathrm{geom}}_R(\eta_2) > 0, \qquad \overline{\Pi}_X(\eta_2) = (0,1,1),$$

and a computation

$$P^{\Pi,+}_{R,(0,1,1)} = H^0 \operatorname{Prim}_{\mathrm{prot}}(\overline{\Pi}^{\Theta}_{R,*} Q_{E,R} \mathcal{G}^{\mathrm{wr}}_{\eta_2,R})$$

with parity dimensions

$$\dim P^{\Pi,+}_{R,(0,1,1),\bar{0}} = 1, \qquad \dim P^{\Pi,+}_{R,(0,1,1),\bar{1}} = 0.$$

Equivalently, the source exponent is $\operatorname{sdim} P^{\Pi,+}_{R,(0,1,1)} = f(0,1) = 1$. A protected primitive generator v_{δ_2} must satisfy the trace normalization

$$I^{\mathrm{prot}}_R(Q_{E,R} v_{\delta_2}) = q^0 r^1 s^1,$$

with no quotient-orientation or finite-stabilizer sign changing it. The corresponding product factor is $(1 - rs)^{+1}$, so its linear product coefficient is $-rs$; the exponent and the product coefficient are not the same datum. A projection-finite source has $b^{\mathrm{geom}}_R = 0$, hence cannot produce this positive trace s-degree $m_R = 1$. A detached one-dimensional K_0 shadow in degree $(0,1,1)$ has the right signed dimension but supplies no wrapped moduli, vanishing cycles, quotient descent, protected primitive, or Hall bracket. The present datum only tests a supplied source object against the target exponents.

Definition 10.60 *(Finite Hall source kernel and downstream normal-ordering package).* Fix $R \in \mathcal{R}_{\mathrm{HN}}$. A finite Hall source kernel at height R is first the following source-side geometric datum, before normal-ordering, protected integration, or comparison with the Borcherds target:

$$\mathfrak{S}^{\mathrm{geom}}_R = (\mathfrak{M}_{R,\hat{c}}, \Phi_{R,\hat{c}}, \mathfrak{E}_{R,\hat{c},\hat{c}'}, p_{R,\hat{c},\hat{c}'}, q_{R,\hat{c},\hat{c}'}, o_{R,\hat{c}}, \mu^{or}_R)_{\hat{c},\hat{c}' \in \hat{\Gamma}_R}.$$

The finite source package used by the D_0-datum is

$$\mathfrak{S}_R = \big(\mathfrak{S}_R^{\mathrm{geom}}, I_R^{\mathrm{prot}}, \Theta_{\Pi,R}, \Theta_{\Pi,R}^{-}\big).$$

Only $\mathfrak{S}_R^{\mathrm{geom}}$ is the common finite geometric kernel. The protected integration map and the normal-ordering cochains are downstream finite data; they are part of the datum, not consequences of the geometric kernel alone. The requirements are finite and source-internal. All pull-push operations below are taken in one fixed constructible compact-support six-functor theory on the finite-type Artin stacks under discussion. The symbol $f_!$ means exceptional direct image; proper pushforward is only the special case $f_! = f_*$ on a proper protected support. On an affine Ext stratum of rank d, the normalization includes

$$R\Gamma_c(\mathbb{A}^d, \mathbb{C}) = \mathbb{C}(-d)[-2d],$$

and this shift/Tate twist is absorbed into the orientation transport.

(S0) The finite direct sum of protected cohomology groups has a supplied unit, counit, multiplication, and coproduct, making it a finite graded bialgebra object before primitives are taken. The counit is denoted ϵ_R, the coproduct Δ_R, and

$$\mathrm{Prim}_R := \ker\big(\Delta_R - (1 \otimes \mathrm{id}) - (\mathrm{id} \otimes 1)\big) \cap \ker(\epsilon_R)$$

is the finite primitive subobject.

(S1) Each $\mathfrak{M}_{R,\widehat{c}}$ is a finite-type d-critical Artin stack with finite residual inertia after the prescribed scalar, E-translation, and wrapped rigidifications, with compactly supported BBDJS vanishing-cycle complex $\Phi_{R,\widehat{c}}$, orientation line $o_{R,\widehat{c}}$, and finite-dimensional protected cohomology. The finite-inertia clause is load-bearing: a residual $\mathbb{G}_m$-stabilizer already gives

$$H^\bullet(B\mathbb{G}_m, \mathbb{C}) \simeq \mathbb{C}[u],$$

so finite type of the coarse support and E-rigidification alone do not imply finite protected cohomology. Finite stabilizers are harmless for characteristic-zero constructible cohomology, while their mod-2 orientation characters remain part of the quotient-orientation obstruction data.

(S2) For every product whose target charge remains in $\widehat{\Gamma}_R$, the extension correspondence

$$\mathfrak{M}_{R,\widehat{c}} \times \mathfrak{M}_{R,\widehat{c}'} \xleftarrow{p_{R,\widehat{c},\widehat{c}'}} \mathfrak{E}_{R,\widehat{c},\widehat{c}'} \xrightarrow{q_{R,\widehat{c},\widehat{c}'}} \mathfrak{M}_{R,\widehat{c}\star\widehat{c}'}$$

is admissible for vanishing-cycle pull-push in that six-functor theory, has either proper support for q or a specified exceptional pushforward $q_!$ with finite compact-support realization, has the dual $p_!$ needed for the coproduct, and is equipped with the oriented Thom–Sebastiani transport μ_R^{or}. The notation p^* below denotes the supplied pullback or Gysin-normalized pullback appropriate to the Borel–Moore convention. If $\widehat{c} \star \widehat{c}' \notin \widehat{\Gamma}_R$, the projected correspondence is declared zero in the finite quotient.

(S3) The two-step extension stacks carry the usual flag associativity isomorphisms, and the coproduct correspondences are admissible in the same Borel–Moore or compact-support formalism.

(S3') The multiplication and coproduct satisfy the finite bialgebra compatibility diagram after vanishing-cycle pull-push and orientation transport. Consequently the supercommutator of two primitives is again primitive at this finite stage.

(S4) The added normal-ordering cochains $\Theta_{\Pi,R}$, $\Theta^{-}_{\Pi,R}$ act on the actual finite correspondence targets, not merely on the lattice labels.

For a successor arrow $R^{+} \to R$, a transition-compatible family of kernels also includes truncation morphisms on stacks, extension stacks, coefficient complexes, orientations, and protected integration maps, with non-successor maps defined as their composites.

Proposition 10.61 *(Finite D_0 Hall source package from a supplied kernel).* A supplied finite Hall source kernel $\mathfrak{S}_R$ determines the finite state-side Hall object, its primitive bracket, and the justified first-order differential shadow as follows.

First set

$$\mathcal{H}^{or}_{R,\widehat{c}} := \mathrm{H}^{\bullet}_{c}(\mathfrak{M}_{R,\widehat{c}}, \Phi_{R,\widehat{c}} \otimes o_{R,\widehat{c}}), \qquad \mathcal{H}^{or}_{R} := \bigoplus_{\widehat{c}\in\widehat{\Gamma}_R} \mathcal{H}^{or}_{R,\widehat{c}}.$$

This is finite-dimensional because $\widehat{\Gamma}_R$ is finite and (S1) imposes finite protected cohomology on each summand. The Hall product and coproduct are the finite pull-push maps

$$m_{R,\widehat{c},\widehat{c}'} = q_{R,\widehat{c},\widehat{c}',!}\, \mathrm{TS}^{or}_{R,\widehat{c},\widehat{c}'}\, p^{*}_{R,\widehat{c},\widehat{c}'},$$

$$\Delta_{R,\widehat{c},\widehat{c}'} = p_{R,\widehat{c},\widehat{c}',!}\, (\mathrm{TS}^{or}_{R,\widehat{c},\widehat{c}'})^{-1}\, q^{*}_{R,\widehat{c},\widehat{c}'},$$

with zero output when the target charge leaves $\widehat{\Gamma}_R$. The two-step flag isomorphisms in (S3) give associativity and coassociativity at this finite height; their failure is exactly the finite flag component of the D_0 residual.

Let $P_R \subset \mathcal{H}^{or}_R$ be the primitive part for Δ_R, and put

$$P^{\Pi}_{R} := \overline{\Pi}^{\Theta_R}_{R,*} P_R.$$

The normal-ordering cochain in (S4) transports the supercommutator of m_R to a finite Γ^{Π}_R-graded bracket

$$[\,\cdot\,,\,\cdot\,]_{\Theta,R} : P^{\Pi}_{R,\gamma} \otimes P^{\Pi}_{R,\gamma'} \longrightarrow P^{\Pi}_{R,\gamma+\gamma'}.$$

If the finite Jacobi diagram (NO5) holds, this is a finite Lie superalgebra bracket. Equivalently, it determines the Chevalley–Eilenberg first-order differential

$$d_{\mathrm{CE},R} : C^{\bullet}_{\mathrm{CE}}(P^{\Pi}_{R}) \longrightarrow C^{\bullet+1}_{\mathrm{CE}}(P^{\Pi}_{R}),$$

the unique degree-one derivation whose quadratic part is dual to $[\,\cdot\,,\,\cdot\,]_{\Theta,R}$. The identity $d^{2}_{\mathrm{CE},R} = 0$ is exactly the finite Jacobi identity with the Koszul signs. This is the first-order differential justified by the finite Hall source core. It is not yet the protected Dirac operator $\mathfrak{D}_{X,R}$, and it carries no Pfaffian line, unless the extra Dirac/Pfaffian data of Proposition 10.63 are supplied.

For a transition-compatible family $\mathfrak{S}_{R^{+}} \to \mathfrak{S}_R$, the specified truncation morphisms induce

$$\rho^{H}_{R^{+}R} : \mathcal{H}^{or}_{R^{+}} \to \mathcal{H}^{or}_{R}, \qquad \rho^{P}_{R^{+}R} : P^{\Pi}_{R^{+}} \to P^{\Pi}_{R}.$$

They are morphisms of finite Hall objects precisely when the product, coproduct, pairing, integration, and normal-ordering squares (NO1)–(NO7) commute. Thus the source kernel constructs the finite Hall source and names the exact residuals obstructing its passage to a $D_{0,R}$-stage.

Proof. The direct sum is finite by finiteness of $\widehat{\Gamma}_R$ and the finite-dimensionality imposed in (S1). The BBDJS Thom–Sebastiani isomorphism, transported through the orientation isomorphisms, identifies the vanishing-cycle coefficients on the source of the correspondence with the coefficient system on the target. The six-functor admissibility in (S2) therefore makes the displayed pull-push maps defined; properness is only one way to identify the exceptional pushforward with ordinary pushforward. The zero convention for charges outside $\widehat{\Gamma}_R$ is exactly the finite charge-support Hall quotient convention. The two-step flag isomorphisms in (S3) identify the two parenthesized pull-push composites, hence give associativity and the dual coproduct coherence at height R. Clause (S0) supplies the finite unit, counit, multiplication, and coproduct, and defines primitives as the displayed kernel. Clause (S3') supplies the finite bialgebra compatibility diagram, so the supercommutator of two primitives is again primitive. The normal-ordering cochain acts on the actual correspondence targets by (S4), so it transports the bracket to the additive Γ_R^{Π}-degree. Finally, the Chevalley–Eilenberg differential is the standard derivation dual to a Lie superalgebra bracket; its square is the Jacobiator with the Koszul signs. The transition assertion is the same pull-push naturality applied to the supplied truncation morphisms. □

Proposition 10.62 *(Cofinal target-window K_0 parity shadow).* For every $R \in \mathcal{R}_{\mathrm{HN}}$, the target exponents construct a canonical minimal finite super-vector-space model on the test window:

$$\mathsf{P}^{\min}_{\Delta,R,\gamma} := \begin{cases} \mathbb{C}^{a_\Delta(\gamma)} \text{ in parity } \bar{0}, & a_\Delta(\gamma) > 0, \\ \mathbb{C}^{-a_\Delta(\gamma)} \text{ in parity } \bar{1}, & a_\Delta(\gamma) < 0, \\ 0, & a_\Delta(\gamma) = 0, \end{cases} \qquad \gamma \in \Gamma_R^{\mathrm{test},+}.$$

It satisfies

$$[\mathsf{P}^{\min}_{\Delta,R}] = \sum_{\gamma \in \Gamma_R^{\mathrm{test},+}} a_\Delta(\gamma)[\mathbb{C}_\gamma] = [\mathsf{P}^{K_0}_{\Delta,R}]$$

and gives the formal Pfaffian section $\mathrm{pf}^{K_0}_{\Delta,R}$ of Proposition 10.58.

Let $P_R^{\Pi,+} := P_R^{\Pi}|_{\Gamma_R^{\Pi,+}}$ be the cusp-positive primitive source half constructed from a finite Hall source kernel. On $\Gamma_R^{\mathrm{test},+}$, vanishing of $\mathfrak{o}^{\mathrm{supp}}_{R,\gamma}$ and $\mathfrak{o}^{\mathrm{par}}_{R,\gamma}$ is equivalent to the Grothendieck equality

$$[P_R^{\Pi,+}|_{\Gamma_R^{\mathrm{test},+}}] = [\mathsf{P}^{\min}_{\Delta,R}]$$

together with active-window source coverage

$$\{\gamma \in \Gamma_R^{\mathrm{test},+} \mid a_\Delta(\gamma) \neq 0\} \subset \Gamma_R^{\Pi,+}$$

and actual vanishing of source primitive spaces in all target-zero degrees. This equality upgrades to an isomorphism of graded super vector spaces

$$P^{\Pi,+}_{R,\gamma} \simeq \mathsf{P}^{\min}_{\Delta,R,\gamma} \qquad (\gamma \in \Gamma_R^{\mathrm{test},+})$$

if and only if the no-hidden-cancellation condition

$$\min\{\dim P^{\Pi,+}_{R,\gamma,\bar{0}}, \dim P^{\Pi,+}_{R,\gamma,\bar{1}}\} = 0$$

holds for every active γ in the test window. Without that extra source condition, one may add $W_\gamma \oplus \Pi W_\gamma$ in any active degree without changing the Pfaffian product, the signed parity residual, or the leading coefficient.

Proof. The definition of $\mathsf{P}^{\min}_{\Delta,R,\gamma}$ gives signed dimension $a_\Delta(\gamma)$ in every test degree and no vector space when $a_\Delta(\gamma) = 0$. Hence its Grothendieck class is the displayed finite K_0-Pfaffian shadow. The equivalence between vanishing residuals, active-window source coverage, and the stated K_0-equality is exactly Proposition 10.58, with the support residual remembering actual source spaces in target-zero degrees. For an active γ, two finite super vector spaces of signed dimension $a_\Delta(\gamma)$ are isomorphic to the minimal model precisely when one parity has dimension zero. Adding $W_\gamma \oplus \Pi W_\gamma$ changes neither signed dimension nor any superdeterminant depending only on signed dimension. This proves the last assertion. □

Proposition 10.63 *(Finite Hall–Dirac atlas and parity-lift obstruction).* Fix $R \in \mathcal{R}_{\mathrm{HN}}$. The finite Hall source core is constructed from the source kernel of Definition 10.60 by Proposition 10.61. In this manuscript, a full finite stage $D_{0,R}$, including the Dirac operator and Pfaffian line, is represented by the following supplied finite Hall–Dirac atlas data:

$$\mathfrak{A}_R = \big(\mathfrak{M}_{R,\widehat{c}}, \mathfrak{E}_{R,\widehat{c},\widehat{c}'}, \Phi_{R,\widehat{c}}, o_{R,\widehat{c}}, \mu^{or}_R, \mathcal{F}^{D_0}_R, \Theta_{\Pi,R}, \Theta^-_{\Pi,R}, \iota^{obs}_R, K^\Pi_R, \mathfrak{D}_{X,R}, \mathscr{L}_{\mathrm{Pf},R}, \ell_R\big)$$

with the following data.

(A1) *Finite extension-closed d-critical source.* For each $\widehat{c} = (c, T) \in \widehat{\Gamma}_R$ there is a finite-type d-critical stack $\mathfrak{M}_{R,\widehat{c}}$ with compactly supported BBDJS vanishing-cycle coefficient complex $\Phi_{R,\widehat{c}}$. The finite union $\mathfrak{M}_R = \coprod_{\widehat{c}\in\widehat{\Gamma}_R} \mathfrak{M}_{R,\widehat{c}}$ is closed under those extensions whose charge remains in $\widehat{\Gamma}_R$. Thus for $\widehat{c}, \widehat{c}'$, with $\widehat{c} \star \widehat{c}' \in \widehat{\Gamma}_R$, there is a finite-type extension correspondence

$$\mathfrak{M}_{R,\widehat{c}} \times \mathfrak{M}_{R,\widehat{c}'} \xleftarrow{p_{R,\widehat{c},\widehat{c}'}} \mathfrak{E}_{R,\widehat{c},\widehat{c}'} \xrightarrow{q_{R,\widehat{c},\widehat{c}'}} \mathfrak{M}_{R,\widehat{c}\star\widehat{c}'},$$

and the correspondence is empty after projection to the quotient when the target charge leaves $\widehat{\Gamma}_R$. The two-step extension stacks for triples in $\widehat{\Gamma}_R$ satisfy the usual flag associativity isomorphisms.

(A2) *Orientation square and quotient linearization.* There are orientation lines

$$o^{\otimes 2}_{R,\widehat{c}} \simeq \det \mathrm{RHom}_{\mathrm{red}}(C_{\widehat{c}}, C_{\widehat{c}})$$

on the finite strata, Thom–Sebastiani orientation isomorphisms $\mu^{or}_{R,\widehat{c},\widehat{c}'}$ over the extension correspondences, and zero quotient-linearization characters for every finite stabilizer appearing in the reduced E-quotient. Equivalently the classes $\mathfrak{o}^{or}_{R,C}$ of (Do-Q) vanish with chosen finite trivializations.

(A3) *Finite Hall operations, normal ordering, and protected integration.* The maps $p_{R,\widehat{c},\widehat{c}'}$ and $q_{R,\widehat{c},\widehat{c}'}$ are admissible for vanishing-cycle pull-push and proper on the support used by the protected trace. The resulting Joyce–Kontsevich–Soibelman pull-push operations define

$$m_R, \qquad \Delta_R, \qquad [\,\cdot\,,\,\cdot\,]_R, \qquad I^{\mathrm{prot}}_R,$$

and the source observable shadow $\mathcal{F}^{D_0}_R$, normal-ordering cochains $\Theta_{\Pi,R}$, $\Theta^-_{\Pi,R}$, and comparison map ι^{obs}_R are defined source-first. They satisfy the finite flag, coproduct, trace, observable-comparison, Hochschild, and negative-cyclic diagrams required in (Do-F), (Do-O), and (Do-N).

(A4) *Finite first-order complex.* There is a perfect Γ^Π_R-graded supercomplex K^Π_R and an odd skew-adjoint first-order endomorphism

$$\mathfrak{D}_{X,R} : K^\Pi_R \to K^\Pi_R$$

whose primitive symbol on

$$P_R^{\Pi} = H^0 \operatorname{Prim}_{\mathrm{prot}}(\overline{\Pi}^{\Theta}_{R,*}\mathcal{F}_R^{D_0})$$

is the Hall bracket of (A3). Here "primitive symbol" means the principal symbol for the finite CE/Koszul filtration on K_R^{Π}, with the Koszul signs and Γ_R^{Π}-degree convention induced by the finite Hall product.

(A5) *Pfaffian square-root datum.* On the reduced finite quotient $\mathfrak{Q}_R := (\mathfrak{M}_R/E)^{\mathrm{red}}$ there is a determinant line $\mathscr{L}_{\det,R}$ attached to K_R^{Π}, a square root

$$\eta_R : \mathscr{L}_{\mathrm{Pf},R}^{\otimes 2} \xrightarrow{\sim} \mathscr{L}_{\det,R},$$

and a finite section $\mathrm{pf}_{X,R}$ whose leading coefficient at the Borcherds cusp is 64.

(A6) *Parity lift on the positive Pfaffian test window.* For every $\gamma \in \Gamma_R^{\mathrm{test},+}$ there are actual compact primitive super-vector spaces

$$P_{R,\gamma}^{\Pi,+} = P_{R,\gamma,\bar{0}}^{\Pi,+} \oplus P_{R,\gamma,\bar{1}}^{\Pi,+}$$

with

$$\ell_R(\gamma) = \big(\dim P_{R,\gamma,\bar{0}}^{\Pi,+}, \dim P_{R,\gamma,\bar{1}}^{\Pi,+}\big),$$

such that $P_{R,\gamma}^{\Pi,+} = 0$ for $a_\Delta(\gamma) = 0$, and

$$\dim P_{R,\gamma,\bar{0}}^{\Pi,+} - \dim P_{R,\gamma,\bar{1}}^{\Pi,+} = a_\Delta(\gamma)$$

for $a_\Delta(\gamma) \neq 0$. This includes the active-window source coverage $A_R \subset \Gamma_R^{\Pi,+}$.

(A7) *Successor transition diagrams.* For $R^+ \to R$ there are truncation maps on the stacks, extension stacks, coefficient complexes, first-order complexes, and Pfaffian lines such that

$$\begin{tikzcd}
\mathfrak{E}_{R^+,\widehat{c},\widehat{c}'} \arrow[r, "q_{R^+}"] \arrow[d, "\tau^E_{R^+R}"'] & \mathfrak{M}_{R^+,\widehat{c}\star\widehat{c}'} \arrow[d, "\tau^M_{R^+R}"] \\
\mathfrak{E}_{R,\widehat{c},\widehat{c}'} \arrow[r, "q_R"'] & \mathfrak{M}_{R,\widehat{c}\star\widehat{c}'}
\end{tikzcd}
\qquad
\begin{tikzcd}
\mathfrak{E}_{R^+,\widehat{c},\widehat{c}'} \arrow[r, "p_{R^+}"] \arrow[d, "\tau^E_{R^+R}"'] & \mathfrak{M}_{R^+,\widehat{c}} \times \mathfrak{M}_{R^+,\widehat{c}'} \arrow[d, "\tau^M_{R^+R}\times\tau^M_{R^+R}"] \\
\mathfrak{E}_{R,\widehat{c},\widehat{c}'} \arrow[r, "p_R"'] & \mathfrak{M}_{R,\widehat{c}} \times \mathfrak{M}_{R,\widehat{c}'}
\end{tikzcd}$$

commute after projection to the finite quotient, and the induced diagrams for $\rho^K_{R^+R}\mathfrak{D}_{X,R^+}$, $\rho^{\mathrm{Pf}}_{R^+R}\eta_{R^+}$, and $\rho^{\mathrm{Pf}}_{R^+R}(\mathrm{pf}_{X,R^+})$ commute. The finite tuple of failures of these diagrams is denoted $\mathfrak{o}^{\mathrm{atlas}}_{R^+R}$; it is one source-level realization of the transition residual $\mathfrak{o}^{\mathrm{tr}}_{R^+R}$.

If $\mathfrak{A}_R$ is supplied and $\mathfrak{o}^{\mathrm{atlas}}_{R^+R} = 0$, then the pull-push operations, orientation square, first-order operator, Pfaffian line, and parity lift construct the finite package $D_{0,R}$ of Definition 10.54. Conversely, every constructed $D_{0,R}$ gives such finite atlas data after forgetting to the finite support and reduced quotient. Thus the missing object is not the target determinant; it is the finite atlas $\mathfrak{A}_R$ and its successor-compatible trivializations.

Moreover the two remaining Pfaffian ambiguities are exact. The Pfaffian square-root choices for a fixed determinant line are empty unless $[\mathscr{L}_{\det,R}]$ is divisible by 2 in $\operatorname{Pic}(\mathfrak{Q}_R)$; when nonempty, their isomorphism classes form a torsor under $\operatorname{Pic}(\mathfrak{Q}_R)[2]$, with the $H^1(\mathfrak{Q}_R;\mathbb{F}_2)$ local-system shadow used

in the finite-stabilizer orientation tests. For an active degree $\gamma \in A_R$, the compact parity lifts form the affine monoid

$$\mathscr{P}_R(\gamma) = \{(u,v) \in \mathbb{Z}^2_{\geq 0} \mid u - v = a_\Delta(\gamma)\}, \qquad (u,v) \mapsto (u+t, v+t).$$

The finite K_0-Pfaffian shadow fixes only the difference $u - v$. The no-hidden-cancellation condition

$$\min(u,v) = 0$$

would select the unique minimal lift

$$(u,v) = \begin{cases} (a_\Delta(\gamma), 0), & a_\Delta(\gamma) > 0, \\ (0, -a_\Delta(\gamma)), & a_\Delta(\gamma) < 0, \end{cases}$$

but this is an extra source axiom. It is not implied by Δ_5, by the denominator algebra, by the OP scalar square, or by equality in K_0.

Proof. Given $\mathfrak{A}_R$, clauses (A1)–(A3) are exactly the finite Joyce–Kontsevich–Soibelman Hall pull-push construction with BBDJS vanishing-cycle coefficients and Joyce–Upmeier orientation transport. Finiteness of the index set makes all sums and products finite, and the two-step flag isomorphisms are precisely the associativity and coproduct-coherence diagrams. Clause (A4) gives (Do-D). Clause (A5) gives the finite Pfaffian square-root datum and the leading coefficient condition in (Do-Pf). Clause (A6) is exactly the vanishing of the support and parity residuals on $\Gamma_R^{\mathrm{test},+}$, with active-window coverage as in Proposition 10.58. Clause (A7) gives the successor compatibility diagrams; their failure is the displayed finite residual $\mathfrak{o}^{\mathrm{atlas}}_{R^+R}$, which maps into $\mathfrak{o}^{\mathrm{tr}}_{R^+R}$. Hence the data assemble to $D_{0,R}$. The converse is tautological: a constructed finite $D_{0,R}$ already contains the finite source stacks, correspondences, coefficient systems, operator, Pfaffian line, and parity pieces named above, after restriction to its finite support.

The assertion about square roots is the elementary Picard-group classification: a line L admits a square root iff its class lies in $2\,\mathrm{Pic}(\mathfrak{Q}_R)$; the set of square roots, when nonempty, is acted on simply transitively by 2-torsion line bundles. Passing to $\mathbb{F}_2$-local systems gives the $H^1(\mathfrak{Q}_R; \mathbb{F}_2)$ shadow. Finally $\mathscr{P}_R(\gamma)$ is the set of nonnegative integer solutions to one linear equation $u - v = a_\Delta(\gamma)$. Adding (t,t) preserves the K_0 class and all product exponents. Therefore signed dimension does not determine compact parity pieces unless the extra condition $\min(u,v) = 0$ is supplied. □

Theorem 10.64 *(Finite D_0-datum).* This first-order datum separates finite objects obtained from supplied kernels and atlases from the open existence of those kernels and atlases.

(from supplied HN support) For every $R \in \mathcal{R}_{\mathrm{HN}}$, the normal-ordered support $\widehat{\Gamma}_R$, its image Γ_R^{Π}, the target window $\Gamma_{\Delta,R}$, the ambient test window Γ_R^{test}, and the positive Pfaffian test window $\Gamma_R^{\mathrm{test},+}$ of Definition 10.54 are finite; by cofinality of the target-window schedule, the windows $\Gamma_{\Delta,R}$ exhaust $\Gamma_{\mathrm{eff}} \setminus \{0\}$. The lattice charge completion $\widehat{\Gamma}_X$ and the normal-ordered homomorphism $\overline{\Pi}_X$ exist and are additive, by Lemma 10.53 and Lemma 6.9. The exhaustion statement is target-side only; it does not assert that every target degree lies in the image of the supplied finite Hall support.

(imported) The target observable algebra $\mathtt{Den}_{\Delta_5,E}$, its constant-current denominator algebra $\mathfrak{g}_{\Delta_5}$, and the product

$$64 q^{1/2} r^{1/2} s^{1/2} \prod_{(n,l,m)\in\Gamma_{\mathrm{eff}}} (1 - q^n r^l s^m)^{f(nm,l)} = \Delta_5$$

are supplied by the Borcherds–Gritsenko–Nikulin theorem and Proposition 10.2.

ckage from supplied kernel) If, at height R, a finite Hall source kernel $\mathfrak{S}_R$ in the sense of Definition 10.60 is supplied, then Proposition 10.61 determines the finite state object $\mathcal{H}_R^{or}$, the Hall product, coproduct, primitive bracket, normal-ordered finite primitive algebra P_R^{Π}, successor transition maps, and the Chevalley–Eilenberg first-order differential $d_{\mathrm{CE},R}$ justified by the finite bracket. This is finite algebra obtained from supplied source data at height R. It is not yet the protected Dirac operator $\mathfrak{D}_{X,R}$ and carries no Pfaffian line unless the extra Dirac/Pfaffian atlas data are supplied. Independently, the cofinal target exponents construct the minimal finite target-window parity model $\mathsf{P}^{\min}_{\Delta,R}$ of Proposition 10.62; comparison with actual compact source primitives is exactly the support/parity and no-hidden-cancellation test recorded there.

(finite residual) Once the finite Hall–Dirac atlas $\mathfrak{A}_R$ of Proposition 10.63 is supplied, including the finite semistable stacks, extension correspondences, vanishing-cycle pull-push maps, protected integration maps, source comparison, first-order complex, Pfaffian square-root datum, and compact parity lifts at height R, the obstruction tuple recorded by the finite-stage D_0 datum is the vanishing, with compatible trivializations, of

$$\mathfrak{O}_{D_0,R} := \Big(\mathfrak{o}^{\mathrm{tr}}_{R^+R},\ \{\mathfrak{o}^{or}_{R,C}\}_{C\in\mathsf{St}_R},\ \mathfrak{o}^{obs}_R,\ \mathfrak{o}^{\mathrm{top}}_{\Theta,R^+R},\ \mathfrak{o}^{\mathrm{Hoch}}_{\Theta,R},\ \mathfrak{o}^{\mathrm{tri}}_{\Theta,R},\ \mathfrak{o}^{\mathrm{Jac}}_{\Theta,R},\ \mathfrak{o}_{F,R},\ \mathfrak{o}^{\mathrm{cyc}}_{\Theta,R},\ \mathfrak{o}^{\mathrm{rad}}_{\Theta,R},\ \{\mathfrak{o}^{\mathrm{no}}_{R,c,T}\}_{(c,T)\in\widehat{\Gamma}_R},\ \{\mathfrak{o}^{\mathrm{supp}}_{R,\gamma}\}_{\gamma\in\Gamma_R^{\mathrm{tes}}}$$

These classes are typed in Definition 10.54, the no-overcount label paragraph following $\widehat{\Gamma}_R$, and Theorem 10.39. They are not determined by Δ_5, by the denominator algebra, or by the OP scalar square. Vanishing here means chosen null-trivializations in the displayed finite obstruction groups or complexes, compatible with the successor transition $R^+ \to R$ and hence with its composites; a numerical zero in a Grothendieck group is not such a trivialization. The Pfaffian part has the sharp finite consequence recorded in Proposition 10.58: active target degrees in $\Gamma_R^{\mathrm{test},+}$ must lie in the source image, but the compact parity dimensions themselves remain extra source data.

(consequence) If a tuple satisfying Definition 10.54 is supplied, then it supplies hypothesis (D0) of Theorem 3.15. Its primitive bracket is compatible with normal-ordered descent on the nose:

$$[P^{\Pi}_{X,\gamma}, P^{\Pi}_{X,\gamma'}]_{\Theta} \subset P^{\Pi}_{X,\gamma+\gamma'}, \qquad \gamma,\gamma' \in \Gamma_{\mathrm{gram}},$$

and its Pfaffian section has the formal cusp expansion

$$\mathrm{pf}_X|_{\mathfrak{c}_\infty} = 64 q^{1/2} r^{1/2} s^{1/2} \prod_{\gamma=(n,l,m)\in\Gamma_{\mathrm{eff}}} (1-q^n r^l s^m)^{f(nm,l)},$$

which the imported Borcherds–Gritsenko–Nikulin theorem identifies with the normalized Δ_5 product at $\mathfrak{c}_\infty$. This is not a section-level equality in the automorphic line; that conclusion requires the supplied Pfaffian-to-automorphic comparison ι_{aut} of Theorem 3.15.

(use of D_0) Every use of (D0) in Theorem 3.15 factors through one of the displayed clauses of Definition 10.54: finite state and transition data through (D0-F)–(D0-S), reduced orientation descent through (D0-Q), source/observable comparison through (D0-O), normal-ordered grading through (D0-C) and (D0-N), the protected first-order complex through (D0-D), and the Pfaffian target windows, support residuals, signed parity exponents, and leading coefficient through (D0-Pf).

(open) The compact state-side construction of $\widehat{\mathcal{H}}^{or}_{\sigma,S}$, the finite Hall source kernels $\mathfrak{S}_R$, the finite Hall–Dirac atlases $\mathfrak{A}_R$, the finite-stage hybrid wrapped object $\mathcal{F}_R^{D_0}$, the obstruction trivializations $\mathfrak{O}_{D_0,R} = 0$ and $\mathfrak{o}^{\mathrm{atlas}}_{R^+R} = 0$, the compatible inverse system, the Pfaffian line on the reduced compact

quotient, and the compact parity pieces $P^{\Pi,+}_{R,\gamma,\bar{0}}$, $P^{\Pi,+}_{R,\gamma,\bar{1}}$ on $\Gamma^{\mathrm{test},+}_R$, including vanishing in target-zero degrees, are not constructed in this paper. The target K_0-shadow supplies only the signed differences; it does not trivialize the Pfaffian square-root torsor or select a compact parity lift.

Proof. For fixed R, the set $F_{\sigma,S}(R)$ is finite by Lemma 10.53. Since h_S is positive on active semistable charges, the length of a decomposition $c = c_1 + \cdots + c_k$ with $\sum_i h_S(c_i) \leq R$ is bounded. Hence only finitely many sums $\sum_{i<j} B(c_i, c_j)$ occur, so $\widehat{\Gamma}_R$ and Γ^{Π}_R are finite. The function $h_\Delta(n, l, m) = |n| + |l| + |m|$ is proper on the lattice Γ_{gram}, so $\Gamma_{\Delta,R}$ is finite. Since $\sup_R N^\Delta_R = \infty$, the union over all R is $\Gamma_{\mathrm{eff}} \setminus \{0\}$. Thus Γ^{test}_R is finite. If $R' \geq R$, then $F_{\sigma,S}(R) \subset F_{\sigma,S}(R')$ and every decomposition of height $\leq R$ is also a decomposition of height $\leq R'$; this gives the displayed inclusions of $\widehat{\Gamma}_R$, Γ^{Π}_R, and $\Gamma_{\Delta,R}$. Lemma 6.9 gives associativity of $\star$ and $\overline{\Pi}_X((c,T) \star (c',T')) = \overline{\Pi}_X(c,T) + \overline{\Pi}_X(c',T')$. The imported item is Proposition 4.5 together with Theorem 9.2 and Proposition 10.2.

Proposition 10.61 is the finite source package determined by the displayed source kernel: the Hall vector spaces are compactly supported vanishing-cycle cohomologies, the product and coproduct are the finite pull-push maps, the primitive bracket is the normal-ordered supercommutator, and the first-order CE differential is the standard derivation dual to that finite Lie bracket. Proposition 10.62 constructs the minimal target-window parity model directly from the finite exponent function a_Δ, and identifies the exact extra condition needed to compare it with the compact source primitives.

The residual tuple $\mathfrak{O}_{D_0,R}$ is a collection of already typed obstructions. The transition component records compatibility in the finite inverse system. The quotient-orientation component is the criterion of Lemma 5.8. The observable component records that the source/target comparison respects the finite product, coproduct, bracket, trace, and primitive comparison. The next seven are the finite-topology, Hochschild, triple, Jacobi, Frobenius, negative-cyclic, and Hopf-radical ideal/coideal obstruction classes of Theorem 10.39; the final three state exactly the finite target coverage, absence of invisible zero-target primitive support, signed Pfaffian multiplicity, and cusp-leading normalization required in the Pfaffian line. Their vanishing is therefore necessary for the finite stage to be a D_0-stage, and, once the finite stacks, correspondences, first-order complex, comparison map, and Pfaffian line have been supplied, is precisely the extra trivialization data demanded by Definition 10.54.

Assume the D_0-datum. Clause (D0-F) gives the finite stages and transition maps. Clause (D0-S) gives the HN-completed state object on which the Hall operations are defined. Clause (D0-Q) supplies the reduced quotient and orientation line used in the Pfaffian construction. Clause (D0-C) places operations in the corrected charge extension. Clause (D0-N) supplies the chain-level shift categorifying the additivity of $\overline{\Pi}_X$; hence the descended primitive bracket is Γ_{gram}-graded. Clause (D0-O) supplies the source observable shadow and comparison map. Clauses (D0-D) and (D0-Pf) supply the first-order object and its Pfaffian line. The support, signed-superdimension, target-window exhaustion, and leading-coefficient conditions in (D0-Pf), combined with the imported Borcherds product, give the normalized Δ_5 cusp expansion for pf_X.

It remains to check the stated use of (D0) in Theorem 3.15. Step 1 uses only the supplied first-order object, the normal-ordered primitive multiplicities, and the leading Pfaffian normalization; these are (D0-D) and (D0-Pf). Step 2 uses the imported Borcherds–Gritsenko–Nikulin product, not a compact construction. Step 3 uses the Pfaffian cusp normalization from (D0-Pf) and the imported Oberdieck–Pixton scalar normalization. Step 5 uses only the supplied chiral protected first-order theory and its decategorification, supplied by (D0-F)–(D0-Pf). Step 4 is independent of D_0 except for the existence of the Pfaffian object; its signs come from (O1), (O1)$^+$, and (O2). Thus the datum implies every use of (D0) and no stronger compact existence claim is made.

The final item follows because none of the source-side objects or trivializations named there is constructed by the lattice calculation, by the denominator algebra, or by the scalar OP square. □

Definition 10.65 *(Dirac–Igusa realization datum).* A compact realization of the Igusa square root is a Dirac–Igusa realization datum

$$\mathfrak{K}_X = (L_X, H_X, O_X, D_X, R_X).$$

Its first-order truncation is the D_0-datum of Definition 10.54. The full datum adds the orientation-monodromy, chiral Koszul, and primitive-recognition data needed to identify the compact state-side algebra with the Gritsenko–Nikulin target. It is not a consequence of the Borcherds product, the scalar Oberdieck–Pixton square, the target current envelope, the target bar–cobar counit, or the existence of orientation lines alone. It is the list of data whose construction would turn the virtual determinant $\mathcal{D}_X = \Delta_5$ into a compact first-order protected realization. The five entries have the following meanings.

(L_X) *Local charge and E_3 host.* This entry consists of the formal dyonic Mukai lattice Γ_X^{form}, the algebraic/effective Hall support inside it, the Gram map $\Pi_X : \Gamma_X^{\mathrm{form}} \to \Gamma_{\mathrm{gram}}$, and the cocycle B of Definition 6.5 and Lemma 6.8. The Hall category is graded by the supplied algebraic/effective charge support, not by Γ_{gram} itself. (L_X) also includes the normal-ordered extension

$$\widehat{\Gamma}_X = \Gamma_X^{\mathrm{form}} \oplus_B \Gamma_{\mathrm{gram}}, \qquad \overline{\Pi}_X(c, T) = \Pi_X(c) - T,$$

and the primitive saturated lifts of Lemma 6.6. Finally, it includes a supplied charge-completed holomorphic E_3-factorization algebra, denoted $\mathcal{A}_X^{E_3}$, on $X = K3 \times E$, finite-first after the sectorial HN quotients of Lemma 10.53. A compact holomorphic-Chern–Simons/BV construction, if it satisfies descent, the quantum master equation, and anomaly cancellation in the completed charge category, is a candidate source for this datum. The datum also includes a fixed E_3-formality point, a chain-level homotopy/framing datum for the chosen operadic model, compact-support control, and continuity of the E_3 operations across the finite HN inverse system.

(H_X) *Hybrid $K3$-to-E factorization shadow.* This entry consists of a supplied finite-first $K3$-to-E source interface in the sense of Definition 10.8, a chain-level specialization $\mathrm{Sp}^{\mathrm{ch}}_{K3,E}$, and named comparison maps κ_E (chiral/factorization rectification comparison) and π_*^{ch} (factorization pushforward identification), with the required rectified finite-chain zigzag

$$\mathcal{A}_X^{E_3} \xrightarrow{\mathrm{Sp}^{\mathrm{ch}}_{K3,E}} \mathcal{F}_X^{(0)} \xrightarrow{\kappa_E} \mathcal{F}_X^{(1)} \xrightarrow{\pi_*^{\mathrm{ch}}} (p_E)_!^{\mathrm{fact}} \mathcal{A}_X^{E_3}$$

in the projection-finite sectors. These arrows are supplied finite-chain comparison data; they are not constructed by the determinant, by the Francis–Gaitsgory chiral Koszul theorem, or by ordinary properness of p_E. They become strict equivalences only after the finite obstruction tuple $\mathfrak{O}_R^{K3\to E}$ of the finite $K3$-to-E square above vanishes compatibly in R. In the wrapped elliptic-degree sectors, the required filling is the finite-HN inverse system whose limit is the hybrid object $\mathcal{F}_X^{\mathrm{hyb}}$ of Definition 10.27. Thus (H_X) includes the hybrid Ran base $\mathrm{Ran}^{\mathrm{hyb}}(E)$, projection-finite local sectors $\mathcal{F}_{\alpha,R}^{\mathrm{loc}}(U)$, rigidified wrapped prequotient sectors $\mathcal{G}_{\eta,R}^{\mathrm{wr}}$ with anchors $\lambda_{\eta,R}$, finite correspondence categories $\mathsf{Corr}_R^{E,\mathrm{hyb}}$, and the three finite operation families

$$\mu_R^{\mathrm{loc}}, \qquad \mu_R^{\mathrm{hyb}}, \qquad \mu_R^{\mathrm{wr}},$$

where μ_R^{hyb} includes the ordered two-sided mixed maps $\mu^{\mathrm{hyb}}_{\alpha_-;\eta;\beta_+,R}$. It also includes the eight-word two-step binary flag atlas in the finite HN quotients, the higher colored residual $\mathfrak{o}_R^{\mathrm{col}}$, the properness/admissibility residuals which make $\mathcal{B}_R^{E,\mathrm{hyb}}$ an actual correspondence-module functor, the quotient operation $Q_{E,R}$ applied only after the extension correspondences are formed, the quotient comparison isomorphisms $\theta^Q_{\mu,R}$, and protected integration maps I_R^{prot} whose homogeneous s-degree is the Gram trace exponent $m_R = \mathrm{pr}_3\,\overline{\Pi}_X$, while b_R^{geom} records geometric wrapped support. Equivalently,

$$\mathfrak{O}_{\mathrm{hyb},R} = 0$$

for every finite height R, compatibly with transition maps. The orientation lines used in these hybrid pull-push operations are the local coefficient part of (O_X); the Weyl-equivariant monodromy package $(\mathrm{O1})^+$ and the type-II wall package $(\mathrm{O2})$ are additional data. A global Fock/Hecke sector is acceptable here only if it is represented by the rigidified wrapped prequotients and acts through the mixed maps μ_R^{hyb} before the reduced E-quotient. These equivalences and wrapped-sector data are part of the datum, not operations constructed from the determinant.

(O_X) *Oriented compact Hall object and Weyl orientation character.* This entry consists of an oriented compact Hall factorization object $\mathcal{H}_X$ graded by supplied effective charges c in the formal dyonic lattice, with finite-type sectorial moduli stacks and extension correspondences

$$\mathfrak{M}_c^\sigma \times \mathfrak{M}_{c'}^\sigma \xleftarrow{p_{c,c'}} \mathfrak{E}_{c,c'}^\sigma \xrightarrow{q_{c,c'}} \mathfrak{M}_{c+c'}^\sigma,$$

perfect reduced obstruction theories or critical (-1)-shifted symplectic structures, vanishing-cycle coefficients, orientation lines with chosen square roots, Thom–Sebastiani orientation transport coherent for triple extensions, and protected integration maps compatible with HN completion and wall crossing. The orientation entry also includes the exact quotient-descent package. For every charge stratum C used by the Hall correspondences one must specify

$$\alpha_C^{\mathrm{red}} = w_2(\det \mathrm{RHom}_{\mathrm{red}}(C,C)) + \mathrm{cs}^* w_2(\det(\operatorname{coker}\mathrm{cs})) \in H^2(\mathfrak{M}_C^{\mathrm{red}};\mathbb{F}_2),$$

$$\alpha_C^{E,\mathrm{free}} = a_1(C)u_1 + a_2(C)u_2 \in H^2(BE;\mathbb{F}_2),$$

and, for every finite-stabilizer stratum S with stabilizer $E[N]$,

$$\widetilde{\beta}_{C,S}^{E,N} \in H^2_{E[N]}(S;\mathbb{F}_2).$$

The point-stabilizer class

$$\beta_{C,S}^{E,N} \in H^2(BE[N];\mathbb{F}_2)$$

is used only after the mixed Borel terms, the stabilizer action on $H^\bullet(S;\mathbb{F}_2)$, and the Borel spectral-sequence differentials have vanished. One must prove these Borel classes, and then their point-stabilizer restrictions, vanish on the corresponding free and small-orbit strata. At $N = 2$, after this restriction,

$$\beta_{C,S}^{E,2} = b_{20}x_1^2 + b_{11}x_1x_2 + b_{02}x_2^2 \in H^2(BE[2];\mathbb{F}_2),$$

with $E[2] = \langle e_1, e_2\rangle$, $e_3 = e_1 + e_2$, and

$$\mathrm{res}_{\langle e_i\rangle}\,\beta_{C,S}^{E,2} = r_i(C,S)\tau_i^2, \qquad (b_{20}, b_{11}, b_{02}) = (r_1, r_1 + r_2 + r_3, r_2).$$

Thus the finite two-torsion residual vanishes exactly when $r_1 = r_2 = r_3 = 0$, but the full quotient descent also requires $\alpha_C^{\mathrm{red}} = 0$ and $a_1(C) = a_2(C) = 0$. After this degree-two two-torsion class vanishes, the chosen $E[2]$-linearization has a degree-one character

$$\lambda_C^{E[2]} = \lambda_1 x_1 + \lambda_2 x_2 \in H^1(BE[2]; \mathbb{F}_2),$$

and the quotient orientation requires $\lambda_1 = \lambda_2 = 0$. Odd N finite stabilizers have no positive-degree mod-2 obstruction; even $N \geq 4$ is detected on the 2-primary stabilizer $(\mathbb{Z}/2^a)^2$, $2^a \parallel N$, where for $a \geq 2$

$$H^*(B(\mathbb{Z}/2^a)^2; \mathbb{F}_2) = \Lambda(x_1, x_2) \otimes \mathbb{F}_2[y_1, y_2], \qquad |x_i| = 1, \quad |y_i| = 2, \quad x_i^2 = 0,$$

$$\beta_{C,(2)}^{E,N} = A_1^{(N)} y_1 + A_{12}^{(N)} x_1 x_2 + A_2^{(N)} y_2.$$

The mixed coefficient $A_{12}^{(N)}$ is not detected by cyclic restrictions; after this degree-two class vanishes, the residual $H^1(B(\mathbb{Z}/2^a)^2; \mathbb{F}_2)$-linearization character must also vanish. Thus the even $N \geq 4$ residual remains part of the descent datum; it is not implied by the $E[2]$ vector. The Hall atlas must include local-local, wrapped-wrapped, and mixed local/wrapped extension correspondences before quotienting by the translation action of E; quotienting first forgets the relative E-position needed by mixed convolution. If a lifted action of $W^{(2)}(\Lambda_{II}^{2,1})$ on the oriented completed Hall object is supplied, its finite-stage obstruction is first the projective determinant-line cocycle on the retained action groupoid $\mathcal{G}_R$:

$$[c_{o,R}] \in H^2(\mathcal{G}_R; \underline{\mathbb{F}}_2).$$

Only on a complete $W^{(2)}(\Lambda_{II}^{2,1})$-orbit with constant coefficients does this reduce to

$$[c_o] \in H^2\Big(W^{(2)}(\Lambda_{II}^{2,1}); \mathbb{F}_2\Big).$$

The relevant groupoid or group class must vanish, the generator lifts must satisfy $s_{\partial_i}^2 = 1$, and the full quotient package, including the zero finite characters $\lambda_C^{E,N} = 0$, must be transported with zero

$$H^1(\mathfrak{M}_{s_{\partial_i} C}^{\mathrm{red}}; \mathbb{F}_2) \oplus \bigoplus_{N \geq 2} H^1(BE[N]; \mathbb{F}_2)$$

torsor defect, as in Lemma 3.24. Only then does the induced orientation character ϵ_o satisfy

$$\epsilon_o|_{W^{(2)}(\Lambda_{II}^{2,1})} = \nu_{\Delta_5}|_{W^{(2)}(\Lambda_{II}^{2,1})}.$$

Constructing the domain of this comparison and the character ϵ_o is part of the datum. The comparison is a geometric orientation-monodromy condition on the Hall side; it requires computing the determinant-line signs on the three type-II simple reflections and checking the reflection relations. It is not obtained by choosing the scalar normalization of Δ_5.

(D_X) *Normal-ordered descent and chiral Koszul comparison.* This entry consists of the chain-level normal-ordering package

$$\mathsf{Ch}_{k,\widehat{\mathrm{HN}},E}^{\Gamma_{\mathrm{gram}}}, \quad \{\mathsf{T}_\eta\}_{\eta \in \Gamma_{\mathrm{gram}}}, \quad B_{\mathrm{ch}}, \quad \Theta_\Pi, \Theta_\Pi^-, \quad \overline{\Pi}_{X,*}^{\Theta} \mathcal{F}_X,$$

as in Definition 10.35 and Theorem 10.39. The protected primitives are Γ_{gram}-graded only after applying $\overline{\Pi}^{\Theta}_{X,*}$. Their bracket is induced from the upstairs Γ^{phys}_X-graded Hall bracket by translating the raw target degree

$$\Pi_X(c+c') = \Pi_X(c) + \Pi_X(c') + B(c,c')$$

through $\mathsf{T}_{B(c,c')}$, hence by shifting the displayed Gram degree by $-B(c,c')$. The datum includes the triple-coherence pentagon over the two-step Hall flag stacks, Jacobi compatibility of the transported bracket, the S^1-equivariant negative-cyclic lift

$$(b + uB_{\mathrm{C}})\Theta^-_{\Pi} = B^-_{\mathrm{ch}},$$

and compatibility with the Joyce–Upmeier orientation cocycle, the graded Frobenius identity, the completed Hopf pairing, and the reduced E-quotient. Equivalently, the seven obstruction classes

$$\mathfrak{o}^{\mathrm{top}}_{\Theta}, \quad \mathfrak{o}^{\mathrm{Hoch}}_{\Theta}, \quad \mathfrak{o}^{\mathrm{tri}}_{\Theta}, \quad \mathfrak{o}^{\mathrm{Jac}}_{\Theta}, \quad \mathfrak{o}_F, \quad \mathfrak{o}^{\mathrm{cyc}}_{\Theta}, \quad \mathfrak{o}^{\mathrm{rad}}_{\Theta}$$

of Theorem 10.39 vanish on every finite HN quotient and compatibly in the inverse limit, with $\mathfrak{o}^{\mathrm{top}}_{\Theta}$ enforcing finite-fibre descent and completed tensor/coproduct continuity, and with $\mathfrak{o}^{\mathrm{rad}}_{\Theta}$ read as the Lie-ideal and coideal pair. The finite-stage tests are the seven diagrams (NO1)–(NO7) of Proposition 10.40. The finite closure criterion Proposition 10.48 separates the formal Picard normal-ordering lift from the non-formal Hall correspondence, cyclic trace, and Hopf-radical compatibilities. Before this descent, the protected primitives of $\mathcal{F}_X$ are graded by the upstairs additive normal-ordered labels (c,T), not by a Gram triple. The comparison with $\mathfrak{g}_{\Delta_5,\alpha(n,l,m)}$ is made after passing to $(n,l,m) = \overline{\Pi}_X(c,T)$; the shorthand $\Pi_X(c)$ is legitimate only on labels with $T = 0$. (D_X) also includes the full chiral Koszul source datum of Definition 10.19. Thus, at every finite HN height R, one must supply a conilpotent chiral coalgebra $C_{X,R}$ attached to the hybrid source $\mathcal{F}^{\mathrm{hyb}}_{X,\sigma,S,\le R}$, a finite source charge filtration $F^{\mathrm{co}}_{R,\bullet}$, the collision coproduct Δ^{ch}_R, a charge-preserving bar comparison

$$b_{X,R} : C_{X,R} \xrightarrow{\simeq} \bar{B}^{\mathrm{ch}}_E(\mathcal{F}^{\mathrm{hyb}}_{X,\sigma,S,\le R}),$$

and a chiral cobar quasi-isomorphism

$$\Theta_{\mathrm{Kos},R} : \Omega^{\mathrm{ch}}_E C_{X,R} \xrightarrow{\simeq} \mathsf{Den}_{\Delta_5,E,\le R},$$

compatibly in R, and the Koszul Mittag–Leffler exactness hypothesis of Proposition 10.22. In the inverse limit this gives

$$b_X : C_X \xrightarrow{\simeq} \bar{B}^{\mathrm{ch}}_E(\mathcal{F}^{\mathrm{hyb}}_X), \qquad \Theta_{\mathrm{Kos}} : \Omega^{\mathrm{ch}}_E C_X \xrightarrow{\simeq} \mathsf{Den}_{\Delta_5,E}.$$

The target-internal counit $\Omega^{\mathrm{ch}}_E \bar{B}^{\mathrm{ch}}_E(\mathsf{Den}_{\Delta_5,E}) \to \mathsf{Den}_{\Delta_5,E}$ does not supply C_X, b_X, or Θ_{Kos}. These are source-side compact data. Their finite residual is the vanishing of $\mathfrak{O}_{\mathrm{Kos},R}$ in every finite HN quotient, including $\mathfrak{o}^{\mathrm{hyb}}_R$, the assertion that the finite hybrid source has been built with all correspondences, quotient, and integration laws, $\mathfrak{o}^{\Delta,\mathrm{ch}}_R$ and $\mathfrak{o}^{C,\mathrm{tr}}_{R'R}$, the finite collision-coproduct and transition-continuity assertions, the Koszul $R^1\varprojlim = 0$ condition for the source coalgebras, completed bar/cobar systems, target truncations, and cones of $b_{X,R}$ and $\Theta_{\mathrm{Kos},R}$, $\mathfrak{o}^{\Omega}_R$, the split cobar residual

$$\mathfrak{o}^{\Omega}_R = (\mathfrak{o}^{\varepsilon,\mathrm{src}}_R, \mathfrak{o}^{\vartheta}_R)$$

of Corollary 10.21, and $\mathfrak{o}_R^{\mathrm{sep}}$, the assertion that the source coalgebra, filtration, collision coproduct, bar comparison, and primitive restriction were supplied independently of the target counit.

(R_X) *Primitive recognition.* This entry is a primitive recognition datum Π_{rec} for

$$P_X^{\Pi} := H^0 \operatorname{Prim}_{\mathrm{prot}}(\overline{\Pi}^{\Theta}_{X,*}\mathcal{F}_X).$$

This datum is not a determinant datum and is not the Gritsenko–Nikulin target itself. It consists of a source-side Gram-descended Lie superalgebra structure on P_X^{Π}, the parity split of each homogeneous primitive piece, compact representatives for the real and imaginary simple-root fibres of the Gritsenko–Nikulin presentation, verification of the Borcherds–Kac defining relations in the protected Hall bracket, the completed Hopf pairing, generation by the simple primitives and the Cartan, a no-extra-relations theorem, completed PBW compatibility, and identification of the source Hopf radical with the standard Gritsenko–Nikulin/Kac radical after completion.

Theorem 10.66 *(Primitive recognition from finite source representatives).* Let $\mathscr{B}_{\Delta_5} = (H, I, p, \alpha_i, (\ ,\ ))$ be the imported Gritsenko–Nikulin Borcherds–Cartan superdatum for the automorphic correction. Thus

$$H = \Lambda_{II}^{2,1} \otimes \mathbb{C}, \qquad \{\delta_1, \delta_2, \delta_3\} \subset \Lambda_{II}^{2,1} \subset H,$$

the three δ_i are even real simple roots, and

$$((\delta_i, \delta_j))_{1\le i,j\le 3} = \begin{pmatrix} 2 & -2 & -2 \\ -2 & 2 & -2 \\ -2 & -2 & 2 \end{pmatrix}.$$

The remaining fibres of $I \to H^*$ are the imaginary simple-root fibres with the Gritsenko–Nikulin parity split $\mu_{\bar{0}}(a), \mu_{\bar{1}}(a)$ of Definition 10.86. The degree map is

$$\alpha(n, l, m) = 2nf_2 - lf_3 + 2mf_{-2} = n\delta_1 + m\delta_2 + (n + m - l)\delta_3.$$

Let $\widehat{\mathfrak{F}}(\mathscr{B}_{\Delta_5})$ be the completed free Lie superalgebra generated by the even Cartan H and by parity-homogeneous generators $e_i, f_i, i \in I$, with $|e_i| = |f_i| = p(i)$ and $\deg e_i = \alpha_i$, $\deg f_i = -\alpha_i$. Let J_{BK} be the closed Borcherds–Kac ideal generated by the Cartan, Chevalley, real Serre, orthogonality, and super sign relations. After quotienting by the standard radical of the completed invariant pairing one obtains the target denominator algebra

$$G(\mathscr{B}_{\Delta_5}) = \widehat{\mathfrak{F}}(\mathscr{B}_{\Delta_5})/\overline{J_{\mathrm{BK}} + \mathrm{Rad}_{\mathrm{GN}}} = \mathcal{A}_{\mathrm{den}} = \mathfrak{g}_{\Delta_5}.$$

Here $\mathrm{Rad}_{\mathrm{GN}}$ denotes the preimage in the completed presentation of the standard radical. This target-side statement is imported from the Gritsenko–Nikulin/Kac construction [37–39, 48]; it is not a compact Hall construction.

Let a compact Hall/factorization construction give a completed $\widehat{\Gamma}_X$-graded primitive Hall Lie superalgebra

$$\widetilde{P}_X = \prod_{\widehat{c}\in\widehat{\Gamma}_X} \widetilde{P}_{X,\widehat{c}}, \qquad [\widetilde{P}_{X,\widehat{c}}, \widetilde{P}_{X,\widehat{c}'}] \subset \widetilde{P}_{X,\widehat{c}\star\widehat{c}'},$$

and a completed Hopf pairing. Its normal-ordered Gram descent is

$$P_X^{\Pi,\mathrm{raw}} = H^0 \operatorname{Prim}_{\mathrm{prot}}(\overline{\Pi}^{\Theta}_{X,*}\mathcal{F}_X) = \overline{\Pi}^{\Theta}_{X,*}\widetilde{P}_X$$

and

$$P_X^{\Pi,\mathrm{red}} = P_X^{\Pi,\mathrm{raw}}/\overline{\mathrm{Rad}\langle\,\cdot\,,\,\cdot\,\rangle_X}.$$

Construct the following source-side presentation data. None of these data is inferred from the Borcherds product, from the determinant, or from signed dimensions alone.

(S1) *Charge and normal-ordered descent.* The chain-level descent $\overline{\Pi}^{\Theta}_{X,*}$ categorifies the normal-ordered homomorphism

$$\overline{\Pi}_X : \widehat{\Gamma}_X \longrightarrow \Gamma_{\mathrm{gram}}, \qquad \overline{\Pi}_X(c,T) = \Pi_X(c) - T,$$

of Definition 6.1. For $\widehat{c} = (c,T)$ and $\widehat{c}' = (c',T')$,

$$\overline{\Pi}_X(\widehat{c} \star \widehat{c}') = \overline{\Pi}_X(\widehat{c}) + \overline{\Pi}_X(\widehat{c}'),$$

so the descended bracket is a Γ_{gram}-graded bracket:

$$[\widetilde{P}_{X,\widehat{c}}, \widetilde{P}_{X,\widehat{c}'}] \subset \widetilde{P}_{X,\widehat{c}\star\widehat{c}'} \xrightarrow{\overline{\Pi}^{\Theta}_{X,*}} P^{\Pi,\mathrm{raw}}_{X,\overline{\Pi}_X(\widehat{c})+\overline{\Pi}_X(\widehat{c}')}.$$

Equivalently, the raw quadratic shadow $\Pi_X(c+c') = \Pi_X(c) + \Pi_X(c') + B(c,c')$ has been corrected by the Θ-degree shift before comparison with the denominator algebra. Each descended homogeneous parity piece is finite dimensional. The construction of (Θ), the orientation datum (O), and the chain-level Frobenius compatibility is exactly the compact content of Theorem 10.39, Open Problem 10.34, and Open Problem 5.7.

(S2) *Cartan and root-degree matching.* There is an even Cartan identification

$$\iota_H : P^{\Pi,\mathrm{red}}_{X,0} \xrightarrow{\sim} \Lambda^{2,1}_{II} \otimes \mathbb{C}.$$

The degree map $\alpha(n,l,m) = 2nf_2 - lf_3 + 2mf_{-2}$ identifies the descended Gram grading with the Gritsenko–Nikulin root grading after applying $\overline{\Pi}^{\Theta}_{X,*}$. The upstairs Hall bracket remains $\widehat{\Gamma}_X$-graded; only the descended object is Γ_{gram}-graded.

(S3) *Simple primitive representatives and parities.* For every simple-root index $i \in I$ of $\mathscr{B}_{\Delta_5}$, the compact source supplies parity-homogeneous primitive representatives

$$e_i^X \in P^{\Pi,\mathrm{red}}_{X,\alpha_i}, \qquad f_i^X \in P^{\Pi,\mathrm{red}}_{X,-\alpha_i}, \qquad h_i^X \in P^{\Pi,\mathrm{red}}_{X,0}$$

with $|e_i^X| = |f_i^X| = p(i)$ and $h_i^X = \iota_H^{-1}(\alpha_i)$. The representatives preserve the real/imaginary distinction, the multiplicity fibres, and the parity map. The real representatives for $\delta_1, \delta_2, \delta_3$ are even. For each imaginary simple root a, the supplied representatives have dimensions

$$\dim\langle e_i^X : \alpha_i = a,\ p(i) = 0\rangle = \mu_{\bar{0}}(a), \qquad \dim\langle e_i^X : \alpha_i = a,\ p(i) = 1\rangle = \mu_{\bar{1}}(a)$$

with the same convention for the negative representatives. The negative representatives belong to the supplied BKM double and its independent negative primitive half. They are not constructed or tested by the cusp-positive Pfaffian support residuals of (Do-Pf).

(S4) *Borcherds–Kac relations in the Hall bracket.* In the protected Hall/factorization bracket these representatives satisfy the Borcherds–Kac defining relations for $\mathscr{B}_{\Delta_5}$:

$$[h, h'] = 0, \qquad [h, e_i^X] = (h, \alpha_i) e_i^X, \qquad [h, f_i^X] = -(h, \alpha_i) f_i^X,$$
$$[e_i^X, f_j^X] = \delta_{ij} h_i^X,$$

the real Serre relations

$$(\operatorname{ad} e_i^X)^{1-2(\alpha_i,\alpha_j)/(\alpha_i,\alpha_i)} e_j^X = (\operatorname{ad} f_i^X)^{1-2(\alpha_i,\alpha_j)/(\alpha_i,\alpha_i)} f_j^X = 0$$

for real i and $i \neq j$, the Borcherds orthogonality relations

$$(\alpha_i, \alpha_j) = 0 \quad \Longrightarrow \quad [e_i^X, e_j^X] = [f_i^X, f_j^X] = 0,$$

and the super sign rule. In the real rank-three subsystem $(\delta_i, \delta_j) = -2$ for $i \neq j$, so the real Serre exponent is 3; the first brackets $[e_i^X, e_j^X]$ are not forced to vanish. For the primitive isotropic simple roots

$$a_{ij} := \delta_i + \delta_j, \qquad i < j,$$

the source representatives $u_{ij,r}^{+,X} \in P_{X,a_{ij}}^{\Pi,\mathrm{red}}$ and $u_{ij,r}^{-,X} \in P_{X,-a_{ij}}^{\Pi,\mathrm{red}}$, $1 \leq r \leq 9$, satisfy the orthogonality and Chevalley zero relations with the two adjacent real roots,

$$[e_i^X, u_{ij,r}^{+,X}] = [e_j^X, u_{ij,r}^{+,X}] = [f_i^X, u_{ij,r}^{+,X}] = [f_j^X, u_{ij,r}^{+,X}] = 0,$$

with the analogous negative-root relations, and for the complementary root $\{i, j, k\} = \{1, 2, 3\}$ the real-string relations

$$(\operatorname{ad} e_k^X)^5 u_{ij,r}^{+,X} = 0, \qquad (\operatorname{ad} f_k^X)^5 u_{ij,r}^{-,X} = 0.$$

These are relation checks in the compact bracket. They are not consequences of the equality sdim $= f(nm, l)$. The number 9 is the Gritsenko–Nikulin/Kac target multiplicity in the first finite window computed in Proposition 10.82; the compact source contains representatives with these ranks; the determinant records only the resulting signed trace.

(S5) *Completed Hopf pairing and radical quotient.* The completed Hopf pairing on the positive and negative primitive halves is continuous for the Hall completion, homogeneous for the Γ_{gram}-grading, invariant for the bracket and coproduct, and has closed homogeneous radical

$$\mathrm{Rad}_X = \{x \mid \langle x, y \rangle_X = 0 \text{ for all } y\}$$

which is a Lie ideal and a coideal. After quotienting by $\overline{\mathrm{Rad}_X}$, identify the source radical quotient with the Gritsenko–Nikulin/Kac quotient by $\overline{\mathrm{Rad}_{\mathrm{GN}}}$. Equivalently, on every finite-height saturated set of roots the positive-negative pairing matrices are supplied with the same parity-block ranks as the target invariant pairing matrices, their kernels are carried by the transition maps, and the inverse-limit kernel is exactly Rad_X. At every finite stage one has

$$[P_R^\Pi, \mathrm{Rad}_{X,R}] \subset \mathrm{Rad}_{X,R}, \qquad \Delta_R(\mathrm{Rad}_{X,R}) \subset \mathrm{Rad}_{X,R} \otimes P_R^\Pi + P_R^\Pi \otimes \mathrm{Rad}_{X,R},$$

and the completed statement is obtained by inverse limit and closure. The radical rank is computed from these matrices and kernel maps; it is not read from the signed determinant. Equality of ranks alone is not a Hopf quotient theorem without the Lie-ideal, coideal, closedness, and transition-compatibility assertions.

(S6) *No extra relations.* The Hall primitive algebra satisfies a no-extra-relations theorem: if

$$\pi : \widehat{\mathfrak{F}}(\mathscr{B}_{\Delta_5}) \longrightarrow P_X^{\Pi,\mathrm{red}}$$

is the continuous map determined by the displayed representatives, then

$$\ker \pi = \overline{J_{\mathrm{BK}} + \mathrm{Rad}_{\mathrm{GN}}}.$$

Equivalently, the induced map $G(\mathscr{B}_{\Delta_5}) \to P_X^{\Pi,\mathrm{red}}$ is injective. This is required finite stage by finite stage: for every saturated finite root window W, the restricted map π_W has kernel

$$\ker \pi_W = \left(\overline{J_{\mathrm{BK}} + \mathrm{Rad}_{\mathrm{GN}}}\right)_W$$

after projection to W, and these kernel identifications commute with the HN transition maps. A comparison of signed dimensions or a determinant identity is not this theorem.

(S7) *Generation.* $P_X^{\Pi,\mathrm{red}}$ is topologically generated by $P_{X,0}^{\Pi,\mathrm{red}}$ and the simple primitive representatives e_i^X, f_i^X.

(S8) *Completed PBW.* A completed PBW theorem holds for $P_X^{\Pi,\mathrm{red}}$: the PBW filtration on its enveloping algebra is transported to the PBW filtration on $U(\mathfrak{g}_{\Delta_5})$, the associated graded finite-height pieces agree with the target pieces, and the finite-height PBW bases form a closed inverse system in the chosen completion. Concretely, for every finite saturated window W, the associated graded map

$$\mathrm{gr}_{\mathrm{PBW}}\, U(P_X^{\Pi,\mathrm{red}})_W \longrightarrow \mathrm{gr}_{\mathrm{PBW}}\, U(\mathfrak{g}_{\Delta_5})_W$$

is an isomorphism, the transition maps preserve the PBW filtration strictly, and inverse limits are taken in the same HN topology as Definition 10.35.

(S9) *Exact completion.* The inverse limit is exact on the saturated finite root windows used above: the transition maps are strict for the HN topology, images are closed, and passage to the completed quotient by the radical commutes with finite-window kernels, cokernels, PBW associated gradeds, and root-space parity pieces. Equivalently, no new relation, generator, radical element, or parity-cancelling summand appears only after completion; equivalently, the relevant $\lim^1$ terms vanish on these finite root-window systems.

(S10) *Full parity dimensions.* The full root-space parity dimensions in $P_X^{\Pi,\mathrm{red}}$ agree with those of $G(\mathscr{B}_{\Delta_5})$, not only their differences. For every nonzero γ with $\alpha(\gamma) \in \mathcal{R}$,

$$\dim P_{X,\gamma,\overline{p}}^{\Pi,\mathrm{red}} = d_{\gamma,\overline{p}}^{\mathrm{GN}}, \qquad d_{\gamma,\overline{p}}^{\mathrm{GN}} := \mathrm{mult}_{\overline{p}}(\alpha(\gamma)) \qquad (p = 0, 1).$$

Thus the source matches two integers in every root degree; the determinant records their difference. In the first timelike channel $\gamma = (1, 1, 1)$, Proposition 10.81 gives

$$d_{(1,1,1),\overline{0}}^{\mathrm{GN}} = 29, \qquad d_{(1,1,1),\overline{1}}^{\mathrm{GN}} = 93.$$

These are presentation-level target dimensions, not an inference from $29 - 93 = -64$. If $\alpha(\gamma)$ is not a target root and $\gamma \neq 0$, the corresponding source piece is zero after the radical quotient. On active positive Gram degrees this implies

$$\mathrm{sdim}\, P_{X,(n,l,m)}^{\Pi,\mathrm{red}} = f(nm, l).$$

Then

$$P_X^{\Pi,\mathrm{red}} \cong \mathcal{A}_{\mathrm{den}} = \mathfrak{g}_{\Delta_5}$$

as Gram-graded Lie superalgebras, and its Weyl–Kac–Borcherds denominator is $64^{-1}\Delta_5(2Z)$. If in addition the chiral Koszul source datum of Definition 10.19 is constructed, including the Koszul Mittag–Leffler exactness hypothesis of Proposition 10.22, then

$$\Omega_E^{\mathrm{ch}} C_X \simeq \mathsf{Den}_{\Delta_5,E}.$$

Proof. The simple primitive representatives define a continuous homomorphism

$$\pi : \widehat{\mathfrak{F}}(\mathscr{B}_{\Delta_5}) \longrightarrow P_X^{\Pi,\mathrm{red}}.$$

(S1) and (S2) make this a homomorphism of the descended Gram-graded completions. (S3) fixes the real and imaginary simple generators with their parities and multiplicity fibres. (S4) puts J_{BK} in the kernel. (S5) identifies the compact Hopf radical with the Gritsenko–Nikulin/Kac radical, so π descends to

$$G(\mathscr{B}_{\Delta_5}) \longrightarrow P_X^{\Pi,\mathrm{red}}.$$

(S7) gives surjectivity. (S6) gives injectivity after the completed radical quotient. (S8) gives the PBW-compatible finite-height identifications, and (S9) says that the completed inverse limit is exact for the kernels, radicals, and PBW associated gradeds under discussion. (S10) rules out hidden zero-superdimension root spaces and cancelling ideals invisible to the determinant. Hence the descended map is an isomorphism of Gram-graded Lie superalgebras. The denominator statement is Theorem 9.2. The chiral statement is Proposition 10.22; it uses C_X, b_X, Θ_{Kos}, and the Koszul Mittag–Leffler exactness hypothesis of that proposition, not the target-internal bar–cobar counit. □

Definition 10.67 *(Cofinal finite-window primitive-recognition datum).* A finite set $W \subset Q_+ \setminus \{0\}$ is *downward saturated* if

$$0 < \beta' \le \beta, \qquad \beta \in W, \qquad \beta - \beta' \in Q_+ \quad \Longrightarrow \quad \beta' \in W.$$

Thus every positive presentation word whose total degree lies in W uses only positive degrees already in W. For the full BKM double, each finite datum below is triangular: positive and negative letters are separately restricted to W_ν and $-W_\nu$, mixed Chevalley and pairing checks are made only for the supplied letters in $W_\nu \cup (-W_\nu) \cup \{0\}$, and words whose total degree enters the window only by cancellation are deferred to the exact-completion clause below. A cofinal finite-window primitive-recognition datum for $P_X^{\Pi,\mathrm{red}}$ is an increasing sequence of finite downward-saturated windows

$$W_1 \subset W_2 \subset \cdots, \qquad \bigcup_\nu W_\nu = \mathcal{R}_+,$$

together with the following finite data for every W_ν, compatible under $W_\nu \subset W_{\nu+1}$.

(R1) *Representatives and parities.* All simple-root representatives e_i^X, f_i^X, h_i^X whose roots lie in W_ν are supplied in the compact source, with the Gritsenko–Nikulin parity fibres $\mu_{\bar{0}}(a)$, $\mu_{\bar{1}}(a)$ for imaginary simple roots and even real representatives for δ_1, δ_2, δ_3.

(R2) *Relations.* The Cartan, Chevalley, real Serre, Borcherds orthogonality, and super sign relations are checked for every relation whose displayed source and target degrees lie in $W_\nu \cup (-W_\nu) \cup \{0\}$. Downward saturation makes this a finite relation check for all positive presentation words of total degree in W_ν.

(R3) *Full finite parity dimensions.* For every $\beta \in W_\nu$, the compact root space after the source radical quotient has the two target dimensions

$$\dim P^{\Pi,\mathrm{red}}_{X,\beta,\bar{p}} = \mathrm{mult}^{\mathrm{GN}}_{\bar{p}}(\beta), \qquad p = 0, 1,$$

and the corresponding non-root degrees in the finite test window have zero source piece. This is a two-integer equality in each degree, not a signed-superdimension equality.

(R4) *Hopf pairing and radical.* The positive-negative pairing matrices between W_ν and $-W_\nu$ are written in parity blocks. Their kernels agree with the Gritsenko–Nikulin/Kac invariant-pairing kernels, are stable under all finite brackets whose degrees remain in the window, are coideals for the finite coproduct, and are carried to the next window by the transition maps. The inverse-limit radical is the closure of these finite kernels.

(R5) *No extra relations.* Let

$$\pi_\nu : \widehat{\mathfrak{F}}(\mathscr{B}_{\Delta_5})_{\leq W_\nu} \longrightarrow P^{\Pi,\mathrm{red}}_{X,\leq W_\nu}$$

be the finite presentation map obtained by projecting to words whose letters lie in the triangular finite span generated by $W_\nu \cup (-W_\nu) \cup \{0\}$. The datum requires the finite kernel equality

$$\ker \pi_\nu = \left(\overline{J_{\mathrm{BK}} + \mathrm{Rad}_{\mathrm{GN}}}\right)_{\leq W_\nu}.$$

This is the no-extra-relations theorem in finite form.

(R6) *Generation.* Every source primitive root space in $W_\nu \cup (-W_\nu)$, after the finite radical quotient, is generated by the Cartan and the supplied simple-root representatives through brackets whose intermediate positive degrees stay in W_ν.

(R7) *Completed PBW compatibility.* The PBW filtrations on the finite enveloping spans agree:

$$\mathrm{gr}_{\mathrm{PBW}}\, U(P^{\Pi,\mathrm{red}}_X)_{\leq W_\nu} \xrightarrow{\sim} \mathrm{gr}_{\mathrm{PBW}}\, U(\mathfrak{g}_{\Delta_5})_{\leq W_\nu},$$

and the transition maps $W_{\nu+1} \to W_\nu$ preserve the filtration strictly. The completed PBW statement is the inverse-limit statement for this compatible finite system.

(R8) *Exact triangular completion.* Words whose total degree lies in the finite test window only through positive-negative cancellation are not added as extra finite generators. They are controlled by exact completion: the transition maps are strict, images are closed, and passage to the inverse limit commutes with the finite kernels, cokernels, radicals, PBW associated gradeds, and parity pieces supplied in (R3)–(R7). Equivalently, the relevant $\lim^1$ terms vanish for these triangular finite-window systems.

The datum is finite at each ν, but it is not numerical: it contains representatives, relation verifications, pairing matrices, kernel equalities, and PBW transition data. The Borcherds product and the OP scalar square provide none of these source-side objects.

Proposition 10.68 *(Global recognition is exactly the cofinal datum).* Assume (S1)–(S4) of Theorem 10.66. Then the remaining global source-side assertions (S5)–(S10) are supplied by a cofinal finite-window primitive-recognition datum in the sense of Definition 10.67. Under that datum,

$$P^{\Pi,\mathrm{red}}_X \cong \mathfrak{g}_{\Delta_5}$$

as completed Gram-graded Lie superalgebras. Conversely, any proposed global proof of this isomorphism, with the chosen Gritsenko–Nikulin presentation and source Hopf pairing, must restrict to such finite data on every downward-saturated window.

Proof. For a fixed W_ν, (R1) and (R2) define the finite presentation map and put the Borcherds–Kac relation ideal in its kernel. (R4) identifies the compact finite Hopf radical with the target radical and makes the quotient a finite Hopf Lie-superalgebra quotient. (R5) gives injectivity of the induced target map on the window, while (R6) gives surjectivity. Thus the source and target primitive Lie superalgebras agree on W_ν.

(R3) rules out zero-superdimension source summands and wrong parity lifts invisible to the determinant. (R7) says that the finite isomorphisms are compatible with PBW associated gradeds and with the chosen HN/window transition maps. (R8) is the exact triangular completion clause. Since the windows are cofinal and the radical in (R4) is the closure of the finite kernels, passing to the inverse limit gives (S5)–(S10) and hence the isomorphism of Theorem 10.66.

The converse is restriction: a completed isomorphism preserving the chosen representatives, presentation map, Hopf pairing, radical quotient, and PBW filtration yields the displayed finite representatives, kernels, pairing matrices, parity dimensions, and PBW maps after projection to each W_ν. The finite restriction is the construction visible in the window; the determinant is its signed trace. □

Theorem 10.69 *(Finite retained $K3 \times E$ Hall stacks from a sector datum).* Let $X = S \times E$, where S is a smooth projective $K3$ surface, and fix Liu's product stability condition $\sigma_{s,t}$ on the Abramovich–Polishchuk heart in $D^b\,\mathrm{Coh}(X)$ [1, 53]. Fix a finite HN height R. Let C_R be a finite set of full effective Liu classes, closed under every retained Hall sum, and let

$$\ell_R : \Gamma_R^{\Pi} \longrightarrow C_R$$

be a lift with $\Pi_X(\ell_R(\gamma)) = \gamma$ on the retained active root degrees. For $\widehat{c} = (c, T)$, with $c \in C_R$, put

$$\mathfrak{M}^{ss}_{R,c} := \mathfrak{M}^{ss}_{\sigma_{s,t}}(X, c)_{\leq R}$$

for the retained semistable sector of full Liu numerical class c and HN height $\leq R$.

A finite retained $K3 \times E$ sector datum $\mathfrak{C}_R^{K3\times E}$ consists of the following data.

(Bnd)	$\mathfrak{M}^{ss}_{R,c}$, the retained extension stacks, and the retained two-step flag stacks are finite-type quasi-smooth derived Art
(Rig)	the scalar and connected translation stabilizers are rigidified, and the wrapped anchors leave only finite residual in
(dCrit)	the retained stacks carry the PTVV (-1)-shifted symplectic forms and BBDJS d-critical charts;
(RedOr)	trace and K3 semiregularity cosections, reduced orientation lines, and multiplicative Joyce–Upmeier transports ar
(Six)	$p^*, q_!, p_!$ are admissible in a fixed compact-support six-functor theory, and reduced Thom–Sebastiani is coherent

The finite stabilizer linearization characters in (Rig) and (RedOr) are those of Theorem 10.71; they are chosen to be zero after the Borel class is killed. Under $\mathfrak{C}_R^{K3\times E}$, the following stacks and correspondences form an actual finite Hall source kernel on $K3 \times E$.

Its tangent complex at A is

$$T_A\mathfrak{M}^{ss}_{R,c} \simeq \mathrm{RHom}_X(A, A)[1],$$

and the PTVV 3-Calabi–Yau pairing gives a (-1)-shifted symplectic form. The reduced tangent complex is the cone

$$\mathrm{RHom}_{\mathrm{red}}(A, A) := \mathrm{Cone}\Big(\mathrm{RHom}_X(A, A) \xrightarrow{(\mathrm{tr},\,\mathrm{cs})} R\Gamma(X, \mathcal{O}_X) \oplus H^2(S, \mathcal{O}_S)[-1]\Big)[-1],$$

where cs is the $K3$ semiregularity cosection pulled back along $p_S : X \to S$. Let

$$\mathfrak{M}^{\mathrm{sc}}_{R,c}$$

be the scalar-rigidified truncation of the corresponding oriented d-critical stack, with BBDJS vanishing-cycle complex $\Phi_{R,c}$ and reduced orientation line

$$o_{R,c} := \det \mathrm{Ext}^1_{\mathrm{red}}(A, A)^{-1} \quad \text{on Darboux charts.}$$

The 3-Calabi–Yau duality $\mathrm{Ext}^2_{\mathrm{red}}(A, A) \simeq \mathrm{Ext}^1_{\mathrm{red}}(A, A)^\vee$ identifies

$$o_{R,c}^{\otimes 2} \simeq \det \mathrm{RHom}_{\mathrm{red}}(A, A).$$

For projection-finite charge c, define the actual local stack over closed configurations by

$$\mathfrak{M}^{\mathrm{loc}}_{R,\widehat{c},I} := \left\{ (A, \mathbf{e}) \in \mathfrak{M}^{\mathrm{sc}}_{R,c} \times E^I \;\middle|\; p_E(\mathrm{Supp}\, A) \subset \bigcup_{i \in I} S \times \{e_i\} \right\} \times \{T\}.$$

For wrapped charge c, define the rigidified wrapped stack by

$$\mathfrak{M}^{\mathrm{wr}}_{R,\widehat{c}} := \left\{ (A, \ell) \;\middle|\; A \in \mathfrak{M}^{\mathrm{sc}}_{R,c},\ \ell : \det R p_{E,*} A \otimes \mathcal{O}_E(-\chi(A) \cdot \mathrm{o}_E) \xrightarrow{\sim} \mathcal{O}_E \right\} \times \{T\}.$$

The first construction remembers finite support over E; the second fixes the translation anchor of a full-support wrapped object. Put

$$\mathfrak{M}^{\mathrm{geom}}_{R,\widehat{c}} := \left(\coprod_I \mathfrak{M}^{\mathrm{loc}}_{R,\widehat{c},I} \right) \sqcup \mathfrak{M}^{\mathrm{wr}}_{R,\widehat{c}},$$

with the summand selected by the elliptic degree of c.

For $\widehat{c}, \widehat{c}' \in \widehat{\Gamma}_R$, define

$$\mathfrak{E}^{\mathrm{geom}}_{R,\widehat{c},\widehat{c}'}$$

to be the derived stack of exact triangles

$$A \longrightarrow C \longrightarrow B \longrightarrow A[1]$$

with

$$A \in \mathfrak{M}^{\mathrm{geom}}_{R,\widehat{c}}, \qquad B \in \mathfrak{M}^{\mathrm{geom}}_{R,\widehat{c}'}, \qquad C \in \mathfrak{M}^{\mathrm{geom}}_{R,\widehat{c}\star\widehat{c}'}.$$

The maps

$$\mathfrak{M}^{\mathrm{geom}}_{R,\widehat{c}} \times \mathfrak{M}^{\mathrm{geom}}_{R,\widehat{c}'} \xleftarrow{p} \mathfrak{E}^{\mathrm{geom}}_{R,\widehat{c},\widehat{c}'} \xrightarrow{q} \mathfrak{M}^{\mathrm{geom}}_{R,\widehat{c}\star\widehat{c}'}$$

send a triangle to (A, B) and C. The two-step flag stack $\mathfrak{F}^{(2),\mathrm{geom}}_{R,\widehat{c},\widehat{c}',\widehat{c}''}$ classifies filtrations with three associated graded factors. All maps are finite type in the finite Liu-bounded sector, and the six-functor operation $q_!$ is the exceptional compact-support pushforward.

The finite geometric Hall object is

$$\mathcal{H}^{\mathrm{geom}}_{R,\widehat{c}} := H^\bullet_c\Big(\mathfrak{M}^{\mathrm{geom}}_{R,\widehat{c}}, \Phi_{R,c} \otimes o_{R,c}\Big), \qquad \mathcal{H}^{\mathrm{geom}}_R := \bigoplus_{\widehat{c} \in \widehat{\Gamma}_R} \mathcal{H}^{\mathrm{geom}}_{R,\widehat{c}}.$$

Its product and coproduct are

$$m_R = q_! \, \mathrm{TS}^{\mathrm{red}} \, p^*, \qquad \Delta_R = p_! (\mathrm{TS}^{\mathrm{red}})^{-1} q^*.$$

The two-step flag stack identifies the two parenthesized products, hence $(\mathcal{H}_R^{\mathrm{geom}}, m_R, \Delta_R)$ is a finite Hall bialgebra. The primitive part

$$P_R^{\mathrm{geom}} := \ker\big(\Delta_R - 1 \otimes \mathrm{id} - \mathrm{id} \otimes 1\big) \cap \ker \epsilon_R$$

has normal-ordered Gram descent

$$P_R^{\Pi,\mathrm{geom}} := \overline{\Pi}_{X,*}^{\Theta_R} P_R^{\mathrm{geom}},$$

and its supercommutator is a finite Γ_R^{Π}-graded Hall Lie bracket.

Proof. The PTVV theorem gives the ambient (-1)-shifted symplectic structure on the derived stack of perfect complexes on the Calabi–Yau threefold X [69]. Liu's construction supplies the product stability condition, not the finite retained Hall sector by itself. The finite-type assertion, the finite residual inertia, and the admissible compact-support operations are the contents of $\mathfrak{C}_R^{K3\times E}$. Piyaratne– Toda boundedness for their threefold Bridgeland/BMT setting is not used as a substitute for (Bnd).

The trace and $K3$ semiregularity maps are additive in exact triangles by (RedOr), so the displayed cone is compatible with extension and flag stacks. On Darboux charts, the reduced obstruction theory has tangent half $\mathrm{Ext}^1_{\mathrm{red}}$ and cotangent half $\mathrm{Ext}^2_{\mathrm{red}}$; Serre duality identifies these halves as duals and gives the displayed square root of the determinant line. The BBDJS vanishing-cycle complex and the reduced Thom–Sebastiani transports are those of (dCrit) and (Six) [11]. Multiplicativity of the orientation line is the Joyce–Upmeier/Joyce–Tanaka–Upmeier transport in (RedOr) [45, 46].

The local stacks are closed incidence substacks over E^I, hence finite type in the retained finite sector. The wrapped stack is the fibre of the determinant anchor over 0_E, with a chosen trivialization of the anchor line; by (Rig) this removes the free translation direction and leaves only finite torsion stabilizers. The extension and flag stacks are the certified retained stacks of exact triangles and filtered objects. Their tangent complexes are computed by filtered endomorphism complexes, and the cosection equation follows from additivity of trace and semiregularity on filtrations.

The BBDJS Thom–Sebastiani isomorphism transports $\Phi_{R,c} \boxtimes \Phi_{R,c'}$ to the vanishing-cycle complex on the extension stack with the reduced orientation factors included. Therefore $q_! \, \mathrm{TS}^{\mathrm{red}} \, p^*$ and $p_!(\mathrm{TS}^{\mathrm{red}})^{-1} q^*$ are defined in the compact-support six-functor theory. The two parenthesized products are pull-pushes from the same two-step flag stack, so associativity follows from base change and Thom–Sebastiani coherence. Coassociativity is the dual statement. The supercommutator of primitives is primitive by the bialgebra identity, and normal-ordering additivity of $\overline{\Pi}_X$ gives the displayed Gram grading. □

Proposition 10.70 *(Type-II rank-one normal self-Ext criterion).* Let $\delta \in \{\delta_1, \delta_2, \delta_3\}$. Suppose the retained finite $K3 \times E$ sector contains charges

$$\widehat{a}_\delta, \widehat{b}_\delta, \widehat{c}_\delta \in \widehat{\Gamma}_R, \qquad \widehat{a}_\delta \star \widehat{b}_\delta = \widehat{c}_\delta, \qquad \overline{\Pi}_X(\widehat{c}_\delta) = \delta,$$

and a generic type-II wall object

$$C = A \oplus B$$

with

$$A \in \mathfrak{M}^{\mathrm{geom}}_{R,\widehat{a}_\partial}, \qquad B \in \mathfrak{M}^{\mathrm{geom}}_{R,\widehat{b}_\partial}, \qquad C \in \mathfrak{M}^{\mathrm{geom}}_{R,\widehat{c}_\partial}.$$

Assume A and B are stable factors in the two adjacent chambers and the reduced normal Ext groups satisfy

$$L_\partial := \mathrm{Ext}^1_{\mathrm{red}}(B, A), \qquad L_\partial^\vee := \mathrm{Ext}^1_{\mathrm{red}}(A, B), \qquad \dim L_\partial = 1.$$

Assume the remaining off-diagonal reduced Ext groups vanish in degree 1 at the generic wall point. Then the normal slice to the wall in $\mathfrak{M}^{\mathrm{geom}}_{R,\widehat{c}_\partial}$ is the critical chart

$$\mathrm{Crit}(uv) \subset L_\partial \oplus L_\partial^\vee,$$

with u the extension coordinate and v the coextension coordinate. After the reduced quotient and the wall orientation convention, the normal Dirac block is

$$D_\partial(u) = \begin{pmatrix} 0 & u \\ -u & 0 \end{pmatrix}, \qquad K_\partial^{\mathrm{cross}} = [\mathbb{C} \xrightarrow{u} \mathbb{C}], \qquad \mathrm{pf}_\partial = u.$$

The wall reflection interchanges the two chambers and sends

$$u \longmapsto -u, \qquad \mathrm{pf}_\partial \longmapsto -\mathrm{pf}_\partial.$$

Thus the rank-one type-II Pfaffian crossing of Definition 3.11 is obtained from the reduced normal self-Ext complex of every retained type-II wall object with the displayed one-dimensional normal splitting.

Proof. The reduced self-Ext complex of $C = A \oplus B$ splits into diagonal and off-diagonal blocks:

$$\mathrm{RHom}_{\mathrm{red}}(C, C) \simeq \mathrm{RHom}_{\mathrm{red}}(A, A) \oplus \mathrm{RHom}_{\mathrm{red}}(B, B) \oplus \mathrm{RHom}_{\mathrm{red}}(A, B) \oplus \mathrm{RHom}_{\mathrm{red}}(B, A).$$

The diagonal summands are tangent to the wall stratum. The normal deformation directions are $L_\partial \oplus L_\partial^\vee$. The 3-Calabi–Yau pairing identifies the two factors as duals and the Kuranishi potential has quadratic term uv; higher terms have order at least three and do not alter the rank-one Pfaffian normal form at the generic crossing. The linearized odd operator in the normal direction is therefore the skew matrix $D_\partial(u)$. Its Pfaffian is u. Crossing the wall swaps extension and coextension chambers; with the chosen orientation of L_∂, this sends the oriented coordinate to $-u$. The sign calculation is exactly the local sign used in Proposition 3.12. □

Theorem 10.71 *(Global quotient orientation descent).* Fix a cofinal finite HN subsystem. For every finite stage R, let Str^{or}_R be the finite set of object, extension, mixed, wrapped, and two-step flag strata entering the geometric Hall source. For $S \in \mathsf{Str}^{or}_R$, let $G = E[N]$ be a finite stabilizer on a connected stabilizer component, and let $\mathscr{D}_{R,S}$ be the reduced determinant line of the perfect complex controlling the stratum. For object strata,

$$\mathscr{D}_{R,S} = \det \mathrm{RHom}_{\mathrm{red}}(A, A);$$

for a correspondence $e = (Y \xleftarrow{p} \mathfrak{E}_e \xrightarrow{q} Z)$,

$$\mathscr{D}_{R,e} = q^* \mathscr{D}_{R,Z} \otimes p^* \mathscr{D}_{R,Y}^{-1};$$

and for a two-step flag word $w \in \{\mathrm{L}, \mathrm{W}\}^3$,

$$\mathscr{D}^{w}_{R,\mathrm{flag}} = \mathscr{D}^{(12)_3}_{R,w} \otimes (\mathscr{D}^{1(23)}_{R,w})^{-1}.$$

The quotient orientation descends on the finite stage if the following classes vanish with compatible null-trivializations:

$$\alpha^{\mathrm{red}}_{R,S} = w_2(\mathscr{D}_{R,S}) = c_1(\mathscr{D}_{R,S}) \bmod 2 \in H^2(S;\mathbb{F}_2), \qquad \alpha^{E,\mathrm{free}}_{R,S} = \mathrm{edge}_{2,0}\big(w^E_2(\mathscr{D}_{R,S})\big) = a_1u_1 + a_2u_2 \in H^2(BE;\mathbb{F}_2),$$

$$\widetilde{\beta}^{E,N}_{R,S} = w^G_2(\mathscr{D}_{R,S}) \in H^2_G(S;\mathbb{F}_2), \qquad \lambda^{E,N}_{R,S} \in H^1_G(S;\mathbb{F}_2).$$

Here

$$E_2^{p,q} = H^p(BG; H^q(S;\mathbb{F}_2)) \Rightarrow H_G^{p+q}(S;\mathbb{F}_2)$$

is the Borel spectral sequence. The point-stabilizer edge class in $H^2(BG;\mathbb{F}_2)$ is used only after the ordinary term $E_\infty^{0,2}$, the mixed term $E_\infty^{1,1}$, the stabilizer action on $H^\bullet(S;\mathbb{F}_2)$, and all total-degree-two differentials have been killed.

The edge calculation is:

$$N \text{ odd} \quad \Longrightarrow \quad H^{>0}(BE[N];\mathbb{F}_2) = 0,$$

$$N = 2: \quad \beta = b_{20}x_1^2 + b_{11}x_1x_2 + b_{02}x_2^2, \qquad (b_{20}, b_{11}, b_{02}) = (r_1, r_1 + r_2 + r_3, r_2),$$

so degree-two vanishing is equivalent to

$$r_1 = r_2 = r_3 = 0.$$

The residual point-stabilizer character is

$$\lambda = \lambda_1x_1 + \lambda_2x_2, \qquad \rho_1 = \lambda_1, \quad \rho_2 = \lambda_2, \quad \rho_3 = \lambda_1 + \lambda_2,$$

so the $N = 2$ character vanishes exactly when

$$\rho_1 = \rho_2 = \rho_3 = 0.$$

If $2^a \parallel N$, $a \geq 2$, then

$$H^*((B(\mathbb{Z}/2^a)^2);\mathbb{F}_2) = \Lambda(x_1, x_2) \otimes \mathbb{F}_2[y_1, y_2], \qquad |x_i| = 1, \quad |y_i| = 2, \quad x_i^2 = 0,$$

and hence

$$H^2(B(\mathbb{Z}/2^a)^2;\mathbb{F}_2) = \mathbb{F}_2\langle y_1, x_1x_2, y_2\rangle$$

and

$$\beta = A_1^{(N)}y_1 + A_{12}^{(N)}x_1x_2 + A_2^{(N)}y_2;$$

the coefficient $A_{12}^{(N)}$ is checked on the full rank-two two-primary stabilizer. The residual point-stabilizer character is

$$\lambda = \lambda_1^{(N)}x_1 + \lambda_2^{(N)}x_2.$$

After $\widetilde{\beta}^{E,N}_{R,S} = 0$, the linearizations form an $H^1_G(S;\mathbb{F}_2)$-torsor, and the quotient orientation chooses the zero character

$$\lambda^{E,N}_{R,S} = 0.$$

The null-trivializations are required to agree on overlaps, on Hall correspondence pullbacks, on two-step flag stacks, and under successor maps $R' \to R$. Then the finite-stage quotient orientation lines glue to a quotient orientation system

$$\mathfrak{Q}_R^{or}$$

and a compatible cofinal family gives the strong reduced orientation datum $(O1)$ on the HN-completed compact/wrapped Hall sector.

In the geometric source above, the stabilizer action on a self-Ext complex is by conjugation:

$$x \longmapsto \phi_g \circ t_g^* x \circ \phi_g^{-1}.$$

Thus, after the Borel class itself has been killed, the determinant character of the square-root line is trivial on the inner-conjugation part. The remaining anchor, framing, Thom–Sebastiani, and flag coherence signs are exactly the class

$$\lambda_{R,S}^{E,N}(g) = \begin{cases} 0, & g \text{ acts by } +1 \text{ on } o_{R,S}, \\ 1, & g \text{ acts by } -1 \text{ on } o_{R,S}. \end{cases}$$

They vanish precisely by the zero linearization above.

Proof. The equivariant square-root gerbe of the reduced determinant complex is a class in $H_G^2(S;\mathbb{F}_2)$. The Borel spectral sequence filters this class by ordinary, mixed, and point-stabilizer pieces. Vanishing of α^{red}, $\alpha^{E,\mathrm{free}}$, the mixed Borel term, and the total-degree-two differentials identifies the remaining edge with a class in $H^2(BG;\mathbb{F}_2)$. The displayed cohomology calculations for odd N, $N = 2$, and $2^a \parallel N$ are the standard mod-two group cohomology of $E[N]$ and its two-primary subgroup. Vanishing of this edge class gives an equivariant square root. The possible linearizations of that square root form an $H_G^1(S;\mathbb{F}_2)$-torsor. After the $E_\infty^{0,1}$ stratum term is killed, the point-stabilizer character lies in $H^1(BG;\mathbb{F}_2)$; the formulas for $N = 2$ and $2^a \parallel N$ are the two-primary edge coordinates of this character. Choosing the zero element gives $\lambda^{E,N} = 0$.

The overlap, correspondence, flag, and transition conditions say exactly that these local equivariant square roots form a cocycle in the finite Cech–Borel total complex. Since the total degree-one defect vanishes, the local lines glue. Thom–Sebastiani is functorial on extension and flag stacks, so the glued quotient orientation is multiplicative for the Hall product and compatible with the HN transition maps. Finally, on self-Ext determinant factors a stabilizer acts by conjugation, and conjugation has determinant 1; the trace and semiregularity summands are trivial representations. Hence inner conjugation contributes no further H^1-term after the Borel gerbe has been null-trivialized. The anchor, framing, Thom–Sebastiani, and flag signs are the displayed λ-class and are killed by the zero-character choice. □

Proposition 10.72 *(Geometric hybrid local/wrapped/mixed atlas).* The finite geometric stacks of Theorem 10.69 define a hybrid correspondence functor

$$\mathcal{B}_R^{E,\mathrm{geom}} : \mathsf{Corr}_{R,\mathrm{geom}}^{E,\mathrm{hyb}} \longrightarrow \mathrm{Ch}_{\mathbb{C}}.$$

Its objects are the closed-configuration local stacks $\mathfrak{M}_{R,\widehat{c},I}^{\mathrm{loc}}$ and the anchored wrapped stacks $\mathfrak{M}_{R,\widehat{c}}^{\mathrm{wr}}$. Its generating arrows are the exact-triangle stacks

$$\mathfrak{E}_{\alpha,\beta,R}^{\mathrm{loc}}, \qquad \mathfrak{E}_{\alpha_-;\eta;\beta_+,R}^{\mathrm{mix}}, \qquad \mathfrak{E}_{\eta_1,\eta_2,R}^{\mathrm{wr}},$$

formed before quotienting by E. For a generator

$$e = (Y \xleftarrow{p} \mathfrak{E}_e \xrightarrow{q} Z)$$

the operation is

$$\mathcal{B}_R^{E,\mathrm{geom}}(e) = q_! \, \mathrm{TS}_e^{\mathrm{red}} \, p^*.$$

The eight two-step words

$$\mathrm{LLL}, \mathrm{LLW}, \mathrm{LWL}, \mathrm{WLL}, \mathrm{LWW}, \mathrm{WLW}, \mathrm{WWL}, \mathrm{WWW}$$

are represented by the corresponding geometric flag stacks of filtered objects. These flag stacks identify the two pull-push composites, so the local, mixed, and wrapped products are associative at height R. The quotient-after-correspondence functor

$$Q_{E,R} : \mathcal{B}_R^{E,\mathrm{geom}} \longrightarrow \mathcal{B}_R^{\mathrm{red,geom}}$$

is obtained by applying E-equivariant compactly supported vanishing-cycle chains and the orientation descent of Theorem 10.71.

Proof. All correspondences are formed before the reduced E-quotient. In the local-local closed-configuration model the output fibre of q over a fixed total object is a closed relative Quot scheme, and for an iterated product it is a closed relative flag-Quot scheme. Thus q is projective in that abelian closed-filtration compactification, and $q_! = q_*$ on the protected closed support.

For mixed and wrapped generators, p records the ordered local inputs and anchored wrapped inputs, while q records the anchored wrapped output. Properness is again a statement about the abelian closed-filtration compactification: over a fixed output the fibre is a closed moduli of filtrations inside that output, with the determinant anchor imposed as a closed condition on the rigidified wrapped stratum. For the exact-triangle stack itself the operation is the exceptional compact-support pushforward $q_!$ supplied by (Six) in Theorem 10.69; ordinary properness is not asserted for raw exact triangles.

For a word $w \in \{\mathrm{L}, \mathrm{W}\}^3$, the corresponding flag stack maps to the two iterated correspondence fibre products and to the total filtered output. In the abelian closed-filtration model this output map is a projective flag-Quot map. In the geometric exact-triangle model the two parenthesized products are the two exceptional pull-pushes from the same finite-type, finite-inertia, oriented, Thom–Sebastiani coherent flag stack. Base change identifies them in the compact-support six-functor theory. Quotienting after the correspondence preserves these equalities because the orientation gerbes and finite stabilizer characters vanish by Theorem 10.71.

The restriction to exceptional compact support is necessary. If $x \in S \times \{e\}$ and R_0 is a DVR with uniformizer t, the map

$$t : \mathcal{O}_x \otimes R_0 \longrightarrow \mathcal{O}_x \otimes R_0$$

is an isomorphism on the generic fibre, while its special cone is $\mathcal{O}_x \oplus \mathcal{O}_x[1]$. Thus a raw exact-triangle q-map need not satisfy the valuative criterion when zero-charge cones are not retained. The finite atlas is therefore a compact-support admissible atlas, with ordinary properness only in the closed Quot/flag compactifications. □

Proposition 10.73 *(Source chiral Koszul coalgebra from the geometric atlas).* Let

$$\mathcal{F}^{\mathrm{geom}}_{X,\sigma,S,\leq R} := \mathcal{Q}_{E,R}\mathcal{B}^{E,\mathrm{geom}}_{R}$$

be the reduced geometric hybrid factorization algebra on E, placed in the Beilinson–Drinfeld/Francis–Gaitsgory conilpotent completed domain. Its augmentation is the projection to the vacuum charge. Define

$$C^{\mathrm{geom}}_{X,R} := \bar{B}^{\mathrm{ch}}_{E}(\mathcal{F}^{\mathrm{geom}}_{X,\sigma,S,\leq R}).$$

Let

$$\overline{\mathcal{F}}^{\mathrm{geom}}_{X,R} := \ker(\varepsilon^{\mathrm{hyb}}_{X,R})$$

be the augmentation ideal. If N_R is the maximal length of a non-vacuum word of total HN height $\leq R$, then

$$C^{\mathrm{geom}}_{X,R} = \mathbb{C}\mathbf{1} \oplus \bigoplus_{1\leq p\leq N_R} \bar{B}^{\mathrm{ch},p}_{E}(\overline{\mathcal{F}}^{\mathrm{geom}}_{X,R}).$$

The bar-length filtration is

$$F^{\mathrm{co}}_{R,p}C^{\mathrm{geom}}_{X,R} = \mathbb{C}\mathbf{1} \oplus \bigoplus_{1\leq j\leq p} \bar{B}^{\mathrm{ch},j}_{E}(\overline{\mathcal{F}}^{\mathrm{geom}}_{X,R}),$$

and the reduced coproduct satisfies

$$(\overline{\Delta}^{\mathrm{bar}}_{R})^{N_R+1} = 0.$$

Its collision coproduct is the cut map over Ran collision strata:

$$\Delta^{\mathrm{ch}}[x_1|\cdots|x_k] = \mathbf{1}\otimes[x_1|\cdots|x_k] + [x_1|\cdots|x_k]\otimes\mathbf{1} + \sum_{i=1}^{k-1}[x_1|\cdots|x_i]\otimes[x_{i+1}|\cdots|x_k],$$

with the usual Koszul signs. Each cut is represented by the finite collision/flag pull-push map in $\mathsf{Corr}^{E,\mathrm{hyb}}_{R,\mathrm{geom}}$. The chiral cobar counit gives a source-internal quasi-isomorphism

$$\varepsilon^{\mathrm{geom}}_{R} : \Omega^{\mathrm{ch}}_{E}C^{\mathrm{geom}}_{X,R} \xrightarrow{\simeq} \mathcal{F}^{\mathrm{geom}}_{X,\sigma,S,\leq R}.$$

This is the Francis–Gaitsgory chiral Koszul equivalence applied to the geometric source; no denominator-algebra presentation is used in the definition or proof of $\varepsilon^{\mathrm{geom}}_{R}$.

If a source-to-target quasi-isomorphism

$$\vartheta_R : \mathcal{F}^{\mathrm{geom}}_{X,\sigma,S,\leq R} \xrightarrow{\simeq} \mathsf{Den}_{\Delta_S,E,\leq R}$$

is constructed from the geometric primitive representatives, Hall products, coproducts, pairings, and transition maps, then the finite Koszul comparison is the composite

$$\Theta_{\mathrm{Kos},R} = \vartheta_R \circ \varepsilon^{\mathrm{geom}}_{R}.$$

Proof. The HN window is finite, and every non-vacuum surviving color has positive HN height. Hence a nonzero bar word of total height $\leq R$ has length at most N_R. The reduced coproduct lowers bar length on each branch, so the (N_R+1)-fold reduced coproduct vanishes. Coassociativity is the equality of the two iterated cut maps on the finite collision flag stack; the four-input pentagon supplies the higher coherence, and the vacuum collision flags give the counit. The Francis–Gaitsgory theorem applies in the stated conilpotent completed domain and gives the source cobar counit quasi-isomorphism. The map ϑ_R, when present, is a separate geometric comparison; it is not defined by the target bar–cobar counit or by the Borcherds–Kac presentation. □

Proposition 10.74 *(Primitive matrices from the compact Hall source).* Let

$$P_R^{\Pi,\mathrm{geom}} = H^0\,\mathrm{Prim}_{\mathrm{prot}}\big(\overline{\Pi}_{X,*}^{\Theta_R}\mathcal{F}_{X,\sigma,S,\le R}^{\mathrm{geom}}\big).$$

For every $\gamma \in \Gamma_R^{\Pi}$, choose a homogeneous vanishing-cycle Borel–Moore basis

$$\mathcal{E}_{\gamma,\bar{p}} = \{e_{\gamma,\bar{p},a}\}_a \subset P_{R,\gamma,\bar{p}}^{\Pi,\mathrm{geom}}$$

represented by oriented irreducible critical cycles or IC summands of $\mathfrak{M}_{R,\widehat{c}}^{\mathrm{geom}}$ with $\overline{\Pi}_X(\widehat{c}) = \gamma$. Write

$$P_\gamma := \bigoplus_{\bar{p}} P_{R,\gamma,\bar{p}}^{\Pi,\mathrm{geom}}.$$

The Hall product matrices are the compact-support integrals

$$e_{\alpha,a} * e_{\beta,b} = \sum_c M_{\alpha,a;\beta,b}^{\alpha+\beta,c}\, e_{\alpha+\beta,c},$$

$$M_{\alpha,a;\beta,b}^{\alpha+\beta,c} = \int_{\mathfrak{E}_{\alpha,\beta;\alpha+\beta}^{\mathrm{geom}}} q^!(e_{\alpha+\beta,c}^\vee) \cap \mathrm{TS}^{\mathrm{red}}\big(p^*(e_{\alpha,a} \boxtimes e_{\beta,b})\big).$$

Write $M_{\alpha,\beta} : P_\alpha \otimes P_\beta \to P_{\alpha+\beta}$ for this matrix and let τ be the ordinary flip. On homogeneous tensors the primitive bracket matrix is

$$B_{\alpha,\beta}(x \otimes y) = M_{\alpha,\beta}(x \otimes y) - (-1)^{|x||y|} M_{\beta,\alpha}\tau(x \otimes y).$$

The coproduct matrices are computed from the source splitting correspondences:

$$\Delta(e_{\rho,d}) = \sum_{\mu+\nu=\rho} \sum_{a,b} D_{\rho,d}^{\mu,a;\nu,b}\, e_{\mu,a} \otimes e_{\nu,b},$$

$$D_{\rho,d}^{\mu,a;\nu,b} = \int_{\mathfrak{C}_{\rho;\mu,\nu}^{\mathrm{geom}}} (e_{\mu,a}^\vee \boxtimes e_{\nu,b}^\vee) \cap \mathrm{TS}^{\mathrm{red}}\big(q^*(e_{\rho,d})\big),$$

where $\mathfrak{C}_{\rho;\mu,\nu}^{\mathrm{geom}}$ is the compact geometric coproduct correspondence dual to the extension stack. Let

$$\mathfrak{P}_{\gamma,-\gamma;0}^{\mathrm{geom}}$$

be the positive-negative pairing correspondence with output the vacuum charge, and let tr_0 be the compact-support trace on the vacuum summand. The Hopf pairing matrix is

$$G_{\gamma,ab} = \mathrm{tr}_0\Big(q_!\,\mathrm{TS}^{\mathrm{red}}\, p^*(e_{\gamma,a} \boxtimes e_{-\gamma,b})\Big),$$

and, in each parity block, the finite radical is

$$K_{\gamma,\bar{p}} := \ker G_{\gamma,\bar{p}} \cap \ker G_{-\gamma,\bar{p}}^t.$$

Choose splittings

$$P_{R,\gamma,\bar{p}}^{\Pi,\mathrm{geom}} = K_{\gamma,\bar{p}} \oplus L_{\gamma,\bar{p}}, \qquad Q_{\gamma,\bar{p}} = [\,0 \;\; I\,] : P_{R,\gamma,\bar{p}}^{\Pi,\mathrm{geom}} \twoheadrightarrow L_{\gamma,\bar{p}}.$$

Put $K_\gamma = \bigoplus_{\bar p} K_{\gamma,\bar p}$, $L_\gamma = \bigoplus_{\bar p} L_{\gamma,\bar p}$, and $Q_\gamma = \bigoplus_{\bar p} Q_{\gamma,\bar p}$. The quotient bracket and coproduct are the source matrices

$$[\,\bar x, \bar y\,]_{\mathrm{red}} = Q_{\alpha+\beta} B_{\alpha,\beta}(x \otimes y), \qquad \Delta_{\mathrm{red}}(\bar z) = (Q_\mu \otimes Q_\nu) D^{\mu,\nu}_\rho(z),$$

and they are independent of the lifts because

$$Q_{\alpha+\beta} B_{\alpha,\beta}\big(P_\alpha \otimes K_\beta\big) = 0, \qquad Q_{\alpha+\beta} B_{\alpha,\beta}\big(K_\alpha \otimes P_\beta\big) = 0,$$

$$(Q_\mu \otimes Q_\nu) D^{\mu,\nu}_\rho K_\rho = 0.$$

Product, coproduct, and pairing are adjoint in the source:

$$M^t_{\alpha,\beta} G_{\alpha+\beta} = (G_\alpha \otimes G_\beta) D^{\alpha,\beta}_{\alpha+\beta}.$$

The reduced primitive block is

$$P^{\Pi,\mathrm{red},\mathrm{geom}}_{R,\gamma} = L_\gamma \simeq P^{\Pi,\mathrm{geom}}_{R,\gamma}/K_\gamma.$$

The PBW filtration on

$$U(P^{\Pi,\mathrm{red},\mathrm{geom}}_R)$$

is tensor length:

$$F^d_{\mathrm{PBW}} U(P^{\Pi,\mathrm{red},\mathrm{geom}}_R) = \mathrm{im}\left(\bigoplus_{0\le k\le d} (P^{\Pi,\mathrm{red},\mathrm{geom}}_R)^{\otimes k} \longrightarrow U(P^{\Pi,\mathrm{red},\mathrm{geom}}_R)\right).$$

Its associated graded is the super-symmetric algebra on the reduced primitive blocks:

$$\mathrm{gr}_{\mathrm{PBW}}\, U(P^{\Pi,\mathrm{red},\mathrm{geom}}_R) \simeq \mathrm{Sym}\big(P^{\Pi,\mathrm{red},\mathrm{geom}}_{R,\bar 0}\big) \otimes \Lambda\big(P^{\Pi,\mathrm{red},\mathrm{geom}}_{R,\bar 1}\big).$$

Thus primitive representatives, pairing, radical, PBW filtration, and structure constants are computed from the compact $K3 \times E$ Hall correspondences themselves.

Proof. The chosen basis lies in compactly supported vanishing-cycle Borel–Moore homology of the geometric moduli stacks. The bracket is the supercommutator of the Hall product, and the Hall product is the pull-push along $\mathfrak{E}^{\mathrm{geom}}$; pairing with the dual basis gives the displayed M-matrix. The coproduct matrix is the same construction on the dual flag correspondence. Verdier duality and the orientation line give the Hopf pairing integral, and base change on the product-pairing square gives the displayed adjunction identity. Hence the two-sided Hopf radical is a Lie ideal and a coideal; the three vanishing identities are exactly this statement after projection by Q. Quotienting by the kernel gives the reduced primitive blocks and well-defined source bracket and coproduct matrices. PBW is the ordinary PBW theorem for the resulting finite-dimensional Lie superalgebra over $\mathbb{C}$, with parity inherited from the vanishing-cycle grading. □

Proposition 10.75 *(Geometric construction of the finite comparison maps).* Fix R and the downward-saturated window $W_{\nu(R)}$. Let

$$P^{\Pi,\mathrm{geom}}_{R,\gamma,\bar p} = K_{\gamma,\bar p} \oplus L_{\gamma,\bar p}$$

be the source splitting obtained from the compact Hall pairing matrices, with quotient $Q_{\gamma,\bar{p}}$, source bracket matrices $B^X_{\alpha,\beta}$, source coproduct matrices $D^{X,\mu,\nu}_\rho$, and source pairing matrices G^X_γ of Proposition 10.74. Suppose there is a parity-preserving family of finite linear isomorphisms

$$A_{\gamma,\bar{p}} : L_{\gamma,\bar{p}} \xrightarrow{\sim} \mathfrak{g}_{\Delta_s,\gamma,\bar{p}}, \qquad A_0 = \iota_H,$$

compatible with HN transitions and satisfying, for every retained degree,

$$A_{\alpha+\beta} Q_{\alpha+\beta} B^X_{\alpha,\beta} = B^\Delta_{\alpha,\beta}(A_\alpha Q_\alpha \otimes A_\beta Q_\beta),$$

$$(A_\mu \otimes A_\nu)(Q_\mu \otimes Q_\nu) D^{X,\mu,\nu}_\rho = D^{\Delta,\mu,\nu}_\rho A_\rho Q_\rho,$$

and

$$G^\Delta_\gamma(A_\gamma x, A_{-\gamma} y) = G^X_\gamma(x, y).$$

Then

$$\vartheta^{\mathrm{Prim}}_R := \bigoplus_{\gamma,\bar{p}} A_{\gamma,\bar{p}} Q_{\gamma,\bar{p}} : P^{\Pi,\mathrm{geom}}_R\big|_{W_{\nu(R)}} \longrightarrow \mathfrak{g}_{\Delta_s}\big|_{W_{\nu(R)}}$$

is the finite primitive source-to-target comparison. It is computed from the geometric representatives and compact Hall matrices. The Gritsenko–Nikulin presentation is only the comparison object.

If the finite hybrid source is generated by these reduced primitive representatives and satisfies the PBW and no-extra-relations clauses of Definition 10.67, define ϑ_R on source PBW words by

$$\vartheta_R\big(\mu^{\mathrm{hyb}}_R(x_1, \dots, x_k)\big) = \mu_{\Delta,R}\big(\vartheta^{\mathrm{Prim}}_R x_1, \dots, \vartheta^{\mathrm{Prim}}_R x_k\big), \qquad \vartheta_R(\mathbf{1}) = \mathbf{1}.$$

The bracket, coproduct, pairing, radical, and PBW equalities above make this independent of the chosen source PBW representative. Its associated graded map is

$$\mathrm{Sym}(A_{\bar{0}}) \otimes \Lambda(A_{\bar{1}}),$$

so it is an isomorphism on the finite PBW filtration. Therefore

$$\vartheta_R : \mathcal{F}^{\mathrm{geom}}_{X,\sigma,S,\leq R} \xrightarrow{\simeq} \mathsf{Den}_{\Delta_s,E,\leq R}$$

is a charge-preserving finite chiral algebra quasi-isomorphism. The finite Koszul comparison is

$$\Theta_{\mathrm{Kos},R} = \vartheta_R \circ \varepsilon^{\mathrm{geom}}_R,$$

where $\varepsilon^{\mathrm{geom}}_R$ is the source-internal chiral cobar counit.

Proof. The matrices B^X, D^X, G^X are compact-support vanishing-cycle pull-push integrals on the geometric extension, coproduct, and positive-negative pairing correspondences. Their kernels $K_{\gamma,\bar{p}}$ are the source Hopf-radical blocks. The displayed equations say that the quotient source bracket, coproduct, Cartan action, pairing, and transition maps coincide with the corresponding Gritsenko–Nikulin finite window after applying A. Hence $\vartheta^{\mathrm{Prim}}_R$ is a morphism of finite Gram-graded primitive Lie superalgebras after the source radical quotient.

The extension to ϑ_R uses the source PBW presentation. It does not use the target bar–cobar counit. The PBW filtration is finite in the retained window; on associated gradeds the displayed super-symmetric

algebra map is an isomorphism by the full parity-dimension and no-extra-relations clauses. The finite filtration spectral sequence gives the quasi-isomorphism. Finally,

$$C^{\mathrm{geom}}_{X,R} = \bar{B}^{\mathrm{ch}}_E(\mathcal{F}^{\mathrm{geom}}_{X,\sigma,S,\leq R})$$

and the Francis–Gaitsgory source counit give

$$\varepsilon^{\mathrm{geom}}_R : \Omega^{\mathrm{ch}}_E C^{\mathrm{geom}}_{X,R} \xrightarrow{\simeq} \mathcal{F}^{\mathrm{geom}}_{X,\sigma,S,\leq R}.$$

Composing with the geometric comparison ϑ_R gives $\Theta_{\mathrm{Kos},R}$. Neither factor is obtained from $\Omega^{\mathrm{ch}}_E \bar{B}^{\mathrm{ch}}_E(\mathsf{Den}_{\Delta_5,E,\leq R}) \to \mathsf{Den}_{\Delta_5,E,\leq R}$. □

Theorem 10.76 *(Universal finite Dirac–Igusa normal form on a root window).* Let $W \subset Q_+ \setminus \{0\}$ be finite and downward saturated. Put

$$\mathfrak{n}^+_W := \bigoplus_{\beta\in W} \mathfrak{g}_{\Delta_5,\beta}, \qquad \mathfrak{n}^-_W := \bigoplus_{\beta\in W} \mathfrak{g}_{\Delta_5,-\beta}, \qquad P_W := \mathfrak{n}^-_W \oplus (\Lambda^{2,1}_{II} \otimes \mathbb{C}) \oplus \mathfrak{n}^+_W.$$

The bracket on P_W is the Gritsenko–Nikulin/Kac bracket whenever the resulting root degree belongs to $\{\pm\beta : \beta \in W\} \cup \{0\}$, and is zero otherwise. The standard invariant form gives a finite Hopf pairing

$$\langle\,\cdot\,,\,\cdot\,\rangle_W : \mathfrak{n}^+_W \otimes \mathfrak{n}^-_W \longrightarrow \mathbb{C}$$

with homogeneous radical Rad_W; put

$$P^{\mathrm{red}}_W := P_W/\mathrm{Rad}_W\,.$$

Let $U(\mathfrak{n}^+_W)$ be graded by positive root degree and define

$$\mathcal{H}^+_W := \bigoplus_{\beta\in W\cup\{0\}} U(\mathfrak{n}^+_W)_\beta.$$

Multiplication is the product of $U(\mathfrak{n}^+_W)$ followed by projection to the displayed direct sum:

$$x \cdot_W y = \begin{cases} xy, & \deg x + \deg y \in W \cup \{0\}, \\ 0, & \deg x + \deg y \notin W \cup \{0\}. \end{cases}$$

The coproduct is the restriction of the usual enveloping-algebra coproduct. Since W is downward saturated, every coproduct summand of a homogeneous element of degree $\beta \in W$ has both degrees in $W \cup \{0\}$. Hence $\mathcal{H}^+_W$ is a finite cocommutative bialgebra, and

$$\mathrm{Prim}(\mathcal{H}^+_W) = \mathfrak{n}^+_W.$$

The same construction applied to $\mathfrak{n}^-_W$ gives $\mathcal{H}^-_W$. Together with the Cartan and the pairing above, these two halves form the finite algebraic primitive double P_W.

Choose a homogeneous PBW basis $\mathcal{B}_W$ of $\mathcal{H}^+_W$. For $a, b, c \in \mathcal{B}_W$, write

$$a \cdot_W b = \sum_c m^c_{ab} c, \qquad \Delta_W(c) = \sum_{a,b} d^{ab}_c\, a \otimes b.$$

The finite algebraic atlas $\mathfrak{S}_W$ is the $\mathbb{C}$-linear finite correspondence atlas with one oriented point $M_a = \operatorname{Spec}\mathbb{C}$ for each $a \in \mathcal{B}_W$, coefficient sheaf $\Phi_a = \mathbb{C}_{M_a}$, orientation line $o_a = \mathbb{C}$, and product and coproduct correspondences the weighted finite spans with matrices m^c_{ab} and d^{ab}_c. The flag identities for the spans are exactly associativity of $\cdot_W$ and coassociativity of Δ_W; the primitive bracket is the supercommutator.

Let $\beta \in W$ correspond to the Gram monomial $x^\beta = q^n r^l s^m$. For each homogeneous basis vector $v \in \mathfrak{g}_{\Delta_5,\beta,\bar{p}}$, define the rank-one skew block

$$D_{\beta,v}(x) = \begin{pmatrix} 0 & 1 - x^\beta \\ -(1 - x^\beta) & 0 \end{pmatrix}.$$

The finite Dirac object is the virtual skew family

$$\mathfrak{D}_W := \bigoplus_{\beta \in W} \left(\bigoplus_{v \in \mathfrak{g}_{\Delta_5,\beta,\bar{0}}} D_{\beta,v} - \bigoplus_{v \in \mathfrak{g}_{\Delta_5,\beta,\bar{1}}} D_{\beta,v} \right),$$

with super-Pfaffian line

$$\mathrm{Pf}^{\mathrm{vir}}(\mathfrak{D}_W) = \bigotimes_{\beta \in W} \det(\mathfrak{g}_{\Delta_5,\beta,\bar{0}}) \otimes \det(\mathfrak{g}_{\Delta_5,\beta,\bar{1}})^{-1}$$

and Pfaffian section

$$\mathrm{pf}_W = 64 q^{1/2} r^{1/2} s^{1/2} \prod_{\beta \in W} (1 - x^\beta)^{\dim \mathfrak{g}_{\Delta_5,\beta,\bar{0}} - \dim \mathfrak{g}_{\Delta_5,\beta,\bar{1}}}.$$

On the three type-II simple walls δ_i, replace the factor $1 - x^{\delta_i}$ by the local coordinate u_i. The resulting wall chart is precisely the rank-one crossing

$$K^{\mathrm{cross}}_{\delta_i} = [\mathbb{C} \xrightarrow{u_i} \mathbb{C}], \qquad \mathrm{pf}_{\delta_i} = u_i,$$

and the reflection $u_i \mapsto -u_i$ changes the Pfaffian sign. Thus the local Weyl transports satisfy

$$\epsilon^{\mathrm{loc}}_o(s_{\delta_i}) = -1 \qquad (i = 1, 2, 3),$$

and extend to the type-II Weyl group by Proposition 3.23.

The finite colored normal form is the following factorization object over E. A positive degree $\beta = \alpha(n, l, m)$ has local color when $m = 0$ and wrapped color when $m > 0$. Local colors are placed on ordinary finite configurations of E; wrapped colors are represented by a rigidified full-support symbol $w_\beta = (E, o_E, \beta)$. Products are the matrices m^c_{ab} above, with color rule local-local, local-wrapped, wrapped-local, and wrapped-wrapped according to addition of the third Gram coordinate. The protected trace is

$$I^{\mathrm{prot}}_W(a_\beta) = \mathrm{str}(a_\beta)\, q^n r^l s^m, \qquad m_W(\beta) = \mathrm{pr}_3\, \overline{\Pi}_X(\beta) = m.$$

Finally set

$$C_W := \bar{B}^{\mathrm{ch}}_E(\mathcal{F}^{\mathrm{hyb}}_W), \qquad b_W = \mathrm{id}_{C_W},$$

with the bar coalgebra filtration by total positive root degree and the usual deconcatenation/collision coproduct. The map

$$\Theta_{\mathrm{Kos},W} : \Omega^{\mathrm{ch}}_E C_W \longrightarrow \mathsf{Den}_{\Delta_5,E,W}$$

sends each primitive generator $v \in \mathfrak{g}_{\Delta_s,\beta}$ to the corresponding constant current of degree β. Since both sides are generated by the same finite Borcherds–Kac presentation modulo the same invariant-pairing radical on W, this is a quasi-isomorphism. Consequently

$$P_W^{\mathrm{red}} \cong \mathfrak{g}_{\Delta_s}\big|_{\{\pm\beta:\beta\in W\}\cup\{0\}}$$

as a finite Gram-graded Lie superalgebra with parity, pairing, PBW filtration, and Weyl action on the type-II overlap.

Proof. The truncated product is associative. If $\deg x + \deg y \notin W \cup \{0\}$, then for every homogeneous z of positive degree the sum $\deg x + \deg y + \deg z$ cannot lie in W, for otherwise downward saturation would force $\deg x + \deg y \in W$. Thus both parenthesizations give zero outside the window and agree inside the window because they are the enveloping-algebra product. The same downward-saturation argument shows that the enveloping coproduct of an element of degree $\beta \in W$ has only tensor degrees in $(W \cup \{0\})^2$, so Δ_W is coassociative. PBW identifies the primitive subspace of the truncated coalgebra with the degree-positive Lie generators $\mathfrak{n}_W^+$.

The finite correspondence atlas is just the matrix form of this finite bialgebra. Matrix multiplication of the product spans is associativity; matrix multiplication of the coproduct spans is coassociativity; and the primitive supercommutator is the restriction of the Lie bracket.

The skew block $D_{\beta,v}$ has Pfaffian $1 - x^\beta$. Placing even primitive blocks in the numerator and odd primitive blocks in the Berezinian denominator gives the displayed exponent $\dim \bar{o} - \dim \bar{\imath}$. For $\beta = \delta_i$, the local coordinate substitution $1 - x^{\delta_i} = u_i$ is the rank-one type-II crossing, and the involution $u_i \mapsto -u_i$ gives the sign -1. The Weyl transport assertion is then exactly Proposition 3.23.

The colored normal form is closed under the four displayed color products because the third Gram coordinate is additive under $\overline{\Pi}_X$. Hence the trace exponent of a product is the sum of the trace exponents, and I_W^{prot} is multiplicative on the finite Hall product. The bar construction is conilpotent because the positive root-degree filtration is finite on W. The cobar map is the identity on the finite primitive presentation and carries the Borcherds–Kac relations and the invariant-pairing radical to the same relations on the denominator current algebra; hence it is a quasi-isomorphism. Primitive recognition is therefore the identity identification of the algebraic normal-form quotient with the Gritsenko–Nikulin window. □

Proposition 10.77 *(Central normal-ordering point normal form).* Let $W \subset Q_+ \setminus \{0\}$ be finite and downward saturated. For every $\beta \in W$, use the multiplicative normal-ordering lift

$$\widehat{c}^{\,\circ}_\beta := (0, -\beta) \in \widehat{\Gamma}_X, \qquad \overline{\Pi}_X(\widehat{c}^{\,\circ}_\beta) = \beta,$$

where 0 is the zero formal Mukai charge and $-\beta$ lies in the central Gram summand. For a PBW basis word $a = v_{\beta_1} \cdots v_{\beta_k} \in \mathcal{B}_W$, set

$$\widehat{c}^{\,\circ}_a := \widehat{c}^{\,\circ}_{\beta_1} \star \cdots \star \widehat{c}^{\,\circ}_{\beta_k} = (0, -\deg a) \in \widehat{\Gamma}_X.$$

Then the universal normal form of Theorem 10.76 is realized by the following finite central-label point atlas. For each $\widehat{c} \in \widehat{\Gamma}_X$, define

$$\mathfrak{M}_{W,\widehat{c}} := \coprod_{\substack{a\in\mathcal{B}_W \\ \widehat{c}^{\,\circ}_a = \widehat{c}}} \operatorname{Spec} \mathbb{C}, \qquad \Phi_{W,\widehat{c}} := \mathbb{C}_{\mathfrak{M}_{W,\widehat{c}}}, \qquad o_{W,\widehat{c}} := \mathbb{C}.$$

The product correspondence over $(\widehat{c}, \widehat{c}')$ is the finite $\mathbb{C}$-linear span

$$\mathfrak{M}_{W,\widehat{c}} \times \mathfrak{M}_{W,\widehat{c}'} \xleftarrow{p} \mathfrak{E}_{W,\widehat{c},\widehat{c}'} \xrightarrow{q} \mathfrak{M}_{W,\widehat{c}\star\widehat{c}'}$$

whose points are triples (a, b, c) with $\widehat{c}^{\circ}_a = \widehat{c}, \widehat{c}^{\circ}_b = \widehat{c}', \widehat{c}^{\circ}_c = \widehat{c} \star \widehat{c}'$, and product coefficient $m^c_{ab} \neq 0$. The map p sends (a, b, c) to (a, b), the map q sends it to c, and the coefficient morphism on the constant sheaf is multiplication by m^c_{ab}. The coproduct correspondence is defined by the same finite points with coefficient d^{ab}_c. If the target charge leaves the lifted window, the correspondence is empty.

The resulting compactly supported cohomology is

$$\mathrm{H}^{\bullet}_c(\mathfrak{M}_{W,\widehat{c}}, \Phi_{W,\widehat{c}} \otimes o_{W,\widehat{c}}) = \bigoplus_{\widehat{c}^{\circ}_a = \widehat{c}} \mathbb{C}a,$$

and the pull-push product and coproduct are exactly the matrices

$$a \otimes b \longmapsto \sum_c m^c_{ab} c, \qquad c \longmapsto \sum_{a,b} d^{ab}_c \, a \otimes b.$$

Thus the finite point atlas realizes the algebraic normal form $\mathfrak{S}_W$, its primitive bracket, and the normal-ordered central charge map $\overline{\Pi}_X$. A geometric $K3 \times E$ realization is a morphism from the retained Hall stacks of Theorem 10.69 to this normal form, preserving products, coproducts, orientations, Pfaffian blocks, bar coalgebras, pairings, and primitive matrices.

Proof. The element $\widehat{c}^{\circ}_{\beta} = (0, -\beta)$ belongs to the central normal-ordering summand of Definition 6.11. Associativity of $\star$, $B(0, 0) = 0$, and additivity of $\overline{\Pi}_X$ give

$$\overline{\Pi}_X(\widehat{c}^{\circ}_a) = \deg a, \qquad \widehat{c}^{\circ}_a \star \widehat{c}^{\circ}_b = (0, -\deg a - \deg b), \qquad \overline{\Pi}_X(\widehat{c}^{\circ}_a \star \widehat{c}^{\circ}_b) = \deg a + \deg b.$$

Hence the target of every nonempty product correspondence has exactly the charge prescribed by the normal-ordered product. The stacks are finite disjoint unions of points, so their compactly supported cohomology is the free vector space on the displayed basis and their stabilizers are trivial. Pullback along p, multiplication by the coefficient m^c_{ab}, and exceptional pushforward along the finite map q give the product matrix. The coproduct is identical with the coefficients d^{ab}_c. The two-step flag identities are matrix associativity and coassociativity from Theorem 10.76; therefore the point atlas realizes the same finite algebraic bialgebra and primitive supercommutator with central normal-ordering labels. □

Theorem 10.78 *(Compact Dirac–Igusa source from geometric finite stages).* Fix a cofinal HN schedule $R \in \mathcal{R}_{\mathrm{HN}}$ and a cofinal downward-saturated root-window schedule $W_{\nu(R)}$. For every height R, assume the finite hybrid atlas has verified properness or exceptional compact-support admissibility for every local, mixed, wrapped, and flag correspondence; equivalently the entries of $\mathfrak{O}_{\mathrm{hyb},R}$ vanish with the stated null-trivializations. Take the retained geometric $K3 \times E$ Hall stacks of Theorem 10.69, the quotient orientation system of Theorem 10.71, the type-II normal self-Ext charts of Proposition 10.70, the hybrid correspondence functor of Proposition 10.72, the source chiral coalgebra of Proposition 10.73, and the primitive matrices of Proposition 10.74, together with the finite comparison maps of Proposition 10.75. Denote the resulting finite geometric objects by

$$\mathfrak{S}_R, \qquad \mathfrak{A}_R, \qquad \mathcal{F}^{\mathrm{hyb}}_{X,\sigma,S,\leq R}, \qquad O^{\mathrm{quot}}_R, \qquad \{K^{\mathrm{cross}}_{\delta_i}, \mathrm{pf}_{\delta_i}\}^3_{i=1}, \qquad C_{X,R}, \qquad \Pi_{\mathrm{rec},R}.$$

Concretely,

$$\mathfrak{S}_R := \mathfrak{S}_R^{\mathrm{geom}}, \qquad \mathcal{F}^{\mathrm{hyb}}_{X,\sigma,S,\le R} := \mathcal{F}^{\mathrm{geom}}_{X,\sigma,S,\le R}, \qquad C_{X,R} := C^{\mathrm{geom}}_{X,R},$$

$$\mathfrak{A}_R := \big(\mathfrak{S}_R^{\mathrm{geom}}, \mathfrak{D}^{\mathrm{geom}}_{X,R}, \mathrm{Pf}(\mathfrak{D}^{\mathrm{geom}}_{X,R}), \mathrm{pf}^{\mathrm{geom}}_{X,R}\big),$$

$$O_R^{\mathrm{quot}} := \big(o_{R,c}, \mathfrak{Q}_R^{or}, \{\tau_i\}_{i=1}^3\big), \qquad \Pi_{\mathrm{rec},R} := P_R^{\Pi,\mathrm{red},\mathrm{geom}}\big|_{W_{\nu(R)}} \xrightarrow{\vartheta_R^{\mathrm{Prim}}} \mathfrak{g}_{\Delta_5}\big|_{W_{\nu(R)}}.$$

The successor map $R' \to R$ is induced by the HN truncation functor on geometric object, extension, flag, quotient, bar, Dirac, Pfaffian, and primitive matrix terms. The map $\vartheta_R^{\mathrm{Prim}}$ is the source-to-target primitive comparison computed from the geometric Hall matrices, not from the target presentation.

Define the finite compact source at height R by

$$\mathfrak{K}_{X,R} = \big(L_{X,R}, H_{X,R}, O_{X,R}, D_{X,R}, R_{X,R}\big)$$

with

$$L_{X,R} = \big(\Gamma_X^{\mathrm{form}}, \widehat{\Gamma}_R, \overline{\Pi}_X, \mathcal{A}^{E_3}_{X,\le R}\big),$$

$$H_{X,R} = \big(\mathrm{Ran}^{\mathrm{hyb}}(E)_{\le R}, \mathcal{F}^{\mathrm{hyb}}_{X,\sigma,S,\le R}, \mathsf{Corr}_R^{E,\mathrm{hyb}}, \mu_R^{\mathrm{loc}}, \mu_R^{\mathrm{hyb}}, \mu_R^{\mathrm{wr}}, Q_{E,R}, I_R^{\mathrm{prot}}\big),$$

$$O_{X,R} = \big(o_{R,\widehat{c}}, \mu_R^{or}, O_R^{\mathrm{quot}}, \{\mathscr{L}^{\mathrm{tan}}_{\partial_i}, K^{\mathrm{cross}}_{\partial_i}, \mathrm{pf}_{\partial_i}, \tau_i\}_{i=1}^3\big),$$

$$D_{X,R} = \big(\mathfrak{A}_R, \Theta_{\Pi,R}, \Theta^-_{\Pi,R}, d_{\mathrm{CE},R}, \mathfrak{D}_{X,R}, \mathrm{Pf}(\mathfrak{D}_{X,R}), \mathrm{pf}_{X,R}, C_{X,R}, b_{X,R}, \Theta_{\mathrm{Kos},R}\big),$$

and

$$R_{X,R} = \big(W_{\nu(R)}, \Pi_{\mathrm{rec},R}, P_R^{\Pi}, \langle\,\cdot\,,\,\cdot\,\rangle_{X,R}, \mathrm{Rad}_{X,R}\big).$$

Then $\mathfrak{K}_{X,R}$ is a finite Dirac–Igusa source over the height-R charge quotient. Its Hall product, coproduct, primitive bracket, and first-order Chevalley–Eilenberg differential are the maps constructed from $\mathfrak{S}_R$. Its protected Dirac operator and Pfaffian line are the finite Dirac–Pfaffian objects carried by $\mathfrak{A}_R$. Its local type-II wall charts are the three rank-one Pfaffian crossings, and the induced local Weyl orientation character satisfies

$$\epsilon^{\mathrm{loc}}_{o,R}(s_{\partial_i}) = -1 = \nu_{\Delta_5}(s_{\partial_i}) \qquad (i = 1, 2, 3).$$

The protected integration on the hybrid wrapped source has trace degree

$$m_R = \mathrm{pr}_3\,\overline{\Pi}_X$$

and the finite source Koszul comparison is

$$b_{X,R} : C_{X,R} \xrightarrow{\simeq} \bar{B}_E^{\mathrm{ch}}(\mathcal{F}^{\mathrm{hyb}}_{X,\sigma,S,\le R}), \qquad \Theta_{\mathrm{Kos},R} : \Omega_E^{\mathrm{ch}} C_{X,R} \xrightarrow{\simeq} \mathsf{Den}_{\Delta_5,E,\le R}.$$

Here

$$\Theta_{\mathrm{Kos},R} = \vartheta_R \circ \varepsilon_R^{\mathrm{geom}}$$

with $\varepsilon_R^{\mathrm{geom}}$ the source-internal cobar counit and ϑ_R the geometric source-to-target quasi-isomorphism. The finite primitive quotient satisfies

$$P_R^{\Pi,\mathrm{red}}\big|_{W_{\nu(R)}} \cong \mathfrak{g}_{\Delta_5}\big|_{W_{\nu(R)}}$$

as a Gram-graded Lie superalgebra with parity.

If the transition maps are strict for the HN topology and the source coalgebras, radical kernels, PBW filtrations, and root-space parity pieces have exact inverse limits, then

$$\mathfrak{K}_X := \varprojlim_R \mathfrak{K}_{X,R}$$

is a Dirac–Igusa realization datum in the sense of Definition 10.65. In this limit

$$\epsilon_0|_{W^{(2)}(\Lambda^{2,1}_{II})} = \nu_{\Delta_5}|_{W^{(2)}(\Lambda^{2,1}_{II})},$$

$$\Omega^{\mathrm{ch}}_E C_X \simeq \mathsf{Den}_{\Delta_5,E}, \qquad P^{\Pi,\mathrm{red}}_X \cong \mathfrak{g}_{\Delta_5},$$

and the Pfaffian cusp expansion is

$$\mathrm{pf}_X|_{\mathfrak{c}_\infty} = 64 q^{1/2} r^{1/2} s^{1/2} \prod_{(n,l,m)\in\Gamma_{\mathrm{eff}}} (1 - q^n r^l s^m)^{f(nm,l)} = \Delta_5(Z).$$

Proof. The finite charge term $L_{X,R}$ is closed under the normal-ordered product because the central extension $\widehat{\Gamma}_R \subset \widehat{\Gamma}_X$ is finite and $\overline{\Pi}_X$ is additive on it. By Proposition 10.61, $\mathfrak{S}_R$ gives the finite oriented Hall vector space $\mathcal{H}^{or}_R$, the pull-push product, the coproduct, the primitive part P_R, the normal-ordered primitive object P^{Π}_R, and the finite Lie bracket on primitives. Its Chevalley–Eilenberg derivation is the displayed $d_{\mathrm{CE},R}$.

The finite hybrid term $H_{X,R}$ is the geometric correspondence functor of Proposition 10.72. The local, wrapped, and mixed products are the pull-push maps $q_!\,\mathrm{TS}^{\mathrm{red}}\,p^*$ on the verified geometric extension stacks; the reduced E-quotient is applied by $Q_{E,R}$ after those correspondences are formed; and the protected integration I^{prot}_R records the normal-ordered trace degree $\mathrm{pr}_3\,\overline{\Pi}_X$.

The quotient-orientation part O^{quot}_R gives orientation descent through the reduced quotient. For the three type-II simple walls, Proposition 10.70 gives the reduced normal self-Ext complexes $K^{\mathrm{cross}}_{\partial_i}$ and Pfaffian sections pf_{∂_i} from the retained wall objects. Proposition 3.12 computes their local sign as -1, and Proposition 3.23 extends the three sign-changing transports to the type-II Weyl group. This is the displayed equality with the Maass character on $W^{(2)}(\Lambda^{2,1}_{II})$.

The finite Hall–Dirac atlas $\mathfrak{A}_R$, together with the normal-ordering cochains, gives the protected first-order object $\mathfrak{D}_{X,R}$, its Pfaffian line, and its finite Pfaffian section. The source coalgebra $C_{X,R}$, the bar comparison $b_{X,R}$, and the source cobar counit $\varepsilon^{\mathrm{geom}}_R$ are constructed in Proposition 10.73. The geometric source-to-target map ϑ_R and the primitive map $\vartheta^{\mathrm{Prim}}_R$ are the maps of Proposition 10.75; hence $\Theta_{\mathrm{Kos},R}$ is the composite $\vartheta_R \circ \varepsilon^{\mathrm{geom}}_R$.

The primitive-recognition object $R_{X,R}$ is the finite matrix system of Proposition 10.74. The map $\vartheta^{\mathrm{Prim}}_R$ identifies the geometric radical quotient with the corresponding Gritsenko–Nikulin window exactly when the displayed source matrices satisfy the Chevalley, Serre, orthogonality, Hopf-radical, comparison, and PBW identities. Passing to the inverse limit gives $P^{\Pi,\mathrm{red}}_X \cong \mathfrak{g}_{\Delta_5}$.

Strictness of the HN transitions and exactness of the listed inverse systems let products, coproducts, brackets, Pfaffian lines, bar/cobar maps, radical quotients, and PBW associated gradeds commute with $\varprojlim_R$. Hence the inverse limit has precisely the five entries $(L_X, H_X, O_X, D_X, R_X)$ of Definition 10.65. The Pfaffian cusp expansion is the finite-window expansion of Theorem 3.15, and the last equality is the Borcherds–Gritsenko–Nikulin product identification for Δ_5. □

Proposition 10.79 *(Low-degree bracket information).* The denominator algebra gives the following target-side low-degree checks. With

$$\alpha(n,l,m) = n\delta_1 + m\delta_2 + (n+m-l)\delta_3,$$

the active positive rows through height three are

$\mathrm{ht}\,\beta$	β	$\gamma(\beta)$	(β,β)	$\mathrm{smult}(\beta)$
1	δ_1	$(1,1,0)$	2	1
1	δ_2	$(0,1,1)$	2	1
1	δ_3	$(0,-1,0)$	2	1
2	$\delta_1+\delta_2$	$(1,2,1)$	0	10
2	$\delta_1+\delta_3$	$(1,0,0)$	0	10
2	$\delta_2+\delta_3$	$(0,0,1)$	0	10
3	$2\delta_1+\delta_2$	$(2,3,1)$	2	1
3	$\delta_1+2\delta_2$	$(1,3,2)$	2	1
3	$2\delta_1+\delta_3$	$(2,1,0)$	2	1
3	$\delta_1+2\delta_3$	$(1,-1,0)$	2	1
3	$2\delta_2+\delta_3$	$(0,1,2)$	2	1
3	$\delta_2+2\delta_3$	$(0,-1,1)$	2	1
3	$\delta_1+\delta_2+\delta_3$	$(1,1,1)$	-6	-64.

After choosing even Chevalley generators e_i, f_i, h_i for the real simple roots and setting

$$A_{ij} = (\delta_i,\delta_j), \qquad E_{ij} := [e_i, e_j] \quad (i \neq j),$$

so $E_{ji} = -E_{ij}$, one has

$$[h_i,h_j] = 0, \qquad [h_i,e_j] = A_{ij}e_j, \qquad [h_i,f_j] = -A_{ij}f_j, \qquad [e_i,f_j] = \delta_{ij}h_i,$$

and

$$(\mathrm{ad}\,e_i)^3 e_j = 0 \quad (i \neq j).$$

If $\{i,j,k\} = \{1,2,3\}$, then

$$[h_i,E_{ij}] = [h_j,E_{ij}] = 0, \qquad [h_k,E_{ij}] = -4E_{ij},$$

$$[f_i,E_{ij}] = 2e_j, \qquad [f_j,E_{ij}] = -2e_i, \qquad [f_k,E_{ij}] = 0.$$

For ordered $i \neq j$, put $E_{iij} := [e_i, E_{ij}]$. Then

$$[e_i,E_{iij}] = 0, \qquad [f_i,E_{iij}] = 2E_{ij}, \qquad [f_j,E_{iij}] = 0.$$

For $i < j$, the isotropic primitive degree $a_{ij} := \delta_i + \delta_j$ has

$$\mathrm{smult}(a_{ij}) = 10 = 1 + 9 :$$

one even real-real commutator direction E_{ij} and nine even Gritsenko–Nikulin isotropic simple directions

$$u_{ij,1},\dots,u_{ij,9} \in \mathfrak{g}_{a_{ij}}, \qquad \tau(ta_{ij}) = 9 \quad (t \geq 1).$$

The orthogonality relations give

$$[e_i, u_{ij,r}] = [e_j, u_{ij,r}] = 0, \qquad [f_i, u_{ij,r}] = [f_j, u_{ij,r}] = 0.$$

Thus the target splits the signed multiplicity 10 at a_{ij} into one real-real commutator direction and nine isotropic simple directions. A compact source must realize this splitting by actual protected Hall primitives; the equality sdim = 10 alone does not choose the summands or the orthogonality constants. In degree $\delta_{123} := \delta_1 + \delta_2 + \delta_3$, the real bracketings

$$T_1 = [e_1, E_{23}], \qquad T_2 = [e_2, E_{31}], \qquad T_3 = [e_3, E_{12}]$$

satisfy $T_1 + T_2 + T_3 = 0$. The brackets $[e_k, u_{ij,r}] \in \mathfrak{g}_{\delta_{123}}$ for $\{i, j, k\} = \{1, 2, 3\}$ are not numerical constants until a basis and parity split of $\mathfrak{g}_{\delta_{123}}$ are chosen. Since $(\delta_k, a_{ij}) = -4$, the same Borcherds–Kac presentation also imposes the real-string relation

$$(\operatorname{ad} e_k)^5 u_{ij,r} = 0$$

and the analogous relation for the negative generators. The first bracket $[e_k, u_{ij,r}]$ is only the first visible map in this string. The product exponent alone gives only

$$\operatorname{smult}(\delta_{123}) = f(1, 1) = -64.$$

Let

$$d^{\mathrm{GN}}_{p,123} := \operatorname{mult}_{\overline{p}}(\delta_{123}), \qquad p = 0, 1.$$

Then

$$d^{\mathrm{GN}}_{0,123} - d^{\mathrm{GN}}_{1,123} = -64,$$

but the product exponent alone does not give the two integers $d^{\mathrm{GN}}_{0,123}$, $d^{\mathrm{GN}}_{1,123}$, the bracket coordinates, or the radical. Proposition 10.81 computes the target presentation values 29 and 93 from the Weyl-sum coefficient and the Borcherds–Kac presentation; the compact source must still reproduce them by protected primitives and the Hopf pairing. The first unresolved compact test channel is therefore the following: produce parity-homogeneous compact primitives

$$v^{\overline{0}}_1, \ldots, v^{\overline{0}}_{29}, \qquad v^{\overline{1}}_1, \ldots, v^{\overline{1}}_{93},$$

in the descended degree (1, 1, 1), compute the Hall constants

$$[e^X_k, u^X_{ij,r}] = \sum_{a=1}^{29} C^{\overline{0}}_{k;ijr,a} v^{\overline{0}}_a, \qquad \{i, j, k\} = \{1, 2, 3\},$$

the Jacobi relation $T^X_1 + T^X_2 + T^X_3 = 0$, the real-string relations $(\operatorname{ad} e^X_k)^5 u^X_{ij,r} = 0$, the completed Hopf-pairing matrix between degrees δ_{123} and $-\delta_{123}$, with even block size 29 and odd block size 93, and the closed radical quotient of Theorem 10.66. The bracket above is even, but the odd basis is still load-bearing: it is part of the full parity-dimension, PBW, and radical comparison. This channel is the first place where the compact Hall source must supply more than the GN signed multiplicity. These are denominator-algebra constants. They are not compact Hall constants.

Proof. The table is obtained by writing $\beta = c_1\delta_1 + c_2\delta_2 + c_3\delta_3$ and solving

$$\gamma(\beta) = (c_1, c_1 + c_2 - c_3, c_2).$$

Then $(\beta, \beta) = -2(4nm - l^2)$, and $\mathrm{smult}(\beta) = f(nm, l)$ follows from Theorem 9.2. The displayed bracket constants are the normalized Chevalley, Jacobi and Serre constants for the symmetric Cartan matrix $A = ((\delta_i, \delta_j))$. The signs $E_{ji} = -E_{ij}$, $[f_i, E_{ij}] = 2e_j$, and $[f_j, E_{ij}] = -2e_i$ follow from the graded Jacobi identity and the normalization $[e_i, f_i] = h_i$. The relation $T_1 + T_2 + T_3 = 0$ is the ordinary Jacobi identity for the even real generators e_1, e_2, e_3. For $a_{ij} = \delta_i + \delta_j$ and $\{i, j, k\} = \{1, 2, 3\}$, one has $(\delta_k, a_{ij}) = -4$, hence the real Borcherds–Kac Serre exponent is $1 - 2(\delta_k, a_{ij})/(\delta_k, \delta_k) = 5$. The equality $\tau(ta_{ij}) = 9$ on primitive isotropic rays is the Gritsenko–Nikulin isotropic simple-root datum recorded above. No compact Hall structure constant is inferred: a compact computation would have to construct the relevant extension correspondences, protected primitive bases, orientation signs, the δ_{123} channel constants, and the radical quotient. □

Proposition 10.80 *(The determinant does not determine compact Hall constants).* The scalar OP square, the Borcherds product for Δ_5, and the Gritsenko–Nikulin denominator algebra determine the visible signed exponents

$$\mathrm{smult}(\alpha) = \dim \mathfrak{g}_{\alpha,\bar{0}} - \dim \mathfrak{g}_{\alpha,\bar{1}}.$$

They do not determine a compact Hall bracket, compact primitive representatives, full source parity dimensions, the positive-negative Hopf pairing, its closed radical quotient, the no-extra-relations theorem, or completed PBW compatibility.

Proof. A Γ_{gram}-graded super vector space with the same signed dimensions gives the same determinant. Adding a cancelling pair $W_\gamma \oplus \Pi W_\gamma$ in any active degree leaves every signed exponent unchanged, while changing the full parity dimensions. Equipping the same graded vector space with the zero bracket also leaves the determinant unchanged, but it cannot be the denominator algebra because the target has nonzero real-root brackets, for example $[e_i, e_j] \neq 0$ for $i \neq j$, and the first timelike channel requires the 29|93 presentation split of Proposition 10.81. A compact Hall realization must therefore construct extension correspondences, orientation signs, protected primitive bases, Hall structure constants, pairing matrices, radical kernels, and PBW finite-window isomorphisms. The determinant is a necessary decategorified shadow of these data, not a source for them. □

Proposition 10.81 *(First timelike presentation count).* Let

$$\delta_{123} := \delta_1 + \delta_2 + \delta_3 = 2f_2 - f_3 + 2f_{-2}.$$

In the Gritsenko–Nikulin additive correction coefficient one has

$$m(\delta_{123}) = -93.$$

Consequently the target denominator algebra has full parity dimensions

$$\mathrm{mult}_{\bar{0}}(\delta_{123}) = 29, \qquad \mathrm{mult}_{\bar{1}}(\delta_{123}) = 93.$$

The 29 even directions are the two real-real-real Borcherds–Kac presentation words

$$T_1 = [e_1, [e_2, e_3]], \qquad T_2 = [e_2, [e_3, e_1]], \qquad T_3 = [e_3, [e_1, e_2]], \qquad T_1 + T_2 + T_3 = 0,$$

together with the 27 mixed real-isotropic words

$$[e_k, u_{ij,r}], \qquad \{i, j, k\} = \{1, 2, 3\}, \quad 1 \le r \le 9.$$

The 93 odd directions are the negative-norm imaginary simple generators in degree δ_{123}. This is a target-side presentation count. It is not a compact Hall computation and is not deduced from the signed equality $29 - 93 = -64$.

Proof. The root δ_{123} corresponds in the notation of the Weyl-sum coefficient formula to

$$\delta_{123} = (3-1)f_2 - \frac{3-1}{2} f_3 + (3-1)f_{-2}.$$

Hence $m(\delta_{123}) = -64^{-1} c_\Delta(3,3,3)$. After removing the leading monomial from $D_5 = 64^{-1}\Delta_5$,

$$D_5 = q^{1/2} r^{1/2} s^{1/2} \prod_{\gamma \in \Gamma_{\mathrm{eff}}} (1 - q^n r^l s^m)^{f(nm,l)},$$

the finite coefficient calculation in q- and s-degree one gives

$$[qrs] \prod_{\gamma \in \Gamma_{\mathrm{eff}}} (1 - q^n r^l s^m)^{f(nm,l)} = 93.$$

Explicitly, only the q-linear, s-linear, and qs-linear factors enter. With

$$A(r) = 1 - r^{-1}, \qquad B(r) = -(r^{-1} + 10 + r),$$

$$C(r) = -10r^{-2} + 64r^{-1} - 108 + 64r - 10r^2,$$

one has

$$[qrs] \prod_{\gamma \in \Gamma_{\mathrm{eff}}} (1 - q^n r^l s^m)^{f(nm,l)} = [r]\, A(r)\big(B(r)^2 + C(r)\big)$$

and

$$A(r)\big(B(r)^2 + C(r)\big) = 9r^{-3} - 93r^{-2} + 90r^{-1} - 90 + 93r - 9r^2.$$

Equivalently $c_\Delta(3,3,3) = 64 \cdot 93$, so $m(\delta_{123}) = -93$. Since this coefficient is negative, the Gritsenko–Nikulin parity convention makes these 93 simple generators odd.

It remains to count the lower presentation words in the same total degree. The multidegree $(1,1,1)$ part of the free Lie algebra on the three even real generators has dimension 2; the three displayed bracketings satisfy exactly the Jacobi relation $T_1 + T_2 + T_3 = 0$. For each primitive isotropic root $a_{ij} = \delta_i + \delta_j$, the Gritsenko–Nikulin isotropic multiplicity is $\tau(a_{ij}) = 9$, and the complementary real root δ_k has $(\delta_k, a_{ij}) = -4$. Thus the first mixed brackets $[e_k, u_{ij,r}]$ are allowed presentation words; the orthogonality relations kill only the adjacent brackets with δ_i and δ_j, and the real-string relation occurs first at $(\operatorname{ad} e_k)^5 u_{ij,r}$. There are $3 \cdot 9 = 27$ such even words. No other simple-root decomposition has total degree δ_{123}. The Borcherds–Kac presentation and PBW therefore give $2 + 27 = 29$ even and 93 odd target directions in this degree. Their signed difference is

$$29 - 93 = -64 = f(1,1),$$

which checks the denominator exponent but does not determine the compact source parity split, bracket constants, pairing matrix, or radical quotient. □

Proposition 10.82 *(First saturated primitive-recognition window).* Let

$$W_{\leq 3} := \{\beta = c_1\delta_1 + c_2\delta_2 + c_3\delta_3 \in \mathcal{R}_+ \ : \ c_i \geq 0, \quad 1 \leq c_1 + c_2 + c_3 \leq 3\}.$$

This is the first downward-saturated finite target window containing the real simple roots, the primitive isotropic simple-root fibres, and the first timelike degree. In $G(\mathscr{B}_{\Delta_5})$ the full parity dimensions in this window are

β	$\mathrm{mult}_{\bar{0}}(\beta)$	$\mathrm{mult}_{\bar{1}}(\beta)$
δ_i $(1 \leq i \leq 3)$	1	0
$\delta_i + \delta_j$ $(i < j)$	10	0
$2\delta_i + \delta_j$ $(i \neq j)$	1	0
δ_{123}	29	93.

The parity entries use the imported Gritsenko–Nikulin/Kac parity convention; the target computation checks signed coefficients and presentation counts, not source parity bases. Writing $a_{ij} = \delta_i + \delta_j$ and $\tau = \delta_1 + \delta_2 + \delta_3$, the target-side ordered binary decompositions in $W_{\leq 3}$ are exactly

$$a_{ij} = \delta_i + \delta_j = \delta_j + \delta_i,$$

$$2\delta_i + \delta_j = \delta_i + a_{ij} = a_{ij} + \delta_i, \qquad \delta_i + 2\delta_j = \delta_j + a_{ij} = a_{ij} + \delta_j,$$

and

$$\tau = \delta_1 + a_{23} = a_{23} + \delta_1 = \delta_2 + a_{13} = a_{13} + \delta_2 = \delta_3 + a_{12} = a_{12} + \delta_3.$$

Choose target bases

$$e_i \in \mathfrak{g}_{\delta_i}, \qquad E_{ij} = [e_i, e_j], \quad u_{ij,1}, \ldots, u_{ij,9} \in \mathfrak{g}_{a_{ij}},$$

$$Y_{i;ij} = [e_i, E_{ij}], \qquad Y_{j;ij} = [e_j, E_{ij}],$$

and, in the even part of $\mathfrak{g}_\tau$,

$$T_1 = [e_1, E_{23}], \qquad T_2 = [e_2, E_{31}], \qquad M_{12,r} = [e_3, u_{12,r}], \quad M_{13,r} = [e_2, u_{13,r}], \quad M_{23,r} = [e_1, u_{23,r}],$$

with $E_{31} = -E_{13}$ and $T_3 = [e_3, E_{12}] = -T_1 - T_2$; let $w_1, \ldots, w_{93}$ be odd simple generators in degree τ. With $c_0 = (1, 0, \ldots, 0)^t \in \mathbb{C}^{10}$, the target bracket matrices start as

$$B_{\delta_i,\delta_j} = c_0, \qquad B_{\delta_j,\delta_i} = -c_0,$$

and the adjacent simple-isotropic maps $B_{\delta_p,a_{ij}} : \mathbb{C}^{10} \to \mathbb{C}$, $p = i, j$, are $(1, 0, \ldots, 0)$, killing the $u_{ij,r}$-columns. The complementary simple-isotropic maps to the even τ-block are

$$B_{\delta_1,a_{23}}(E_{23}) = T_1, \quad B_{\delta_1,a_{23}}(u_{23,r}) = M_{23,r},$$

$$B_{\delta_2,a_{13}}(E_{13}) = -T_2, \quad B_{\delta_2,a_{13}}(u_{13,r}) = M_{13,r},$$

$$B_{\delta_3,a_{12}}(E_{12}) = -T_1 - T_2, \quad B_{\delta_3,a_{12}}(u_{12,r}) = M_{12,r},$$

and their odd τ-rows are zero. With dual positive-negative target pairings, the target coproduct templates are the transposes of these matrices. A compact Hall source must reproduce these templates after

quotienting by its own radical; the target package with zero radical is only a consistency check. Consequently a compact source passes primitive recognition on $W_{\leq 3}$ only after supplying the following finite datum: parity-homogeneous representatives with the displayed dimensions; the Chevalley, real-Serre, orthogonality, and super-sign relations restricted to $W_{\leq 3}$; the Jacobi relation

$$T_1^X + T_2^X + T_3^X = 0;$$

the 27 mixed maps $[e_k^X, u_{ij,r}^X]$ into the even δ_{123}-block; positive-negative Hopf-pairing matrices in every degree of $W_{\leq 3}$ with closed radical kernels stable under the finite transition maps; the finite kernel equality

$$\ker \pi_{W_{\leq 3}} = \left(\overline{J_{\mathrm{BK}} + \mathrm{Rad}_{\mathrm{GN}}}\right)_{W_{\leq 3}};$$

and the associated-graded PBW isomorphism on the truncated enveloping algebras. If this finite datum is supplied, the restricted map

$$G(\mathscr{B}_{\Delta_5})_{W_{\leq 3}} \longrightarrow P^{\Pi,\mathrm{red}}_{X,W_{\leq 3}}$$

is an isomorphism of finite Gram-graded Lie superalgebras. Without the kernel equality, the Hopf-radical ideal/coideal assertion, and the PBW associated-graded isomorphism, equality of the displayed dimensions is not recognition. In the notation of (NO7), this first window requires, at minimum, explicit compact primitive bases for $W_{\leq 3} \cup (-W_{\leq 3}) \cup \{0\}$. Equivalently, for each retained parity block one must exhibit a source splitting

$$P^X_{R,\beta,\bar{p}} = K_{\beta,\bar{p}} \oplus L_{\beta,\bar{p}}, \qquad Q_{\beta,\bar{p}} = [0\ I],$$

where $K_{\beta,\bar{p}}$ is the radical block and $L_{\beta,\bar{p}}$ maps to the corresponding target basis above. The datum further requires positive-negative pairing matrices in every displayed degree, radical kernel bases and quotient maps, the Chevalley, real-Serre, isotropic, Jacobi, and 27 mixed bracket constants, coproduct matrices for every retained decomposition, product matrices or equivalent Hopf-adjointness data, and successor transition matrices preserving the kernels. The 29|93 block in degree δ_{123} is the first place where signed dimension alone is especially weak: the compact source must exhibit the two parity vector spaces, the pairing/radical matrices, and the bracket/coproduct operations on them.

The next coefficient checks already lie outside this window. The exact finite product computation in `compute/verify_square_root.py` is target-only: it verifies Borcherds/Gritsenko–Nikulin arithmetic, not compact Hall representatives, Hopf-pairing matrices, coproduct matrices, or radical kernels. It gives

$$[q^2r^2s]\frac{D_5}{(qrs)^{1/2}} = [qr^2s^2]\frac{D_5}{(qrs)^{1/2}} = [qs]\frac{D_5}{(qrs)^{1/2}} = -90,$$

and

$$[q^2r^2s^2]\frac{D_5}{(qrs)^{1/2}} = 540.$$

Thus the Gritsenko–Nikulin additive coefficients are

$$m(2\delta_1 + \delta_2 + \delta_3) = m(\delta_1 + 2\delta_2 + \delta_3) = m(\delta_1 + \delta_2 + 2\delta_3) = 90,$$

$$m(2\delta_{123}) = -540.$$

These are target simple-root coefficients. They do not count the lower PBW words in those degrees and do not supply compact Hall representatives, pairing matrices, radicals, or no-extra-relations data.

The first obstruction to extending the displayed finite recognition statement from $W_{\leq 3}$ to height four is already numerical. The full signed target root supermultiplicities in the three height-four timelike degrees are

$$\mathrm{smult}(2\delta_1 + \delta_2 + \delta_3) = \mathrm{smult}(\delta_1 + 2\delta_2 + \delta_3) = \mathrm{smult}(\delta_1 + \delta_2 + 2\delta_3) = 108,$$

whereas the new negative-norm simple-root coefficients in those same degrees are 90. Thus each of these degrees carries a signed height-four residual

$$108 - 90 = 18$$

coming from lower presentation words after the Borcherds–Kac relations and radical quotient are imposed. This number is a signed residual, not a count of compact PBW basis vectors. It says precisely that the height-four window cannot be obtained by appending the additive coefficients alone.

The same phenomenon is visible on the doubled isotropic boundary rays: for $i < j$,

$$\mathrm{smult}(2\delta_i + 2\delta_j) = 10, \qquad \tau(2(\delta_i + \delta_j)) = 9,$$

so the signed non-simple residual is 1 on each doubled isotropic ray. These doubled-ray checks test the isotropic multiplicity convention $\tau(t(\delta_i + \delta_j)) = 9$ against the product exponents; they do not identify compact primitives.

The height-four real-string obstruction is also explicit. If w_s, $1 \leq s \leq 93$, denotes an odd negative-norm simple generator in degree δ_{123}, then

$$(\delta_i, \delta_{123}) = -2, \qquad 1 - 2(\delta_i, \delta_{123})/(\delta_i, \delta_i) = 3,$$

and the target presentation contains

$$(\mathrm{ad}\, e_i)^3 w_s = 0, \qquad (\mathrm{ad}\, f_i)^3 w_s^- = 0.$$

The first brackets $[e_i, w_s]$ land in the three height-four timelike degrees above and are odd. A height-four compact recognition datum must therefore supply those odd maps, the even simple generators of multiplicity 90, the lower-word relations producing the signed residual 18, the positive-negative pairing matrices, the radical kernels, and PBW associated-graded compatibility. Dimension equality alone still does not prove recognition.

Proof. The table is exactly the target data of Propositions 10.79 and 10.81: the three real simple roots are even one-dimensional, each primitive isotropic degree has one real-real commutator and nine even isotropic simple directions, the six height-three real roots are even one-dimensional, and δ_{123} has presentation dimensions 29|93. The finite datum is the restriction of clauses (S3)–(S10) of Theorem 10.66 to the downward-saturated window $W_{\leq 3}$. The same proof as the theorem then gives the restricted isomorphism. The height-four additive coefficients are computed from the monic Borcherds product and converted to Gritsenko–Nikulin $m(a)$ by the coefficient formula $m(a) = -64^{-1} c_\Delta(n, l, m)$ of [37, Theorem 2.1]; the script performs this conversion with $\alpha(n, l, m) = n\delta_1 + m\delta_2 + (n + m - l)\delta_3$. The full signed height-four target multiplicities are read from the same denominator theorem:

$$\mathrm{smult}(\beta) = f(nm, l) \quad \text{for} \quad \gamma(\beta) = (n, l, m).$$

For $\beta = 2\delta_1 + \delta_2 + \delta_3$, $\delta_1 + 2\delta_2 + \delta_3$, $\delta_1 + \delta_2 + 2\delta_3$ this gives respectively $(n, l, m) = (2, 2, 1), (1, 2, 2), (1, 0, 1)$, hence the coefficients $f(2, 2) = 108$, $f(2, 2) = 108$, $f(1, 0) = 108$. Subtracting the additive coefficients 90 gives the displayed signed residual 18. For the doubled isotropic degrees $2(\delta_i + \delta_j)$, the denominator exponent is $f(4, 4) = f(0, 0) = 10$ in the three symmetric chambers, while the isotropic simple-root multiplicity is the Gritsenko–Nikulin constant $\tau(2(\delta_i + \delta_j)) = 9$. Finally, the real-string exponents follow directly from the Gram matrix: $(\delta_i, \delta_{123}) = -2$, $(\delta_i, \delta_i) = 2$, and therefore the exponent is 3. All displayed numerical checks are asserted in `compute/verify_square_root.py`. □

Theorem 10.83 *(Executable finite source matrix criterion).* Let $W \subset Q_+ \setminus \{0\}$ be finite and downward saturated. A compact Hall stage R evaluates primitive recognition on W exactly when it supplies, for every

$$\beta \in W \cup (-W) \cup \{0\},$$

parity-homogeneous compact source bases of $P^X_{R,\beta,\bar{p}}$, product matrices $M_{\alpha,\beta}$, coproduct matrices $D^{\mu,\nu}_{\rho}$, positive-negative pairing matrices $G_{\beta,\bar{p}}$, HN transition matrices, and geometric provenance for every basis vector. Define

$$B_{\alpha,\beta} = M_{\alpha,\beta} - (-1)^{|x||y|} M_{\beta,\alpha}\tau, \qquad K_{\beta,\bar{p}} = \ker G_{\beta,\bar{p}} \cap \ker G^t_{-\beta,\bar{p}}.$$

Choose splittings

$$P^X_{R,\beta,\bar{p}} = K_{\beta,\bar{p}} \oplus L_{\beta,\bar{p}}, \qquad Q_{\beta,\bar{p}} = [\ 0\ \ I\ ].$$

The data pass the finite source test if and only if the following matrix identities hold:

$$Q_{\alpha+\beta} B_{\alpha,\beta}(P^X_{R,\alpha} \otimes K_\beta) = 0, \qquad Q_{\alpha+\beta} B_{\alpha,\beta}(K_\alpha \otimes P^X_{R,\beta}) = 0,$$

$$(Q_\mu \otimes Q_\nu) D^{\mu,\nu}_{\rho} K_\rho = 0, \qquad M^t_{\alpha,\beta} G_{\alpha+\beta} = (G_\alpha \otimes G_\beta) D^{\alpha,\beta}_{\alpha+\beta}.$$

The same matrices satisfy the Chevalley, real-Serre, orthogonality, super-sign, Jacobi, and real-string relation rows in the retained window. The finite presentation map

$$\pi_W : \widehat{\mathfrak{F}}(\mathscr{B}_{\Delta_5})_{\leq W} \longrightarrow P^{\Pi,\mathrm{red}}_{X,\leq W}$$

satisfies

$$\ker \pi_W = \left(\overline{J_{\mathrm{BK}} + \mathrm{Rad}_{\mathrm{GN}}}\right)_{\leq W},$$

and the PBW associated-graded comparison

$$\mathrm{gr}_{\mathrm{PBW}}\, U(P^{\Pi,\mathrm{red}}_X)_{\leq W} \longrightarrow \mathrm{gr}_{\mathrm{PBW}}\, U(\mathfrak{g}_{\Delta_5})_{\leq W}$$

is an isomorphism. The transition matrices preserve radical kernels and PBW filtrations strictly and satisfy the finite-slice Mittag–Leffler stable-image rank condition. Then radical descent, no-extra-relations, PBW, and exact inverse limits hold on W.

For $W = W_{\leq 3}$, this demands compact bases on

$$W_{\leq 3} \cup (-W_{\leq 3}) \cup \{0\},$$

including 29 even and 93 odd source primitives in degree δ_{123}, together with all M, D, B, G, K, Q matrices, the 27 real-isotropic mixed maps, Jacobi, real-string, Hopf-radical, no-extra-relations, PBW, and transition matrices.

Proof. The two QB-identities make the quotient bracket independent of lifts. The QD-identity makes the quotient coproduct independent of lifts. Hopf adjointness and nondegeneracy of the quotient tensor pairing give the coideal half of the radical descent; Frobenius and Jacobi give the Lie-ideal half. The finite kernel equality is the no-extra-relations assertion. PBW is checked on associated gradeds after the radical quotient. The stable-image Mittag–Leffler condition gives $R^1 \varprojlim = 0$, so kernels, radicals, PBW associated gradeds, and parity pieces commute with the inverse limit.

The first window statement is the specialization of the criterion to Proposition 10.82. The scripts `compute/verify_lattice.py` and `compute/verify_square_root.py` verify the target lattice and Borcherds–Gritsenko–Nikulin arithmetic in this window; they do not provide compact source bases, source pairings, radicals, coproducts, or PBW matrices. □

Proposition 10.84 *(Finite $W_{\leq 3}$ NO7 source protocol).* For a finite Hall stage R whose source image contains $W_{\leq 3} \cup (-W_{\leq 3}) \cup \{0\}$, a compact source verifies the (NO7) part of the $W_{\leq 3}$-recognition datum only after supplying parity-homogeneous source splittings

$$P^X_{R,\beta,\bar{p}} = K_{\beta,\bar{p}} \oplus L_{\beta,\bar{p}}, \qquad Q_{\beta,\bar{p}} = [0\ I],$$

positive-negative pairing matrices G_β, radical bases

$$K_\beta = \ker G_\beta \cap \ker G^t_{-\beta},$$

product matrices $M_{\alpha,\beta}$, coproduct matrices $D^{\mu,\nu}_\rho$, bracket matrices $B_{\alpha,\beta}$, and successor matrices $\rho^P_{R^+R}$ preserving K_β. Let $\mathrm{Pair}^{\pm}_{\leq 3}(R)$ be the finite set of ordered pairs (α, β) with $\alpha, \beta, \alpha + \beta \in W_{\leq 3} \cup (-W_{\leq 3}) \cup \{0\}$ whose product component is not killed by the height-R quotient, and let $\mathrm{Dec}^{\pm}_{\leq 3}(R)$ be the finite set of ordered triples $(\rho; \mu, \nu)$ with $\rho, \mu, \nu \in W_{\leq 3} \cup (-W_{\leq 3}) \cup \{0\}$, $\rho = \mu + \nu$, and whose reduced coproduct component is not killed by the height-R quotient. For every $(\alpha, \beta) \in \mathrm{Pair}^{\pm}_{\leq 3}(R)$ one must verify

$$Q_{\alpha+\beta} B_{\alpha,\beta}(P^X_{R,\alpha} \otimes K_\beta) = 0, \qquad Q_{\alpha+\beta} B_{\alpha,\beta}(K_\alpha \otimes P^X_{R,\beta}) = 0,$$

and, for every $(\rho; \mu, \nu) \in \mathrm{Dec}^{\pm}_{\leq 3}(R)$,

$$(Q_\mu \otimes Q_\nu) D^{\mu,\nu}_\rho K_\rho = 0.$$

For every retained pair whose product degree also carries the displayed reduced coproduct component, one must verify Frobenius compatibility

$$\langle M_{\alpha,\beta}(x \otimes y), z\rangle_R = \langle x \otimes y, D^{\alpha,\beta}_{\alpha+\beta} z\rangle_R,$$

and the quotient-tensor pairing must be non-degenerate. These are source-side matrix equalities. The target denominator algebra and the script `compute/verify_square_root.py` do not supply them. Indeed a source can have the same quotient bracket templates as $G(\mathscr{B}_{\Delta_5})_{W_{\leq 3}}$ and still fail (NO7): there may be $k \in K_\rho$ with

$$(Q_\mu \otimes Q_\nu) D^{\mu,\nu}_\rho (k) \neq 0.$$

Then the quotient target table sees no discrepancy, but the radical is not a coideal and the Hopf quotient is not defined. This is the finite tensor-Hopf obstruction hidden by signed determinant data.

Proof. This is (NO7) and Lemma 10.42 restricted to the first downward-saturated window. The two QB-containments are the finite Lie-ideal tests for the radical, the QD-containment is the finite coideal square, and Hopf adjointness together with quotient-tensor non-degeneracy is the finite algebraic condition forcing the coproduct to descend through the radical quotient. Transition preservation of K_β is the finite part of the closed radical condition in the HN inverse system. □

Definition 10.85 *(State-side construction of D_0 and the Dirac–Igusa datum).* Construct first the D_0-datum of Definition 10.54, and then the full datum $\mathfrak{K}_X$ of Definition 10.65. The construction is the following compatible finite inverse system:

(D_0-state) for every $R \in \mathcal{R}_{\mathrm{HN}}$, the finite Hall source kernel $\mathfrak{S}_R$ of Definition 10.60, so that Proposition 10.61 produces the finite state object $\mathcal{H}_R^{or}$, pull-push Hall product, coproduct, primitive bracket, normal-ordered primitive algebra, and finite CE differential. Then construct the full finite stage $D_{0,R}$ on $\widehat{\Gamma}_R$, equivalently the finite Hall–Dirac atlas $\mathfrak{A}_R$ of Proposition 10.63, including the Dirac complex, source observable shadow, Pfaffian square-root datum, compact parity lifts, protected integration, and successor transition maps with $\mathfrak{o}^{\mathrm{atlas}}_{R^+R} = \mathfrak{o}^{\mathrm{tr}}_{R^+R} = 0$, whose composites give the non-successor transitions and whose inverse limit is $\widehat{\mathcal{H}}^{or}_{\sigma,S}$;

(D_0-quotient) compute and trivialize the reduced orientation and quotient obstruction tuple $\mathfrak{o}^{or}_{R,C}$ for every stable charge stratum and finite stabilizer appearing at height R, including the zero $H^1(BE[N];\mathbb{F}_2)$-linearization characters after finite gerbe null-trivialization, compatibly in R;

(D_0-observable) construct the chain-level $K3$-to-E specialization, the projection-finite chiral shadow, and the hybrid wrapped sector $\mathcal{F}_X^{D_0}$ that recovers positive elliptic degree before quotienting by translation in E, together with a source-to-target finite comparison map satisfying $\mathfrak{o}^{obs}_R = 0$ on Γ^{test}_R;

(D_0-descent) construct the finite-topology, Hochschild, triple-coherence, Jacobi, Frobenius, negative-cyclic, and Hopf-radical ideal/coideal cochains $\Theta_{\Pi,R}$, $\Theta^-_{\Pi,R}$ with

$$\mathfrak{o}^{\mathrm{top}}_{\Theta,R^+R} = \mathfrak{o}^{\mathrm{Hoch}}_{\Theta,R} = \mathfrak{o}^{\mathrm{tri}}_{\Theta,R} = \mathfrak{o}^{\mathrm{Jac}}_{\Theta,R} = \mathfrak{o}_{F,R} = \mathfrak{o}^{\mathrm{cyc}}_{\Theta,R} = \mathfrak{o}^{\mathrm{rad}}_{\Theta,R} = 0,$$

giving exact compatibility with the normal-ordered homomorphism $\overline{\Pi}_X$ on primitive brackets, equivalently the finite diagrams (NO1)–(NO7) of Proposition 10.40, with compatibility against the Hopf radical quotients at every transition. The Picard and finite-fibre subclaims are reduced to Lemma 10.36; the surviving geometric work is exactly the cyclic/flag/Frobenius/Hopf-adjointness package of Proposition 10.48;

(D_0-Pfaffian) construct the Pfaffian line and finite first-order operator/algebra $\mathfrak{D}_{X,R}$ on the reduced compact quotient $\mathfrak{Q}_R = (\mathfrak{M}_R/E)^{\mathrm{red}}$, compatible under $R^+ \to R$, including a trivialization of the Pfaffian square-root torsor in $\mathrm{Pic}(\mathfrak{Q}_R)[2]$, with compact parity lifts $\ell_R(\gamma) \in \mathscr{P}_R(\gamma)$ on every $\gamma \in \Gamma^{\mathrm{test},+}_R$ satisfying

$$\mathfrak{o}^{\mathrm{supp}}_{R,\gamma} = 0, \qquad \mathfrak{o}^{\mathrm{par}}_{R,\gamma} = 0,$$

including absence of actual primitive support in target-zero degrees, coverage of every nonzero Borcherds target degree in the exhausting windows $\Gamma_{\Delta,R}$, finite Pfaffian square-root datum $\mathscr{L}^{\otimes 2}_{\mathrm{Pf},R} \simeq \mathscr{L}_{\det,R}$, and normalized leading coefficient $\mathfrak{o}^{\mathrm{lead}}_R = 0$. Negative representatives are supplied only in the primitive-recognition datum, through the full positive-negative pairing and Chevalley data. The minimal target-window parity model $\mathsf{P}^{\min}_{\Delta,R}$ is already constructed in Proposition 10.62; the compact identification is to identify it, or a non-minimal lift in $\mathscr{P}_R(\gamma)$,

with actual source primitive spaces and to prove the no-hidden-cancellation condition if the minimal lift is used;

(full datum) construct the strong orientation monodromy and the chiral Koszul source datum

$$\mathfrak{K}^{\mathrm{Kos}}_{X,R} = (\mathcal{F}^{\mathrm{hyb}}_{X,\sigma,S,\leq R}, C_{X,R}, F^{\mathrm{co}}_{R,\bullet}, \Delta^{\mathrm{ch}}_{R}, b_{X,R}, \Theta_{\mathrm{Kos},R})$$

of Definition 10.19, with

$$\mathfrak{D}_{\mathrm{Kos},R} = 0$$

compatibly in R, together with the Koszul $R^1 \varprojlim = 0$ condition for the source coalgebras, completed bar/cobar systems, target truncations, and cones of $b_{X,R}$ and $\Theta_{\mathrm{Kos},R}$, including $\mathfrak{D}_{\mathrm{hyb},R} = 0$ for the hybrid source. Equivalently, construct the full finite local/wrapped correspondence atlas, the properness/admissibility residuals for its pull-push operations, the quotient-after-correspondence functor, protected integration, the conilpotent compact source coalgebra, the finite conilpotent filtration, the source collision coproduct with finite coassociativity, counit, and transition-continuity diagrams, the bar comparison to $\bar{B}^{\mathrm{ch}}_{E}(\mathcal{F}^{\mathrm{hyb}}_{X})$, and the cobar quasi-isomorphism to $\mathsf{Den}_{\Delta_5,E}$, rather than using the target-internal bar–cobar counit. The cobar residual includes both the source-internal bar–cobar admissibility $\mathfrak{o}^{\varepsilon,\mathrm{src}}_{R}$ and the source-to-target quasi-isomorphism defect $\mathfrak{o}^{\vartheta}_{R}$. The residual components $\mathfrak{o}^{\mathrm{hyb}}_{R}$ and $\mathfrak{o}^{\mathrm{sep}}_{R}$ record, respectively, the finite hybrid construction and this source/target separation. Then construct the compact Hall bracket constants, the simple primitive representatives, the Borcherds–Kac relation verification, the full compact parity dimensions, the completed Hopf pairing with its closed radical quotient, the no-extra-relations theorem, and completed PBW compatibility of Theorem 10.66; equivalently, construct the cofinal finite-window datum of Definition 10.67. The first finite bracket test is the $\delta_1 + \delta_2 + \delta_3$ channel of Propositions 10.79 and 10.81: the signed dimension -64 must be refined to compact $29|93$ bases, parity split, structure constants, Jacobi/Serre checks, pairing matrices, and radical quotient.

The proved lattice normal-ordering theorem and the imported denominator algebra are inputs to this construction. The scalar square is an orientation-forgetful check. The compact state-side datum is the finite-inverse-system object assembled from the entries above and in Theorem 10.78.

Definition 10.86 *(Imaginary simple roots).* For $a \in \Lambda^{2,1}_{II} \cap \mathbb{R}_{>0}\mathcal{P}_{II}$ set

$$\mu_{\bar{0}}(a) = \begin{cases} m(a), & (a,a) < 0 \text{ and } m(a) > 0, \\ \tau(a), & (a,a) = 0 \text{ and } \tau(a) > 0, \\ 0, & \text{otherwise}, \end{cases}$$

and

$$\mu_{\bar{1}}(a) = \begin{cases} -m(a), & (a,a) < 0 \text{ and } m(a) < 0, \\ -\tau(a), & (a,a) = 0 \text{ and } \tau(a) < 0, \\ 0, & \text{otherwise}. \end{cases}$$

This sign split is the Gritsenko–Nikulin BKM parity datum. The product statement alone fixes only a Grothendieck representative for the signed exponent. Define multisets by multiplicity:

$$a \in \Delta^{im}_{\bar{0}} \quad \text{with multiplicity } \mu_{\bar{0}}(a),$$

$$a \in \Delta^{im}_{\bar{1}} \quad \text{with multiplicity } \mu_{\bar{1}}(a).$$

Put

$$\Delta^{im} = \Delta^{im}_{\bar{0}} \cup \Delta^{im}_{\bar{1}}.$$

$$\mathcal{O}^{im}_{\mathrm{den}} = \bigoplus_{a \in \Delta^{im}} \left(\mathfrak{g}_{a,\bar{0}} \ominus \mathfrak{g}_{a,\bar{1}}\right).$$

This is notation inside the denominator algebra; it is not a BPS state space.

The real simple roots are

$$\Delta^{re,\mathrm{simp}}_{\bar{0}} = \Delta^{re}_{\mathrm{simp}} = \mathcal{P}_{II,prim} = \{\delta_1, \delta_2, \delta_3\}, \qquad \Delta^{re,\mathrm{simp}}_{\bar{1}} = \varnothing.$$

Put

$$\Delta_{\bar{p}} = \Delta^{re,\mathrm{simp}}_{\bar{p}} \cup \Delta^{im}_{\bar{p}} \qquad (p = 0, 1), \qquad \Delta = \Delta_{\bar{0}} \cup \Delta_{\bar{1}}.$$

The Cartan data satisfy $(\alpha, \alpha) > 0$ for $\alpha \in \Delta^{re}$, $(\alpha, \alpha) \leq 0$ for $\alpha \in \Delta^{im}$, and $(\alpha, \alpha') \leq 0$ for all distinct $\alpha, \alpha' \in \Delta$. If $(\alpha, \alpha) > 0$, then

$$2\frac{(\alpha, \alpha')}{(\alpha, \alpha)} \in \mathbb{Z},$$

since $(\alpha, \alpha) = 2$ and $\Lambda^{2,1}_{II} = \mathbb{Z}\delta_1 + \mathbb{Z}\delta_2 + \mathbb{Z}\delta_3$.

Definition 10.87 *(Generators and relations).* Let I be an index set equipped with a parity map $p : I \to \mathbb{Z}/2\mathbb{Z}$ and a root map $i \mapsto \alpha_i \in \Delta$, with fibres over an imaginary root of cardinality $\mu_{\bar{p}}(a)$ in parity p. If $i \neq j$ and $\alpha_i = \alpha_j$, the two indices are distinct simple roots.

$$\mathrm{ch} : I \to \Lambda^{2,1}_{II}, \qquad i \mapsto \alpha_i.$$

The superalgebra $\mathfrak{g}_{\Delta_5}$ is generated by

$$h_i,\ e_i,\ f_i, \qquad i \in I.$$

The Cartan subalgebra is identified with $\Lambda^{2,1}_{II} \otimes \mathbb{C}$ by $h_i \mapsto \alpha_i$. Relations:

- $[h_i, e_j] = (\alpha_i, \alpha_j)e_j$ and $[h_i, f_j] = -(\alpha_i, \alpha_j)f_j$
- $[e_i, f_j] = \delta_{ij}h_i$
- if $\alpha_i \in \Delta^{re}_{\mathrm{simp}}$ and $i \neq j$, then

$$(\mathrm{ad}\, e_i)^{1-2(\alpha_i,\alpha_j)/(\alpha_i,\alpha_i)} e_j = (\mathrm{ad}\, f_i)^{1-2(\alpha_i,\alpha_j)/(\alpha_i,\alpha_i)} f_j = 0$$

- if $(\alpha_i, \alpha_j) = 0$, then $[e_i, e_j] = [f_i, f_j] = 0$

The grading is by $\Lambda^{2,1}_{II}$. Put

$$Q_+ = \sum_{\alpha \in \Delta} \mathbb{Z}_{\geq 0}\alpha \subset \Lambda^{2,1}_{II}.$$

The triangular decomposition is

$$\mathfrak{g}_{\Delta_5} = \left(\bigoplus_{\alpha \in Q_+ \setminus \{0\}} \mathfrak{g}_\alpha\right) \oplus (\Lambda^{2,1}_{II} \otimes \mathbb{C}) \oplus \left(\bigoplus_{\alpha \in Q_+ \setminus \{0\}} \mathfrak{g}_{-\alpha}\right).$$

For $i \in I$,

$$\deg e_i = \alpha_i, \qquad \deg f_i = -\alpha_i, \qquad \deg h_i = 0.$$

$$[\mathfrak{g}_\alpha, \mathfrak{g}_\beta] \subset \mathfrak{g}_{\alpha+\beta}.$$
$$\mathfrak{g}_0 = \Lambda^{2,1}_{II} \otimes \mathbb{C}.$$

The positive and negative roots are

$$\mathcal{R}_+ = \{\alpha \in Q_+ \setminus \{0\} \mid \mathfrak{g}_\alpha \neq 0\}, \qquad \mathcal{R}_- = -\mathcal{R}_+.$$

For every root α, set

$$\mathrm{mult}_{\bar{0}}(\alpha) = \dim \mathfrak{g}_{\alpha,\bar{0}}, \qquad \mathrm{mult}_{\bar{1}}(\alpha) = \dim \mathfrak{g}_{\alpha,\bar{1}}.$$

The signed root superdimension is

$$\mathrm{smult}(\alpha) = \mathrm{mult}_{\bar{0}}(\alpha) - \mathrm{mult}_{\bar{1}}(\alpha).$$

The Weyl–Kac–Borcherds denominator is

$$\begin{aligned}\Phi_{\Delta_5}(z) &= \sum_{w \in W^{(2)}(\Lambda^{2,1}_{II})} \det(w) \left(e^{-2\pi i (w(\rho), z)} - \sum_{a \in \Lambda^{2,1}_{II} \cap \mathbb{R}_{>0}\mathcal{P}_{II}} m(a) e^{-2\pi i (w(\rho+a), z)} \right) \\ &= e^{-2\pi i (\rho, z)} \prod_{\alpha \in \mathcal{R}_+} \left(1 - e^{-2\pi i (\alpha, z)}\right)^{\mathrm{smult}(\alpha)}.\end{aligned}$$

$$\Omega(\Lambda^{2,1}_{II}) = \Lambda^{2,1}_{II} \otimes \mathbb{R} + iC(\Lambda^{2,1}_{II})_+.$$

Theorem 10.88 *(Denominator identity).*

$$\frac{1}{64}\Delta_5(2Z) = \Phi_{\Delta_5}(z).$$

$$\mathrm{den}(\mathfrak{g}_{\Delta_5}) = \frac{1}{64}\Delta_5(2Z).$$

Proof.

$$\frac{1}{64}\Delta_5(Z) = \sum_{w \in W^{(2)}(\Lambda^{2,1}_{II})} \det(w) \left(e^{-\pi i (w(\rho), z)} - \sum_{a \in \Lambda^{2,1}_{II} \cap \mathbb{R}_{>0}\mathcal{P}_{II}} m(a) e^{-\pi i (w(\rho+a), z)} \right).$$

$$Z \mapsto 2Z : \qquad e^{-\pi i (w(\rho+a), z)} \mapsto e^{-2\pi i (w(\rho+a), z)}.$$

Weyl–Kac–Borcherds:

$$e^{-2\pi i (\rho, z)} \prod_{\alpha \in \mathcal{R}_+} \left(1 - e^{-2\pi i (\alpha, z)}\right)^{\mathrm{smult}(\alpha)}.$$

□

Proposition 10.89 *(Gram-index cone and denominator cone).*

$$\alpha(\Gamma_{\mathrm{act}}) = \{\alpha(n, l, m) \mid (n, l, m) \in \Gamma_{\mathrm{eff}},\ f(nm, l) \neq 0\} \subset \Lambda^{2,1}_{II}.$$

$$\alpha(\Gamma_{\mathrm{act}}) \subset \mathcal{R}^{\mathrm{vis}}_+ := \{\alpha \in \mathcal{R}_+ \mid \mathrm{smult}(\alpha) \neq 0\}.$$

$$\mathrm{smult}(\alpha(n, l, m)) = \dim \mathfrak{g}_{\alpha(n,l,m),\bar{0}} - \dim \mathfrak{g}_{\alpha(n,l,m),\bar{1}} = f(nm, l) \qquad ((n, l, m) \in \Gamma_{\mathrm{act}}).$$

Proof.

$$\alpha : \Gamma_{\mathrm{ind}} \to \Lambda^{2,1}_{II}, \qquad (n,l,m) \mapsto 2nf_2 - lf_3 + 2mf_{-2}.$$

$$q^n r^l s^m = \exp(-\pi i(\alpha(n,l,m), z))$$

$$Z \mapsto 2Z.$$

$$\frac{1}{64}\Delta_5(2Z) = e^{-2\pi i(\rho,z)} \prod_{\alpha\in\mathcal{R}_+} \left(1 - e^{-2\pi i(\alpha,z)}\right)^{\mathrm{smult}(\alpha)}.$$

$$\frac{1}{64}\mathcal{D}_X(2Z) = e^{-2\pi i(\rho,z)} \prod_{(n,l,m)\in\Gamma_{\mathrm{eff}}} \left(1 - e^{-2\pi i(\alpha(n,l,m),z)}\right)^{f(nm,l)}.$$

The triples in $\Gamma_{\mathrm{eff}} \setminus \Gamma_{\mathrm{act}}$ have exponent zero and do not contribute visible product factors. The denominator product alone does not prove absence of root spaces with zero signed superdimension.

$$\mathcal{D}_X = \Delta_5 \quad \Longrightarrow \quad \mathrm{smult}(\alpha(n,l,m)) = f(nm,l) \quad ((n,l,m) \in \Gamma_{\mathrm{act}}).$$

□

Proposition 10.90 *(What the denominator determines).* The identity

$$\frac{1}{64}\Delta_5(2Z) = e^{-2\pi i(\rho,z)} \prod_{\alpha\in\mathcal{R}_+} \left(1 - e^{-2\pi i(\alpha,z)}\right)^{\mathrm{smult}(\alpha)}$$

determines the signed integers $\mathrm{smult}(\alpha)$, once the Gritsenko–Nikulin root set and positive root monoid have been fixed. It does not determine invisible zero-superdimension root spaces, the two parity dimensions

$$\mathrm{mult}_{\bar{0}}(\alpha), \qquad \mathrm{mult}_{\bar{1}}(\alpha),$$

and it does not determine the structure constants

$$[\mathfrak{g}_\alpha, \mathfrak{g}_\beta] \to \mathfrak{g}_{\alpha+\beta}.$$

It also does not determine simple primitive representatives, the no-extra-relations statement, completed PBW compatibility, or the closed radical of the positive-negative Hopf pairing. Thus signed root multiplicity is necessary input for primitive recognition, but it is not a recognition datum.

Proof. Write $X^\alpha = e^{-2\pi i(\alpha,z)}$. In the completed positive root monoid algebra,

$$\log \prod_{\alpha\in\mathcal{R}_+} (1 - X^\alpha)^{\mathrm{smult}(\alpha)} = -\sum_{\alpha\in\mathcal{R}_+} \sum_{k\geq 1} \frac{\mathrm{smult}(\alpha)}{k} X^{k\alpha}.$$

Thus a logarithmic coefficient mixes divisor terms $k\alpha = \lambda$. The exponents $\mathrm{smult}(\alpha)$ are recovered recursively by height, equivalently by Mobius inversion on each primitive ray. The recovered exponent is only

$$\mathrm{smult}(\alpha) = \mathrm{mult}_{\bar{0}}(\alpha) - \mathrm{mult}_{\bar{1}}(\alpha).$$

The difference $a - b$ does not determine the pair (a, b), and no bilinear bracket, PBW filtration, or Hopf-pairing radical appears in the product. □

Proposition 10.91 *(What the determinant forgets).* Let

$$\mathcal{B} = \bigoplus_{\gamma \in \Gamma_{\mathrm{eff}}} \mathcal{B}_\gamma$$

be any charge-graded super vector space with finite-dimensional homogeneous pieces and

$$\operatorname{sdim} \mathcal{B}_{(n,l,m)} = f(nm, l).$$

Then its Gram-graded Euler/Fock determinant is

$$64 q^{1/2} r^{1/2} s^{1/2} \prod_{\gamma \in \Gamma_{\mathrm{eff}}} (1 - x^\gamma)^{\operatorname{sdim} \mathcal{B}_\gamma} = \Delta_5(Z).$$

If $\mathcal{B}$ carries a bracket compatible with charge addition, a parity split, simple-root representatives, or a Hopf pairing, the determinant forgets those data and remembers the class of $\mathcal{B}$ in

$$K_0(\Gamma_{\mathrm{eff}}\text{-graded finite super vector spaces}).$$

Thus the determinant of $\mathcal{A}_{\mathrm{den}} = \mathfrak{g}_{\Delta_5}$ is the Grothendieck shadow of its denominator bracket. The Borcherds denominator bracket exists inside $\mathcal{A}_{\mathrm{den}}$ by the Gritsenko–Nikulin construction, but the determinant sees only its charge-graded Grothendieck class.

Proof. The Fock determinant lemma gives

$$\operatorname{sdet}_{\mathcal{B}}(1 - x) = \prod_{\gamma \in \Gamma_{\mathrm{eff}}} (1 - x^\gamma)^{\operatorname{sdim} \mathcal{B}_\gamma}.$$

The hypothesis $\operatorname{sdim} \mathcal{B}_{(n,l,m)} = f(nm, l)$ identifies this product with the Borcherds product for $64^{-1}\Delta_5$. Multiplying by the vacuum factor gives Δ_5. A bracket

$$[\mathcal{B}_\gamma, \mathcal{B}_{\gamma'}] \to \mathcal{B}_{\gamma+\gamma'}$$

does not occur in the determinant formula. Neither do simple primitive representatives, parity dimensions, relation constants, completed PBW data, or a Hopf-pairing radical. The determinant is therefore invariant under replacing $\mathcal{B}$ by any other graded super vector space with the same Grothendieck class. In particular the zero bracket on such a space gives the same determinant and the wrong Lie algebra, and adjoining a root-graded cancelling pair $M \oplus \Pi M$ changes possible brackets and radicals without changing the signed determinant. □

11 The scalar square

Unitarity is used to localize the right-moving K3 supertrace before the index is taken; the index itself is a charge-graded virtual object that no longer requires Hilbert positivity, so the OP/DT scalar series below records charge-graded coefficients of a virtual character, not dimensions of a positive Hilbert space.

$$Z^X_{\mathrm{DT}}(q, t, p) = \sum_{h=0}^{\infty} \sum_{d=0}^{\infty} \sum_{n \in \mathbb{Z}} \mathbf{DT}^X_{n,(\beta_h,d)} q^{d-1} t^{\frac{1}{2}\langle \beta_h, \beta_h \rangle} (-p)^n.$$

role	OP variable	DT variable	Borcherds variable
$K3$ norm $h-1$	q_{OP}	t_{DT}	q_{Bch}
Borcherds/Gram exponent $m = d - 1$ from elliptic degree d	$\widetilde{q}_{\mathrm{OP}}$	q_{DT}	s_{Bch}
Do/Fourier index n	$p_{\mathrm{OP}} = e^{iu}$	p_{DT} with $(-p_{\mathrm{DT}})^n$	r_{Bch}

Proposition 11.1 *(Oberdieck–Pixton normalization).* Let the Oberdieck–Pixton variables be

$$p_{\mathrm{OP}} = e^{iu}, \qquad q_{\mathrm{OP}}, \qquad \widetilde{q}_{\mathrm{OP}}.$$

The variable map is

$$q = q_{\mathrm{OP}}, \qquad r = p_{\mathrm{OP}}, \qquad s = \widetilde{q}_{\mathrm{OP}}.$$

The product index set is

$$h, d \geq 0, \qquad h > 0 \text{ or } d > 0, \qquad \text{or} \qquad h = d = 0,\ k < 0.$$

The Laurent expansion is the Oberdieck–Pixton expansion in its specified chamber. The reciprocal p_{OP}-direction used to compare it with a Borcherds cusp expansion is the formal re-expansion convention

$$0 < |\widetilde{q}_{\mathrm{OP}}| \ll |q_{\mathrm{OP}}| \ll |p_{\mathrm{OP}}|^{-1} \ll 1, \qquad r = p_{\mathrm{OP}}.$$

This convention is recorded separately from the imported OP scalar identity. The enumerative identity is the Laurent expansion obtained after $p_{\mathrm{OP}} = e^{iu}$ and $y = -e^{iu}$. For the displayed DT variables,

$$q_{\mathrm{DT}} = \widetilde{q}_{\mathrm{OP}} = s, \qquad t_{\mathrm{DT}} = q_{\mathrm{OP}} = q, \qquad p_{\mathrm{DT}} = p_{\mathrm{OP}} = r.$$

The product denoted χ_{10} by Oberdieck–Pixton is

$$\chi_{10}^{\mathrm{OP}} = D_5^2 = \left(64^{-1}\Delta_5\right)^2 = 4096^{-1}\Delta_5^2.$$

Consequently their primitive reduced partition function is

$$Z_{\mathrm{OP}}^X = -\left(\chi_{10}^{\mathrm{OP}}\right)^{-1} = -4096\,\Delta_5^{-2}.$$

Proof. Oberdieck–Pixton use the primitive expansion [66, Theorem 1]

$$Z_{\mathrm{OP}}^X = \sum_{g,h,d} N_{g,h,d} u^{2g-2} q_{\mathrm{OP}}^{h-1} \widetilde{q}_{\mathrm{OP}}^{d-1}$$

and the stable-pairs substitution, in the coordinate convention of Oberdieck–Pandharipande [65, Section 2],

$$y = -p_{\mathrm{OP}}, \qquad p_{\mathrm{OP}} = e^{iu}.$$

There

$$\Omega = \begin{pmatrix} \tau & z \\ z & \widetilde{\tau} \end{pmatrix}, \qquad p_{\mathrm{OP}} = e^{2\pi i z} = e^{iu}, \qquad q_{\mathrm{OP}} = e^{2\pi i \tau}, \qquad \widetilde{q}_{\mathrm{OP}} = e^{2\pi i \widetilde{\tau}}.$$

The displayed DT convention uses

$$q_{\mathrm{DT}}^{d-1} t_{\mathrm{DT}}^{h-1} (-p_{\mathrm{DT}})^n,$$

and Bryan records that the DT convention interchanges the q and $\widetilde{q}$ variables relative to Oberdieck–Pandharipande [13, Definition 2.1 and Conjecture 2.3]. This gives the stated variable map.

Oberdieck–Pixton define the Igusa product by [66, (3)]

$$\chi_{10}^{\mathrm{OP}}(p, q, \widetilde{q}) = pq\widetilde{q} \prod (1 - p^k q^h \widetilde{q}^d)^{c(4hd-k^2)}$$

in the chamber $h > 0$ or $d > 0$, together with $h = d = 0$, $k < 0$. The exponents are the K3 elliptic-genus coefficients. Since

$$Z_{\mathrm{K3}} = 2\phi_{0,1}, \qquad \phi_{0,1} = \sum f(n,l) q^n r^l,$$

those exponents are $2f(nm, l)$. Therefore the Oberdieck–Pixton monic product is exactly the square of

$$D_5 = 64^{-1}\Delta_5 = q^{1/2} r^{1/2} s^{1/2} \prod_{(n,l,m)\in\Gamma_{\mathrm{eff}}} (1 - q^n r^l s^m)^{f(nm,l)}.$$

Their Theorem 1, equivalently the Laurent expansion in [66, (56)], gives $Z^X_{\mathrm{OP}} = -1/\chi^{\mathrm{OP}}_{10}$. □

Theorem 11.2 *(Reduced DT/PT/OP scalar square).* Let $\beta_h \in H_2(S, \mathbb{Z})$ be primitive and non-zero, with

$$\langle \beta_h, \beta_h \rangle = 2h - 2.$$

Under the variable map

$$p = P = e^{iu}, \qquad t = Q, \qquad q = T, \qquad (q_{\mathrm{Bch}}, r_{\mathrm{Bch}}, s_{\mathrm{Bch}}) = (Q, P, T),$$

one has

$$Z^X_{\mathrm{DT}}(T, Q, P) = Z^X_{\mathrm{OP}}(P, Q, T) = -4096\, \Delta_5(Z)^{-2}.$$

The equality is an equality of Laurent expansions in the OP chamber of Proposition 11.1. It is the OP/DT/PT scalar branch attached to the Behrend-weighted quotient integration of Proposition 5.2. The physical protected-index identification is Conjecture 5.3. Imprimitive K3 curve classes require the separate imprimitive multiple-cover datum; the displayed OP scalar-square theorem is the primitive-class branch used for the present square-root comparison.

Proof. Oberdieck's reduced DT/PT theorem for $S \times E$, [64, Theorem 3(i)], gives, for every non-zero $\beta \in H_2(S, \mathbb{Z})$ and every $d \geq 0$,

$$\sum_n \mathbf{DT}^X_{n,(\beta,d)} x^n = \sum_n \mathsf{P}^X_{n,(\beta,d)} x^n.$$

Set $x = -P$. The degree-zero factor is

$$M(-P)^{\chi(S\times E)} = M(-P)^0 = 1.$$

For primitive β_h, the reduced GW/PT comparison of Oberdieck–Pandharipande [65, Proposition 5] gives

$$\sum_n \mathsf{P}^X_{n,(\beta_h,d)} (-P)^n = \sum_g \mathsf{N}^X_{g,h,d} u^{2g-2}, \qquad P = e^{iu}.$$

Multiplying by $Q^{h-1}T^{d-1}$ and summing over h, d gives

$$Z^X_{\mathrm{DT}}(T, Q, P) = Z^X_{\mathrm{OP}}(P, Q, T).$$

Oberdieck–Pixton [66, Theorem 1] and Proposition 11.1 give

$$Z^X_{\mathrm{OP}}(P, Q, T) = -4096\, \Delta_5(Z)^{-2}.$$

□

Definition 11.3 *(OP normalization branch).* The OP/DT normalization branch is the proved enumerative series of Theorem 11.2, displayed as a square in the OP convention:

$$Z^X_{\square,\mathrm{OP/DT}} := Z^X_{\mathrm{DT}} = Z^X_{\mathrm{OP}} = -4096\,\Delta_5^{-2}, \qquad C_{\square,\mathrm{OP/DT}} := -4096.$$

The numerical constant $C_{\square,\mathrm{OP/DT}} = -4096$ is a *normalization* datum, not an enumerative invariant beyond the identification of the series itself: it factors as $-4096 = (-1) \times 64^2$, where the 64^2 is the squared theta-leading coefficient $([q^{1/2}r^{1/2}s^{1/2}]\Delta_5)^2$ and the leading -1 is the OP scalar convention $Z^X_{\mathrm{OP}} = -(\chi_{10}^{\mathrm{OP}})^{-1}$. No additional enumerative content beyond the identification of Z^X_{OP} with D_5^{-2} is carried by the constant. An external scalar-square series, if introduced later, is a Laurent expansion of the form

$$Z^X_{\square,\mathrm{ext}}(Z) = \frac{C_{\square,\mathrm{ext}}}{\Delta_5(Z)^2}, \qquad C_{\square,\mathrm{ext}} \in \mathbb{C}^\times,$$

and carries no OP/DT enumerative content unless its source is specified.

Proposition 11.4 *(Square-root consequence).* For $\mathfrak{s} \in \{\mathrm{OP}, \mathrm{DT}\}$, with variables normalised as in Theorem 11.2,

$$Z^X_{\mathfrak{s}}(Z) = -4096\,\Delta_5(Z)^{-2}.$$

Thus

$$\mathcal{D}_X(Z)^2 = \frac{-4096}{Z^X_{\mathfrak{s}}(Z)}, \qquad \mathcal{D}_X(Z) = \Delta_5(Z).$$

The virtual inverse is fixed by the normalized determinant:

$$\mathcal{Z}^{\mathrm{vir}}_{\mathrm{Borch}} := \mathcal{D}_X^{-1}.$$

A supplied Pfaffian square root may differ from this chosen cusp trivialization by a sign character; that sign is the orientation datum sought in Open Problem 5.7 and is not determined by the OP scalar. The squared scalar $Z^X_{\mathfrak{s}} = -4096(\mathcal{D}_X^{-1})^2$ does not fix an orientation-lifted Pfaffian sign because squaring kills it. With the fixed virtual inverse,

$$Z^X_{\mathfrak{s}} = -4096\,(\mathcal{Z}^{\mathrm{vir}}_{\mathrm{Borch}})^2,$$

$$|\mathcal{Z}^{\mathrm{vir}}_{\mathrm{Borch}}| = \frac{1}{64}|q|^{-1/2}|r|^{-1/2}|s|^{-1/2}|\operatorname{sch}\mathcal{F}(\mathcal{V})|.$$

Proof. Theorem 11.2 gives the first display for $\mathfrak{s} = \mathrm{OP}$ and $\mathfrak{s} = \mathrm{DT}$. Hence

$$-4096/Z^X_{\mathfrak{s}} = \Delta_5^2.$$

The Borcherds determinant $\mathcal{D}_X = \Delta_5$ fixes the automorphic square-root section in the chosen cusp trivialization. A supplied Pfaffian orientation lift may carry an additional sign character, but that sign is not part of the notation $\mathcal{Z}^{\mathrm{vir}}_{\mathrm{Borch}}$. Squaring any supplied sign choice ε, with $\varepsilon^2 = 1$, gives

$$Z^X_{\mathfrak{s}} = -4096(\mathcal{D}_X^{-1})^2 = -4096(\varepsilon\mathcal{D}_X^{-1})^2,$$

the square-root sign being invisible to the scalar. Selecting the sign requires an orientation datum, deferred to Open Problem 5.7. □

Remark 11.5 *(Constants are not orientation data).* The number 64 fixes the theta-normalized Igusa section and hence the Weyl monomial in the determinant:

$$\Delta_5 = 64q^{1/2}r^{1/2}s^{1/2}\prod(1-q^n r^l s^m)^{f(nm,l)}.$$

The monic denominator section is $D_5 = 64^{-1}\Delta_5$. The number 4096 is 64^2, appearing because Oberdieck–Pixton use

$$\chi_{10}^{\mathrm{OP}} = D_5^2 = 4096^{-1}\Delta_5^2.$$

Read as a normalization conversion, the OP scalar constant factors as

$$-4096 \;=\; (-1)_{\mathrm{OP}}\cdot 64^2,$$

where the leading -1 is the OP scalar convention and 64^2 is the squared theta-leading coefficient $([q^{1/2}r^{1/2}s^{1/2}]\Delta_5)^2$. It is not the square of an orientation sign; every square-root sign squares to $+1$. The minus sign in

$$Z_{\mathrm{OP}}^X = -(\chi_{10}^{\mathrm{OP}})^{-1}$$

is the OP scalar convention. None of these constants is the Hall orientation character. The automorphic character ν_{Δ_5} is fixed by the Maass transformation law, and the geometric character ϵ_0, if it exists, must be produced by orientation-line monodromy as in Open Problem 5.7.

12 Coefficient dictionary and main synthesis

The coefficient convention is fixed once and for all by Eichler–Zagier normalization:

$$Z_{\mathrm{K3}} = 2\phi_{0,1}, \qquad \phi_{0,1}(\tau,z) = \sum_{n\geq 0,\, l\in\mathbb{Z}} f(n,l)q^n r^l.$$

Equivalently, if $\phi_{12,1}$ is the first Fourier–Jacobi coefficient of Igusa's weight-12 form F_{12} [42, 43], equivalently in the Gritsenko–Nikulin normalization [39, Example 2.4, (2.15)], and if

$$\delta_{12}(\tau) = q\prod_{n\geq 1}(1-q^n)^{24},$$

then

$$\phi_{0,1} = \frac{\phi_{12,1}}{\delta_{12}}.$$

The first terms are

$$\phi_{0,1} = (r^{-1}+10+r) + q(10r^{-2}-64r^{-1}+108-64r+10r^2)$$
$$+q^2(r^{-3}+108r^{-2}-513r^{-1}+808-513r+108r^2+r^3)+O(q^3).$$

Thus

$$f(0,-1)=1, \qquad f(0,0)=10, \qquad f(1,1)=-64, \qquad f(1,0)=108.$$

The weak Jacobi support condition is

$$f(n,l)=0 \quad \text{unless} \quad n\geq 0, \quad 4n-l^2\geq -1,$$

and $f(n,l) = f(n,-l)$. These integers have three distinct readings in the paper:

object	meaning of $f(nm,l)$
$\mathcal{V}_{(n,l,m)}$	virtual K_0-superdimension
$\mathcal{D}_X$	Borcherds product exponent
$\mathfrak{g}_{\Delta_5,\alpha(n,l,m)}$	signed root supermultiplicity on active support

The first reading is a protected Euler characteristic. The second and third readings are automorphic and denominator-algebra incarnations of the same integer. This is not an equality of Hilbert spaces, brackets, or Hall structure constants.

The synthesis is the following spine:

$$\text{virtual determinant} \longrightarrow \text{formal charge and normal ordering} \longrightarrow \text{denominator algebra} \longrightarrow \text{reduced scalar square}$$
$$\longrightarrow \text{compact realization problem} \longrightarrow \text{eight-row datum extension.}$$

Theorem 12.1 *(Coefficient dictionary and compact realization).* Let $X = S \times E$, where S is a smooth complex K3 surface and E is a smooth elliptic curve. Boundary-degeneration cases–singular K3 limits, nodal or cuspidal elliptic limits, imprimitive classes, the compact Hall $s = 0$ cusp, multi-component charge closures, and boundary closures of Hall correspondences–are not hypotheses of this theorem. Let

$$\mathbb{U}_{n,l} \in K_0(\mathrm{sVect}_{\mathrm{fd}}), \qquad \mathrm{sdim}\, \mathbb{U}_{n,l} = f(n,l),$$

and define the virtual K_0-determinant object in the cusp chamber by

$$\mathcal{V}_{(n,l,m)} = \mathbb{U}_{nm,l}, \qquad (n,l,m) \in \Gamma_{\mathrm{eff}}.$$

First, the virtual determinant is the theta-normalized Igusa square root:

$$\mathcal{D}_X(Z) = 64 q^{1/2} r^{1/2} s^{1/2} \prod_{(n,l,m)\in\Gamma_{\mathrm{eff}}} (1 - q^n r^l s^m)^{f(nm,l)} = \Delta_5(Z).$$

The numerical factor 64 is the theta-constant leading coefficient $[q^{1/2} r^{1/2} s^{1/2}]\Delta_5$; the monic Borcherds product is

$$D_5 = 64^{-1}\Delta_5.$$

Second, the formal dyonic Mukai lattice is additive but its Igusa Gram shadow is quadratic. For

$$\Gamma_X^{\mathrm{form}} = \Lambda_S \oplus \Lambda_S, \qquad c = (Q,P),$$

the map

$$\Pi_X(c) = \left(\frac{Q^2}{2}, Q\cdot P, \frac{P^2}{2}\right)$$

satisfies

$$\Pi_X(c+c') = \Pi_X(c) + \Pi_X(c') + B(c,c').$$

The normal-ordered extension

$$\widehat{\Gamma}_X = \Gamma_X^{\mathrm{form}} \oplus_B \Gamma_{\mathrm{gram}}, \qquad \overline{\Pi}_X(c,T) = \Pi_X(c) - T$$

is additive, and every formal Gram triple has a primitive saturated formal lift when Λ_S contains two orthogonal hyperbolic planes. This does not prove algebraic/effective compact Hall support. Strict fixed-lift raw Π_X-descent cannot realize the type-II BKM real-root strings; Theorem 6.15 is the obstruction.

Third, under the Lorentzian degree map

$$\alpha(n,l,m) = 2nf_2 - lf_3 + 2mf_{-2}, \qquad q^n r^l s^m = \exp(-\pi i(\alpha(n,l,m),z)),$$

the same active coefficients are the visible signed root supermultiplicities of the Gritsenko–Nikulin denominator algebra:

$$\operatorname{smult} \mathfrak{g}_{\Delta_5,\alpha(n,l,m)} = f(nm,l) \qquad ((n,l,m) \in \Gamma_{\mathrm{act}}),$$

and

$$\operatorname{den}(\mathfrak{g}_{\Delta_5}) = 64^{-1}\Delta_5(2Z).$$

This is the Gritsenko–Nikulin denominator theorem. It determines the signed root supermultiplicities in the fixed Borcherds datum; it does not determine compact Hall constants or microscopic brackets. The normal-ordered charge extension is the necessary degree-level bridge from an additive Hall source to these Gram degrees.

Fourth, for the reduced Hilbert-scheme DT/PT/OP scalar branch,

$$Z^X_{\mathrm{DT}}(T,Q,P) = Z^X_{\mathrm{OP}}(P,Q,T) = -4096\,\Delta_5(Z)^{-2}.$$

Here

$$\chi^{\mathrm{OP}}_{10} = D_5^2 = (64^{-1}\Delta_5)^2 = 4096^{-1}\Delta_5^2, \qquad Z^X_{\mathrm{OP}} = -(\chi^{\mathrm{OP}}_{10})^{-1};$$

hence $-4096 = -(64)^2$ is the OP scalar sign together with the theta-to-monic normalization. It is not an orientation character. Thus, with

$$\mathcal{Z}^{\mathrm{vir}}_{\mathrm{Borch}} = \mathcal{D}_X^{-1},$$

one has

$$Z^X_{\mathrm{DT}} = Z^X_{\mathrm{OP}} = -4096\,(\mathcal{Z}^{\mathrm{vir}}_{\mathrm{Borch}})^2.$$

Fifth, the stronger microscopic comparison

$$H^0\operatorname{Prim}_{\mathrm{prot}}(\overline{\Pi}^{\Theta}_{X,*}\mathcal{F}_X)/\overline{\operatorname{Rad}\langle\,\cdot\,,\,\cdot\,\rangle_X} \cong \mathfrak{g}_{\Delta_5}$$

is the primitive part of a Dirac–Igusa realization datum in the sense of Definition 10.65. The cofinal source construction of Theorem 10.78, together with Theorem 10.66, identifies its primitive denominator with $64^{-1}\Delta_5(2Z)$.

Sixth, the compact realization satisfies three consequences.

(i) A naive support-local factorization object on $\operatorname{Ran}(E)$ sees only projection-finite sectors. If a one-dimensional subscheme of $K3 \times E$ has positive elliptic degree, its projection to E is all of E. Thus the full s-direction of the Igusa product requires finite-stage local/wrapped data: a geometric elliptic-degree map b^{geom}_R, a Gram trace exponent $m_R = \mathrm{pr}_3\,\overline{\Pi}_X$, rigidified wrapped prequotients, mixed extension correspondences in both local/wrapped orders, wrapped/wrapped correspondences, the eight-word two-step binary flag atlas in $\mathsf{Corr}^{E,\mathrm{hyb}}_R$, the higher colored residual $\mathfrak{o}^{\mathrm{col}}_R$, properness/admissibility residuals for the pull-push operations, quotient-after-correspondence $Q_{E,R}$-compatibility, and protected integration with trace degree s^{m_R}. Equivalently, $\mathfrak{D}_{\mathrm{hyb},R} = 0$ for every finite R. A separate global Fock/Hecke mode is only determinant-level unless it is coupled to the local sector by mixed Hall actions before the reduced E-quotient.

(ii) The automorphic character ν_{Δ_5} belongs to Δ_5, and Δ_5^2 forgets it. A compact Hall orientation constructs a strong orientation character ϵ_o and prove

$$\epsilon_o|_{W^{(2)}(\Lambda_{II}^{2,1})} = \nu_{\Delta_5}|_{W^{(2)}(\Lambda_{II}^{2,1})}.$$

The constants 64, 4096, and the OP minus sign do not contain this orientation-monodromy datum. The Hall-side proof computes the three simple type-II determinant-line signs and checks the reflection relations.

(iii) A primitive comparison with $\mathfrak{g}_{\Delta_5}$ is a presentation-level recognition datum, not a consequence of signed dimensions. The source contains primitive representatives for the real and imaginary simple roots, their parities, the Chevalley, real Serre and orthogonality relations, generation, full root-space parity dimensions, the completed Hopf pairing and the same closed radical quotient, a no-extra-relations theorem, and completed PBW compatibility in the finite-height inverse system, including exact triangular completion/$\lim^1$-vanishing and Koszul Mittag–Leffler exactness for the chiral cobar comparison. These data make the Borcherds–Kac universal presentation give the isomorphism with $\mathfrak{g}_{\Delta_5}$, while the determinant records the resulting Grothendieck class.

Finally, the diagonal-divisor rows extend this same logic. A row is promoted beyond its automorphic product only by a row datum fixing the modular group or cover, character, divisor, Jacobi seed, product lattice, Weyl chamber, reflection character, scalar theorem if one is claimed, orientation data, one-particle index, and primitive algebra comparison.

Proof. The Fock trace calculation gives

$$\mathcal{D}_X = 64q^{1/2}r^{1/2}s^{1/2}\,\mathrm{sdet}_{\mathcal{V}}(1-x).$$

The Borcherds–Gritsenko–Nikulin product formula for the input $\phi_{0,1}$ identifies this product with Δ_5, as recorded in Proposition 4.5. The leading coefficient 64 is fixed in the theta-normalized Fourier expansion of Δ_5, hence $D_5 = 64^{-1}\Delta_5$ is the monic product.

The formal charge and normal-ordering statements are Definition 6.5, Lemma 6.6, Lemma 6.8, Lemma 6.9, and Definition 6.11. They are direct lattice calculations in the Mukai pairing. The raw-descent obstruction is Theorem 6.15; it uses the non-zero second real-root strings recorded in Proposition 10.79.

The Lorentzian monomial identity and pairing computation are in Section 6 and the Weyl-chamber section. The root-supermultiplicity and denominator identity are Theorem 9.2; the argument $2Z$ comes from passing from $\exp(-\pi i(\alpha, z))$ to the usual denominator character $\exp(-2\pi i(\alpha, z))$.

The reduced scalar identity is Theorem 11.2, with the normalization conversion of Proposition 11.1. Since Oberdieck–Pixton use $\chi_{10}^{\mathrm{OP}} = D_5^2$ and prove $Z_{\mathrm{OP}}^X = -(\chi_{10}^{\mathrm{OP}})^{-1}$, the theta normalization gives

$$Z_{\mathrm{OP}}^X = -4096\Delta_5^{-2}.$$

Oberdieck's reduced DT/PT comparison and the primitive Oberdieck–Pandharipande comparison give the same branch for Z_{DT}^X.

The compact comparison is not used in these proofs. It is precisely Open Problem 10.85. The $\mathrm{Ran}(E)$ obstruction follows because a positive elliptic-degree curve has one-dimensional image in the irreducible curve E, hence image E; support separation on disjoint opens of $\mathrm{Ran}(E)$ cannot see such a sector. The automorphic character computation gives the target values of ν_{Δ_5}; rescaling Δ_5 by 64^{-1} changes no character. A compact orientation statement follows only after (O1) constructs the reduced quotient

orientation, $(O1)^+$ supplies the honest Weyl transport, and (O2) supplies the local Pfaffian signs. The primitive recognition statement is the universal property of the Borcherds–Kac presentation recorded in the denominator-algebra section, together with the closed Hopf-radical and completed PBW hypotheses in Theorem 10.66. The determinant and scalar square do not contain the missing presentation, orientation, radical, PBW, or Hall correspondence data. The row datum condition is the same separation applied to the Cléry–Gritsenko catalogue in the next part. □

Part IV

Diagonal-divisor rows and CHL boundary physics

13 Eight diagonal-divisor rows: automorphic products and physical hosts

The Cléry–Gritsenko catalogue gives eight diagonal-divisor automorphic products. The Gritsenko–Nikulin and Govindarajan theorems promote these products to denominator identities in the rows where the Weyl data are known. A CHL dyon square, a reduced CY/DT scalar host, and a compact Hall/chiral host are further structures, recorded only where their data have been specified.

$$g = 2.$$

$$t, N \in \mathbb{Z}_+ : \quad \Gamma_t(N) = \left\{ \begin{pmatrix} * & *t & * & * \\ & * & * & *t^{-1} \\ N & *Nt & * & * \\ Nt & *Nt & *t & * \end{pmatrix} \in \mathbf{Sp}_4(\mathbb{Q}) \mid * \in \mathbb{Z} \right\}.$$

$$\mathbb{H}_2/\Gamma_t \longleftrightarrow \{(1, t)\text{-polarized abelian surfaces}\}.$$

$$\mathcal{H}_{\mathrm{diag}} = \left\{ \begin{pmatrix} a & 0 \\ 0 & b \end{pmatrix} \mid a, b \in \mathbb{H}_1 \right\} \subset \mathbb{H}_2.$$

Definition 13.1 *(Physical-host datum).* For a diagonal-divisor row F_j, a physical-host datum is a tuple

$$\mathfrak{C}_j = \left(\mathscr{A}_j, \mathscr{D}_j, \mathscr{S}_j, \mathscr{Z}_j, \mathscr{H}_j \right).$$

The entries have the following meanings.

(i) $\mathscr{A}_j$ is the automorphic product datum: the form F_j, the group or multiplier cover, the automorphy cocycle, the character or multiplier with exact order, the diagonal divisor with multiplicities, the product chamber and branch, the Jacobi input at every cusp, the product-degree map, the product normalization, and the source theorem.

(ii) $\mathscr{D}_j$ is the denominator datum: a Weyl–Kac–Borcherds denominator algebra with root lattice, Weyl vector, simple real roots, Weyl chamber, Weyl group, reflection character, imaginary simple-root coefficients, product side, and Weyl alternating side.

(iii) $\mathscr{S}_j$ is a CHL/dyon-square datum: a sourced dyon-counting scalar form Φ_j, a row map to the same automorphic row and cover, and an identity $F_j^2 = \Phi_j$ in the stated chamber. This is not a denominator theorem and not a CY/DT partition theorem.

(iv) $\mathscr{Z}_j$ is a reduced CY/DT scalar-host datum: a compact Calabi–Yau threefold, orbifold Calabi–Yau, or CHL background with oriented reduced obstruction theory, fixed variables and chamber, and a proved identity

$$Z_{\mathrm{red}}^{X_j}(Z) = C_j F_j(Z)^{-2}.$$

(v) $\mathscr{H}_j$ is a compact Hall/chiral datum: an oriented compact Hall, Drinfeld, or factorization object whose protected one-particle index gives the product exponents and whose protected primitive bracket identifies the denominator algebra with the same orientation/reflection character.

The product row, denominator row, dyon square, reduced scalar host, and compact Hall/chiral host are separate structures. None of the five entries implies another unless a cited theorem constructs the implication. In particular, matching a weight, a square root, or a comparison symbol does not fill a datum. A row-map theorem must name the host, charge lattice, effective chamber, modular group

or cover, product lattice, orientation character, one-particle index, scalar identity when claimed, and primitive Hall/chiral comparison.

Theorem 13.2 *(Cléry–Gritsenko).* In the class of Siegel modular forms with character or multiplier system for the Hecke-type congruence subgroups $\Gamma_t(N)$ whose divisor is the union of the $\Gamma_t(N)$-translates of $\mathcal{H}_{\mathrm{diag}}$ with multiplicity one, there are exactly eight rows [35, Theorem 1.4]. The nine-candidate list is [35, Proposition 1.2]; the proof of Theorem 1.4 excludes only the ninth candidate $(t, N; k) = (2, 4; \frac{1}{2})$. The congruence Borcherds product with all cusp coefficients is [35, Theorem 3.1], with the final Section 3 product displays for ∇_3, ∇_2, $\nabla_{3/2}$, and Q_1.

j	F_j	Γ	$\mathrm{wt}(F_j)$	character/multiplier
1	Δ_5	Γ_1	5	$v_\eta^{12} \times v_H$
2	Δ_2	Γ_2	2	$v_\eta^{6} \times v_H$
3	Δ_1	Γ_3	1	$v_\eta^{4} \times v_H$
4	$\Delta_{1/2}$	Γ_4	$\frac{1}{2}$	$v_\eta^{3} \times v_H$ (degree 8)
5	∇_3	$\Gamma_0^{(2)}(2)$	3	$\chi_2^{(2)} \times v_H$
6	∇_2	$\Gamma_0^{(2)}(3)$	2	$\chi_2^{(3)} \times v_H$
7	$\nabla_{3/2}$	$\Gamma_0^{(2)}(4)$	$\frac{3}{2}$	order-four multiplier $\chi_4^{(4)} \times v_H$
8	Q_1	$\Gamma_2(2)$	1	$\chi_4^{(2)} \times v_H$

$$\operatorname{supp} \operatorname{div}_{\mathbb{H}_2}(F_j) = \bigcup_{\gamma \in \Gamma_j} \gamma \langle \mathcal{H}_{\mathrm{diag}} \rangle, \qquad \mathrm{mult}_{\mathcal{H}_{\mathrm{diag}}}(F_j) = 1.$$

$$F_1 = \Delta_5, \qquad \frac{1}{64} F_1(2Z) = \mathrm{den}(\mathfrak{g}_{\Delta_5}).$$

row	automorphic product	denominator theorem	CHL/dyon square	reduced CY/DT scalar host	compact Hall/chiral host
1	$F_1 = \Delta_5$; Cléry–Gritsenko product and Borcherds–Gritsenko–Nikulin lift of $\phi_{0,1}$	Gritsenko–Nikulin denominator algebra $\mathfrak{g}_{\Delta_5}$; $64^{-1}\Delta_5(2Z)$ normalization	no CHL row map used; Δ_5^2 is the Igusa scalar square, not a boundary dyon datum	proved OP/DT $K3 \times E$ scalar host: $-4096\Delta_5^{-2}$	open; the formal determinant is not a compact Hall/chiral source
2	$F_2 = \Delta_2$; Cléry–Gritsenko product	Gritsenko–Nikulin denominator row $\mathfrak{g}_2^{\mathrm{GN}}$	no CHL/dyon row map	no curve-counting scalar theorem	no Hall or factorization host
3	$F_3 = \Delta_1$; Cléry–Gritsenko product	Gritsenko–Nikulin denominator row $\mathfrak{g}_3^{\mathrm{GN}}$	no CHL/dyon row map	no curve-counting scalar theorem	no Hall or factorization host
4	$F_4 = \Delta_{1/2}$; Cléry–Gritsenko product on the degree-eight multiplier cover	Gritsenko–Nikulin denominator row on the same cover $\mathfrak{g}_4^{\mathrm{GN}}$	no CHL/dyon row map	no curve-counting scalar theorem	no Hall or factorization host
5	$F_5 = \nabla_3$; Cléry–Gritsenko product	Govindarajan $\mathcal{G}_1(2)$ CHL/WKB denominator theorem	sourced scalar dyon square $\nabla_3^2 = \Phi_6^{(1,2)}$	no reduced CY/DT datum	no compact Hall/chiral datum
6	$F_6 = \nabla_2$; Cléry–Gritsenko product	Govindarajan $\mathcal{G}_1(3)$ CHL/WKB denominator theorem	sourced scalar dyon square $\nabla_2^2 = \Phi_4^{(1,3)}$	no reduced CY/DT datum	no compact Hall/chiral datum
7	$F_7 = \nabla_{3/2}$; Cléry–Gritsenko product on the order-four multiplier cover	Govindarajan $\mathcal{G}_1(4)$ CHL/WKB denominator theorem on the μ_4-cover	sourced scalar dyon square $F_7^2 = \Phi_3$ for the composite $N = 4$ scalar form, only after fixing the cover and branch	no reduced CY/DT datum	no compact Hall/chiral datum
8	$F_8 = Q_1$; Cléry–Gritsenko $(t, N) = (2, 2)$ product	Govindarajan $\mathcal{G}_2(4)$ CHL/WKB denominator theorem	sourced twisted dyon square $Q_1^2 = \Phi_2^{(2,4)}$; Q_1 is the special $\mathcal{G}_2(4)$ row	no reduced CY/DT datum	no compact Hall/chiral datum

By Theorem B.7 the full-paramodular rows F_2, F_3, F_4 carry Gritsenko–Nikulin denominator identities. The cited sources do not construct a reduced curve-counting theorem, Hall object, or factorization host for these three rows. The congruence rows F_5, F_6, F_7, F_8 have product-degree lattices in Proposition B.9. The additional denominator data (Weyl chamber, real roots, reflection character, Weyl sum) are supplied by the CHL denominator theorem of Govindarajan for F_5, F_6, F_7, F_8. For F_7 the theorem lives on the μ_4-multiplier cover determined by the Cléry–Gritsenko order-four multiplier. For Q_1 the row is the special $\mathcal{G}_2(4)$ twisted CHL denominator row. Neither statement supplies a reduced CY/DT scalar host or compact Hall/chiral datum.

Proposition 13.3 *(Extension separation).* No statement in this section is used in the $N = 1$ theorem for Δ_5. For every row the product formula itself gives a Gram-indexed primitive product. It gives a Gram-graded Euler/Fock determinant only after a one-particle index and a charge-invariant interpretation have been supplied. A nonabelian denominator theorem is stronger: it additionally requires a Weyl chamber, real simple roots, a reflection character, and a Weyl alternating sum.

Open Problem 13.4 *(Data beyond the known rows).* For any diagonal-divisor or boundary product not already covered by the Gritsenko–Nikulin denominator theorems or by the Govindarajan CHL/WKB denominator theorem, first construct the product datum

$$\mathscr{P}_j = (\phi_j, \Gamma^+_{j,\mathrm{prod}}, \Gamma_{j,\mathrm{mod}}, \rho_j, \nu_j, L_j, \alpha_j)$$

with

$$\phi^{(j)}(\tau, z) = \sum c_j(n, l) q^n r^l,$$

and then construct, if possible, the additional denominator data: positive product cone, Weyl chamber, simple real roots, reflection character, Weyl vector, Weyl alternating sum, imaginary simple-root coefficients, and a generalized Borcherds–Kac–Moody superalgebra $\mathfrak{g}_j$ such that

$$F_j(Z) = C_j e^{2\pi i(\rho_j, Z)} \prod_{\gamma \in \Gamma^+_{j,\mathrm{prod}}} (1 - e^{2\pi i(\gamma, Z)})^{c_j(\gamma)},$$

$$\mathrm{den}(\mathfrak{g}_j) = C'_j F_j(a_j Z), \qquad \mathrm{sdim}(\mathfrak{g}_{j,\gamma}) = c_j(\gamma).$$

If a CHL/K3 sector supplies a dyon-counting scalar square, it fills only the $\mathscr{S}_j$ level of Definition 13.1. If a reduced CY/DT scalar identity is claimed, one must separately construct X_j, the reduced obstruction theory, orientation, Fourier chamber, variables, and the theorem

$$Z^{X_j}_{\mathrm{red}}(Z) = C_j F_j(Z)^{-2}.$$

If a compact Hall/chiral host is claimed, one must separately construct the protected one-particle index and primitive bracket comparison.

Proposition 13.5 *(CHL squares and denominator signs).* The identities

$$\nabla_3^2 = \Phi_6^{(1,2)}, \qquad \nabla_2^2 = \Phi_4^{(1,3)}$$

which occur in the CHL dyon literature do not by themselves give denominator identities for the square-root rows $F_5 = \nabla_3$ and $F_6 = \nabla_2$. They give scalar Siegel modular forms on the CHL side. To descend from a scalar square to a denominator theorem for a root form, one must also construct the reflection character, the parity, and the Weyl anti-invariant additive side. Govindarajan's twisted CHL theorem supplies exactly this missing structure for

$$F_5 = \nabla_3, \qquad F_6 = \nabla_2, \qquad F_7 = \nabla_{3/2}, \qquad F_8 = Q_1,$$

as recorded in Theorem B.13.

Proof. Cléry–Gritsenko record the dd-forms ∇_3 and ∇_2 and note that their squares were constructed in the CHL papers as $\Phi_6^{(1,2)}$ and $\Phi_4^{(1,3)}$. The square forgets the sign character which a denominator identity needs on simple real reflections. Thus the scalar CHL product is evidence for the automorphic product, not by itself a proof of the BKM datum $(P_j, \mathcal{M}_j, W_j, \epsilon_j, \rho_j, N_j(a))$ for the root form. The additional datum is supplied by [31], building on [32] and [17]. □

A separate comparison tuple has weights

$$(5, 2, 1, 1, \tfrac{1}{2}, 1, \tfrac{1}{4}, 0).$$

It is not the Cléry–Gritsenko table. A comparison with the Cléry–Gritsenko forms requires the relevant entries of Definition 13.1. Thus

$$\text{Cléry–Gritsenko classification} \neq \text{CHL realisation theorem.}$$

$$F_1: \text{ theorem}, \qquad F_2, F_3, F_4: \text{ GN denominator theorem},$$

$$F_5, F_6, F_7, F_8: \text{ Govindarajan CHL/WKB denominator theorem},$$

with F_7 understood on the μ_4-multiplier cover determined by its order-four multiplier. Dijkgraaf–Verlinde–Verlinde and Borcherds give the $N = 1$ physical and automorphic background [9, 26]; they do not construct physical-host data for the other seven rows.

A Spin L-factor normalisation for Δ_5^2

Set the theta-constant normalisation

$$\Delta_{10}^{\theta} := \Delta_5^2.$$

The Oberdieck–Pixton monic normalisation is

$$\chi_{10}^{\mathrm{OP}} := 2^{-12}\Delta_{10}^{\theta} = 4096^{-1}\Delta_5^2.$$

The spinor L-function is attached to the Hecke eigenline

$$\mathbb{C}\Delta_{10}^{\theta} = \mathbb{C}\chi_{10}^{\mathrm{OP}};$$

it is independent of the non-zero scalar used to name its generator. Put

$$\Delta(q) = q\prod_{n\geq 1}(1-q^n)^{24}, \qquad E_6(q) = 1 - 504\sum_{n\geq 1}\sigma_5(n)q^n,$$

$$g := \Delta E_6 = \sum_{n\geq 1} a_n(g)q^n \in S_{18}(\mathbf{SL}_2(\mathbb{Z})).$$

Theorem A.1 *(Andrianov factorisation).* For the Saito–Kurokawa representation generated by Δ_{10}^{θ},

$$L_{\mathrm{spin}}(s, \Delta_{10}^{\theta}) = \zeta(s-8)\zeta(s-9)L(s, g).$$

Here L_{spin} is Andrianov's uncompleted classical spinor L-function. For a genus-two form of weight 10, the completed function is

$$L^*_{\mathrm{spin}}(s, F) = (2\pi)^{-2s}\Gamma(s)\Gamma(s-8)L_{\mathrm{spin}}(s, F).$$

Proof. The Igusa cusp form Δ_{10}^{θ} spans the weight-10 Maass Spezialschar line attached by the Saito–Kurokawa correspondence to the unique normalized form $g = \Delta E_6 \in S_{18}(\mathbf{SL}_2(\mathbb{Z}))$ [81]. This is Andrianov's spinor L-factor formula for the degree-2 Saito–Kurokawa lift attached to the weight-18 elliptic cusp form g [2]. Since $\dim S_{18}(\mathbf{SL}_2(\mathbb{Z})) = 1$, the source form is ΔE_6. □

Proposition A.2 *(Central-zero cancellation).* With

$$\Lambda(g, s) := (2\pi)^{-s}\Gamma(s)L(s, g),$$

one has

$$\Lambda(g, s) = -\Lambda(g, 18-s), \qquad L(9, g) = 0.$$

Hence the possible pole of $\zeta(s-8)L(s, g)$ at $s = 9$ is cancelled. At $s = 10$ the residue is

$$\operatorname*{Res}_{s=10} L_{\mathrm{spin}}(s, \Delta_{10}^{\theta}) = \zeta(2)L(10, g) = \frac{\pi^2}{6}L(10, g),$$

with $L(10, g) \neq 0$, so $L_{\mathrm{spin}}(s, \Delta_{10}^{\theta})$ has a simple pole at $s = 10$ with non-zero residue.

Proof. For a level-one cusp form of weight 18, the root number is $(-1)^{18/2} = -1$. Thus the completed L-function is odd about $s = 9$, and the central value vanishes. The factor $\zeta(s-8)$ has its only possible pole at $s = 9$, so the product $\zeta(s-8)L(s, g)$ is regular there. At $s = 10$ the factor $\zeta(s-9)$ has residue 1, while $\zeta(s-8) = \zeta(2)$. Write

$$L_p(s, g) = (1 - \alpha_p p^{-s})^{-1}(1 - \beta_p p^{-s})^{-1}.$$

Deligne's Ramanujan–Petersson theorem gives

$$|\alpha_p| = |\beta_p| = p^{17/2}.$$

At $s = 10$,

$$|\alpha_p p^{-10}| = |\beta_p p^{-10}| = p^{-3/2},$$

so the Euler product converges absolutely and has no zero. □

Proposition A.3 *(Manin–Deligne period).* There are Manin periods $\Omega^{\pm}(g)$ such that

$$\varepsilon(m) = \begin{cases} +, & m \equiv 0 \pmod 2, \\ -, & m \equiv 1 \pmod 2, \end{cases} \qquad \frac{L(m, g)}{(2\pi i)^m \Omega^{\varepsilon(m)}(g)} \in \mathbb{Q}, \qquad 1 \le m \le 17.$$

In particular

$$L(10, g) \in \mathbb{Q}^{\times}(2\pi i)^{10}\Omega^{+}(g).$$

Proof. This is the Eichler–Shimura–Manin period theorem for the rational newform g [57], [49], in Deligne's period normalisation [24]. The nonvanishing at $m = 10$ is supplied by the Euler product in Proposition A.2. No identity of the form

$$L(10, g) = q\, a_{10}(g)\Omega^{-}(g), \qquad q \in \mathbb{Q}^{\times},$$

is used. □

Proposition A.4 *(Coefficient check).*

$$a_{10}(g) = 541648800 = 2^5 \cdot 3^2 \cdot 5^2 \cdot 7 \cdot 11 \cdot 977.$$

Proof.

$$a_{10}(g) = \tau(10) - 504 \sum_{k=1}^{9} \tau(k)\sigma_5(10-k).$$

$$(\tau(1), \dots, \tau(10)) = (1, -24, 252, -1472, 4830, -6048, -16744, 84480, -113643, -115920),$$

$$(\sigma_5(1), \dots, \sigma_5(9)) = (1, 33, 244, 1057, 3126, 8052, 16808, 33825, 59293).$$

Substitution gives $a_{10}(g) = 541648800$. Equivalently, since g is a Hecke eigenform,

$$a_{10}(g) = a_2(g)a_5(g) = (-528)(-1025850) = 541648800.$$

□

Remark A.5 *(Scalar-square separations).* The spinor L-factor calculation is independent of the Dirac–Igusa construction. The OP normalization is $\chi_{10}^{\mathrm{OP}} = 2^{-12}\Delta_5^2$. The Hecke eigenline, not the scalar normalization of its generator, determines L_{spin}. The pole of $\zeta(s-8)$ at $s = 9$ is cancelled by $L(9, g) = 0$, while the pole at $s = 10$ has residue $\zeta(2)L(10, g)$ with $L(10, g) \in \mathbb{Q}^{\times}(2\pi i)^{10}\Omega^{+}(g)$. None of these facts is a BKM denominator proof or a compact Hall/Pfaffian construction.

B Eight diagonal-divisor boundary rows

Let F_j denote the diagonal-divisor forms in the present F-order. The Cléry–Gritsenko source order is

$$\Delta_5, \Delta_2, \nabla_3, \Delta_1, \nabla_2, \Delta_{1/2}, \nabla_{3/2}, Q_1.$$

The present order is [35]

j	F_j	(t, N)	Γ_j	wt
1	Δ_5	$(1, 1)$	Γ_1	5
2	Δ_2	$(2, 1)$	Γ_2	2
3	Δ_1	$(3, 1)$	Γ_3	1
4	$\Delta_{1/2}$	$(4, 1)$	Γ_4	$\frac{1}{2}$
5	∇_3	$(1, 2)$	$\Gamma_0^{(2)}(2)$	3
6	∇_2	$(1, 3)$	$\Gamma_0^{(2)}(3)$	2
7	$\nabla_{3/2}$	$(1, 4)$	$\Gamma_0^{(2)}(4)$	$\frac{3}{2}$
8	Q_1	$(2, 2)$	$\Gamma_2(2)$	1

The plus-groups Γ_t^+ used below for $t = 2, 3, 4$ are the Fricke/Gritsenko–Nikulin denominator normalizations; they are not the source-group labels in the Cléry–Gritsenko table.

$$\operatorname{supp} \operatorname{div}_{\mathbb{H}_2}(F_j) = \bigcup_{\gamma \in \Gamma_j} \gamma \langle \mathcal{H}_{\mathrm{diag}} \rangle, \qquad \operatorname{mult}_{\mathcal{H}_{\mathrm{diag}}}(F_j) = 1.$$

Here $F_7 = \nabla_{3/2}$ is the Cléry–Gritsenko row at $(t, N) = (1, 4)$. It is not a physical comparison slot until a separate row map has been constructed.

The characters and multiplier systems are

j	ν_j
1	$v_\eta^{12} \times v_H$
2	$v_\eta^{6} \times v_H$
3	$v_\eta^{4} \times v_H$
4	$v_\eta^{3} \times v_H$ (degree 8)
5	$\chi_2^{(2)} \times v_H$
6	$\chi_2^{(3)} \times v_H$
7	$\chi_4^{(4)} \times v_H$ (order 4)
8	$\chi_4^{(2)} \times v_H$

Definition B.1 *(Candidate denominator proof datum).* For a row j, a candidate denominator proof datum consists of

$$\mathscr{P}_j = (\phi_j, \Gamma_{j,\mathrm{prod}}^+, \Gamma_{j,\mathrm{mod}}, \rho_j, \nu_j, L_j, \alpha_j),$$

where

$$\phi_j(\tau, z) = \sum c_j(n, l) q^n r^l,$$

$$F_j(Z) = C_j e^{2\pi i (\rho_j, Z)} \prod_{\gamma \in \Gamma_{j,\mathrm{prod}}^+} (1 - e^{2\pi i (\gamma, Z)})^{c_j(\gamma)},$$

$$\nu_j : \Gamma_{j,\mathrm{mod}} \to \mathbb{C}^\times, \qquad \operatorname{supp} \operatorname{div}(F_j) = \Gamma_{j,\mathrm{mod}} \cdot \mathcal{H}_{\mathrm{diag}}, \qquad \operatorname{mult}_{\mathcal{H}_{\mathrm{diag}}}(F_j) = 1,$$

and a degree map

$$\alpha_j : \Gamma^+_{j,\mathrm{prod}} \to L_j.$$

Only after a Weyl–Kac–Borcherds denominator theorem has been supplied does this datum include a Lie superalgebra $\mathfrak{g}_j$ with

$$\mathrm{sdim}(\mathfrak{g}_{j,\alpha_j(\gamma)}) = c_j(\gamma)$$

on the corresponding product side. For a product with several cusps this single-source notation is only a template. The congruence-row product datum is the cusp-labelled datum of Proposition B.9.

Definition B.2 *(Gritsenko–Nikulin degree lattice).* For $t \in \mathbb{Z}_{>0}$ set

$$M_t := U(4t) \oplus \langle 2 \rangle = \mathbb{Z}f_2 \oplus \mathbb{Z}\widehat{f_3} \oplus \mathbb{Z}f_{-2},$$

with Gram matrix

$$\begin{pmatrix} 0 & 0 & -4t \\ 0 & 2 & 0 \\ -4t & 0 & 0 \end{pmatrix}.$$

For $(n, l, m) \in \mathbb{Z}^3$ put

$$\alpha_t(n, l, m) := nf_2 + l\widehat{f_3} + mf_{-2} \in M_t, \qquad e_t^{(n,l,m)} := q^n r^l s^{tm}.$$

Then

$$(\alpha_t(n, l, m), \alpha_t(n, l, m)) = -2(4tnm - l^2).$$

Let Γ_t^{prod} be the Gritsenko–Nikulin product cone: $(n, l, m) > 0$ if $m > 0$, or if $m = 0, n > 0$, or if $m = n = 0, l < 0$.

Lemma B.3 *(Two exponential normalizations).* Let

$$D_5(Z) := 64^{-1}\Delta_5(Z).$$

Gritsenko–Nikulin's Siegel normalization gives

$$D_5(Z) = q^{1/2}r^{1/2}s^{1/2} \prod_{(n,l,m)>0} (1 - q^n r^l s^m)^{f(nm,l)}$$

with leading monomial $q^{1/2}r^{1/2}s^{1/2}$. The denominator normalization used for the main BKM algebra is

$$q^n r^l s^m = \exp(-\pi i(\alpha(n, l, m), z)).$$

Thus replacing Z by $2Z$ converts the product character to the usual $\exp(-2\pi i(\alpha, z))$ denominator character, and

$$\mathrm{den}(\mathfrak{g}_{\Delta_5}) = D_5(2Z) = 64^{-1}\Delta_5(2Z).$$

For rows F_2, F_3, F_4, Appendix B uses the Gritsenko–Nikulin exponential

$$e_t^{(n,l,m)} = q^n r^l s^{tm}$$

directly. Hence

$$C'_j = 1, \qquad a_j = 1 \qquad (j = 2, 3, 4)$$

is a statement in the e_t normalization. It is not the doubled $N = 1$ normalization of the main Δ_5 denominator.

Proof. The first product is the singular-theta lift of Borcherds [8] together with the explicit paramodular product identity of Gritsenko–Nikulin [39, Theorem 2.1, Example 2.4, (2.16)]; the Lorentzian Kac–Moody correction and the identification of $D_5(2Z)$ with the denominator function of the corrected Lorentzian BKM superalgebra is the content of [37, Proposition 3.1], in particular the corrected denominator identity for $\mathfrak{g}_{\Delta_5}$. The rows F_2, F_3, F_4 are the separate $e_t^{(n,l,m)} = q^n r^l s^{tm}$ denominator identities of Gritsenko–Nikulin [39, Theorem 2.1, (2.3), (2.11), Lemma 2.5, (2.18), (2.19), Theorem 2.6, (2.20), (2.21), Section 5.1.1]. □

Proposition B.4 *(Full-paramodular rows).* Let

$$\phi_{0,2} = \sum f_2(n,l) q^n r^l, \qquad \phi_{0,3} = \sum f_3(n,l) q^n r^l, \qquad \phi_{0,4} = \sum f_4(n,l) q^n r^l,$$

where [39]

$$\phi_{0,2} = \frac{1}{2}\eta^{-4} \sum_{m,n\in\mathbb{Z}} (3m-n)\left(\frac{-4}{m}\right)\left(\frac{12}{n}\right) q^{(3m^2+n^2)/24} r^{(m+n)/2},$$

$$\phi_{0,3} = \left(\frac{\vartheta(\tau,2z)}{\vartheta(\tau,z)}\right)^2, \qquad \phi_{0,4} = \frac{\vartheta(\tau,3z)}{\vartheta(\tau,z)}.$$

Then the rows F_2, F_3, F_4 have the following product data:

j	F_j	ϕ_j	L_j	ρ_j
2	Δ_2	$\phi_{0,2}$	M_2	$(\frac{1}{4}, \frac{1}{2}, \frac{1}{4})$
3	Δ_1	$\phi_{0,3}$	M_3	$(\frac{1}{6}, \frac{1}{2}, \frac{1}{6})$
4	$\Delta_{1/2}$	$\phi_{0,4}$	M_4	$(\frac{1}{8}, \frac{1}{2}, \frac{1}{8})$

and

$$\alpha_j^{\mathrm{GN}}(n,l,m) := \alpha_t(n,l,m) \quad \text{for } (j,t) = (2,2), (3,3), (4,4).$$

The product identities are

$$\Delta_2(Z) = q^{1/4} r^{1/2} s^{1/2} \prod_{(n,l,m)\in\Gamma_2^{\mathrm{prod}}} (1 - q^n r^l s^{2m})^{f_2(nm,l)},$$

$$\Delta_1(Z) = q^{1/6} r^{1/2} s^{1/2} \prod_{(n,l,m)\in\Gamma_3^{\mathrm{prod}}} (1 - q^n r^l s^{3m})^{f_3(nm,l)},$$

$$\Delta_{1/2}(Z) = q^{1/8} r^{1/2} s^{1/2} \prod_{(n,l,m)\in\Gamma_4^{\mathrm{prod}}} (1 - q^n r^l s^{4m})^{f_4(nm,l)}.$$

The characters are respectively induced by $v_\eta^6 \times v_H$, $v_\eta^4 \times v_H$, and $v_\eta^3 \times v_H$. Equivalently, on the extended paramodular groups,

$$\nu_2 = \chi_4 : \Gamma_2^+ \to \mu_4, \qquad \nu_3 = \chi_6 : \Gamma_3^+ \to \mu_6, \qquad \nu_4 = v_8 : \Gamma_4^+ \to \mu_8.$$

Thus for $(j,t) = (2,2), (3,3), (4,4)$,

$$\Gamma_{j,\mathrm{prod}}^+ = \Gamma_t^{\mathrm{prod}}, \qquad \Gamma_{j,\mathrm{mod}} = \Gamma_t^+.$$

Proof. The product formulae are Gritsenko–Nikulin [39, Theorem 2.1, (2.21) for Δ_2, (2.20) for Δ_1, and (2.11) for $\Delta_{1/2}$]. Their Section 5 puts all formulae of Lie type on the hyperbolic lattice M_t and writes the exponent as $e^{(n,l,m)} = q^n r^l s^{tm}$. The Weyl vector is

$$\rho_{t,II,\bar{0}} = \left(\frac{1}{2t}, \frac{1}{2}, \frac{1}{2t}\right),$$

which gives the three displayed prefactors for $t = 2, 3, 4$. □

Proposition B.5 *(Gritsenko–Nikulin chambers for F_2, F_3, F_4).* For $t = 2, 3, 4$ the Weyl group is

$$W_{t,II,\bar{0}} = W^{(2)}(M_t),$$

and all simple real roots in this chamber are even. The Weyl vector is

$$\rho_t = \left(\frac{1}{2t}, \frac{1}{2}, \frac{1}{2t}\right).$$

The simple real roots are

$$P_2 = \{(0,-1,0),\ (1,1,0),\ (1,3,1),\ (0,1,1)\},$$

$$P_3 = \{(0,-1,0),\ (1,1,0),\ (2,5,1),\\ (2,7,2),\ (1,5,2),\ (0,1,1)\},$$

and

$$P_4 = \{\delta \in M_4 \mid (\delta,\delta) = 2,\ (\delta,\rho_4) = -1\}.$$

Thus the automorphic denominator chamber for F_j is the chamber in $M_t \otimes \mathbb{R}$ cut out by $(x,\delta) \le 0$ for $\delta \in P_t$, where $(j,t) = (2,2), (3,3), (4,4)$.

Proof. This is Gritsenko–Nikulin [39], Section 5.1.1, cases $(t, II, \bar{0})$. The displayed sets are exactly $P_{t,II,\bar{0}}(\mathcal{M}_{t,II,\bar{0}})$ for $t = 2, 3, 4$. □

Lemma B.6 *(Product cone and chamber compatibility).* Let $t = 2, 3, 4$. If

$$f_t(nm, l) \neq 0, \qquad (n,l,m) \in \Gamma_t^{\mathrm{prod}},$$

then

$$(\alpha_t(n,l,m), \rho_t) \le 0.$$

For $t = 2, 3$ the inequality is strict. For $t = 4$, equality occurs only on the parabolic ray

$$\alpha_t(n,l,m) \in \mathbb{Q}_{>0}\rho_4.$$

Proof. The Jacobi support satisfies

$$4tnm - l^2 \ge -1.$$

Since

$$(\alpha_t(n,l,m), \rho_t) = -2n + l - 2m,$$

the chamber inequality is $l \le 2n + 2m$. For $t = 2, 3$, Gritsenko– Nikulin Lemma 2.5 gives the required strict chamber condition on the active support. For $t = 4$, the proof of Gritsenko–Nikulin (1.33) gives

the same condition, with equality only in the parabolic case. Equivalently, equality and $4tnm - l^2 \geq -1$ force

$$n = m, \qquad l = 4n,$$

so

$$\alpha_4(n, 4n, n) = 8n\rho_4.$$

□

Theorem B.7 *(Gritsenko–Nikulin denominator rows).* Let $t = 2, 3, 4$ and put

t	$F(t)$	ϕ_t
2	Δ_2	$\phi_{0,2}$
3	Δ_1	$\phi_{0,3}$
4	$\Delta_{1/2}$	$\phi_{0,4}$

Let

$$\mathfrak{g}_t^{\mathrm{GN}} \quad (t = 2, 3, 4)$$

be the generalized Borcherds–Kac–Moody superalgebra whose Lorentzian root lattice is M_t, whose Weyl group is $W_t = W^{(2)}(M_t)$, whose Weyl vector is

$$\rho_t = \left(\frac{1}{2t}, \frac{1}{2}, \frac{1}{2t}\right),$$

and whose simple real roots are $P_t = P_{t,II,\bar{0}}$ from Proposition B.5. Let $\mathcal{M}_t$ denote the corresponding chamber. Then

$$P_{t,\bar{1}} = \varnothing, \qquad P_{t,\bar{0}} = P_t,$$

so every simple real root is even and $\epsilon_t(s_\delta) = -1$ for every $\delta \in P_t$; equivalently $\epsilon_t(w) = \det(w)$ on W_t.

Let

$$\phi_t(\tau, z) = \sum f_t(n, l) q^n r^l.$$

The integers $N_t(a)$ are the additive-side coefficients in Gritsenko–Nikulin (5.1), equivalently the signed supermultiplicities of imaginary simple roots. They are not the product exponents $f_t(nm, l)$. In the Gritsenko–Nikulin exponential normalization $e_t^{(n,l,m)} = q^n r^l s^{tm}$, the Weyl–Kac–Borcherds denominator identity is

$$\begin{aligned}\operatorname{den}(\mathfrak{g}_t^{\mathrm{GN}}) &= \sum_{w \in W_t} \epsilon_t(w) \left(e_t^{w(\rho_t)} + \sum_{a \in \mathbb{R}_{++} \mathcal{M}_t \cap M_t} N_t(a) e_t^{w(\rho_t + a)} \right) \\ &= e_t^{\rho_t} \prod_{(n,l,m) \in \Gamma_t^{\mathrm{prod}}} \left(1 - e_t^{(n,l,m)}\right)^{f_t(nm,l)} = F(t)(Z).\end{aligned}$$

Equivalently,

$$\operatorname{den}(\mathfrak{g}_2^{\mathrm{GN}}) = \Delta_2(Z), \qquad \operatorname{den}(\mathfrak{g}_3^{\mathrm{GN}}) = \Delta_1(Z), \qquad \operatorname{den}(\mathfrak{g}_4^{\mathrm{GN}}) = \Delta_{1/2}(Z).$$

Thus, for rows $j = 2, 3, 4$,

$$C'_j = 1, \qquad a_j = 1, \qquad L_j = M_j, \qquad \alpha_j = \alpha_j^{\mathrm{GN}}$$

in this normalization. The imaginary root spaces are determined here only as signed multiplicity spaces:

$$\operatorname{sdim}(\mathfrak{g}_{t,\alpha_t(n,l,m)}) = f_t(nm, l)$$

on the product side, with imaginary simple-root superdimensions $N_t(a)$ on the Weyl-sum side.

Proof. Gritsenko–Nikulin [39, Section 5.1] state that almost all identities proved earlier in their paper give denominator formulae for Lorentzian Kac–Moody algebras. Their formula (5.1) is exactly the displayed Weyl–Kac–Borcherds identity on $M_t = U(4t) \oplus \langle 2 \rangle$, with $e^{(n,l,m)} = q^n r^l s^{tm}$.

For the cases $(t, II, \overline{0})$, $t = 2, 3, 4$, they give

$$W_{t,II,\overline{0}} = W^{(2)}(M_t), \qquad \rho_{t,II,\overline{0}} = \left(\frac{1}{2t}, \frac{1}{2}, \frac{1}{2t}\right),$$

and

$$P_{t,II,\overline{0}}(\mathcal{M}_{t,II,\overline{0}}) = P_{t,II,\overline{0}}(\mathcal{M}_{t,II,\overline{0}})_{\overline{0}}.$$

Hence the odd simple-real-root set is empty and the Weyl character is the ordinary reflection sign on the simple real reflections. The roots in P_t have square 2, and the generalized Cartan matrix is $((\delta, \delta'))_{\delta,\delta' \in P_t}$; for $t = 4$ this is the infinite symmetric parabolic matrix recorded by Gritsenko–Nikulin. For real roots $4tn - l^2 = -1$, the relevant Jacobi coefficient is 1: this is Gritsenko–Nikulin Lemma 2.5 for $t = 2, 3$ and the proof of Gritsenko–Nikulin (1.33) for $t = 4$.

The product factors are the full-paramodular products [39, (2.21), (2.20), (2.11)], already displayed in Proposition B.4. Their prefactors are exactly $e_t^{\rho_t}$. Therefore the denominator functions are Δ_2, Δ_1, and $\Delta_{1/2}$ with no further scalar or argument rescaling in the e_t normalization. □

Proposition B.8 *(Congruence products).* For $j = 5, 6, 7, 8$ Cléry–Gritsenko construct the following Borcherds products [35]. The weak Jacobi inputs are

j	ϕ_j	ρ_j^{pref}
5	$4(\xi_{1,0}^{(2)})^2 \in J_{0,1}^w(\Gamma_0(2))$	$(\frac{1}{2}, \frac{1}{2}, \frac{1}{2})$
6	$3\xi_{3,1}^{(6)}\xi_{3,5}^{(6)} \in J_{0,1}^w(\Gamma_0(3))$	$(\frac{1}{2}, \frac{1}{2}, \frac{1}{2})$
7	$2\xi_{2,1}^{(4)}\xi_{2,3}^{(4)} \in J_{0,1}^w(\Gamma_0(4))$	$(\frac{1}{2}, \frac{1}{2}, \frac{1}{2})$
8	$2\xi_{1,0}^{(2)}(\tau, 2z) \in J_{0,2}^w(\Gamma_0(2))$	$(\frac{1}{4}, \frac{1}{2}, \frac{1}{2})$

where ρ_j^{pref} is read from the literal product prefactor $q^{\rho_\tau} r^{\rho_z} s^{\rho_\omega}$. At this stage it is a prefactor exponent, not a proved Weyl vector. In particular, for F_8 the M_2 product coordinate corresponding to $s^{1/2}$ is $(1/4, 1/2, 1/4)$ because $e^{\alpha_2(a,b,c)} = q^a r^b s^{2c}$. More precisely, if

$$S = \left(\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right),$$

and

$$\phi_5 = \sum c_5(n,l) q^n r^l, \qquad \phi_5|_{0,1} S = \sum c_{5,S}(n,l) q^n r^l,$$

then

$$\begin{aligned} F_5(Z) &= \nabla_3(Z) \\ &= q^{1/2} r^{1/2} s^{1/2} \prod_{(n,l,m) \in \Gamma_1^{\mathrm{prod}}} (1 - q^n r^l s^m)^{c_5(nm,l)} (1 - q^{2n} r^{2l} s^{2m})^{c_{5,S}(nm,l)}. \end{aligned}$$

If

$$\phi_6 = \sum c_6(n,l)q^n r^l, \qquad \phi_6|_{0,1}S = \sum c_{6,S}(n,l)q^n r^l,$$

then

$$\begin{aligned} F_6(Z) &= \nabla_2(Z) \\ &= q^{1/2}r^{1/2}s^{1/2} \prod_{(n,l,m)\in\Gamma_1^{\mathrm{prod}}} (1-q^n r^l s^m)^{c_6(nm,l)}(1-q^{3n}r^{3l}s^{3m})^{c_{6,S}(nm,l)}. \end{aligned}$$

If

$$\phi_7 = \sum c_7(n,l)q^n r^l, \qquad \phi_7|_{0,1}S = \sum c_{7,S}(n,l)q^n r^l, \qquad \phi_7|_{0,1}M = \sum c_{7,M}(n,l)q^n r^l,$$

where $M = \left(\begin{smallmatrix} 1 & -1 \\ 2 & -1 \end{smallmatrix}\right)$, then

$$\begin{aligned} F_7(Z) &= \nabla_{3/2}(Z) \\ &= q^{1/2}r^{1/2}s^{1/2} \prod_{(n,l,m)\in\Gamma_1^{\mathrm{prod}}} (1-q^n r^l s^m)^{c_7(nm,l)} \\ &\qquad \cdot (1-q^{2n}r^{2l}s^{2m})^{c_{7,M}(nm,l)/2}(1-q^{4n}r^{4l}s^{4m})^{c_{7,S}(nm,l)}. \end{aligned}$$

The exponent $c_{7,M}/2$ is the literal middle-cusp exponent in the μ_4-cover normalization; it is not a scalar-character exponent on the original congruence group. If

$$\phi_8 = \sum c_8(n,l)q^n r^l, \qquad \phi_8|_{0,2}S = \sum c_{8,S}(n,l)q^n r^l,$$

then

$$\begin{aligned} F_8(Z) &= Q_1(Z) \\ &= q^{1/4}r^{1/2}s^{1/2} \prod_{(n,l,m)\in\Gamma_2^{\mathrm{prod}}} (1-q^n r^l s^{2m})^{c_8(nm,l)}(1-q^{2n}r^{2l}s^{4m})^{c_{8,S}(nm,l)}. \end{aligned}$$

These four displays pin the automorphic products. By themselves they do not construct a BKM chamber, parity function, or Weyl-sum identity.

Proof. The formulae are Cléry–Gritsenko [35, Theorem 3.1] and the four product displays for $\nabla_3, \nabla_2, \nabla_{3/2}$, and Q_1 at the end of Section 3. The second factors in F_5, F_6, F_8 and the second and third factors in F_7 come from the non-infinity cusps of $\Gamma_0(N)$ in their congruence Borcherds product theorem. □

Proposition B.9 *(Product degree lattices for F_5, F_6, F_7, F_8).* For the product rows $j = 5, 6, 7, 8$ set

$$(t_5, N_5) = (1,2), \qquad (t_6, N_6) = (1,3), \qquad (t_7, N_7) = (1,4), \qquad (t_8, N_8) = (2,2).$$

The Cléry–Gritsenko modular host is the orthogonal lattice

$$O_j = U \oplus U(N_j) \oplus \langle -2t_j \rangle$$

of signature $(2,3)$. The single product-degree lattice for the monomial denominator is

$$L_j^{\mathrm{prod}} := M_{t_j}.$$

Define the cusp-labelled degree maps by

$$\alpha_5^{\mathrm{prod}}(n,l,m;\infty)=\alpha_1(n,l,m),\qquad \alpha_5^{\mathrm{prod}}(n,l,m;S)=\alpha_1(2n,2l,2m),$$

$$\alpha_6^{\mathrm{prod}}(n,l,m;\infty)=\alpha_1(n,l,m),\qquad \alpha_6^{\mathrm{prod}}(n,l,m;S)=\alpha_1(3n,3l,3m),$$

$$\alpha_7^{\mathrm{prod}}(n,l,m;\infty)=\alpha_1(n,l,m),\qquad \alpha_7^{\mathrm{prod}}(n,l,m;M)=\alpha_1(2n,2l,2m),$$

$$\alpha_7^{\mathrm{prod}}(n,l,m;S)=\alpha_1(4n,4l,4m),$$

and

$$\alpha_8^{\mathrm{prod}}(n,l,m;\infty)=\alpha_2(n,l,m),\qquad \alpha_8^{\mathrm{prod}}(n,l,m;S)=\alpha_2(2n,2l,2m).$$

The corresponding product-prefactor degrees are

$$\rho_5^{\mathrm{prod}}=\rho_6^{\mathrm{prod}}=\rho_7^{\mathrm{prod}}=\alpha_1(\tfrac12,\tfrac12,\tfrac12),\qquad \rho_8^{\mathrm{prod}}=\alpha_2(\tfrac14,\tfrac12,\tfrac14).$$

For $\lambda\in L_j^{\mathrm{prod}}$ let $\mu_j(\lambda)$ be the sum of all cusp coefficients whose labelled degree is λ:

$$\mu_j(\lambda)=\sum_{\alpha_j^{\mathrm{prod}}(n,l,m;\kappa)=\lambda}\widetilde{c}_{j,\kappa}(nm,l),$$

where

$$\widetilde{c}_{j,\infty}=c_j,\qquad \widetilde{c}_{5,S}=c_{5,S},\qquad \widetilde{c}_{6,S}=c_{6,S},$$

$$\widetilde{c}_{7,M}=\frac12 c_{7,M},\qquad \widetilde{c}_{7,S}=c_{7,S},\qquad \widetilde{c}_{8,S}=c_{8,S}.$$

For $j=7$ this is a product datum on the μ_4-multiplier cover; the integrality of the displayed half-exponents is part of the Cléry–Gritsenko product theorem in that cover. Until one proves that integrality directly from the Jacobi seed, the row is used only with this sourced cover datum. The product is therefore used as an integral automorphic product only in the sourced multiplier-cover normalization. Let

$$\mathcal{K}_5=\mathcal{K}_6=\mathcal{K}_8=\{\infty,S\},\qquad \mathcal{K}_7=\{\infty,M,S\},$$

and

$$\mathcal{D}_j:=\Gamma_{t_j}^{\mathrm{prod}}\times\mathcal{K}_j.$$

The cusp-labelled product datum is

$$\mathscr{E}_j^{\mathrm{prod}}:=(\mathcal{D}_j,\alpha_j^{\mathrm{prod}},\mu_j).$$

Then

$$F_j(Z)=e^{\rho_j^{\mathrm{prod}}}\prod_{\lambda\in\alpha_j^{\mathrm{prod}}(\mathcal{D}_j)}\left(1-e^{\lambda}\right)^{\mu_j(\lambda)}\qquad (j=5,6,7,8),$$

where $e^{\alpha_1(a,b,c)}=q^a r^b s^c$ and $e^{\alpha_2(a,b,c)}=q^a r^b s^{2c}$.

Proof. The displayed maps are the unique monomial-degree maps forced by the three products in Proposition B.8. For F_5 the non-infinity cusp contributes the monomials $q^{2n}r^{2l}s^{2m}$, hence degree $\alpha_1(2n, 2l, 2m)$. For F_6 the corresponding monomial is $q^{3n}r^{3l}s^{3m}$, hence degree $\alpha_1(3n, 3l, 3m)$. For F_7 the two non-infinity factors have monomials $q^{2n}r^{2l}s^{2m}$ and $q^{4n}r^{4l}s^{4m}$; the first has half exponent $c_{7,M}/2$ exactly as in the Cléry–Gritsenko product. For F_8 the product has index $t = 2$: $e^{\alpha_2(n,l,m)} = q^n r^l s^{2m}$ and the non-infinity cusp contributes $e^{\alpha_2(2n,2l,2m)} = q^{2n}r^{2l}s^{4m}$. If two labelled cusp contributions land on the same lattice degree, exponents add; this is precisely the definition of μ_j.

Cléry–Gritsenko [35, Theorem 3.1] realize these products as orthogonal Borcherds products for $U \oplus U(N_j) \oplus \langle -2t_j \rangle$. That theorem supplies the automorphic product and the divisor. It does not supply a Weyl chamber, simple real roots, a parity function, or a Weyl-sum identity on M_{t_j}. Thus L_j^{prod} and α_j^{prod} are product data, not a completed BKM denominator datum. The completed denominator datum for F_5, F_6, F_7, F_8 comes from the CHL theorem in Theorem B.13. □

Definition B.10 *(Formal product algebra of a row).* For a row j with product-degree function μ_j, let

$$\mathcal{V}_j^{\mathrm{prod}} = \bigoplus_{\lambda \in L_j^{\mathrm{prod}},\, \lambda > 0} \mathcal{V}_{j,\lambda}$$

be any graded super vector space with finite-dimensional homogeneous pieces and

$$\operatorname{sdim} \mathcal{V}_{j,\lambda} = \mu_j(\lambda).$$

Choose a Grothendieck representative: the even part has dimension $\mu_j(\lambda)$ when $\mu_j(\lambda) > 0$, the odd part of dimension $-\mu_j(\lambda)$ when $\mu_j(\lambda) < 0$, and the zero space when $\mu_j(\lambda) = 0$. Give it the abelian Lie superalgebra bracket:

$$[\mathcal{V}_{j,\lambda}, \mathcal{V}_{j,\lambda'}] = 0.$$

Its formal current envelope on E is

$$\mathsf{Prod}_{j,E} := U_E^{\mathrm{ch}}\Big(\mathcal{V}_j^{\mathrm{prod}} \otimes \mathcal{O}_E\Big).$$

Proposition B.11 *(Eight formal current rows).* Every Cléry–Gritsenko diagonal-divisor row has a formal determinant object. For $j = 1$ this is $\mathsf{Den}_{\Delta_5,E}$ from Proposition 10.2. For $j = 2, 3, 4$ it is the current envelope of the Gritsenko–Nikulin BKM algebra $\mathfrak{g}_j^{\mathrm{GN}}$. For $j = 5, 6, 7, 8$ it is the formal product algebra $\mathsf{Prod}_{j,E}$. For $j = 5, 6, 7, 8$, Theorem B.13 also supplies a nonabelian CHL/WKB denominator algebra; the abelian product algebra in this theorem is the part constructed directly from the Cléry–Gritsenko product exponents. For $j = 5, 6, 7, 8$ this is only the tautological virtual product object realizing the Cléry–Gritsenko exponents. Its Beilinson–Drinfeld envelope is abelian; it carries no Weyl-sum side, no BKM bracket, and no physical protected BPS interpretation.

In all eight cases the formal determinant is the corresponding monic automorphic product:

$$e^{\rho_j} \operatorname{sdet}_{\mathcal{V}_j}(1 - e) = D_j(Z),$$

where $D_1 = 64^{-1}\Delta_5$ and $D_j = F_j$ for the displayed Cléry–Gritsenko product normalizations $j > 1$, with the normalizations stated in Propositions B.4, B.8, and B.9.

Proof. For $j = 1$, Theorem 9.2 gives the Borcherds denominator algebra and Proposition 10.2 gives the formal current envelope. For $j = 2, 3, 4$, Theorem B.7 gives the BKM brackets and their denominator identities.

For $j = 5, 6, 7, 8$, Proposition B.9 constructs the degree lattice, the positive product support, the prefactor, and the signed exponent μ_j. The super vector space $\mathcal{V}_j^{\mathrm{prod}}$ realizes these signed dimensions. The Fock determinant lemma gives

$$e^{\rho_j^{\mathrm{prod}}} \prod_{\lambda} (1 - e^{\lambda})^{\mathrm{sdim}\, \mathcal{V}_{j,\lambda}} = e^{\rho_j^{\mathrm{prod}}} \prod_{\lambda} (1 - e^{\lambda})^{\mu_j(\lambda)} = F_j(Z).$$

The formal current enveloping construction gives a chiral algebra on E from this abelian Lie superalgebra. It is not a compact BPS factorization algebra. □

Proposition B.12 *(Nonabelian promotion test for F_5, F_6, F_7, F_8).* Suppose that one of the congruence product rows

$$j \in \{5, 6, 7, 8\}$$

is the denominator of a generalized Borcherds–Kac–Moody superalgebra on the product lattice L_j^{prod}. Then the following additional data must exist:

$$\Big(P_j, \mathcal{M}_j, W_j, \epsilon_j, \rho_j, N_j(a)\Big).$$

They must satisfy

$$W_j = \langle s_\delta \mid \delta \in P_j \rangle, \qquad \mathcal{M}_j = \{x \mid (x, \delta) \le 0 \text{ for all } \delta \in P_j\},$$

$$(\rho_j, \delta) = -\frac{(\delta, \delta)}{2} \qquad (\delta \in P_j),$$

and the Weyl–Kac–Borcherds identity

$$\begin{aligned} F_j(Z) &= \sum_{w \in W_j} \epsilon_j(w) \left(e^{w(\rho_j)} + \sum_{a \in L_j^{\mathrm{prod}} \cap \mathbb{R}_{>0} \mathcal{M}_j} N_j(a) e^{w(\rho_j + a)} \right) \\ &= e^{\rho_j^{\mathrm{prod}}} \prod_{\lambda \in \alpha_j^{\mathrm{prod}}(\mathcal{D}_j)} \left(1 - e^{\lambda}\right)^{\mu_j(\lambda)}. \end{aligned}$$

In particular the product already fixes the signed root multiplicity function μ_j and the product-prefactor degree

$$\rho_5^{\mathrm{prod}} = \rho_6^{\mathrm{prod}} = \rho_7^{\mathrm{prod}} = \alpha_1(\tfrac{1}{2}, \tfrac{1}{2}, \tfrac{1}{2}), \qquad \rho_8^{\mathrm{prod}} = \alpha_2(\tfrac{1}{4}, \tfrac{1}{2}, \tfrac{1}{4}),$$

but it does not fix $P_j, \mathcal{M}_j, W_j, \epsilon_j$, or the imaginary simple-root coefficients $N_j(a)$.

Proof. The equalities in Proposition B.9 construct only the monomial degree and the total exponent $\mu_j(\lambda)$. A denominator theorem has a second side: the Weyl alternating sum. That side requires a reflection chamber, a set of simple real roots, a reflection character, and the imaginary simple-root contribution.

The Gritsenko–Nikulin chambers for M_1 and M_2 do not follow by notation. The forms $\nabla_3, \nabla_2, \nabla_{3/2}$, and Q_1 are congruence products with non-infinity cusp factors, whereas Δ_5 and Δ_2 are full-paramodular

denominator forms. Cléry–Gritsenko construct the automorphic product, character, and diagonal divisor on

$$U \oplus U(N_j) \oplus \langle -2t_j \rangle,$$

but their product theorem does not identify the simple real roots or prove a Weyl anti-invariant additive expansion on M_{t_j}. Therefore the displayed six-tuple is an additional theorem to be proved, not a consequence of the product formula. □

Theorem B.13 *(Govindarajan CHL/WKB denominator rows).* In the CHL/twisted-dyon normalization of Cheng–Dabholkar and Govindarajan, the rows

$$F_5 = \nabla_3, \qquad F_6 = \nabla_2, \qquad F_7 = \nabla_{3/2}, \qquad F_8 = Q_1$$

are Weyl–Kac–Borcherds denominator forms. The row F_7 is taken on the μ_4-multiplier cover determined by the Cléry–Gritsenko order-four multiplier. Thus F_7 is a denominator on the μ_4-cover, with no physical row map, metaplectic boundary theorem, or CY/DT/chiral host. In the product-degree notation of this appendix the real-root data are

$$L_5 = L_6 = L_7 = M_1, \qquad P_5 = P_6 = P_7 = P^{(1)}, \qquad \rho_5 = \rho_6 = \rho_7 = \alpha_1(\tfrac{1}{2}, \tfrac{1}{2}, \tfrac{1}{2}),$$

where

$$P^{(1)} = \{(0,-1,0),\ (1,1,0),\ (0,1,1)\},$$

and

$$L_8 = M_2, \qquad P_8 = P^{(2)}, \qquad \rho_8 = \alpha_2(\tfrac{1}{4}, \tfrac{1}{2}, \tfrac{1}{4}),$$

where

$$P^{(2)} = \{(0,-1,0),\ (1,1,0),\ (1,3,1),\ (0,1,1)\}.$$

The Weyl chambers and Weyl groups are

$$\mathcal{M}_j = \{x \in L_j \otimes \mathbb{R} \mid (x,\delta) \le 0 \text{ for every } \delta \in P_j\}, \qquad W_j = \langle s_\delta \mid \delta \in P_j \rangle.$$

Every simple real root is even and the reflection character is

$$\epsilon_j(w) = \det(w).$$

For $j = 5, 6, 7, 8$ the denominator product side is expressed through the cusp-labelled product datum $\mathscr{E}_j^{\mathrm{prod}}$, with root multiplicity function μ_j. In Govindarajan's CHL normalization the corresponding square-root form is anti-invariant under the Weyl reflections, with character $\epsilon_j(w) = \det(w)$. This anti-invariance is part of the Govindarajan WKB denominator theorem, not a consequence of the Cléry–Gritsenko product alone; this manuscript does not construct an independent row-by-row embedding of the CHL Weyl group into each Cléry–Gritsenko modular group or multiplier cover. The Cartan matrices are, in the CHL/WKB BKM normalization of [17, 31],

$$A^{(1)} = \begin{pmatrix} 2 & -2 & -2 \\ -2 & 2 & -2 \\ -2 & -2 & 2 \end{pmatrix},$$

for F_5, F_6, F_7, and

$$A^{(2)} = \begin{pmatrix} 2 & -2 & -6 & -2 \\ -2 & 2 & -2 & -6 \\ -6 & -2 & 2 & -2 \\ -2 & -6 & -2 & 2 \end{pmatrix},$$

for F_8.

Equivalently, for $x = (u, v, w)$, the two chambers are

$$\mathcal{M}_1 = \{x \mid v \geq 0,\ v \leq 2w,\ v \leq 2u\},$$

and

$$\mathcal{M}_2 = \{x \mid v \geq 0,\ v \leq 4w,\ 3v \leq 4(u + w),\ v \leq 4u\}.$$

There are integral imaginary simple-root coefficients $N_j(a)$, depending on the CHL twist [17, 31], such that

$$\begin{aligned} F_j(Z) &= \sum_{w \in W_j} \det(w) \left(e^{w(\rho_j)} + \sum_{a \in L_j \cap \mathbb{R}_{>0} \mathcal{M}_j} N_j(a) e^{w(\rho_j + a)} \right) \\ &= e^{\rho_j} \prod_{\lambda \in \alpha_j^{\mathrm{prod}}(\mathcal{D}_j)} \left(1 - e^{\lambda}\right)^{\mu_j(\lambda)} \qquad (j = 5, 6, 7, 8). \end{aligned}$$

Thus the six-tuples

$$(P_j, \mathcal{M}_j, W_j, \epsilon_j, \rho_j, N_j(a))$$

are constructed for $j = 5, 6, 7, 8$ by the CHL denominator theorem. They are not consequences of the Cléry–Gritsenko product formula alone. In the monic Cléry–Gritsenko/Govindarajan product variables of Proposition B.8, the identities have

$$C'_j = 1, \qquad a_j = 1 \qquad (j = 5, 6, 7, 8).$$

This is not the doubled $N = 1$ normalization of Lemma B.3.

Remark B.14 *(F_7 on the μ_4-multiplier cover).* The row $F_7 = \nabla_{3/2}$ is the Cléry–Gritsenko diagonal-divisor form of type $(t, N) = (1, 4)$, weight $3/2$, on $\Gamma_0^{(2)}(4)$ with an order-four multiplier. After pullback to the associated μ_4-multiplier cover, Govindarajan's CHL/WKB theorem identifies it with the Weyl–Kac–Borcherds denominator of $\mathcal{G}_1(4)$. It is the μ_4-cover denominator identity, with no physical comparison-row identification, no metaplectic boundary host, and no CY/DT or chiral/Hall theorem.

The square identity $F_7^2 = \Phi_3$ in the composite $N = 4$ normalization is only a scalar square identity. It does not carry the reflection sign, parity, Weyl chamber, or Weyl-alternating side. The coefficient convention for the additive side is fixed in Lemma B.16.

Lemma B.15 *(Coefficient normalisation for ∇_3 and ∇_2).* Let

$$\psi_5(\tau, z) = \vartheta(\tau, z)\eta(\tau)\eta(2\tau)^4 = \sum g_5(u, v) q^{u/2} r^{v/2},$$

and

$$\psi_6(\tau, z) = \vartheta(\tau, z)\eta(3\tau)^3 = \sum g_6(u, v) q^{u/2} r^{v/2}.$$

For $a = \alpha_1(A, B, C) \in M_1 \cap \mathbb{R}_{\geq 0}\mathcal{M}_1$, put

$$n = 2A + 1, \qquad l = 2B + 1, \qquad m = 2C + 1.$$

In the plus-sign convention of Theorem B.13, the imaginary simple-root coefficients for the rows F_5 and F_6 are

$$N_5(a) = \sum_{d|(n,l,m)} d^2\, g_5\left(\frac{nm}{d^2}, \frac{l}{d}\right),$$

and

$$N_6(a) = \sum_{d|(n,l,m)} \left(\frac{-3}{d}\right) d\, g_6\left(\frac{nm}{d^2}, \frac{l}{d}\right).$$

Govindarajan–Gopala Krishna write the same coefficients with the opposite sign in the imaginary-root correction term:

$$m_j(a) = -N_j(a) \qquad (j = 5, 6).$$

Proof. Govindarajan–Gopala Krishna construct the additive lift whose Weyl alternating side gives the denominator identity for the corresponding CHL square roots [32, (4.2), (4.4), (4.5), (4.8)]. Their Maass lift gives the displayed divisor sums from the Jacobi seeds ψ_5 and ψ_6, and their Weyl–Kac–Borcherds sum writes these coefficients as the imaginary simple-root correction. The sign difference is only the convention:

$$e^\rho + \sum_a N_j(a) e^{\rho + a},$$

whereas [32] writes the correction with a minus sign. □

Lemma B.16 *(Coefficient normalisation for $\nabla_{3/2}$).* Let

$$\nabla_{3/2}(Z) = \Delta_{3/2}^{(4)}(Z) = \frac{1}{8}\theta_5(2Z)\theta_7(2Z)\theta_9(2Z) = \sum_{n,l,m} A_7(n, l, m) q^{n/2} r^{l/2} s^{m/2},$$

with $A_7(1, 1, 1) = 1$. The first term is

$$\left(\sqrt{r} - \frac{1}{\sqrt{r}}\right)\sqrt{q}\,\sqrt{s} + \cdots,$$

so $A_7(1, -1, 1) = -1$. For $a = \alpha_1(A, B, C) \in M_1 \cap \mathbb{R}_{\geq 0}\mathcal{M}_1$, put

$$n = 2A + 1, \qquad l = 2B + 1, \qquad m = 2C + 1.$$

Then the imaginary simple-root coefficient in the plus-sign convention of Theorem B.13 is

$$N_7(\alpha_1(A, B, C)) = A_7(2A + 1, 2B + 1, 2C + 1) = [q^{A+1/2} r^{B+1/2} s^{C+1/2}]\nabla_{3/2}(Z).$$

In Govindarajan–Gopala Krishna's WKB sign convention,

$$m_7(a) = -N_7(a).$$

No Maass divisor-sum formula for N_7 is asserted here.

Proof. Cléry–Gritsenko construct $\nabla_{3/2}$ as the order-four multiplier row. Govindarajan–Gopala Krishna identify the square-root theta product and the square relation

$$F_7^2 = \Phi_3$$

in the composite $N = 4$ CHL case [33]. Factoring the Weyl vector gives

$$e^{\rho_7 + a} = q^{A+1/2} r^{B+1/2} s^{C+1/2},$$

which is exactly the displayed coefficient extraction. The sourced divisor-sum formula belongs to the square F_7^2, with its composite $N = 4$ Jacobi seed. A direct prime-type half-index formula for F_7 introduces fractional divisor powers at non-primitive chamber points, whereas the theta product has integral coefficients. This is a theta-product normalization identification. It is not a constructed physical host and not a physical row-map equivalence. □

Lemma B.17 *(Q_1 in M_2 notation).* In M_2,

$$e^{\alpha_2(n,l,m)} = q^n r^l s^{2m}, \qquad (\alpha_2(n,l,m), \alpha_2(n',l',m')) = 2ll' - 8(nm' + n'm).$$

The four simple real roots of the Q_1 row are

$$\delta_1 = (0,-1,0), \quad \delta_2 = (1,1,0), \quad \delta_3 = (1,3,1), \quad \delta_4 = (0,1,1),$$

with Gram matrix $A^{(2)}$. The Weyl vector is

$$\rho_8 = \alpha_2(\tfrac{1}{4}, \tfrac{1}{2}, \tfrac{1}{4}), \qquad e^{\rho_8} = q^{1/4} r^{1/2} s^{1/2},$$

and

$$(\rho_8, \delta_i) = -1 \qquad (1 \le i \le 4).$$

Let $A_8(n,l,m)$ be defined by

$$Q_1(Z) = \sum_{n,l,m} A_8(n,l,m) q^{n+1/4} r^{l+1/2} s^{2m+1/2}.$$

Govindarajan's theta-product formula, in his characteristic notation, is

$$Q_1(Z) = \frac{1}{4} \vartheta\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}(Z') \vartheta\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}(Z'), \qquad Z' = \begin{pmatrix} \tau & 2z \\ 2z & 4\sigma \end{pmatrix}.$$

Equivalently,

$$A_8(n,l,m) = \frac{1}{4} \sum_{\mathcal{S}(n,l,m)} (-1)^{j+k+h},$$

where $\mathcal{S}(n,l,m)$ is the set of $(i,j,k,h) \in \mathbb{Z}^4$ satisfying

$$\frac{1}{2}\big((i+\tfrac{1}{2})^2 + (k+\tfrac{1}{2})^2\big) = n + \tfrac{1}{4},$$

$$2(i+\tfrac{1}{2})j + 2(k+\tfrac{1}{2})(h+\tfrac{1}{2}) = l + \tfrac{1}{2},$$

and

$$2j^2 + 2(h+\tfrac{1}{2})^2 = 2m + \tfrac{1}{2}.$$

Then the imaginary simple-root coefficient in the plus-sign convention is

$$N_8(\alpha_2(n,l,m)) = A_8(n,l,m)$$

for the chamber-supported imaginary shifts. Govindarajan's expansion has coefficient -1 at the monomials

$$r^{-1}, \qquad qr, \qquad rs^2, \qquad qr^3s^2.$$

After factoring e^{ρ_8} these are precisely the four simple real root monomials for

$$\delta_1, \quad \delta_2, \quad \delta_4, \quad \delta_3,$$

so the signed real-simple-root supermultiplicity is $+1$ in each case. Their even parity and multiplicity-one real-root data are the Cheng–Dabholkar/Govindarajan CHL denominator data.

Proof. The first display is the definition of α_2 and the Gram matrix on M_2. The simple roots and Weyl vector are Cheng–Dabholkar's $N = 2$ chamber data translated through

$$\alpha_2(n,l,m) \longleftrightarrow \begin{pmatrix} 2n & l \\ l & 4m \end{pmatrix}.$$

Govindarajan identifies Q_1 with the row $\mathcal{G}_2(4)$ and gives both the product formula and the first Fourier terms. In that CHL normalization the square-root form is anti-invariant under the Weyl reflections. This is the sourced denominator anti-invariance, not an independently constructed restriction of the Cléry–Gritsenko character through a physical row map. □

Proof of Theorem B.13. Cheng–Dabholkar identify, for CHL level $N = 1, 2, 3$, the Weyl group, simple real roots, Weyl chamber, and Weyl denominator identity of the dyon algebra [17, (2.10)–(2.16), (3.16), (3.22), (3.23), (3.26), (3.42), (5.1), (5.2), (5.8), (5.10), (5.12)–(5.15), Appendix A]. For $N = 1$ the simple real roots are $P^{(1)}$ and the Cartan matrix is $A^{(1)}$; for $N = 2$ the simple real roots are $P^{(2)}$ and the Cartan matrix is $A^{(2)}$. Govindarajan then treats twisted dyons and proves that, for fixed N, the algebras $\mathcal{G}_N(M)$ have the same real simple roots, Weyl vector, and Cartan matrix, while the imaginary simple roots depend on M [31, Section 4.2, Section 4.3, Table 3].

Govindarajan's Table 3 gives

$$\nabla_3 \longleftrightarrow \mathcal{G}_1(2), \qquad \nabla_2 \longleftrightarrow \mathcal{G}_1(3), \qquad \nabla_{3/2} \longleftrightarrow \mathcal{G}_1(4), \qquad Q_1 \longleftrightarrow \mathcal{G}_2(4).$$

Section 4.2 of [31] checks the required ingredients: sum and product representations, integrality, correct multiplicity of the simple real roots, and the sign change of the square root under a Weyl reflection. Section 4.3 states the Weyl–Kac–Borcherds denominator formula for $\mathcal{G}_N(M)$. For $F_7 = \nabla_{3/2}$ the theorem is on the μ_4-multiplier cover, with no physical boundary partition identity and no CY/DT/chiral host. For Q_1, Govindarajan supplies the WKB denominator data [31, Appendix C, Appendix D]. The dyon-counting additive-seed issue for the $(2,4)$ and $(4,2)$ examples plays no role. The later hyperbolic and umbral extensions of [34] are also outside this proof: for $N = 5, 6$ they replace W by W_+, leave the sum side incompletely characterized, and treat some rows as product-only or candidate BKM data [34, Sections 3.3 and 4].

The real-root data translate to this notation as follows. On M_t,

$$((n,l,m),(n',l',m')) = 2ll' - 4t(nm' + n'm).$$

With $t = 1$, the set $P^{(1)}$ has Gram matrix $A^{(1)}$ and $\rho = \alpha_1(1/2, 1/2, 1/2)$ satisfies $(\rho, \delta) = -1$ for every $\delta \in P^{(1)}$. With $t = 2$, the set $P^{(2)}$ has Gram matrix $A^{(2)}$ and $\rho = \alpha_2(1/4, 1/2, 1/4)$ satisfies $(\rho, \delta) = -1$ for every $\delta \in P^{(2)}$. These are exactly the Weyl-vector equations $(\rho, \delta) = -(\delta, \delta)/2$ because all displayed real roots have square 2. □

Remark B.18 *(Nonabelian row data).* The row data are:

j	automorphic data $(F_j, \Gamma_j, \nu_j, \rho_j^{\mathrm{pref}})$	lattice and degree datum
1	computed/imported (Δ_5 product)	formal lattice computed; GN denominator imported
2, 3, 4	GN denominator theorem	$L_j = M_j,\ \alpha_j = \alpha_j^{\mathrm{GN}}$
5, 6, 7	CG product; Govindarajan denominator	$P^{(1)}, \mathcal{M}_1, W_1, \epsilon = \det$ (CHL)
8	CG product; Govindarajan denominator	$P^{(2)}, \mathcal{M}_2, W_2, \epsilon = \det$ (CHL)

Thus all eight rows have formal product/current-envelope objects. Rows 1–4 have the Gritsenko–Nikulin denominator algebras. Rows 5–8 have the Govindarajan CHL/WKB denominator algebras in the CHL normalization. The row F_7 lives on the μ_4-multiplier cover and fills no physical-host datum beyond the sourced denominator level.

Open Problem B.19 *(Physical-host data and row maps).* Construct, for every claimed physical realization of a Cléry–Gritsenko row, a datum

$$(\iota_j, \mathscr{R}_j, \mathscr{A}_j, \mathscr{D}_j, \mathscr{S}_j, \mathscr{Z}_j, \mathscr{H}_j)$$

such that

$$\iota_j : \Phi_j^{\mathrm{phys}} \longleftrightarrow F_i^{\mathrm{CG}}$$

fixes the exact row, parent form when a root is used, group or cover, automorphy cocycle, multiplier and its order, divisor multiplicities, Jacobi seed at every cusp, product-degree lattice, chamber, branch, normalization, and source theorem. The entries $(\mathscr{A}_j, \mathscr{D}_j, \mathscr{S}_j, \mathscr{Z}_j, \mathscr{H}_j)$ are the five levels of Definition 13.1. For rows $F_5, \ldots, F_8$, the Govindarajan CHL/WKB theorem fills $\mathscr{D}_j$, and the sourced scalar square fills $\mathscr{S}_j$ where stated. It does not fill $\mathscr{Z}_j$ or $\mathscr{H}_j$. A boundary partition identity and a compact Hall/chiral source remain open until these datum entries are constructed.

Proposition B.20 *(Comparison row systems).* The Cléry–Gritsenko rows in the present F_i-order have weights

$$(5, 2, 1, \tfrac{1}{2}, 3, 2, \tfrac{3}{2}, 1).$$

In triple order

$$(t, N) = (1, 1), (2, 1), (1, 2), (3, 1), (1, 3), (4, 1), (1, 4), (2, 2),$$

the weights are

$$(5, 2, 3, 1, 2, \tfrac{1}{2}, \tfrac{3}{2}, 1),$$

and the permutation from triple order to present order is

$$(1, 2, 5, 3, 6, 4, 7, 8).$$

Consequently

$$\{j : \mathrm{wt}(F_j) \in \mathbb{Z}\} = \{1, 2, 3, 5, 6, 8\}, \qquad \{j : \mathrm{wt}(F_j) \notin \mathbb{Z}\} = \{4, 7\}.$$

The comparison notation

$$\Phi_1^{\mathrm{cmp}}, \dots, \Phi_8^{\mathrm{cmp}}, \qquad \mathrm{wt}(\Phi_j^{\mathrm{cmp}}) = (5, 2, 1, 1, \tfrac{1}{2}, 1, \tfrac{1}{4}, 0),$$

does not define the Cléry–Gritsenko row system and defines no physical atlas in this paper. A row map

$$\Phi_j^{\mathrm{cmp}} \longleftrightarrow F_i$$

requires the relevant entries of Definition 13.1. Consequently the direct identifications

$$\Phi_5^{\mathrm{cmp}} = F_5, \qquad \Phi_7^{\mathrm{cmp}} = F_7, \qquad \Phi_8^{\mathrm{cmp}} = F_8$$

are false already by weight. The eight-row Cléry–Gritsenko catalogue contains no weight-1/4 row and no weight-0 row. The CHL square identities, such as $\nabla_3^2 = \Phi_6^{(1,2)}$ and $\nabla_2^2 = \Phi_4^{(1,3)}$, are scalar square identities; they are not row maps and do not by themselves carry the denominator sign, parity, or Weyl alternating side.

Proof. The Cléry–Gritsenko theorem gives the exact weights, groups, characters, and divisors of the eight diagonal-divisor rows [35]. The displayed comparison tuple is not that theorem table. The Borcea–Voisin route [7], [78] and higher-dimensional curve-counting routes [68], [67], [14] require separate construction of X_j, the reduced theory, the protected index, and the product identity. None is constructed here. Thus the comparison symbols remain external labels until a datum identifies them with Cléry–Gritsenko rows. □

Definition B.21 *(Canonical fractional branch).* Let

$$F(Z) = a\, q^{\alpha} r^{\beta} s^{\gamma} (1 + O(q, r, s)), \qquad a \in \mathbb{R}_{>0},$$

be a product expansion in a fixed positive Weyl chamber. Its canonical k-th root in that chamber is

$$F(Z)^{1/k} = a^{1/k} q^{\alpha/k} r^{\beta/k} s^{\gamma/k} \exp\Big(\frac{1}{k} \log(1 + O(q, r, s))\Big),$$

where $a^{1/k}$ is the positive real root and the logarithm is the formal logarithm in the completed product algebra. This defines a formal branch. It does not define a modular form until the relevant metaplectic or multiplier-cover character has been constructed.

Proposition B.22 *(Fractional-power comparison test).* The formal fractional-power comparison symbols

$$\Phi_5^{\mathrm{cmp}}, \qquad \Phi_7^{\mathrm{cmp}}, \qquad \Phi_8^{\mathrm{cmp}}$$

do not label Cléry–Gritsenko rows. The source-compatible statement is the following test.

The displayed comparison weight of Φ_5^{cmp} is 1/2. If it is presented as a square root, the parent form must have weight 1, and the root must live on the pullback of the parent group to the genus-two metaplectic cover with a genuine half-integral multiplier. A weight-5 parent would produce a weight-5/2 square root and does not produce this displayed comparison slot.

If the comparison symbol Φ_7^{cmp} is supplied with its displayed comparison weight 1/4, then it must be a genuine quarter-integral-weight form on a multiplier cover. If it is presented as a fourth root of a parent form with exact order-four character, an order-eight cover is not forced by character theory. If a character root ψ with $\psi^4 = \chi$ exists after pullback and χ has exact order 4, then the character part has

exact order 16, unless an explicit automorphy-cocycle relation reduces it. No such relation is constructed here. The μ_4-cover of the Cléry–Gritsenko row $F_7 = \nabla_{3/2}$, whose weight is $3/2$, is not a modular-root datum for a comparison object Φ_7^{cmp} of weight $1/4$. That comparison object needs its own parent, cover, cocycle, and character-root computation.

If a comparison form Φ_8^{cmp} has weight 0, the canonical branch of Definition B.21 is only a formal logarithmic branch. By itself it gives no finite-order modular multiplier and no chiral square-root theorem.

Proof. The first two assertions are necessary metaplectic-cover conditions: a fractional power of an automorphic product is modular only after the automorphy factor, character, and divisor multiplicities lift to a genuine character on a cover; Weil and Shimura give the classical background for such metaplectic phenomena [80], [72]. The stated character-order obstruction is elementary: if χ has exact order 4 and $\psi^4 = \chi$, then ψ cannot have order 8, since ψ^4 would have order at most 2; absent a further cocycle relation, order 16 is the first possible character order. In the weight-zero case the formal branch supplies no metaplectic weight contribution; modularity would have to come entirely from divisor divisibility and a character or automorphy cocycle root. A formal root can be chosen in a chamber, but this is not a modular or chiral construction. □

Open Problem B.23 *(Fractional comparison-row physical-host datum).* For a fractional comparison row, construct a modular-root datum $\mathscr{R}_j$ and the five-level physical-host datum of Definition 13.1. The root datum must fix the parent form, root exponent, exact group or cover, multiplier character and its order, automorphy cocycle, divisor multiplicities, branch, product normalization, and source theorem. The reduced scalar level, if claimed, must produce a compact Calabi–Yau, orbifold Calabi–Yau, or CHL background with oriented reduced obstruction theory and prove

$$Z_j^{\mathrm{red}} = C_j(\Phi_j^{\mathrm{cmp}})^{-2}$$

in the fixed chamber and cover. The compact Hall/chiral level must construct the protected one-particle index and primitive bracket comparison for the same row.

Borcea–Voisin and higher-dimensional curve-counting constructions [7], [78], [68], [67], [14] are possible sources for such a datum. This paper does not assert that they supply it. Until $\mathscr{R}_j$ and the relevant datum entries are constructed, the symbols $\Phi_5^{\mathrm{cmp}}, \Phi_7^{\mathrm{cmp}}, \Phi_8^{\mathrm{cmp}}$ are formal comparison symbols only: no Cléry–Gritsenko row, finite-order multiplier, denominator identity, CY/DT partition theorem, or chiral power follows.

The following comparison table is restricted to the $N = 1$ Igusa boundary. It contains no row map

$$\Phi_j^{\mathrm{cmp}} \longleftrightarrow F_i$$

and no physical row-map denominator theorem for any congruence row. The separate Govindarajan CHL/WKB denominator theorem remains Theorem B.13.

For $X = K3 \times E$ the protected K3 index input and the virtual K_0-determinant identify Δ_5 in the theta normalization, and the same active half-coefficients are the visible signed root supermultiplicities of the Gritsenko–Nikulin denominator algebra $\mathfrak{g}_{\Delta_5}$; the primitive reduced scalar branch is $-4096\,\Delta_5^{-2}$ by Oberdieck–Pixton. The open recognition target is a compact protected one-particle object and observable algebra realizing these on $K3 \times E$, with orientation character $\epsilon_0 = \nu_{\Delta_5}$ on $W^{(2)}(\Lambda_{II}^{2,1})$ after the finite orientation, Weyl transport, and type-II Pfaffian wall data are supplied. The seven other Cléry–Gritsenko diagonal-divisor rows extend the same separation: each row carries its automorphic product, and the levels $\mathscr{A}_j, \mathscr{D}_j, \mathscr{S}_j, \mathscr{Z}_j, \mathscr{H}_j$ of physical-host datum are filled only where the cited literature supplies them.

C $N = 1$ Boundary-Compatibility Conditions

Let a chambered Hall-descent datum supply the compact comparison data of Definition 10.65, so that the criterion of Theorem 10.66 applies. This appendix uses the same smooth compact-realization standing datum: S is a smooth complex K3 surface and E is a smooth elliptic curve. Singular K3 limits, nodal or cuspidal elliptic limits, imprimitive multiple-cover contributions, the $s = 0$ compact Hall cusp, and multi-component charge closures require separate boundary-degeneration data. Its $N = 1$ decategorified boundary would then be constrained by the Igusa datum

$$\begin{aligned}\Gamma_{\mathrm{eff}} = \{(n,l,m) \in \mathbb{Z}^3 \mid m > 0,\ n \geq 0,\ l \in \mathbb{Z}\}\\ \cup \{(n,l,0) \in \mathbb{Z}^3 \mid n > 0,\ l \in \mathbb{Z}\}\\ \cup \{(0,l,0) \in \mathbb{Z}^3 \mid l < 0\} \subset \mathbb{Z}^3.\end{aligned}$$

comparison Hall descent	Igusa boundary
Gram index lattice	$\Gamma_{\mathrm{ind}} = \mathbb{Z}^3$
sector completion	Γ_{eff}
active denominator support	$\Gamma_{\mathrm{act}} = \{\gamma \in \Gamma_{\mathrm{eff}} \mid f(nm,l) \neq 0\}$
Lorentzian degree	$\alpha(n,l,m) = 2nf_2 - lf_3 + 2mf_{-2}$
orientation character	required geometric monodromy comparison $\epsilon_0\vert_{W^{(2)}} = \nu_{\Delta_5}\vert_{W^{(2)}}$
automorphic boundary	$\mathrm{Borch}_\theta(\phi_{0,1}) = \Delta_5$
BKM denominator	$\mathrm{den}(\mathfrak{g}_{\Delta_5}) = 64^{-1}\Delta_5(2Z)$
denominator algebra	$\mathcal{A}_{\mathrm{den}} = \mathfrak{g}_{\Delta_5}$
formal current envelope	$\mathsf{Den}_{\Delta_5,E} = U_E^{\mathrm{ch}}(\mathfrak{g}_{\Delta_5} \otimes \mathcal{O}_E)$
scalar square	$Z^X_{\mathrm{OP}} = Z^X_{\mathrm{DT}} = -4096\Delta_5^{-2}$ (proved for the smooth primitive nonzero $K3$-class branch).

The degree map satisfies

$$(\alpha(\gamma), \alpha(\eta)) = -2\langle\gamma,\eta\rangle_{\mathrm{ind}}, \qquad (\alpha(n,l,m), \alpha(n,l,m)) = -2(4nm - l^2).$$

The real walls are

$$\delta_1 = 2f_2 - f_3, \qquad \delta_2 = 2f_{-2} - f_3, \qquad \delta_3 = f_3.$$

The Weyl vector is

$$\rho = f_2 - \frac{1}{2}f_3 + f_{-2}, \qquad (\rho, \delta_i) = -1.$$

These equations give the denominator polarisation of the same Gram-index lattice whose cusp polarisation gives the Fock determinant. The cone is generated by the active support. Triples in Γ_{eff} with zero Jacobi coefficient fix ordering and contribute no visible product factors. The product alone does not exclude invisible zero-superdimension root spaces.

The comparison is the $N = 1$ normalization

$$\mathrm{Borch}_\theta(\phi_{0,1}) = \Delta_5, \qquad \mathrm{den}(\mathfrak{g}_{\Delta_5}) = 64^{-1}\Delta_5(2Z), \qquad \kappa_{\mathrm{BKM}}(\mathfrak{g}_{\Delta_5}) = \mathrm{wt}(\Delta_5) = 5.$$

It contains no denominator theorem beyond the $N = 1$ theorem and no row map

$$\Phi_j^{\mathrm{cmp}} \longleftrightarrow F_i.$$

The protected scalar has the form

$$Z_{\square}^{X} = C_{\square}\Delta_5^{-2}.$$

The Oberdieck–Pixton and reduced Hilbert-scheme DT branches have constant -4096 in the quotient-by-E normalization of Theorem 11.2. The chiral/denominator half is Δ_5.

A compact Hall–Drinfeld double would realize a protected bracket from the compactification only when the comparison package of Definition 10.65 and Theorem 10.66 is supplied: finite-first oriented Hall descent, protected integration, charge additive Hall primitives, Borcherds relations inside the Hall super-commutator, and the completed Hopf pairing with orientation character required to restrict to ν_{Δ_5}. Homotopy descent must produce the positive half, its completion data, and the Hall-side domain on which the comparison character is defined. This orientation character is independent of rescaling Δ_5; it comes from monodromy of the chosen square root of the Hall determinant line. The Igusa boundary identifies the denominator algebra $\mathfrak{g}_{\Delta_5}$ and its protected decategorified multiplicities before those comparison data are proved.

For the Cléry–Gritsenko rows F_5, F_6, F_7, F_8, this table contains no nonabelian chamber, no simple real roots, no reflection character, and no Weyl-sum side. Rows F_5, F_6, F_7, F_8 use the separate Govindarajan CHL/WKB denominator theorem when a nonabelian bracket is needed. For F_7 this is a theorem on the μ_4-multiplier cover determined by the Cléry–Gritsenko order-four multiplier. None of these denominator rows supplies a comparison row map or a CY/DT/chiral boundary host.

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